Supplementary material

Learning to Fix Retiming and Reordering Decisions in Railway Timetable Rescheduling

This document accompanies the manuscript and its reproducibility repository. Sections S1 and S4 together define all 267 candidate catalog items: S1 covers the 191 items with at least one estimable, nonconstant numerical form, while S4 records the remaining items and why no association is reported. Section S3 preserves all 6,402 estimable association responses without a significance filter, together with prediction-input selection frequencies. Section S2 gives the complete computation-time accounting. Section S5 gives the prediction architectures and key parameter settings. The archived machine-readable results accompany this document; unavailable or constant indicators are not assigned fabricated correlations. The study uses 75 training and test scenarios and 36 transfer scenarios on one railway network and one planned timetable.

Contents

S1 Computable, nonconstant indicator definitions **2**
S1.1 Retrieved studies used to develop the catalog . . . 6

S2 Computation time by comparison setting **8**

S3 Complete indicator associations and input selection **9**
S3.1 Availability of numerical forms . . . 9
S3.2 Stability of prediction-input selection . . . 9
S3.3 Strategy-2 dwell durations . . . 11
S3.4 Strategy-2 running durations . . . 37
S3.5 Strategy-3 dwell durations . . . 64
S3.6 Strategy-3 running durations . . . 90
S3.7 Strategy-3 train orders . . . 117

S4 Other catalog items and association availability **226**

S5 Prediction architectures and key parameter settings **230**

S1 Computable, nonconstant indicator definitions

The register lists the 191 computable, nonconstant indicator items used in the association analysis in Section 4.2 of the main paper. Items are grouped into the same five categories as the method. An item is retained when at least one numerical form has two distinct finite values and positive scenario-weighted variation; individual forms may be constant in other decision types. Inclusion does not imply statistical significance or selection by every prediction model. Section S4 defines the remaining catalog items and records why no association is available.

Arrival and departure descriptors, train-pair contrasts, and summaries over different time windows provide the numerical forms. Here W denotes a time interval centered on a reference departure, with a total width of 5, 10, or 20 minutes. Blockage start and reopening use τ_b and τ_e, as in the formulation. Summary measures include counts, totals, means, extrema, dispersion, and quantiles. Missing measurements remain distinct from zero.

Definitions reflect the information available in the reference timetable. H17–H19 measure competition within a time window, while P15 uses track count as an overtaking-opportunity proxy. Identifiers are used only to match records and resolve tied prediction scores.

Table S1: Computable, nonconstant indicator catalog

Code	Indicator	Operational definition
Disruption conditions		
D01	Blockage start time	Blockage start τ_b, in minutes after 08:00.
D02	Blockage end	Known reopening time τ_e, in minutes after 08:00.
D03	Blockage duration	τ_e minus τ_b, in minutes.
D04	Start-time wave phase	τ_b modulo the 40-minute departure-wave period.
D05	Reopening wave phase	τ_e modulo the 40-minute departure-wave period.
D08	Remaining blockage	Known remaining blockage duration at the initial decision.
D09	Time from start time	Reference or planned departure time minus τ_b.
D10	Time from reopening	Reference or planned departure time minus τ_e.
D11	Blocked section	Separate indicators for H1–C, U13–H1 and U11–U12; categorical location, not a continuous distance.
D20	Lost resource time	Two blocked directional running resources times blockage duration.
E01	Train state at start time	State indicators derived from the original timetable and committed history.
E02	Executed route fraction	Fraction of station departures already executed by start time.
E03	Current route position	Route index of the emergency stopping position or completed route end.
E04	Committed running remainder	Remaining time of an irrevocable run to the emergency station.
E05	Emergency stopping position	Route index of the first station reached under the emergency stopping rule.
E06	Time to emergency station	Known time from start time to the emergency station.
E07	Emergency arrival time	Known emergency-station arrival time, using τ_b for an origin hold.
E08	Minimum emergency hold	Required hold from emergency arrival or start time until τ_e.
E09	Already stopped at start time	Indicator that the train is at a station when blockage begins.
E12	Origin waiting count	Network-wide number of emergency origin holds.
E13	Station waiting count	Network-wide number of emergency holds away from origins.
E16	Emergency backlog	Number of trains subject to the emergency stopping rule.
E17	Early release concentration	Fraction of emergency trains released before τ_e plus five minutes.
E19	Committed event count	Number of arrival and departure events fixed by operational history.
E20	Planned starts before release	Number of trains planned to start strictly between τ_b and τ_e.
E21	Train-state counts at blockage start	Counts of origin holds, committed-run completions, station holds and completed trains.
E26	Committed movement count	Recorded runs that must finish despite the emergency rule.
I05	Predicted remaining duration	Known remaining blockage duration under the deterministic assumption.
I06	Remaining-duration bounds	Lower and upper predicted remaining durations; both equal the known duration here.
I09	Remaining-duration quantiles	The 0.1, 0.5 and 0.9 quantiles of remaining duration; equal under deterministic recovery.
X01	Blocked-section exposure	Whether the train's fixed route includes the blocked section.
X02	Already past blocked section	Whether the planned exit from the blocked section is no later than τ_b.
X03	Planned clearance before start time	Whether the train was planned to clear the blocked section by τ_b.

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Table S1 continued

Code	Indicator	Operational definition
X04	Planned blockage overlap	Length of the planned blocked-section occupation intersecting $[\tau_b, \tau_e)$.
X05	Planned overlap fraction	Blockage overlap divided by planned blocked-section running duration.
X06	Planned entry from start time	Planned entry to the blocked section minus τ_b.
X07	Planned exit from start time	Planned exit from the blocked section minus τ_b.
X08	Sections to blockage entry	Remaining section count from the emergency position to blockage entry.
X09	Minimum time to entry	Sum of minimum running times from the emergency position to blockage entry.
X10	Signed blockage distance	Blockage-entry route index minus the candidate station index.
X11	At blockage entrance	Whether the candidate station is the blocked-section entrance.
X12	Can clear before start time	Single-train minimum-running estimate of clearing the blocked section before τ_b.
X13	Arrival before reopening	Whether the emergency waiting station can be reached before reopening under the recorded operating state.
X14	Earliest-entry estimate	Single-train entry-time estimate, ignoring other trains.
X15	Earliest entry from release	Earliest-entry estimate minus τ_e.
X16	Blockage-only waiting	Nonnegative τ_e minus earliest-entry estimate for a train not cleared by start time.
X17	Remaining route fraction	Remaining section count divided by total route section count.
X18	Emergency origin hold	Whether the emergency stopping rule holds this train at its origin.
X19	Emergency restriction type	Recorded category: no emergency hold, origin hold, committed-run completion, or station hold.
Timetable characteristics		
F12	Minimum dwell	Station-specific minimum dwell in the operational input.
F13	Minimum running time	Train- and section-specific minimum running duration.
F14	Allowed stopping mode	Required-stop and unfixed optional-stop indicators in the operational input.
M03	Original domain width	Raw operational upper minus lower bound before common propagation.
M04	Propagated domain	Lower bound, upper bound, domain width or order-choice count after common preprocessing.
M05	Reference adjustment room	Reference value minus lower bound, upper bound minus reference, or reference order.
M06	Remaining candidate decisions	Counts of the train's remaining stopping, duration, precedence and track candidates.
T01	Planned arrival time	Original arrival time at the station.
T02	Planned departure time	Original visit departure time.
T03	Planned dwell	Original departure minus original arrival at a station.
T04	Planned running duration	Original duration of the outgoing running activity.
T05	Planned stopping mode	Original planned stopping indicator, without treating required stops as cancellable.
T06	Planned track label	Recorded numeric track-label suffix; an assignment descriptor, not a cardinal capacity.
T09	Planned entry buffer	Gap to the preceding planned entry minus the minimum entry headway.
T10	Planned exit buffer	Gap to the preceding planned exit minus the minimum exit headway.
T11	Planned track buffer	Planned arrival minus the latest preceding release on the same planned track.
T12	Planned local flow	Planned running-resource entries in W, divided by window width.
T13	Planned train-type pattern	Fast-train share and consecutive type-alternation fraction in W.
T14	Departure wave	Integer origin departure time divided by the 40-minute wave period.
T15	Fast-train indicator	Recorded fast/slow class, distinguished by minimum running duration.
T17	Total planned journey time	Planned terminal arrival minus origin departure.
T18	Remaining planned journey	Planned terminal arrival minus the candidate planned departure.
T19	Remaining minimum journey	Remaining minimum runs plus subsequent mandatory minimum dwells.
T20	Remaining scheduled stops	Count of subsequent stops required by the original timetable.
T21	Nearby scheduled-stop distance	Route-index distance to the preceding and following stops in the original timetable.
T22	Route extent	Total and remaining route section counts.
T24	Planned terminal position	Planned terminal arrival and its difference from τ_e.

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Table S1 continued

Code	Indicator	Operational definition
Train interactions		
B17	Relation-layer overlap	Jaccard overlap of train-pair sets in entry, exit and station-track layers.
E11	Co-located emergency trains	Number of same-direction emergency trains waiting at the same station.
E14	Reference release rank	Rank in the emergency-station queue ordered by reference release time.
E15	Queue fast-train share	Fast-train fraction in the emergency-station queue.
H17	Window competition size	Distinct trains occupying the station in W; a window-based proxy.
H18	Co-window occupations	Number of train occupations at the station intersecting W; a competition-membership proxy.
H20	Support distance	Nearest optional-stop route-index distance and finite active distance for an order pair.
H21	Shared-visit coupling	Number of candidate decision classes and order candidates incident on the visit.
H22	Shared downstream support	Jaccard overlap of downstream event sets for endpoints or train pairs.
P02	Planned rank distance	Following-minus-leading rank on the common running resource.
P03	Planned entry gap	Following-minus-leading planned resource-entry time.
P04	Planned exit gap	Following-minus-leading planned resource-exit time.
P05	Running-duration contrast	Following-minus-leading planned running duration on the common resource.
P06	Train-type pair	Fast/fast, fast/slow, slow/fast and slow/slow category indicators.
P07	Remaining-journey contrast	Following-minus-leading remaining planned and minimum journey times.
P08	Terminal-time contrast	Following-minus-leading planned terminal arrival.
P09	Stop-pattern disagreement	Number of shared stations at which planned stopping indicators differ.
P10	Shared route extent	Common section count and the leading train's summed minimum time on those sections.
P11	Shared station pools	Number of stations common to both routes.
P12	Merge and split positions	First and last shared-station indices relative to the candidate visit.
P13	Catch-up proxy	Positive minimum-running-time advantage over common sections minus planned entry gap.
P14	Potential order positions	Number of free precedence candidates involving this train pair.
P15	Parallel common stations	Number of common stations with at least two same-direction tracks.
P16	Reference-delay contrast	Following-minus-leading reference departure delay.
P17	Same reference track	Whether both trains use the same nonempty track label in the reference.
P18	Reference interval overlap	Overlap of the two reference running intervals on the common resource.
X20	Same exposure category	Whether the two precedence trains have the same emergency restriction category.
X21	Pair entry-time contrast	Following-minus-leading earliest-entry time relative to τ_e.
X22	Pair waiting contrast	Following-minus-leading blockage-only waiting estimate.
X23	Pair blockage-rank gap	Planned rank difference of the two trains on their common running resource.
Infrastructure utilization		
B07	Regional saturation	Station saturation fraction in W.
B13	Adjacent-window difference	Following minus preceding window mean occupation.
E10	Emergency pool capacity	Number of same-direction tracks at the assigned emergency station.
E18	Parking concentration	Herfindahl concentration of emergency holds across stations.
E23	Initial station occupation	Same-direction station occupation at τ_b in the recorded reference.
F01	Facility role	Hub and common-corridor indicators for the candidate station.
F02	Physical degree	Degree in the undirected physical station graph.
F03	Branch-count proxy	Physical station degree minus one, bounded below by zero.
F04	Physical betweenness	Unweighted betweenness in the physical station graph.
F05	Physical articulation point	Whether the station is an articulation point of that graph.
F06	Directional pool size	Number of same-direction station tracks.
F07	Compatible tracks	Number of tracks allowed for the visit; zero at external boundaries.
F15	Parallel-capacity proxy	Same-direction track-pool size as a proxy for parallel reception.
F16	Downstream capacity gradient	Next station's directional capacity minus current capacity.
F17	Distance to a hub	Minimum physical section count to H1 or H2.
F18	Next stopping opportunity	Route-section distance to the next station at which stopping is permitted.
H19	Peak demand/capacity	Peak same-direction occupation in W divided by track-pool size.

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Table S1 continued

Code	Indicator	Operational definition
R01	Mean occupation	Time-average number of occupied same-direction station tracks in *W*.
R02	Peak occupation	Maximum same-direction station occupation in *W*.
R03	Minimum free capacity	Directional capacity minus peak occupation in *W*.
R04	Utilization	Mean occupation divided by directional track-pool capacity.
R05	Saturated fraction	Fraction of *W* at which occupation reaches directional capacity.
R06	Saturation persistence	Longest consecutive saturated interval within *W*.
R07	Entry load	Reference entries in *W* times minimum entry headway, divided by width.
R08	Exit load	Reference exits in *W* times minimum exit headway, divided by width.
R09	Reference headway buffer	Reference entry/exit gap to the preceding train minus minimum headway.
R10	Intersecting dwell distribution	Durations of train occupations at the station whose occupations intersect *W*; not passenger waiting.
R11	Flow imbalance	Station arrivals minus departures within *W*.
R12	Demand/capacity proxy	Mean station occupation divided by directional capacity.
R13	Adjacent-station occupation	Upstream/downstream occupation at reference departure and their difference.
R14	Window competitors	Distinct trains with station occupations intersecting *W*.
R15	Free windows	Count, total duration and maximum duration of unsaturated intervals in *W*.
R16	Planned-arrival overlap	Planned station arrivals in *W* that overlap a reference occupation.
R17	Occupation change rate	Difference of following and preceding window means divided by twice the width.
R18	Opposite-direction utilization	Opposite-direction mean occupation normalized by the corresponding capacity here.
Delay propagation patterns		
B01	Local region size	Same-station, same-direction event and distinct-train counts in *W*.
B02	Local edges and density	Internal edge count and directed density in the local region.
B03	Active local density	Active internal edge count divided by possible directed event pairs.
B04	Boundary connections	Numbers of edges entering and leaving the local region.
B05	Regional delay distribution	Reference event-delay summaries for the local region.
B06	Regional delay variance	Variance of reference event delays in the local region.
B08	Community structure	Greedy-modularity community size and links to other communities.
B09	Boundary pressure imbalance	Outgoing minus incoming delay support across the region boundary.
B10	Affected-train share	Fraction of trains with positive reference terminal delay.
B11	Global path distribution	Summaries of longest active path lengths.
B12	Delay concentration	Entropy and Herfindahl indices of delay totals by station and 20-minute time bin.
B15	Global degree distribution	Incoming/outgoing active-degree summaries over graph events.
B16	Global reachability	Reachable ordered-event-pair fraction and mean finite active distance.
G01	Reference arrival delay	Reference arrival minus original planned arrival.
G02	Reference departure delay	Reference departure minus original planned departure.
G03	Activity delay increment	Difference of reference endpoint delays for dwell or running activities.
G04	Activity-duration increment	Reference dwell/running duration minus its planned duration.
G05	Event degree	Incoming and outgoing edge counts in the reference event graph.
G06	Cross-train degree	Incoming and outgoing edges linking different trains.
G07	Degree by relation	Incoming/outgoing counts for dwell, run, entry, exit, track and release relations.
G08	Active degree	Numbers of active incoming and outgoing edges.
G09	Active cross-train links	Count and fraction of active links between different trains.
G10	Edge-slack distribution	Incoming/outgoing edge slack summaries.
G11	Near-tight links	Incident edges with slack at most one or two minutes.
G12	Delay-support distribution	Incoming/outgoing positive delay-support summaries.
G13	Upstream-neighbor delay	Delay summaries of directly upstream different trains, deduplicated by train.
G14	Downstream-neighbor delay	Delay summaries of directly downstream different trains, deduplicated by train.
G15	Active-neighbor dispersion	Range, variance and coefficient of variation of active-neighbor delays.
G16	Incident-link imbalance	Outgoing minus incoming degree and delay-support sums.
G17	Relation-type entropy	Entropy of incident edge-type counts.
G18	External-source share	Fraction of incoming edges from the merged external release source.

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Table S1 continued

Code	Indicator	Operational definition
G19	Endpoint contrasts	Departure/arrival feature differences and recorded running-endpoint contrasts.
H01	Shortest source distance	Shortest active-source edge count and recorded cumulative minimum-lag distance.
H02	Longest active path	Longest active directed path ending at the event.
H03	Event time from release	Reference event time minus reopening time τ_e.
H04	Upstream reachability	Number of active upstream events, excluding the merged source.
H05	Downstream reachability	Number of active downstream events.
H06	Upstream trains	Distinct trains among active upstream events.
H07	Downstream trains	Distinct trains among active downstream events.
H08	Spatial reachability	Distinct stations and branches reachable upstream/downstream.
H09	Reachable delay totals	Event-delay sums and sums of per-train maximum delays upstream/downstream.
H10	Active betweenness	Normalized unweighted betweenness in the active directed graph.
H11	Harmonic closeness	Normalized sums of reciprocal finite active distances, incoming and outgoing.
H12	PageRank	Active-graph PageRank with damping 0.85.
H13	Weak-component size	Event and distinct-train counts in the active weak component.
H15	Chain and crossing length	Longest active chain and maximum number of cross-train edges along a chain.
H16	Local motifs	Fan-in pairs, fan-out pairs and directed diamonds at their source.
H23	Local clustering	Local clustering coefficient in the undirected projection of the active propagation graph.
H24	Projection core number	Core number in the active graph's undirected projection.
H25	Local assortativity	Neighbor same-speed-class fraction and mean absolute delay difference.

Notes: Codes identify catalog items. Only computable items with nonconstant numerical forms in the association analysis of Section 4.2 of the main paper are listed. The symbols τ_b and τ_e denote blockage start and reopening. The interval W is centered on a reference departure, with a total width of 5, 10, or 20 minutes as specified by the numerical form.

S1.1 Retrieved studies used to develop the catalog

The 22 studies below supplied the literature context for developing the candidate indicator catalog. This list records the consulted studies; it does not imply that every candidate indicator was proposed in a cited study or that full texts were available for every study. Study identifiers and the original metadata are also provided in the accompanying retrieved-studies CSV and JSON files.

Table S2: Retrieved studies used to develop the indicator catalog

No.	Study	Title and source
1	Goverde (2007)	Railway timetable stability analysis using max-plus system theory. Source
2	Goverde (2010)	A delay propagation algorithm for large-scale railway traffic networks. Source
3	Büker and Seybold (2012)	Stochastic modelling of delay propagation in large networks. Source
4	Harrod, Cerreto and Nielsen (2019)	A closed form railway line delay propagation model. Source
5	Andersson, Peterson and Törnquist Krasemann (2013; online 2014)	Quantifying railway timetable robustness in critical points. Source
6	Ma et al. (2021)	Scheduling-measure dependent modelling of delay propagation on a single railway line: A matrix transformation approach. Source
7	Meng et al. (2019)	Increasing Robustness by Reallocating the Margins in the Timetable. Source
8	Wu et al. (2023)	Cascading Delays for the High-Speed Rail Network Under Different Emergencies: A Double Layer Network Approach. Source
9	Lian et al. (2024)	Bi-Directional Delay Propagation Analysis and Modeling for High-Speed Railway Networks Under Disturbance. Source
10	Tessitore et al. (2025)	On the fragility of a train timetable. Source
11	Corman et al. (2014)	Evaluating Disturbance Robustness of Railway Schedules. Source
12	Zhang et al. (2024)	Propagation-Based Train Rescheduling under Recoverable Delay Disturbances. Source
13	Huang et al. (2026; online 2025)	Solving the railway timetable rescheduling problem with graph neural networks. Source

No.	Study	Title and source
14	Acuña-Agost et al. (2011)	SAPI: Statistical Analysis of Propagation of Incidents. A new approach for rescheduling trains after disruptions. Source
15	Tessitore et al. (2025)	An iterative fragility-based approach to timetable design. Source
16	Hua et al. (2024)	Improving timetable resilience for high-speed railways through time supplement allocation. Source
17	Corman and Kecman (2018)	Stochastic prediction of train delays in real-time using Bayesian networks. Source
18	Ulak et al. (2020)	Analyzing network-wide patterns of rail transit delays using Bayesian network learning. Source
19	Dekker et al. (2022)	Modelling railway delay propagation as diffusion-like spreading. Source
20	Ge et al. (2024; online 2023)	Modelling the cascading effects of train delay patterns and inter-train control actions with Bayesian networks. Source
21	Daniotti et al. (2024)	Systemic risk approach to mitigate delay cascading in railway networks. Source
22	Xu et al. (2026)	Multivariate event hypergraph diffusion model for train delay prediction. Source

S2 Computation time by comparison setting

Table S3 reports total computation times for the training and test scenarios and the transfer scenarios. Solver time includes the main optimization, auxiliary optimization, and necessary continuations. Prediction-plus-solver time adds model inference. Complete online time additionally includes indicator calculation, model loading, fixing preparation, model construction, and timetable checks, with Strategy-1 generation counted once for each comparison setting. Enclosing process times are not added to the components they already contain. The full-reference and chained settings for Strategy-3 include the costs of obtaining full and restricted Strategy-2 timetables, respectively; FULL Strategy-3 starts from Strategy-1 and does not require Strategy-2. Offline training and label generation are excluded from complete online time and reported separately.

Table S3: Computation time by scenario group and comparison setting

Strategy / setting / method	Solver time (s)	Prediction-plus-solver time (s)	Complete online time (s)	Solver computational speedup ratio	Complete online computational speedup ratio
Training and test scenarios					
Strategy-2 / FULL	34.30	34.30	394.02	1.00	1.00
Strategy-2 / STAT	25.52	27.27	803.62	1.34	0.49
Strategy-2 / ML	26.12	27.43	808.49	1.31	0.49
Strategy-3 / FULL	24642.53	24642.53	25245.97	1.00	1.00
Full-reference / STAT	269.77	272.90	1623.23	91.35	15.55
Full-reference / ML	355.04	357.37	1708.79	69.41	14.77
Chained / STAT	226.96	231.43	2007.86	108.58	12.57
Chained / ML	349.11	352.49	2135.52	70.59	11.82
Transfer scenarios					
Strategy-2 / FULL	19.17	19.17	192.79	1.00	1.00
Strategy-2 / STAT	14.54	15.31	398.79	1.32	0.48
Strategy-2 / ML	14.41	15.03	400.89	1.33	0.48
Strategy-3 / FULL	194.07	194.07	493.12	1.00	1.00
Full-reference / STAT	68.67	69.99	737.24	2.83	0.67
Full-reference / ML	64.66	65.79	732.89	3.00	0.67
Chained / STAT	63.48	65.56	943.78	3.06	0.52
Chained / ML	57.78	59.37	938.85	3.36	0.53

Notes: There are 75 training and test scenarios and 36 transfer scenarios. Full-reference and chained settings concern Strategy-3. Each computational speedup ratio divides the corresponding FULL total by the setting's total, using solver time or complete online time as specified in the column heading. Values above one indicate faster computation. FULL denotes the full rescheduling model; STAT and ML denote statistical prediction and machine learning. All times are summed over the indicated scenario group and reported in seconds.

Offline preparation. The five-fold training runs, including internal model selection and calibration, require 1,061.09 seconds for STAT and 2,249.27 seconds for ML across the two strategy expansions. Fitting the final models to all training and test scenarios requires another 272.43 seconds for STAT and 592.44 seconds for ML; these models are then frozen for transfer. These training times are excluded from complete online time. The full-model solutions for the training and test scenarios also supply the representative labels; their measured computation times appear in Table S3 and are not charged again as a second set of optimizations.

S3 Complete indicator associations and input selection

This section reports every estimable numerical form of the indicators in Section S1: 4,191 associations with a decision change and 2,211 associations with signed duration change. The latter equals the full rescheduling solution minus its reference duration; negative values indicate shortening. All estimates use the 75 training and test scenarios. Strategy-2 uses Strategy-1 as its reference; Strategy-3 uses the full Strategy-2 reference.

The five catalog categories are retained. Within each category and decision type, numerical forms are ordered by the absolute scenario-weighted Spearman correlation with the change indicator, retaining the correlation sign. This is a descriptive ordering of association strength, not a ranking of causal effects or importance inside STAT or ML. The standardized regression effect answers a different question: the change in the response associated with one standard deviation of the indicator. Numerical forms that coincide under this design remain separately identified; their repeated associations are not independent evidence.

In duration tables, the upper line in each numerical cell concerns the probability of a change, and the lower line concerns its signed size. Effects are in percentage points on the upper line and minutes on the lower line. Order tables have one line only. Confidence intervals account for dependence within scenarios and repeated trains, including order pairs sharing a train. Adjusted p-values use the Benjamini–Yekutieli correction separately for each strategy, decision type and response. Scientific e notation is used for small p-values: 1.1e-7 denotes 1.1×10^{-7}. Direction stability reports the percentages of estimates obtained by omitting one start-time group and one blockage location having the full-sample coefficient sign, in that order. Missing estimates are shown as dashes. No significance filter is applied to these tables.

S3.1 Availability of numerical forms

Availability is assessed separately for each strategy and decision type. Of the catalog items, 26 have no calculable form in the available model and data. The register includes only the 191 items with at least one estimable, nonconstant form. A missing form may be inapplicable to that decision type or undefined for all its records; missingness is not replaced by zero. Constant forms have no estimable association.

Some catalog entries are equivalent in this deterministic design. D03, D08, I05, the two I06 bounds and the three I09 quantiles all equal the known blockage duration; D20 is twice that duration. R04 and R12 give the same normalized occupation in each time window. These entries remain identifiable in the complete tables but do not represent separate sources of information.

Table S4: Availability of numerical forms in the training and test scenario sample.

Strategy	Decision	Nonconstant	Constant	Missing
2	Dwell	541	80	2,131
2	Running	561	74	2,117
3	Dwell	544	77	2,131
3	Running	565	70	2,117
3	Order	1,980	231	541

Note: Counts refer to numerical forms within each decision type, not independent indicators. “Missing” means no finite measurement in that class; “nonconstant” forms all appear below.

S3.2 Stability of prediction-input selection

Table S5 links the descriptive analysis to the actual prediction inputs. It reports whether at least one numerical form of a catalog item was selected in each outer training fold and in the final model fitted to all training and test scenarios. STAT and ML share these selections. The 33 listed items were selected at least once; the remaining 158 nonconstant items were not selected in any of these fits. Reference values and operational bounds are retained separately as common controls. Selection frequency measures reproducibility of input selection, not a coefficient magnitude, causal contribution or neural feature attribution.

Table S5: Catalog-item selection across training folds and in the frozen transfer models.

Code	Indicator	Strategy-2		Strategy-3		
		Dwell	Running	Dwell	Running	Order
Disruption conditions						
X04	Planned blockage overlap	0/5; N	0/5; N	0/5; N	0/5; N	5/5; Y
X13	Arrival before reopening	0/5; N	0/5; N	0/5; N	0/5; N	5/5; Y
X05	Planned overlap fraction	0/5; N	0/5; N	0/5; N	0/5; N	5/5; Y
E08	Minimum emergency hold	0/5; N	0/5; N	0/5; N	0/5; N	5/5; Y
X16	Blockage-only waiting	0/5; N	0/5; N	0/5; N	0/5; N	5/5; Y
X18	Emergency origin hold	0/5; N	0/5; N	0/5; N	0/5; N	5/5; Y
X19	Emergency restriction type	0/5; N	0/5; N	0/5; N	0/5; N	5/5; Y
Timetable characteristics						
M05	Reference adjustment room	5/5; Y	0/5; N	5/5; Y	0/5; N	0/5; N
Train interactions						
H22	Shared downstream support	5/5; Y	5/5; Y	0/5; N	0/5; N	0/5; N
E11	Co-located emergency trains	0/5; N	0/5; N	0/5; N	0/5; N	5/5; Y
Infrastructure utilization						
R10	Intersecting dwell distribution	5/5; Y	5/5; Y	5/5; Y	0/5; N	0/5; N
R13	Adjacent-station occupation	5/5; Y	5/5; Y	0/5; N	4/5; Y	0/5; N
R04	Utilization	5/5; Y	2/5; N	0/5; N	0/5; N	0/5; N
R01	Mean occupation	1/5; N	0/5; N	0/5; N	4/5; Y	0/5; N
R08	Exit load	0/5; N	5/5; Y	0/5; N	0/5; N	0/5; N
Delay propagation patterns						
G03	Activity delay increment	5/5; Y	5/5; Y	5/5; Y	5/5; Y	0/5; N
G19	Endpoint contrasts	5/5; Y	5/5; Y	5/5; Y	5/5; Y	0/5; N
G10	Edge-slack distribution	5/5; Y	5/5; Y	5/5; Y	5/5; Y	0/5; N
H25	Local assortativity	0/5; N	5/5; Y	0/5; N	5/5; Y	0/5; N
G15	Active-neighbor dispersion	0/5; N	5/5; Y	0/5; N	5/5; Y	0/5; N
B05	Regional delay distribution	0/5; N	5/5; Y	0/5; N	5/5; Y	0/5; N
B06	Regional delay variance	0/5; N	5/5; Y	0/5; N	5/5; Y	0/5; N
H12	PageRank	0/5; N	5/5; Y	0/5; N	0/5; N	0/5; N
H07	Downstream trains	0/5; N	0/5; N	0/5; N	0/5; N	5/5; Y
H09	Reachable delay totals	0/5; N	0/5; N	0/5; N	0/5; N	4/5; Y
H10	Active betweenness	0/5; N	0/5; N	0/5; N	0/5; N	3/5; N
H11	Harmonic closeness	0/5; N	0/5; N	0/5; N	0/5; N	1/5; N
G09	Active cross-train links	0/5; N	0/5; N	0/5; N	1/5; N	0/5; N
H16	Local motifs	0/5; N	0/5; N	0/5; N	5/5; Y	0/5; N
B10	Affected-train share	5/5; Y	5/5; Y	5/5; Y	5/5; Y	3/5; Y
B15	Global degree distribution	5/5; Y	5/5; Y	5/5; Y	5/5; Y	5/5; Y
B11	Global path distribution	5/5; Y	5/5; Y	5/5; Y	5/5; Y	5/5; Y
B16	Global reachability	0/5; N	0/5; N	0/5; N	5/5; Y	0/5; N

Notes: Each cell gives the number of outer training folds (out of five) in which any form of the item was selected, followed by Y/N for selection in the final frozen model. STAT and ML use the same selected indicators. Categories match Table S1; within a category, items are ordered by their largest absolute change-response correlation across the five decision types.

S3.3 Strategy-2 dwell durations

Table S6: Complete indicator associations: Strategy-2 dwell durations.

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted p	Direction stable (%)
Disruption conditions				
E05 Emergency stopping position	-0.315	-7.27 [-11.49, -3.05]	0.014	100.00 / 100.00
first stop index (2,830 records)	0.301	0.28 [0.10, 0.45]	0.039	100.00 / 100.00
E20 Planned starts before release	0.295	9.37 [6.53, 12.21]	5.0e-6	100.00 / 100.00
planned starts before reopen (9,227 records)	-0.260	-0.86 [-1.41, -0.30]	0.047	100.00 / 100.00
X06 Planned entry from start time	0.252	7.11 [2.53, 11.68]	0.034	100.00 / 100.00
planned entry from start time (5,062 records)	-0.232	-0.35 [-0.57, -0.13]	0.039	100.00 / 100.00
E02 Executed route fraction	-0.251	-7.17 [-9.46, -4.88]	9.6e-6	100.00 / 100.00
executed fraction (9,227 records)	0.225	0.57 [0.29, 0.84]	0.006	100.00 / 100.00
E03 Current route position	-0.251	-7.17 [-9.46, -4.88]	9.6e-6	100.00 / 100.00
current route index (9,227 records)	0.225	0.57 [0.29, 0.84]	0.006	100.00 / 100.00
E01 Train state at start time	0.248	7.86 [4.88, 10.84]	1.2e-4	100.00 / 100.00
state 0 (9,227 records)	-0.220	-0.36 [-0.81, 0.09]	0.986	100.00 / 66.67
E17 Early release concentration	-0.243	-7.18 [-9.95, -4.40]	1.5e-4	100.00 / 100.00
release first5 fraction (9,227 records)	0.211	0.73 [0.27, 1.19]	0.036	100.00 / 100.00
X07 Planned exit from start time	0.243	6.89 [2.43, 11.36]	0.035	100.00 / 100.00
planned exit from start time (5,062 records)	-0.222	-0.34 [-0.55, -0.12]	0.039	100.00 / 100.00
E12 Origin waiting count	0.242	6.26 [3.34, 9.19]	0.002	100.00 / 100.00
origin waiting (9,227 records)	-0.217	-0.26 [-0.66, 0.15]	1.000	100.00 / 100.00
E21 Train-state counts at blockage start	0.242	6.26 [3.34, 9.19]	0.002	100.00 / 100.00
origin hold (9,227 records)	-0.217	-0.26 [-0.66, 0.15]	1.000	100.00 / 100.00
D09 Time from start time	0.229	7.13 [4.20, 10.05]	3.4e-4	100.00 / 100.00
reference from start (9,227 records)	-0.199	-0.79 [-1.16, -0.42]	0.004	100.00 / 100.00
X17 Remaining route fraction	0.227	7.08 [5.30, 8.85]	2.6e-7	100.00 / 100.00
remaining route fraction (9,227 records)	-0.216	-1.64 [-2.32, -0.97]	0.001	100.00 / 100.00
E16 Emergency backlog	0.222	5.59 [3.22, 7.96]	4.9e-4	100.00 / 100.00
backlog (9,227 records)	-0.201	-0.05 [-0.36, 0.26]	1.000	86.67 / 66.67
E07 Emergency arrival time	-0.211	-5.60 [-10.06, -1.13]	0.143	100.00 / 100.00
emergency arrival (2,830 records)	0.203	0.24 [0.04, 0.44]	0.222	100.00 / 100.00
E06 Time to emergency station	-0.204	-4.53 [-7.47, -1.59]	0.036	100.00 / 100.00
time to stop (2,830 records)	0.194	0.18 [0.05, 0.31]	0.104	100.00 / 100.00
D01 Blockage start time	-0.203	-6.44 [-9.39, -3.48]	0.001	100.00 / 100.00
start (9,227 records)	0.178	0.21 [-0.38, 0.80]	1.000	100.00 / 66.67
E19 Committed event count	-0.203	-6.37 [-9.25, -3.49]	0.001	100.00 / 100.00
fixed events (9,227 records)	0.178	0.20 [-0.38, 0.79]	1.000	100.00 / 66.67
E01 Train state at start time	-0.189	-6.00 [-8.57, -3.43]	5.6e-4	100.00 / 100.00
state 2 (9,227 records)	0.166	0.09 [-0.40, 0.59]	1.000	93.33 / 33.33
E04 Committed running remainder	-0.188	-5.10 [-7.25, -2.95]	4.7e-4	100.00 / 100.00
committed remaining (9,227 records)	0.166	0.18 [-0.17, 0.53]	1.000	100.00 / 66.67
D03 Blockage duration	0.188	5.85 [3.47, 8.24]	3.0e-4	100.00 / 100.00
duration (9,227 records)	-0.168	-1.22 [-1.73, -0.72]	0.001	100.00 / 100.00
D08 Remaining blockage	0.188	5.85 [3.47, 8.24]	3.0e-4	100.00 / 100.00
remaining (9,227 records)	-0.168	-1.22 [-1.73, -0.72]	0.001	100.00 / 100.00
D20 Lost resource time	0.188	5.85 [3.47, 8.24]	3.0e-4	100.00 / 100.00
lost resource minutes (9,227 records)	-0.168	-1.22 [-1.73, -0.72]	0.001	100.00 / 100.00
I05 Predicted remaining duration	0.188	5.85 [3.47, 8.24]	3.0e-4	100.00 / 100.00
predicted remaining (9,227 records)	-0.168	-1.22 [-1.73, -0.72]	0.001	100.00 / 100.00
I06 Remaining-duration bounds	0.188	5.85 [3.47, 8.24]	3.0e-4	100.00 / 100.00
lower (9,227 records)	-0.168	-1.22 [-1.73, -0.72]	0.001	100.00 / 100.00

Continued on the next page.

Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
I06 Remaining-duration bounds	0.188	5.85 [3.47, 8.24]	3.0e-4	100.00 / 100.00
upper (9,227 records)	-0.168	-1.22 [-1.73, -0.72]	0.001	100.00 / 100.00
I09 Remaining-duration quantiles	0.188	5.85 [3.47, 8.24]	3.0e-4	100.00 / 100.00
10th percentile (9,227 records)	-0.168	-1.22 [-1.73, -0.72]	0.001	100.00 / 100.00
I09 Remaining-duration quantiles	0.188	5.85 [3.47, 8.24]	3.0e-4	100.00 / 100.00
median (9,227 records)	-0.168	-1.22 [-1.73, -0.72]	0.001	100.00 / 100.00
I09 Remaining-duration quantiles	0.188	5.85 [3.47, 8.24]	3.0e-4	100.00 / 100.00
90th percentile (9,227 records)	-0.168	-1.22 [-1.73, -0.72]	0.001	100.00 / 100.00
E21 Train-state counts at blockage start	-0.187	-5.79 [-8.40, -3.17]	0.001	100.00 / 100.00
completed (9,227 records)	0.168	0.11 [-0.50, 0.72]	1.000	86.67 / 66.67
D09 Time from start time	0.176	4.74 [1.61, 7.87]	0.041	100.00 / 100.00
planned from start (9,227 records)	-0.148	0.42 [0.00, 0.83]	0.456	100.00 / 66.67
E26 Committed movement count	-0.175	-5.46 [-8.58, -2.34]	0.012	100.00 / 100.00
committed runs (9,227 records)	0.154	0.34 [-0.21, 0.89]	1.000	100.00 / 100.00
X18 Emergency origin hold	0.148	4.70 [1.90, 7.50]	0.018	100.00 / 100.00
origin emergency (9,227 records)	-0.141	0.27 [0.02, 0.51]	0.328	100.00 / 100.00
D04 Start-time wave phase	-0.140	-4.63 [-7.42, -1.83]	0.020	100.00 / 100.00
start phase40 (9,227 records)	0.115	0.32 [-0.11, 0.75]	1.000	100.00 / 100.00
X08 Sections to blockage entry	0.118	3.42 [-0.83, 7.67]	0.882	100.00 / 100.00
intervals to entry (5,062 records)	-0.112	-0.21 [-0.43, 0.00]	0.436	100.00 / 100.00
E01 Train state at start time	-0.113	-3.60 [-4.79, -2.40]	1.7e-5	100.00 / 100.00
state 1 (9,227 records)	0.104	0.45 [0.27, 0.63]	0.001	100.00 / 100.00
E09 Already stopped at start time	-0.113	-3.60 [-4.79, -2.40]	1.7e-5	100.00 / 100.00
already stopped (9,227 records)	0.104	0.45 [0.27, 0.63]	0.001	100.00 / 100.00
X02 Already past blocked section	-0.108	-3.20 [-4.80, -1.60]	0.003	100.00 / 100.00
passed at start time (5,062 records)	0.101	0.13 [0.06, 0.21]	0.011	100.00 / 100.00
X03 Planned clearance before start time	-0.108	-3.20 [-4.80, -1.60]	0.003	100.00 / 100.00
planned clear before (5,062 records)	0.101	0.13 [0.06, 0.21]	0.011	100.00 / 100.00
X12 Can clear before start time	-0.108	-3.20 [-4.80, -1.60]	0.003	100.00 / 100.00
can clear without emergency (5,062 records)	0.101	0.13 [0.06, 0.21]	0.011	100.00 / 100.00
X10 Signed blockage distance	0.095	2.75 [-0.60, 6.10]	0.838	100.00 / 100.00
signed distance to blockage (5,062 records)	-0.096	-0.26 [-0.46, -0.06]	0.124	100.00 / 100.00
E21 Train-state counts at blockage start	0.079	2.63 [0.44, 4.82]	0.184	100.00 / 100.00
finish committed run then hold (9,227 records)	-0.078	0.35 [0.05, 0.66]	0.232	100.00 / 66.67
D10 Time from reopening	0.075	3.14 [0.75, 5.53]	0.107	100.00 / 100.00
reference from release (9,227 records)	-0.053	0.19 [-0.10, 0.49]	1.000	100.00 / 66.67
X09 Minimum time to entry	0.062	1.49 [-2.03, 5.01]	1.000	100.00 / 100.00
minrun to entry (5,062 records)	-0.058	-0.13 [-0.29, 0.04]	1.000	100.00 / 100.00
D10 Time from reopening	-0.053	-1.56 [-3.47, 0.34]	0.843	100.00 / 100.00
planned from release (9,227 records)	0.054	1.21 [0.61, 1.81]	0.006	100.00 / 100.00
E13 Station waiting count	0.050	2.35 [0.01, 4.69]	0.410	100.00 / 100.00
station waiting (9,227 records)	-0.052	0.36 [0.06, 0.67]	0.211	100.00 / 66.67
X01 Blocked-section exposure	-0.049	-1.54 [-3.97, 0.88]	1.000	100.00 / 100.00
crosses closed (9,227 records)	0.043	1.21 [0.75, 1.67]	0.001	100.00 / 100.00
X19 Emergency restriction type	-0.048	-2.91 [-4.33, -1.48]	0.003	100.00 / 100.00
direct stop type (9,227 records)	0.042	0.77 [0.44, 1.10]	0.002	100.00 / 100.00
D11 Blocked section	0.048	1.52 [-0.96, 4.01]	1.000	100.00 / 100.00
U13 H1 (9,227 records)	-0.045	-0.67 [-1.23, -0.12]	0.187	100.00 / 100.00
X15 Earliest entry from release	0.043	1.24 [-0.81, 3.30]	1.000	100.00 / 100.00
earliest entry minus release (5,062 records)	-0.036	-0.08 [-0.18, 0.03]	1.000	100.00 / 100.00
X04 Planned blockage overlap	-0.041	-1.48 [-3.28, 0.31]	0.820	100.00 / 100.00
planned blockage overlap (9,227 records)	0.035	0.77 [0.43, 1.11]	0.002	100.00 / 100.00

Continued on the next page.

Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
X14 Earliest-entry estimate	0.032	1.17 [-2.31, 4.65]	1.000	100.00 / 100.00
earliest entry no other conflicts (5,062 records)	-0.024	-0.08 [-0.23, 0.07]	1.000	100.00 / 100.00
D02 Blockage end	0.030	0.30 [-1.96, 2.56]	1.000	80.00 / 66.67
end (9,227 records)	-0.028	-0.80 [-1.41, -0.19]	0.128	100.00 / 100.00
X11 At blockage entrance	0.028	0.88 [-1.23, 2.99]	1.000	100.00 / 66.67
at closed entry (9,227 records)	-0.029	0.30 [0.15, 0.46]	0.008	100.00 / 100.00
X16 Blockage-only waiting	-0.026	-1.69 [-3.41, 0.03]	0.446	100.00 / 100.00
blockage only wait (9,227 records)	0.021	0.64 [0.37, 0.92]	0.002	100.00 / 100.00
D11 Blocked section	-0.025	-0.78 [-3.15, 1.58]	1.000	93.33 / 100.00
H1 C (9,227 records)	0.018	0.78 [0.39, 1.17]	0.006	100.00 / 100.00
D11 Blocked section	-0.023	-0.74 [-3.30, 1.81]	1.000	93.33 / 50.00
U11 U12 (9,227 records)	0.027	-0.10 [-0.57, 0.36]	1.000	100.00 / 50.00
X05 Planned overlap fraction	-0.014	-0.46 [-2.66, 1.75]	1.000	86.67 / 66.67
planned overlap fraction (9,227 records)	0.010	0.75 [0.40, 1.09]	0.004	100.00 / 100.00
X13 Arrival before reopening	-0.013	-0.42 [-2.62, 1.78]	1.000	86.67 / 66.67
reaches stop before reopen (9,227 records)	0.009	0.75 [0.40, 1.10]	0.004	100.00 / 100.00
E08 Minimum emergency hold	0.012	1.09 [-1.46, 3.64]	1.000	100.00 / 100.00
minimum emergency hold (9,227 records)	-0.015	0.60 [0.28, 0.93]	0.012	100.00 / 100.00
D05 Reopening wave phase	-0.012	-0.42 [-2.53, 1.69]	1.000	100.00 / 33.33
end phase40 (9,227 records)	0.005	0.03 [-0.36, 0.42]	1.000	66.67 / 33.33
E21 Train-state counts at blockage start	0.007	0.18 [-2.42, 2.78]	1.000	86.67 / 33.33
hold current station (9,227 records)	-0.011	0.24 [-0.15, 0.62]	1.000	100.00 / 100.00
Timetable characteristics				
M05 Reference adjustment room	0.810	11.70 [8.96, 14.44]	1.1e-7	100.00 / 100.00
compressible (9,227 records)	-0.784	-7.85 [-9.51, -6.19]	4.5e-8	100.00 / 100.00
M06 Remaining candidate decisions	0.303	9.72 [6.94, 12.50]	2.2e-6	100.00 / 100.00
remaining order decisions (9,227 records)	-0.275	-1.32 [-1.97, -0.67]	0.006	100.00 / 100.00
M06 Remaining candidate decisions	0.251	8.24 [6.10, 10.38]	4.1e-7	100.00 / 100.00
remaining stopping decisions (9,227 records)	-0.252	-0.72 [-1.21, -0.22]	0.067	100.00 / 100.00
T21 Nearby scheduled-stop distance	0.241	7.66 [5.33, 9.98]	5.1e-6	100.00 / 100.00
next scheduled-stop distance (9,227 records)	-0.240	-0.28 [-0.64, 0.07]	0.972	100.00 / 100.00
T21 Nearby scheduled-stop distance	0.241	7.66 [5.33, 9.98]	5.1e-6	100.00 / 100.00
previous scheduled-stop distance (9,227 records)	-0.240	-0.28 [-0.64, 0.07]	0.972	100.00 / 100.00
T10 Planned exit buffer	-0.238	-7.12 [-8.97, -5.27]	5.2e-7	100.00 / 100.00
previous buffer (8,952 records)	0.203	0.53 [0.03, 1.03]	0.359	100.00 / 100.00
T15 Fast-train indicator	0.235	7.47 [5.14, 9.80]	7.2e-6	100.00 / 100.00
fast (9,227 records)	-0.235	-0.26 [-0.62, 0.09]	1.000	100.00 / 100.00
F13 Minimum running time	-0.235	-7.47 [-9.80, -5.14]	7.2e-6	100.00 / 100.00
minimum run (9,227 records)	0.235	0.26 [-0.09, 0.62]	1.000	100.00 / 100.00
T04 Planned running duration	-0.235	-7.47 [-9.80, -5.14]	7.2e-6	100.00 / 100.00
planned run next (9,227 records)	0.235	0.26 [-0.09, 0.62]	1.000	100.00 / 100.00
T17 Total planned journey time	-0.235	-7.47 [-9.80, -5.14]	7.2e-6	100.00 / 100.00
planned total (9,227 records)	0.235	0.26 [-0.09, 0.62]	1.000	100.00 / 100.00
M06 Remaining candidate decisions	0.232	7.24 [5.41, 9.07]	2.8e-7	100.00 / 100.00
remaining track-assignment decisions (9,227 records)	-0.221	-1.66 [-2.35, -0.98]	0.001	100.00 / 100.00
T12 Planned local flow	-0.229	-6.36 [-8.59, -4.13]	3.7e-5	100.00 / 100.00
planned flow 5 min (9,227 records)	0.210	0.99 [0.59, 1.39]	0.001	100.00 / 100.00

Continued on the next page.

Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
M06 Remaining candidate decisions	0.227	7.08 [5.30, 8.85]	2.6e-7	100.00 / 100.00
remaining duration decisions (9,227 records)	-0.216	-1.64 [-2.32, -0.97]	0.001	100.00 / 100.00
T22 Route extent	0.227	7.08 [5.30, 8.85]	2.6e-7	100.00 / 100.00
remaining edges (9,227 records)	-0.216	-1.64 [-2.32, -0.97]	0.001	100.00 / 100.00
T12 Planned local flow	-0.193	-4.98 [-7.12, -2.83]	6.1e-4	100.00 / 100.00
planned flow 10 min (9,227 records)	0.175	0.90 [0.49, 1.30]	0.003	100.00 / 100.00
T09 Planned entry buffer	-0.192	-6.27 [-7.91, -4.64]	5.6e-7	100.00 / 100.00
previous buffer (8,952 records)	0.156	0.49 [-0.07, 1.05]	0.746	100.00 / 100.00
T12 Planned local flow	-0.156	-3.62 [-6.26, -0.99]	0.081	100.00 / 100.00
planned flow 20 min (9,227 records)	0.139	0.85 [0.38, 1.31]	0.013	100.00 / 100.00
T06 Planned track label	-0.154	-3.63 [-6.29, -0.97]	0.085	100.00 / 100.00
planned track index (9,227 records)	0.157	0.41 [0.17, 0.65]	0.018	100.00 / 100.00
T18 Remaining planned journey	0.138	4.19 [2.18, 6.21]	0.002	100.00 / 100.00
remaining planned (9,227 records)	-0.124	-1.41 [-2.08, -0.74]	0.004	100.00 / 100.00
T19 Remaining minimum journey	0.138	4.19 [2.18, 6.21]	0.002	100.00 / 100.00
remaining minimum (9,227 records)	-0.124	-1.41 [-2.08, -0.74]	0.004	100.00 / 100.00
T14 Departure wave	0.107	3.49 [0.66, 6.31]	0.155	100.00 / 100.00
origin wave40 (9,227 records)	-0.099	-0.12 [-0.62, 0.38]	1.000	93.33 / 100.00
M05 Reference adjustment room	-0.098	-1.99 [-4.52, 0.53]	0.932	100.00 / 100.00
extendable (9,227 records)	0.103	1.45 [0.31, 2.59]	0.150	100.00 / 100.00
T20 Remaining scheduled stops	0.097	3.12 [0.72, 5.52]	0.116	100.00 / 100.00
remaining scheduled stops (9,227 records)	-0.085	-1.29 [-1.95, -0.62]	0.008	100.00 / 100.00
T13 Planned train-type pattern	-0.070	-2.04 [-4.05, -0.04]	0.391	100.00 / 100.00
alternation 20 min (1,885 records)	0.059	-0.08 [-0.74, 0.58]	1.000	86.67 / 66.67
M04 Propagated domain	-0.069	-1.87 [-2.53, -1.21]	3.9e-5	100.00 / 100.00
domain lower (9,227 records)	0.063	0.26 [0.16, 0.36]	0.001	100.00 / 100.00
T01 Planned arrival time	-0.061	-2.00 [-4.63, 0.62]	1.000	100.00 / 100.00
planned arrival (9,227 records)	0.066	0.64 [0.13, 1.14]	0.152	100.00 / 100.00
T02 Planned departure time	-0.061	-2.00 [-4.62, 0.63]	1.000	100.00 / 100.00
planned departure (9,227 records)	0.066	0.63 [0.13, 1.14]	0.152	100.00 / 100.00
T11 Planned track buffer	-0.056	-1.65 [-3.12, -0.18]	0.258	100.00 / 100.00
planned track buffer (8,007 records)	0.023	-0.43 [-0.73, -0.13]	0.073	100.00 / 100.00
T13 Planned train-type pattern	-0.036	-0.61 [-1.73, 0.51]	1.000	100.00 / 100.00
fast fraction 5 min (3,846 records)	0.038	0.05 [-0.07, 0.17]	1.000	100.00 / 100.00
F12 Minimum dwell	0.025	0.81 [0.54, 1.07]	1.7e-5	100.00 / 100.00
minimum dwell (9,227 records)	-0.023	-0.10 [-0.15, -0.06]	0.001	100.00 / 100.00
F14 Allowed stopping mode	0.025	0.81 [0.54, 1.07]	1.7e-5	100.00 / 100.00
required (9,227 records)	-0.023	-0.10 [-0.15, -0.06]	0.001	100.00 / 100.00
T03 Planned dwell	0.025	0.81 [0.54, 1.07]	1.7e-5	100.00 / 100.00
planned dwell (9,227 records)	-0.023	-0.10 [-0.15, -0.06]	0.001	100.00 / 100.00
T05 Planned stopping mode	0.025	0.81 [0.54, 1.07]	1.7e-5	100.00 / 100.00
planned stop (9,227 records)	-0.023	-0.10 [-0.15, -0.06]	0.001	100.00 / 100.00
T13 Planned train-type pattern	-0.023	-0.54 [-1.94, 0.86]	1.000	100.00 / 66.67
fast fraction 20 min (5,314 records)	0.019	0.02 [-0.18, 0.23]	1.000	80.00 / 33.33
T24 Planned terminal position	0.023	-0.09 [-2.41, 2.22]	1.000	66.67 / 33.33
planned terminal (9,227 records)	-0.016	-0.01 [-0.53, 0.52]	1.000	46.67 / 66.67
T24 Planned terminal position	-0.016	-0.36 [-2.26, 1.55]	1.000	100.00 / 66.67
terminal from release (9,227 records)	0.021	0.80 [0.36, 1.24]	0.014	100.00 / 100.00
T13 Planned train-type pattern	-0.016	-0.31 [-1.79, 1.16]	1.000	100.00 / 66.67
fast fraction 10 min (4,464 records)	0.016	-0.04 [-0.29, 0.21]	1.000	93.33 / 66.67
M04 Propagated domain	-0.012	0.35 [-1.77, 2.47]	1.000	86.67 / 100.00
domain upper (9,227 records)	0.008	-0.44 [-0.78, -0.09]	0.149	100.00 / 100.00
T13 Planned train-type pattern	-0.011	-0.28 [-0.42, -0.13]	0.004	100.00 / 100.00
alternation 10 min (1,128 records)	0.010	0.02 [0.00, 0.03]	0.577	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
M04 Propagated domain	0.007	0.94 [-1.16, 3.03]	1.000	93.33 / 100.00
domain width (9,227 records)	-0.009	-0.51 [-0.86, -0.17]	0.056	100.00 / 100.00
Train interactions				
H22 Shared downstream support	-0.526	-20.98 [-26.04, -15.93]	1.5e-7	100.00 / 100.00
arrival departure downstream Jaccard (5,879 records)	0.514	2.67 [1.50, 3.84]	0.002	100.00 / 100.00
H17 Window competition size	0.264	10.01 [6.78, 13.24]	1.1e-5	100.00 / 100.00
competitor trains 20 min (9,227 records)	-0.215	-0.15 [-0.45, 0.15]	1.000	100.00 / 100.00
H18 Co-window occupations	0.264	10.01 [6.78, 13.24]	1.1e-5	100.00 / 100.00
competition set memberships 20 min (9,227 records)	-0.215	-0.15 [-0.45, 0.15]	1.000	100.00 / 100.00
H17 Window competition size	0.205	6.91 [4.64, 9.17]	1.4e-5	100.00 / 100.00
competitor trains 10 min (9,227 records)	-0.159	-0.25 [-0.56, 0.06]	0.911	100.00 / 100.00
H18 Co-window occupations	0.205	6.91 [4.64, 9.17]	1.4e-5	100.00 / 100.00
competition set memberships 10 min (9,227 records)	-0.159	-0.25 [-0.56, 0.06]	0.911	100.00 / 100.00
H21 Shared-visit coupling	0.197	6.75 [4.40, 9.10]	3.3e-5	100.00 / 100.00
same visit classes (9,227 records)	-0.145	0.33 [-0.05, 0.72]	0.794	100.00 / 100.00
E14 Reference release rank	0.173	5.27 [2.07, 8.48]	0.021	100.00 / 100.00
release queue rank (2,830 records)	-0.158	-0.21 [-0.36, -0.05]	0.133	100.00 / 100.00
E15 Queue fast-train share	0.169	4.03 [1.13, 6.93]	0.076	100.00 / 100.00
queue fast fraction (2,830 records)	-0.164	-0.16 [-0.27, -0.04]	0.104	100.00 / 100.00
H17 Window competition size	0.134	4.47 [3.05, 5.88]	8.5e-6	100.00 / 100.00
competitor trains 5 min (9,227 records)	-0.101	-0.40 [-0.79, -0.01]	0.411	100.00 / 100.00
H18 Co-window occupations	0.134	4.47 [3.05, 5.88]	8.5e-6	100.00 / 100.00
competition set memberships 5 min (9,227 records)	-0.101	-0.40 [-0.79, -0.01]	0.411	100.00 / 100.00
B17 Relation-layer overlap	0.087	3.25 [0.40, 6.10]	0.238	100.00 / 100.00
entry headway station track Jaccard (9,227 records)	-0.067	0.05 [-0.48, 0.58]	1.000	80.00 / 66.67
B17 Relation-layer overlap	0.087	3.25 [0.40, 6.10]	0.238	100.00 / 100.00
exit headway station track Jaccard (9,227 records)	-0.067	0.05 [-0.48, 0.58]	1.000	80.00 / 66.67
H20 Support distance	-0.074	-3.07 [-3.91, -2.24]	1.7e-5	100.00 / 100.00
nearest optional stop index (2,432 records)	0.073	0.25 [0.15, 0.35]	0.003	100.00 / 100.00
E11 Co-located emergency trains	0.004	1.11 [-1.78, 4.00]	1.000	100.00 / 100.00
same stop count (9,227 records)	-0.007	0.59 [0.25, 0.92]	0.019	100.00 / 100.00
Infrastructure utilization				
R10 Intersecting dwell distribution	0.601	3.44 [0.79, 6.09]	0.116	100.00 / 100.00
waiting duration 5 min minimum (9,227 records)	-0.593	-2.85 [-4.86, -0.85]	0.078	100.00 / 100.00
R10 Intersecting dwell distribution	0.584	5.04 [2.23, 7.86]	0.010	100.00 / 100.00
waiting duration 5 min median (9,227 records)	-0.564	-3.93 [-6.12, -1.73]	0.015	100.00 / 100.00
R10 Intersecting dwell distribution	0.560	5.10 [2.27, 7.92]	0.009	100.00 / 100.00
waiting duration 10 min median (9,227 records)	-0.543	-3.97 [-6.19, -1.76]	0.015	100.00 / 100.00
R10 Intersecting dwell distribution	0.554	5.11 [2.26, 7.96]	0.010	100.00 / 100.00
waiting duration 5 min mean (9,227 records)	-0.530	-3.86 [-6.03, -1.69]	0.015	100.00 / 100.00
R10 Intersecting dwell distribution	0.542	5.75 [3.20, 8.30]	8.6e-4	100.00 / 100.00
waiting duration 5 min 90th percentile (9,227 records)	-0.514	-4.10 [-6.04, -2.15]	0.004	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R10 Intersecting dwell distribution	0.531	5.85 [3.39, 8.31]	4.6e-4	100.00 / 100.00
waiting duration 5 min maximum (9,227 records)	-0.504	-4.06 [-5.92, -2.20]	0.003	100.00 / 100.00
R10 Intersecting dwell distribution	0.520	5.15 [2.33, 7.97]	0.008	100.00 / 100.00
waiting duration 10 min mean (9,227 records)	-0.501	-3.86 [-5.97, -1.74]	0.013	100.00 / 100.00
R10 Intersecting dwell distribution	0.516	5.88 [3.37, 8.40]	5.5e-4	100.00 / 100.00
waiting duration 10 min 90th percentile (9,227 records)	-0.488	-4.08 [-5.96, -2.20]	0.003	100.00 / 100.00
R10 Intersecting dwell distribution	0.511	6.00 [3.57, 8.44]	3.0e-4	100.00 / 100.00
waiting duration 10 min maximum (9,227 records)	-0.482	-3.98 [-5.76, -2.21]	0.003	100.00 / 100.00
R13 Adjacent-station occupation	0.504	16.70 [14.64, 18.75]	8.1e-15	100.00 / 100.00
downstream occupancy (9,227 records)	-0.466	-1.92 [-2.52, -1.33]	1.1e-4	100.00 / 100.00
R10 Intersecting dwell distribution	0.473	6.28 [3.61, 8.95]	5.2e-4	100.00 / 100.00
waiting duration 20 min maximum (9,227 records)	-0.444	-3.74 [-5.42, -2.07]	0.003	100.00 / 100.00
R10 Intersecting dwell distribution	0.471	6.11 [3.39, 8.83]	8.8e-4	100.00 / 100.00
waiting duration 20 min 90th percentile (9,227 records)	-0.444	-3.96 [-5.77, -2.16]	0.003	100.00 / 100.00
R04 Utilization	0.450	15.80 [11.84, 19.76]	2.6e-7	100.00 / 100.00
utilization 20 min (9,227 records)	-0.425	-2.21 [-2.95, -1.46]	3.0e-4	100.00 / 100.00
R12 Demand/capacity proxy	0.450	15.80 [11.84, 19.76]	2.6e-7	100.00 / 100.00
demand capacity 20 min (9,227 records)	-0.425	-2.21 [-2.95, -1.46]	3.0e-4	100.00 / 100.00
R10 Intersecting dwell distribution	0.447	5.23 [2.12, 8.35]	0.018	100.00 / 100.00
waiting duration 20 min mean (9,227 records)	-0.432	-3.72 [-5.88, -1.57]	0.019	100.00 / 100.00
R10 Intersecting dwell distribution	0.433	4.65 [2.75, 6.55]	3.2e-4	100.00 / 100.00
waiting duration 10 min interquartile range (9,227 records)	-0.393	-2.96 [-4.31, -1.60]	0.003	100.00 / 100.00
R10 Intersecting dwell distribution	0.432	6.35 [3.42, 9.28]	0.001	100.00 / 100.00
waiting duration 20 min sum (9,227 records)	-0.402	-3.03 [-4.65, -1.41]	0.011	100.00 / 100.00
R01 Mean occupation	0.429	13.14 [9.00, 17.28]	8.2e-6	100.00 / 100.00
mean occupancy 20 min (9,227 records)	-0.394	-1.78 [-2.51, -1.05]	0.001	100.00 / 100.00
R10 Intersecting dwell distribution	0.428	5.15 [2.48, 7.81]	0.005	100.00 / 100.00
waiting duration 10 min sum (9,227 records)	-0.400	-3.12 [-4.88, -1.36]	0.016	100.00 / 100.00
R10 Intersecting dwell distribution	0.423	5.03 [2.74, 7.32]	0.001	100.00 / 100.00
waiting duration 20 min interquartile range (9,227 records)	-0.388	-3.14 [-4.64, -1.64]	0.005	100.00 / 100.00
R10 Intersecting dwell distribution	0.418	4.76 [2.27, 7.25]	0.005	100.00 / 100.00
waiting duration 5 min sum (9,227 records)	-0.394	-3.27 [-5.09, -1.45]	0.015	100.00 / 100.00
R10 Intersecting dwell distribution	0.409	5.14 [3.10, 7.19]	2.4e-4	100.00 / 100.00
waiting duration 10 min standard deviation (9,227 records)	-0.368	-3.07 [-4.54, -1.60]	0.005	100.00 / 100.00
R08 Exit load	0.403	14.28 [11.98, 16.58]	1.8e-11	100.00 / 100.00
exit load 5 min (9,227 records)	-0.337	-0.06 [-0.32, 0.19]	1.000	100.00 / 66.67
R10 Intersecting dwell distribution	0.392	5.51 [3.01, 8.01]	0.001	100.00 / 100.00
waiting duration 20 min standard deviation (9,227 records)	-0.356	-3.07 [-4.55, -1.59]	0.005	100.00 / 100.00
R04 Utilization	0.389	13.60 [9.95, 17.24]	6.8e-7	100.00 / 100.00
utilization 10 min (9,227 records)	-0.368	-1.80 [-2.45, -1.14]	7.4e-4	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R12 Demand/capacity proxy	0.389	13.60 [9.95, 17.24]	6.8e-7	100.00 / 100.00
demand capacity 10 min (9,227 records)	-0.368	-1.80 [-2.45, -1.14]	7.4e-4	100.00 / 100.00
R10 Intersecting dwell distribution	0.376	2.94 [0.60, 5.27]	0.140	100.00 / 100.00
waiting duration 10 min minimum (9,227 records)	-0.404	-2.53 [-4.21, -0.84]	0.054	100.00 / 100.00
R10 Intersecting dwell distribution	0.374	5.01 [1.94, 8.08]	0.022	100.00 / 100.00
waiting duration 20 min median (9,227 records)	-0.370	-3.79 [-6.07, -1.50]	0.027	100.00 / 100.00
R10 Intersecting dwell distribution	0.372	4.30 [2.65, 5.95]	1.4e-4	100.00 / 100.00
waiting duration 5 min interquartile range (9,227 records)	-0.333	-2.71 [-3.99, -1.44]	0.004	100.00 / 100.00
R08 Exit load	0.370	13.64 [11.01, 16.26]	1.3e-9	100.00 / 100.00
exit load 10 min (9,227 records)	-0.312	-0.09 [-0.30, 0.12]	1.000	100.00 / 66.67
R11 Flow imbalance	-0.370	-12.53 [-15.67, -9.38]	2.6e-7	100.00 / 100.00
arrivals minus departures 5 min (9,227 records)	0.346	1.58 [0.80, 2.37]	0.006	100.00 / 100.00
R10 Intersecting dwell distribution	0.366	4.91 [3.11, 6.70]	7.2e-5	100.00 / 100.00
waiting duration 5 min standard deviation (9,227 records)	-0.328	-2.93 [-4.37, -1.49]	0.006	100.00 / 100.00
R01 Mean occupation	0.349	11.02 [7.61, 14.43]	6.8e-6	100.00 / 100.00
mean occupancy 10 min (9,227 records)	-0.313	-1.42 [-2.05, -0.78]	0.003	100.00 / 100.00
R06 Saturation persistence	0.342	12.68 [9.40, 15.95]	3.7e-7	100.00 / 100.00
longest saturation 20 min (9,227 records)	-0.333	-2.18 [-3.07, -1.29]	0.001	100.00 / 100.00
B13 Adjacent-window difference	-0.334	-10.69 [-13.60, -7.78]	8.5e-7	100.00 / 100.00
adjacent occupancy difference 5 min (9,227 records)	0.312	1.51 [0.64, 2.38]	0.018	100.00 / 100.00
R17 Occupation change rate	-0.334	-10.69 [-13.60, -7.78]	8.5e-7	100.00 / 100.00
occupancy rate 5 min (9,227 records)	0.312	1.51 [0.64, 2.38]	0.018	100.00 / 100.00
R11 Flow imbalance	-0.333	-10.35 [-13.47, -7.23]	4.7e-6	100.00 / 100.00
arrivals minus departures 10 min (9,227 records)	0.309	1.41 [0.54, 2.28]	0.031	100.00 / 100.00
B07 Regional saturation	0.325	12.66 [9.41, 15.92]	3.6e-7	100.00 / 100.00
saturated fraction 20 min (9,227 records)	-0.317	-1.79 [-2.55, -1.03]	0.002	100.00 / 100.00
R05 Saturated fraction	0.325	12.66 [9.41, 15.92]	3.6e-7	100.00 / 100.00
saturated fraction 20 min (9,227 records)	-0.317	-1.79 [-2.55, -1.03]	0.002	100.00 / 100.00
R15 Free windows	-0.325	-12.66 [-15.92, -9.41]	3.6e-7	100.00 / 100.00
free window total 20 min (9,227 records)	0.317	1.79 [1.03, 2.55]	0.002	100.00 / 100.00
R07 Entry load	0.315	12.54 [9.57, 15.51]	1.2e-7	100.00 / 100.00
entry load 20 min (9,227 records)	-0.261	-0.11 [-0.33, 0.11]	1.000	100.00 / 100.00
R04 Utilization	0.307	10.14 [6.92, 13.37]	8.9e-6	100.00 / 100.00
utilization 5 min (9,227 records)	-0.298	-1.34 [-1.98, -0.70]	0.004	100.00 / 100.00
R12 Demand/capacity proxy	0.307	10.14 [6.92, 13.37]	8.9e-6	100.00 / 100.00
demand capacity 5 min (9,227 records)	-0.298	-1.34 [-1.98, -0.70]	0.004	100.00 / 100.00
R13 Adjacent-station occupation	0.296	10.32 [7.60, 13.03]	5.2e-7	100.00 / 100.00
occupancy difference (9,227 records)	-0.278	-1.57 [-2.07, -1.07]	1.7e-4	100.00 / 100.00
H19 Peak demand/capacity	0.295	9.30 [6.50, 12.09]	4.4e-6	100.00 / 100.00
peak demand capacity 20 min (9,227 records)	-0.286	-0.76 [-1.13, -0.38]	0.006	100.00 / 100.00
R09 Reference headway buffer	-0.290	-4.87 [-6.90, -2.84]	4.6e-4	100.00 / 100.00
1 previous buffer (8,952 records)	0.243	-0.84 [-1.54, -0.14]	0.198	100.00 / 100.00
R01 Mean occupation	0.288	8.29 [5.81, 10.78]	4.3e-6	100.00 / 100.00
mean occupancy 5 min (9,227 records)	-0.257	-1.05 [-1.62, -0.49]	0.011	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R06 Saturation persistence	0.281	13.14 [9.92, 16.37]	2.0e-7	100.00 / 100.00
longest saturation 10 min (9,227 records)	-0.275	-1.90 [-2.68, -1.11]	0.001	100.00 / 100.00
R15 Free windows	-0.280	-8.95 [-12.10, -5.80]	4.0e-5	100.00 / 100.00
free window maximum 20 min (9,227 records)	0.272	0.75 [0.21, 1.28]	0.086	100.00 / 100.00
R08 Exit load	0.279	11.09 [8.82, 13.36]	5.5e-9	100.00 / 100.00
exit load 20 min (9,227 records)	-0.222	0.15 [-0.09, 0.40]	1.000	100.00 / 100.00
R09 Reference headway buffer	-0.272	-4.53 [-6.40, -2.66]	4.1e-4	100.00 / 100.00
0 previous buffer (8,952 records)	0.223	-0.87 [-1.58, -0.16]	0.181	100.00 / 100.00
R03 Minimum free capacity	-0.268	-8.13 [-11.10, -5.17]	6.9e-5	100.00 / 100.00
minimum free 20 min (9,227 records)	0.275	0.76 [0.42, 1.10]	0.003	100.00 / 100.00
F18 Next stopping opportunity	0.267	8.84 [6.43, 11.26]	8.9e-7	100.00 / 100.00
next stop distance (7,132 records)	-0.264	-0.56 [-1.06, -0.05]	0.302	100.00 / 100.00
R10 Intersecting dwell distribution	0.264	10.01 [6.78, 13.24]	1.1e-5	100.00 / 100.00
waiting duration 20 min count (9,227 records)	-0.215	-0.15 [-0.45, 0.15]	1.000	100.00 / 100.00
R14 Window competitors	0.264	10.01 [6.78, 13.24]	1.1e-5	100.00 / 100.00
window competitors 20 min (9,227 records)	-0.215	-0.15 [-0.45, 0.15]	1.000	100.00 / 100.00
B07 Regional saturation	0.258	11.11 [8.03, 14.19]	1.2e-6	100.00 / 100.00
saturated fraction 10 min (9,227 records)	-0.253	-1.55 [-2.27, -0.82]	0.004	100.00 / 100.00
R05 Saturated fraction	0.258	11.11 [8.03, 14.19]	1.2e-6	100.00 / 100.00
saturated fraction 10 min (9,227 records)	-0.253	-1.55 [-2.27, -0.82]	0.004	100.00 / 100.00
R15 Free windows	-0.258	-11.11 [-14.19, -8.03]	1.2e-6	100.00 / 100.00
free window total 10 min (9,227 records)	0.253	1.55 [0.82, 2.27]	0.004	100.00 / 100.00
R10 Intersecting dwell distribution	0.254	2.50 [0.07, 4.94]	0.377	100.00 / 100.00
waiting duration 20 min minimum (9,227 records)	-0.298	-2.19 [-3.91, -0.47]	0.148	100.00 / 100.00
R07 Entry load	0.245	8.35 [6.20, 10.51]	3.7e-7	100.00 / 100.00
entry load 10 min (9,227 records)	-0.191	-0.09 [-0.39, 0.21]	1.000	93.33 / 100.00
B13 Adjacent-window difference	-0.244	-9.47 [-12.28, -6.65]	4.0e-6	100.00 / 100.00
adjacent occupancy difference 20 min (9,227 records)	0.220	1.63 [0.71, 2.56]	0.017	100.00 / 100.00
R17 Occupation change rate	-0.244	-9.47 [-12.28, -6.65]	4.0e-6	100.00 / 100.00
occupancy rate 20 min (9,227 records)	0.220	1.63 [0.71, 2.56]	0.017	100.00 / 100.00
H19 Peak demand/capacity	0.242	7.51 [5.08, 9.93]	1.1e-5	100.00 / 100.00
peak demand capacity 10 min (9,227 records)	-0.239	-0.67 [-1.03, -0.30]	0.013	100.00 / 100.00
R16 Planned-arrival overlap	-0.237	-6.56 [-9.03, -4.09]	1.1e-4	100.00 / 100.00
planned arrival release overlap 5 min (9,227 records)	0.216	0.96 [0.51, 1.41]	0.004	100.00 / 100.00
R02 Peak occupation	0.231	8.63 [6.36, 10.91]	5.2e-7	100.00 / 100.00
peak occupancy 20 min (9,227 records)	-0.198	-0.73 [-1.20, -0.25]	0.049	100.00 / 100.00
R15 Free windows	-0.219	-6.68 [-9.55, -3.82]	5.7e-4	100.00 / 100.00
free window maximum 10 min (9,227 records)	0.215	0.73 [0.17, 1.30]	0.131	100.00 / 100.00
B13 Adjacent-window difference	-0.214	-8.36 [-11.10, -5.61]	1.4e-5	100.00 / 100.00
adjacent occupancy difference 10 min (9,227 records)	0.198	1.30 [0.47, 2.14]	0.043	100.00 / 100.00
R17 Occupation change rate	-0.214	-8.36 [-11.10, -5.61]	1.4e-5	100.00 / 100.00
occupancy rate 10 min (9,227 records)	0.198	1.30 [0.47, 2.14]	0.043	100.00 / 100.00
R03 Minimum free capacity	-0.212	-6.51 [-9.27, -3.76]	4.9e-4	100.00 / 100.00
minimum free 10 min (9,227 records)	0.227	0.69 [0.35, 1.02]	0.005	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R06 Saturation persistence	0.209	11.03 [8.33, 13.74]	2.0e-7	100.00 / 100.00
longest saturation 5 min (9,227 records)	-0.207	-1.51 [-2.25, -0.77]	0.006	100.00 / 100.00
R10 Intersecting dwell distribution	0.205	6.91 [4.64, 9.17]	1.4e-5	100.00 / 100.00
waiting duration 10 min count (9,227 records)	-0.159	-0.25 [-0.56, 0.06]	0.911	100.00 / 100.00
R14 Window competitors	0.205	6.91 [4.64, 9.17]	1.4e-5	100.00 / 100.00
window competitors 10 min (9,227 records)	-0.159	-0.25 [-0.56, 0.06]	0.911	100.00 / 100.00
R07 Entry load	0.203	6.42 [4.33, 8.51]	1.2e-5	100.00 / 100.00
entry load 5 min (9,227 records)	-0.158	-0.38 [-0.76, 0.01]	0.506	100.00 / 100.00
R02 Peak occupation	0.191	7.23 [5.26, 9.19]	8.5e-7	100.00 / 100.00
peak occupancy 10 min (9,227 records)	-0.160	-0.67 [-1.16, -0.19]	0.092	100.00 / 100.00
R16 Planned-arrival overlap	-0.191	-5.25 [-7.37, -3.12]	3.0e-4	100.00 / 100.00
planned arrival release overlap 10 min (9,227 records)	0.173	0.90 [0.49, 1.31]	0.003	100.00 / 100.00
B07 Regional saturation	0.187	9.14 [6.29, 12.00]	7.2e-6	100.00 / 100.00
saturated fraction 5 min (9,227 records)	-0.186	-1.27 [-1.96, -0.57]	0.014	100.00 / 100.00
R05 Saturated fraction	0.187	9.14 [6.29, 12.00]	7.2e-6	100.00 / 100.00
saturated fraction 5 min (9,227 records)	-0.186	-1.27 [-1.96, -0.57]	0.014	100.00 / 100.00
R15 Free windows	-0.187	-9.14 [-12.00, -6.29]	7.2e-6	100.00 / 100.00
free window total 5 min (9,227 records)	0.186	1.27 [0.57, 1.96]	0.014	100.00 / 100.00
R11 Flow imbalance	-0.181	-6.21 [-9.66, -2.76]	0.009	100.00 / 100.00
arrivals minus departures 20 min (9,227 records)	0.174	1.35 [0.52, 2.18]	0.031	100.00 / 100.00
R15 Free windows	-0.173	-5.30 [-6.99, -3.61]	9.3e-6	100.00 / 100.00
free window count 5 min (9,227 records)	0.157	0.93 [0.55, 1.31]	0.001	100.00 / 100.00
H19 Peak demand/capacity	0.166	5.11 [2.92, 7.30]	5.6e-4	100.00 / 100.00
peak demand capacity 5 min (9,227 records)	-0.164	-0.53 [-0.95, -0.12]	0.139	100.00 / 100.00
E10 Emergency pool capacity	-0.160	-4.38 [-6.82, -1.95]	0.010	100.00 / 100.00
emergency pool capacity (2,830 records)	0.153	0.17 [0.06, 0.27]	0.041	100.00 / 100.00
R15 Free windows	0.158	6.49 [2.76, 10.22]	0.013	100.00 / 100.00
free window count 20 min (9,227 records)	-0.149	0.45 [0.16, 0.75]	0.047	100.00 / 100.00
R15 Free windows	-0.157	-5.55 [-8.34, -2.77]	0.003	100.00 / 100.00
free window maximum 5 min (9,227 records)	0.158	0.75 [0.17, 1.34]	0.143	100.00 / 100.00
R16 Planned-arrival overlap	-0.151	-3.65 [-6.19, -1.12]	0.059	100.00 / 100.00
planned arrival release overlap 20 min (9,227 records)	0.134	0.78 [0.36, 1.21]	0.013	100.00 / 100.00
R13 Adjacent-station occupation	0.141	3.28 [0.21, 6.35]	0.321	100.00 / 100.00
upstream occupancy (9,227 records)	-0.124	0.17 [-0.07, 0.41]	1.000	100.00 / 100.00
R03 Minimum free capacity	-0.137	-4.24 [-7.02, -1.45]	0.039	100.00 / 100.00
minimum free 5 min (9,227 records)	0.155	0.52 [0.08, 0.96]	0.222	100.00 / 100.00
R02 Peak occupation	0.136	5.25 [3.65, 6.85]	5.4e-6	100.00 / 100.00
peak occupancy 5 min (9,227 records)	-0.106	-0.54 [-0.99, -0.08]	0.222	100.00 / 100.00
R10 Intersecting dwell distribution	0.134	4.47 [3.05, 5.88]	8.5e-6	100.00 / 100.00
waiting duration 5 min count (9,227 records)	-0.101	-0.40 [-0.79, -0.01]	0.411	100.00 / 100.00
R14 Window competitors	0.134	4.47 [3.05, 5.88]	8.5e-6	100.00 / 100.00
window competitors 5 min (9,227 records)	-0.101	-0.40 [-0.79, -0.01]	0.411	100.00 / 100.00
R18 Opposite-direction utilization	0.132	3.69 [1.58, 5.80]	0.012	100.00 / 100.00
opposite utilization 5 min (9,227 records)	-0.100	0.40 [0.15, 0.65]	0.035	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R18 Opposite-direction utilization	0.129	4.01 [1.65, 6.37]	0.016	100.00 / 100.00
opposite utilization 10 min (9,227 records)	-0.099	0.44 [0.20, 0.68]	0.013	100.00 / 100.00
R18 Opposite-direction utilization	0.114	4.47 [1.69, 7.25]	0.025	100.00 / 100.00
opposite utilization 20 min (9,227 records)	-0.082	0.52 [0.25, 0.79]	0.008	100.00 / 100.00
F16 Downstream capacity gradient	0.100	2.87 [0.04, 5.70]	0.397	100.00 / 100.00
next capacity delta (9,227 records)	-0.131	-0.58 [-1.02, -0.14]	0.122	100.00 / 100.00
F17 Distance to a hub	-0.098	-3.15 [-4.14, -2.15]	8.3e-6	100.00 / 100.00
hub distance (9,227 records)	0.062	0.22 [-0.08, 0.51]	1.000	93.33 / 66.67
E18 Parking concentration	-0.088	-2.12 [-5.20, 0.97]	1.000	100.00 / 100.00
parking concentration (9,227 records)	0.086	0.22 [-0.20, 0.64]	1.000	100.00 / 66.67
F04 Physical betweenness	0.085	1.75 [0.81, 2.69]	0.007	100.00 / 100.00
betweenness (9,227 records)	-0.046	0.28 [-0.05, 0.62]	0.854	100.00 / 100.00
R10 Intersecting dwell distribution	-0.074	-2.51 [-3.89, -1.13]	0.009	100.00 / 100.00
waiting duration 20 min zero fraction (9,227 records)	0.109	0.33 [0.10, 0.57]	0.086	100.00 / 100.00
F01 Facility role	0.037	1.18 [0.30, 2.07]	0.096	100.00 / 100.00
trunk (9,227 records)	0.001	0.46 [0.13, 0.80]	0.091	100.00 / 100.00
R10 Intersecting dwell distribution	-0.037	-1.41 [-2.42, -0.40]	0.074	100.00 / 100.00
waiting duration 10 min zero fraction (9,227 records)	0.074	0.26 [0.13, 0.40]	0.006	100.00 / 100.00
F01 Facility role	0.021	0.66 [-1.42, 2.74]	1.000	100.00 / 100.00
hub (9,227 records)	0.020	0.00 [-0.38, 0.39]	1.000	46.67 / 33.33
F02 Physical degree	0.021	0.66 [-1.42, 2.74]	1.000	100.00 / 100.00
undirected degree (9,227 records)	0.020	0.00 [-0.38, 0.39]	1.000	46.67 / 33.33
F03 Branch-count proxy	0.021	0.66 [-1.42, 2.74]	1.000	100.00 / 100.00
branch count (9,227 records)	0.020	0.00 [-0.38, 0.39]	1.000	46.67 / 33.33
F06 Directional pool size	0.021	0.66 [-1.42, 2.74]	1.000	100.00 / 100.00
direction pool (9,227 records)	0.020	0.00 [-0.38, 0.39]	1.000	46.67 / 33.33
F07 Compatible tracks	0.021	0.66 [-1.42, 2.74]	1.000	100.00 / 100.00
compatible tracks (9,227 records)	0.020	0.00 [-0.38, 0.39]	1.000	46.67 / 33.33
F15 Parallel-capacity proxy	0.021	0.66 [-1.42, 2.74]	1.000	100.00 / 100.00
parallel capacity (9,227 records)	0.020	0.00 [-0.38, 0.39]	1.000	46.67 / 33.33
E23 Initial station occupation	-0.018	-0.57 [-1.71, 0.57]	1.000	100.00 / 100.00
initial station occupied (9,227 records)	0.012	0.41 [0.17, 0.65]	0.019	100.00 / 100.00
R15 Free windows	0.017	-0.60 [-2.60, 1.39]	1.000	100.00 / 100.00
free window count 10 min (9,227 records)	-0.019	0.67 [0.37, 0.98]	0.003	100.00 / 100.00
R10 Intersecting dwell distribution	-0.012	-0.36 [-1.41, 0.69]	1.000	100.00 / 100.00
waiting duration 5 min zero fraction (9,227 records)	0.033	0.10 [0.00, 0.21]	0.425	100.00 / 100.00
Delay propagation patterns				
G03 Activity delay increment	0.697	5.83 [2.97, 8.69]	0.003	100.00 / 100.00
dwell delay increment (9,227 records)	-0.676	-4.70 [-6.97, -2.44]	0.005	100.00 / 100.00
G04 Activity-duration increment	0.697	5.83 [2.97, 8.69]	0.003	100.00 / 100.00
dwell increment (9,227 records)	-0.676	-4.70 [-6.97, -2.44]	0.005	100.00 / 100.00
G10 Edge-slack distribution	0.656	5.18 [2.29, 8.07]	0.010	100.00 / 100.00
outgoing slack minimum arrival (9,227 records)	-0.637	-4.20 [-6.48, -1.93]	0.012	100.00 / 100.00
G10 Edge-slack distribution	0.641	9.88 [8.10, 11.65]	2.9e-10	100.00 / 100.00
incoming slack minimum departure (9,227 records)	-0.654	-6.80 [-7.60, -6.00]	4.2e-15	100.00 / 100.00
G19 Endpoint contrasts	0.605	19.29 [14.92, 23.65]	5.6e-8	100.00 / 100.00
G08 active outgoing degree endpoint contrast (9,227 records)	-0.564	-2.59 [-3.51, -1.68]	6.0e-4	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution	-0.531	-18.34 [-21.83, -14.84]	1.2e-9	100.00 / 100.00
outgoing slack zero fraction arrival (9,227 records)	0.511	2.58 [1.67, 3.49]	6.0e-4	100.00 / 100.00
G19 Endpoint contrasts	0.506	12.41 [8.66, 16.15]	4.8e-6	100.00 / 100.00
H05 downstream reachability endpoint contrast (9,227 records)	-0.483	-1.52 [-2.31, -0.72]	0.009	100.00 / 100.00
H25 Local assortativity	0.429	13.67 [10.32, 17.03]	2.0e-7	100.00 / 100.00
local delay absolute difference arrival (5,187 records)	-0.430	-0.81 [-1.03, -0.59]	1.5e-5	100.00 / 100.00
G10 Edge-slack distribution	-0.416	-14.40 [-18.30, -10.50]	8.1e-7	100.00 / 100.00
incoming slack zero fraction departure (9,227 records)	0.434	2.28 [1.53, 3.02]	2.2e-4	100.00 / 100.00
G15 Active-neighbor dispersion	0.408	10.70 [7.95, 13.46]	3.7e-7	100.00 / 100.00
delay variance arrival (5,187 records)	-0.409	-0.78 [-1.01, -0.55]	3.7e-5	100.00 / 100.00
G15 Active-neighbor dispersion	0.401	11.69 [7.25, 16.14]	1.2e-4	100.00 / 100.00
delay range arrival (5,187 records)	-0.402	-0.68 [-0.94, -0.42]	0.001	100.00 / 100.00
B05 Regional delay distribution	0.400	12.73 [8.34, 17.12]	2.9e-5	100.00 / 100.00
region delay 20 min standard deviation (9,227 records)	-0.366	-1.29 [-2.11, -0.46]	0.041	100.00 / 100.00
B06 Regional delay variance	0.400	11.19 [6.95, 15.42]	1.2e-4	100.00 / 100.00
delay variance 20 min (9,227 records)	-0.366	-1.57 [-2.56, -0.57]	0.039	100.00 / 100.00
H12 PageRank	-0.397	-7.69 [-11.39, -3.99]	0.002	100.00 / 100.00
pagerank departure (9,227 records)	0.394	0.93 [0.51, 1.35]	0.003	100.00 / 100.00
G19 Endpoint contrasts	-0.395	-14.29 [-18.76, -9.81]	7.4e-6	100.00 / 100.00
G08 active incoming degree endpoint contrast (9,227 records)	0.386	-0.22 [-0.91, 0.47]	1.000	93.33 / 66.67
B05 Regional delay distribution	0.393	12.56 [7.96, 17.16]	7.3e-5	100.00 / 100.00
region delay 20 min interquartile range (9,227 records)	-0.351	-0.93 [-1.63, -0.23]	0.119	100.00 / 100.00
G15 Active-neighbor dispersion	0.382	8.27 [4.47, 12.07]	0.001	100.00 / 100.00
delay coefficient of variation arrival (5,187 records)	-0.384	-0.47 [-0.70, -0.24]	0.006	100.00 / 100.00
B05 Regional delay distribution	0.379	11.06 [6.75, 15.37]	1.7e-4	100.00 / 100.00
region delay 10 min standard deviation (9,227 records)	-0.338	-1.59 [-2.54, -0.64]	0.025	100.00 / 100.00
B06 Regional delay variance	0.379	8.66 [5.30, 12.02]	1.6e-4	100.00 / 100.00
delay variance 10 min (9,227 records)	-0.338	-2.00 [-3.22, -0.78]	0.029	100.00 / 100.00
B05 Regional delay distribution	0.366	8.08 [3.91, 12.24]	0.005	100.00 / 100.00
region delay 10 min interquartile range (9,227 records)	-0.327	-1.19 [-2.04, -0.35]	0.082	100.00 / 100.00
G09 Active cross-train links	0.360	11.97 [9.21, 14.74]	8.3e-8	100.00 / 100.00
active cross fraction departure (9,227 records)	-0.305	-1.25 [-1.77, -0.74]	0.001	100.00 / 100.00
G09 Active cross-train links	0.352	11.59 [8.84, 14.33]	1.2e-7	100.00 / 100.00
active cross departure (9,227 records)	-0.299	-1.14 [-1.60, -0.68]	0.001	100.00 / 100.00
H13 Weak-component size	0.352	11.07 [7.90, 14.24]	2.2e-6	100.00 / 100.00
component trains departure (9,227 records)	-0.318	-0.93 [-1.37, -0.50]	0.004	100.00 / 100.00
H13 Weak-component size	0.349	10.93 [7.76, 14.11]	2.9e-6	100.00 / 100.00
component events departure (9,227 records)	-0.315	-0.89 [-1.34, -0.44]	0.007	100.00 / 100.00
H25 Local assortativity	0.342	9.60 [6.96, 12.24]	1.0e-6	100.00 / 100.00
local delay absolute difference departure (5,879 records)	-0.323	-2.63 [-3.90, -1.36]	0.005	100.00 / 100.00
G08 Active degree	0.341	11.37 [8.01, 14.73]	3.7e-6	100.00 / 100.00
outgoing active departure (9,227 records)	-0.310	-1.39 [-1.95, -0.83]	0.001	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G15 Active-neighbor dispersion	0.339	6.38 [4.11, 8.65]	4.9e-5	100.00 / 100.00
delay variance departure (5,879 records)	-0.319	-2.20 [-3.33, -1.07]	0.008	100.00 / 100.00
G15 Active-neighbor dispersion	0.338	7.86 [5.14, 10.57]	3.0e-5	100.00 / 100.00
delay coefficient of variation departure (5,879 records)	-0.318	-1.05 [-1.77, -0.34]	0.062	100.00 / 100.00
G15 Active-neighbor dispersion	0.338	8.91 [6.25, 11.56]	4.0e-6	100.00 / 100.00
delay range departure (5,879 records)	-0.316	-2.32 [-3.46, -1.19]	0.006	100.00 / 100.00
G19 Endpoint contrasts	-0.333	-15.88 [-17.83, -13.93]	8.1e-15	100.00 / 100.00
H04 upstream reachability endpoint contrast (9,227 records)	0.342	0.19 [-0.09, 0.47]	1.000	100.00 / 66.67
H25 Local assortativity	-0.329	-11.90 [-16.93, -6.88]	4.7e-4	100.00 / 100.00
local same type fraction arrival (5,187 records)	0.335	0.44 [0.21, 0.67]	0.010	100.00 / 100.00
B05 Regional delay distribution	0.324	6.79 [3.52, 10.06]	0.002	100.00 / 100.00
region delay 5 min interquartile range (9,227 records)	-0.299	-1.64 [-2.68, -0.60]	0.039	100.00 / 100.00
B10 Affected-train share	0.318	9.94 [7.01, 12.88]	3.7e-6	100.00 / 100.00
affected fraction (9,227 records)	-0.287	-0.77 [-1.25, -0.28]	0.038	100.00 / 100.00
B08 Community structure	0.318	9.77 [7.72, 11.83]	1.0e-8	100.00 / 100.00
community events departure (9,227 records)	-0.281	-1.11 [-1.59, -0.64]	0.002	100.00 / 100.00
B15 Global degree distribution	0.313	9.97 [7.03, 12.90]	3.6e-6	100.00 / 100.00
outgoing degree mean (9,227 records)	-0.282	-0.75 [-1.24, -0.26]	0.047	100.00 / 100.00
B15 Global degree distribution	0.313	9.97 [7.03, 12.90]	3.6e-6	100.00 / 100.00
outgoing degree sum (9,227 records)	-0.282	-0.75 [-1.24, -0.26]	0.047	100.00 / 100.00
B15 Global degree distribution	0.313	9.95 [7.02, 12.88]	3.6e-6	100.00 / 100.00
incoming degree mean (9,227 records)	-0.281	-0.74 [-1.23, -0.25]	0.050	100.00 / 100.00
B15 Global degree distribution	0.313	9.95 [7.02, 12.88]	3.6e-6	100.00 / 100.00
incoming degree sum (9,227 records)	-0.281	-0.74 [-1.23, -0.25]	0.050	100.00 / 100.00
B15 Global degree distribution	-0.309	-9.85 [-12.78, -6.92]	4.0e-6	100.00 / 100.00
outgoing degree zero fraction (9,227 records)	0.278	0.75 [0.27, 1.23]	0.041	100.00 / 100.00
H08 Spatial reachability	0.309	8.36 [5.24, 11.49]	1.0e-4	100.00 / 100.00
downstream branches departure (9,227 records)	-0.273	-1.09 [-1.69, -0.49]	0.014	100.00 / 100.00
B11 Global path distribution	-0.307	-9.82 [-12.75, -6.89]	4.0e-6	100.00 / 100.00
root longest paths zero fraction (9,227 records)	0.277	0.74 [0.26, 1.21]	0.047	100.00 / 100.00
B15 Global degree distribution	-0.307	-9.82 [-12.75, -6.89]	4.0e-6	100.00 / 100.00
incoming degree zero fraction (9,227 records)	0.277	0.74 [0.26, 1.21]	0.047	100.00 / 100.00
B15 Global degree distribution	0.306	8.65 [5.87, 11.44]	1.0e-5	100.00 / 100.00
outgoing degree standard deviation (9,227 records)	-0.277	-0.75 [-1.17, -0.33]	0.015	100.00 / 100.00
B05 Regional delay distribution	0.306	7.48 [4.11, 10.84]	0.001	100.00 / 100.00
region delay 5 min standard deviation (9,227 records)	-0.277	-1.73 [-2.76, -0.69]	0.026	100.00 / 100.00
B06 Regional delay variance	0.306	6.34 [3.91, 8.77]	1.4e-4	100.00 / 100.00
delay variance 5 min (9,227 records)	-0.277	-1.95 [-3.15, -0.76]	0.030	100.00 / 100.00
H16 Local motifs	0.305	9.77 [4.68, 14.87]	0.005	100.00 / 100.00
fan-in pairs arrival (9,227 records)	-0.287	0.18 [-0.16, 0.52]	1.000	100.00 / 66.67
G13 Upstream-neighbor delay	-0.303	-9.76 [-13.02, -6.49]	2.1e-5	100.00 / 100.00
neighbor delay zero fraction departure (8,952 records)	0.267	0.26 [-0.01, 0.54]	0.538	100.00 / 100.00
B16 Global reachability	0.301	9.66 [6.78, 12.54]	4.0e-6	100.00 / 100.00
reachable pair fraction (9,227 records)	-0.269	-0.78 [-1.29, -0.28]	0.043	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B11 Global path distribution root longest paths mean (9,227 records)	0.300 -0.269	9.48 [6.56, 12.39] -0.77 [-1.27, -0.27]	6.0e-6 0.043	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths sum (9,227 records)	0.300 -0.269	9.48 [6.56, 12.39] -0.77 [-1.27, -0.27]	6.0e-6 0.043	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths interquartile range (9,227 records)	0.295 -0.262	9.30 [6.38, 12.23] -0.77 [-1.30, -0.24]	7.9e-6 0.066	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution incoming degree standard deviation (9,227 records)	0.294 -0.265	8.35 [5.63, 11.08] -0.75 [-1.16, -0.34]	1.3e-5 0.013	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration time20 concentration (9,227 records)	-0.293 0.260	-7.28 [-10.56, -4.00] 0.79 [0.43, 1.14]	0.001 0.003	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration time20 entropy (9,227 records)	0.292 -0.260	8.01 [5.07, 10.95] -0.88 [-1.22, -0.53]	7.5e-5 0.001	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution outgoing degree 90th percentile (9,227 records)	0.291 -0.264	9.06 [6.23, 11.89] -0.90 [-1.25, -0.55]	7.2e-6 0.001	100.00 / 100.00 100.00 / 100.00
H16 Local motifs fan-out pairs departure (9,227 records)	0.288 -0.253	9.65 [6.71, 12.59] -1.04 [-1.64, -0.45]	5.3e-6 0.017	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum departure (9,227 records)	0.288 -0.251	7.86 [4.56, 11.17] -0.47 [-0.89, -0.05]	4.6e-4 0.294	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum departure (8,952 records)	0.287 -0.250	7.96 [4.62, 11.29] -0.48 [-0.89, -0.06]	4.7e-4 0.246	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay mean departure (8,952 records)	0.287 -0.250	7.96 [4.62, 11.29] -0.48 [-0.89, -0.06]	4.7e-4 0.246	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay median departure (8,952 records)	0.287 -0.250	7.96 [4.62, 11.29] -0.48 [-0.89, -0.06]	4.7e-4 0.246	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum departure (8,952 records)	0.287 -0.250	7.96 [4.62, 11.29] -0.48 [-0.89, -0.06]	4.7e-4 0.246	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay 90th percentile departure (8,952 records)	0.287 -0.250	7.96 [4.62, 11.29] -0.48 [-0.89, -0.06]	4.7e-4 0.246	100.00 / 100.00 100.00 / 100.00
H25 Local assortativity local same type fraction departure (5,879 records)	-0.285 0.270	-10.79 [-13.39, -8.19] 2.21 [1.03, 3.39]	1.5e-7 0.011	100.00 / 100.00 100.00 / 100.00
H24 Projection core number core departure (9,227 records)	0.284 -0.245	8.95 [6.23, 11.66] -0.80 [-1.24, -0.35]	5.0e-6 0.015	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution incoming degree 90th percentile (9,227 records)	0.283 -0.258	8.76 [6.04, 11.48] -0.83 [-1.18, -0.49]	7.2e-6 0.001	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths median (9,227 records)	0.283 -0.255	8.61 [5.56, 11.66] -0.60 [-1.12, -0.07]	4.4e-5 0.257	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum arrival (9,227 records)	0.275 -0.229	8.30 [4.83, 11.77] 0.52 [0.10, 0.95]	4.2e-4 0.172	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution incoming degree median (9,227 records)	0.274 -0.248	8.70 [5.72, 11.68] -0.63 [-1.14, -0.13]	2.7e-5 0.159	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay interquartile range arrival (8,947 records)	0.271 -0.238	4.00 [1.67, 6.34] 0.13 [-0.10, 0.35]	0.015 1.000	100.00 / 100.00 100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G13 Upstream-neighbor delay	0.271	4.00 [1.67, 6.34]	0.015	100.00 / 100.00
neighbor delay standard deviation arrival (8,947 records)	-0.238	0.13 [-0.10, 0.35]	1.000	100.00 / 100.00
B05 Regional delay distribution	0.270	7.83 [4.61, 11.06]	3.5e-4	100.00 / 100.00
region delay 20 min sum (9,227 records)	-0.226	-0.23 [-0.47, 0.02]	0.611	100.00 / 100.00
B05 Regional delay distribution	-0.270	-8.69 [-11.40, -5.98]	7.2e-6	100.00 / 100.00
region delay 5 min zero fraction (9,227 records)	0.248	1.14 [0.70, 1.58]	0.001	100.00 / 100.00
B15 Global degree distribution	0.264	8.38 [5.50, 11.25]	2.7e-5	100.00 / 100.00
outgoing degree median (9,227 records)	-0.238	-0.64 [-1.14, -0.14]	0.142	100.00 / 100.00
B11 Global path distribution	0.264	8.10 [5.47, 10.72]	1.2e-5	100.00 / 100.00
root longest paths 90th percentile (9,227 records)	-0.237	-0.79 [-1.19, -0.38]	0.008	100.00 / 100.00
B01 Local region size	0.263	10.00 [6.77, 13.23]	1.1e-5	100.00 / 100.00
region trains 20 min (9,227 records)	-0.213	-0.10 [-0.40, 0.20]	1.000	93.33 / 100.00
H08 Spatial reachability	-0.260	-6.87 [-9.40, -4.34]	8.1e-5	100.00 / 100.00
downstream stations arrival (9,227 records)	0.262	0.99 [0.54, 1.44]	0.003	100.00 / 100.00
H11 Harmonic closeness	-0.260	-7.16 [-9.72, -4.59]	5.2e-5	100.00 / 100.00
outgoing harmonic arrival (9,227 records)	0.260	1.03 [0.57, 1.49]	0.003	100.00 / 100.00
H05 Downstream reachability	-0.259	-5.77 [-8.07, -3.47]	2.4e-4	100.00 / 100.00
downstream arrival (9,227 records)	0.261	0.82 [0.42, 1.22]	0.005	100.00 / 100.00
G10 Edge-slack distribution	0.259	2.99 [0.25, 5.72]	0.293	100.00 / 100.00
outgoing slack mean arrival (9,227 records)	-0.270	-3.68 [-5.58, -1.77]	0.008	100.00 / 100.00
G10 Edge-slack distribution	0.259	2.99 [0.25, 5.72]	0.293	100.00 / 100.00
outgoing slack median arrival (9,227 records)	-0.270	-3.68 [-5.58, -1.77]	0.008	100.00 / 100.00
G16 Incident-link imbalance	-0.258	-5.92 [-9.15, -2.69]	0.008	100.00 / 100.00
pressure balance arrival (9,227 records)	0.213	-0.27 [-0.53, -0.01]	0.375	100.00 / 100.00
H10 Active betweenness	-0.258	-6.14 [-8.44, -3.83]	1.1e-4	100.00 / 100.00
betweenness arrival (9,227 records)	0.256	0.86 [0.47, 1.26]	0.003	100.00 / 100.00
G12 Delay-support distribution	-0.257	-8.27 [-10.91, -5.62]	1.0e-5	100.00 / 100.00
outgoing support zero fraction departure (9,227 records)	0.235	1.17 [0.70, 1.65]	0.001	100.00 / 100.00
H09 Reachable delay totals	-0.256	-4.81 [-6.75, -2.88]	2.7e-4	100.00 / 100.00
downstream event delay arrival (9,227 records)	0.257	0.69 [0.37, 1.00]	0.004	100.00 / 100.00
G10 Edge-slack distribution	-0.254	-5.09 [-7.05, -3.13]	1.5e-4	100.00 / 100.00
incoming slack median arrival (9,227 records)	0.226	0.32 [0.20, 0.44]	0.001	100.00 / 100.00
G13 Upstream-neighbor delay	0.254	7.55 [4.12, 10.97]	0.001	100.00 / 100.00
neighbor delay maximum arrival (8,947 records)	-0.211	0.57 [0.12, 1.02]	0.153	100.00 / 100.00
B05 Regional delay distribution	-0.254	-8.38 [-11.05, -5.71]	9.3e-6	100.00 / 100.00
region delay 10 min zero fraction (9,227 records)	0.233	1.12 [0.67, 1.56]	0.001	100.00 / 100.00
G12 Delay-support distribution	-0.253	-8.01 [-11.05, -4.96]	1.3e-4	100.00 / 100.00
incoming support zero fraction departure (9,227 records)	0.213	-0.39 [-0.79, 0.01]	0.521	100.00 / 100.00
G13 Upstream-neighbor delay	0.252	7.49 [4.08, 10.90]	0.001	100.00 / 100.00
neighbor delay 90th percentile arrival (8,947 records)	-0.210	0.57 [0.12, 1.02]	0.148	100.00 / 100.00
G08 Active degree	-0.252	-7.73 [-10.34, -5.12]	2.1e-5	100.00 / 100.00
outgoing active arrival (9,227 records)	0.246	1.18 [0.69, 1.67]	0.001	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H08 Spatial reachability downstream stations departure (9,227 records)	0.252 -0.216	6.21 [3.50, 8.92] -0.78 [-1.37, -0.19]	7.1e-4 0.116	100.00 / 100.00 100.00 / 100.00
H07 Downstream trains downstream departure (9,227 records)	0.252 -0.224	6.43 [2.58, 10.28] -0.71 [-1.22, -0.20]	0.019 0.091	100.00 / 100.00 100.00 / 100.00
H07 Downstream trains downstream arrival (9,227 records)	-0.250 0.250	-6.45 [-8.90, -3.99] 0.96 [0.50, 1.42]	1.3e-4 0.005	100.00 / 100.00 100.00 / 100.00
H08 Spatial reachability downstream branches arrival (9,227 records)	-0.250 0.251	-6.86 [-9.26, -4.47] 1.03 [0.57, 1.48]	3.5e-5 0.003	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support minimum departure (9,227 records)	0.250 -0.203	6.76 [3.61, 9.91] 0.56 [0.16, 0.96]	0.002 0.082	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay mean arrival (8,947 records)	0.249 -0.207	7.09 [3.76, 10.43] 0.58 [0.14, 1.02]	0.002 0.120	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay median arrival (8,947 records)	0.249 -0.207	7.09 [3.76, 10.43] 0.58 [0.14, 1.02]	0.002 0.120	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay zero fraction arrival (8,947 records)	-0.247 0.208	-7.94 [-11.29, -4.60] -0.67 [-1.17, -0.17]	4.9e-4 0.110	100.00 / 100.00 100.00 / 100.00
H09 Reachable delay totals downstream train maximum delay sum arrival (9,227 records)	-0.246 0.245	-5.32 [-7.36, -3.29] 0.79 [0.42, 1.15]	1.3e-4 0.003	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack interquartile range departure (9,227 records)	-0.245 0.204	-5.03 [-7.23, -2.84] -0.41 [-0.94, 0.11]	7.0e-4 1.000	100.00 / 100.00 100.00 / 66.67
G10 Edge-slack distribution incoming slack standard deviation departure (9,227 records)	-0.244 0.203	-5.02 [-7.24, -2.81] -0.43 [-0.96, 0.10]	7.8e-4 0.919	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min zero fraction (9,227 records)	-0.242 0.221	-8.20 [-10.89, -5.50] 1.12 [0.66, 1.57]	1.4e-5 0.001	100.00 / 100.00 100.00 / 100.00
G08 Active degree incoming active arrival (9,227 records)	0.241 -0.208	8.28 [3.87, 12.69] 0.72 [0.17, 1.27]	0.006 0.124	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support sum departure (9,227 records)	0.238 -0.214	5.25 [2.47, 8.02] -1.03 [-1.56, -0.51]	0.006 0.007	100.00 / 100.00 100.00 / 100.00
H13 Weak-component size component events arrival (9,227 records)	0.238 -0.196	7.90 [4.73, 11.08] 0.68 [0.16, 1.20]	2.6e-4 0.124	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution outgoing slack sum arrival (9,227 records)	0.237 -0.248	2.03 [-0.08, 4.15] -3.10 [-4.59, -1.61]	0.481 0.005	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths standard deviation (9,227 records)	0.237 -0.213	7.57 [5.00, 10.14] -0.72 [-1.16, -0.28]	2.3e-5 0.029	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution outgoing degree maximum (9,227 records)	0.236 -0.213	7.20 [4.46, 9.93] -0.63 [-1.05, -0.21]	1.2e-4 0.053	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration station entropy (9,227 records)	0.234 -0.209	4.55 [1.83, 7.27] -0.54 [-0.85, -0.23]	0.019 0.019	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min maximum (9,227 records)	0.232 -0.210	6.49 [3.72, 9.25] -1.29 [-1.91, -0.67]	5.3e-4 0.005	100.00 / 100.00 100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H13 Weak-component size	0.232	7.77 [4.53, 11.01]	4.2e-4	100.00 / 100.00
component trains arrival (9,227 records)	-0.194	0.72 [0.19, 1.25]	0.101	100.00 / 100.00
B02 Local edges and density	-0.232	-6.85 [-9.48, -4.23]	1.4e-4	100.00 / 100.00
density 20 min (9,125 records)	0.182	0.03 [-0.24, 0.30]	1.000	86.67 / 66.67
H11 Harmonic closeness	0.231	6.70 [3.73, 9.67]	8.4e-4	100.00 / 100.00
outgoing harmonic departure (9,227 records)	-0.196	-0.65 [-1.12, -0.18]	0.091	100.00 / 100.00
B05 Regional delay distribution	0.230	6.37 [3.59, 9.14]	7.0e-4	100.00 / 100.00
region delay 20 min 90th percentile (9,227 records)	-0.208	-1.26 [-1.88, -0.64]	0.006	100.00 / 100.00
H09 Reachable delay totals	0.229	4.35 [1.32, 7.37]	0.061	100.00 / 100.00
downstream train maximum delay sum departure (9,227 records)	-0.203	-0.90 [-1.53, -0.27]	0.076	100.00 / 100.00
B05 Regional delay distribution	0.228	6.10 [3.40, 8.80]	8.4e-4	100.00 / 100.00
region delay 10 min maximum (9,227 records)	-0.206	-1.35 [-2.01, -0.70]	0.005	100.00 / 100.00
B12 Delay concentration	-0.228	-3.11 [-5.59, -0.63]	0.143	100.00 / 100.00
station concentration (9,227 records)	0.202	0.36 [0.09, 0.64]	0.114	100.00 / 100.00
G12 Delay-support distribution	0.224	5.73 [2.83, 8.64]	0.004	100.00 / 100.00
outgoing support standard deviation departure (9,227 records)	-0.217	-1.02 [-1.70, -0.33]	0.058	100.00 / 100.00
B05 Regional delay distribution	0.221	5.79 [3.17, 8.40]	0.001	100.00 / 100.00
region delay 10 min 90th percentile (9,227 records)	-0.200	-1.34 [-2.00, -0.68]	0.006	100.00 / 100.00
G12 Delay-support distribution	0.221	5.45 [2.64, 8.26]	0.005	100.00 / 100.00
outgoing support interquartile range departure (9,227 records)	-0.214	-1.02 [-1.71, -0.33]	0.059	100.00 / 100.00
G13 Upstream-neighbor delay	0.221	6.14 [3.01, 9.27]	0.004	100.00 / 100.00
neighbor delay minimum arrival (8,947 records)	-0.185	0.56 [0.16, 0.95]	0.080	100.00 / 100.00
G10 Edge-slack distribution	0.220	1.43 [-1.07, 3.93]	1.000	100.00 / 100.00
outgoing slack 90th percentile arrival (9,227 records)	-0.231	-2.91 [-4.40, -1.41]	0.008	100.00 / 100.00
B05 Regional delay distribution	0.219	5.59 [3.03, 8.16]	0.001	100.00 / 100.00
region delay 5 min maximum (9,227 records)	-0.199	-1.37 [-2.05, -0.70]	0.006	100.00 / 100.00
H05 Downstream reachability	0.219	4.84 [1.77, 7.91]	0.030	100.00 / 100.00
downstream departure (9,227 records)	-0.183	-0.49 [-0.98, 0.01]	0.494	100.00 / 100.00
G12 Delay-support distribution	0.217	5.72 [2.58, 8.85]	0.008	100.00 / 100.00
incoming support median arrival (9,227 records)	-0.174	0.63 [0.21, 1.05]	0.053	100.00 / 100.00
B05 Regional delay distribution	0.216	5.47 [2.91, 8.03]	0.002	100.00 / 100.00
region delay 5 min 90th percentile (9,227 records)	-0.196	-1.35 [-2.02, -0.68]	0.006	100.00 / 100.00
B05 Regional delay distribution	0.214	4.19 [1.82, 6.56]	0.011	100.00 / 100.00
region delay 10 min sum (9,227 records)	-0.176	-0.24 [-0.52, 0.05]	0.870	100.00 / 100.00
G02 Reference departure delay	0.213	5.29 [2.68, 7.89]	0.003	100.00 / 100.00
departure delay (9,227 records)	-0.194	-1.34 [-2.01, -0.67]	0.006	100.00 / 100.00
G12 Delay-support distribution	0.213	5.29 [2.69, 7.89]	0.003	100.00 / 100.00
outgoing support maximum departure (9,227 records)	-0.194	-1.34 [-2.01, -0.67]	0.006	100.00 / 100.00
B15 Global degree distribution	0.212	6.61 [4.19, 9.02]	7.2e-5	100.00 / 100.00
incoming degree maximum (9,227 records)	-0.189	-0.60 [-0.93, -0.27]	0.013	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution outgoing support 90th percentile departure (9,227 records)	0.211 -0.192	5.17 [2.58, 7.76] -1.33 [-2.01, -0.65]	0.003 0.007	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution outgoing slack maximum arrival (9,227 records)	0.210 -0.221	1.13 [-1.30, 3.57] -2.74 [-4.15, -1.33]	1.000 0.008	100.00 / 66.67 100.00 / 100.00
H09 Reachable delay totals downstream event delay departure (9,227 records)	0.209 -0.175	3.31 [0.83, 5.79] -0.74 [-1.42, -0.06]	0.098 0.318	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction arrival (9,227 records)	-0.209 0.168	-6.96 [-10.27, -3.65] -0.81 [-1.37, -0.25]	0.002 0.068	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min mean (9,227 records)	0.209 -0.189	5.04 [2.50, 7.58] -1.26 [-1.92, -0.59]	0.004 0.011	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support mean departure (9,227 records)	0.208 -0.188	4.74 [2.17, 7.31] -1.27 [-1.97, -0.57]	0.008 0.014	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min mean (9,227 records)	0.208 -0.188	5.00 [2.49, 7.51] -1.25 [-1.91, -0.58]	0.003 0.011	100.00 / 100.00 100.00 / 100.00
G19 Endpoint contrasts H10 active betweenness endpoint contrast (9,227 records)	0.207 -0.169	4.66 [2.91, 6.41] -0.51 [-0.78, -0.25]	1.1e-4 0.009	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min median (9,227 records)	0.207 -0.188	5.01 [2.48, 7.53] -1.25 [-1.91, -0.59]	0.004 0.010	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min median (9,227 records)	0.207 -0.187	4.96 [2.48, 7.45] -1.26 [-1.92, -0.59]	0.003 0.011	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack mean arrival (9,227 records)	-0.207 0.181	-4.14 [-5.58, -2.70] 0.28 [0.09, 0.47]	3.4e-5 0.070	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack sum arrival (9,227 records)	-0.206 0.181	-4.07 [-5.51, -2.63] 0.28 [0.08, 0.47]	4.3e-5 0.077	100.00 / 100.00 100.00 / 100.00
B02 Local edges and density edges 20 min (9,227 records)	0.205 -0.152	8.92 [5.38, 12.46] 0.44 [0.07, 0.81]	2.3e-4 0.203	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution incoming degree interquartile range (9,227 records)	0.205 -0.182	6.51 [3.70, 9.32] -0.61 [-1.09, -0.13]	6.2e-4 0.148	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution outgoing degree interquartile range (9,227 records)	0.205 -0.182	6.51 [3.70, 9.32] -0.61 [-1.09, -0.13]	6.2e-4 0.148	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min mean (9,227 records)	0.205 -0.184	4.93 [2.43, 7.43] -1.19 [-1.84, -0.53]	0.004 0.014	100.00 / 100.00 100.00 / 100.00
B04 Boundary connections entering 5 min (9,227 records)	-0.204 0.224	-6.45 [-8.77, -4.14] 1.24 [0.61, 1.87]	5.4e-5 0.007	100.00 / 100.00 100.00 / 100.00
B01 Local region size region events 20 min (9,227 records)	0.203 -0.148	8.47 [5.08, 11.85] 0.46 [0.08, 0.85]	2.5e-4 0.200	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min count (9,227 records)	0.203 -0.148	8.47 [5.08, 11.85] 0.46 [0.08, 0.85]	2.5e-4 0.200	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min median (9,227 records)	0.203 -0.182	4.90 [2.38, 7.43] -1.17 [-1.83, -0.52]	0.005 0.015	100.00 / 100.00 100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support minimum arrival (9,227 records)	0.202 -0.166	5.36 [2.50, 8.22] 0.54 [0.19, 0.90]	0.006 0.049	100.00 / 100.00 100.00 / 100.00
B01 Local region size region trains 10 min (9,227 records)	0.201 -0.154	6.82 [4.57, 9.07] -0.11 [-0.38, 0.15]	1.5e-5 1.000	100.00 / 100.00 93.33 / 100.00
G12 Delay-support distribution incoming support median departure (9,227 records)	0.201 -0.165	5.38 [2.54, 8.21] 0.26 [-0.05, 0.57]	0.006 0.875	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths maximum (9,227 records)	0.200 -0.184	6.30 [3.70, 8.90] -0.52 [-0.91, -0.13]	3.5e-4 0.122	100.00 / 100.00 100.00 / 100.00
H15 Chain and crossing length longest crossings arrival (9,227 records)	0.198 -0.173	6.41 [1.81, 11.02] 0.55 [0.12, 0.98]	0.075 0.145	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support median departure (9,227 records)	0.196 -0.176	4.60 [2.07, 7.14] -1.26 [-1.96, -0.56]	0.009 0.015	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min minimum (9,227 records)	0.196 -0.177	4.51 [1.97, 7.05] -1.13 [-1.80, -0.46]	0.011 0.024	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support mean departure (9,227 records)	0.195 -0.159	5.37 [2.57, 8.17] 0.28 [-0.04, 0.59]	0.005 0.752	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack 90th percentile arrival (9,227 records)	-0.193 0.169	-3.36 [-4.63, -2.08] 0.24 [0.00, 0.47]	1.2e-4 0.440	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support sum departure (9,227 records)	0.192 -0.156	5.25 [2.37, 8.13] 0.29 [-0.03, 0.60]	0.008 0.655	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction departure (6,869 records)	-0.189 0.170	-6.17 [-8.62, -3.71] 0.84 [0.47, 1.20]	2.6e-4 0.002	100.00 / 100.00 100.00 / 100.00
H06 Upstream trains upstream arrival (9,227 records)	0.188 -0.159	6.59 [2.21, 10.98] 0.72 [0.21, 1.23]	0.043 0.081	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack zero fraction arrival (9,227 records)	0.186 -0.179	5.57 [1.46, 9.67] 0.30 [-0.13, 0.73]	0.089 1.000	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack standard deviation arrival (9,227 records)	-0.186 0.163	-2.83 [-4.04, -1.61] 0.24 [-0.01, 0.49]	5.8e-4 0.505	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack interquartile range arrival (9,227 records)	-0.186 0.163	-2.86 [-4.10, -1.61] 0.21 [-0.05, 0.47]	7.1e-4 0.897	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack maximum arrival (9,227 records)	-0.186 0.163	-2.86 [-4.10, -1.61] 0.21 [-0.05, 0.47]	7.1e-4 0.897	100.00 / 100.00 100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum arrival (9,227 records)	0.181 -0.148	6.16 [2.05, 10.27] 0.64 [0.20, 1.08]	0.044 0.069	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support minimum departure (9,227 records)	0.181 -0.159	4.17 [1.64, 6.71] -1.17 [-1.90, -0.44]	0.021 0.035	100.00 / 100.00 100.00 / 100.00
G09 Active cross-train links active cross arrival (9,227 records)	0.176 -0.168	4.51 [0.29, 8.74] 0.50 [0.16, 0.83]	0.321 0.056	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min minimum (9,227 records)	0.175 -0.157	3.92 [1.54, 6.30] -1.10 [-1.78, -0.42]	0.021 0.034	100.00 / 100.00 100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G09 Active cross-train links	0.175	4.58 [0.56, 8.61]	0.240	100.00 / 100.00
active cross fraction arrival (9,227 records)	-0.167	0.49 [0.16, 0.83]	0.059	100.00 / 100.00
G12 Delay-support distribution	0.173	4.99 [1.87, 8.10]	0.026	100.00 / 100.00
incoming support sum arrival (9,227 records)	-0.133	0.76 [0.29, 1.22]	0.033	100.00 / 100.00
H16 Local motifs	0.171	5.41 [3.17, 7.65]	3.8e-4	100.00 / 100.00
diamonds at source departure (9,227 records)	-0.136	-0.06 [-0.46, 0.34]	1.000	93.33 / 66.67
G12 Delay-support distribution	0.169	4.84 [1.87, 7.80]	0.023	100.00 / 100.00
incoming support mean arrival (9,227 records)	-0.130	0.76 [0.29, 1.23]	0.031	100.00 / 100.00
H11 Harmonic closeness	-0.165	-5.87 [-8.88, -2.86]	0.004	100.00 / 100.00
incoming harmonic departure (9,227 records)	0.183	0.90 [0.51, 1.29]	0.002	100.00 / 100.00
B05 Regional delay distribution	0.164	1.80 [0.02, 3.58]	0.398	100.00 / 100.00
region delay 5 min sum (9,227 records)	-0.136	-0.35 [-0.68, -0.03]	0.328	100.00 / 100.00
B08 Community structure	0.163	4.80 [2.41, 7.19]	0.003	100.00 / 100.00
community events arrival (9,227 records)	-0.126	0.78 [0.29, 1.27]	0.038	100.00 / 100.00
B03 Active local density	0.163	4.12 [2.60, 5.65]	8.5e-5	100.00 / 100.00
active density 5 min (8,544 records)	-0.154	-0.93 [-1.40, -0.47]	0.006	100.00 / 100.00
H02 Longest active path	-0.162	-6.36 [-9.16, -3.55]	8.2e-4	100.00 / 100.00
longest path departure (9,227 records)	0.174	0.86 [0.51, 1.21]	0.001	100.00 / 100.00
H15 Chain and crossing length	-0.162	-6.36 [-9.16, -3.55]	8.2e-4	100.00 / 100.00
longest chain departure (9,227 records)	0.174	0.86 [0.51, 1.21]	0.001	100.00 / 100.00
B04 Boundary connections	0.161	5.59 [3.04, 8.14]	0.001	100.00 / 100.00
entering 20 min (9,227 records)	-0.107	0.61 [0.13, 1.10]	0.154	100.00 / 100.00
G12 Delay-support distribution	0.159	5.54 [3.03, 8.04]	0.001	100.00 / 100.00
outgoing support interquartile range arrival (9,227 records)	-0.143	0.35 [0.02, 0.68]	0.358	100.00 / 100.00
G12 Delay-support distribution	0.159	5.54 [3.03, 8.04]	0.001	100.00 / 100.00
outgoing support standard deviation arrival (9,227 records)	-0.143	0.35 [0.02, 0.68]	0.358	100.00 / 100.00
G14 Downstream-neighbor delay	0.158	3.14 [0.87, 5.41]	0.078	100.00 / 100.00
neighbor delay sum departure (9,227 records)	-0.136	-0.47 [-0.86, -0.08]	0.190	100.00 / 100.00
H16 Local motifs	-0.157	-5.00 [-6.65, -3.35]	1.5e-5	100.00 / 100.00
fan-out pairs arrival (9,227 records)	0.145	0.65 [0.40, 0.90]	0.001	100.00 / 100.00
H08 Spatial reachability	-0.155	-6.27 [-9.01, -3.52]	7.3e-4	100.00 / 100.00
upstream stations departure (9,227 records)	0.169	0.87 [0.51, 1.23]	0.001	100.00 / 100.00
G12 Delay-support distribution	0.151	3.74 [1.30, 6.19]	0.037	100.00 / 100.00
incoming support 90th percentile departure (9,227 records)	-0.123	0.06 [-0.31, 0.42]	1.000	86.67 / 66.67
G16 Incident-link imbalance	0.151	2.74 [0.65, 4.83]	0.111	100.00 / 100.00
pressure balance departure (9,227 records)	-0.147	-2.19 [-3.28, -1.09]	0.006	100.00 / 100.00
B05 Regional delay distribution	0.150	3.08 [0.84, 5.32]	0.082	100.00 / 100.00
region delay 20 min minimum (9,227 records)	-0.133	-1.04 [-1.75, -0.34]	0.058	100.00 / 100.00
G13 Upstream-neighbor delay	0.150	4.71 [2.47, 6.94]	0.002	100.00 / 100.00
neighbor delay count arrival (9,227 records)	-0.129	-0.03 [-0.45, 0.40]	1.000	53.33 / 66.67
H04 Upstream reachability	-0.150	-6.23 [-8.78, -3.67]	3.4e-4	100.00 / 100.00
upstream departure (9,227 records)	0.165	0.88 [0.54, 1.23]	0.001	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G11 Near-tight links	-0.148	-4.10 [-6.76, -1.44]	0.036	100.00 / 100.00
slack at most 1 minute arrival (9,227 records)	0.161	2.02 [1.22, 2.83]	0.001	100.00 / 100.00
G12 Delay-support distribution	-0.145	-4.64 [-7.50, -1.77]	0.024	100.00 / 100.00
outgoing support zero fraction arrival (9,227 records)	0.110	-0.99 [-1.62, -0.37]	0.037	100.00 / 100.00
H11 Harmonic closeness	0.144	4.70 [0.55, 8.86]	0.247	100.00 / 100.00
incoming harmonic arrival (9,227 records)	-0.126	0.88 [0.32, 1.45]	0.041	100.00 / 100.00
G10 Edge-slack distribution	-0.144	-2.68 [-4.70, -0.66]	0.102	100.00 / 100.00
incoming slack maximum departure (9,227 records)	0.095	-1.97 [-3.12, -0.83]	0.019	100.00 / 100.00
B01 Local region size	-0.143	-4.32 [-5.65, -2.99]	6.0e-6	100.00 / 100.00
region events 5 min (9,227 records)	0.168	0.99 [0.51, 1.46]	0.005	100.00 / 100.00
B05 Regional delay distribution	-0.143	-4.32 [-5.65, -2.99]	6.0e-6	100.00 / 100.00
region delay 5 min count (9,227 records)	0.168	0.99 [0.51, 1.46]	0.005	100.00 / 100.00
G12 Delay-support distribution	0.143	3.33 [0.99, 5.67]	0.065	100.00 / 100.00
incoming support maximum departure (9,227 records)	-0.116	0.01 [-0.37, 0.39]	1.000	60.00 / 33.33
H10 Active betweenness	-0.142	-4.54 [-6.77, -2.31]	0.003	100.00 / 100.00
betweenness departure (9,227 records)	0.164	0.70 [0.35, 1.06]	0.007	100.00 / 100.00
B02 Local edges and density	-0.142	-3.84 [-4.87, -2.81]	6.6e-7	100.00 / 100.00
edges 5 min (9,227 records)	0.164	0.88 [0.47, 1.29]	0.004	100.00 / 100.00
B02 Local edges and density	-0.139	-4.50 [-6.11, -2.89]	5.2e-5	100.00 / 100.00
density 10 min (8,935 records)	0.096	0.05 [-0.18, 0.29]	1.000	86.67 / 66.67
G10 Edge-slack distribution	-0.136	-2.42 [-4.44, -0.40]	0.185	100.00 / 100.00
incoming slack 90th percentile departure (9,227 records)	0.087	-2.12 [-3.33, -0.91]	0.017	100.00 / 100.00
H09 Reachable delay totals	-0.131	-5.12 [-7.31, -2.92]	5.6e-4	100.00 / 100.00
upstream event delay departure (9,227 records)	0.151	0.66 [0.40, 0.93]	0.001	100.00 / 100.00
G12 Delay-support distribution	0.130	3.28 [0.57, 5.99]	0.175	100.00 / 100.00
incoming support 90th percentile arrival (9,227 records)	-0.095	0.87 [0.37, 1.36]	0.018	100.00 / 100.00
G14 Downstream-neighbor delay	-0.129	-3.90 [-6.40, -1.40]	0.033	100.00 / 100.00
neighbor delay zero fraction arrival (6,812 records)	0.115	0.02 [-0.27, 0.31]	1.000	80.00 / 66.67
H09 Reachable delay totals	0.127	2.85 [-0.71, 6.42]	0.890	100.00 / 100.00
upstream event delay arrival (9,227 records)	-0.104	0.65 [0.27, 1.03]	0.020	100.00 / 100.00
B09 Boundary pressure imbalance	-0.124	-4.35 [-7.46, -1.23]	0.075	100.00 / 100.00
cross boundary pressure balance 20 min (9,227 records)	0.081	-1.39 [-2.16, -0.62]	0.015	100.00 / 100.00
G12 Delay-support distribution	0.122	2.45 [-0.30, 5.20]	0.642	100.00 / 100.00
outgoing support sum arrival (9,227 records)	-0.092	0.83 [0.37, 1.29]	0.015	100.00 / 100.00
B01 Local region size	0.122	3.89 [2.48, 5.29]	5.9e-5	100.00 / 100.00
region trains 5 min (9,227 records)	-0.088	-0.19 [-0.50, 0.11]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.121	-3.63 [-4.99, -2.28]	9.9e-5	100.00 / 100.00
outgoing slack median departure (9,227 records)	0.109	0.51 [0.32, 0.69]	8.5e-4	100.00 / 100.00
H08 Spatial reachability	0.121	3.42 [0.57, 6.27]	0.184	100.00 / 100.00
upstream branches arrival (9,227 records)	-0.096	0.89 [0.38, 1.40]	0.017	100.00 / 100.00
G01 Reference arrival delay	0.116	2.62 [0.06, 5.18]	0.387	100.00 / 100.00
arrival delay (9,227 records)	-0.082	0.88 [0.39, 1.38]	0.015	100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support maximum arrival (9,227 records)	0.116 -0.082	2.62 [0.06, 5.18] 0.88 [0.39, 1.38]	0.387 0.015	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support maximum arrival (9,227 records)	0.115 -0.082	2.62 [0.06, 5.18] 0.88 [0.39, 1.38]	0.387 0.015	100.00 / 100.00 100.00 / 100.00
H08 Spatial reachability upstream branches departure (9,227 records)	-0.115 0.138	-4.25 [-6.91, -1.60] 0.62 [0.26, 0.98]	0.026 0.019	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support 90th percentile arrival (9,227 records)	0.114 -0.080	2.53 [-0.01, 5.08] 0.88 [0.39, 1.38]	0.425 0.015	100.00 / 100.00 100.00 / 100.00
H04 Upstream reachability upstream arrival (9,227 records)	0.112 -0.094	2.66 [-1.10, 6.42] 0.78 [0.33, 1.24]	1.000 0.020	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution outgoing slack sum departure (9,227 records)	-0.111 0.098	-3.84 [-5.32, -2.37] 0.52 [0.32, 0.72]	1.4e-4 0.001	100.00 / 100.00 100.00 / 100.00
H24 Projection core number core arrival (9,227 records)	0.110 -0.079	3.22 [0.24, 6.20] 0.95 [0.39, 1.52]	0.308 0.024	100.00 / 100.00 100.00 / 100.00
B04 Boundary connections leaving 20 min (9,227 records)	0.110 -0.069	3.67 [1.29, 6.05] 0.36 [-0.08, 0.81]	0.036 0.908	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay mean departure (6,869 records)	0.110 -0.093	2.28 [0.15, 4.42] -0.80 [-1.28, -0.32]	0.321 0.027	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay median departure (6,869 records)	0.110 -0.093	2.28 [0.15, 4.42] -0.80 [-1.28, -0.32]	0.321 0.027	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support mean arrival (9,227 records)	0.110 -0.076	2.18 [-0.30, 4.65] 0.88 [0.40, 1.37]	0.670 0.014	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support median arrival (9,227 records)	0.110 -0.076	2.18 [-0.30, 4.65] 0.88 [0.40, 1.37]	0.670 0.014	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution outgoing slack mean departure (9,227 records)	-0.109 0.097	-3.83 [-5.30, -2.36] 0.52 [0.31, 0.72]	1.4e-4 0.001	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay maximum departure (6,869 records)	0.109 -0.094	2.38 [0.30, 4.47] -0.87 [-1.35, -0.39]	0.238 0.014	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay 90th percentile departure (6,869 records)	0.109 -0.093	2.37 [0.27, 4.47] -0.86 [-1.34, -0.38]	0.251 0.015	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution outgoing slack 90th percentile departure (9,227 records)	-0.108 0.096	-3.85 [-5.36, -2.34] 0.51 [0.30, 0.72]	1.9e-4 0.001	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay minimum departure (6,869 records)	0.108 -0.091	2.14 [-0.03, 4.31] -0.72 [-1.21, -0.23]	0.440 0.063	100.00 / 100.00 100.00 / 100.00
H08 Spatial reachability upstream stations arrival (9,227 records)	0.107 -0.088	2.59 [-0.54, 5.73] 0.87 [0.37, 1.38]	0.823 0.019	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution outgoing slack interquartile range departure (9,227 records)	-0.106 0.093	-3.83 [-5.35, -2.30] 0.50 [0.29, 0.71]	2.4e-4 0.002	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution outgoing slack maximum departure (9,227 records)	-0.106 0.093	-3.83 [-5.35, -2.30] 0.50 [0.29, 0.71]	2.4e-4 0.002	100.00 / 100.00 100.00 / 100.00

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Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution	-0.105	-3.79 [-5.29, -2.28]	2.3e-4	100.00 / 100.00
outgoing slack standard deviation departure (9,227 records)	0.092	0.50 [0.28, 0.71]	0.002	100.00 / 100.00
G14 Downstream-neighbor delay	0.101	1.14 [-0.56, 2.85]	1.000	100.00 / 100.00
neighbor delay interquartile range departure (6,869 records)	-0.098	-0.65 [-1.20, -0.09]	0.230	100.00 / 100.00
G14 Downstream-neighbor delay	0.101	1.14 [-0.56, 2.85]	1.000	100.00 / 100.00
neighbor delay standard deviation departure (6,869 records)	-0.098	-0.65 [-1.20, -0.09]	0.230	100.00 / 100.00
H01 Shortest source distance	0.098	3.03 [0.03, 6.04]	0.403	100.00 / 100.00
source edges arrival (3,406 records)	-0.090	-0.01 [-0.09, 0.08]	1.000	73.33 / 66.67
G07 Degree by relation	0.096	3.04 [0.78, 5.30]	0.095	100.00 / 100.00
outgoing entry headway departure (9,227 records)	-0.082	-0.01 [-0.47, 0.45]	1.000	60.00 / 33.33
H02 Longest active path	0.094	1.85 [-1.37, 5.07]	1.000	100.00 / 100.00
longest path arrival (9,227 records)	-0.078	0.87 [0.38, 1.36]	0.016	100.00 / 100.00
H15 Chain and crossing length	0.094	1.85 [-1.37, 5.07]	1.000	100.00 / 100.00
longest chain arrival (9,227 records)	-0.078	0.87 [0.38, 1.36]	0.016	100.00 / 100.00
G12 Delay-support distribution	0.094	1.68 [-0.68, 4.04]	1.000	100.00 / 100.00
outgoing support minimum arrival (9,227 records)	-0.062	0.87 [0.40, 1.35]	0.012	100.00 / 100.00
G10 Edge-slack distribution	-0.092	-0.83 [-2.87, 1.20]	1.000	100.00 / 66.67
incoming slack sum departure (9,227 records)	0.042	-2.79 [-4.35, -1.24]	0.015	100.00 / 100.00
B09 Boundary pressure imbalance	0.092	2.35 [0.81, 3.89]	0.039	100.00 / 100.00
cross boundary pressure balance 5 min (9,227 records)	-0.114	-1.74 [-2.61, -0.87]	0.006	100.00 / 100.00
G10 Edge-slack distribution	-0.090	-0.42 [-2.60, 1.75]	1.000	93.33 / 66.67
incoming slack mean departure (9,227 records)	0.039	-3.24 [-4.83, -1.66]	0.006	100.00 / 100.00
H16 Local motifs	-0.089	-2.84 [-3.78, -1.89]	1.6e-5	100.00 / 100.00
diamonds at source arrival (9,227 records)	0.082	0.37 [0.23, 0.51]	0.001	100.00 / 100.00
G08 Active degree	-0.087	-2.92 [-5.67, -0.16]	0.333	100.00 / 100.00
incoming active departure (9,227 records)	0.118	0.69 [0.29, 1.09]	0.019	100.00 / 100.00
H06 Upstream trains	-0.085	-3.10 [-5.42, -0.78]	0.098	100.00 / 100.00
upstream departure (9,227 records)	0.115	0.74 [0.40, 1.09]	0.004	100.00 / 100.00
G14 Downstream-neighbor delay	0.085	2.72 [0.57, 4.87]	0.135	100.00 / 100.00
neighbor delay count departure (9,227 records)	-0.072	0.00 [-0.46, 0.47]	1.000	53.33 / 66.67
G10 Edge-slack distribution	-0.084	-0.49 [-2.55, 1.56]	1.000	93.33 / 66.67
incoming slack median departure (9,227 records)	0.034	-3.00 [-4.58, -1.41]	0.010	100.00 / 100.00
B16 Global reachability	0.080	3.20 [0.89, 5.51]	0.078	100.00 / 100.00
mean reachable distance (9,227 records)	-0.070	-0.37 [-0.76, 0.01]	0.526	100.00 / 100.00
G10 Edge-slack distribution	0.079	2.16 [-0.24, 4.57]	0.623	100.00 / 100.00
outgoing slack zero fraction departure (9,227 records)	-0.071	-0.58 [-1.10, -0.06]	0.289	100.00 / 100.00
H16 Local motifs	-0.078	-2.49 [-3.59, -1.38]	8.5e-4	100.00 / 100.00
fan-in pairs departure (9,227 records)	0.106	0.42 [0.26, 0.58]	0.001	100.00 / 100.00
H09 Reachable delay totals	-0.078	-2.78 [-4.86, -0.71]	0.095	100.00 / 100.00
upstream train maximum delay sum departure (9,227 records)	0.109	0.52 [0.23, 0.81]	0.016	100.00 / 100.00
B04 Boundary connections	0.077	2.46 [0.37, 4.56]	0.204	100.00 / 100.00
leaving 10 min (9,227 records)	-0.046	0.40 [-0.02, 0.81]	0.576	100.00 / 100.00

Continued on the next page.

Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H03 Event time from release	0.075	3.14 [0.75, 5.53]	0.107	100.00 / 100.00
time from release departure (9,227 records)	-0.053	0.19 [-0.10, 0.49]	1.000	100.00 / 66.67
G14 Downstream-neighbor delay	0.074	1.33 [-0.71, 3.37]	1.000	100.00 / 100.00
neighbor delay maximum arrival (6,812 records)	-0.064	-0.08 [-0.39, 0.22]	1.000	93.33 / 66.67
G14 Downstream-neighbor delay	0.074	1.33 [-0.71, 3.37]	1.000	100.00 / 100.00
neighbor delay mean arrival (6,812 records)	-0.064	-0.08 [-0.39, 0.22]	1.000	93.33 / 66.67
G14 Downstream-neighbor delay	0.074	1.33 [-0.71, 3.37]	1.000	100.00 / 100.00
neighbor delay median arrival (6,812 records)	-0.064	-0.08 [-0.39, 0.22]	1.000	93.33 / 66.67
G14 Downstream-neighbor delay	0.074	1.33 [-0.71, 3.37]	1.000	100.00 / 100.00
neighbor delay minimum arrival (6,812 records)	-0.064	-0.08 [-0.39, 0.22]	1.000	93.33 / 66.67
G14 Downstream-neighbor delay	0.074	1.33 [-0.71, 3.37]	1.000	100.00 / 100.00
neighbor delay 90th percentile arrival (6,812 records)	-0.064	-0.08 [-0.39, 0.22]	1.000	93.33 / 66.67
G14 Downstream-neighbor delay	0.071	1.28 [-1.10, 3.66]	1.000	100.00 / 100.00
neighbor delay sum arrival (9,227 records)	-0.067	0.06 [-0.36, 0.48]	1.000	93.33 / 33.33
G07 Degree by relation	-0.069	-2.18 [-2.93, -1.44]	2.5e-5	100.00 / 100.00
incoming emergency departure (9,227 records)	0.063	0.30 [0.18, 0.41]	0.001	100.00 / 100.00
G18 External-source share	-0.069	-2.15 [-2.88, -1.43]	2.0e-5	100.00 / 100.00
source fraction departure (9,227 records)	0.063	0.30 [0.18, 0.41]	0.001	100.00 / 100.00
G11 Near-tight links	-0.068	-2.27 [-4.06, -0.47]	0.137	100.00 / 100.00
slack at most 1 minute departure (9,227 records)	0.114	0.96 [0.48, 1.44]	0.006	100.00 / 100.00
H01 Shortest source distance	0.068	1.86 [-1.29, 5.00]	1.000	100.00 / 100.00
source minimum time lag arrival (3,406 records)	-0.062	0.04 [-0.07, 0.14]	1.000	100.00 / 100.00
B02 Local edges and density	-0.066	-2.36 [-4.12, -0.60]	0.095	100.00 / 100.00
density 5 min (8,544 records)	0.036	-0.16 [-0.38, 0.06]	1.000	100.00 / 100.00
G19 Endpoint contrasts	0.065	2.31 [0.19, 4.43]	0.297	100.00 / 100.00
G05 outgoing degree endpoint contrast (9,227 records)	-0.042	0.03 [-0.37, 0.43]	1.000	86.67 / 66.67
H01 Shortest source distance	-0.063	-0.97 [-1.79, -0.15]	0.195	100.00 / 100.00
source edges departure (3,303 records)	0.065	0.33 [-0.07, 0.73]	0.895	100.00 / 100.00
G19 Endpoint contrasts	-0.063	-2.02 [-3.12, -0.92]	0.008	100.00 / 100.00
G05 incoming degree endpoint contrast (9,227 records)	0.053	0.08 [-0.22, 0.39]	1.000	100.00 / 100.00
H01 Shortest source distance	-0.062	-1.08 [-1.95, -0.22]	0.145	100.00 / 100.00
source minimum time lag departure (3,303 records)	0.064	0.34 [-0.05, 0.74]	0.737	100.00 / 100.00
G17 Relation-type entropy	-0.060	-1.44 [-2.91, 0.02]	0.443	100.00 / 100.00
type entropy departure (9,227 records)	0.056	0.41 [-0.08, 0.90]	0.833	100.00 / 100.00
G11 Near-tight links	0.053	2.65 [0.30, 5.00]	0.252	100.00 / 100.00
slack at most 2 minutes departure (9,227 records)	-0.002	0.93 [0.43, 1.42]	0.011	100.00 / 100.00
B08 Community structure	-0.051	-1.55 [-3.23, 0.14]	0.579	100.00 / 100.00
cross community edges departure (9,227 records)	0.063	0.28 [0.04, 0.53]	0.250	100.00 / 100.00
B01 Local region size	0.049	1.47 [-0.02, 2.97]	0.443	100.00 / 100.00
region events 10 min (9,227 records)	-0.001	0.82 [0.40, 1.23]	0.007	100.00 / 100.00

Continued on the next page.

Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution	0.049	1.47 [-0.02, 2.97]	0.443	100.00 / 100.00
region delay 10 min count (9,227 records)	-0.001	0.82 [0.40, 1.23]	0.007	100.00 / 100.00
B08 Community structure	-0.048	-1.64 [-2.61, -0.68]	0.016	100.00 / 100.00
cross community edges arrival (9,227 records)	0.059	0.43 [0.26, 0.60]	0.001	100.00 / 100.00
B03 Active local density	0.048	-1.10 [-2.71, 0.51]	1.000	100.00 / 66.67
active density 20 min (9,125 records)	-0.054	-0.55 [-1.03, -0.07]	0.241	100.00 / 100.00
G16 Incident-link imbalance	0.043	1.81 [-0.13, 3.75]	0.546	100.00 / 100.00
degree balance departure (9,227 records)	-0.036	0.08 [-0.41, 0.56]	1.000	93.33 / 66.67
B02 Local edges and density	0.042	1.85 [-0.12, 3.83]	0.537	100.00 / 100.00
edges 10 min (9,227 records)	0.004	0.79 [0.40, 1.18]	0.006	100.00 / 100.00
G07 Degree by relation	0.038	1.22 [0.52, 1.93]	0.014	100.00 / 100.00
incoming exit headway arrival (9,227 records)	-0.033	0.03 [-0.30, 0.36]	1.000	86.67 / 66.67
G06 Cross-train degree	0.038	1.20 [0.50, 1.90]	0.015	100.00 / 100.00
incoming cross departure (9,227 records)	-0.033	0.03 [-0.30, 0.37]	1.000	86.67 / 66.67
G07 Degree by relation	0.038	1.20 [0.50, 1.90]	0.015	100.00 / 100.00
incoming entry headway departure (9,227 records)	-0.033	0.03 [-0.30, 0.37]	1.000	86.67 / 66.67
G13 Upstream-neighbor delay	0.038	1.20 [0.50, 1.90]	0.015	100.00 / 100.00
neighbor delay count departure (9,227 records)	-0.033	0.03 [-0.30, 0.37]	1.000	86.67 / 66.67
G12 Delay-support distribution	0.036	-2.02 [-3.12, -0.93]	0.007	100.00 / 100.00
incoming support standard deviation arrival (9,227 records)	-0.013	0.61 [0.33, 0.89]	0.003	100.00 / 100.00
G12 Delay-support distribution	0.036	-2.02 [-3.12, -0.92]	0.008	100.00 / 100.00
incoming support interquartile range arrival (9,227 records)	-0.013	0.61 [0.33, 0.89]	0.003	100.00 / 100.00
G05 Event degree	0.035	1.60 [-0.38, 3.58]	0.876	100.00 / 100.00
outgoing departure (9,227 records)	-0.028	0.15 [-0.35, 0.65]	1.000	93.33 / 66.67
G06 Cross-train degree	0.035	1.60 [-0.38, 3.58]	0.876	100.00 / 100.00
outgoing cross departure (9,227 records)	-0.028	0.15 [-0.35, 0.65]	1.000	93.33 / 66.67
G10 Edge-slack distribution	0.035	1.60 [-0.38, 3.58]	0.876	100.00 / 100.00
outgoing slack count departure (9,227 records)	-0.028	0.15 [-0.35, 0.65]	1.000	93.33 / 66.67
G12 Delay-support distribution	0.035	1.60 [-0.38, 3.58]	0.876	100.00 / 100.00
outgoing support count departure (9,227 records)	-0.028	0.15 [-0.35, 0.65]	1.000	93.33 / 66.67
G05 Event degree	-0.035	-1.10 [-2.17, -0.03]	0.375	100.00 / 100.00
incoming departure (9,227 records)	0.033	0.28 [0.05, 0.50]	0.172	100.00 / 100.00
G10 Edge-slack distribution	-0.035	-1.10 [-2.17, -0.03]	0.375	100.00 / 100.00
incoming slack count departure (9,227 records)	0.033	0.28 [0.05, 0.50]	0.172	100.00 / 100.00
G12 Delay-support distribution	-0.035	-1.10 [-2.17, -0.03]	0.375	100.00 / 100.00
incoming support count departure (9,227 records)	0.033	0.28 [0.05, 0.50]	0.172	100.00 / 100.00
G05 Event degree	0.033	1.23 [0.01, 2.45]	0.398	100.00 / 100.00
incoming arrival (9,227 records)	-0.024	0.09 [-0.28, 0.47]	1.000	93.33 / 66.67
G06 Cross-train degree	0.033	1.23 [0.01, 2.45]	0.398	100.00 / 100.00
incoming cross arrival (9,227 records)	-0.024	0.09 [-0.28, 0.47]	1.000	93.33 / 66.67
G10 Edge-slack distribution	0.033	1.23 [0.01, 2.45]	0.398	100.00 / 100.00
incoming slack count arrival (9,227 records)	-0.024	0.09 [-0.28, 0.47]	1.000	93.33 / 66.67

Continued on the next page.

Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution	0.033	1.23 [0.01, 2.45]	0.398	100.00 / 100.00
incoming support count arrival (9,227 records)	-0.024	0.09 [-0.28, 0.47]	1.000	93.33 / 66.67
G07 Degree by relation	0.032	1.00 [-0.48, 2.49]	1.000	100.00 / 100.00
incoming station track arrival (9,227 records)	-0.023	0.11 [-0.28, 0.50]	1.000	93.33 / 66.67
G10 Edge-slack distribution	0.031	-2.55 [-3.75, -1.35]	0.002	100.00 / 100.00
outgoing slack interquartile range arrival (9,227 records)	-0.037	-0.18 [-0.55, 0.19]	1.000	100.00 / 100.00
G10 Edge-slack distribution	0.031	-2.55 [-3.75, -1.35]	0.002	100.00 / 100.00
outgoing slack standard deviation arrival (9,227 records)	-0.037	-0.18 [-0.55, 0.19]	1.000	100.00 / 100.00
B03 Active local density	0.026	-0.21 [-1.93, 1.51]	1.000	73.33 / 66.67
active density 10 min (8,935 records)	-0.033	-0.78 [-1.27, -0.29]	0.035	100.00 / 100.00
G11 Near-tight links	-0.024	0.31 [-3.59, 4.21]	1.000	100.00 / 66.67
slack at most 2 minutes arrival (9,227 records)	0.052	1.91 [1.08, 2.74]	0.002	100.00 / 100.00
H15 Chain and crossing length	0.021	0.06 [-1.73, 1.85]	1.000	60.00 / 66.67
longest crossings departure (9,227 records)	0.015	0.38 [0.10, 0.67]	0.104	100.00 / 100.00
B04 Boundary connections	-0.018	-0.87 [-2.15, 0.40]	1.000	100.00 / 100.00
entering 10 min (9,227 records)	0.063	0.95 [0.42, 1.49]	0.015	100.00 / 100.00
G05 Event degree	0.009	0.27 [-2.18, 2.73]	1.000	80.00 / 66.67
outgoing arrival (9,227 records)	-0.020	0.25 [-0.40, 0.91]	1.000	93.33 / 66.67
G06 Cross-train degree	0.009	0.27 [-2.18, 2.73]	1.000	80.00 / 66.67
outgoing cross arrival (9,227 records)	-0.020	0.25 [-0.40, 0.91]	1.000	93.33 / 66.67
G07 Degree by relation	0.009	0.27 [-2.18, 2.73]	1.000	80.00 / 66.67
outgoing exit headway arrival (9,227 records)	-0.020	0.25 [-0.40, 0.91]	1.000	93.33 / 66.67
G10 Edge-slack distribution	0.009	0.27 [-2.18, 2.73]	1.000	80.00 / 66.67
outgoing slack count arrival (9,227 records)	-0.020	0.25 [-0.40, 0.91]	1.000	93.33 / 66.67
G12 Delay-support distribution	0.009	0.27 [-2.18, 2.73]	1.000	80.00 / 66.67
outgoing support count arrival (9,227 records)	-0.020	0.25 [-0.40, 0.91]	1.000	93.33 / 66.67
G14 Downstream-neighbor delay	0.009	0.27 [-2.18, 2.73]	1.000	80.00 / 66.67
neighbor delay count arrival (9,227 records)	-0.020	0.25 [-0.40, 0.91]	1.000	93.33 / 66.67
B09 Boundary pressure imbalance	-0.006	-0.17 [-1.69, 1.35]	1.000	73.33 / 66.67
cross boundary pressure balance 10 min (9,227 records)	-0.024	-1.56 [-2.42, -0.71]	0.013	100.00 / 100.00
G16 Incident-link imbalance	-0.006	-0.50 [-2.46, 1.45]	1.000	100.00 / 100.00
degree balance arrival (9,227 records)	-0.006	0.13 [-0.47, 0.73]	1.000	93.33 / 66.67
G12 Delay-support distribution	-0.005	-2.95 [-4.73, -1.18]	0.019	100.00 / 100.00
incoming support interquartile range departure (9,227 records)	-0.003	-0.59 [-1.26, 0.07]	0.693	100.00 / 100.00
G12 Delay-support distribution	-0.005	-2.93 [-4.71, -1.15]	0.021	100.00 / 100.00
incoming support standard deviation departure (9,227 records)	-0.003	-0.61 [-1.28, 0.06]	0.661	100.00 / 100.00
H12 PageRank	-0.004	-1.63 [-3.37, 0.12]	0.543	100.00 / 100.00
pagerank arrival (9,227 records)	0.017	0.94 [0.46, 1.42]	0.007	100.00 / 100.00
G17 Relation-type entropy	0.004	0.75 [-0.92, 2.42]	1.000	100.00 / 100.00
type entropy arrival (9,227 records)	0.008	0.03 [-0.44, 0.49]	1.000	66.67 / 66.67
B04 Boundary connections	0.004	0.56 [-1.29, 2.41]	1.000	100.00 / 100.00
leaving 5 min (9,227 records)	0.018	0.41 [-0.04, 0.86]	0.657	100.00 / 100.00

Continued on the next page.

Table S6 continued: Strategy-2 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G07 Degree by relation	0.002	0.05 [-1.68, 1.78]	1.000	73.33 / 33.33
outgoing station track departure (9,227 records)	0.001	0.28 [-0.23, 0.79]	1.000	93.33 / 100.00
H03 Event time from release	0.001	0.48 [-2.00, 2.96]	1.000	93.33 / 33.33
time from release arrival (9,227 records)	0.025	2.04 [1.08, 2.99]	0.004	100.00 / 100.00

Notes: Codes and definitions follow Table S1. Rank corr.: scenario-weighted Spearman correlation. Upper/lower lines: change probability/signed duration change. Effects per indicator standard deviation are in percentage points/minutes, respectively. Brackets contain 95% confidence intervals. Adjusted *p*: Benjamini–Yekutieli p-value; 1.1e-7 denotes 1.1×10^{-7}. Direction stable: unchanged coefficient sign after omitting one start-time group/one blockage location (percent). Pair differences are the following train value minus the leading train value. Record counts include only finite values of the indicated form.

S3.4 Strategy-2 running durations

Table S7: Complete indicator associations: Strategy-2 running durations.

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
Disruption conditions				
E05 Emergency stopping position	-0.310	-7.39 [-11.77, -3.00]	0.024	100.00 / 100.00
first stop index (4,061 records)	-0.309	-0.25 [-0.38, -0.11]	0.020	100.00 / 100.00
E20 Planned starts before release	0.289	8.81 [5.35, 12.27]	5.5e-4	100.00 / 100.00
planned starts before reopen (13,490 records)	0.285	0.83 [0.34, 1.31]	0.029	100.00 / 100.00
E12 Origin waiting count	0.241	6.33 [3.10, 9.55]	0.007	100.00 / 100.00
origin waiting (13,490 records)	0.238	0.26 [-0.11, 0.63]	1.000	100.00 / 100.00
E21 Train-state counts at blockage start	0.241	6.33 [3.10, 9.55]	0.007	100.00 / 100.00
origin hold (13,490 records)	0.238	0.26 [-0.11, 0.63]	1.000	100.00 / 100.00
E17 Early release concentration	-0.238	-6.85 [-9.92, -3.78]	0.002	100.00 / 100.00
release first5 fraction (13,490 records)	-0.235	-0.74 [-1.16, -0.32]	0.026	100.00 / 100.00
E02 Executed route fraction	-0.235	-6.45 [-8.76, -4.13]	2.2e-4	100.00 / 100.00
executed fraction (13,490 records)	-0.231	-0.52 [-0.75, -0.28]	0.005	100.00 / 100.00
E03 Current route position	-0.235	-6.45 [-8.76, -4.13]	2.2e-4	100.00 / 100.00
current route index (13,490 records)	-0.231	-0.52 [-0.75, -0.28]	0.005	100.00 / 100.00
E01 Train state at start time	0.232	7.12 [3.90, 10.33]	0.002	100.00 / 100.00
state 0 (13,490 records)	0.228	0.31 [-0.07, 0.70]	1.000	100.00 / 100.00
E16 Emergency backlog	0.221	5.71 [2.98, 8.45]	0.004	100.00 / 100.00
backlog (13,490 records)	0.217	0.07 [-0.20, 0.35]	1.000	100.00 / 66.67
X06 Planned entry from start time	0.220	6.01 [0.84, 11.18]	0.223	100.00 / 100.00
planned entry from start time (7,340 records)	0.222	0.33 [0.12, 0.53]	0.044	100.00 / 100.00
D09 Time from start time	0.220	6.67 [2.65, 10.70]	0.026	100.00 / 100.00
reference from start (13,490 records)	0.213	0.11 [-0.24, 0.45]	1.000	100.00 / 66.67
E07 Emergency arrival time	-0.218	-5.93 [-10.69, -1.17]	0.159	100.00 / 100.00
emergency arrival (4,061 records)	-0.218	-0.22 [-0.39, -0.05]	0.164	100.00 / 100.00
X07 Planned exit from start time	0.210	5.76 [0.65, 10.86]	0.253	100.00 / 100.00
planned exit from start time (7,340 records)	0.212	0.32 [0.12, 0.52]	0.050	100.00 / 100.00
E06 Time to emergency station	-0.205	-4.66 [-8.04, -1.28]	0.090	100.00 / 100.00
time to stop (4,061 records)	-0.204	-0.16 [-0.27, -0.05]	0.096	100.00 / 100.00
D01 Blockage start time	-0.195	-6.01 [-9.16, -2.87]	0.009	100.00 / 100.00
start (13,490 records)	-0.191	-0.25 [-0.76, 0.25]	1.000	100.00 / 100.00
E19 Committed event count	-0.195	-5.96 [-9.05, -2.87]	0.008	100.00 / 100.00
fixed events (13,490 records)	-0.191	-0.25 [-0.75, 0.26]	1.000	100.00 / 100.00
D09 Time from start time	0.191	5.09 [0.84, 9.33]	0.191	100.00 / 100.00
planned from start (13,490 records)	0.184	-0.23 [-0.62, 0.16]	1.000	100.00 / 66.67
E21 Train-state counts at blockage start	-0.185	-5.48 [-8.34, -2.63]	0.008	100.00 / 100.00
completed (13,490 records)	-0.181	-0.16 [-0.68, 0.36]	1.000	93.33 / 66.67
D03 Blockage duration	0.180	5.43 [2.85, 8.01]	0.004	100.00 / 100.00
duration (13,490 records)	0.181	1.08 [0.64, 1.52]	0.003	100.00 / 100.00
D08 Remaining blockage	0.180	5.43 [2.85, 8.01]	0.004	100.00 / 100.00
remaining (13,490 records)	0.181	1.08 [0.64, 1.52]	0.003	100.00 / 100.00
D20 Lost resource time	0.180	5.43 [2.85, 8.01]	0.004	100.00 / 100.00
lost resource minutes (13,490 records)	0.181	1.08 [0.64, 1.52]	0.003	100.00 / 100.00
I05 Predicted remaining duration	0.180	5.43 [2.85, 8.01]	0.004	100.00 / 100.00
predicted remaining (13,490 records)	0.181	1.08 [0.64, 1.52]	0.003	100.00 / 100.00
I06 Remaining-duration bounds	0.180	5.43 [2.85, 8.01]	0.004	100.00 / 100.00
lower (13,490 records)	0.181	1.08 [0.64, 1.52]	0.003	100.00 / 100.00
I06 Remaining-duration bounds	0.180	5.43 [2.85, 8.01]	0.004	100.00 / 100.00
upper (13,490 records)	0.181	1.08 [0.64, 1.52]	0.003	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
I09 Remaining-duration quantiles	0.180	5.43 [2.85, 8.01]	0.004	100.00 / 100.00
10th percentile (13,490 records)	0.181	1.08 [0.64, 1.52]	0.003	100.00 / 100.00
I09 Remaining-duration quantiles	0.180	5.43 [2.85, 8.01]	0.004	100.00 / 100.00
median (13,490 records)	0.181	1.08 [0.64, 1.52]	0.003	100.00 / 100.00
I09 Remaining-duration quantiles	0.180	5.43 [2.85, 8.01]	0.004	100.00 / 100.00
90th percentile (13,490 records)	0.181	1.08 [0.64, 1.52]	0.003	100.00 / 100.00
E01 Train state at start time	-0.180	-5.51 [-8.32, -2.70]	0.007	100.00 / 100.00
state 2 (13,490 records)	-0.175	-0.09 [-0.52, 0.33]	1.000	93.33 / 33.33
E04 Committed running remainder	-0.179	-4.59 [-7.01, -2.17]	0.009	100.00 / 100.00
committed remaining (13,490 records)	-0.174	-0.13 [-0.44, 0.18]	1.000	100.00 / 66.67
E26 Committed movement count	-0.167	-4.99 [-7.99, -1.98]	0.026	100.00 / 100.00
committed runs (13,490 records)	-0.164	-0.36 [-0.83, 0.12]	1.000	100.00 / 100.00
X18 Emergency origin hold	0.160	4.90 [2.02, 7.79]	0.022	100.00 / 100.00
origin emergency (13,490 records)	0.154	-0.25 [-0.47, -0.03]	0.302	100.00 / 100.00
X17 Remaining route fraction	0.130	3.75 [2.48, 5.03]	1.1e-4	100.00 / 100.00
remaining route fraction (13,490 records)	0.138	1.18 [0.67, 1.70]	0.004	100.00 / 100.00
D04 Start-time wave phase	-0.124	-4.00 [-6.69, -1.32]	0.055	100.00 / 100.00
start phase40 (13,490 records)	-0.122	-0.33 [-0.71, 0.04]	0.800	100.00 / 100.00
X02 Already past blocked section	-0.114	-3.37 [-5.37, -1.37]	0.024	100.00 / 100.00
passed at start time (7,340 records)	-0.114	-0.13 [-0.20, -0.06]	0.010	100.00 / 100.00
X03 Planned clearance before start time	-0.114	-3.37 [-5.37, -1.37]	0.024	100.00 / 100.00
planned clear before (7,340 records)	-0.114	-0.13 [-0.20, -0.06]	0.010	100.00 / 100.00
X12 Can clear before start time	-0.114	-3.37 [-5.37, -1.37]	0.024	100.00 / 100.00
can clear without emergency (7,340 records)	-0.114	-0.13 [-0.20, -0.06]	0.010	100.00 / 100.00
E01 Train state at start time	-0.106	-3.25 [-4.54, -1.96]	6.0e-4	100.00 / 100.00
state 1 (13,490 records)	-0.106	-0.38 [-0.54, -0.23]	0.003	100.00 / 100.00
E09 Already stopped at start time	-0.106	-3.25 [-4.54, -1.96]	6.0e-4	100.00 / 100.00
already stopped (13,490 records)	-0.106	-0.38 [-0.54, -0.23]	0.003	100.00 / 100.00
D10 Time from reopening	0.103	2.89 [-0.19, 5.98]	0.548	100.00 / 100.00
reference from release (13,490 records)	0.093	-0.86 [-1.43, -0.29]	0.072	100.00 / 100.00
X08 Sections to blockage entry	0.096	2.77 [-1.93, 7.47]	1.000	100.00 / 100.00
intervals to entry (7,340 records)	0.098	0.20 [0.03, 0.37]	0.292	100.00 / 100.00
E21 Train-state counts at blockage start	0.085	2.79 [0.37, 5.21]	0.230	100.00 / 100.00
finish committed run then hold (13,490 records)	0.082	-0.30 [-0.57, -0.03]	0.352	100.00 / 66.67
E13 Station waiting count	0.057	2.53 [0.02, 5.04]	0.415	100.00 / 100.00
station waiting (13,490 records)	0.055	-0.32 [-0.60, -0.04]	0.304	100.00 / 66.67
X10 Signed blockage distance	0.048	1.12 [-2.22, 4.46]	1.000	100.00 / 100.00
signed distance to blockage (7,340 records)	0.052	0.19 [0.04, 0.34]	0.177	100.00 / 100.00
E08 Minimum emergency hold	0.046	2.00 [-0.86, 4.85]	1.000	100.00 / 100.00
minimum emergency hold (13,490 records)	0.041	-0.49 [-0.78, -0.21]	0.028	100.00 / 100.00
D10 Time from reopening	-0.044	-1.19 [-3.00, 0.61]	1.000	100.00 / 100.00
planned from release (13,490 records)	-0.049	-0.99 [-1.50, -0.49]	0.009	100.00 / 100.00
D11 Blocked section	0.038	1.16 [-1.03, 3.34]	1.000	100.00 / 100.00
U13 H1 (13,490 records)	0.041	0.56 [0.08, 1.04]	0.291	100.00 / 100.00
X09 Minimum time to entry	0.032	0.54 [-3.75, 4.83]	1.000	93.33 / 66.67
minrun to entry (7,340 records)	0.035	0.12 [-0.03, 0.27]	1.000	100.00 / 100.00
D02 Blockage end	0.030	0.25 [-1.86, 2.36]	1.000	80.00 / 66.67
end (13,490 records)	0.033	0.66 [0.15, 1.17]	0.168	100.00 / 100.00
X01 Blocked-section exposure	-0.027	-0.84 [-3.46, 1.78]	1.000	100.00 / 66.67
crosses closed (13,490 records)	-0.033	-1.02 [-1.43, -0.62]	0.003	100.00 / 100.00
D11 Blocked section	-0.025	-0.77 [-3.04, 1.51]	1.000	93.33 / 100.00
U11 U12 (13,490 records)	-0.025	0.11 [-0.30, 0.52]	1.000	93.33 / 50.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
X05 Planned overlap fraction	0.023	0.69 [-1.75, 3.12]	1.000	93.33 / 100.00
planned overlap fraction (13,490 records)	0.018	-0.61 [-0.91, -0.31]	0.009	100.00 / 100.00
X13 Arrival before reopening	0.023	0.70 [-1.71, 3.12]	1.000	93.33 / 100.00
reaches stop before reopen (13,490 records)	0.018	-0.61 [-0.92, -0.31]	0.009	100.00 / 100.00
D11 Blocked section	-0.013	-0.39 [-2.67, 1.89]	1.000	80.00 / 50.00
H1 C (13,490 records)	-0.016	-0.67 [-1.00, -0.33]	0.009	100.00 / 100.00
X15 Earliest entry from release	0.012	0.45 [-1.59, 2.49]	1.000	93.33 / 100.00
earliest entry minus release (7,340 records)	0.013	0.07 [-0.01, 0.16]	0.850	100.00 / 100.00
E21 Train-state counts at blockage start	0.011	0.33 [-2.06, 2.73]	1.000	93.33 / 66.67
hold current station (13,490 records)	0.010	-0.25 [-0.61, 0.12]	1.000	100.00 / 100.00
X19 Emergency restriction type	-0.011	-1.87 [-3.25, -0.49]	0.101	100.00 / 100.00
direct stop type (13,490 records)	-0.015	-0.63 [-0.90, -0.35]	0.004	100.00 / 100.00
X16 Blockage-only waiting	0.010	-0.73 [-2.55, 1.08]	1.000	100.00 / 100.00
blockage only wait (13,490 records)	0.005	-0.53 [-0.77, -0.30]	0.004	100.00 / 100.00
X04 Planned blockage overlap	-0.010	-0.65 [-2.56, 1.25]	1.000	93.33 / 100.00
planned blockage overlap (13,490 records)	-0.014	-0.63 [-0.92, -0.35]	0.005	100.00 / 100.00
D05 Reopening wave phase	-0.008	-0.29 [-2.23, 1.65]	1.000	93.33 / 33.33
end phase40 (13,490 records)	-0.009	-0.04 [-0.37, 0.30]	1.000	66.67 / 66.67
X11 At blockage entrance	-0.002	-0.06 [-1.03, 0.90]	1.000	66.67 / 66.67
at closed entry (13,490 records)	-0.005	-0.29 [-0.43, -0.16]	0.005	100.00 / 100.00
X14 Earliest-entry estimate	0.000	0.23 [-3.64, 4.10]	1.000	66.67 / 66.67
earliest entry no other conflicts (7,340 records)	0.002	0.07 [-0.05, 0.20]	1.000	100.00 / 100.00
Timetable characteristics				
M06 Remaining candidate decisions	0.260	7.40 [5.48, 9.33]	2.0e-6	100.00 / 100.00
remaining stopping decisions (13,490 records)	0.252	0.33 [-0.02, 0.68]	0.688	100.00 / 100.00
F13 Minimum running time	-0.249	-7.63 [-9.98, -5.28]	2.6e-5	100.00 / 100.00
minimum run (13,490 records)	-0.239	-0.04 [-0.38, 0.30]	1.000	93.33 / 33.33
M04 Propagated domain	-0.249	-7.63 [-9.98, -5.28]	2.6e-5	100.00 / 100.00
domain lower (13,490 records)	-0.239	-0.04 [-0.38, 0.30]	1.000	93.33 / 33.33
T04 Planned running duration	-0.249	-7.63 [-9.98, -5.28]	2.6e-5	100.00 / 100.00
planned run next (13,490 records)	-0.239	-0.04 [-0.38, 0.30]	1.000	93.33 / 33.33
T15 Fast-train indicator	0.249	7.63 [5.28, 9.98]	2.6e-5	100.00 / 100.00
fast (13,490 records)	0.239	0.04 [-0.30, 0.38]	1.000	93.33 / 33.33
T17 Total planned journey time	-0.249	-7.63 [-9.98, -5.28]	2.6e-5	100.00 / 100.00
planned total (13,490 records)	-0.239	-0.04 [-0.38, 0.30]	1.000	93.33 / 33.33
M06 Remaining candidate decisions	0.204	6.00 [3.95, 8.04]	1.1e-4	100.00 / 100.00
remaining order decisions (13,490 records)	0.208	1.00 [0.50, 1.51]	0.009	100.00 / 100.00
T21 Nearby scheduled-stop distance	0.199	6.10 [4.47, 7.73]	3.0e-6	100.00 / 100.00
next scheduled-stop distance (13,490 records)	0.199	0.48 [0.13, 0.83]	0.124	100.00 / 100.00
T12 Planned local flow	-0.192	-5.25 [-7.96, -2.55]	0.008	100.00 / 100.00
planned flow 5 min (13,490 records)	-0.190	-0.35 [-0.55, -0.14]	0.030	100.00 / 100.00
T21 Nearby scheduled-stop distance	0.189	5.67 [4.30, 7.03]	5.7e-7	100.00 / 100.00
previous scheduled-stop distance (13,490 records)	0.189	0.55 [0.22, 0.87]	0.031	100.00 / 100.00
T12 Planned local flow	-0.175	-4.40 [-6.94, -1.86]	0.020	100.00 / 100.00
planned flow 10 min (13,490 records)	-0.173	-0.40 [-0.60, -0.21]	0.007	100.00 / 100.00
T12 Planned local flow	-0.164	-4.26 [-6.97, -1.56]	0.038	100.00 / 100.00
planned flow 20 min (13,490 records)	-0.163	-0.50 [-0.73, -0.26]	0.006	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
M06 Remaining candidate decisions	0.135	4.23 [2.91, 5.54]	2.8e-5	100.00 / 100.00
remaining track-assignment decisions (13,490 records)	0.143	1.28 [0.73, 1.84]	0.003	100.00 / 100.00
T10 Planned exit buffer	-0.134	-3.91 [-5.62, -2.20]	0.002	100.00 / 100.00
previous buffer (12,890 records)	-0.127	0.05 [-0.22, 0.32]	1.000	100.00 / 100.00
M06 Remaining candidate decisions	0.130	3.94 [2.65, 5.23]	6.6e-5	100.00 / 100.00
remaining duration decisions (13,490 records)	0.138	1.23 [0.70, 1.76]	0.003	100.00 / 100.00
T22 Route extent	0.130	3.75 [2.48, 5.03]	1.1e-4	100.00 / 100.00
remaining edges (13,490 records)	0.138	1.18 [0.67, 1.70]	0.004	100.00 / 100.00
T06 Planned track label	-0.129	-3.64 [-5.99, -1.29]	0.043	100.00 / 100.00
planned track index (12,290 records)	-0.127	-0.32 [-0.59, -0.06]	0.241	100.00 / 100.00
F14 Allowed stopping mode	0.108	3.30 [1.69, 4.91]	0.005	100.00 / 100.00
optional (13,490 records)	0.096	-0.41 [-0.65, -0.17]	0.030	100.00 / 100.00
T14 Departure wave	0.102	3.25 [0.30, 6.20]	0.280	100.00 / 100.00
origin wave40 (13,490 records)	0.100	0.09 [-0.29, 0.46]	1.000	93.33 / 100.00
T09 Planned entry buffer	-0.087	-2.90 [-4.36, -1.45]	0.006	100.00 / 100.00
previous buffer (12,890 records)	-0.080	0.04 [-0.27, 0.36]	1.000	100.00 / 66.67
F12 Minimum dwell	-0.074	-2.27 [-3.55, -1.00]	0.015	100.00 / 100.00
minimum dwell (13,490 records)	-0.064	0.47 [0.20, 0.75]	0.026	100.00 / 100.00
F14 Allowed stopping mode	-0.074	-2.27 [-3.55, -1.00]	0.015	100.00 / 100.00
required (13,490 records)	-0.064	0.47 [0.20, 0.75]	0.026	100.00 / 100.00
T03 Planned dwell	-0.074	-2.27 [-3.55, -1.00]	0.015	100.00 / 100.00
planned dwell (13,490 records)	-0.064	0.47 [0.20, 0.75]	0.026	100.00 / 100.00
T05 Planned stopping mode	-0.074	-2.27 [-3.55, -1.00]	0.015	100.00 / 100.00
planned stop (13,490 records)	-0.064	0.47 [0.20, 0.75]	0.026	100.00 / 100.00
T18 Remaining planned journey	0.074	1.65 [0.24, 3.05]	0.213	100.00 / 100.00
remaining planned next endpoint (13,490 records)	0.084	1.16 [0.58, 1.74]	0.009	100.00 / 100.00
T13 Planned train-type pattern	-0.069	-1.46 [-3.31, 0.39]	0.958	100.00 / 100.00
alternation 20 min (3,241 records)	-0.064	0.54 [0.10, 0.97]	0.227	100.00 / 100.00
T13 Planned train-type pattern	-0.058	-1.29 [-2.74, 0.16]	0.654	100.00 / 100.00
fast fraction 20 min (8,008 records)	-0.059	-0.33 [-0.87, 0.21]	1.000	93.33 / 100.00
T13 Planned train-type pattern	-0.056	-1.26 [-2.67, 0.16]	0.655	100.00 / 100.00
fast fraction 10 min (6,806 records)	-0.057	-0.52 [-1.07, 0.02]	0.632	100.00 / 100.00
M03 Original domain width	0.055	1.71 [-0.79, 4.20]	1.000	100.00 / 100.00
raw domain width (13,490 records)	0.057	0.49 [0.13, 0.85]	0.126	100.00 / 100.00
T13 Planned train-type pattern	-0.053	-1.14 [-2.47, 0.20]	0.756	100.00 / 100.00
fast fraction 5 min (5,952 records)	-0.054	-0.57 [-1.17, 0.04]	0.688	100.00 / 100.00
T02 Planned departure time	-0.048	-1.48 [-4.04, 1.08]	1.000	100.00 / 100.00
planned departure (13,490 records)	-0.050	-0.49 [-0.85, -0.14]	0.122	100.00 / 100.00
T01 Planned arrival time	-0.046	-1.43 [-4.01, 1.16]	1.000	100.00 / 100.00
planned arrival (13,490 records)	-0.048	-0.51 [-0.87, -0.15]	0.110	100.00 / 100.00
T18 Remaining planned journey	0.043	0.92 [-0.61, 2.44]	1.000	100.00 / 100.00
remaining planned (13,490 records)	0.053	1.10 [0.52, 1.67]	0.012	100.00 / 100.00
T19 Remaining minimum journey	0.043	0.92 [-0.61, 2.44]	1.000	100.00 / 100.00
remaining minimum (13,490 records)	0.053	1.10 [0.52, 1.67]	0.012	100.00 / 100.00
T24 Planned terminal position	-0.038	-0.87 [-2.62, 0.88]	1.000	100.00 / 100.00
terminal from release (13,490 records)	-0.039	-0.62 [-0.97, -0.28]	0.021	100.00 / 100.00
T11 Planned track buffer	0.023	1.20 [-0.70, 3.11]	1.000	100.00 / 100.00
planned track buffer (10,690 records)	0.032	0.75 [0.39, 1.12]	0.008	100.00 / 100.00
T13 Planned train-type pattern	0.019	0.41 [-0.84, 1.65]	1.000	93.33 / 66.67
alternation 10 min (1,438 records)	0.021	0.01 [-0.03, 0.05]	1.000	93.33 / 66.67
T20 Remaining scheduled stops	0.009	0.07 [-1.71, 1.85]	1.000	66.67 / 66.67
remaining scheduled stops (13,490 records)	0.019	1.02 [0.43, 1.60]	0.028	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
T24 Planned terminal position	-0.007	-1.03 [-3.38, 1.32]	1.000	100.00 / 100.00
planned terminal (13,490 records)	-0.004	0.07 [-0.36, 0.49]	1.000	93.33 / 66.67
M04 Propagated domain	0.006	0.90 [-1.31, 3.10]	1.000	100.00 / 100.00
domain width (13,490 records)	0.007	0.38 [0.11, 0.65]	0.113	100.00 / 100.00
M05 Reference adjustment room	0.006	0.90 [-1.31, 3.10]	1.000	100.00 / 100.00
extendable (13,490 records)	0.007	0.38 [0.11, 0.65]	0.113	100.00 / 100.00
M04 Propagated domain	0.003	0.74 [-1.50, 2.98]	1.000	93.33 / 100.00
domain upper (13,490 records)	0.004	0.38 [0.11, 0.65]	0.108	100.00 / 100.00
Train interactions				
H22 Shared downstream support	-0.377	-13.21 [-17.54, -8.88]	6.6e-5	100.00 / 100.00
arrival departure downstream Jaccard (8,228 records)	-0.376	-1.39 [-1.95, -0.83]	0.003	100.00 / 100.00
H17 Window competition size	0.208	8.34 [4.24, 12.45]	0.005	100.00 / 100.00
competitor trains 20 min (13,490 records)	0.195	-0.29 [-0.63, 0.04]	0.807	100.00 / 100.00
H18 Co-window occupations	0.208	8.34 [4.24, 12.45]	0.005	100.00 / 100.00
competition set memberships 20 min (13,490 records)	0.195	-0.29 [-0.63, 0.04]	0.807	100.00 / 100.00
E14 Reference release rank	0.167	5.02 [-0.60, 10.64]	0.651	100.00 / 100.00
release queue rank (4,061 records)	0.168	0.20 [0.02, 0.37]	0.322	100.00 / 100.00
H17 Window competition size	0.163	5.77 [2.04, 9.51]	0.043	100.00 / 100.00
competitor trains 10 min (13,490 records)	0.152	-0.31 [-0.64, 0.03]	0.720	100.00 / 100.00
H18 Co-window occupations	0.163	5.77 [2.04, 9.51]	0.043	100.00 / 100.00
competition set memberships 10 min (13,490 records)	0.152	-0.31 [-0.64, 0.03]	0.720	100.00 / 100.00
H21 Shared-visit coupling	0.148	4.36 [1.29, 7.42]	0.077	100.00 / 100.00
same visit classes (13,490 records)	0.136	-0.47 [-0.85, -0.09]	0.213	100.00 / 100.00
H17 Window competition size	0.100	3.74 [0.71, 6.76]	0.166	100.00 / 100.00
competitor trains 5 min (13,490 records)	0.093	-0.23 [-0.53, 0.08]	1.000	100.00 / 66.67
H18 Co-window occupations	0.100	3.74 [0.71, 6.76]	0.166	100.00 / 100.00
competition set memberships 5 min (13,490 records)	0.093	-0.23 [-0.53, 0.08]	1.000	100.00 / 66.67
E15 Queue fast-train share	0.095	2.76 [0.08, 5.44]	0.378	100.00 / 100.00
queue fast fraction (4,061 records)	0.095	0.10 [0.01, 0.19]	0.295	100.00 / 100.00
B17 Relation-layer overlap	0.073	2.81 [0.17, 5.45]	0.331	100.00 / 100.00
entry headway station track Jaccard (13,490 records)	0.070	0.01 [-0.46, 0.47]	1.000	53.33 / 33.33
B17 Relation-layer overlap	0.073	2.81 [0.17, 5.45]	0.331	100.00 / 100.00
exit headway station track Jaccard (13,490 records)	0.070	0.01 [-0.46, 0.47]	1.000	53.33 / 33.33
E11 Co-located emergency trains	0.043	2.38 [-1.06, 5.82]	1.000	100.00 / 100.00
same stop count (13,490 records)	0.037	-0.47 [-0.77, -0.17]	0.050	100.00 / 100.00
H20 Support distance	-0.027	-1.04 [-2.61, 0.54]	1.000	100.00 / 100.00
nearest optional stop index (5,871 records)	-0.044	-0.69 [-1.07, -0.30]	0.031	100.00 / 100.00
Infrastructure utilization				
R13 Adjacent-station occupation	0.385	12.51 [9.90, 15.12]	3.3e-8	100.00 / 100.00
downstream occupancy (13,490 records)	0.386	1.53 [0.90, 2.15]	0.003	100.00 / 100.00
R08 Exit load	0.356	12.41 [8.98, 15.84]	5.4e-6	100.00 / 100.00
exit load 5 min (13,490 records)	0.340	-0.09 [-0.29, 0.11]	1.000	93.33 / 66.67
R10 Intersecting dwell distribution	0.334	1.61 [0.65, 2.58]	0.025	100.00 / 100.00
waiting duration 10 min interquartile range (12,290 records)	0.329	0.69 [0.12, 1.27]	0.245	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R10 Intersecting dwell distribution waiting duration 10 min maximum (12,290 records)	0.326 0.327	2.81 [1.52, 4.11] 1.59 [0.85, 2.32]	0.003 0.005	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min standard deviation (12,290 records)	0.322 0.318	2.08 [0.91, 3.25] 0.85 [0.15, 1.55]	0.016 0.241	100.00 / 100.00 100.00 / 100.00
R04 Utilization utilization 20 min (13,490 records)	0.320 0.316	11.50 [7.30, 15.69] 0.92 [0.53, 1.30]	2.8e-4 0.003	100.00 / 100.00 100.00 / 100.00
R12 Demand/capacity proxy demand capacity 20 min (13,490 records)	0.320 0.316	11.50 [7.30, 15.69] 0.92 [0.53, 1.30]	2.8e-4 0.003	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min 90th percentile (12,290 records)	0.318 0.320	2.56 [1.21, 3.91] 1.65 [0.85, 2.45]	0.009 0.007	100.00 / 100.00 100.00 / 100.00
R01 Mean occupation mean occupancy 10 min next endpoint (13,490 records)	0.317 0.321	11.10 [8.49, 13.71] 2.22 [1.41, 3.03]	4.6e-7 0.002	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min 90th percentile (12,290 records)	0.316 0.318	2.64 [1.33, 3.95] 1.65 [0.86, 2.44]	0.006 0.006	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min maximum (12,290 records)	0.313 0.315	2.70 [1.40, 4.00] 1.61 [0.86, 2.37]	0.004 0.005	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min median (12,290 records)	0.309 0.312	2.06 [0.58, 3.54] 1.71 [0.76, 2.67]	0.084 0.022	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min maximum (12,290 records)	0.306 0.307	3.22 [1.83, 4.62] 1.53 [0.79, 2.27]	0.001 0.007	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min median (12,290 records)	0.304 0.307	1.96 [0.49, 3.42] 1.66 [0.72, 2.60]	0.106 0.025	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min 90th percentile (12,290 records)	0.302 0.304	2.94 [1.57, 4.30] 1.62 [0.84, 2.40]	0.003 0.007	100.00 / 100.00 100.00 / 100.00
R08 Exit load exit load 10 min (13,490 records)	0.301 0.287	10.77 [6.42, 15.12] -0.24 [-0.57, 0.09]	6.9e-4 1.000	100.00 / 100.00 100.00 / 66.67
R10 Intersecting dwell distribution waiting duration 5 min mean (12,290 records)	0.288 0.291	2.24 [0.68, 3.80] 1.70 [0.72, 2.68]	0.071 0.027	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min interquartile range (12,290 records)	0.287 0.283	1.49 [0.53, 2.45] 0.71 [0.10, 1.32]	0.041 0.280	100.00 / 100.00 100.00 / 100.00
R01 Mean occupation mean occupancy 20 min (13,490 records)	0.286 0.281	8.97 [4.91, 13.03] 0.41 [0.16, 0.67]	0.002 0.041	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min mean (12,290 records)	0.286 0.289	2.35 [0.77, 3.93] 1.78 [0.77, 2.80]	0.055 0.025	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min standard deviation (12,290 records)	0.285 0.281	2.15 [1.09, 3.21] 0.91 [0.16, 1.65]	0.005 0.233	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min sum (13,490 records)	0.279 0.277	3.50 [1.55, 5.45] 0.86 [0.40, 1.32]	0.015 0.017	100.00 / 100.00 100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R10 Intersecting dwell distribution	0.277	1.71 [0.61, 2.81]	0.041	100.00 / 100.00
waiting duration 20 min interquartile range (12,290 records)	0.274	0.82 [0.14, 1.51]	0.252	100.00 / 100.00
R10 Intersecting dwell distribution	0.271	2.37 [1.07, 3.67]	0.013	100.00 / 100.00
waiting duration 20 min standard deviation (12,290 records)	0.269	0.92 [0.16, 1.68]	0.237	100.00 / 100.00
R04 Utilization	0.269	9.87 [5.86, 13.88]	7.3e-4	100.00 / 100.00
utilization 10 min (13,490 records)	0.266	0.74 [0.37, 1.12]	0.009	100.00 / 100.00
R12 Demand/capacity proxy	0.269	9.87 [5.86, 13.88]	7.3e-4	100.00 / 100.00
demand capacity 10 min (13,490 records)	0.266	0.74 [0.37, 1.12]	0.009	100.00 / 100.00
R09 Reference headway buffer	-0.264	-4.67 [-7.19, -2.14]	0.012	100.00 / 100.00
1 previous buffer (12,890 records)	-0.251	0.44 [0.04, 0.84]	0.383	100.00 / 100.00
R10 Intersecting dwell distribution	0.264	2.35 [0.72, 3.99]	0.071	100.00 / 100.00
waiting duration 10 min sum (13,490 records)	0.263	0.88 [0.38, 1.38]	0.025	100.00 / 100.00
R10 Intersecting dwell distribution	0.261	2.54 [0.81, 4.26]	0.060	100.00 / 100.00
waiting duration 20 min mean (12,290 records)	0.264	1.81 [0.75, 2.86]	0.029	100.00 / 100.00
R06 Saturation persistence	0.254	8.62 [5.31, 11.94]	4.7e-4	100.00 / 100.00
longest saturation 20 min (13,490 records)	0.249	0.90 [0.34, 1.46]	0.042	100.00 / 100.00
R07 Entry load	0.253	10.20 [5.60, 14.80]	0.002	100.00 / 100.00
entry load 20 min (13,490 records)	0.240	-0.28 [-0.61, 0.05]	0.881	100.00 / 100.00
R15 Free windows	-0.246	-9.01 [-12.61, -5.41]	6.4e-4	100.00 / 100.00
free window total 20 min (13,490 records)	-0.241	-0.70 [-1.16, -0.23]	0.071	100.00 / 100.00
B07 Regional saturation	0.246	9.01 [5.41, 12.61]	6.4e-4	100.00 / 100.00
saturated fraction 20 min (13,490 records)	0.241	0.70 [0.23, 1.16]	0.071	100.00 / 100.00
R05 Saturated fraction	0.246	9.01 [5.41, 12.61]	6.4e-4	100.00 / 100.00
saturated fraction 20 min (13,490 records)	0.241	0.70 [0.23, 1.16]	0.071	100.00 / 100.00
R10 Intersecting dwell distribution	0.237	1.54 [-0.06, 3.14]	0.499	100.00 / 100.00
waiting duration 5 min minimum (12,290 records)	0.243	1.46 [0.37, 2.55]	0.144	100.00 / 100.00
R15 Free windows	-0.234	-7.38 [-10.92, -3.84]	0.004	100.00 / 100.00
free window maximum 20 min (13,490 records)	-0.228	-0.25 [-0.58, 0.08]	1.000	100.00 / 100.00
B13 Adjacent-window difference	-0.232	-5.08 [-8.01, -2.14]	0.020	100.00 / 100.00
adjacent occupancy difference 5 min (13,490 records)	-0.231	-0.58 [-0.88, -0.28]	0.010	100.00 / 100.00
R17 Occupation change rate	-0.232	-5.08 [-8.01, -2.14]	0.020	100.00 / 100.00
occupancy rate 5 min (13,490 records)	-0.231	-0.58 [-0.88, -0.28]	0.010	100.00 / 100.00
R06 Saturation persistence	0.231	9.80 [6.39, 13.22]	1.4e-4	100.00 / 100.00
longest saturation 10 min (13,490 records)	0.225	0.78 [0.31, 1.24]	0.035	100.00 / 100.00
R01 Mean occupation	0.230	7.64 [3.62, 11.66]	0.009	100.00 / 100.00
mean occupancy 10 min (13,490 records)	0.225	0.27 [-0.01, 0.54]	0.614	100.00 / 100.00
R10 Intersecting dwell distribution	0.230	1.93 [0.53, 3.34]	0.092	100.00 / 100.00
waiting duration 5 min sum (13,490 records)	0.231	0.93 [0.42, 1.45]	0.020	100.00 / 100.00
R10 Intersecting dwell distribution	0.227	2.07 [0.49, 3.65]	0.118	100.00 / 100.00
waiting duration 20 min median (12,290 records)	0.233	1.70 [0.69, 2.71]	0.031	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
F18 Next stopping opportunity	0.222	6.93 [5.14, 8.72]	2.0e-6	100.00 / 100.00
next stop distance (10,415 records)	0.220	0.72 [0.20, 1.23]	0.119	100.00 / 100.00
R04 Utilization	0.220	7.48 [4.07, 10.88]	0.002	100.00 / 100.00
utilization 5 min (13,490 records)	0.219	0.64 [0.26, 1.01]	0.030	100.00 / 100.00
R12 Demand/capacity proxy	0.220	7.48 [4.07, 10.88]	0.002	100.00 / 100.00
demand capacity 5 min (13,490 records)	0.219	0.64 [0.26, 1.01]	0.030	100.00 / 100.00
B07 Regional saturation	0.217	8.62 [5.10, 12.14]	7.4e-4	100.00 / 100.00
saturated fraction 10 min (13,490 records)	0.211	0.59 [0.16, 1.02]	0.120	100.00 / 100.00
R05 Saturated fraction	0.217	8.62 [5.10, 12.14]	7.4e-4	100.00 / 100.00
saturated fraction 10 min (13,490 records)	0.211	0.59 [0.16, 1.02]	0.120	100.00 / 100.00
R15 Free windows	-0.217	-8.62 [-12.14, -5.10]	7.4e-4	100.00 / 100.00
free window total 10 min (13,490 records)	-0.211	-0.59 [-1.02, -0.16]	0.120	100.00 / 100.00
R09 Reference headway buffer	-0.215	-4.35 [-6.74, -1.97]	0.013	100.00 / 100.00
0 previous buffer (12,890 records)	-0.204	0.44 [0.04, 0.84]	0.380	100.00 / 100.00
R08 Exit load	0.213	7.18 [2.87, 11.49]	0.026	100.00 / 100.00
exit load 20 min (13,490 records)	0.202	-0.26 [-0.48, -0.04]	0.278	100.00 / 100.00
R13 Adjacent-station occupation	0.209	7.45 [5.15, 9.75]	2.7e-5	100.00 / 100.00
occupancy difference (13,490 records)	0.216	1.36 [0.80, 1.92]	0.003	100.00 / 100.00
R10 Intersecting dwell distribution	0.208	8.34 [4.24, 12.45]	0.005	100.00 / 100.00
waiting duration 20 min count (13,490 records)	0.195	-0.29 [-0.63, 0.04]	0.807	100.00 / 100.00
R14 Window competitors	0.208	8.34 [4.24, 12.45]	0.005	100.00 / 100.00
window competitors 20 min (13,490 records)	0.195	-0.29 [-0.63, 0.04]	0.807	100.00 / 100.00
R11 Flow imbalance	-0.207	-4.61 [-7.55, -1.67]	0.039	100.00 / 100.00
arrivals minus departures 10 min (13,490 records)	-0.207	-0.47 [-0.75, -0.19]	0.032	100.00 / 100.00
R15 Free windows	-0.204	-6.10 [-9.44, -2.76]	0.013	100.00 / 100.00
free window maximum 10 min (13,490 records)	-0.199	-0.24 [-0.60, 0.11]	1.000	100.00 / 100.00
H19 Peak demand/capacity	0.199	6.01 [3.14, 8.88]	0.004	100.00 / 100.00
peak demand capacity 20 min (13,490 records)	0.196	0.24 [-0.02, 0.51]	0.688	100.00 / 100.00
R11 Flow imbalance	-0.196	-5.39 [-8.72, -2.07]	0.031	100.00 / 100.00
arrivals minus departures 5 min (13,490 records)	-0.196	-0.53 [-0.80, -0.26]	0.011	100.00 / 100.00
R07 Entry load	0.195	5.67 [1.55, 9.80]	0.092	100.00 / 100.00
entry load 10 min (13,490 records)	0.183	-0.38 [-0.69, -0.06]	0.245	100.00 / 100.00
R03 Minimum free capacity	-0.194	-5.79 [-8.00, -3.58]	4.7e-4	100.00 / 100.00
minimum free 20 min (13,490 records)	-0.193	-0.48 [-0.70, -0.27]	0.005	100.00 / 100.00
R01 Mean occupation	0.185	5.99 [2.31, 9.67]	0.030	100.00 / 100.00
mean occupancy 5 min (13,490 records)	0.182	0.17 [-0.12, 0.47]	1.000	100.00 / 100.00
F06 Directional pool size	0.181	5.53 [4.12, 6.95]	1.9e-6	100.00 / 100.00
direction pool next endpoint (13,490 records)	0.190	1.59 [0.87, 2.30]	0.005	100.00 / 100.00
H19 Peak demand/capacity	0.178	5.38 [2.55, 8.21]	0.009	100.00 / 100.00
peak demand capacity 10 min (13,490 records)	0.174	0.23 [-0.04, 0.51]	0.910	100.00 / 100.00
R03 Minimum free capacity	-0.174	-5.22 [-7.35, -3.10]	7.4e-4	100.00 / 100.00
minimum free 10 min (13,490 records)	-0.173	-0.49 [-0.71, -0.27]	0.005	100.00 / 100.00
R06 Saturation persistence	0.173	8.84 [5.81, 11.86]	1.1e-4	100.00 / 100.00
longest saturation 5 min (13,490 records)	0.168	0.62 [0.17, 1.06]	0.119	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R16 Planned-arrival overlap	-0.166	-4.54 [-6.83, -2.25]	0.007	100.00 / 100.00
planned arrival release overlap 5 min (13,490 records)	-0.164	-0.32 [-0.50, -0.14]	0.025	100.00 / 100.00
R10 Intersecting dwell distribution	0.163	5.77 [2.04, 9.51]	0.043	100.00 / 100.00
waiting duration 10 min count (13,490 records)	0.152	-0.31 [-0.64, 0.03]	0.720	100.00 / 100.00
R14 Window competitors	0.163	5.77 [2.04, 9.51]	0.043	100.00 / 100.00
window competitors 10 min (13,490 records)	0.152	-0.31 [-0.64, 0.03]	0.720	100.00 / 100.00
R15 Free windows	-0.162	-7.54 [-10.66, -4.41]	8.7e-4	100.00 / 100.00
free window total 5 min (13,490 records)	-0.157	-0.49 [-0.91, -0.06]	0.296	100.00 / 100.00
B07 Regional saturation	0.162	7.54 [4.41, 10.66]	8.7e-4	100.00 / 100.00
saturated fraction 5 min (13,490 records)	0.157	0.49 [0.06, 0.91]	0.296	100.00 / 100.00
R05 Saturated fraction	0.162	7.54 [4.41, 10.66]	8.7e-4	100.00 / 100.00
saturated fraction 5 min (13,490 records)	0.157	0.49 [0.06, 0.91]	0.296	100.00 / 100.00
R15 Free windows	0.158	5.93 [2.75, 9.12]	0.011	100.00 / 100.00
free window count 20 min (13,490 records)	0.151	-0.27 [-0.51, -0.03]	0.322	100.00 / 100.00
E10 Emergency pool capacity	-0.157	-4.41 [-7.14, -1.69]	0.031	100.00 / 100.00
emergency pool capacity (4,061 records)	-0.157	-0.15 [-0.23, -0.06]	0.029	100.00 / 100.00
R15 Free windows	-0.150	-5.26 [-8.33, -2.20]	0.021	100.00 / 100.00
free window maximum 5 min (13,490 records)	-0.146	-0.26 [-0.64, 0.12]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution	0.149	1.55 [-0.16, 3.26]	0.622	100.00 / 100.00
waiting duration 10 min minimum (12,290 records)	0.158	1.56 [0.36, 2.75]	0.165	100.00 / 100.00
R02 Peak occupation	0.147	4.68 [1.37, 7.99]	0.078	100.00 / 100.00
peak occupancy 20 min (13,490 records)	0.140	-0.17 [-0.46, 0.11]	1.000	100.00 / 100.00
R16 Planned-arrival overlap	-0.143	-3.79 [-5.88, -1.69]	0.014	100.00 / 100.00
planned arrival release overlap 10 min (13,490 records)	-0.141	-0.38 [-0.56, -0.20]	0.006	100.00 / 100.00
R03 Minimum free capacity	-0.142	-4.32 [-6.26, -2.38]	0.002	100.00 / 100.00
minimum free 5 min (13,490 records)	-0.142	-0.50 [-0.74, -0.26]	0.006	100.00 / 100.00
R10 Intersecting dwell distribution	0.137	1.52 [-0.29, 3.33]	0.788	100.00 / 100.00
waiting duration 20 min minimum (12,290 records)	0.147	1.51 [0.27, 2.75]	0.237	100.00 / 100.00
H19 Peak demand/capacity	0.137	4.13 [1.62, 6.64]	0.028	100.00 / 100.00
peak demand capacity 5 min (13,490 records)	0.134	0.24 [-0.04, 0.51]	0.908	100.00 / 100.00
R13 Adjacent-station occupation	0.137	3.10 [0.11, 6.09]	0.371	100.00 / 100.00
upstream occupancy (13,490 records)	0.128	-0.27 [-0.46, -0.07]	0.130	100.00 / 100.00
R02 Peak occupation	0.130	4.29 [0.96, 7.61]	0.132	100.00 / 100.00
peak occupancy 10 min (13,490 records)	0.123	-0.18 [-0.47, 0.12]	1.000	100.00 / 100.00
R16 Planned-arrival overlap	-0.128	-3.27 [-5.60, -0.95]	0.080	100.00 / 100.00
planned arrival release overlap 20 min (13,490 records)	-0.127	-0.43 [-0.63, -0.22]	0.007	100.00 / 100.00
F16 Downstream capacity gradient	0.124	3.84 [2.71, 4.97]	1.4e-5	100.00 / 100.00
next capacity delta (13,490 records)	0.134	1.33 [0.73, 1.93]	0.005	100.00 / 100.00
B13 Adjacent-window difference	-0.117	-4.36 [-7.95, -0.77]	0.179	100.00 / 100.00
adjacent occupancy difference 20 min (13,490 records)	-0.119	-0.56 [-0.88, -0.24]	0.026	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R17 Occupation change rate	-0.117	-4.36 [-7.95, -0.77]	0.179	100.00 / 100.00
occupancy rate 20 min (13,490 records)	-0.119	-0.56 [-0.88, -0.24]	0.026	100.00 / 100.00
R15 Free windows	-0.106	-3.28 [-4.72, -1.85]	0.002	100.00 / 100.00
free window count 5 min (13,490 records)	-0.105	-0.48 [-0.69, -0.26]	0.004	100.00 / 100.00
R10 Intersecting dwell distribution	-0.106	-4.02 [-5.97, -2.08]	0.004	100.00 / 100.00
waiting duration 20 min zero fraction (12,290 records)	-0.109	-0.60 [-0.86, -0.34]	0.003	100.00 / 100.00
R10 Intersecting dwell distribution	0.100	3.74 [0.71, 6.76]	0.166	100.00 / 100.00
waiting duration 5 min count (13,490 records)	0.093	-0.23 [-0.53, 0.08]	1.000	100.00 / 66.67
R14 Window competitors	0.100	3.74 [0.71, 6.76]	0.166	100.00 / 100.00
window competitors 5 min (13,490 records)	0.093	-0.23 [-0.53, 0.08]	1.000	100.00 / 66.67
R02 Peak occupation	0.090	3.48 [0.44, 6.51]	0.237	100.00 / 100.00
peak occupancy 5 min (13,490 records)	0.084	-0.17 [-0.46, 0.12]	1.000	100.00 / 100.00
E18 Parking concentration	-0.087	-2.05 [-4.91, 0.81]	1.000	100.00 / 100.00
parking concentration (13,490 records)	-0.086	-0.14 [-0.54, 0.25]	1.000	100.00 / 66.67
B13 Adjacent-window difference	-0.085	-3.46 [-6.58, -0.35]	0.270	100.00 / 100.00
adjacent occupancy difference 10 min (13,490 records)	-0.089	-0.54 [-0.85, -0.22]	0.030	100.00 / 100.00
R17 Occupation change rate	-0.085	-3.46 [-6.58, -0.35]	0.270	100.00 / 100.00
occupancy rate 10 min (13,490 records)	-0.089	-0.54 [-0.85, -0.22]	0.030	100.00 / 100.00
R18 Opposite-direction utilization	0.075	2.44 [0.70, 4.18]	0.083	100.00 / 100.00
opposite utilization 10 min (13,490 records)	0.070	-0.28 [-0.45, -0.12]	0.032	100.00 / 100.00
R07 Entry load	0.074	1.56 [-0.85, 3.97]	1.000	100.00 / 100.00
entry load 5 min (13,490 records)	0.066	-0.34 [-0.59, -0.09]	0.126	100.00 / 100.00
R18 Opposite-direction utilization	0.072	3.16 [1.09, 5.24]	0.047	100.00 / 100.00
opposite utilization 20 min (13,490 records)	0.066	-0.30 [-0.49, -0.11]	0.047	100.00 / 100.00
R11 Flow imbalance	-0.072	-2.80 [-5.78, 0.19]	0.548	100.00 / 100.00
arrivals minus departures 20 min (13,490 records)	-0.075	-0.54 [-0.85, -0.23]	0.025	100.00 / 100.00
R15 Free windows	0.071	0.86 [-0.99, 2.71]	1.000	100.00 / 100.00
free window count 10 min (13,490 records)	0.065	-0.42 [-0.65, -0.19]	0.020	100.00 / 100.00
R10 Intersecting dwell distribution	-0.070	-3.19 [-5.09, -1.30]	0.024	100.00 / 100.00
waiting duration 10 min zero fraction (12,290 records)	-0.073	-0.55 [-0.79, -0.31]	0.004	100.00 / 100.00
R18 Opposite-direction utilization	0.057	1.88 [0.28, 3.48]	0.214	100.00 / 100.00
opposite utilization 5 min (13,490 records)	0.050	-0.38 [-0.59, -0.16]	0.026	100.00 / 100.00
R10 Intersecting dwell distribution	-0.046	-2.13 [-3.94, -0.31]	0.214	100.00 / 100.00
waiting duration 5 min zero fraction (12,290 records)	-0.050	-0.50 [-0.74, -0.27]	0.006	100.00 / 100.00
F17 Distance to a hub	-0.046	-1.46 [-2.88, -0.04]	0.378	100.00 / 100.00
hub distance (13,490 records)	-0.043	0.00 [-0.13, 0.13]	1.000	33.33 / 33.33
F05 Physical articulation point	0.036	1.09 [0.30, 1.88]	0.088	100.00 / 100.00
articulation (13,490 records)	0.036	0.17 [0.05, 0.28]	0.097	100.00 / 100.00
F04 Physical betweenness	0.035	0.16 [-1.58, 1.90]	1.000	100.00 / 100.00
betweenness (13,490 records)	0.030	-0.60 [-0.90, -0.30]	0.009	100.00 / 100.00
E23 Initial station occupation	-0.030	-0.93 [-1.74, -0.12]	0.233	100.00 / 100.00
initial station occupied (13,490 records)	-0.032	-0.37 [-0.56, -0.18]	0.012	100.00 / 100.00
F01 Facility role	-0.028	-0.85 [-1.98, 0.28]	1.000	100.00 / 100.00
hub (13,490 records)	-0.035	-0.61 [-0.90, -0.32]	0.006	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
F06 Directional pool size	-0.028	-0.85 [-1.98, 0.28]	1.000	100.00 / 100.00
direction pool (13,490 records)	-0.035	-0.61 [-0.90, -0.32]	0.006	100.00 / 100.00
F15 Parallel-capacity proxy	-0.028	-0.85 [-1.98, 0.28]	1.000	100.00 / 100.00
parallel capacity (13,490 records)	-0.035	-0.61 [-0.90, -0.32]	0.006	100.00 / 100.00
F01 Facility role	-0.013	-0.39 [-2.20, 1.43]	1.000	100.00 / 100.00
trunk (13,490 records)	-0.023	-0.83 [-1.22, -0.43]	0.006	100.00 / 100.00
F02 Physical degree	-0.009	-0.48 [-1.69, 0.72]	1.000	100.00 / 100.00
undirected degree (13,490 records)	-0.015	-0.51 [-0.76, -0.26]	0.008	100.00 / 100.00
F03 Branch-count proxy	-0.009	-0.48 [-1.69, 0.72]	1.000	100.00 / 100.00
branch count (13,490 records)	-0.015	-0.51 [-0.76, -0.26]	0.008	100.00 / 100.00
F07 Compatible tracks	-0.009	-0.17 [-1.37, 1.04]	1.000	93.33 / 66.67
compatible tracks (13,490 records)	-0.015	-0.41 [-0.62, -0.19]	0.012	100.00 / 100.00
Delay propagation patterns				
G10 Edge-slack distribution	0.565	6.99 [5.12, 8.87]	3.0e-6	100.00 / 100.00
incoming slack minimum departure (13,490 records)	0.567	4.34 [3.37, 5.31]	4.0e-7	100.00 / 100.00
G19 Endpoint contrasts	0.548	5.53 [3.67, 7.39]	9.1e-5	100.00 / 100.00
run G02 departure delay difference (13,490 records)	0.551	3.76 [2.55, 4.96]	5.4e-4	100.00 / 100.00
G03 Activity delay increment	0.466	2.77 [1.16, 4.38]	0.020	100.00 / 100.00
dwell delay increment (13,490 records)	0.463	1.78 [0.86, 2.71]	0.011	100.00 / 100.00
G04 Activity-duration increment	0.466	2.77 [1.16, 4.38]	0.020	100.00 / 100.00
dwell increment (13,490 records)	0.463	1.78 [0.86, 2.71]	0.011	100.00 / 100.00
G19 Endpoint contrasts	0.466	2.77 [1.16, 4.38]	0.020	100.00 / 100.00
run G01 arrival delay difference (13,490 records)	0.463	1.78 [0.86, 2.71]	0.011	100.00 / 100.00
H25 Local assortativity	0.411	12.34 [8.76, 15.92]	1.3e-5	100.00 / 100.00
local delay absolute difference arrival (7,429 records)	0.415	0.89 [0.49, 1.29]	0.005	100.00 / 100.00
G15 Active-neighbor dispersion	0.398	9.59 [6.24, 12.94]	1.5e-4	100.00 / 100.00
delay variance arrival (7,429 records)	0.401	0.85 [0.42, 1.28]	0.009	100.00 / 100.00
G19 Endpoint contrasts	0.395	12.27 [7.84, 16.69]	2.4e-4	100.00 / 100.00
G08 active outgoing degree endpoint contrast (13,490 records)	0.392	1.16 [0.71, 1.60]	0.003	100.00 / 100.00
G10 Edge-slack distribution	0.394	2.07 [0.61, 3.54]	0.079	100.00 / 100.00
outgoing slack minimum arrival (13,490 records)	0.396	1.61 [0.70, 2.53]	0.025	100.00 / 100.00
G15 Active-neighbor dispersion	0.392	11.46 [7.15, 15.76]	4.3e-4	100.00 / 100.00
delay range arrival (7,429 records)	0.395	0.80 [0.42, 1.19]	0.007	100.00 / 100.00
B05 Regional delay distribution	0.381	11.97 [7.25, 16.69]	5.7e-4	100.00 / 100.00
region delay 20 min standard deviation (13,490 records)	0.372	0.51 [0.08, 0.94]	0.265	100.00 / 100.00
B06 Regional delay variance	0.381	10.69 [6.44, 14.93]	6.1e-4	100.00 / 100.00
delay variance 20 min (13,490 records)	0.372	0.56 [0.11, 1.00]	0.207	100.00 / 100.00
G10 Edge-slack distribution	-0.380	-12.43 [-16.41, -8.45]	4.9e-5	100.00 / 100.00
incoming slack zero fraction departure (13,490 records)	-0.380	-1.44 [-1.96, -0.92]	0.002	100.00 / 100.00
B05 Regional delay distribution	0.377	11.78 [6.87, 16.69]	9.2e-4	100.00 / 100.00
region delay 20 min interquartile range (13,490 records)	0.366	0.26 [-0.09, 0.60]	1.000	100.00 / 100.00
G15 Active-neighbor dispersion	0.375	7.59 [4.62, 10.56]	5.5e-4	100.00 / 100.00
delay coefficient of variation arrival (7,429 records)	0.378	0.39 [0.18, 0.60]	0.017	100.00 / 100.00
B05 Regional delay distribution	0.369	10.30 [5.64, 14.96]	0.002	100.00 / 100.00
region delay 10 min standard deviation (13,490 records)	0.361	0.69 [0.22, 1.16]	0.086	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B06 Regional delay variance	0.369	7.54 [3.93, 11.15]	0.004	100.00 / 100.00
delay variance 10 min (13,490 records)	0.361	0.72 [0.20, 1.25]	0.120	100.00 / 100.00
G19 Endpoint contrasts	-0.364	-12.65 [-16.37, -8.93]	1.4e-5	100.00 / 100.00
G08 active incoming degree endpoint contrast (13,490 records)	-0.356	-0.49 [-0.74, -0.25]	0.009	100.00 / 100.00
B05 Regional delay distribution	0.356	8.05 [3.41, 12.68]	0.019	100.00 / 100.00
region delay 10 min interquartile range (13,490 records)	0.349	0.43 [0.06, 0.81]	0.300	100.00 / 100.00
H25 Local assortativity	-0.350	-11.93 [-15.87, -7.99]	7.1e-5	100.00 / 100.00
local same type fraction arrival (7,429 records)	-0.355	-0.52 [-0.78, -0.26]	0.008	100.00 / 100.00
H12 PageRank	-0.337	-6.06 [-9.21, -2.92]	0.008	100.00 / 100.00
pagerank departure (13,490 records)	-0.338	-0.85 [-1.23, -0.48]	0.004	100.00 / 100.00
G10 Edge-slack distribution	-0.337	-11.10 [-14.76, -7.45]	7.0e-5	100.00 / 100.00
outgoing slack zero fraction arrival (13,490 records)	-0.335	-1.19 [-1.58, -0.80]	5.7e-4	100.00 / 100.00
G19 Endpoint contrasts	-0.312	-13.47 [-15.90, -11.04]	1.4e-9	100.00 / 100.00
H04 upstream reachability endpoint contrast (13,490 records)	-0.310	-0.35 [-0.50, -0.20]	0.003	100.00 / 100.00
H13 Weak-component size	0.312	9.72 [5.76, 13.69]	7.4e-4	100.00 / 100.00
component trains departure (13,490 records)	0.305	0.30 [-0.04, 0.64]	0.822	100.00 / 100.00
H13 Weak-component size	0.312	9.70 [5.79, 13.62]	6.9e-4	100.00 / 100.00
component events departure (13,490 records)	0.305	0.30 [-0.05, 0.65]	0.852	100.00 / 100.00
B10 Affected-train share	0.309	9.35 [5.75, 12.95]	4.8e-4	100.00 / 100.00
affected fraction (13,490 records)	0.305	0.73 [0.31, 1.15]	0.028	100.00 / 100.00
B15 Global degree distribution	0.306	9.41 [5.83, 12.98]	4.7e-4	100.00 / 100.00
outgoing degree mean (13,490 records)	0.303	0.72 [0.29, 1.15]	0.032	100.00 / 100.00
B15 Global degree distribution	0.306	9.41 [5.83, 12.98]	4.7e-4	100.00 / 100.00
outgoing degree sum (13,490 records)	0.303	0.72 [0.29, 1.15]	0.032	100.00 / 100.00
B15 Global degree distribution	0.306	9.39 [5.82, 12.97]	4.7e-4	100.00 / 100.00
incoming degree mean (13,490 records)	0.302	0.72 [0.29, 1.14]	0.034	100.00 / 100.00
B15 Global degree distribution	0.306	9.39 [5.82, 12.97]	4.7e-4	100.00 / 100.00
incoming degree sum (13,490 records)	0.302	0.72 [0.29, 1.14]	0.034	100.00 / 100.00
H16 Local motifs	0.304	9.34 [4.64, 14.05]	0.006	100.00 / 100.00
fan-in pairs arrival (13,490 records)	0.296	0.08 [-0.21, 0.37]	1.000	93.33 / 100.00
B15 Global degree distribution	-0.302	-9.30 [-12.86, -5.74]	4.7e-4	100.00 / 100.00
outgoing degree zero fraction (13,490 records)	-0.299	-0.72 [-1.14, -0.29]	0.030	100.00 / 100.00
B11 Global path distribution	-0.301	-9.27 [-12.84, -5.71]	4.7e-4	100.00 / 100.00
root longest paths zero fraction (13,490 records)	-0.298	-0.70 [-1.12, -0.28]	0.034	100.00 / 100.00
B15 Global degree distribution	-0.301	-9.27 [-12.84, -5.71]	4.7e-4	100.00 / 100.00
incoming degree zero fraction (13,490 records)	-0.298	-0.70 [-1.12, -0.28]	0.034	100.00 / 100.00
B15 Global degree distribution	0.295	8.04 [4.70, 11.37]	8.8e-4	100.00 / 100.00
outgoing degree standard deviation (13,490 records)	0.292	0.69 [0.32, 1.07]	0.016	100.00 / 100.00
B05 Regional delay distribution	0.294	5.51 [1.88, 9.15]	0.049	100.00 / 100.00
region delay 5 min interquartile range (13,490 records)	0.288	0.59 [0.14, 1.05]	0.167	100.00 / 100.00
B16 Global reachability	0.294	9.10 [5.62, 12.58]	4.7e-4	100.00 / 100.00
reachable pair fraction (13,490 records)	0.291	0.75 [0.32, 1.19]	0.028	100.00 / 100.00
B11 Global path distribution	0.292	8.94 [5.45, 12.43]	5.4e-4	100.00 / 100.00
root longest paths mean (13,490 records)	0.289	0.73 [0.30, 1.17]	0.031	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B11 Global path distribution root longest paths sum (13,490 records)	0.292 0.289	8.94 [5.45, 12.43] 0.73 [0.30, 1.17]	5.4e-4 0.031	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min standard deviation (13,490 records)	0.288 0.282	6.25 [2.66, 9.83] 0.75 [0.21, 1.30]	0.019 0.119	100.00 / 100.00 100.00 / 100.00
B06 Regional delay variance delay variance 5 min (13,490 records)	0.288 0.282	4.02 [1.78, 6.27] 0.73 [0.17, 1.28]	0.015 0.164	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths interquartile range (13,490 records)	0.286 0.283	8.72 [5.28, 12.16] 0.74 [0.28, 1.20]	5.7e-4 0.043	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum departure (12,890 records)	0.286 0.276	8.12 [3.81, 12.44] -0.03 [-0.37, 0.31]	0.010 1.000	100.00 / 100.00 73.33 / 66.67
G13 Upstream-neighbor delay neighbor delay mean departure (12,890 records)	0.286 0.276	8.12 [3.81, 12.44] -0.03 [-0.37, 0.31]	0.010 1.000	100.00 / 100.00 73.33 / 66.67
G13 Upstream-neighbor delay neighbor delay median departure (12,890 records)	0.286 0.276	8.12 [3.81, 12.44] -0.03 [-0.37, 0.31]	0.010 1.000	100.00 / 100.00 73.33 / 66.67
G13 Upstream-neighbor delay neighbor delay minimum departure (12,890 records)	0.286 0.276	8.12 [3.81, 12.44] -0.03 [-0.37, 0.31]	0.010 1.000	100.00 / 100.00 73.33 / 66.67
G13 Upstream-neighbor delay neighbor delay 90th percentile departure (12,890 records)	0.286 0.276	8.12 [3.81, 12.44] -0.03 [-0.37, 0.31]	0.010 1.000	100.00 / 100.00 73.33 / 66.67
G13 Upstream-neighbor delay neighbor delay sum departure (13,490 records)	0.286 0.276	7.94 [3.69, 12.19] -0.05 [-0.36, 0.27]	0.011 1.000	100.00 / 100.00 93.33 / 66.67
G13 Upstream-neighbor delay neighbor delay zero fraction departure (12,890 records)	-0.285 -0.275	-8.91 [-13.51, -4.32] 0.13 [-0.23, 0.49]	0.008 1.000	100.00 / 100.00 93.33 / 66.67
B15 Global degree distribution outgoing degree 90th percentile (13,490 records)	0.282 0.279	8.44 [5.04, 11.83] 0.81 [0.50, 1.13]	6.8e-4 0.003	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution incoming degree standard deviation (13,490 records)	0.281 0.278	7.74 [4.49, 10.99] 0.69 [0.33, 1.05]	9.9e-4 0.013	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration time20 concentration (13,490 records)	-0.281 -0.277	-6.68 [-10.21, -3.15] -0.71 [-1.04, -0.38]	0.010 0.005	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum arrival (13,490 records)	0.280 0.269	8.15 [3.75, 12.55] -0.13 [-0.43, 0.16]	0.011 1.000	100.00 / 100.00 100.00 / 66.67
B11 Global path distribution root longest paths median (13,490 records)	0.279 0.275	8.26 [4.94, 11.59] 0.60 [0.14, 1.06]	6.9e-4 0.160	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration time20 entropy (13,490 records)	0.279 0.276	7.34 [4.05, 10.63] 0.79 [0.47, 1.10]	0.002 0.003	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support minimum departure (13,490 records)	0.278 0.268	7.24 [3.49, 11.00] -0.17 [-0.46, 0.13]	0.008 1.000	100.00 / 100.00 100.00 / 66.67
B15 Global degree distribution incoming degree 90th percentile (13,490 records)	0.276 0.273	8.21 [5.00, 11.43] 0.76 [0.45, 1.06]	5.5e-4 0.003	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum arrival (11,730 records)	0.275 0.264	8.37 [3.95, 12.79] -0.13 [-0.49, 0.23]	0.010 1.000	100.00 / 100.00 100.00 / 66.67

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G13 Upstream-neighbor delay neighbor delay 90th percentile arrival (11,730 records)	0.275 0.264	8.36 [3.94, 12.77] -0.14 [-0.50, 0.22]	0.010 1.000	100.00 / 100.00 100.00 / 66.67
G13 Upstream-neighbor delay neighbor delay mean arrival (11,730 records)	0.274 0.263	8.10 [3.77, 12.42] -0.17 [-0.53, 0.20]	0.011 1.000	100.00 / 100.00 100.00 / 66.67
G13 Upstream-neighbor delay neighbor delay median arrival (11,730 records)	0.274 0.263	8.10 [3.77, 12.42] -0.17 [-0.53, 0.20]	0.011 1.000	100.00 / 100.00 100.00 / 66.67
B15 Global degree distribution incoming degree median (13,490 records)	0.269 0.265	8.22 [4.86, 11.59] 0.61 [0.18, 1.04]	7.5e-4 0.102	100.00 / 100.00 100.00 / 100.00
H13 Weak-component size component events arrival (13,490 records)	0.267 0.256	8.54 [4.66, 12.43] -0.22 [-0.59, 0.16]	0.002 1.000	100.00 / 100.00 100.00 / 66.67
G08 Active degree outgoing active departure (13,490 records)	0.264 0.257	8.49 [4.27, 12.70] 0.32 [-0.06, 0.70]	0.006 0.935	100.00 / 100.00 100.00 / 100.00
H13 Weak-component size component trains arrival (13,490 records)	0.264 0.253	8.45 [4.49, 12.41] -0.25 [-0.64, 0.13]	0.003 1.000	100.00 / 100.00 100.00 / 66.67
B15 Global degree distribution outgoing degree median (13,490 records)	0.264 0.260	8.07 [4.90, 11.24] 0.63 [0.20, 1.06]	5.5e-4 0.086	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support standard deviation departure (13,490 records)	0.263 0.255	6.48 [3.39, 9.57] 0.18 [-0.14, 0.51]	0.004 1.000	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support interquartile range departure (13,490 records)	0.260 0.252	6.17 [3.17, 9.17] 0.18 [-0.14, 0.51]	0.005 1.000	100.00 / 100.00 100.00 / 100.00
G19 Endpoint contrasts H05 downstream reachability endpoint contrast (13,490 records)	0.260 0.260	5.99 [2.87, 9.11] 0.64 [0.26, 1.03]	0.009 0.035	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction departure (13,490 records)	-0.259 -0.247	-7.89 [-12.14, -3.63] 0.39 [-0.05, 0.84]	0.011 0.814	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths 90th percentile (13,490 records)	0.258 0.256	7.60 [4.53, 10.67] 0.72 [0.36, 1.08]	6.9e-4 0.009	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay zero fraction arrival (11,730 records)	-0.257 -0.245	-8.19 [-12.67, -3.71] 0.31 [-0.08, 0.70]	0.013 1.000	100.00 / 100.00 100.00 / 66.67
B05 Regional delay distribution region delay 20 min sum (13,490 records)	0.254 0.243	7.73 [3.89, 11.58] -0.20 [-0.49, 0.08]	0.006 1.000	100.00 / 100.00 100.00 / 66.67
G12 Delay-support distribution outgoing support interquartile range arrival (13,490 records)	0.253 0.243	6.68 [4.12, 9.25] -0.09 [-0.33, 0.15]	4.7e-4 1.000	100.00 / 100.00 100.00 / 66.67
G12 Delay-support distribution outgoing support standard deviation arrival (13,490 records)	0.253 0.243	6.68 [4.12, 9.25] -0.09 [-0.33, 0.15]	4.7e-4 1.000	100.00 / 100.00 100.00 / 66.67
H25 Local assortativity local delay absolute difference departure (8,228 records)	0.251 0.250	6.70 [3.93, 9.47] 1.39 [0.52, 2.25]	8.7e-4 0.047	100.00 / 100.00 100.00 / 100.00
G08 Active degree incoming active arrival (13,490 records)	0.248 0.238	8.23 [3.68, 12.77] -0.28 [-0.70, 0.13]	0.014 1.000	100.00 / 100.00 100.00 / 66.67

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G15 Active-neighbor dispersion delay variance departure (8,228 records)	0.246 0.245	3.57 [1.75, 5.39] 0.91 [0.24, 1.59]	0.007 0.138	100.00 / 100.00 100.00 / 100.00
G15 Active-neighbor dispersion delay range departure (8,228 records)	0.244 0.243	5.56 [3.01, 8.10] 1.05 [0.37, 1.74]	0.003 0.063	100.00 / 100.00 100.00 / 100.00
G15 Active-neighbor dispersion delay coefficient of variation departure (8,228 records)	0.242 0.241	4.24 [1.40, 7.08] 0.48 [0.06, 0.89]	0.054 0.296	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum arrival (11,730 records)	0.242 0.230	7.23 [3.26, 11.19] -0.19 [-0.54, 0.15]	0.013 1.000	100.00 / 100.00 100.00 / 66.67
G10 Edge-slack distribution incoming slack interquartile range departure (13,490 records)	-0.236 -0.230	-5.07 [-7.31, -2.84] -0.37 [-0.52, -0.21]	0.002 0.003	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack standard deviation departure (13,490 records)	-0.236 -0.230	-5.08 [-7.31, -2.84] -0.36 [-0.52, -0.21]	0.002 0.003	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support median arrival (12,610 records)	0.236 0.225	6.18 [2.69, 9.67] -0.26 [-0.57, 0.06]	0.016 1.000	100.00 / 100.00 100.00 / 100.00
G16 Incident-link imbalance pressure balance arrival (13,490 records)	-0.232 -0.224	-5.25 [-8.86, -1.64] 0.04 [-0.14, 0.22]	0.065 1.000	100.00 / 100.00 93.33 / 66.67
G12 Delay-support distribution incoming support zero fraction arrival (12,610 records)	-0.231 -0.219	-7.46 [-11.69, -3.24] 0.45 [0.03, 0.88]	0.017 0.401	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack median arrival (12,610 records)	-0.230 -0.222	-4.57 [-6.68, -2.45] -0.27 [-0.40, -0.14]	0.003 0.008	100.00 / 100.00 100.00 / 100.00
G02 Reference departure delay departure delay next endpoint (13,490 records)	0.230 0.230	6.02 [3.14, 8.91] 1.26 [0.69, 1.83]	0.004 0.005	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths standard deviation (13,490 records)	0.229 0.228	7.06 [4.06, 10.06] 0.66 [0.28, 1.05]	0.001 0.029	100.00 / 100.00 100.00 / 100.00
B08 Community structure community events departure (13,490 records)	0.229 0.226	6.37 [3.38, 9.35] 0.36 [0.03, 0.69]	0.003 0.380	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration station entropy (13,490 records)	0.229 0.227	4.08 [1.23, 6.93] 0.48 [0.20, 0.75]	0.072 0.030	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution outgoing degree maximum (13,490 records)	0.229 0.227	6.73 [3.59, 9.86] 0.57 [0.20, 0.94]	0.003 0.059	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration station concentration (13,490 records)	-0.224 -0.221	-2.70 [-5.21, -0.20] -0.32 [-0.57, -0.07]	0.312 0.167	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min maximum (13,490 records)	0.217 0.211	6.07 [2.70, 9.44] 0.31 [0.03, 0.60]	0.014 0.380	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min zero fraction (13,490 records)	-0.217 -0.212	-6.71 [-10.11, -3.32] -0.19 [-0.48, 0.10]	0.007 1.000	100.00 / 100.00 100.00 / 66.67
B01 Local region size region trains 20 min (13,490 records)	0.215 0.201	8.71 [4.39, 13.03] -0.38 [-0.75, -0.01]	0.006 0.499	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support minimum arrival (12,610 records)	0.214 0.203	5.51 [2.38, 8.63] -0.33 [-0.60, -0.06]	0.017 0.237	100.00 / 100.00 100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G09 Active cross-train links	0.214	5.81 [1.49, 10.12]	0.105	100.00 / 100.00
active cross arrival (13,490 records)	0.207	-0.19 [-0.47, 0.08]	1.000	100.00 / 66.67
G13 Upstream-neighbor delay	0.214	3.59 [1.34, 5.83]	0.035	100.00 / 100.00
neighbor delay interquartile range arrival (11,730 records)	0.208	0.09 [-0.12, 0.30]	1.000	100.00 / 100.00
G13 Upstream-neighbor delay	0.214	3.59 [1.34, 5.83]	0.035	100.00 / 100.00
neighbor delay standard deviation arrival (11,730 records)	0.208	0.09 [-0.12, 0.30]	1.000	100.00 / 100.00
B05 Regional delay distribution	-0.213	-6.68 [-10.11, -3.26]	0.008	100.00 / 100.00
region delay 10 min zero fraction (13,490 records)	-0.208	-0.19 [-0.49, 0.10]	1.000	100.00 / 66.67
G12 Delay-support distribution	0.213	4.91 [1.96, 7.86]	0.026	100.00 / 100.00
outgoing support sum departure (13,490 records)	0.207	0.20 [-0.12, 0.51]	1.000	100.00 / 66.67
H06 Upstream trains	0.212	7.16 [2.92, 11.39]	0.023	100.00 / 100.00
upstream arrival (13,490 records)	0.202	-0.38 [-0.75, 0.00]	0.544	100.00 / 100.00
B05 Regional delay distribution	0.212	5.86 [2.53, 9.20]	0.017	100.00 / 100.00
region delay 20 min 90th percentile (13,490 records)	0.206	0.29 [0.01, 0.57]	0.468	100.00 / 100.00
B05 Regional delay distribution	-0.212	-6.75 [-10.27, -3.24]	0.009	100.00 / 100.00
region delay 20 min zero fraction (13,490 records)	-0.206	-0.19 [-0.49, 0.11]	1.000	100.00 / 66.67
G12 Delay-support distribution	-0.211	-6.52 [-10.06, -2.99]	0.012	100.00 / 100.00
outgoing support zero fraction departure (13,490 records)	-0.206	-0.23 [-0.53, 0.08]	1.000	100.00 / 66.67
H16 Local motifs	0.210	7.07 [3.48, 10.67]	0.007	100.00 / 100.00
fan-out pairs departure (13,490 records)	0.203	0.27 [-0.07, 0.62]	1.000	100.00 / 100.00
B02 Local edges and density	0.209	9.32 [4.92, 13.72]	0.004	100.00 / 100.00
edges 20 min (13,490 records)	0.192	-0.50 [-0.90, -0.10]	0.202	100.00 / 100.00
H24 Projection core number	0.208	6.23 [2.66, 9.79]	0.018	100.00 / 100.00
core departure (13,490 records)	0.201	-0.04 [-0.34, 0.26]	1.000	86.67 / 33.33
H09 Reachable delay totals	0.207	6.64 [2.72, 10.55]	0.023	100.00 / 100.00
upstream train maximum delay sum arrival (13,490 records)	0.197	-0.30 [-0.63, 0.03]	0.740	100.00 / 100.00
G09 Active cross-train links	0.207	5.68 [1.50, 9.86]	0.099	100.00 / 100.00
active cross fraction arrival (13,490 records)	0.200	-0.19 [-0.46, 0.08]	1.000	100.00 / 66.67
H15 Chain and crossing length	0.207	6.98 [2.89, 11.08]	0.022	100.00 / 100.00
longest crossings arrival (13,490 records)	0.198	-0.29 [-0.61, 0.02]	0.710	100.00 / 100.00
B05 Regional delay distribution	0.206	5.50 [2.28, 8.73]	0.022	100.00 / 100.00
region delay 10 min maximum (13,490 records)	0.201	0.32 [0.04, 0.61]	0.310	100.00 / 100.00
H25 Local assortativity	-0.204	-7.45 [-9.87, -5.03]	6.3e-5	100.00 / 100.00
local same type fraction departure (8,228 records)	-0.207	-1.23 [-1.99, -0.46]	0.047	100.00 / 100.00
G12 Delay-support distribution	0.204	5.65 [2.16, 9.14]	0.031	100.00 / 100.00
incoming support median departure (13,490 records)	0.194	-0.33 [-0.69, 0.03]	0.715	100.00 / 100.00
G19 Endpoint contrasts	0.203	4.13 [2.74, 5.53]	9.1e-5	100.00 / 100.00
run H10 betweenness departure difference (13,490 records)	0.209	1.10 [0.60, 1.61]	0.005	100.00 / 100.00
B05 Regional delay distribution	0.203	4.36 [1.11, 7.62]	0.105	100.00 / 100.00
region delay 10 min sum (13,490 records)	0.194	-0.27 [-0.55, 0.01]	0.610	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support sum arrival (13,490 records)	0.202 0.192	5.74 [2.17, 9.30] -0.33 [-0.67, 0.00]	0.032 0.552	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support sum departure (13,490 records)	0.200 0.190	5.68 [2.08, 9.28] -0.34 [-0.71, 0.02]	0.037 0.674	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support mean departure (13,490 records)	0.200 0.190	5.66 [2.20, 9.12] -0.34 [-0.71, 0.02]	0.029 0.674	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution incoming degree maximum (13,490 records)	0.200 0.198	6.03 [3.39, 8.67] 0.54 [0.25, 0.83]	0.002 0.016	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths maximum (13,490 records)	0.200 0.197	5.84 [2.88, 8.81] 0.47 [0.13, 0.82]	0.007 0.126	100.00 / 100.00 100.00 / 100.00
H07 Downstream trains downstream departure (13,490 records)	0.199 0.194	4.60 [0.87, 8.34] 0.10 [-0.23, 0.42]	0.167 1.000	100.00 / 100.00 100.00 / 66.67
G19 Endpoint contrasts run T18 remaining planned difference (13,490 records)	0.197 0.190	6.07 [4.05, 8.09] 0.00 [-0.40, 0.39]	8.0e-5 1.000	100.00 / 100.00 86.67 / 66.67
B05 Regional delay distribution region delay 10 min 90th percentile (13,490 records)	0.197 0.192	5.16 [2.08, 8.24] 0.31 [0.03, 0.59]	0.025 0.377	100.00 / 100.00 100.00 / 100.00
G09 Active cross-train links active cross departure (13,490 records)	0.194 0.186	5.67 [1.90, 9.43] 0.11 [-0.19, 0.41]	0.052 1.000	100.00 / 100.00 100.00 / 66.67
G12 Delay-support distribution incoming support mean arrival (12,610 records)	0.193 0.182	5.37 [2.07, 8.68] -0.39 [-0.74, -0.05]	0.031 0.322	100.00 / 100.00 100.00 / 100.00
B08 Community structure community events arrival (13,490 records)	0.192 0.183	5.48 [2.32, 8.64] -0.35 [-0.71, 0.02]	0.019 0.663	100.00 / 100.00 100.00 / 100.00
G09 Active cross-train links active cross fraction departure (13,490 records)	0.191 0.184	5.25 [1.50, 9.01] 0.12 [-0.18, 0.42]	0.083 1.000	100.00 / 100.00 100.00 / 66.67
B15 Global degree distribution incoming degree interquartile range (13,490 records)	0.191 0.189	5.83 [2.81, 8.86] 0.56 [0.14, 0.98]	0.008 0.149	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution outgoing degree interquartile range (13,490 records)	0.191 0.189	5.83 [2.81, 8.86] 0.56 [0.14, 0.98]	0.008 0.149	100.00 / 100.00 100.00 / 100.00
H09 Reachable delay totals downstream train maximum delay sum departure (13,490 records)	0.190 0.185	3.14 [0.22, 6.05] 0.18 [-0.14, 0.50]	0.317 1.000	100.00 / 100.00 100.00 / 100.00
B01 Local region size region events 20 min (13,490 records)	0.188 0.171	8.40 [3.99, 12.82] -0.60 [-1.05, -0.16]	0.009 0.130	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min count (13,490 records)	0.188 0.171	8.40 [3.99, 12.82] -0.60 [-1.05, -0.16]	0.009 0.130	100.00 / 100.00 100.00 / 100.00
H08 Spatial reachability downstream branches departure (13,490 records)	0.188 0.184	3.82 [0.77, 6.88] 0.24 [-0.11, 0.60]	0.156 1.000	100.00 / 100.00 100.00 / 100.00
G01 Reference arrival delay arrival delay next endpoint (13,490 records)	0.186 0.182	4.66 [1.73, 7.59] 0.31 [0.02, 0.60]	0.035 0.417	100.00 / 100.00 100.00 / 100.00
G02 Reference departure delay departure delay (13,490 records)	0.186 0.182	4.66 [1.73, 7.59] 0.31 [0.02, 0.60]	0.035 0.417	100.00 / 100.00 100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution region delay 5 min maximum (13,490 records)	0.186 0.182	4.69 [1.80, 7.59] 0.32 [0.03, 0.61]	0.031 0.337	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support maximum departure (13,490 records)	0.186 0.182	4.66 [1.73, 7.59] 0.31 [0.02, 0.60]	0.035 0.417	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min mean (13,490 records)	0.185 0.180	4.48 [1.58, 7.38] 0.25 [-0.02, 0.52]	0.043 0.720	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min mean (13,490 records)	0.184 0.180	4.40 [1.55, 7.26] 0.26 [-0.01, 0.54]	0.043 0.634	100.00 / 100.00 100.00 / 100.00
H11 Harmonic closeness incoming harmonic arrival (13,490 records)	0.184 0.175	5.68 [1.74, 9.62] -0.50 [-0.91, -0.08]	0.070 0.253	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min median (13,490 records)	0.184 0.179	4.50 [1.60, 7.41] 0.24 [-0.03, 0.51]	0.042 0.763	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min 90th percentile (13,490 records)	0.184 0.180	4.60 [1.72, 7.48] 0.31 [0.02, 0.59]	0.034 0.389	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support 90th percentile departure (13,490 records)	0.183 0.179	4.51 [1.58, 7.44] 0.31 [0.02, 0.60]	0.044 0.439	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min mean (13,490 records)	0.179 0.175	4.27 [1.46, 7.08] 0.27 [-0.01, 0.54]	0.047 0.622	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min median (13,490 records)	0.179 0.174	4.29 [1.49, 7.09] 0.25 [-0.02, 0.53]	0.045 0.694	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support mean departure (13,490 records)	0.179 0.175	3.97 [1.08, 6.86] 0.30 [0.01, 0.59]	0.092 0.499	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction departure (10,772 records)	-0.178 -0.176	-5.59 [-8.32, -2.86] -0.43 [-0.70, -0.17]	0.005 0.039	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min median (13,490 records)	0.177 0.173	4.23 [1.44, 7.02] 0.26 [-0.02, 0.53]	0.049 0.692	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support zero fraction arrival (13,490 records)	-0.177 -0.167	-5.45 [-9.16, -1.75] 0.54 [0.08, 0.99]	0.060 0.277	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support sum arrival (13,490 records)	0.174 0.164	4.00 [1.07, 6.93] -0.42 [-0.76, -0.07]	0.095 0.238	100.00 / 100.00 100.00 / 100.00
B01 Local region size region trains 10 min (13,490 records)	0.172 0.160	6.01 [1.97, 10.05] -0.44 [-0.82, -0.06]	0.056 0.298	100.00 / 100.00 100.00 / 100.00
H09 Reachable delay totals upstream event delay arrival (13,490 records)	0.170 0.160	4.18 [0.95, 7.42] -0.41 [-0.70, -0.11]	0.130 0.119	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support median departure (13,490 records)	0.169 0.165	3.81 [0.93, 6.69] 0.29 [-0.01, 0.58]	0.113 0.586	100.00 / 100.00 100.00 / 100.00
B02 Local edges and density density 20 min (13,387 records)	-0.169 -0.155	-4.77 [-8.33, -1.20] 0.58 [0.21, 0.96]	0.106 0.058	100.00 / 100.00 100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution	0.168	3.87 [1.14, 6.59]	0.077	100.00 / 100.00
region delay 5 min minimum (13,490 records)	0.163	0.22 [-0.05, 0.49]	0.998	100.00 / 100.00
H10 Active betweenness	-0.166	-4.87 [-6.98, -2.76]	0.001	100.00 / 100.00
betweenness departure (13,490 records)	-0.169	-0.66 [-0.97, -0.35]	0.006	100.00 / 100.00
H04 Upstream reachability	0.163	4.31 [0.91, 7.70]	0.144	100.00 / 100.00
upstream arrival (13,490 records)	0.153	-0.50 [-0.85, -0.15]	0.110	100.00 / 100.00
G10 Edge-slack distribution	-0.162	-3.58 [-5.65, -1.52]	0.020	100.00 / 100.00
incoming slack maximum departure (13,490 records)	-0.153	0.52 [0.10, 0.94]	0.209	100.00 / 100.00
H24 Projection core number	0.161	4.71 [1.38, 8.05]	0.079	100.00 / 100.00
core arrival (13,490 records)	0.152	-0.57 [-0.99, -0.14]	0.147	100.00 / 100.00
G10 Edge-slack distribution	0.161	4.21 [1.00, 7.41]	0.118	100.00 / 100.00
incoming slack zero fraction arrival (12,610 records)	0.155	-0.10 [-0.44, 0.24]	1.000	93.33 / 66.67
H02 Longest active path	-0.159	-5.47 [-7.78, -3.17]	0.001	100.00 / 100.00
longest path departure (13,490 records)	-0.162	-0.85 [-1.21, -0.49]	0.003	100.00 / 100.00
H15 Chain and crossing length	-0.159	-5.47 [-7.78, -3.17]	0.001	100.00 / 100.00
longest chain departure (13,490 records)	-0.162	-0.85 [-1.21, -0.49]	0.003	100.00 / 100.00
G14 Downstream-neighbor delay	0.157	3.04 [0.74, 5.34]	0.113	100.00 / 100.00
neighbor delay sum departure (13,490 records)	0.151	0.08 [-0.26, 0.43]	1.000	100.00 / 66.67
H11 Harmonic closeness	-0.157	-5.27 [-7.65, -2.88]	0.002	100.00 / 100.00
incoming harmonic departure (13,490 records)	-0.160	-0.94 [-1.34, -0.53]	0.003	100.00 / 100.00
H08 Spatial reachability	0.157	4.26 [0.94, 7.57]	0.134	100.00 / 100.00
upstream branches arrival (13,490 records)	0.147	-0.56 [-0.94, -0.17]	0.089	100.00 / 100.00
G12 Delay-support distribution	0.156	4.03 [1.02, 7.04]	0.105	100.00 / 100.00
incoming support 90th percentile arrival (12,610 records)	0.146	-0.46 [-0.83, -0.09]	0.224	100.00 / 100.00
B05 Regional delay distribution	0.156	3.45 [0.87, 6.02]	0.106	100.00 / 100.00
region delay 10 min minimum (13,490 records)	0.151	0.21 [-0.06, 0.48]	1.000	100.00 / 100.00
G12 Delay-support distribution	0.155	3.63 [0.73, 6.52]	0.155	100.00 / 100.00
outgoing support maximum arrival (13,490 records)	0.146	-0.46 [-0.82, -0.10]	0.182	100.00 / 100.00
G10 Edge-slack distribution	-0.154	-3.41 [-5.46, -1.36]	0.026	100.00 / 100.00
incoming slack 90th percentile departure (13,490 records)	-0.144	0.61 [0.15, 1.07]	0.149	100.00 / 100.00
G12 Delay-support distribution	0.154	3.90 [0.87, 6.92]	0.132	100.00 / 100.00
incoming support 90th percentile departure (13,490 records)	0.145	-0.43 [-0.80, -0.06]	0.290	100.00 / 100.00
B05 Regional delay distribution	0.154	2.05 [-0.24, 4.34]	0.650	100.00 / 100.00
region delay 5 min sum (13,490 records)	0.147	-0.25 [-0.51, 0.01]	0.622	100.00 / 100.00
G12 Delay-support distribution	0.153	3.52 [0.65, 6.40]	0.173	100.00 / 100.00
outgoing support 90th percentile arrival (13,490 records)	0.144	-0.46 [-0.82, -0.10]	0.173	100.00 / 100.00
H08 Spatial reachability	0.153	3.90 [0.69, 7.10]	0.179	100.00 / 100.00
upstream stations arrival (13,490 records)	0.144	-0.52 [-0.90, -0.15]	0.116	100.00 / 100.00
G10 Edge-slack distribution	-0.152	-3.17 [-4.60, -1.75]	0.002	100.00 / 100.00
incoming slack mean arrival (12,610 records)	-0.143	-0.11 [-0.26, 0.04]	1.000	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G01 Reference arrival delay	0.151	3.57 [0.70, 6.44]	0.161	100.00 / 100.00
arrival delay (13,490 records)	0.142	-0.46 [-0.82, -0.10]	0.180	100.00 / 100.00
G12 Delay-support distribution	0.151	3.28 [0.50, 6.05]	0.206	100.00 / 100.00
outgoing support minimum departure (13,490 records)	0.147	0.29 [0.00, 0.59]	0.575	100.00 / 100.00
B04 Boundary connections	0.148	5.32 [1.54, 9.10]	0.080	100.00 / 100.00
entering 20 min (13,490 records)	0.134	-0.65 [-1.09, -0.22]	0.068	100.00 / 100.00
G12 Delay-support distribution	0.148	3.08 [0.29, 5.87]	0.277	100.00 / 100.00
outgoing support mean arrival (13,490 records)	0.139	-0.47 [-0.82, -0.12]	0.143	100.00 / 100.00
G12 Delay-support distribution	0.148	3.08 [0.29, 5.87]	0.277	100.00 / 100.00
outgoing support median arrival (13,490 records)	0.139	-0.47 [-0.82, -0.12]	0.143	100.00 / 100.00
G14 Downstream-neighbor delay	-0.148	-4.70 [-7.44, -1.96]	0.022	100.00 / 100.00
neighbor delay zero fraction arrival (9,875 records)	-0.142	0.08 [-0.22, 0.38]	1.000	100.00 / 33.33
H04 Upstream reachability	-0.145	-5.01 [-7.15, -2.86]	0.001	100.00 / 100.00
upstream departure (13,490 records)	-0.147	-0.78 [-1.11, -0.46]	0.003	100.00 / 100.00
G12 Delay-support distribution	0.145	3.46 [0.57, 6.35]	0.192	100.00 / 100.00
incoming support maximum departure (13,490 records)	0.136	-0.44 [-0.81, -0.08]	0.237	100.00 / 100.00
H02 Longest active path	0.144	3.55 [0.52, 6.58]	0.214	100.00 / 100.00
longest path arrival (13,490 records)	0.135	-0.57 [-0.95, -0.20]	0.063	100.00 / 100.00
H15 Chain and crossing length	0.144	3.55 [0.52, 6.58]	0.214	100.00 / 100.00
longest chain arrival (13,490 records)	0.135	-0.57 [-0.95, -0.20]	0.063	100.00 / 100.00
H08 Spatial reachability	-0.143	-5.15 [-7.31, -2.99]	9.9e-4	100.00 / 100.00
upstream stations departure (13,490 records)	-0.146	-0.85 [-1.21, -0.49]	0.003	100.00 / 100.00
G12 Delay-support distribution	0.142	3.41 [0.57, 6.25]	0.189	100.00 / 100.00
incoming support maximum arrival (12,610 records)	0.133	-0.48 [-0.85, -0.11]	0.162	100.00 / 100.00
H01 Shortest source distance	0.142	4.55 [1.64, 7.46]	0.039	100.00 / 100.00
source edges arrival (4,835 records)	0.134	0.05 [-0.02, 0.13]	1.000	100.00 / 100.00
B05 Regional delay distribution	0.140	2.71 [0.32, 5.09]	0.245	100.00 / 100.00
region delay 20 min minimum (13,490 records)	0.136	0.19 [-0.08, 0.47]	1.000	100.00 / 66.67
G10 Edge-slack distribution	-0.137	-2.26 [-3.49, -1.02]	0.013	100.00 / 100.00
incoming slack 90th percentile arrival (12,610 records)	-0.127	-0.02 [-0.19, 0.15]	1.000	80.00 / 66.67
H09 Reachable delay totals	-0.135	-4.36 [-6.22, -2.50]	0.001	100.00 / 100.00
upstream event delay departure (13,490 records)	-0.138	-0.66 [-0.92, -0.40]	0.003	100.00 / 100.00
G14 Downstream-neighbor delay	0.134	2.69 [0.47, 4.92]	0.184	100.00 / 100.00
neighbor delay sum arrival (13,490 records)	0.126	-0.14 [-0.47, 0.19]	1.000	93.33 / 66.67
H08 Spatial reachability	0.132	1.62 [-0.86, 4.10]	1.000	100.00 / 100.00
downstream stations departure (13,490 records)	0.128	0.06 [-0.26, 0.38]	1.000	100.00 / 66.67
H08 Spatial reachability	-0.131	-4.45 [-6.52, -2.37]	0.003	100.00 / 100.00
upstream branches departure (13,490 records)	-0.135	-0.86 [-1.23, -0.49]	0.003	100.00 / 100.00
G10 Edge-slack distribution	-0.130	-1.73 [-2.96, -0.51]	0.079	100.00 / 100.00
incoming slack interquartile range arrival (12,610 records)	-0.121	0.03 [-0.15, 0.22]	1.000	93.33 / 66.67
G10 Edge-slack distribution	-0.130	-1.70 [-2.91, -0.49]	0.080	100.00 / 100.00
incoming slack standard deviation arrival (12,610 records)	-0.120	0.00 [-0.17, 0.18]	1.000	46.67 / 66.67

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G13 Upstream-neighbor delay	0.130	3.84 [1.45, 6.23]	0.033	100.00 / 100.00
neighbor delay count arrival (13,490 records)	0.127	0.22 [-0.10, 0.53]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.127	-1.70 [-2.92, -0.48]	0.086	100.00 / 100.00
incoming slack maximum arrival (12,610 records)	-0.118	0.03 [-0.15, 0.22]	1.000	93.33 / 66.67
B09 Boundary pressure imbalance	-0.126	-4.20 [-7.91, -0.49]	0.248	100.00 / 100.00
cross boundary pressure balance 20 min (13,490 records)	-0.117	0.33 [0.06, 0.59]	0.209	100.00 / 100.00
H11 Harmonic closeness	0.126	2.78 [-0.11, 5.67]	0.500	100.00 / 100.00
outgoing harmonic departure (13,490 records)	0.120	-0.04 [-0.36, 0.27]	1.000	93.33 / 33.33
H10 Active betweenness	-0.124	-3.82 [-5.31, -2.33]	5.4e-4	100.00 / 100.00
betweenness arrival (13,490 records)	-0.127	-0.62 [-0.90, -0.34]	0.005	100.00 / 100.00
G19 Endpoint contrasts	0.124	3.84 [2.71, 4.97]	1.4e-5	100.00 / 100.00
run F06 direction pool difference (13,490 records)	0.134	1.33 [0.73, 1.93]	0.005	100.00 / 100.00
H01 Shortest source distance	0.124	3.84 [0.53, 7.15]	0.224	100.00 / 100.00
source minimum time lag arrival (4,835 records)	0.117	0.03 [-0.06, 0.12]	1.000	100.00 / 100.00
H05 Downstream reachability	-0.124	-3.74 [-5.13, -2.34]	4.0e-4	100.00 / 100.00
downstream arrival (13,490 records)	-0.127	-0.56 [-0.83, -0.29]	0.006	100.00 / 100.00
G10 Edge-slack distribution	-0.124	-2.20 [-4.12, -0.28]	0.238	100.00 / 100.00
incoming slack sum departure (13,490 records)	-0.113	1.12 [0.39, 1.85]	0.063	100.00 / 100.00
G10 Edge-slack distribution	-0.123	-2.84 [-4.13, -1.56]	0.002	100.00 / 100.00
incoming slack sum arrival (13,490 records)	-0.115	-0.09 [-0.24, 0.06]	1.000	100.00 / 66.67
G10 Edge-slack distribution	-0.122	-2.05 [-4.06, -0.04]	0.394	100.00 / 100.00
incoming slack mean departure (13,490 records)	-0.112	1.31 [0.55, 2.06]	0.028	100.00 / 100.00
H11 Harmonic closeness	-0.122	-4.16 [-5.73, -2.60]	4.3e-4	100.00 / 100.00
outgoing harmonic arrival (13,490 records)	-0.125	-0.74 [-1.09, -0.40]	0.005	100.00 / 100.00
G12 Delay-support distribution	0.122	2.47 [-0.17, 5.10]	0.548	100.00 / 100.00
outgoing support minimum arrival (13,490 records)	0.114	-0.48 [-0.82, -0.14]	0.111	100.00 / 100.00
G10 Edge-slack distribution	0.121	-0.02 [-1.80, 1.76]	1.000	60.00 / 66.67
outgoing slack mean arrival (13,490 records)	0.124	1.28 [0.57, 1.99]	0.021	100.00 / 100.00
G10 Edge-slack distribution	0.121	-0.02 [-1.80, 1.76]	1.000	60.00 / 66.67
outgoing slack median arrival (13,490 records)	0.124	1.28 [0.57, 1.99]	0.021	100.00 / 100.00
H08 Spatial reachability	-0.120	-4.01 [-5.57, -2.45]	5.4e-4	100.00 / 100.00
downstream stations arrival (13,490 records)	-0.123	-0.68 [-1.01, -0.36]	0.006	100.00 / 100.00
G10 Edge-slack distribution	0.119	-0.26 [-1.88, 1.37]	1.000	86.67 / 66.67
outgoing slack sum arrival (13,490 records)	0.122	1.05 [0.46, 1.65]	0.024	100.00 / 100.00
H09 Reachable delay totals	0.119	0.43 [-1.29, 2.14]	1.000	100.00 / 66.67
downstream event delay departure (13,490 records)	0.114	0.07 [-0.23, 0.38]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.118	-2.00 [-3.92, -0.08]	0.364	100.00 / 100.00
incoming slack median departure (13,490 records)	-0.108	1.22 [0.47, 1.96]	0.039	100.00 / 100.00
H16 Local motifs	-0.118	-3.61 [-5.10, -2.12]	8.5e-4	100.00 / 100.00
fan-out pairs arrival (13,490 records)	-0.117	-0.47 [-0.65, -0.29]	0.003	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H06 Upstream trains	-0.117	-3.55 [-5.64, -1.47]	0.022	100.00 / 100.00
upstream departure (13,490 records)	-0.121	-0.77 [-1.11, -0.42]	0.005	100.00 / 100.00
B03 Active local density	0.115	2.73 [0.91, 4.55]	0.052	100.00 / 100.00
active density 5 min (12,601 records)	0.114	0.12 [-0.24, 0.49]	1.000	93.33 / 66.67
H09 Reachable delay totals	-0.114	-3.23 [-4.48, -1.98]	5.0e-4	100.00 / 100.00
downstream event delay arrival (13,490 records)	-0.118	-0.48 [-0.70, -0.26]	0.005	100.00 / 100.00
B02 Local edges and density	-0.110	-3.36 [-6.28, -0.43]	0.237	100.00 / 100.00
density 10 min (13,191 records)	-0.100	0.48 [0.15, 0.81]	0.083	100.00 / 100.00
H09 Reachable delay totals	-0.108	-3.25 [-5.10, -1.40]	0.017	100.00 / 100.00
upstream train maximum delay sum departure (13,490 records)	-0.112	-0.65 [-0.94, -0.37]	0.003	100.00 / 100.00
G14 Downstream-neighbor delay	0.108	2.08 [-0.12, 4.28]	0.532	100.00 / 100.00
neighbor delay mean departure (10,772 records)	0.106	0.39 [0.06, 0.73]	0.281	100.00 / 100.00
G14 Downstream-neighbor delay	0.108	2.08 [-0.12, 4.28]	0.532	100.00 / 100.00
neighbor delay median departure (10,772 records)	0.106	0.39 [0.06, 0.73]	0.281	100.00 / 100.00
G14 Downstream-neighbor delay	0.108	2.18 [0.02, 4.34]	0.415	100.00 / 100.00
neighbor delay maximum departure (10,772 records)	0.106	0.43 [0.10, 0.76]	0.168	100.00 / 100.00
B04 Boundary connections	0.107	3.16 [0.54, 5.78]	0.185	100.00 / 100.00
leaving 10 min (13,490 records)	0.093	-0.61 [-1.06, -0.17]	0.118	100.00 / 100.00
G14 Downstream-neighbor delay	0.107	2.16 [-0.01, 4.34]	0.436	100.00 / 100.00
neighbor delay 90th percentile departure (10,772 records)	0.106	0.42 [0.09, 0.75]	0.185	100.00 / 100.00
G08 Active degree	-0.106	-3.23 [-5.47, -0.98]	0.071	100.00 / 100.00
incoming active departure (13,490 records)	-0.111	-0.87 [-1.28, -0.47]	0.005	100.00 / 100.00
G14 Downstream-neighbor delay	0.106	1.95 [-0.27, 4.16]	0.681	100.00 / 100.00
neighbor delay minimum departure (10,772 records)	0.104	0.35 [0.01, 0.70]	0.503	100.00 / 100.00
G14 Downstream-neighbor delay	0.105	1.09 [-0.31, 2.50]	0.993	100.00 / 100.00
neighbor delay interquartile range departure (10,772 records)	0.100	0.33 [-0.03, 0.68]	0.716	100.00 / 100.00
G14 Downstream-neighbor delay	0.105	1.09 [-0.31, 2.50]	0.993	100.00 / 100.00
neighbor delay standard deviation departure (10,772 records)	0.100	0.33 [-0.03, 0.68]	0.716	100.00 / 100.00
H03 Event time from release	0.103	2.89 [-0.19, 5.98]	0.548	100.00 / 100.00
time from release departure (13,490 records)	0.093	-0.86 [-1.43, -0.29]	0.072	100.00 / 100.00
G08 Active degree	-0.102	-3.45 [-4.95, -1.96]	0.001	100.00 / 100.00
outgoing active arrival (13,490 records)	-0.106	-0.84 [-1.22, -0.45]	0.005	100.00 / 100.00
H05 Downstream reachability	0.101	1.01 [-1.26, 3.27]	1.000	100.00 / 100.00
downstream departure (13,490 records)	0.096	-0.04 [-0.32, 0.23]	1.000	93.33 / 66.67
G11 Near-tight links	0.098	3.66 [0.12, 7.19]	0.373	100.00 / 100.00
slack at most 2 minutes departure (13,490 records)	0.084	-0.79 [-1.27, -0.31]	0.039	100.00 / 100.00
B04 Boundary connections	0.098	3.65 [0.90, 6.39]	0.110	100.00 / 100.00
leaving 20 min (13,490 records)	0.084	-0.61 [-1.05, -0.18]	0.111	100.00 / 100.00
H07 Downstream trains	-0.098	-3.24 [-4.49, -1.98]	5.2e-4	100.00 / 100.00
downstream arrival (13,490 records)	-0.101	-0.67 [-1.01, -0.34]	0.008	100.00 / 100.00
G10 Edge-slack distribution	0.097	-0.99 [-2.87, 0.89]	1.000	100.00 / 66.67
outgoing slack 90th percentile arrival (13,490 records)	0.100	0.92 [0.36, 1.49]	0.036	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H08 Spatial reachability	-0.096	-3.21 [-4.61, -1.82]	0.001	100.00 / 100.00
downstream branches arrival (13,490 records)	-0.099	-0.70 [-1.04, -0.36]	0.008	100.00 / 100.00
B02 Local edges and density	-0.094	-2.64 [-3.95, -1.32]	0.006	100.00 / 100.00
edges 5 min (13,490 records)	-0.104	-0.85 [-1.24, -0.46]	0.005	100.00 / 100.00
G05 Event degree	0.093	2.86 [0.73, 4.99]	0.105	100.00 / 100.00
outgoing arrival (13,490 records)	0.087	-0.17 [-0.70, 0.36]	1.000	93.33 / 66.67
G06 Cross-train degree	0.093	2.86 [0.73, 4.99]	0.105	100.00 / 100.00
outgoing cross arrival (13,490 records)	0.087	-0.17 [-0.70, 0.36]	1.000	93.33 / 66.67
G07 Degree by relation	0.093	2.86 [0.73, 4.99]	0.105	100.00 / 100.00
outgoing exit headway arrival (13,490 records)	0.087	-0.17 [-0.70, 0.36]	1.000	93.33 / 66.67
G10 Edge-slack distribution	0.093	2.86 [0.73, 4.99]	0.105	100.00 / 100.00
outgoing slack count arrival (13,490 records)	0.087	-0.17 [-0.70, 0.36]	1.000	93.33 / 66.67
G12 Delay-support distribution	0.093	2.86 [0.73, 4.99]	0.105	100.00 / 100.00
outgoing support count arrival (13,490 records)	0.087	-0.17 [-0.70, 0.36]	1.000	93.33 / 66.67
G14 Downstream-neighbor delay	0.093	2.86 [0.73, 4.99]	0.105	100.00 / 100.00
neighbor delay count arrival (13,490 records)	0.087	-0.17 [-0.70, 0.36]	1.000	93.33 / 66.67
H09 Reachable delay totals	-0.093	-2.87 [-4.03, -1.71]	6.9e-4	100.00 / 100.00
downstream train maximum delay sum arrival (13,490 records)	-0.097	-0.57 [-0.83, -0.30]	0.006	100.00 / 100.00
G14 Downstream-neighbor delay	0.093	2.87 [0.64, 5.11]	0.134	100.00 / 100.00
neighbor delay count departure (13,490 records)	0.084	-0.29 [-0.77, 0.18]	1.000	100.00 / 100.00
G10 Edge-slack distribution	0.092	-1.15 [-3.02, 0.73]	1.000	100.00 / 100.00
outgoing slack maximum arrival (13,490 records)	0.095	0.86 [0.32, 1.39]	0.044	100.00 / 100.00
G10 Edge-slack distribution	-0.091	-3.22 [-4.98, -1.46]	0.012	100.00 / 100.00
outgoing slack median departure (13,490 records)	-0.091	-0.18 [-0.45, 0.09]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.091	-3.23 [-5.17, -1.28]	0.026	100.00 / 100.00
outgoing slack 90th percentile departure (13,490 records)	-0.091	0.05 [-0.29, 0.40]	1.000	93.33 / 66.67
B01 Local region size	0.091	2.61 [-0.47, 5.69]	0.774	100.00 / 100.00
region events 10 min (13,490 records)	0.076	-0.85 [-1.32, -0.38]	0.020	100.00 / 100.00
B05 Regional delay distribution	0.091	2.61 [-0.47, 5.69]	0.774	100.00 / 100.00
region delay 10 min count (13,490 records)	0.076	-0.85 [-1.32, -0.38]	0.020	100.00 / 100.00
G10 Edge-slack distribution	-0.090	-3.18 [-5.14, -1.21]	0.031	100.00 / 100.00
outgoing slack interquartile range departure (13,490 records)	-0.090	0.11 [-0.26, 0.47]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.090	-3.18 [-5.14, -1.21]	0.031	100.00 / 100.00
outgoing slack maximum departure (13,490 records)	-0.090	0.11 [-0.26, 0.47]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.090	-3.26 [-5.14, -1.38]	0.020	100.00 / 100.00
outgoing slack sum departure (13,490 records)	-0.090	-0.03 [-0.34, 0.27]	1.000	86.67 / 100.00
G10 Edge-slack distribution	-0.090	-3.19 [-5.13, -1.24]	0.028	100.00 / 100.00
outgoing slack standard deviation departure (13,490 records)	-0.090	0.08 [-0.27, 0.44]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.089	-3.27 [-5.17, -1.37]	0.020	100.00 / 100.00
outgoing slack mean departure (13,490 records)	-0.090	-0.03 [-0.35, 0.29]	1.000	80.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G07 Degree by relation	0.089	2.73 [0.60, 4.86]	0.135	100.00 / 100.00
outgoing entry headway departure (13,490 records)	0.082	-0.24 [-0.82, 0.33]	1.000	93.33 / 66.67
G14 Downstream-neighbor delay	0.087	1.62 [-0.58, 3.82]	1.000	100.00 / 100.00
neighbor delay maximum arrival (9,875 records)	0.082	-0.07 [-0.33, 0.19]	1.000	100.00 / 33.33
G14 Downstream-neighbor delay	0.087	1.62 [-0.58, 3.82]	1.000	100.00 / 100.00
neighbor delay mean arrival (9,875 records)	0.082	-0.07 [-0.33, 0.19]	1.000	100.00 / 33.33
G14 Downstream-neighbor delay	0.087	1.62 [-0.58, 3.82]	1.000	100.00 / 100.00
neighbor delay median arrival (9,875 records)	0.082	-0.07 [-0.33, 0.19]	1.000	100.00 / 33.33
G14 Downstream-neighbor delay	0.087	1.62 [-0.58, 3.82]	1.000	100.00 / 100.00
neighbor delay minimum arrival (9,875 records)	0.082	-0.07 [-0.33, 0.19]	1.000	100.00 / 33.33
G14 Downstream-neighbor delay	0.087	1.62 [-0.58, 3.82]	1.000	100.00 / 100.00
neighbor delay 90th percentile arrival (9,875 records)	0.082	-0.07 [-0.33, 0.19]	1.000	100.00 / 33.33
B02 Local edges and density	0.085	2.87 [-0.23, 5.97]	0.572	100.00 / 100.00
edges 10 min (13,490 records)	0.070	-0.77 [-1.18, -0.35]	0.018	100.00 / 100.00
H15 Chain and crossing length	-0.083	-2.45 [-4.48, -0.42]	0.185	100.00 / 100.00
longest crossings departure (13,490 records)	-0.087	-0.58 [-0.87, -0.30]	0.008	100.00 / 100.00
G16 Incident-link imbalance	0.083	1.65 [0.02, 3.29]	0.413	100.00 / 100.00
pressure balance departure (13,490 records)	0.082	0.80 [0.27, 1.33]	0.069	100.00 / 100.00
B01 Local region size	-0.081	-2.30 [-3.70, -0.91]	0.027	100.00 / 100.00
region events 5 min (13,490 records)	-0.091	-0.93 [-1.37, -0.48]	0.007	100.00 / 100.00
B05 Regional delay distribution	-0.081	-2.30 [-3.70, -0.91]	0.027	100.00 / 100.00
region delay 5 min count (13,490 records)	-0.091	-0.93 [-1.37, -0.48]	0.007	100.00 / 100.00
G12 Delay-support distribution	0.078	-1.24 [-2.26, -0.23]	0.173	100.00 / 100.00
incoming support standard deviation arrival (12,610 records)	0.073	-0.30 [-0.52, -0.07]	0.149	100.00 / 100.00
G12 Delay-support distribution	0.078	-1.24 [-2.26, -0.23]	0.173	100.00 / 100.00
incoming support interquartile range arrival (12,610 records)	0.073	-0.30 [-0.53, -0.08]	0.144	100.00 / 100.00
B16 Global reachability	0.077	2.92 [0.70, 5.13]	0.115	100.00 / 100.00
mean reachable distance (13,490 records)	0.077	0.34 [0.01, 0.68]	0.495	100.00 / 100.00
B09 Boundary pressure imbalance	-0.075	-1.69 [-3.79, 0.41]	0.899	100.00 / 100.00
cross boundary pressure balance 10 min (13,490 records)	-0.070	0.38 [0.12, 0.64]	0.090	100.00 / 100.00
B02 Local edges and density	-0.073	-2.46 [-4.87, -0.04]	0.398	100.00 / 100.00
density 5 min (12,601 records)	-0.067	0.50 [0.18, 0.81]	0.055	100.00 / 100.00
H23 Local clustering	0.073	2.29 [0.86, 3.72]	0.033	100.00 / 100.00
undirected clustering departure (13,490 records)	0.069	0.00 [-0.03, 0.04]	1.000	86.67 / 33.33
B01 Local region size	0.072	2.20 [-0.51, 4.92]	0.884	100.00 / 100.00
region trains 5 min (13,490 records)	0.063	-0.43 [-0.79, -0.08]	0.220	100.00 / 100.00
H16 Local motifs	-0.071	-2.18 [-3.03, -1.32]	5.5e-4	100.00 / 100.00
diamonds at source arrival (13,490 records)	-0.071	-0.27 [-0.37, -0.17]	0.002	100.00 / 100.00
G11 Near-tight links	-0.061	-1.83 [-3.91, 0.24]	0.670	100.00 / 100.00
slack at most 1 minute departure (13,490 records)	-0.071	-0.84 [-1.27, -0.42]	0.010	100.00 / 100.00

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Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B04 Boundary connections	-0.059	-1.78 [-3.20, -0.36]	0.152	100.00 / 100.00
entering 5 min (13,490 records)	-0.068	-0.78 [-1.16, -0.40]	0.007	100.00 / 100.00
G19 Endpoint contrasts	-0.059	-1.72 [-2.78, -0.66]	0.031	100.00 / 100.00
G05 incoming degree endpoint contrast (13,490 records)	-0.058	-0.19 [-0.33, -0.04]	0.167	100.00 / 100.00
H03 Event time from release	0.058	1.72 [-1.23, 4.67]	1.000	100.00 / 100.00
time from release arrival (13,490 records)	0.047	-1.44 [-2.19, -0.70]	0.010	100.00 / 100.00
G18 External-source share	-0.058	-1.75 [-2.45, -1.06]	6.2e-4	100.00 / 100.00
source fraction departure (13,490 records)	-0.058	-0.22 [-0.31, -0.14]	0.002	100.00 / 100.00
G07 Degree by relation	-0.058	-1.77 [-2.48, -1.05]	7.3e-4	100.00 / 100.00
incoming emergency departure (13,490 records)	-0.058	-0.22 [-0.31, -0.14]	0.002	100.00 / 100.00
G11 Near-tight links	0.057	2.52 [-0.15, 5.20]	0.539	100.00 / 100.00
slack at most 2 minutes arrival (13,490 records)	0.046	-0.84 [-1.26, -0.42]	0.009	100.00 / 100.00
G07 Degree by relation	0.056	1.73 [0.70, 2.76]	0.025	100.00 / 100.00
incoming exit headway arrival (13,490 records)	0.055	0.13 [-0.03, 0.29]	1.000	100.00 / 100.00
B08 Community structure	-0.055	-1.68 [-2.77, -0.58]	0.046	100.00 / 100.00
cross community edges departure (13,490 records)	-0.057	-0.32 [-0.46, -0.17]	0.005	100.00 / 100.00
G05 Event degree	0.055	2.00 [0.24, 3.75]	0.243	100.00 / 100.00
outgoing departure (13,490 records)	0.048	-0.26 [-0.78, 0.25]	1.000	93.33 / 66.67
G06 Cross-train degree	0.055	2.00 [0.24, 3.75]	0.243	100.00 / 100.00
outgoing cross departure (13,490 records)	0.048	-0.26 [-0.78, 0.25]	1.000	93.33 / 66.67
G10 Edge-slack distribution	0.055	2.00 [0.24, 3.75]	0.243	100.00 / 100.00
outgoing slack count departure (13,490 records)	0.048	-0.26 [-0.78, 0.25]	1.000	93.33 / 66.67
G12 Delay-support distribution	0.055	2.00 [0.24, 3.75]	0.243	100.00 / 100.00
outgoing support count departure (13,490 records)	0.048	-0.26 [-0.78, 0.25]	1.000	93.33 / 66.67
H16 Local motifs	-0.055	-1.67 [-2.67, -0.67]	0.026	100.00 / 100.00
fan-in pairs departure (13,490 records)	-0.055	-0.30 [-0.44, -0.17]	0.005	100.00 / 100.00
H16 Local motifs	0.054	1.63 [-1.35, 4.61]	1.000	100.00 / 100.00
diamonds at source departure (13,490 records)	0.046	-0.32 [-0.50, -0.15]	0.019	100.00 / 100.00
G16 Incident-link imbalance	0.054	1.94 [0.22, 3.66]	0.253	100.00 / 100.00
degree balance departure (13,490 records)	0.047	-0.20 [-0.70, 0.30]	1.000	93.33 / 66.67
B03 Active local density	0.053	-1.05 [-2.32, 0.22]	0.827	100.00 / 100.00
active density 20 min (13,387 records)	0.053	-0.01 [-0.34, 0.32]	1.000	60.00 / 33.33
G19 Endpoint contrasts	0.052	3.19 [1.21, 5.18]	0.033	100.00 / 100.00
run R01 mean occupancy 10 min difference (13,490 records)	0.068	1.82 [1.04, 2.59]	0.003	100.00 / 100.00
H23 Local clustering	0.052	1.58 [0.19, 2.97]	0.246	100.00 / 100.00
undirected clustering arrival (13,490 records)	0.049	0.00 [-0.03, 0.03]	1.000	46.67 / 33.33
B04 Boundary connections	0.048	1.50 [-1.12, 4.12]	1.000	100.00 / 100.00
entering 10 min (13,490 records)	0.036	-0.77 [-1.20, -0.35]	0.018	100.00 / 100.00
G06 Cross-train degree	0.048	1.46 [0.67, 2.25]	0.011	100.00 / 100.00
incoming cross departure (13,490 records)	0.046	-0.02 [-0.27, 0.23]	1.000	86.67 / 66.67

Continued on the next page.

Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G07 Degree by relation	0.048	1.46 [0.67, 2.25]	0.011	100.00 / 100.00
incoming entry headway departure (13,490 records)	0.046	-0.02 [-0.27, 0.23]	1.000	86.67 / 66.67
G13 Upstream-neighbor delay	0.048	1.46 [0.67, 2.25]	0.011	100.00 / 100.00
neighbor delay count departure (13,490 records)	0.046	-0.02 [-0.27, 0.23]	1.000	86.67 / 66.67
G05 Event degree	0.046	1.60 [0.47, 2.73]	0.079	100.00 / 100.00
incoming arrival (13,490 records)	0.045	0.13 [-0.03, 0.29]	0.986	100.00 / 100.00
G10 Edge-slack distribution	0.046	1.60 [0.47, 2.73]	0.079	100.00 / 100.00
incoming slack count arrival (13,490 records)	0.045	0.13 [-0.03, 0.29]	0.986	100.00 / 100.00
G12 Delay-support distribution	0.046	1.60 [0.47, 2.73]	0.079	100.00 / 100.00
incoming support count arrival (13,490 records)	0.045	0.13 [-0.03, 0.29]	0.986	100.00 / 100.00
G06 Cross-train degree	0.045	1.56 [0.40, 2.71]	0.104	100.00 / 100.00
incoming cross arrival (13,490 records)	0.044	0.13 [-0.05, 0.30]	1.000	100.00 / 100.00
G07 Degree by relation	0.041	1.26 [-0.07, 2.59]	0.528	100.00 / 100.00
incoming station track arrival (13,490 records)	0.040	0.11 [-0.08, 0.30]	1.000	100.00 / 100.00
H01 Shortest source distance	0.038	0.47 [-0.43, 1.37]	1.000	100.00 / 100.00
source edges departure (4,440 records)	0.037	0.00 [-0.02, 0.02]	1.000	100.00 / 66.67
G10 Edge-slack distribution	0.036	0.73 [-1.75, 3.22]	1.000	100.00 / 100.00
outgoing slack zero fraction departure (13,490 records)	0.038	0.35 [-0.09, 0.79]	1.000	100.00 / 100.00
G19 Endpoint contrasts	0.036	-1.58 [-3.60, 0.44]	0.981	100.00 / 100.00
H10 active betweenness endpoint contrast (13,490 records)	0.035	0.00 [-0.06, 0.06]	1.000	46.67 / 33.33
G07 Degree by relation	0.036	1.09 [0.30, 1.88]	0.088	100.00 / 100.00
incoming run arrival (13,490 records)	0.036	0.17 [0.05, 0.28]	0.097	100.00 / 100.00
G07 Degree by relation	0.034	1.03 [-0.75, 2.81]	1.000	100.00 / 100.00
outgoing station track departure (13,490 records)	0.027	-0.24 [-0.63, 0.16]	1.000	93.33 / 66.67
G10 Edge-slack distribution	0.033	-2.61 [-4.17, -1.06]	0.024	100.00 / 100.00
outgoing slack interquartile range arrival (13,490 records)	0.032	-0.01 [-0.31, 0.28]	1.000	73.33 / 33.33
G10 Edge-slack distribution	0.033	-2.61 [-4.17, -1.06]	0.024	100.00 / 100.00
outgoing slack standard deviation arrival (13,490 records)	0.032	-0.01 [-0.31, 0.28]	1.000	73.33 / 33.33
B09 Boundary pressure imbalance	-0.033	-0.75 [-2.36, 0.87]	1.000	100.00 / 100.00
cross boundary pressure balance 5 min (13,490 records)	-0.029	0.46 [0.16, 0.76]	0.063	100.00 / 100.00
G10 Edge-slack distribution	0.030	1.87 [0.75, 2.98]	0.024	100.00 / 100.00
incoming slack minimum arrival (12,610 records)	0.029	-0.03 [-0.08, 0.02]	1.000	100.00 / 66.67
G17 Relation-type entropy	-0.030	1.15 [0.20, 2.10]	0.186	100.00 / 100.00
type entropy arrival (13,490 records)	-0.026	0.16 [0.00, 0.32]	0.578	100.00 / 100.00
B04 Boundary connections	0.028	1.12 [-0.82, 3.06]	1.000	100.00 / 100.00
leaving 5 min (13,490 records)	0.017	-0.61 [-1.06, -0.16]	0.126	100.00 / 100.00
G16 Incident-link imbalance	0.026	0.06 [-1.35, 1.48]	1.000	86.67 / 66.67
degree balance arrival (13,490 records)	0.022	-0.23 [-0.65, 0.19]	1.000	93.33 / 100.00
H01 Shortest source distance	0.023	0.26 [-0.53, 1.05]	1.000	100.00 / 66.67
source minimum time lag departure (4,440 records)	0.023	0.00 [-0.02, 0.02]	1.000	86.67 / 66.67
G17 Relation-type entropy	-0.019	-0.09 [-1.85, 1.66]	1.000	73.33 / 33.33
type entropy departure (13,490 records)	-0.023	-0.28 [-0.62, 0.06]	0.986	100.00 / 100.00
B03 Active local density	0.018	-0.61 [-1.89, 0.67]	1.000	100.00 / 66.67
active density 10 min (13,191 records)	0.018	0.11 [-0.25, 0.48]	1.000	100.00 / 66.67

Continued on the next page.

Table S7 continued: Strategy-2 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G11 Near-tight links	-0.016	-0.18 [-1.83, 1.47]	1.000	93.33 / 66.67
slack at most 1 minute arrival (13,490 records)	-0.026	-0.90 [-1.30, -0.50]	0.004	100.00 / 100.00
H10 Active betweenness	-0.015	-2.33 [-4.14, -0.52]	0.132	100.00 / 100.00
betweenness departure next endpoint (13,490 records)	-0.009	0.03 [-0.18, 0.24]	1.000	93.33 / 66.67
G19 Endpoint contrasts	0.013	0.71 [-1.02, 2.44]	1.000	100.00 / 100.00
G05 outgoing degree endpoint contrast (13,490 records)	0.007	-0.27 [-0.66, 0.12]	1.000	100.00 / 100.00
B08 Community structure	-0.013	-0.57 [-2.04, 0.90]	1.000	100.00 / 100.00
cross community edges arrival (13,490 records)	-0.015	-0.31 [-0.45, -0.18]	0.004	100.00 / 100.00
H12 PageRank	0.011	-0.79 [-2.32, 0.75]	1.000	100.00 / 100.00
pagerank arrival (13,490 records)	0.004	-0.66 [-0.97, -0.35]	0.005	100.00 / 100.00
G05 Event degree	-0.006	-0.22 [-1.39, 0.95]	1.000	93.33 / 66.67
incoming departure (13,490 records)	-0.007	-0.17 [-0.36, 0.02]	0.715	100.00 / 100.00
G10 Edge-slack distribution	-0.006	-0.22 [-1.39, 0.95]	1.000	93.33 / 66.67
incoming slack count departure (13,490 records)	-0.007	-0.17 [-0.36, 0.02]	0.715	100.00 / 100.00
G12 Delay-support distribution	-0.006	-0.22 [-1.39, 0.95]	1.000	93.33 / 66.67
incoming support count departure (13,490 records)	-0.007	-0.17 [-0.36, 0.02]	0.715	100.00 / 100.00
G07 Degree by relation	0.002	0.05 [-0.64, 0.74]	1.000	73.33 / 33.33
incoming emergency arrival (13,490 records)	0.001	-0.15 [-0.22, -0.08]	0.006	100.00 / 100.00
G12 Delay-support distribution	-0.001	-3.42 [-4.87, -1.96]	0.001	100.00 / 100.00
incoming support interquartile range departure (13,490 records)	-0.005	-0.41 [-0.59, -0.24]	0.003	100.00 / 100.00
G12 Delay-support distribution	-0.001	-3.41 [-4.86, -1.96]	0.001	100.00 / 100.00
incoming support standard deviation departure (13,490 records)	-0.005	-0.41 [-0.59, -0.24]	0.003	100.00 / 100.00
G18 External-source share	0.000	-0.01 [-0.71, 0.69]	1.000	60.00 / 66.67
source fraction arrival (12,610 records)	-0.002	-0.15 [-0.23, -0.08]	0.005	100.00 / 100.00

Notes: Codes and definitions follow Table S1. Rank corr.: scenario-weighted Spearman correlation. Upper/lower lines: change probability/signed duration change. Effects per indicator standard deviation are in percentage points/minutes, respectively. Brackets contain 95% confidence intervals. Adjusted *p*: Benjamini–Yekutieli p-value; 1.1e-7 denotes 1.1×10^{-7}. Direction stable: unchanged coefficient sign after omitting one start-time group/one blockage location (percent). Pair differences are the following train value minus the leading train value. Record counts include only finite values of the indicated form.

S3.5 Strategy-3 dwell durations

Table S8: Complete indicator associations: Strategy-3 dwell durations.

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
Disruption conditions				
X17 Remaining route fraction	0.171	4.29 [3.18, 5.40]	1.4e-7	100.00 / 100.00
remaining route fraction (9,227 records)	-0.100	-0.67 [-1.05, -0.29]	0.044	100.00 / 100.00
E20 Planned starts before release	0.151	3.70 [2.48, 4.92]	1.1e-5	100.00 / 100.00
planned starts before reopen (9,227 records)	-0.062	-0.53 [-0.82, -0.24]	0.042	100.00 / 100.00
E16 Emergency backlog	0.149	3.18 [1.96, 4.39]	1.2e-4	100.00 / 100.00
backlog (9,227 records)	0.019	0.12 [-0.05, 0.30]	1.000	100.00 / 66.67
E12 Origin waiting count	0.149	3.04 [1.70, 4.37]	8.4e-4	100.00 / 100.00
origin waiting (9,227 records)	-0.007	-0.02 [-0.25, 0.21]	1.000	60.00 / 33.33
E21 Train-state counts at blockage start	0.149	3.04 [1.70, 4.37]	8.4e-4	100.00 / 100.00
origin hold (9,227 records)	-0.007	-0.02 [-0.25, 0.21]	1.000	60.00 / 33.33
E02 Executed route fraction	-0.105	-2.91 [-3.73, -2.10]	6.5e-7	100.00 / 100.00
executed fraction (9,227 records)	0.106	0.19 [0.00, 0.37]	0.413	100.00 / 100.00
E03 Current route position	-0.105	-2.91 [-3.73, -2.10]	6.5e-7	100.00 / 100.00
current route index (9,227 records)	0.106	0.19 [0.00, 0.37]	0.413	100.00 / 100.00
E07 Emergency arrival time	-0.094	-2.88 [-4.30, -1.46]	0.003	100.00 / 100.00
emergency arrival (2,830 records)	-0.025	-0.05 [-0.11, 0.02]	1.000	100.00 / 100.00
E01 Train state at start time	0.093	2.37 [1.27, 3.46]	0.002	100.00 / 100.00
state 0 (9,227 records)	-0.114	-0.17 [-0.35, 0.01]	0.518	100.00 / 100.00
E17 Early release concentration	-0.091	-1.76 [-2.94, -0.58]	0.053	100.00 / 100.00
release first5 fraction (9,227 records)	0.051	0.37 [0.06, 0.69]	0.200	100.00 / 100.00
E21 Train-state counts at blockage start	-0.090	-2.96 [-3.87, -2.05]	3.7e-6	100.00 / 100.00
completed (9,227 records)	0.034	0.27 [0.08, 0.47]	0.097	100.00 / 100.00
D09 Time from start time	0.087	2.21 [1.42, 3.00]	3.9e-5	100.00 / 100.00
reference from start (9,227 records)	-0.037	-0.29 [-0.52, -0.07]	0.126	100.00 / 100.00
X02 Already past blocked section	-0.082	-1.94 [-2.45, -1.42]	2.6e-7	100.00 / 100.00
passed at start time (5,062 records)	0.000	0.01 [-0.02, 0.03]	1.000	100.00 / 66.67
X03 Planned clearance before start time	-0.082	-1.94 [-2.45, -1.42]	2.6e-7	100.00 / 100.00
planned clear before (5,062 records)	0.000	0.01 [-0.02, 0.03]	1.000	100.00 / 66.67
X12 Can clear before start time	-0.082	-1.94 [-2.45, -1.42]	2.6e-7	100.00 / 100.00
can clear without emergency (5,062 records)	0.000	0.01 [-0.02, 0.03]	1.000	100.00 / 66.67
E13 Station waiting count	0.081	2.34 [1.41, 3.28]	2.1e-4	100.00 / 100.00
station waiting (9,227 records)	0.077	0.36 [0.16, 0.56]	0.040	100.00 / 66.67
D03 Blockage duration	0.078	1.82 [0.75, 2.89]	0.018	100.00 / 100.00
duration (9,227 records)	-0.035	-0.45 [-0.75, -0.16]	0.065	100.00 / 100.00
D08 Remaining blockage	0.078	1.82 [0.75, 2.89]	0.018	100.00 / 100.00
remaining (9,227 records)	-0.035	-0.45 [-0.75, -0.16]	0.065	100.00 / 100.00
D20 Lost resource time	0.078	1.82 [0.75, 2.89]	0.018	100.00 / 100.00
lost resource minutes (9,227 records)	-0.035	-0.45 [-0.75, -0.16]	0.065	100.00 / 100.00
I05 Predicted remaining duration	0.078	1.82 [0.75, 2.89]	0.018	100.00 / 100.00
predicted remaining (9,227 records)	-0.035	-0.45 [-0.75, -0.16]	0.065	100.00 / 100.00
I06 Remaining-duration bounds	0.078	1.82 [0.75, 2.89]	0.018	100.00 / 100.00
lower (9,227 records)	-0.035	-0.45 [-0.75, -0.16]	0.065	100.00 / 100.00
I06 Remaining-duration bounds	0.078	1.82 [0.75, 2.89]	0.018	100.00 / 100.00
upper (9,227 records)	-0.035	-0.45 [-0.75, -0.16]	0.065	100.00 / 100.00
I09 Remaining-duration quantiles	0.078	1.82 [0.75, 2.89]	0.018	100.00 / 100.00
10th percentile (9,227 records)	-0.035	-0.45 [-0.75, -0.16]	0.065	100.00 / 100.00
I09 Remaining-duration quantiles	0.078	1.82 [0.75, 2.89]	0.018	100.00 / 100.00
median (9,227 records)	-0.035	-0.45 [-0.75, -0.16]	0.065	100.00 / 100.00
I09 Remaining-duration quantiles	0.078	1.82 [0.75, 2.89]	0.018	100.00 / 100.00
90th percentile (9,227 records)	-0.035	-0.45 [-0.75, -0.16]	0.065	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
X10 Signed blockage distance	0.077	1.59 [-0.06, 3.23]	0.567	100.00 / 100.00
signed distance to blockage (5,062 records)	0.012	-0.03 [-0.10, 0.03]	1.000	100.00 / 100.00
E21 Train-state counts at blockage start	0.076	2.24 [1.25, 3.23]	9.1e-4	100.00 / 100.00
finish committed run then hold (9,227 records)	0.079	0.40 [0.18, 0.63]	0.043	100.00 / 66.67
E01 Train state at start time	-0.075	-1.93 [-2.41, -1.45]	6.9e-8	100.00 / 100.00
state 1 (9,227 records)	0.037	0.20 [0.06, 0.35]	0.092	100.00 / 100.00
E09 Already stopped at start time	-0.075	-1.93 [-2.41, -1.45]	6.9e-8	100.00 / 100.00
already stopped (9,227 records)	0.037	0.20 [0.06, 0.35]	0.092	100.00 / 100.00
D01 Blockage start time	-0.073	-2.61 [-3.65, -1.58]	2.0e-4	100.00 / 100.00
start (9,227 records)	0.027	0.21 [0.00, 0.43]	0.479	100.00 / 100.00
E19 Committed event count	-0.073	-2.70 [-3.69, -1.71]	6.1e-5	100.00 / 100.00
fixed events (9,227 records)	0.027	0.23 [0.03, 0.44]	0.268	100.00 / 100.00
X07 Planned exit from start time	0.071	1.68 [0.66, 2.70]	0.023	100.00 / 100.00
planned exit from start time (5,062 records)	-0.071	-0.07 [-0.11, -0.04]	0.025	100.00 / 100.00
X06 Planned entry from start time	0.071	1.69 [0.66, 2.71]	0.023	100.00 / 100.00
planned entry from start time (5,062 records)	-0.074	-0.07 [-0.11, -0.04]	0.025	100.00 / 100.00
E05 Emergency stopping position	-0.069	-2.06 [-3.04, -1.08]	0.002	100.00 / 100.00
first stop index (2,830 records)	0.090	0.03 [-0.03, 0.08]	1.000	100.00 / 100.00
E21 Train-state counts at blockage start	0.065	1.64 [0.50, 2.77]	0.068	100.00 / 100.00
hold current station (9,227 records)	0.032	0.02 [-0.35, 0.39]	1.000	73.33 / 33.33
E06 Time to emergency station	0.054	1.46 [-0.67, 3.59]	1.000	100.00 / 100.00
time to stop (2,830 records)	0.142	0.10 [0.05, 0.16]	0.033	100.00 / 100.00
D10 Time from reopening	-0.053	-1.30 [-2.44, -0.15]	0.284	100.00 / 100.00
planned from release (9,227 records)	0.007	0.58 [0.17, 0.99]	0.091	100.00 / 100.00
X11 At blockage entrance	0.053	1.36 [0.03, 2.70]	0.452	100.00 / 100.00
at closed entry (9,227 records)	0.057	0.25 [0.12, 0.38]	0.030	100.00 / 100.00
D11 Blocked section	-0.052	-1.34 [-2.51, -0.16]	0.274	100.00 / 100.00
U11 U12 (9,227 records)	-0.040	-0.16 [-0.46, 0.15]	1.000	100.00 / 50.00
E01 Train state at start time	-0.050	-1.29 [-2.44, -0.13]	0.307	100.00 / 100.00
state 2 (9,227 records)	0.097	0.05 [-0.15, 0.26]	1.000	93.33 / 33.33
E26 Committed movement count	-0.048	-1.18 [-2.57, 0.21]	0.869	100.00 / 100.00
committed runs (9,227 records)	0.037	0.17 [-0.12, 0.46]	1.000	100.00 / 100.00
E04 Committed running remainder	-0.048	-0.88 [-1.88, 0.11]	0.766	100.00 / 100.00
committed remaining (9,227 records)	0.098	0.14 [0.00, 0.28]	0.434	100.00 / 100.00
X09 Minimum time to entry	0.043	0.89 [-0.51, 2.30]	1.000	100.00 / 100.00
minrun to entry (5,062 records)	0.000	-0.05 [-0.09, 0.00]	0.349	100.00 / 100.00
X18 Emergency origin hold	0.038	0.98 [0.12, 1.83]	0.266	100.00 / 66.67
origin emergency (9,227 records)	0.024	0.30 [0.12, 0.47]	0.055	100.00 / 100.00
X08 Sections to blockage entry	0.037	0.80 [-0.58, 2.18]	1.000	100.00 / 100.00
intervals to entry (5,062 records)	-0.021	-0.06 [-0.10, -0.01]	0.110	100.00 / 100.00
D04 Start-time wave phase	-0.032	-1.10 [-2.32, 0.13]	0.746	100.00 / 100.00
start phase40 (9,227 records)	0.034	0.09 [-0.20, 0.38]	1.000	100.00 / 66.67
D11 Blocked section	0.030	0.78 [-0.53, 2.08]	1.000	100.00 / 100.00
U13 H1 (9,227 records)	-0.093	-0.42 [-0.78, -0.05]	0.259	100.00 / 100.00
X15 Earliest entry from release	0.024	0.49 [-0.44, 1.42]	1.000	100.00 / 100.00
earliest entry minus release (5,062 records)	-0.050	-0.06 [-0.09, -0.02]	0.079	100.00 / 100.00
X14 Earliest-entry estimate	-0.022	-0.38 [-1.41, 0.66]	1.000	100.00 / 66.67
earliest entry no other conflicts (5,062 records)	-0.070	-0.08 [-0.12, -0.03]	0.046	100.00 / 100.00
D11 Blocked section	0.022	0.56 [-0.81, 1.93]	1.000	100.00 / 50.00
H1 C (9,227 records)	0.132	0.57 [0.28, 0.86]	0.030	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
D09 Time from start time	0.020	0.19 [-0.83, 1.21]	1.000	80.00 / 66.67
planned from start (9,227 records)	-0.039	0.37 [0.07, 0.67]	0.167	100.00 / 100.00
X04 Planned blockage overlap	0.020	0.48 [-0.73, 1.68]	1.000	100.00 / 66.67
planned blockage overlap (9,227 records)	0.138	0.57 [0.28, 0.86]	0.030	100.00 / 100.00
E08 Minimum emergency hold	0.019	0.51 [-0.53, 1.55]	1.000	100.00 / 66.67
minimum emergency hold (9,227 records)	0.118	0.50 [0.23, 0.76]	0.033	100.00 / 100.00
X13 Arrival before reopening	0.019	0.49 [-0.63, 1.60]	1.000	100.00 / 66.67
reaches stop before reopen (9,227 records)	0.128	0.57 [0.29, 0.86]	0.030	100.00 / 100.00
X05 Planned overlap fraction	0.018	0.43 [-0.69, 1.55]	1.000	86.67 / 66.67
planned overlap fraction (9,227 records)	0.129	0.57 [0.28, 0.85]	0.030	100.00 / 100.00
X01 Blocked-section exposure	-0.015	-0.39 [-1.48, 0.70]	1.000	93.33 / 66.67
crosses closed (9,227 records)	0.155	0.86 [0.46, 1.26]	0.025	100.00 / 100.00
D10 Time from reopening	-0.015	0.92 [0.04, 1.81]	0.402	100.00 / 100.00
reference from release (9,227 records)	0.000	0.07 [-0.06, 0.20]	1.000	100.00 / 100.00
D05 Reopening wave phase	0.014	0.36 [-0.93, 1.65]	1.000	100.00 / 66.67
end phase40 (9,227 records)	-0.026	-0.10 [-0.34, 0.15]	1.000	100.00 / 66.67
X16 Blockage-only waiting	0.012	-0.21 [-1.23, 0.82]	1.000	93.33 / 66.67
blockage only wait (9,227 records)	0.128	0.46 [0.22, 0.70]	0.030	100.00 / 100.00
X19 Emergency restriction type	0.011	-0.14 [-1.32, 1.04]	1.000	86.67 / 66.67
direct stop type (9,227 records)	0.137	0.53 [0.26, 0.81]	0.031	100.00 / 100.00
D02 Blockage end	0.003	-0.30 [-1.61, 1.00]	1.000	80.00 / 100.00
end (9,227 records)	-0.007	-0.21 [-0.49, 0.08]	1.000	100.00 / 100.00
Timetable characteristics				
M05 Reference adjustment room	0.754	10.81 [8.75, 12.87]	4.0e-10	100.00 / 100.00
compressible (9,227 records)	-0.722	-7.07 [-7.09, -7.05]	4.2e-74	100.00 / 100.00
M06 Remaining candidate decisions	0.224	5.80 [4.29, 7.31]	1.7e-7	100.00 / 100.00
remaining order decisions (9,227 records)	-0.120	-0.76 [-1.18, -0.34]	0.043	100.00 / 100.00
T12 Planned local flow	-0.175	-3.99 [-4.91, -3.07]	1.4e-8	100.00 / 100.00
planned flow 5 min (9,227 records)	0.072	0.44 [0.16, 0.72]	0.063	100.00 / 100.00
M06 Remaining candidate decisions	0.175	4.36 [3.24, 5.47]	1.2e-7	100.00 / 100.00
remaining track-assignment decisions (9,227 records)	-0.102	-0.67 [-1.05, -0.30]	0.044	100.00 / 100.00
T10 Planned exit buffer	-0.174	-4.62 [-5.99, -3.25]	2.2e-6	100.00 / 100.00
previous buffer (8,952 records)	0.142	0.74 [0.24, 1.23]	0.074	100.00 / 100.00
M06 Remaining candidate decisions	0.171	4.29 [3.18, 5.40]	1.4e-7	100.00 / 100.00
remaining duration decisions (9,227 records)	-0.100	-0.67 [-1.05, -0.29]	0.044	100.00 / 100.00
T22 Route extent	0.171	4.29 [3.18, 5.40]	1.4e-7	100.00 / 100.00
remaining edges (9,227 records)	-0.100	-0.67 [-1.05, -0.29]	0.044	100.00 / 100.00
T09 Planned entry buffer	-0.171	-4.58 [-5.98, -3.18]	3.8e-6	100.00 / 100.00
previous buffer (8,952 records)	0.125	0.73 [0.23, 1.23]	0.085	100.00 / 100.00
T18 Remaining planned journey	0.169	4.16 [2.94, 5.38]	1.5e-6	100.00 / 100.00
remaining planned (9,227 records)	-0.081	-0.64 [-1.05, -0.22]	0.065	100.00 / 100.00
T19 Remaining minimum journey	0.169	4.16 [2.94, 5.38]	1.5e-6	100.00 / 100.00
remaining minimum (9,227 records)	-0.081	-0.64 [-1.05, -0.22]	0.065	100.00 / 100.00
T20 Remaining scheduled stops	0.159	4.01 [2.75, 5.26]	4.6e-6	100.00 / 100.00
remaining scheduled stops (9,227 records)	-0.066	-0.61 [-1.03, -0.19]	0.084	100.00 / 100.00
T12 Planned local flow	-0.118	-2.37 [-3.26, -1.48]	8.9e-5	100.00 / 100.00
planned flow 10 min (9,227 records)	0.065	0.35 [0.09, 0.61]	0.105	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
T13 Planned train-type pattern	-0.109	-3.35 [-5.60, -1.10]	0.053	100.00 / 100.00
alternation 20 min (1,873 records)	0.038	0.58 [0.04, 1.12]	0.337	100.00 / 100.00
T02 Planned departure time	-0.089	-2.55 [-3.74, -1.36]	0.002	100.00 / 100.00
planned departure (9,227 records)	0.005	0.59 [0.22, 0.96]	0.057	100.00 / 100.00
T01 Planned arrival time	-0.089	-2.55 [-3.74, -1.36]	0.002	100.00 / 100.00
planned arrival (9,227 records)	0.005	0.59 [0.22, 0.96]	0.057	100.00 / 100.00
M05 Reference adjustment room	-0.081	-1.27 [-2.85, 0.32]	1.000	100.00 / 100.00
extendable (9,227 records)	0.011	0.48 [-0.05, 1.01]	0.661	100.00 / 100.00
T11 Planned track buffer	-0.080	-1.71 [-2.79, -0.64]	0.033	100.00 / 100.00
planned track buffer (8,007 records)	0.074	0.37 [0.04, 0.70]	0.289	100.00 / 100.00
T12 Planned local flow	-0.061	-0.61 [-2.08, 0.86]	1.000	100.00 / 100.00
planned flow 20 min (9,227 records)	0.028	0.20 [-0.07, 0.47]	1.000	100.00 / 100.00
T13 Planned train-type pattern	0.055	1.01 [-0.72, 2.75]	1.000	100.00 / 100.00
fast fraction 10 min (4,673 records)	0.000	-0.01 [-0.16, 0.14]	1.000	73.33 / 66.67
T06 Planned track label	0.037	1.84 [0.17, 3.51]	0.314	100.00 / 100.00
planned track index (9,227 records)	0.038	-0.36 [-0.91, 0.20]	1.000	100.00 / 100.00
T13 Planned train-type pattern	0.033	0.66 [-0.91, 2.22]	1.000	100.00 / 100.00
fast fraction 20 min (5,476 records)	0.003	0.06 [-0.13, 0.25]	1.000	100.00 / 100.00
T24 Planned terminal position	-0.033	-0.67 [-1.69, 0.35]	1.000	100.00 / 100.00
planned terminal (9,227 records)	-0.032	0.31 [0.05, 0.56]	0.189	100.00 / 100.00
T14 Departure wave	-0.033	-0.77 [-1.82, 0.28]	1.000	100.00 / 100.00
origin wave40 (9,227 records)	-0.052	0.37 [0.07, 0.67]	0.163	100.00 / 100.00
T13 Planned train-type pattern	0.023	0.35 [-0.91, 1.61]	1.000	100.00 / 100.00
fast fraction 5 min (4,093 records)	0.020	0.01 [-0.15, 0.17]	1.000	66.67 / 33.33
M04 Propagated domain	0.011	-0.18 [-0.88, 0.52]	1.000	93.33 / 66.67
domain lower (9,227 records)	0.058	0.16 [0.08, 0.24]	0.030	100.00 / 100.00
M04 Propagated domain	0.010	0.32 [-0.73, 1.37]	1.000	93.33 / 66.67
domain upper (9,227 records)	-0.067	-0.55 [-0.82, -0.29]	0.025	100.00 / 100.00
M04 Propagated domain	0.009	0.37 [-0.73, 1.47]	1.000	93.33 / 66.67
domain width (9,227 records)	-0.082	-0.59 [-0.87, -0.32]	0.025	100.00 / 100.00
T21 Nearby scheduled-stop distance	-0.005	-0.13 [-1.22, 0.97]	1.000	86.67 / 33.33
next scheduled-stop distance (9,227 records)	-0.066	0.03 [-0.20, 0.26]	1.000	86.67 / 66.67
T21 Nearby scheduled-stop distance	-0.005	-0.13 [-1.22, 0.97]	1.000	86.67 / 33.33
previous scheduled-stop distance (9,227 records)	-0.066	0.03 [-0.20, 0.26]	1.000	86.67 / 66.67
T24 Planned terminal position	-0.005	-0.11 [-1.19, 0.98]	1.000	60.00 / 66.67
terminal from release (9,227 records)	-0.014	0.39 [0.09, 0.69]	0.126	100.00 / 100.00
F13 Minimum running time	0.004	0.11 [-0.99, 1.22]	1.000	86.67 / 33.33
minimum run (9,227 records)	0.064	-0.04 [-0.27, 0.20]	1.000	93.33 / 66.67
T04 Planned running duration	0.004	0.11 [-0.99, 1.22]	1.000	86.67 / 33.33
planned run next (9,227 records)	0.064	-0.04 [-0.27, 0.20]	1.000	93.33 / 66.67
T17 Total planned journey time	0.004	0.11 [-0.99, 1.22]	1.000	86.67 / 33.33
planned total (9,227 records)	0.064	-0.04 [-0.27, 0.20]	1.000	93.33 / 66.67
T15 Fast-train indicator	-0.004	-0.11 [-1.22, 0.99]	1.000	86.67 / 33.33
fast (9,227 records)	-0.064	0.04 [-0.20, 0.27]	1.000	93.33 / 66.67
M06 Remaining candidate decisions	0.004	0.49 [-0.49, 1.47]	1.000	100.00 / 66.67
remaining stopping decisions (9,227 records)	-0.071	-0.12 [-0.37, 0.13]	1.000	100.00 / 66.67
F12 Minimum dwell	-0.003	-0.07 [-0.99, 0.85]	1.000	93.33 / 66.67
minimum dwell (9,227 records)	-0.008	-0.05 [-0.08, -0.02]	0.054	100.00 / 100.00
F14 Allowed stopping mode	-0.003	-0.07 [-0.99, 0.85]	1.000	93.33 / 66.67
required (9,227 records)	-0.008	-0.05 [-0.08, -0.02]	0.054	100.00 / 100.00
T03 Planned dwell	-0.003	-0.07 [-0.99, 0.85]	1.000	93.33 / 66.67
planned dwell (9,227 records)	-0.008	-0.05 [-0.08, -0.02]	0.054	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
T05 Planned stopping mode	-0.003	-0.07 [-0.99, 0.85]	1.000	93.33 / 66.67
planned stop (9,227 records)	-0.008	-0.05 [-0.08, -0.02]	0.054	100.00 / 100.00
Train interactions				
H22 Shared downstream support	-0.319	-12.04 [-16.11, -7.97]	1.8e-5	100.00 / 100.00
arrival departure downstream Jaccard (5,527 records)	0.334	1.85 [0.90, 2.80]	0.030	100.00 / 100.00
H17 Window competition size	0.234	6.53 [5.28, 7.78]	4.0e-10	100.00 / 100.00
competitor trains 20 min (9,227 records)	-0.109	-0.79 [-1.28, -0.30]	0.057	100.00 / 100.00
H18 Co-window occupations	0.234	6.53 [5.28, 7.78]	4.0e-10	100.00 / 100.00
competition set memberships 20 min (9,227 records)	-0.109	-0.79 [-1.28, -0.30]	0.057	100.00 / 100.00
H17 Window competition size	0.211	5.75 [4.60, 6.91]	8.0e-10	100.00 / 100.00
competitor trains 10 min (9,227 records)	-0.105	-0.95 [-1.49, -0.40]	0.049	100.00 / 100.00
H18 Co-window occupations	0.211	5.75 [4.60, 6.91]	8.0e-10	100.00 / 100.00
competition set memberships 10 min (9,227 records)	-0.105	-0.95 [-1.49, -0.40]	0.049	100.00 / 100.00
H21 Shared-visit coupling	0.179	5.00 [3.44, 6.55]	4.2e-6	100.00 / 100.00
same visit classes (9,227 records)	-0.102	-0.62 [-1.03, -0.20]	0.074	100.00 / 100.00
H17 Window competition size	0.172	5.15 [3.78, 6.53]	2.6e-7	100.00 / 100.00
competitor trains 5 min (9,227 records)	-0.084	-1.06 [-1.71, -0.41]	0.057	100.00 / 100.00
H18 Co-window occupations	0.172	5.15 [3.78, 6.53]	2.6e-7	100.00 / 100.00
competition set memberships 5 min (9,227 records)	-0.084	-1.06 [-1.71, -0.41]	0.057	100.00 / 100.00
E14 Reference release rank	0.076	1.80 [0.56, 3.04]	0.064	100.00 / 100.00
release queue rank (2,830 records)	-0.037	-0.04 [-0.09, 0.02]	1.000	100.00 / 100.00
E15 Queue fast-train share	-0.046	-1.32 [-2.99, 0.35]	1.000	100.00 / 100.00
queue fast fraction (2,830 records)	-0.135	-0.10 [-0.15, -0.05]	0.025	100.00 / 100.00
E11 Co-located emergency trains	0.019	0.48 [-0.56, 1.53]	1.000	93.33 / 66.67
same stop count (9,227 records)	0.120	0.48 [0.23, 0.73]	0.033	100.00 / 100.00
B17 Relation-layer overlap	0.016	1.03 [-0.58, 2.63]	1.000	100.00 / 100.00
entry headway station track Jaccard (9,227 records)	-0.046	-0.15 [-0.35, 0.05]	1.000	100.00 / 100.00
B17 Relation-layer overlap	0.016	1.03 [-0.58, 2.63]	1.000	100.00 / 100.00
exit headway station track Jaccard (9,227 records)	-0.046	-0.15 [-0.35, 0.05]	1.000	100.00 / 100.00
H20 Support distance	0.007	0.17 [-1.67, 2.01]	1.000	93.33 / 66.67
nearest optional stop index (2,432 records)	0.031	0.09 [0.03, 0.16]	0.099	100.00 / 100.00
Infrastructure utilization				
R10 Intersecting dwell distribution	0.532	4.99 [2.37, 7.62]	0.006	100.00 / 100.00
waiting duration 5 min maximum (9,227 records)	-0.372	-2.58 [-4.21, -0.95]	0.060	100.00 / 100.00
R10 Intersecting dwell distribution	0.519	3.24 [1.04, 5.44]	0.058	100.00 / 100.00
waiting duration 5 min median (9,227 records)	-0.399	-1.84 [-3.26, -0.42]	0.132	100.00 / 100.00
R10 Intersecting dwell distribution	0.519	5.22 [2.58, 7.86]	0.004	100.00 / 100.00
waiting duration 10 min maximum (9,227 records)	-0.360	-2.54 [-4.14, -0.93]	0.060	100.00 / 100.00
R10 Intersecting dwell distribution	0.511	4.73 [2.12, 7.34]	0.010	100.00 / 100.00
waiting duration 5 min 90th percentile (9,227 records)	-0.368	-2.50 [-4.13, -0.86]	0.065	100.00 / 100.00
R10 Intersecting dwell distribution	0.500	6.13 [3.49, 8.76]	6.2e-4	100.00 / 100.00
waiting duration 20 min maximum (9,227 records)	-0.311	-2.33 [-3.78, -0.88]	0.057	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R10 Intersecting dwell distribution waiting duration 5 min mean (9,227 records)	0.496 -0.362	3.37 [1.05, 5.68] -1.83 [-3.25, -0.42]	0.063 0.134	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min interquartile range (9,227 records)	0.485 -0.281	5.90 [3.71, 8.09] -2.94 [-4.55, -1.33]	7.7e-5 0.040	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min standard deviation (9,227 records)	0.475 -0.269	6.18 [3.71, 8.65] -2.79 [-4.44, -1.14]	2.3e-4 0.053	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min 90th percentile (9,227 records)	0.471 -0.354	4.82 [2.21, 7.42] -2.43 [-4.02, -0.83]	0.008 0.067	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min standard deviation (9,227 records)	0.457 -0.290	6.11 [3.67, 8.55] -3.11 [-4.72, -1.51]	2.3e-4 0.030	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min interquartile range (9,227 records)	0.455 -0.294	5.96 [3.87, 8.04] -3.25 [-4.77, -1.73]	3.0e-5 0.025	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min 90th percentile (9,227 records)	0.452 -0.298	5.50 [2.88, 8.11] -2.30 [-3.80, -0.81]	0.002 0.065	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min interquartile range (9,227 records)	0.450 -0.237	6.33 [3.96, 8.69] -2.65 [-4.26, -1.05]	8.5e-5 0.057	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min median (9,227 records)	0.449 -0.362	2.83 [0.77, 4.89] -1.57 [-2.86, -0.28]	0.093 0.183	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min standard deviation (9,227 records)	0.443 -0.228	6.67 [4.06, 9.28] -2.48 [-4.01, -0.96]	1.7e-4 0.057	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min mean (9,227 records)	0.431 -0.336	3.18 [0.87, 5.48] -1.73 [-3.10, -0.36]	0.091 0.152	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min median (9,227 records)	0.382 -0.312	2.44 [0.40, 4.47] -1.27 [-2.43, -0.11]	0.215 0.304	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min mean (9,227 records)	0.374 -0.276	3.31 [0.87, 5.76] -1.55 [-2.83, -0.27]	0.102 0.189	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min sum (9,227 records)	0.337 -0.200	6.60 [3.06, 10.15] -2.54 [-4.32, -0.76]	0.008 0.091	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min sum (9,227 records)	0.334 -0.218	5.01 [1.64, 8.39] -2.51 [-4.41, -0.61]	0.054 0.121	100.00 / 100.00 100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min sum (9,227 records)	0.326 -0.213	4.92 [1.62, 8.22] -2.65 [-4.63, -0.68]	0.053 0.113	100.00 / 100.00 100.00 / 100.00
R01 Mean occupation mean occupancy 20 min (9,227 records)	0.324 -0.193	10.68 [8.64, 12.72] -2.21 [-3.30, -1.12]	4.0e-10 0.028	100.00 / 100.00 100.00 / 100.00
R04 Utilization utilization 20 min (9,227 records)	0.305 -0.195	8.33 [6.61, 10.05] -1.73 [-2.58, -0.88]	1.4e-9 0.028	100.00 / 100.00 100.00 / 100.00
R12 Demand/capacity proxy demand capacity 20 min (9,227 records)	0.305 -0.195	8.33 [6.61, 10.05] -1.73 [-2.58, -0.88]	1.4e-9 0.028	100.00 / 100.00 100.00 / 100.00
R01 Mean occupation mean occupancy 10 min (9,227 records)	0.293 -0.178	9.55 [7.67, 11.43] -1.97 [-2.99, -0.96]	7.3e-10 0.030	100.00 / 100.00 100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R10 Intersecting dwell distribution	0.275	1.07 [-0.18, 2.32]	0.869	100.00 / 100.00
waiting duration 5 min minimum (9,227 records)	-0.265	-0.63 [-1.43, 0.16]	0.975	100.00 / 100.00
R04 Utilization	0.266	6.88 [5.33, 8.43]	9.1e-9	100.00 / 100.00
utilization 10 min (9,227 records)	-0.175	-1.45 [-2.19, -0.70]	0.030	100.00 / 100.00
R12 Demand/capacity proxy	0.266	6.88 [5.33, 8.43]	9.1e-9	100.00 / 100.00
demand capacity 10 min (9,227 records)	-0.175	-1.45 [-2.19, -0.70]	0.030	100.00 / 100.00
R02 Peak occupation	0.264	9.14 [7.37, 10.91]	5.0e-10	100.00 / 100.00
peak occupancy 20 min (9,227 records)	-0.157	-1.42 [-2.25, -0.59]	0.051	100.00 / 100.00
R07 Entry load	0.258	7.23 [5.56, 8.89]	1.5e-8	100.00 / 100.00
entry load 20 min (9,227 records)	-0.135	-0.75 [-1.23, -0.27]	0.063	100.00 / 100.00
R02 Peak occupation	0.255	8.53 [6.76, 10.30]	1.4e-9	100.00 / 100.00
peak occupancy 10 min (9,227 records)	-0.151	-1.42 [-2.25, -0.58]	0.052	100.00 / 100.00
R11 Flow imbalance	-0.254	-7.59 [-9.73, -5.46]	6.8e-7	100.00 / 100.00
arrivals minus departures 5 min (9,227 records)	0.254	1.18 [0.52, 1.83]	0.044	100.00 / 100.00
R01 Mean occupation	0.253	8.34 [6.41, 10.26]	1.5e-8	100.00 / 100.00
mean occupancy 5 min (9,227 records)	-0.149	-1.78 [-2.79, -0.78]	0.044	100.00 / 100.00
R10 Intersecting dwell distribution	0.234	6.53 [5.28, 7.78]	4.0e-10	100.00 / 100.00
waiting duration 20 min count (9,227 records)	-0.109	-0.79 [-1.28, -0.30]	0.057	100.00 / 100.00
R14 Window competitors	0.234	6.53 [5.28, 7.78]	4.0e-10	100.00 / 100.00
window competitors 20 min (9,227 records)	-0.109	-0.79 [-1.28, -0.30]	0.057	100.00 / 100.00
R07 Entry load	0.233	6.08 [4.68, 7.49]	1.5e-8	100.00 / 100.00
entry load 10 min (9,227 records)	-0.132	-0.80 [-1.29, -0.31]	0.057	100.00 / 100.00
R08 Exit load	0.231	7.00 [5.62, 8.38]	7.3e-10	100.00 / 100.00
exit load 20 min (9,227 records)	-0.137	-0.44 [-0.73, -0.14]	0.077	100.00 / 100.00
R11 Flow imbalance	-0.229	-6.05 [-8.29, -3.81]	7.6e-5	100.00 / 100.00
arrivals minus departures 10 min (9,227 records)	0.233	1.18 [0.47, 1.88]	0.054	100.00 / 100.00
R08 Exit load	0.219	6.41 [4.99, 7.82]	5.8e-9	100.00 / 100.00
exit load 10 min (9,227 records)	-0.189	-0.42 [-0.68, -0.16]	0.057	100.00 / 100.00
R08 Exit load	0.219	6.19 [4.65, 7.72]	6.9e-8	100.00 / 100.00
exit load 5 min (9,227 records)	-0.187	-0.30 [-0.57, -0.03]	0.293	100.00 / 100.00
R13 Adjacent-station occupation	0.214	5.96 [4.54, 7.38]	2.8e-8	100.00 / 100.00
downstream occupancy (9,227 records)	-0.189	-0.49 [-0.76, -0.22]	0.040	100.00 / 100.00
R10 Intersecting dwell distribution	0.211	5.75 [4.60, 6.91]	8.0e-10	100.00 / 100.00
waiting duration 10 min count (9,227 records)	-0.105	-0.95 [-1.49, -0.40]	0.049	100.00 / 100.00
R14 Window competitors	0.211	5.75 [4.60, 6.91]	8.0e-10	100.00 / 100.00
window competitors 10 min (9,227 records)	-0.105	-0.95 [-1.49, -0.40]	0.049	100.00 / 100.00
R02 Peak occupation	0.210	7.07 [5.35, 8.80]	4.9e-8	100.00 / 100.00
peak occupancy 5 min (9,227 records)	-0.108	-1.31 [-2.10, -0.52]	0.057	100.00 / 100.00
B13 Adjacent-window difference	-0.203	-6.43 [-8.66, -4.20]	2.7e-5	100.00 / 100.00
adjacent occupancy difference 5 min (9,227 records)	0.225	0.98 [0.41, 1.55]	0.051	100.00 / 100.00
R17 Occupation change rate	-0.203	-6.43 [-8.66, -4.20]	2.7e-5	100.00 / 100.00
occupancy rate 5 min (9,227 records)	0.225	0.98 [0.41, 1.55]	0.051	100.00 / 100.00
R09 Reference headway buffer	-0.202	-2.56 [-3.42, -1.70]	1.8e-5	100.00 / 100.00
1 previous buffer (8,952 records)	0.171	0.27 [0.01, 0.53]	0.375	100.00 / 100.00
R07 Entry load	0.201	5.31 [3.84, 6.78]	5.4e-7	100.00 / 100.00
entry load 5 min (9,227 records)	-0.128	-0.81 [-1.32, -0.30]	0.058	100.00 / 100.00
F01 Facility role	0.201	5.14 [3.70, 6.58]	6.1e-7	100.00 / 100.00
hub (9,227 records)	-0.081	-0.74 [-1.27, -0.21]	0.097	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
F02 Physical degree	0.201	5.14 [3.70, 6.58]	6.1e-7	100.00 / 100.00
undirected degree (9,227 records)	-0.081	-0.74 [-1.27, -0.21]	0.097	100.00 / 100.00
F03 Branch-count proxy	0.201	5.14 [3.70, 6.58]	6.1e-7	100.00 / 100.00
branch count (9,227 records)	-0.081	-0.74 [-1.27, -0.21]	0.097	100.00 / 100.00
F06 Directional pool size	0.201	5.14 [3.70, 6.58]	6.1e-7	100.00 / 100.00
direction pool (9,227 records)	-0.081	-0.74 [-1.27, -0.21]	0.097	100.00 / 100.00
F07 Compatible tracks	0.201	5.14 [3.70, 6.58]	6.1e-7	100.00 / 100.00
compatible tracks (9,227 records)	-0.081	-0.74 [-1.27, -0.21]	0.097	100.00 / 100.00
F15 Parallel-capacity proxy	0.201	5.14 [3.70, 6.58]	6.1e-7	100.00 / 100.00
parallel capacity (9,227 records)	-0.081	-0.74 [-1.27, -0.21]	0.097	100.00 / 100.00
R09 Reference headway buffer	-0.192	-2.28 [-3.09, -1.47]	4.3e-5	100.00 / 100.00
0 previous buffer (8,952 records)	0.154	0.19 [-0.07, 0.45]	1.000	100.00 / 100.00
B13 Adjacent-window difference	-0.190	-6.47 [-8.57, -4.36]	9.3e-6	100.00 / 100.00
adjacent occupancy difference 10 min (9,227 records)	0.172	0.99 [0.38, 1.59]	0.057	100.00 / 100.00
R17 Occupation change rate	-0.190	-6.47 [-8.57, -4.36]	9.3e-6	100.00 / 100.00
occupancy rate 10 min (9,227 records)	0.172	0.99 [0.38, 1.59]	0.057	100.00 / 100.00
B13 Adjacent-window difference	-0.189	-7.50 [-9.87, -5.13]	5.5e-6	100.00 / 100.00
adjacent occupancy difference 20 min (9,227 records)	0.149	1.42 [0.46, 2.37]	0.076	100.00 / 100.00
R17 Occupation change rate	-0.189	-7.50 [-9.87, -5.13]	5.5e-6	100.00 / 100.00
occupancy rate 20 min (9,227 records)	0.149	1.42 [0.46, 2.37]	0.076	100.00 / 100.00
H19 Peak demand/capacity	0.187	4.43 [3.27, 5.60]	2.0e-7	100.00 / 100.00
peak demand capacity 20 min (9,227 records)	-0.150	-0.67 [-1.05, -0.29]	0.047	100.00 / 100.00
R11 Flow imbalance	-0.187	-5.96 [-8.23, -3.70]	1.1e-4	100.00 / 100.00
arrivals minus departures 20 min (9,227 records)	0.174	1.01 [0.40, 1.62]	0.057	100.00 / 100.00
R16 Planned-arrival overlap	-0.182	-4.08 [-5.21, -2.95]	5.1e-7	100.00 / 100.00
planned arrival release overlap 5 min (9,227 records)	0.079	0.46 [0.17, 0.74]	0.059	100.00 / 100.00
F17 Distance to a hub	-0.176	-4.48 [-5.62, -3.33]	1.3e-7	100.00 / 100.00
hub distance (9,227 records)	0.112	0.67 [0.24, 1.11]	0.064	100.00 / 100.00
H19 Peak demand/capacity	0.173	3.92 [2.80, 5.04]	9.3e-7	100.00 / 100.00
peak demand capacity 10 min (9,227 records)	-0.141	-0.66 [-1.04, -0.28]	0.049	100.00 / 100.00
R10 Intersecting dwell distribution	0.172	5.15 [3.78, 6.53]	2.6e-7	100.00 / 100.00
waiting duration 5 min count (9,227 records)	-0.084	-1.06 [-1.71, -0.41]	0.057	100.00 / 100.00
R14 Window competitors	0.172	5.15 [3.78, 6.53]	2.6e-7	100.00 / 100.00
window competitors 5 min (9,227 records)	-0.084	-1.06 [-1.71, -0.41]	0.057	100.00 / 100.00
R04 Utilization	0.171	4.73 [3.16, 6.30]	1.3e-5	100.00 / 100.00
utilization 5 min (9,227 records)	-0.122	-1.17 [-1.82, -0.51]	0.044	100.00 / 100.00
R12 Demand/capacity proxy	0.171	4.73 [3.16, 6.30]	1.3e-5	100.00 / 100.00
demand capacity 5 min (9,227 records)	-0.122	-1.17 [-1.82, -0.51]	0.044	100.00 / 100.00
R06 Saturation persistence	0.159	7.87 [5.74, 10.01]	3.5e-7	100.00 / 100.00
longest saturation 20 min (9,227 records)	-0.153	-2.13 [-3.24, -1.02]	0.032	100.00 / 100.00
R10 Intersecting dwell distribution	0.154	2.94 [1.44, 4.45]	0.005	100.00 / 100.00
waiting duration 10 min zero fraction (9,227 records)	-0.019	-0.24 [-0.55, 0.07]	1.000	100.00 / 100.00
F04 Physical betweenness	0.153	3.77 [2.71, 4.84]	6.8e-7	100.00 / 100.00
betweenness (9,227 records)	-0.102	-0.57 [-0.98, -0.16]	0.094	100.00 / 100.00
R13 Adjacent-station occupation	0.151	4.37 [3.05, 5.69]	2.5e-6	100.00 / 100.00
occupancy difference (9,227 records)	-0.158	-0.47 [-0.69, -0.24]	0.025	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R10 Intersecting dwell distribution	0.149	2.52 [1.37, 3.67]	0.001	100.00 / 100.00
waiting duration 20 min zero fraction (9,227 records)	-0.001	-0.18 [-0.42, 0.06]	1.000	100.00 / 100.00
R16 Planned-arrival overlap	-0.131	-2.82 [-3.70, -1.94]	4.5e-6	100.00 / 100.00
planned arrival release overlap 10 min (9,227 records)	0.056	0.36 [0.11, 0.61]	0.081	100.00 / 100.00
B07 Regional saturation	0.128	4.24 [2.60, 5.88]	1.5e-4	100.00 / 100.00
saturated fraction 20 min (9,227 records)	-0.135	-1.22 [-1.94, -0.51]	0.052	100.00 / 100.00
R05 Saturated fraction	0.128	4.24 [2.60, 5.88]	1.5e-4	100.00 / 100.00
saturated fraction 20 min (9,227 records)	-0.135	-1.22 [-1.94, -0.51]	0.052	100.00 / 100.00
R15 Free windows	-0.128	-4.24 [-5.88, -2.60]	1.5e-4	100.00 / 100.00
free window total 20 min (9,227 records)	0.135	1.22 [0.51, 1.94]	0.052	100.00 / 100.00
R10 Intersecting dwell distribution	-0.121	-0.04 [-0.87, 0.79]	1.000	66.67 / 66.67
waiting duration 20 min minimum (9,227 records)	-0.023	-0.27 [-0.77, 0.23]	1.000	100.00 / 100.00
R06 Saturation persistence	0.120	7.10 [5.20, 9.01]	2.8e-7	100.00 / 100.00
longest saturation 10 min (9,227 records)	-0.118	-1.96 [-3.00, -0.93]	0.033	100.00 / 100.00
F16 Downstream capacity gradient	-0.120	-3.32 [-4.59, -2.06]	1.1e-4	100.00 / 100.00
next capacity delta (9,227 records)	0.020	0.44 [0.03, 0.86]	0.337	100.00 / 100.00
F01 Facility role	0.114	2.92 [1.93, 3.92]	1.9e-5	100.00 / 100.00
trunk (9,227 records)	-0.068	-0.45 [-0.79, -0.10]	0.132	100.00 / 100.00
R03 Minimum free capacity	-0.108	-2.61 [-3.81, -1.42]	0.001	100.00 / 100.00
minimum free 20 min (9,227 records)	0.122	0.48 [0.18, 0.78]	0.057	100.00 / 100.00
R15 Free windows	-0.106	-2.29 [-3.48, -1.10]	0.006	100.00 / 100.00
free window maximum 20 min (9,227 records)	0.108	0.56 [0.19, 0.93]	0.065	100.00 / 100.00
H19 Peak demand/capacity	0.106	2.30 [1.21, 3.39]	0.002	100.00 / 100.00
peak demand capacity 5 min (9,227 records)	-0.077	-0.53 [-0.87, -0.19]	0.065	100.00 / 100.00
R10 Intersecting dwell distribution	0.105	1.85 [0.28, 3.43]	0.234	100.00 / 100.00
waiting duration 5 min zero fraction (9,227 records)	-0.042	-0.19 [-0.44, 0.06]	1.000	100.00 / 100.00
B07 Regional saturation	0.094	3.56 [1.97, 5.15]	9.5e-4	100.00 / 100.00
saturated fraction 10 min (9,227 records)	-0.103	-1.20 [-1.95, -0.46]	0.057	100.00 / 100.00
R05 Saturated fraction	0.094	3.56 [1.97, 5.15]	9.5e-4	100.00 / 100.00
saturated fraction 10 min (9,227 records)	-0.103	-1.20 [-1.95, -0.46]	0.057	100.00 / 100.00
R15 Free windows	-0.094	-3.56 [-5.15, -1.97]	9.5e-4	100.00 / 100.00
free window total 10 min (9,227 records)	0.103	1.20 [0.46, 1.95]	0.057	100.00 / 100.00
E18 Parking concentration	-0.087	-2.11 [-3.12, -1.10]	0.002	100.00 / 100.00
parking concentration (9,227 records)	0.009	0.12 [-0.08, 0.32]	1.000	100.00 / 66.67
R15 Free windows	-0.082	-2.29 [-2.96, -1.62]	1.5e-6	100.00 / 100.00
free window count 5 min (9,227 records)	0.062	0.39 [0.07, 0.72]	0.195	100.00 / 100.00
R15 Free windows	-0.080	-1.65 [-2.78, -0.51]	0.064	100.00 / 100.00
free window maximum 10 min (9,227 records)	0.084	0.56 [0.15, 0.97]	0.099	100.00 / 100.00
R16 Planned-arrival overlap	-0.077	-1.11 [-2.49, 0.26]	1.000	100.00 / 100.00
planned arrival release overlap 20 min (9,227 records)	0.023	0.21 [-0.04, 0.45]	0.817	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R03 Minimum free capacity	-0.073	-1.76 [-2.97, -0.54]	0.065	100.00 / 100.00
minimum free 10 min (9,227 records)	0.098	0.44 [0.16, 0.72]	0.064	100.00 / 100.00
R06 Saturation persistence	0.071	4.79 [2.99, 6.58]	8.9e-5	100.00 / 100.00
longest saturation 5 min (9,227 records)	-0.070	-1.40 [-2.26, -0.54]	0.057	100.00 / 100.00
B07 Regional saturation	0.053	2.75 [1.18, 4.33]	0.015	100.00 / 100.00
saturated fraction 5 min (9,227 records)	-0.061	-0.97 [-1.64, -0.31]	0.079	100.00 / 100.00
R05 Saturated fraction	0.053	2.75 [1.18, 4.33]	0.015	100.00 / 100.00
saturated fraction 5 min (9,227 records)	-0.061	-0.97 [-1.64, -0.31]	0.079	100.00 / 100.00
R15 Free windows	-0.053	-2.75 [-4.33, -1.18]	0.015	100.00 / 100.00
free window total 5 min (9,227 records)	0.061	0.97 [0.31, 1.64]	0.079	100.00 / 100.00
R15 Free windows	0.051	-0.30 [-0.92, 0.32]	1.000	100.00 / 66.67
free window count 20 min (9,227 records)	-0.065	0.05 [-0.05, 0.15]	1.000	100.00 / 66.67
R15 Free windows	-0.047	-1.35 [-2.54, -0.15]	0.286	100.00 / 100.00
free window maximum 5 min (9,227 records)	0.050	0.57 [0.14, 1.00]	0.122	100.00 / 100.00
E23 Initial station occupation	0.042	1.07 [0.01, 2.13]	0.464	100.00 / 100.00
initial station occupied (9,227 records)	-0.023	-0.16 [-0.52, 0.20]	1.000	100.00 / 66.67
R18 Opposite-direction utilization	0.041	-0.76 [-1.61, 0.10]	0.763	100.00 / 100.00
opposite utilization 5 min (9,227 records)	-0.029	0.24 [0.07, 0.41]	0.090	100.00 / 100.00
R18 Opposite-direction utilization	0.035	-0.58 [-1.55, 0.39]	1.000	100.00 / 100.00
opposite utilization 10 min (9,227 records)	-0.037	0.26 [0.07, 0.44]	0.095	100.00 / 100.00
R18 Opposite-direction utilization	0.023	-0.27 [-1.47, 0.93]	1.000	100.00 / 66.67
opposite utilization 20 min (9,227 records)	-0.044	0.28 [0.07, 0.48]	0.106	100.00 / 100.00
E10 Emergency pool capacity	-0.021	-0.49 [-1.84, 0.86]	1.000	93.33 / 100.00
emergency pool capacity (2,830 records)	0.043	0.01 [-0.03, 0.06]	1.000	93.33 / 100.00
R10 Intersecting dwell distribution	0.020	0.45 [-0.63, 1.53]	1.000	100.00 / 100.00
waiting duration 10 min minimum (9,227 records)	-0.120	-0.52 [-1.30, 0.26]	1.000	100.00 / 100.00
R13 Adjacent-station occupation	0.010	0.00 [-1.03, 1.03]	1.000	46.67 / 33.33
upstream occupancy (9,227 records)	0.032	0.15 [-0.04, 0.35]	0.990	100.00 / 100.00
F18 Next stopping opportunity	0.009	0.24 [-0.84, 1.33]	1.000	86.67 / 66.67
next stop distance (7,132 records)	-0.065	-0.10 [-0.42, 0.22]	1.000	93.33 / 66.67
R03 Minimum free capacity	-0.007	-0.10 [-1.37, 1.16]	1.000	80.00 / 66.67
minimum free 5 min (9,227 records)	0.035	0.28 [0.03, 0.53]	0.299	100.00 / 100.00
R15 Free windows	-0.003	-1.06 [-1.65, -0.48]	0.009	100.00 / 100.00
free window count 10 min (9,227 records)	-0.012	0.16 [-0.01, 0.34]	0.562	100.00 / 100.00
Delay propagation patterns				
G03 Activity delay increment	0.615	4.47 [1.62, 7.32]	0.035	100.00 / 100.00
dwell delay increment (9,227 records)	-0.549	-2.89 [-4.86, -0.92]	0.079	100.00 / 100.00
G04 Activity-duration increment	0.615	4.47 [1.62, 7.32]	0.035	100.00 / 100.00
dwell increment (9,227 records)	-0.549	-2.89 [-4.86, -0.92]	0.079	100.00 / 100.00
G10 Edge-slack distribution	0.583	4.13 [1.38, 6.87]	0.050	100.00 / 100.00
outgoing slack minimum arrival (9,227 records)	-0.531	-2.68 [-4.61, -0.76]	0.097	100.00 / 100.00
G10 Edge-slack distribution	0.488	6.30 [4.24, 8.36]	1.0e-5	100.00 / 100.00
incoming slack minimum departure (9,227 records)	-0.500	-3.36 [-3.48, -3.25]	1.4e-33	100.00 / 100.00
G19 Endpoint contrasts	0.396	11.89 [9.44, 14.33]	1.3e-9	100.00 / 100.00
G08 active outgoing degree endpoint contrast (9,227 records)	-0.320	-2.03 [-3.03, -1.02]	0.030	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution	-0.347	-10.80 [-13.17, -8.43]	5.3e-9	100.00 / 100.00
outgoing slack zero fraction arrival (9,227 records)	0.317	1.79 [0.95, 2.63]	0.025	100.00 / 100.00
B05 Regional delay distribution	0.301	7.01 [5.09, 8.92]	4.2e-7	100.00 / 100.00
region delay 10 min standard deviation (9,227 records)	-0.128	-1.11 [-1.84, -0.38]	0.065	100.00 / 100.00
B06 Regional delay variance	0.301	5.56 [3.16, 7.96]	6.3e-4	100.00 / 100.00
delay variance 10 min (9,227 records)	-0.128	-1.08 [-1.91, -0.26]	0.124	100.00 / 100.00
H25 Local assortativity	0.290	10.95 [9.30, 12.60]	2.7e-12	100.00 / 100.00
local delay absolute difference arrival (5,094 records)	-0.226	-0.64 [-0.89, -0.40]	0.008	100.00 / 100.00
G09 Active cross-train links	0.284	7.86 [6.21, 9.51]	1.9e-9	100.00 / 100.00
active cross fraction departure (9,227 records)	-0.168	-1.17 [-1.84, -0.49]	0.049	100.00 / 100.00
B05 Regional delay distribution	0.278	5.39 [3.61, 7.18]	1.3e-5	100.00 / 100.00
region delay 10 min interquartile range (9,227 records)	-0.122	-1.01 [-1.70, -0.32]	0.078	100.00 / 100.00
G09 Active cross-train links	0.277	7.54 [5.91, 9.18]	4.0e-9	100.00 / 100.00
active cross departure (9,227 records)	-0.160	-1.21 [-1.87, -0.56]	0.039	100.00 / 100.00
B05 Regional delay distribution	0.273	6.40 [4.99, 7.80]	5.4e-9	100.00 / 100.00
region delay 20 min standard deviation (9,227 records)	-0.114	-0.83 [-1.35, -0.31]	0.059	100.00 / 100.00
B06 Regional delay variance	0.273	4.80 [2.87, 6.72]	2.4e-4	100.00 / 100.00
delay variance 20 min (9,227 records)	-0.114	-0.69 [-1.23, -0.16]	0.136	100.00 / 100.00
G19 Endpoint contrasts	0.272	9.04 [6.11, 11.97]	8.8e-6	100.00 / 100.00
H05 downstream reachability endpoint contrast (9,227 records)	-0.322	-1.98 [-3.02, -0.93]	0.033	100.00 / 100.00
H08 Spatial reachability	0.258	6.46 [4.67, 8.25]	5.4e-7	100.00 / 100.00
downstream branches departure (9,227 records)	-0.107	-0.91 [-1.50, -0.31]	0.067	100.00 / 100.00
G08 Active degree	0.257	7.01 [5.71, 8.31]	3.0e-10	100.00 / 100.00
outgoing active departure (9,227 records)	-0.107	-0.94 [-1.46, -0.43]	0.040	100.00 / 100.00
B05 Regional delay distribution	0.255	5.59 [4.03, 7.14]	5.6e-7	100.00 / 100.00
region delay 20 min interquartile range (9,227 records)	-0.113	-0.73 [-1.25, -0.21]	0.097	100.00 / 100.00
B05 Regional delay distribution	0.247	5.49 [3.58, 7.40]	2.8e-5	100.00 / 100.00
region delay 5 min interquartile range (9,227 records)	-0.125	-1.40 [-2.23, -0.57]	0.054	100.00 / 100.00
H08 Spatial reachability	0.244	6.20 [4.27, 8.14]	4.5e-6	100.00 / 100.00
downstream stations departure (9,227 records)	-0.096	-0.87 [-1.48, -0.27]	0.089	100.00 / 100.00
H07 Downstream trains	0.242	5.57 [3.75, 7.39]	9.8e-6	100.00 / 100.00
downstream departure (9,227 records)	-0.074	-0.88 [-1.60, -0.16]	0.177	100.00 / 100.00
H11 Harmonic closeness	0.241	6.47 [4.75, 8.20]	2.6e-7	100.00 / 100.00
outgoing harmonic departure (9,227 records)	-0.086	-0.97 [-1.65, -0.28]	0.093	100.00 / 100.00
B08 Community structure	0.240	5.90 [4.51, 7.30]	2.3e-8	100.00 / 100.00
community events departure (9,227 records)	-0.083	-0.62 [-1.01, -0.24]	0.057	100.00 / 100.00
B05 Regional delay distribution	0.239	6.31 [4.37, 8.25]	3.6e-6	100.00 / 100.00
region delay 5 min standard deviation (9,227 records)	-0.102	-1.40 [-2.25, -0.55]	0.057	100.00 / 100.00
B06 Regional delay variance	0.239	6.04 [3.67, 8.40]	1.7e-4	100.00 / 100.00
delay variance 5 min (9,227 records)	-0.102	-1.60 [-2.68, -0.52]	0.076	100.00 / 100.00
H05 Downstream reachability	0.238	5.55 [3.50, 7.59]	7.1e-5	100.00 / 100.00
downstream departure (9,227 records)	-0.086	-0.86 [-1.62, -0.10]	0.264	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H25 Local assortativity	-0.237	-9.37 [-11.03, -7.70]	1.1e-10	100.00 / 100.00
local same type fraction arrival (5,094 records)	0.162	0.33 [0.15, 0.51]	0.039	100.00 / 100.00
H16 Local motifs	0.236	6.57 [4.85, 8.28]	1.8e-7	100.00 / 100.00
fan-out pairs departure (9,227 records)	-0.120	-1.09 [-1.76, -0.42]	0.057	100.00 / 100.00
H24 Projection core number	0.232	5.95 [4.74, 7.15]	1.0e-9	100.00 / 100.00
core departure (9,227 records)	-0.081	-0.62 [-1.01, -0.23]	0.059	100.00 / 100.00
B01 Local region size	0.231	6.48 [5.24, 7.71]	4.0e-10	100.00 / 100.00
region trains 20 min (9,227 records)	-0.106	-0.75 [-1.22, -0.27]	0.063	100.00 / 100.00
G19 Endpoint contrasts	0.228	5.61 [4.19, 7.03]	1.1e-7	100.00 / 100.00
H10 active betweenness endpoint contrast (9,227 records)	-0.151	-0.89 [-1.44, -0.34]	0.057	100.00 / 100.00
B05 Regional delay distribution	-0.219	-5.67 [-6.82, -4.53]	8.8e-10	100.00 / 100.00
region delay 5 min zero fraction (9,227 records)	0.084	0.61 [0.27, 0.95]	0.043	100.00 / 100.00
G13 Upstream-neighbor delay	-0.218	-5.64 [-6.52, -4.75]	1.7e-11	100.00 / 100.00
neighbor delay zero fraction departure (8,952 records)	0.122	0.55 [0.19, 0.91]	0.065	100.00 / 100.00
B02 Local edges and density	-0.217	-5.41 [-6.22, -4.59]	2.7e-12	100.00 / 100.00
density 20 min (9,163 records)	0.097	0.57 [0.18, 0.95]	0.078	100.00 / 100.00
B02 Local edges and density	-0.208	-5.16 [-6.12, -4.20]	3.0e-10	100.00 / 100.00
density 10 min (9,081 records)	0.086	0.59 [0.21, 0.97]	0.065	100.00 / 100.00
G10 Edge-slack distribution	-0.206	-6.03 [-8.21, -3.86]	5.0e-5	100.00 / 100.00
incoming slack zero fraction departure (9,227 records)	0.188	0.91 [0.49, 1.32]	0.025	100.00 / 100.00
H09 Reachable delay totals	0.205	3.84 [2.12, 5.56]	0.001	100.00 / 100.00
downstream event delay departure (9,227 records)	-0.064	-0.86 [-1.65, -0.07]	0.307	100.00 / 100.00
B01 Local region size	0.205	5.49 [4.40, 6.57]	7.4e-10	100.00 / 100.00
region trains 10 min (9,227 records)	-0.098	-0.79 [-1.26, -0.31]	0.057	100.00 / 100.00
B05 Regional delay distribution	-0.205	-5.48 [-6.60, -4.36]	1.1e-9	100.00 / 100.00
region delay 10 min zero fraction (9,227 records)	0.073	0.61 [0.27, 0.94]	0.044	100.00 / 100.00
G11 Near-tight links	-0.202	-5.01 [-6.38, -3.64]	4.1e-7	100.00 / 100.00
slack at most 1 minute arrival (9,227 records)	0.146	1.02 [0.47, 1.56]	0.039	100.00 / 100.00
B05 Regional delay distribution	0.201	3.90 [2.83, 4.97]	4.2e-7	100.00 / 100.00
region delay 20 min sum (9,227 records)	-0.073	-0.57 [-0.98, -0.16]	0.095	100.00 / 100.00
B04 Boundary connections	0.201	5.85 [5.02, 6.68]	1.3e-12	100.00 / 100.00
entering 20 min (9,227 records)	-0.073	-0.53 [-0.96, -0.09]	0.194	100.00 / 100.00
H13 Weak-component size	0.199	4.86 [3.67, 6.04]	4.9e-8	100.00 / 100.00
component events departure (9,227 records)	-0.028	-0.41 [-0.73, -0.09]	0.152	100.00 / 100.00
H16 Local motifs	0.197	5.04 [3.51, 6.58]	2.8e-6	100.00 / 100.00
diamonds at source departure (9,227 records)	-0.091	-0.38 [-0.88, 0.12]	1.000	100.00 / 100.00
H13 Weak-component size	0.193	5.06 [3.94, 6.18]	5.8e-9	100.00 / 100.00
component trains departure (9,227 records)	-0.036	-0.50 [-0.83, -0.17]	0.069	100.00 / 100.00
B01 Local region size	0.191	4.97 [3.97, 5.96]	8.0e-10	100.00 / 100.00
region events 20 min (9,227 records)	-0.081	-0.39 [-0.72, -0.05]	0.235	100.00 / 100.00
B05 Regional delay distribution	0.191	4.97 [3.97, 5.96]	8.0e-10	100.00 / 100.00
region delay 20 min count (9,227 records)	-0.081	-0.39 [-0.72, -0.05]	0.235	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H09 Reachable delay totals	0.191	3.53 [2.04, 5.02]	4.8e-4	100.00 / 100.00
downstream train maximum delay sum departure (9,227 records)	-0.048	-0.88 [-1.64, -0.12]	0.232	100.00 / 100.00
G12 Delay-support distribution	-0.190	-4.96 [-5.99, -3.93]	1.6e-9	100.00 / 100.00
outgoing support zero fraction departure (9,227 records)	0.067	0.61 [0.25, 0.96]	0.050	100.00 / 100.00
G13 Upstream-neighbor delay	0.188	3.67 [2.62, 4.72]	9.3e-7	100.00 / 100.00
neighbor delay sum departure (9,227 records)	-0.101	-0.67 [-1.15, -0.18]	0.100	100.00 / 100.00
G12 Delay-support distribution	-0.187	-4.79 [-5.59, -3.99]	2.9e-11	100.00 / 100.00
incoming support zero fraction departure (9,227 records)	0.056	0.00 [-0.17, 0.17]	1.000	53.33 / 66.67
G12 Delay-support distribution	0.187	4.48 [2.85, 6.12]	5.6e-5	100.00 / 100.00
outgoing support standard deviation departure (9,227 records)	-0.048	-0.89 [-1.45, -0.32]	0.063	100.00 / 100.00
G12 Delay-support distribution	0.186	4.38 [2.74, 6.01]	8.5e-5	100.00 / 100.00
outgoing support interquartile range departure (9,227 records)	-0.047	-0.89 [-1.46, -0.31]	0.065	100.00 / 100.00
G13 Upstream-neighbor delay	0.184	3.65 [2.59, 4.70]	1.5e-6	100.00 / 100.00
neighbor delay maximum departure (8,952 records)	-0.099	-0.67 [-1.16, -0.19]	0.097	100.00 / 100.00
G13 Upstream-neighbor delay	0.184	3.65 [2.59, 4.70]	1.5e-6	100.00 / 100.00
neighbor delay mean departure (8,952 records)	-0.099	-0.67 [-1.16, -0.19]	0.097	100.00 / 100.00
G13 Upstream-neighbor delay	0.184	3.65 [2.59, 4.70]	1.5e-6	100.00 / 100.00
neighbor delay median departure (8,952 records)	-0.099	-0.67 [-1.16, -0.19]	0.097	100.00 / 100.00
G13 Upstream-neighbor delay	0.184	3.65 [2.59, 4.70]	1.5e-6	100.00 / 100.00
neighbor delay minimum departure (8,952 records)	-0.099	-0.67 [-1.16, -0.19]	0.097	100.00 / 100.00
G13 Upstream-neighbor delay	0.184	3.65 [2.59, 4.70]	1.5e-6	100.00 / 100.00
neighbor delay 90th percentile departure (8,952 records)	-0.099	-0.67 [-1.16, -0.19]	0.097	100.00 / 100.00
B05 Regional delay distribution	0.184	3.16 [2.20, 4.13]	3.0e-6	100.00 / 100.00
region delay 10 min sum (9,227 records)	-0.057	-0.58 [-0.96, -0.19]	0.073	100.00 / 100.00
B05 Regional delay distribution	-0.182	-5.22 [-6.30, -4.14]	1.4e-9	100.00 / 100.00
region delay 20 min zero fraction (9,227 records)	0.059	0.59 [0.25, 0.92]	0.046	100.00 / 100.00
B04 Boundary connections	0.180	5.00 [3.98, 6.02]	1.1e-9	100.00 / 100.00
leaving 20 min (9,227 records)	-0.065	-0.72 [-1.20, -0.25]	0.065	100.00 / 100.00
B02 Local edges and density	0.180	4.16 [3.19, 5.14]	2.2e-8	100.00 / 100.00
edges 20 min (9,227 records)	-0.080	-0.27 [-0.55, 0.01]	0.493	100.00 / 100.00
G10 Edge-slack distribution	0.178	2.46 [0.61, 4.30]	0.113	100.00 / 100.00
outgoing slack mean arrival (9,227 records)	-0.188	-2.24 [-3.69, -0.79]	0.065	100.00 / 100.00
G10 Edge-slack distribution	0.178	2.46 [0.61, 4.30]	0.113	100.00 / 100.00
outgoing slack median arrival (9,227 records)	-0.188	-2.24 [-3.69, -0.79]	0.065	100.00 / 100.00
G12 Delay-support distribution	0.172	2.64 [1.57, 3.71]	2.8e-4	100.00 / 100.00
outgoing support sum departure (9,227 records)	-0.054	-0.76 [-1.24, -0.28]	0.058	100.00 / 100.00
B10 Affected-train share	0.170	4.48 [3.53, 5.43]	2.5e-9	100.00 / 100.00
affected fraction (9,227 records)	-0.051	-0.46 [-0.75, -0.18]	0.057	100.00 / 100.00
B15 Global degree distribution	0.169	4.45 [3.62, 5.28]	3.5e-10	100.00 / 100.00
outgoing degree 90th percentile (9,227 records)	-0.065	-0.49 [-0.76, -0.23]	0.035	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B15 Global degree distribution	-0.169	-4.41 [-5.40, -3.42]	8.5e-9	100.00 / 100.00
outgoing degree zero fraction (9,227 records)	0.033	0.39 [0.10, 0.67]	0.103	100.00 / 100.00
B11 Global path distribution	-0.169	-4.38 [-5.41, -3.35]	2.2e-8	100.00 / 100.00
root longest paths zero fraction (9,227 records)	0.035	0.38 [0.10, 0.67]	0.107	100.00 / 100.00
B15 Global degree distribution	-0.169	-4.38 [-5.41, -3.35]	2.2e-8	100.00 / 100.00
incoming degree zero fraction (9,227 records)	0.035	0.38 [0.10, 0.67]	0.107	100.00 / 100.00
B15 Global degree distribution	0.168	4.32 [3.26, 5.38]	5.4e-8	100.00 / 100.00
incoming degree mean (9,227 records)	-0.035	-0.37 [-0.65, -0.09]	0.131	100.00 / 100.00
B15 Global degree distribution	0.168	4.32 [3.26, 5.38]	5.4e-8	100.00 / 100.00
incoming degree sum (9,227 records)	-0.035	-0.37 [-0.65, -0.09]	0.131	100.00 / 100.00
B15 Global degree distribution	0.167	4.30 [3.24, 5.36]	5.8e-8	100.00 / 100.00
outgoing degree mean (9,227 records)	-0.036	-0.38 [-0.66, -0.09]	0.122	100.00 / 100.00
B15 Global degree distribution	0.167	4.30 [3.24, 5.36]	5.8e-8	100.00 / 100.00
outgoing degree sum (9,227 records)	-0.036	-0.38 [-0.66, -0.09]	0.122	100.00 / 100.00
B15 Global degree distribution	0.167	4.58 [3.81, 5.35]	3.2e-11	100.00 / 100.00
incoming degree standard deviation (9,227 records)	-0.084	-0.56 [-0.85, -0.27]	0.030	100.00 / 100.00
G13 Upstream-neighbor delay	0.166	2.85 [1.67, 4.03]	3.9e-4	100.00 / 100.00
neighbor delay interquartile range arrival (8,947 records)	-0.105	0.03 [-0.11, 0.17]	1.000	93.33 / 66.67
G13 Upstream-neighbor delay	0.166	2.85 [1.67, 4.03]	3.9e-4	100.00 / 100.00
neighbor delay standard deviation arrival (8,947 records)	-0.105	0.03 [-0.11, 0.17]	1.000	93.33 / 66.67
G10 Edge-slack distribution	0.166	2.50 [0.76, 4.25]	0.070	100.00 / 100.00
outgoing slack sum arrival (9,227 records)	-0.174	-2.34 [-3.74, -0.94]	0.056	100.00 / 100.00
B15 Global degree distribution	0.164	4.56 [3.76, 5.36]	8.8e-11	100.00 / 100.00
outgoing degree standard deviation (9,227 records)	-0.074	-0.55 [-0.82, -0.27]	0.030	100.00 / 100.00
B15 Global degree distribution	0.163	4.07 [3.20, 4.94]	2.6e-9	100.00 / 100.00
outgoing degree maximum (9,227 records)	-0.085	-0.48 [-0.74, -0.22]	0.042	100.00 / 100.00
B15 Global degree distribution	0.162	3.97 [2.67, 5.27]	1.0e-5	100.00 / 100.00
outgoing degree interquartile range (9,227 records)	-0.067	-0.52 [-0.78, -0.26]	0.030	100.00 / 100.00
B11 Global path distribution	0.162	3.81 [2.50, 5.12]	2.3e-5	100.00 / 100.00
root longest paths mean (9,227 records)	-0.048	-0.36 [-0.64, -0.09]	0.117	100.00 / 100.00
B11 Global path distribution	0.162	3.81 [2.50, 5.12]	2.3e-5	100.00 / 100.00
root longest paths sum (9,227 records)	-0.048	-0.36 [-0.64, -0.09]	0.117	100.00 / 100.00
B11 Global path distribution	0.161	3.97 [2.74, 5.20]	4.2e-6	100.00 / 100.00
root longest paths interquartile range (9,227 records)	-0.057	-0.48 [-0.75, -0.20]	0.047	100.00 / 100.00
B12 Delay concentration	0.161	2.68 [1.27, 4.09]	0.006	100.00 / 100.00
station entropy (9,227 records)	-0.002	-0.29 [-0.49, -0.08]	0.097	100.00 / 100.00
B05 Regional delay distribution	0.161	3.39 [2.24, 4.53]	1.7e-5	100.00 / 100.00
region delay 20 min maximum (9,227 records)	-0.056	-0.63 [-1.06, -0.21]	0.070	100.00 / 100.00
B16 Global reachability	0.160	3.39 [1.96, 4.83]	4.8e-4	100.00 / 100.00
reachable pair fraction (9,227 records)	-0.048	-0.29 [-0.56, -0.02]	0.338	100.00 / 100.00
B12 Delay concentration	-0.158	-1.90 [-3.29, -0.50]	0.102	100.00 / 100.00
station concentration (9,227 records)	0.011	0.20 [0.02, 0.38]	0.262	100.00 / 100.00
B05 Regional delay distribution	0.157	3.02 [1.82, 4.22]	2.2e-4	100.00 / 100.00
region delay 10 min maximum (9,227 records)	-0.054	-0.68 [-1.12, -0.23]	0.070	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B15 Global degree distribution incoming degree 90th percentile (9,227 records)	0.156 -0.045	4.02 [3.21, 4.82] -0.39 [-0.63, -0.14]	8.0e-10 0.064	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths standard deviation (9,227 records)	0.156 -0.061	4.19 [3.19, 5.19] -0.50 [-0.76, -0.24]	2.8e-8 0.031	100.00 / 100.00 100.00 / 100.00
B01 Local region size region trains 5 min (9,227 records)	0.156 -0.074	4.30 [3.16, 5.44] -0.76 [-1.24, -0.27]	2.5e-7 0.063	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths 90th percentile (9,227 records)	0.155 -0.051	4.38 [3.44, 5.32] -0.49 [-0.75, -0.24]	3.2e-9 0.032	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min 90th percentile (9,227 records)	0.155 -0.052	2.92 [1.72, 4.11] -0.67 [-1.12, -0.22]	3.4e-4 0.073	100.00 / 100.00 100.00 / 100.00
H25 Local assortativity local delay absolute difference departure (5,533 records)	0.155 -0.083	4.96 [2.42, 7.50] -1.43 [-2.33, -0.52]	0.005 0.062	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution incoming degree interquartile range (9,227 records)	0.155 -0.096	3.96 [2.95, 4.98] -0.63 [-0.94, -0.32]	1.3e-7 0.028	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution outgoing slack 90th percentile arrival (9,227 records)	0.155 -0.165	1.29 [-0.04, 2.62] -1.71 [-2.75, -0.66]	0.548 0.057	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay interquartile range departure (6,908 records)	0.154 -0.016	2.43 [1.00, 3.85] -0.64 [-1.05, -0.24]	0.018 0.062	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay standard deviation departure (6,908 records)	0.154 -0.016	2.43 [1.00, 3.85] -0.64 [-1.05, -0.24]	0.018 0.062	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min maximum (9,227 records)	0.154 -0.051	2.80 [1.60, 4.01] -0.69 [-1.15, -0.23]	5.9e-4 0.074	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min 90th percentile (9,227 records)	0.154 -0.052	3.13 [2.02, 4.25] -0.63 [-1.05, -0.21]	4.2e-5 0.073	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min 90th percentile (9,227 records)	0.154 -0.050	2.77 [1.56, 3.98] -0.68 [-1.14, -0.22]	7.2e-4 0.076	100.00 / 100.00 100.00 / 100.00
G02 Reference departure delay departure delay (9,227 records)	0.152 -0.049	2.65 [1.44, 3.86] -0.67 [-1.13, -0.21]	0.001 0.079	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support maximum departure (9,227 records)	0.152 -0.049	2.65 [1.44, 3.87] -0.67 [-1.13, -0.21]	0.001 0.079	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support 90th percentile departure (9,227 records)	0.152 -0.049	2.59 [1.38, 3.81] -0.66 [-1.11, -0.20]	0.002 0.083	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths maximum (9,227 records)	0.150 -0.060	3.84 [2.82, 4.86] -0.44 [-0.70, -0.19]	2.6e-7 0.047	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min sum (9,227 records)	0.149 -0.032	1.99 [1.03, 2.96] -0.47 [-0.83, -0.12]	0.003 0.112	100.00 / 100.00 100.00 / 100.00
G15 Active-neighbor dispersion delay coefficient of variation departure (5,533 records)	0.149 -0.068	4.84 [2.70, 6.99] -0.56 [-1.16, 0.04]	9.1e-4 0.564	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support mean departure (9,227 records)	0.148 -0.049	2.22 [1.01, 3.44] -0.59 [-1.02, -0.16]	0.009 0.100	100.00 / 100.00 100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution outgoing slack maximum arrival (9,227 records)	0.148 -0.158	1.07 [-0.17, 2.31] -1.60 [-2.57, -0.63]	0.833 0.057	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min median (9,227 records)	0.146 -0.043	2.41 [1.25, 3.57] -0.61 [-1.04, -0.18]	0.002 0.092	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min mean (9,227 records)	0.145 -0.046	2.37 [1.22, 3.52] -0.59 [-1.02, -0.17]	0.003 0.094	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min mean (9,227 records)	0.144 -0.047	2.35 [1.19, 3.50] -0.59 [-1.01, -0.17]	0.003 0.092	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min median (9,227 records)	0.144 -0.044	2.33 [1.17, 3.49] -0.60 [-1.03, -0.17]	0.003 0.094	100.00 / 100.00 100.00 / 100.00
B04 Boundary connections entering 10 min (9,227 records)	0.144 -0.021	4.01 [2.99, 5.04] -0.29 [-0.64, 0.06]	1.3e-7 0.834	100.00 / 100.00 100.00 / 100.00
G15 Active-neighbor dispersion delay coefficient of variation arrival (5,094 records)	0.144 -0.073	4.20 [2.08, 6.31] -0.18 [-0.36, 0.01]	0.004 0.504	100.00 / 100.00 100.00 / 100.00
G15 Active-neighbor dispersion delay variance departure (5,533 records)	0.143 -0.069	4.51 [1.98, 7.04] -1.66 [-2.75, -0.58]	0.012 0.065	100.00 / 100.00 100.00 / 100.00
G15 Active-neighbor dispersion delay range departure (5,533 records)	0.143 -0.068	4.67 [2.22, 7.12] -1.44 [-2.32, -0.56]	0.006 0.057	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min mean (9,227 records)	0.141 -0.046	2.36 [1.22, 3.49] -0.55 [-0.94, -0.15]	0.002 0.097	100.00 / 100.00 100.00 / 100.00
G15 Active-neighbor dispersion delay variance arrival (5,094 records)	0.140 -0.080	5.08 [3.21, 6.96] -0.31 [-0.54, -0.09]	7.3e-5 0.099	100.00 / 100.00 100.00 / 100.00
H10 Active betweenness betweenness arrival (9,227 records)	-0.139 0.178	-1.78 [-2.84, -0.71] 0.40 [0.21, 0.58]	0.020 0.025	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration time20 concentration (9,227 records)	-0.139 0.059	-3.51 [-4.70, -2.32] 0.43 [0.19, 0.67]	1.8e-5 0.045	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support sum departure (9,227 records)	0.138 -0.028	2.46 [1.50, 3.42] -0.06 [-0.25, 0.13]	1.6e-4 1.000	100.00 / 100.00 100.00 / 66.67
G12 Delay-support distribution incoming support mean departure (9,227 records)	0.138 -0.029	2.42 [1.50, 3.34] -0.07 [-0.26, 0.13]	1.1e-4 1.000	100.00 / 100.00 100.00 / 66.67
G12 Delay-support distribution incoming support median departure (9,227 records)	0.137 -0.041	2.40 [1.48, 3.33] -0.08 [-0.27, 0.11]	1.3e-4 1.000	100.00 / 100.00 100.00 / 66.67
G12 Delay-support distribution outgoing support median departure (9,227 records)	0.137 -0.040	2.32 [1.09, 3.55] -0.61 [-1.04, -0.17]	0.007 0.097	100.00 / 100.00 100.00 / 100.00
G15 Active-neighbor dispersion delay range arrival (5,094 records)	0.136 -0.076	4.10 [2.33, 5.88] -0.19 [-0.36, -0.02]	6.7e-4 0.244	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min median (9,227 records)	0.135 -0.041	2.21 [1.10, 3.33] -0.53 [-0.92, -0.15]	0.004 0.097	100.00 / 100.00 100.00 / 100.00
B04 Boundary connections leaving 10 min (9,227 records)	0.135 -0.034	3.92 [2.84, 5.01] -0.65 [-1.07, -0.23]	5.3e-7 0.065	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration time20 entropy (9,227 records)	0.134 -0.066	3.65 [2.72, 4.58] -0.48 [-0.73, -0.23]	1.2e-7 0.031	100.00 / 100.00 100.00 / 100.00
B03 Active local density active density 5 min (8,957 records)	0.133 -0.032	2.72 [1.63, 3.81] -0.23 [-0.42, -0.04]	2.4e-4 0.200	100.00 / 100.00 100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B15 Global degree distribution	0.132	3.38 [2.23, 4.54]	2.0e-5	100.00 / 100.00
incoming degree median (9,227 records)	-0.027	-0.25 [-0.56, 0.06]	0.930	100.00 / 100.00
B02 Local edges and density	-0.131	-3.37 [-4.71, -2.03]	2.1e-4	100.00 / 100.00
density 5 min (8,957 records)	0.034	0.46 [0.13, 0.79]	0.095	100.00 / 100.00
G12 Delay-support distribution	0.129	2.31 [1.39, 3.23]	2.1e-4	100.00 / 100.00
incoming support minimum departure (9,227 records)	-0.038	0.35 [0.11, 0.59]	0.078	100.00 / 100.00
G14 Downstream-neighbor delay	-0.129	-3.57 [-4.82, -2.33]	3.4e-5	100.00 / 100.00
neighbor delay zero fraction departure (6,908 records)	0.030	0.58 [0.25, 0.91]	0.047	100.00 / 100.00
B05 Regional delay distribution	0.129	1.77 [0.68, 2.85]	0.026	100.00 / 100.00
region delay 5 min minimum (9,227 records)	-0.038	-0.46 [-0.82, -0.10]	0.146	100.00 / 100.00
B01 Local region size	0.128	3.09 [2.17, 4.01]	2.0e-6	100.00 / 100.00
region events 10 min (9,227 records)	-0.033	-0.26 [-0.54, 0.01]	0.531	100.00 / 100.00
B05 Regional delay distribution	0.128	3.09 [2.17, 4.01]	2.0e-6	100.00 / 100.00
region delay 10 min count (9,227 records)	-0.033	-0.26 [-0.54, 0.01]	0.531	100.00 / 100.00
H15 Chain and crossing length	0.128	2.78 [1.62, 3.95]	4.3e-4	100.00 / 100.00
longest crossings departure (9,227 records)	-0.061	-0.03 [-0.18, 0.11]	1.000	100.00 / 100.00
G10 Edge-slack distribution	0.128	0.26 [-0.54, 1.06]	1.000	100.00 / 66.67
incoming slack minimum arrival (9,227 records)	0.038	0.00 [-0.09, 0.09]	1.000	46.67 / 66.67
G13 Upstream-neighbor delay	0.127	2.74 [1.66, 3.81]	1.7e-4	100.00 / 100.00
neighbor delay sum arrival (9,227 records)	-0.049	0.34 [0.09, 0.59]	0.116	100.00 / 100.00
G14 Downstream-neighbor delay	0.127	1.96 [1.00, 2.93]	0.003	100.00 / 100.00
neighbor delay sum departure (9,227 records)	-0.030	-0.72 [-1.17, -0.28]	0.057	100.00 / 100.00
G12 Delay-support distribution	0.127	2.21 [1.12, 3.30]	0.003	100.00 / 100.00
incoming support 90th percentile departure (9,227 records)	-0.021	-0.34 [-0.70, 0.01]	0.505	100.00 / 100.00
B16 Global reachability	0.126	3.29 [2.26, 4.32]	4.7e-6	100.00 / 100.00
mean reachable distance (9,227 records)	-0.055	-0.39 [-0.62, -0.16]	0.054	100.00 / 100.00
G12 Delay-support distribution	0.124	2.13 [1.00, 3.26]	0.007	100.00 / 100.00
incoming support maximum departure (9,227 records)	-0.019	-0.39 [-0.78, 0.00]	0.434	100.00 / 100.00
B15 Global degree distribution	0.124	3.11 [2.02, 4.20]	2.9e-5	100.00 / 100.00
outgoing degree median (9,227 records)	-0.010	-0.16 [-0.49, 0.17]	1.000	100.00 / 100.00
H25 Local assortativity	-0.122	-4.02 [-6.38, -1.66]	0.018	100.00 / 100.00
local same type fraction departure (5,533 records)	0.061	0.94 [0.33, 1.55]	0.065	100.00 / 100.00
G13 Upstream-neighbor delay	-0.120	-3.08 [-4.25, -1.92]	1.1e-4	100.00 / 100.00
neighbor delay zero fraction arrival (8,947 records)	0.044	-0.42 [-0.76, -0.09]	0.152	100.00 / 100.00
B11 Global path distribution	0.120	2.26 [0.92, 3.59]	0.018	100.00 / 100.00
root longest paths median (9,227 records)	-0.012	-0.06 [-0.35, 0.23]	1.000	100.00 / 66.67
G08 Active degree	-0.118	-2.93 [-4.33, -1.52]	0.002	100.00 / 100.00
outgoing active arrival (9,227 records)	0.206	0.75 [0.39, 1.12]	0.025	100.00 / 100.00
G13 Upstream-neighbor delay	0.115	2.48 [1.35, 3.61]	0.001	100.00 / 100.00
neighbor delay maximum arrival (8,947 records)	-0.043	0.37 [0.09, 0.66]	0.126	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H09 Reachable delay totals	-0.113	-0.50 [-1.77, 0.78]	1.000	100.00 / 66.67
downstream event delay arrival (9,227 records)	0.214	0.38 [0.20, 0.56]	0.025	100.00 / 100.00
B05 Regional delay distribution	0.113	1.45 [0.38, 2.53]	0.104	100.00 / 100.00
region delay 10 min minimum (9,227 records)	-0.030	-0.44 [-0.79, -0.09]	0.163	100.00 / 100.00
H08 Spatial reachability	-0.113	-1.97 [-3.31, -0.62]	0.063	100.00 / 100.00
downstream stations arrival (9,227 records)	0.216	0.63 [0.33, 0.93]	0.025	100.00 / 100.00
G12 Delay-support distribution	0.113	0.31 [-1.05, 1.68]	1.000	100.00 / 66.67
incoming support interquartile range departure (9,227 records)	-0.023	-0.90 [-1.69, -0.12]	0.239	100.00 / 100.00
G12 Delay-support distribution	0.113	0.32 [-1.05, 1.69]	1.000	100.00 / 66.67
incoming support standard deviation departure (9,227 records)	-0.023	-0.92 [-1.71, -0.13]	0.232	100.00 / 100.00
G13 Upstream-neighbor delay	0.112	2.39 [1.26, 3.52]	0.002	100.00 / 100.00
neighbor delay 90th percentile arrival (8,947 records)	-0.042	0.38 [0.09, 0.66]	0.119	100.00 / 100.00
G12 Delay-support distribution	0.112	3.03 [1.54, 4.52]	0.003	100.00 / 100.00
outgoing support interquartile range arrival (9,227 records)	0.015	0.24 [0.06, 0.43]	0.132	100.00 / 100.00
G12 Delay-support distribution	0.112	3.03 [1.54, 4.52]	0.003	100.00 / 100.00
outgoing support standard deviation arrival (9,227 records)	0.015	0.24 [0.06, 0.43]	0.132	100.00 / 100.00
G11 Near-tight links	-0.110	-2.92 [-4.46, -1.37]	0.007	100.00 / 100.00
slack at most 2 minutes arrival (9,227 records)	0.093	0.99 [0.42, 1.56]	0.047	100.00 / 100.00
H05 Downstream reachability	-0.109	-0.42 [-1.80, 0.96]	1.000	100.00 / 33.33
downstream arrival (9,227 records)	0.219	0.44 [0.23, 0.64]	0.025	100.00 / 100.00
H09 Reachable delay totals	-0.108	-0.99 [-2.25, 0.27]	1.000	100.00 / 100.00
downstream train maximum delay sum arrival (9,227 records)	0.212	0.45 [0.24, 0.67]	0.025	100.00 / 100.00
H11 Harmonic closeness	-0.107	-1.58 [-3.00, -0.15]	0.314	100.00 / 100.00
outgoing harmonic arrival (9,227 records)	0.220	0.59 [0.33, 0.86]	0.025	100.00 / 100.00
G13 Upstream-neighbor delay	0.106	1.96 [0.82, 3.11]	0.018	100.00 / 100.00
neighbor delay mean arrival (8,947 records)	-0.037	0.39 [0.11, 0.67]	0.094	100.00 / 100.00
G13 Upstream-neighbor delay	0.106	1.96 [0.82, 3.11]	0.018	100.00 / 100.00
neighbor delay median arrival (8,947 records)	-0.037	0.39 [0.11, 0.67]	0.094	100.00 / 100.00
H08 Spatial reachability	-0.106	-2.08 [-3.44, -0.73]	0.041	100.00 / 100.00
downstream branches arrival (9,227 records)	0.217	0.66 [0.35, 0.98]	0.025	100.00 / 100.00
G12 Delay-support distribution	0.106	1.59 [0.39, 2.78]	0.117	100.00 / 100.00
outgoing support minimum departure (9,227 records)	-0.032	-0.47 [-0.85, -0.09]	0.174	100.00 / 100.00
G16 Incident-link imbalance	0.104	1.64 [0.45, 2.82]	0.089	100.00 / 100.00
pressure balance departure (9,227 records)	-0.046	-1.31 [-2.11, -0.50]	0.057	100.00 / 100.00
B02 Local edges and density	0.104	1.90 [1.04, 2.77]	0.001	100.00 / 100.00
edges 10 min (9,227 records)	-0.025	-0.14 [-0.36, 0.07]	1.000	100.00 / 100.00
H13 Weak-component size	0.100	2.52 [1.50, 3.55]	3.0e-4	100.00 / 100.00
component events arrival (9,227 records)	0.046	0.54 [0.19, 0.89]	0.065	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G13 Upstream-neighbor delay	0.100	2.55 [1.57, 3.53]	1.3e-4	100.00 / 100.00
neighbor delay count arrival (9,227 records)	-0.083	-0.31 [-0.62, 0.01]	0.504	100.00 / 100.00
G11 Near-tight links	0.100	2.02 [1.15, 2.88]	5.7e-4	100.00 / 100.00
slack at most 2 minutes departure (9,227 records)	-0.004	0.02 [-0.23, 0.27]	1.000	86.67 / 66.67
H12 PageRank	-0.098	-2.45 [-3.51, -1.39]	6.6e-4	100.00 / 100.00
pagerank departure (9,227 records)	0.124	0.46 [0.19, 0.73]	0.052	100.00 / 100.00
H07 Downstream trains	-0.098	-0.94 [-2.35, 0.46]	1.000	100.00 / 100.00
downstream arrival (9,227 records)	0.221	0.52 [0.28, 0.76]	0.025	100.00 / 100.00
G14 Downstream-neighbor delay	0.096	2.47 [1.45, 3.50]	3.7e-4	100.00 / 100.00
neighbor delay count departure (9,227 records)	-0.023	-0.53 [-0.87, -0.19]	0.065	100.00 / 100.00
G12 Delay-support distribution	-0.096	-2.57 [-3.63, -1.51]	3.6e-4	100.00 / 100.00
incoming support zero fraction arrival (9,227 records)	-0.003	-0.52 [-0.87, -0.16]	0.079	100.00 / 100.00
G10 Edge-slack distribution	-0.096	-2.30 [-2.95, -1.65]	8.2e-7	100.00 / 100.00
outgoing slack median departure (9,227 records)	0.042	0.25 [0.09, 0.40]	0.059	100.00 / 100.00
G07 Degree by relation	0.094	2.42 [1.28, 3.55]	0.002	100.00 / 100.00
outgoing entry headway departure (9,227 records)	-0.045	-0.42 [-0.64, -0.19]	0.035	100.00 / 100.00
G19 Endpoint contrasts	0.094	1.97 [-2.96, 6.90]	1.000	100.00 / 100.00
H04 upstream reachability endpoint contrast (9,227 records)	0.024	-0.51 [-1.02, 0.00]	0.456	100.00 / 100.00
B05 Regional delay distribution	0.094	1.24 [0.12, 2.35]	0.314	100.00 / 100.00
region delay 20 min minimum (9,227 records)	-0.019	-0.39 [-0.74, -0.05]	0.258	100.00 / 100.00
G12 Delay-support distribution	-0.093	-2.41 [-3.39, -1.42]	3.2e-4	100.00 / 100.00
outgoing support zero fraction arrival (9,227 records)	-0.017	-0.57 [-0.96, -0.17]	0.092	100.00 / 100.00
H10 Active betweenness	0.090	1.89 [0.75, 3.04]	0.023	100.00 / 100.00
betweenness departure (9,227 records)	0.037	-0.18 [-0.49, 0.13]	1.000	100.00 / 66.67
G12 Delay-support distribution	0.090	1.35 [0.32, 2.37]	0.123	100.00 / 100.00
incoming support median arrival (9,227 records)	-0.004	0.41 [0.14, 0.68]	0.065	100.00 / 100.00
H13 Weak-component size	0.089	2.47 [1.42, 3.52]	5.3e-4	100.00 / 100.00
component trains arrival (9,227 records)	0.047	0.54 [0.19, 0.89]	0.065	100.00 / 100.00
B08 Community structure	0.089	2.04 [1.02, 3.06]	0.004	100.00 / 100.00
community events arrival (9,227 records)	0.037	0.49 [0.17, 0.81]	0.068	100.00 / 100.00
G16 Incident-link imbalance	-0.088	-1.26 [-2.46, -0.07]	0.390	100.00 / 100.00
pressure balance arrival (9,227 records)	0.055	-0.18 [-0.31, -0.04]	0.119	100.00 / 100.00
H06 Upstream trains	0.086	1.90 [0.70, 3.11]	0.033	100.00 / 100.00
upstream departure (9,227 records)	0.006	0.12 [-0.04, 0.28]	1.000	100.00 / 100.00
B15 Global degree distribution	0.085	2.19 [1.68, 2.71]	2.3e-8	100.00 / 100.00
incoming degree maximum (9,227 records)	-0.088	-0.48 [-0.78, -0.18]	0.057	100.00 / 100.00
G09 Active cross-train links	0.082	1.43 [0.36, 2.50]	0.113	100.00 / 100.00
active cross fraction arrival (9,227 records)	-0.051	0.27 [0.04, 0.51]	0.245	100.00 / 100.00
G12 Delay-support distribution	0.082	0.81 [-0.31, 1.94]	1.000	100.00 / 100.00
incoming support interquartile range arrival (9,227 records)	0.034	0.33 [0.13, 0.53]	0.054	100.00 / 100.00
H12 PageRank	-0.081	-1.51 [-2.39, -0.63]	0.017	100.00 / 100.00
pagerank arrival (9,227 records)	0.043	0.49 [0.14, 0.84]	0.096	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G14 Downstream-neighbor delay	-0.081	-2.01 [-3.00, -1.01]	0.003	100.00 / 100.00
neighbor delay zero fraction arrival (6,812 records)	-0.012	0.30 [0.03, 0.57]	0.266	100.00 / 100.00
G12 Delay-support distribution	0.081	0.75 [-0.36, 1.86]	1.000	100.00 / 100.00
incoming support standard deviation arrival (9,227 records)	0.034	0.33 [0.13, 0.52]	0.054	100.00 / 100.00
G13 Upstream-neighbor delay	0.081	1.28 [0.15, 2.41]	0.284	100.00 / 100.00
neighbor delay minimum arrival (8,947 records)	-0.020	0.39 [0.14, 0.65]	0.065	100.00 / 100.00
G10 Edge-slack distribution	-0.080	-1.11 [-2.21, -0.02]	0.460	100.00 / 100.00
incoming slack standard deviation departure (9,227 records)	0.053	-0.66 [-1.27, -0.06]	0.302	100.00 / 100.00
B04 Boundary connections	0.080	2.46 [1.27, 3.65]	0.003	100.00 / 100.00
leaving 5 min (9,227 records)	-0.004	-0.48 [-0.82, -0.14]	0.092	100.00 / 100.00
G10 Edge-slack distribution	-0.080	-1.09 [-2.19, 0.00]	0.482	100.00 / 100.00
incoming slack interquartile range departure (9,227 records)	0.053	-0.64 [-1.24, -0.05]	0.316	100.00 / 100.00
H09 Reachable delay totals	0.079	1.51 [0.35, 2.67]	0.130	100.00 / 100.00
upstream train maximum delay sum departure (9,227 records)	0.005	0.04 [-0.11, 0.19]	1.000	100.00 / 100.00
H16 Local motifs	-0.079	-2.02 [-2.57, -1.47]	3.6e-7	100.00 / 100.00
diamonds at source arrival (9,227 records)	0.051	0.27 [0.13, 0.42]	0.038	100.00 / 100.00
G09 Active cross-train links	0.078	1.17 [0.16, 2.17]	0.254	100.00 / 100.00
active cross arrival (9,227 records)	-0.047	0.28 [0.05, 0.51]	0.176	100.00 / 100.00
G14 Downstream-neighbor delay	0.076	1.22 [0.00, 2.45]	0.486	100.00 / 100.00
neighbor delay maximum departure (6,908 records)	-0.009	-0.60 [-1.02, -0.17]	0.094	100.00 / 100.00
G14 Downstream-neighbor delay	0.074	1.17 [-0.05, 2.40]	0.579	100.00 / 100.00
neighbor delay 90th percentile departure (6,908 records)	-0.009	-0.58 [-1.00, -0.17]	0.097	100.00 / 100.00
H16 Local motifs	-0.073	-1.88 [-2.71, -1.04]	9.1e-4	100.00 / 100.00
fan-out pairs arrival (9,227 records)	0.103	0.40 [0.19, 0.61]	0.034	100.00 / 100.00
G14 Downstream-neighbor delay	0.071	0.96 [-0.28, 2.20]	1.000	100.00 / 100.00
neighbor delay mean departure (6,908 records)	-0.009	-0.53 [-0.93, -0.13]	0.117	100.00 / 100.00
G14 Downstream-neighbor delay	0.071	0.96 [-0.28, 2.20]	1.000	100.00 / 100.00
neighbor delay median departure (6,908 records)	-0.009	-0.53 [-0.93, -0.13]	0.117	100.00 / 100.00
G12 Delay-support distribution	0.071	0.95 [0.00, 1.91]	0.493	100.00 / 100.00
incoming support sum arrival (9,227 records)	0.017	0.48 [0.18, 0.78]	0.057	100.00 / 100.00
G19 Endpoint contrasts	0.070	1.87 [0.45, 3.30]	0.122	100.00 / 100.00
G05 outgoing degree endpoint contrast (9,227 records)	-0.025	-0.32 [-0.57, -0.08]	0.114	100.00 / 100.00
G12 Delay-support distribution	0.068	0.87 [-0.11, 1.84]	0.763	100.00 / 100.00
incoming support mean arrival (9,227 records)	0.019	0.49 [0.19, 0.79]	0.057	100.00 / 100.00
H03 Event time from release	-0.068	-0.73 [-1.91, 0.44]	1.000	100.00 / 100.00
time from release arrival (9,227 records)	0.050	1.08 [0.41, 1.76]	0.057	100.00 / 100.00
B09 Boundary pressure imbalance	0.068	1.54 [0.22, 2.87]	0.240	100.00 / 100.00
cross boundary pressure balance 5 min (9,227 records)	-0.061	-1.27 [-2.13, -0.41]	0.078	100.00 / 100.00
G10 Edge-slack distribution	-0.068	-1.35 [-2.33, -0.38]	0.089	100.00 / 100.00
incoming slack median arrival (9,227 records)	0.074	0.20 [0.09, 0.31]	0.047	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H01 Shortest source distance	-0.067	-1.90 [-3.62, -0.19]	0.311	100.00 / 100.00
source minimum time lag departure (5,041 records)	-0.004	0.42 [-0.01, 0.86]	0.501	100.00 / 100.00
G10 Edge-slack distribution	-0.067	-2.19 [-2.82, -1.56]	1.0e-6	100.00 / 100.00
outgoing slack sum departure (9,227 records)	0.037	0.21 [0.06, 0.37]	0.097	100.00 / 100.00
G10 Edge-slack distribution	-0.067	-2.22 [-2.93, -1.51]	6.9e-6	100.00 / 100.00
outgoing slack mean departure (9,227 records)	0.036	0.22 [0.06, 0.38]	0.100	100.00 / 100.00
G08 Active degree	0.064	1.69 [0.36, 3.01]	0.147	100.00 / 100.00
incoming active arrival (9,227 records)	-0.027	0.42 [0.10, 0.74]	0.126	100.00 / 100.00
H01 Shortest source distance	-0.064	-1.58 [-2.92, -0.23]	0.239	100.00 / 100.00
source minimum time lag arrival (4,495 records)	0.017	0.06 [-0.01, 0.12]	0.941	100.00 / 100.00
G12 Delay-support distribution	0.063	0.82 [-0.10, 1.74]	0.763	100.00 / 100.00
outgoing support maximum arrival (9,227 records)	0.029	0.54 [0.21, 0.86]	0.057	100.00 / 100.00
G12 Delay-support distribution	0.063	0.91 [-0.03, 1.84]	0.546	100.00 / 100.00
incoming support 90th percentile arrival (9,227 records)	0.028	0.53 [0.21, 0.86]	0.057	100.00 / 100.00
G01 Reference arrival delay	0.063	0.82 [-0.10, 1.74]	0.763	100.00 / 100.00
arrival delay (9,227 records)	0.029	0.54 [0.21, 0.86]	0.057	100.00 / 100.00
G12 Delay-support distribution	0.062	0.77 [-0.16, 1.69]	0.942	100.00 / 100.00
outgoing support 90th percentile arrival (9,227 records)	0.029	0.54 [0.21, 0.86]	0.057	100.00 / 100.00
G10 Edge-slack distribution	-0.061	-2.17 [-2.85, -1.50]	4.2e-6	100.00 / 100.00
outgoing slack 90th percentile departure (9,227 records)	0.035	0.21 [0.04, 0.37]	0.156	100.00 / 100.00
B03 Active local density	0.060	-0.44 [-1.44, 0.55]	1.000	93.33 / 66.67
active density 20 min (9,163 records)	0.011	-0.04 [-0.24, 0.16]	1.000	86.67 / 66.67
G12 Delay-support distribution	0.060	0.55 [-0.39, 1.49]	1.000	100.00 / 100.00
outgoing support mean arrival (9,227 records)	0.028	0.53 [0.20, 0.86]	0.057	100.00 / 100.00
G12 Delay-support distribution	0.060	0.55 [-0.39, 1.49]	1.000	100.00 / 100.00
outgoing support median arrival (9,227 records)	0.028	0.53 [0.20, 0.86]	0.057	100.00 / 100.00
G12 Delay-support distribution	0.059	0.77 [-0.15, 1.70]	0.926	100.00 / 100.00
incoming support maximum arrival (9,227 records)	0.032	0.54 [0.21, 0.87]	0.057	100.00 / 100.00
H02 Longest active path	0.058	0.75 [-0.63, 2.13]	1.000	100.00 / 100.00
longest path departure (9,227 records)	0.027	0.14 [-0.05, 0.33]	1.000	100.00 / 100.00
H15 Chain and crossing length	0.058	0.75 [-0.63, 2.13]	1.000	100.00 / 100.00
longest chain departure (9,227 records)	0.027	0.14 [-0.05, 0.33]	1.000	100.00 / 100.00
G08 Active degree	0.058	1.13 [-0.41, 2.67]	1.000	100.00 / 100.00
incoming active departure (9,227 records)	0.072	0.07 [-0.12, 0.26]	1.000	100.00 / 66.67
G14 Downstream-neighbor delay	0.058	0.67 [-0.59, 1.93]	1.000	100.00 / 66.67
neighbor delay minimum departure (6,908 records)	-0.005	-0.46 [-0.83, -0.09]	0.168	100.00 / 100.00
G12 Delay-support distribution	0.058	0.34 [-0.48, 1.16]	1.000	100.00 / 66.67
outgoing support sum arrival (9,227 records)	0.035	0.51 [0.22, 0.80]	0.045	100.00 / 100.00
G10 Edge-slack distribution	-0.058	-2.11 [-2.78, -1.44]	6.1e-6	100.00 / 100.00
outgoing slack maximum departure (9,227 records)	0.035	0.19 [0.03, 0.36]	0.222	100.00 / 100.00
H01 Shortest source distance	-0.056	-1.39 [-2.63, -0.14]	0.302	100.00 / 100.00
source edges arrival (4,495 records)	-0.014	0.02 [-0.04, 0.08]	1.000	93.33 / 66.67

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution	0.052	1.15 [-0.22, 2.53]	0.902	100.00 / 100.00
outgoing slack zero fraction departure (9,227 records)	-0.036	-0.09 [-0.38, 0.20]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.050	-1.97 [-2.63, -1.31]	1.4e-5	100.00 / 100.00
outgoing slack interquartile range departure (9,227 records)	0.033	0.18 [0.02, 0.34]	0.284	100.00 / 100.00
H16 Local motifs	-0.049	-1.25 [-2.23, -0.27]	0.142	100.00 / 100.00
fan-in pairs departure (9,227 records)	0.095	0.34 [0.18, 0.49]	0.025	100.00 / 100.00
G10 Edge-slack distribution	-0.049	-1.96 [-2.61, -1.31]	1.3e-5	100.00 / 100.00
outgoing slack standard deviation departure (9,227 records)	0.032	0.18 [0.02, 0.33]	0.264	100.00 / 100.00
B03 Active local density	0.048	-0.08 [-1.08, 0.91]	1.000	80.00 / 66.67
active density 10 min (9,081 records)	0.028	-0.05 [-0.20, 0.10]	1.000	100.00 / 66.67
H08 Spatial reachability	0.047	0.99 [-0.40, 2.39]	1.000	100.00 / 100.00
upstream branches departure (9,227 records)	0.026	-0.08 [-0.31, 0.15]	1.000	100.00 / 66.67
B09 Boundary pressure imbalance	0.046	0.90 [-0.92, 2.73]	1.000	100.00 / 66.67
cross boundary pressure balance 20 min (9,227 records)	-0.048	-1.59 [-2.58, -0.61]	0.057	100.00 / 100.00
H09 Reachable delay totals	0.045	-0.64 [-1.61, 0.34]	1.000	100.00 / 66.67
upstream event delay departure (9,227 records)	0.028	0.18 [0.02, 0.35]	0.262	100.00 / 100.00
H01 Shortest source distance	-0.041	-1.23 [-2.94, 0.48]	1.000	100.00 / 100.00
source edges departure (5,041 records)	-0.038	0.35 [-0.07, 0.77]	0.819	100.00 / 100.00
G10 Edge-slack distribution	-0.041	-0.48 [-1.74, 0.77]	1.000	93.33 / 100.00
incoming slack maximum departure (9,227 records)	0.012	-0.97 [-1.70, -0.23]	0.124	100.00 / 100.00
G12 Delay-support distribution	0.041	0.26 [-0.72, 1.24]	1.000	93.33 / 33.33
outgoing support minimum arrival (9,227 records)	0.036	0.52 [0.20, 0.85]	0.057	100.00 / 100.00
H04 Upstream reachability	0.039	-0.61 [-1.64, 0.43]	1.000	100.00 / 66.67
upstream departure (9,227 records)	0.032	0.25 [0.06, 0.44]	0.117	100.00 / 100.00
G10 Edge-slack distribution	-0.038	-0.42 [-1.68, 0.84]	1.000	93.33 / 100.00
incoming slack 90th percentile departure (9,227 records)	0.009	-0.99 [-1.74, -0.25]	0.119	100.00 / 100.00
G12 Delay-support distribution	0.038	0.21 [-0.81, 1.24]	1.000	93.33 / 100.00
incoming support minimum arrival (9,227 records)	0.006	0.37 [0.15, 0.60]	0.057	100.00 / 100.00
G17 Relation-type entropy	-0.038	-0.47 [-1.66, 0.71]	1.000	100.00 / 100.00
type entropy departure (9,227 records)	0.062	-0.10 [-0.36, 0.16]	1.000	100.00 / 100.00
H16 Local motifs	0.038	1.36 [-0.30, 3.02]	0.981	100.00 / 100.00
fan-in pairs arrival (9,227 records)	-0.057	0.12 [-0.05, 0.29]	1.000	100.00 / 100.00
H06 Upstream trains	0.037	-0.13 [-1.05, 0.79]	1.000	93.33 / 100.00
upstream arrival (9,227 records)	-0.011	0.37 [0.09, 0.64]	0.113	100.00 / 100.00
H15 Chain and crossing length	0.036	-0.20 [-1.06, 0.65]	1.000	100.00 / 100.00
longest crossings arrival (9,227 records)	-0.036	0.30 [0.07, 0.53]	0.123	100.00 / 100.00
G05 Event degree	0.035	1.26 [0.29, 2.24]	0.135	100.00 / 100.00
outgoing departure (9,227 records)	0.001	-0.35 [-0.59, -0.10]	0.092	100.00 / 100.00
G06 Cross-train degree	0.035	1.26 [0.29, 2.24]	0.135	100.00 / 100.00
outgoing cross departure (9,227 records)	0.001	-0.35 [-0.59, -0.10]	0.092	100.00 / 100.00
G10 Edge-slack distribution	0.035	1.26 [0.29, 2.24]	0.135	100.00 / 100.00
outgoing slack count departure (9,227 records)	0.001	-0.35 [-0.59, -0.10]	0.092	100.00 / 100.00
G12 Delay-support distribution	0.035	1.26 [0.29, 2.24]	0.135	100.00 / 100.00
outgoing support count departure (9,227 records)	0.001	-0.35 [-0.59, -0.10]	0.092	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution	-0.034	-1.08 [-2.66, 0.50]	1.000	100.00 / 66.67
incoming slack zero fraction arrival (9,227 records)	-0.036	0.06 [-0.16, 0.27]	1.000	100.00 / 66.67
H24 Projection core number	0.034	0.66 [-0.29, 1.61]	1.000	100.00 / 100.00
core arrival (9,227 records)	0.063	0.60 [0.24, 0.96]	0.057	100.00 / 100.00
B08 Community structure	-0.032	-0.78 [-1.44, -0.12]	0.229	100.00 / 100.00
cross community edges arrival (9,227 records)	0.066	0.26 [0.14, 0.38]	0.025	100.00 / 100.00
H23 Local clustering	0.031	1.22 [0.88, 1.55]	5.5e-7	100.00 / 100.00
undirected clustering arrival (9,227 records)	-0.033	-0.03 [-0.04, -0.01]	0.047	100.00 / 100.00
G10 Edge-slack distribution	-0.031	-0.51 [-1.72, 0.69]	1.000	100.00 / 66.67
incoming slack sum arrival (9,227 records)	0.041	0.11 [-0.01, 0.23]	0.562	100.00 / 100.00
G07 Degree by relation	0.031	0.79 [0.33, 1.24]	0.017	100.00 / 100.00
incoming exit headway arrival (9,227 records)	-0.006	-0.01 [-0.17, 0.15]	1.000	66.67 / 66.67
G06 Cross-train degree	0.030	0.77 [0.31, 1.24]	0.020	100.00 / 100.00
incoming cross departure (9,227 records)	-0.006	-0.01 [-0.17, 0.15]	1.000	66.67 / 66.67
G07 Degree by relation	0.030	0.77 [0.31, 1.24]	0.020	100.00 / 100.00
incoming entry headway departure (9,227 records)	-0.006	-0.01 [-0.17, 0.15]	1.000	66.67 / 66.67
G13 Upstream-neighbor delay	0.030	0.77 [0.31, 1.24]	0.020	100.00 / 100.00
neighbor delay count departure (9,227 records)	-0.006	-0.01 [-0.17, 0.15]	1.000	66.67 / 66.67
G14 Downstream-neighbor delay	0.030	0.02 [-0.86, 0.90]	1.000	66.67 / 33.33
neighbor delay sum arrival (9,227 records)	0.026	-0.33 [-0.66, -0.01]	0.379	100.00 / 100.00
G10 Edge-slack distribution	-0.030	0.05 [-1.26, 1.36]	1.000	53.33 / 33.33
incoming slack median departure (9,227 records)	-0.004	-1.12 [-1.96, -0.29]	0.109	100.00 / 100.00
G10 Edge-slack distribution	-0.029	-0.42 [-1.63, 0.79]	1.000	100.00 / 66.67
incoming slack mean arrival (9,227 records)	0.042	0.12 [0.00, 0.24]	0.486	100.00 / 100.00
H11 Harmonic closeness	0.029	0.08 [-1.22, 1.39]	1.000	80.00 / 33.33
incoming harmonic departure (9,227 records)	0.056	0.18 [0.00, 0.36]	0.461	100.00 / 100.00
G05 Event degree	0.029	0.73 [-0.35, 1.82]	1.000	100.00 / 100.00
incoming departure (9,227 records)	0.046	0.16 [0.05, 0.26]	0.091	100.00 / 100.00
G10 Edge-slack distribution	0.029	0.73 [-0.35, 1.82]	1.000	100.00 / 100.00
incoming slack count departure (9,227 records)	0.046	0.16 [0.05, 0.26]	0.091	100.00 / 100.00
G12 Delay-support distribution	0.029	0.73 [-0.35, 1.82]	1.000	100.00 / 100.00
incoming support count departure (9,227 records)	0.046	0.16 [0.05, 0.26]	0.091	100.00 / 100.00
H09 Reachable delay totals	0.028	-0.50 [-1.26, 0.26]	1.000	100.00 / 100.00
upstream train maximum delay sum arrival (9,227 records)	-0.004	0.35 [0.10, 0.61]	0.100	100.00 / 100.00
G10 Edge-slack distribution	-0.027	0.10 [-1.33, 1.53]	1.000	66.67 / 33.33
incoming slack mean departure (9,227 records)	-0.004	-1.24 [-2.12, -0.36]	0.094	100.00 / 100.00
H23 Local clustering	0.026	0.70 [-0.22, 1.61]	1.000	100.00 / 100.00
undirected clustering departure (9,227 records)	-0.024	-0.10 [-0.32, 0.11]	1.000	100.00 / 100.00

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G11 Near-tight links	0.026	0.23 [-1.19, 1.65]	1.000	93.33 / 100.00
slack at most 1 minute departure (9,227 records)	0.032	0.06 [-0.19, 0.31]	1.000	93.33 / 100.00
H08 Spatial reachability	0.026	0.42 [-0.73, 1.57]	1.000	100.00 / 100.00
upstream branches arrival (9,227 records)	-0.004	0.47 [0.15, 0.80]	0.084	100.00 / 100.00
G10 Edge-slack distribution	-0.026	0.03 [-1.28, 1.35]	1.000	46.67 / 33.33
incoming slack sum departure (9,227 records)	-0.002	-1.03 [-1.83, -0.24]	0.127	100.00 / 100.00
G19 Endpoint contrasts	0.025	0.63 [-0.54, 1.79]	1.000	100.00 / 100.00
G05 incoming degree endpoint contrast (9,227 records)	0.039	0.06 [-0.14, 0.26]	1.000	100.00 / 66.67
G16 Incident-link imbalance	0.024	1.03 [-0.03, 2.09]	0.556	100.00 / 100.00
degree balance departure (9,227 records)	-0.013	-0.37 [-0.61, -0.13]	0.065	100.00 / 100.00
G07 Degree by relation	-0.024	-0.60 [-1.76, 0.56]	1.000	100.00 / 100.00
incoming station track arrival (9,227 records)	-0.013	0.07 [-0.17, 0.30]	1.000	93.33 / 66.67
G10 Edge-slack distribution	-0.023	0.23 [-1.00, 1.47]	1.000	86.67 / 66.67
incoming slack standard deviation arrival (9,227 records)	0.032	0.07 [-0.08, 0.22]	1.000	100.00 / 100.00
B08 Community structure	0.023	0.56 [-0.46, 1.59]	1.000	100.00 / 100.00
cross community edges departure (9,227 records)	0.039	0.03 [-0.21, 0.26]	1.000	93.33 / 66.67
G10 Edge-slack distribution	-0.023	0.03 [-1.26, 1.32]	1.000	53.33 / 33.33
incoming slack 90th percentile arrival (9,227 records)	0.035	0.07 [-0.07, 0.22]	1.000	100.00 / 100.00
G14 Downstream-neighbor delay	0.022	-0.10 [-1.16, 0.96]	1.000	80.00 / 33.33
neighbor delay maximum arrival (6,812 records)	0.027	-0.26 [-0.60, 0.07]	1.000	100.00 / 100.00
G14 Downstream-neighbor delay	0.022	-0.10 [-1.16, 0.96]	1.000	80.00 / 33.33
neighbor delay mean arrival (6,812 records)	0.027	-0.26 [-0.60, 0.07]	1.000	100.00 / 100.00
G14 Downstream-neighbor delay	0.022	-0.10 [-1.16, 0.96]	1.000	80.00 / 33.33
neighbor delay median arrival (6,812 records)	0.027	-0.26 [-0.60, 0.07]	1.000	100.00 / 100.00
G14 Downstream-neighbor delay	0.022	-0.10 [-1.16, 0.96]	1.000	80.00 / 33.33
neighbor delay minimum arrival (6,812 records)	0.027	-0.26 [-0.60, 0.07]	1.000	100.00 / 100.00
G14 Downstream-neighbor delay	0.022	-0.10 [-1.16, 0.96]	1.000	80.00 / 33.33
neighbor delay 90th percentile arrival (6,812 records)	0.027	-0.26 [-0.60, 0.07]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.022	0.26 [-1.01, 1.54]	1.000	86.67 / 66.67
incoming slack interquartile range arrival (9,227 records)	0.031	0.05 [-0.11, 0.20]	1.000	100.00 / 66.67
G05 Event degree	-0.021	-0.12 [-1.19, 0.94]	1.000	80.00 / 66.67
incoming arrival (9,227 records)	-0.013	0.04 [-0.17, 0.26]	1.000	93.33 / 66.67
G06 Cross-train degree	-0.021	-0.12 [-1.19, 0.94]	1.000	80.00 / 66.67
incoming cross arrival (9,227 records)	-0.013	0.04 [-0.17, 0.26]	1.000	93.33 / 66.67
G10 Edge-slack distribution	-0.021	-0.12 [-1.19, 0.94]	1.000	80.00 / 66.67
incoming slack count arrival (9,227 records)	-0.013	0.04 [-0.17, 0.26]	1.000	93.33 / 66.67
G12 Delay-support distribution	-0.021	-0.12 [-1.19, 0.94]	1.000	80.00 / 66.67
incoming support count arrival (9,227 records)	-0.013	0.04 [-0.17, 0.26]	1.000	93.33 / 66.67

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Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B09 Boundary pressure imbalance	0.019	0.38 [-1.35, 2.10]	1.000	100.00 / 66.67
cross boundary pressure balance 10 min (9,227 records)	-0.037	-1.39 [-2.33, -0.46]	0.074	100.00 / 100.00
G10 Edge-slack distribution	-0.018	0.31 [-1.00, 1.61]	1.000	100.00 / 66.67
incoming slack maximum arrival (9,227 records)	0.033	0.04 [-0.11, 0.20]	1.000	100.00 / 66.67
B04 Boundary connections	-0.016	0.03 [-1.52, 1.58]	1.000	66.67 / 66.67
entering 5 min (9,227 records)	0.094	0.04 [-0.23, 0.31]	1.000	93.33 / 66.67
G17 Relation-type entropy	-0.016	-0.70 [-1.85, 0.44]	1.000	100.00 / 100.00
type entropy arrival (9,227 records)	-0.017	0.15 [-0.08, 0.37]	1.000	100.00 / 100.00
H03 Event time from release	-0.015	0.92 [0.04, 1.81]	0.402	100.00 / 100.00
time from release departure (9,227 records)	0.000	0.07 [-0.06, 0.20]	1.000	100.00 / 100.00
H11 Harmonic closeness	0.014	-0.23 [-1.46, 1.00]	1.000	100.00 / 66.67
incoming harmonic arrival (9,227 records)	0.003	0.49 [0.14, 0.83]	0.092	100.00 / 100.00
G16 Incident-link imbalance	0.013	0.19 [-0.92, 1.30]	1.000	86.67 / 66.67
degree balance arrival (9,227 records)	0.017	-0.23 [-0.44, -0.02]	0.325	100.00 / 100.00
B02 Local edges and density	-0.012	-0.69 [-1.55, 0.18]	1.000	100.00 / 100.00
edges 5 min (9,227 records)	0.055	0.06 [-0.10, 0.23]	1.000	100.00 / 100.00
H08 Spatial reachability	0.012	-0.29 [-1.60, 1.02]	1.000	100.00 / 66.67
upstream stations departure (9,227 records)	0.053	0.17 [-0.02, 0.37]	0.651	100.00 / 100.00
G07 Degree by relation	0.012	0.30 [-0.83, 1.43]	1.000	93.33 / 66.67
incoming emergency departure (9,227 records)	0.058	0.19 [0.10, 0.28]	0.025	100.00 / 100.00
G18 External-source share	0.011	0.22 [-0.86, 1.30]	1.000	93.33 / 66.67
source fraction departure (9,227 records)	0.058	0.19 [0.10, 0.28]	0.025	100.00 / 100.00
H04 Upstream reachability	0.008	-1.14 [-2.05, -0.22]	0.173	100.00 / 100.00
upstream arrival (9,227 records)	0.010	0.40 [0.11, 0.68]	0.094	100.00 / 100.00
H09 Reachable delay totals	0.008	-1.32 [-2.10, -0.53]	0.020	100.00 / 100.00
upstream event delay arrival (9,227 records)	0.010	0.37 [0.11, 0.62]	0.084	100.00 / 100.00
G19 Endpoint contrasts	-0.007	-0.88 [-3.94, 2.19]	1.000	100.00 / 66.67
G08 active incoming degree endpoint contrast (9,227 records)	0.098	-0.51 [-1.03, 0.02]	0.504	100.00 / 100.00
B01 Local region size	0.007	0.18 [-0.92, 1.28]	1.000	93.33 / 66.67
region events 5 min (9,227 records)	0.053	-0.03 [-0.25, 0.18]	1.000	93.33 / 66.67
B05 Regional delay distribution	0.007	0.18 [-0.92, 1.28]	1.000	93.33 / 66.67
region delay 5 min count (9,227 records)	0.053	-0.03 [-0.25, 0.18]	1.000	93.33 / 66.67
G05 Event degree	0.006	0.16 [-0.97, 1.30]	1.000	80.00 / 33.33
outgoing arrival (9,227 records)	0.014	-0.28 [-0.51, -0.04]	0.229	100.00 / 100.00
G06 Cross-train degree	0.006	0.16 [-0.97, 1.30]	1.000	80.00 / 33.33
outgoing cross arrival (9,227 records)	0.014	-0.28 [-0.51, -0.04]	0.229	100.00 / 100.00
G07 Degree by relation	0.006	0.16 [-0.97, 1.30]	1.000	80.00 / 33.33
outgoing exit headway arrival (9,227 records)	0.014	-0.28 [-0.51, -0.04]	0.229	100.00 / 100.00
G10 Edge-slack distribution	0.006	0.16 [-0.97, 1.30]	1.000	80.00 / 33.33
outgoing slack count arrival (9,227 records)	0.014	-0.28 [-0.51, -0.04]	0.229	100.00 / 100.00
G12 Delay-support distribution	0.006	0.16 [-0.97, 1.30]	1.000	80.00 / 33.33
outgoing support count arrival (9,227 records)	0.014	-0.28 [-0.51, -0.04]	0.229	100.00 / 100.00

Continued on the next page.

Table S8 continued: Strategy-3 dwell durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G14 Downstream-neighbor delay	0.006	0.16 [-0.97, 1.30]	1.000	80.00 / 33.33
neighbor delay count arrival (9,227 records)	0.014	-0.28 [-0.51, -0.04]	0.229	100.00 / 100.00
H02 Longest active path	0.005	-0.86 [-1.97, 0.25]	1.000	100.00 / 100.00
longest path arrival (9,227 records)	0.013	0.49 [0.16, 0.82]	0.074	100.00 / 100.00
H15 Chain and crossing length	0.005	-0.86 [-1.97, 0.25]	1.000	100.00 / 100.00
longest chain arrival (9,227 records)	0.013	0.49 [0.16, 0.82]	0.074	100.00 / 100.00
G10 Edge-slack distribution	0.004	-1.55 [-2.08, -1.03]	1.8e-5	100.00 / 100.00
outgoing slack interquartile range arrival (9,227 records)	-0.024	-0.07 [-0.28, 0.14]	1.000	100.00 / 100.00
G10 Edge-slack distribution	0.004	-1.55 [-2.08, -1.03]	1.8e-5	100.00 / 100.00
outgoing slack standard deviation arrival (9,227 records)	-0.024	-0.07 [-0.28, 0.14]	1.000	100.00 / 100.00
H08 Spatial reachability	0.004	-0.59 [-1.81, 0.63]	1.000	100.00 / 100.00
upstream stations arrival (9,227 records)	0.011	0.50 [0.17, 0.84]	0.071	100.00 / 100.00
G10 Edge-slack distribution	0.000	-0.72 [-0.98, -0.46]	5.1e-5	100.00 / 100.00
outgoing slack minimum departure (9,227 records)	-0.013	0.08 [0.02, 0.13]	0.078	100.00 / 100.00
G07 Degree by relation	0.000	0.00 [-1.09, 1.10]	1.000	60.00 / 33.33
outgoing station track departure (9,227 records)	0.025	-0.23 [-0.51, 0.04]	0.778	100.00 / 100.00

Notes: Codes and definitions follow Table S1. Rank corr.: scenario-weighted Spearman correlation. Upper/lower lines: change probability/signed duration change. Effects per indicator standard deviation are in percentage points/minutes, respectively. Brackets contain 95% confidence intervals. Adjusted *p*: Benjamini–Yekutieli p-value; 1.1e-7 denotes 1.1×10^{-7}. Direction stable: unchanged coefficient sign after omitting one start-time group/one blockage location (percent). Pair differences are the following train value minus the leading train value. Record counts include only finite values of the indicated form.

S3.6 Strategy-3 running durations

Table S9: Complete indicator associations: Strategy-3 running durations.

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
Disruption conditions				
E05 Emergency stopping position	-0.302	-8.35 [-13.23, -3.46]	0.025	100.00 / 100.00
first stop index (4,061 records)	0.221	0.20 [0.08, 0.33]	0.036	100.00 / 100.00
E20 Planned starts before release	0.286	9.03 [5.64, 12.43]	3.1e-4	100.00 / 100.00
planned starts before reopen (13,490 records)	-0.245	-0.80 [-1.29, -0.31]	0.038	100.00 / 100.00
E12 Origin waiting count	0.245	6.78 [3.85, 9.72]	0.002	100.00 / 100.00
origin waiting (13,490 records)	-0.188	-0.21 [-0.58, 0.16]	1.000	100.00 / 66.67
E21 Train-state counts at blockage start	0.245	6.78 [3.85, 9.72]	0.002	100.00 / 100.00
origin hold (13,490 records)	-0.188	-0.21 [-0.58, 0.16]	1.000	100.00 / 66.67
E02 Executed route fraction	-0.234	-6.77 [-9.11, -4.43]	1.3e-4	100.00 / 100.00
executed fraction (13,490 records)	0.203	0.49 [0.26, 0.73]	0.007	100.00 / 100.00
E03 Current route position	-0.234	-6.77 [-9.11, -4.43]	1.3e-4	100.00 / 100.00
current route index (13,490 records)	0.203	0.49 [0.26, 0.73]	0.007	100.00 / 100.00
E01 Train state at start time	0.229	7.27 [3.99, 10.54]	0.003	100.00 / 100.00
state 0 (13,490 records)	-0.202	-0.29 [-0.67, 0.09]	1.000	100.00 / 100.00
E16 Emergency backlog	0.228	6.18 [3.70, 8.66]	7.4e-4	100.00 / 100.00
backlog (13,490 records)	-0.165	-0.01 [-0.29, 0.26]	1.000	46.67 / 66.67
E07 Emergency arrival time	-0.216	-6.99 [-11.57, -2.40]	0.057	100.00 / 100.00
emergency arrival (4,061 records)	0.148	0.19 [0.02, 0.35]	0.289	100.00 / 100.00
X06 Planned entry from start time	0.204	5.94 [0.77, 11.11]	0.274	100.00 / 100.00
planned entry from start time (7,340 records)	-0.202	-0.31 [-0.51, -0.12]	0.041	100.00 / 100.00
X07 Planned exit from start time	0.194	5.67 [0.57, 10.78]	0.313	100.00 / 100.00
planned exit from start time (7,340 records)	-0.193	-0.31 [-0.50, -0.11]	0.045	100.00 / 100.00
D01 Blockage start time	-0.188	-6.10 [-9.18, -3.02]	0.008	100.00 / 100.00
start (13,490 records)	0.169	0.23 [-0.27, 0.74]	1.000	100.00 / 66.67
E19 Committed event count	-0.188	-6.06 [-9.10, -3.03]	0.008	100.00 / 100.00
fixed events (13,490 records)	0.169	0.23 [-0.28, 0.73]	1.000	100.00 / 66.67
E17 Early release concentration	-0.185	-5.36 [-8.29, -2.44]	0.016	100.00 / 100.00
release first5 fraction (13,490 records)	0.152	0.66 [0.17, 1.14]	0.112	100.00 / 100.00
X18 Emergency origin hold	0.182	5.78 [2.81, 8.75]	0.009	100.00 / 100.00
origin emergency (13,490 records)	-0.091	0.31 [0.09, 0.53]	0.086	100.00 / 100.00
E21 Train-state counts at blockage start	-0.182	-5.70 [-8.56, -2.84]	0.008	100.00 / 100.00
completed (13,490 records)	0.154	0.14 [-0.38, 0.66]	1.000	86.67 / 66.67
E06 Time to emergency station	-0.181	-4.68 [-8.71, -0.66]	0.259	100.00 / 100.00
time to stop (4,061 records)	0.165	0.14 [0.04, 0.24]	0.090	100.00 / 100.00
D03 Blockage duration	0.179	5.55 [2.97, 8.14]	0.004	100.00 / 100.00
duration (13,490 records)	-0.159	-1.06 [-1.51, -0.62]	0.002	100.00 / 100.00
D08 Remaining blockage	0.179	5.55 [2.97, 8.14]	0.004	100.00 / 100.00
remaining (13,490 records)	-0.159	-1.06 [-1.51, -0.62]	0.002	100.00 / 100.00
D20 Lost resource time	0.179	5.55 [2.97, 8.14]	0.004	100.00 / 100.00
lost resource minutes (13,490 records)	-0.159	-1.06 [-1.51, -0.62]	0.002	100.00 / 100.00
I05 Predicted remaining duration	0.179	5.55 [2.97, 8.14]	0.004	100.00 / 100.00
predicted remaining (13,490 records)	-0.159	-1.06 [-1.51, -0.62]	0.002	100.00 / 100.00
I06 Remaining-duration bounds	0.179	5.55 [2.97, 8.14]	0.004	100.00 / 100.00
lower (13,490 records)	-0.159	-1.06 [-1.51, -0.62]	0.002	100.00 / 100.00
I06 Remaining-duration bounds	0.179	5.55 [2.97, 8.14]	0.004	100.00 / 100.00
upper (13,490 records)	-0.159	-1.06 [-1.51, -0.62]	0.002	100.00 / 100.00
I09 Remaining-duration quantiles	0.179	5.55 [2.97, 8.14]	0.004	100.00 / 100.00
10th percentile (13,490 records)	-0.159	-1.06 [-1.51, -0.62]	0.002	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
I09 Remaining-duration quantiles	0.179	5.55 [2.97, 8.14]	0.004	100.00 / 100.00
median (13,490 records)	-0.159	-1.06 [-1.51, -0.62]	0.002	100.00 / 100.00
I09 Remaining-duration quantiles	0.179	5.55 [2.97, 8.14]	0.004	100.00 / 100.00
90th percentile (13,490 records)	-0.159	-1.06 [-1.51, -0.62]	0.002	100.00 / 100.00
E01 Train state at start time	-0.178	-5.65 [-8.55, -2.75]	0.009	100.00 / 100.00
state 2 (13,490 records)	0.156	0.07 [-0.35, 0.50]	1.000	86.67 / 33.33
E04 Committed running remainder	-0.177	-4.73 [-7.25, -2.20]	0.013	100.00 / 100.00
committed remaining (13,490 records)	0.155	0.11 [-0.19, 0.42]	1.000	100.00 / 66.67
D09 Time from start time	0.170	5.35 [1.38, 9.32]	0.120	100.00 / 100.00
reference from start (13,490 records)	-0.144	0.73 [0.20, 1.26]	0.099	100.00 / 100.00
D09 Time from start time	0.170	4.62 [0.51, 8.73]	0.298	100.00 / 100.00
planned from start (13,490 records)	-0.181	0.23 [-0.16, 0.62]	1.000	100.00 / 66.67
E26 Committed movement count	-0.161	-4.95 [-7.96, -1.95]	0.033	100.00 / 100.00
committed runs (13,490 records)	0.149	0.35 [-0.13, 0.83]	1.000	100.00 / 100.00
X17 Remaining route fraction	0.148	4.46 [3.10, 5.81]	1.3e-5	100.00 / 100.00
remaining route fraction (13,490 records)	-0.100	-1.15 [-1.67, -0.63]	0.004	100.00 / 100.00
X02 Already past blocked section	-0.122	-3.81 [-5.88, -1.73]	0.016	100.00 / 100.00
passed at start time (7,340 records)	0.084	0.11 [0.05, 0.18]	0.022	100.00 / 100.00
X03 Planned clearance before start time	-0.122	-3.81 [-5.88, -1.73]	0.016	100.00 / 100.00
planned clear before (7,340 records)	0.084	0.11 [0.05, 0.18]	0.022	100.00 / 100.00
X12 Can clear before start time	-0.122	-3.81 [-5.88, -1.73]	0.016	100.00 / 100.00
can clear without emergency (7,340 records)	0.084	0.11 [0.05, 0.18]	0.022	100.00 / 100.00
D04 Start-time wave phase	-0.111	-3.70 [-6.32, -1.07]	0.094	100.00 / 100.00
start phase40 (13,490 records)	0.122	0.33 [-0.04, 0.71]	0.735	100.00 / 100.00
E01 Train state at start time	-0.103	-3.27 [-4.58, -1.96]	7.1e-4	100.00 / 100.00
state 1 (13,490 records)	0.093	0.38 [0.23, 0.53]	0.002	100.00 / 100.00
E09 Already stopped at start time	-0.103	-3.27 [-4.58, -1.96]	7.1e-4	100.00 / 100.00
already stopped (13,490 records)	0.093	0.38 [0.23, 0.53]	0.002	100.00 / 100.00
X08 Sections to blockage entry	0.102	3.14 [-2.00, 8.27]	1.000	100.00 / 100.00
intervals to entry (7,340 records)	-0.075	-0.18 [-0.34, -0.02]	0.270	100.00 / 100.00
E21 Train-state counts at blockage start	0.095	3.16 [0.94, 5.38]	0.088	100.00 / 100.00
finish committed run then hold (13,490 records)	-0.040	0.35 [0.08, 0.62]	0.133	100.00 / 66.67
E08 Minimum emergency hold	0.069	2.52 [-0.42, 5.47]	0.800	100.00 / 100.00
minimum emergency hold (13,490 records)	0.005	0.53 [0.25, 0.81]	0.015	100.00 / 100.00
X10 Signed blockage distance	0.068	1.77 [-1.69, 5.24]	1.000	100.00 / 100.00
signed distance to blockage (7,340 records)	-0.014	-0.16 [-0.31, -0.02]	0.284	100.00 / 100.00
E13 Station waiting count	0.066	2.85 [0.55, 5.15]	0.188	100.00 / 100.00
station waiting (13,490 records)	-0.015	0.37 [0.09, 0.65]	0.120	100.00 / 66.67
D10 Time from reopening	-0.055	-1.58 [-3.46, 0.31]	0.855	100.00 / 100.00
planned from release (13,490 records)	0.031	0.98 [0.48, 1.49]	0.012	100.00 / 100.00
X13 Arrival before reopening	0.050	1.59 [-1.03, 4.22]	1.000	100.00 / 100.00
reaches stop before reopen (13,490 records)	0.029	0.65 [0.35, 0.96]	0.005	100.00 / 100.00
X05 Planned overlap fraction	0.050	1.55 [-1.11, 4.21]	1.000	100.00 / 100.00
planned overlap fraction (13,490 records)	0.027	0.65 [0.35, 0.95]	0.005	100.00 / 100.00
X09 Minimum time to entry	0.039	0.75 [-3.99, 5.49]	1.000	100.00 / 66.67
minrun to entry (7,340 records)	-0.025	-0.11 [-0.25, 0.03]	1.000	100.00 / 100.00
X16 Blockage-only waiting	0.034	-0.36 [-2.27, 1.55]	1.000	93.33 / 66.67
blockage only wait (13,490 records)	0.034	0.55 [0.31, 0.79]	0.003	100.00 / 100.00
D10 Time from reopening	0.033	1.13 [-1.94, 4.21]	1.000	100.00 / 66.67
reference from release (13,490 records)	-0.045	1.80 [0.92, 2.68]	0.008	100.00 / 100.00

Continued on the next page.

Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
D02 Blockage end	0.031	0.29 [-1.94, 2.52]	1.000	80.00 / 66.67
end (13,490 records)	-0.031	-0.66 [-1.17, -0.15]	0.139	100.00 / 100.00
D11 Blocked section	-0.031	-0.98 [-3.29, 1.33]	1.000	93.33 / 100.00
U11 U12 (13,490 records)	0.005	-0.13 [-0.54, 0.28]	1.000	100.00 / 50.00
D11 Blocked section	0.031	0.97 [-1.28, 3.23]	1.000	100.00 / 100.00
U13 H1 (13,490 records)	-0.058	-0.58 [-1.06, -0.10]	0.207	100.00 / 100.00
X14 Earliest-entry estimate	-0.016	-0.24 [-4.22, 3.74]	1.000	73.33 / 66.67
earliest entry no other conflicts (7,340 records)	-0.018	-0.08 [-0.19, 0.03]	1.000	100.00 / 100.00
X19 Emergency restriction type	0.015	-1.22 [-2.80, 0.36]	1.000	100.00 / 100.00
direct stop type (13,490 records)	0.053	0.65 [0.37, 0.93]	0.003	100.00 / 100.00
X04 Planned blockage overlap	0.013	0.01 [-2.14, 2.16]	1.000	53.33 / 33.33
planned blockage overlap (13,490 records)	0.050	0.66 [0.38, 0.95]	0.003	100.00 / 100.00
E21 Train-state counts at blockage start	0.010	0.32 [-2.03, 2.67]	1.000	86.67 / 66.67
hold current station (13,490 records)	0.014	0.27 [-0.10, 0.64]	1.000	100.00 / 100.00
X01 Blocked-section exposure	-0.010	-0.32 [-2.99, 2.35]	1.000	93.33 / 66.67
crosses closed (13,490 records)	0.065	1.05 [0.65, 1.46]	0.001	100.00 / 100.00
D05 Reopening wave phase	-0.010	-0.34 [-2.40, 1.71]	1.000	93.33 / 33.33
end phase40 (13,490 records)	0.014	0.04 [-0.29, 0.37]	1.000	66.67 / 66.67
X15 Earliest entry from release	0.009	0.43 [-1.73, 2.60]	1.000	80.00 / 100.00
earliest entry minus release (7,340 records)	-0.012	-0.07 [-0.15, 0.00]	0.577	100.00 / 100.00
X11 At blockage entrance	-0.007	-0.23 [-1.23, 0.78]	1.000	100.00 / 66.67
at closed entry (13,490 records)	0.019	0.30 [0.17, 0.43]	0.004	100.00 / 100.00
D11 Blocked section	0.000	0.01 [-2.36, 2.38]	1.000	46.67 / 50.00
H1 C (13,490 records)	0.052	0.70 [0.37, 1.03]	0.007	100.00 / 100.00
Timetable characteristics				
M05 Reference adjustment room	0.925	12.77 [10.40, 15.15]	7.8e-10	100.00 / 100.00
compressible (13,490 records)	-0.926	-8.84 [-8.85, -8.84]	1.0e-91	100.00 / 100.00
M06 Remaining candidate decisions	0.264	7.91 [6.04, 9.77]	1.7e-7	100.00 / 100.00
remaining stopping decisions (13,490 records)	-0.209	-0.30 [-0.65, 0.06]	0.836	100.00 / 100.00
F13 Minimum running time	-0.249	-7.91 [-10.16, -5.65]	4.3e-6	100.00 / 100.00
minimum run (13,490 records)	0.204	0.02 [-0.32, 0.36]	1.000	66.67 / 33.33
M04 Propagated domain	-0.249	-7.91 [-10.16, -5.65]	4.3e-6	100.00 / 100.00
domain lower (13,490 records)	0.204	0.02 [-0.32, 0.36]	1.000	66.67 / 33.33
T04 Planned running duration	-0.249	-7.91 [-10.16, -5.65]	4.3e-6	100.00 / 100.00
planned run next (13,490 records)	0.204	0.02 [-0.32, 0.36]	1.000	66.67 / 33.33
T15 Fast-train indicator	0.249	7.91 [5.65, 10.16]	4.3e-6	100.00 / 100.00
fast (13,490 records)	-0.204	-0.02 [-0.36, 0.32]	1.000	66.67 / 33.33
T17 Total planned journey time	-0.249	-7.91 [-10.16, -5.65]	4.3e-6	100.00 / 100.00
planned total (13,490 records)	0.204	0.02 [-0.32, 0.36]	1.000	66.67 / 33.33
M06 Remaining candidate decisions	0.216	6.57 [4.52, 8.63]	2.2e-5	100.00 / 100.00
remaining order decisions (13,490 records)	-0.162	-0.96 [-1.48, -0.45]	0.015	100.00 / 100.00
T21 Nearby scheduled-stop distance	0.205	6.49 [4.90, 8.08]	3.0e-7	100.00 / 100.00
next scheduled-stop distance (13,490 records)	-0.159	-0.45 [-0.80, -0.10]	0.147	100.00 / 100.00
T21 Nearby scheduled-stop distance	0.188	5.81 [4.50, 7.12]	6.6e-8	100.00 / 100.00
previous scheduled-stop distance (13,490 records)	-0.160	-0.52 [-0.85, -0.20]	0.041	100.00 / 100.00
T12 Planned local flow	-0.157	-4.65 [-7.61, -1.69]	0.047	100.00 / 100.00
planned flow 5 min (13,490 records)	0.114	-0.42 [-0.78, -0.05]	0.270	100.00 / 100.00

Continued on the next page.

Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
M06 Remaining candidate decisions	0.152	4.94 [3.55, 6.32]	4.0e-6	100.00 / 100.00
remaining track-assignment decisions (13,490 records)	-0.105	-1.25 [-1.81, -0.69]	0.004	100.00 / 100.00
M06 Remaining candidate decisions	0.148	4.65 [3.29, 6.01]	7.0e-6	100.00 / 100.00
remaining duration decisions (13,490 records)	-0.100	-1.20 [-1.73, -0.66]	0.004	100.00 / 100.00
T22 Route extent	0.148	4.46 [3.10, 5.81]	1.3e-5	100.00 / 100.00
remaining edges (13,490 records)	-0.100	-1.15 [-1.67, -0.63]	0.004	100.00 / 100.00
T06 Planned track label	-0.137	-3.98 [-6.25, -1.72]	0.020	100.00 / 100.00
planned track index (12,290 records)	0.104	0.31 [0.04, 0.58]	0.249	100.00 / 100.00
T12 Planned local flow	-0.135	-3.61 [-6.29, -0.94]	0.118	100.00 / 100.00
planned flow 10 min (13,490 records)	0.115	-0.21 [-0.48, 0.05]	0.936	100.00 / 100.00
T10 Planned exit buffer	-0.134	-3.98 [-5.58, -2.37]	8.3e-4	100.00 / 100.00
previous buffer (12,890 records)	0.110	-0.05 [-0.32, 0.21]	1.000	100.00 / 100.00
T13 Planned train-type pattern	-0.129	-3.12 [-5.50, -0.73]	0.143	100.00 / 100.00
alternation 20 min (3,252 records)	0.035	-0.23 [-0.64, 0.18]	1.000	100.00 / 100.00
T12 Planned local flow	-0.119	-3.23 [-5.80, -0.66]	0.177	100.00 / 100.00
planned flow 20 min (13,490 records)	0.106	0.01 [-0.19, 0.21]	1.000	66.67 / 66.67
F14 Allowed stopping mode	0.103	3.26 [1.66, 4.86]	0.006	100.00 / 100.00
optional (13,490 records)	-0.092	0.40 [0.17, 0.64]	0.028	100.00 / 100.00
T14 Departure wave	0.093	3.09 [0.09, 6.09]	0.426	100.00 / 100.00
origin wave40 (13,490 records)	-0.100	-0.08 [-0.45, 0.29]	1.000	93.33 / 100.00
T18 Remaining planned journey	0.091	2.22 [0.80, 3.65]	0.048	100.00 / 100.00
remaining planned next endpoint (13,490 records)	-0.056	-1.13 [-1.71, -0.56]	0.011	100.00 / 100.00
M05 Reference adjustment room	-0.090	-1.75 [-4.45, 0.96]	1.000	100.00 / 100.00
extendable (13,490 records)	0.077	1.29 [0.29, 2.29]	0.145	100.00 / 100.00
T09 Planned entry buffer	-0.085	-2.92 [-4.34, -1.49]	0.006	100.00 / 100.00
previous buffer (12,890 records)	0.073	-0.04 [-0.35, 0.27]	1.000	100.00 / 66.67
T13 Planned train-type pattern	-0.077	-2.07 [-3.96, -0.17]	0.335	100.00 / 100.00
fast fraction 20 min (8,230 records)	0.033	0.46 [-0.73, 1.64]	1.000	93.33 / 66.67
F12 Minimum dwell	-0.074	-2.34 [-3.67, -1.02]	0.020	100.00 / 100.00
minimum dwell (13,490 records)	0.059	-0.47 [-0.74, -0.20]	0.025	100.00 / 100.00
F14 Allowed stopping mode	-0.074	-2.34 [-3.67, -1.02]	0.020	100.00 / 100.00
required (13,490 records)	0.059	-0.47 [-0.74, -0.20]	0.025	100.00 / 100.00
T03 Planned dwell	-0.074	-2.34 [-3.67, -1.02]	0.020	100.00 / 100.00
planned dwell (13,490 records)	0.059	-0.47 [-0.74, -0.20]	0.025	100.00 / 100.00
T05 Planned stopping mode	-0.074	-2.34 [-3.67, -1.02]	0.020	100.00 / 100.00
planned stop (13,490 records)	0.059	-0.47 [-0.74, -0.20]	0.025	100.00 / 100.00
M03 Original domain width	0.070	2.26 [-0.27, 4.79]	0.709	100.00 / 100.00
raw domain width (13,490 records)	-0.029	-0.47 [-0.83, -0.11]	0.132	100.00 / 100.00
T13 Planned train-type pattern	-0.066	-1.85 [-3.63, -0.07]	0.417	100.00 / 100.00
fast fraction 10 min (7,036 records)	0.020	0.71 [-0.35, 1.78]	1.000	100.00 / 66.67
T02 Planned departure time	-0.063	-2.03 [-4.63, 0.56]	1.000	100.00 / 100.00
planned departure (13,490 records)	0.023	0.48 [0.11, 0.84]	0.128	100.00 / 100.00
T01 Planned arrival time	-0.061	-1.98 [-4.60, 0.64]	1.000	100.00 / 100.00
planned arrival (13,490 records)	0.021	0.49 [0.13, 0.86]	0.113	100.00 / 100.00
T18 Remaining planned journey	0.059	1.44 [-0.10, 2.98]	0.602	100.00 / 100.00
remaining planned (13,490 records)	-0.030	-1.08 [-1.66, -0.50]	0.016	100.00 / 100.00
T19 Remaining minimum journey	0.059	1.44 [-0.10, 2.98]	0.602	100.00 / 100.00
remaining minimum (13,490 records)	-0.030	-1.08 [-1.66, -0.50]	0.016	100.00 / 100.00
T13 Planned train-type pattern	-0.053	-1.44 [-3.21, 0.32]	0.912	100.00 / 100.00
fast fraction 5 min (6,221 records)	0.013	0.81 [-0.29, 1.91]	1.000	100.00 / 66.67
T24 Planned terminal position	-0.044	-1.08 [-2.92, 0.76]	1.000	100.00 / 100.00
terminal from release (13,490 records)	0.027	0.62 [0.27, 0.96]	0.023	100.00 / 100.00

Continued on the next page.

Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
T11 Planned track buffer	0.031	1.47 [-0.42, 3.36]	1.000	100.00 / 100.00
planned track buffer (10,690 records)	-0.027	-0.74 [-1.11, -0.38]	0.010	100.00 / 100.00
T20 Remaining scheduled stops	0.024	0.52 [-1.27, 2.30]	1.000	93.33 / 100.00
remaining scheduled stops (13,490 records)	-0.003	-1.00 [-1.59, -0.41]	0.028	100.00 / 100.00
T24 Planned terminal position	-0.014	-1.32 [-3.72, 1.09]	1.000	100.00 / 100.00
planned terminal (13,490 records)	-0.009	-0.08 [-0.50, 0.35]	1.000	93.33 / 66.67
T13 Planned train-type pattern	0.013	0.36 [-0.94, 1.65]	1.000	93.33 / 66.67
alternation 10 min (1,454 records)	-0.022	0.05 [-0.09, 0.19]	1.000	93.33 / 100.00
M04 Propagated domain	-0.005	0.52 [-1.87, 2.90]	1.000	93.33 / 100.00
domain upper (13,490 records)	-0.017	-0.40 [-0.66, -0.13]	0.067	100.00 / 100.00
M04 Propagated domain	-0.002	0.68 [-1.66, 3.03]	1.000	93.33 / 100.00
domain width (13,490 records)	-0.019	-0.39 [-0.66, -0.12]	0.072	100.00 / 100.00
Train interactions				
H22 Shared downstream support	-0.212	-8.62 [-11.36, -5.88]	2.8e-5	100.00 / 100.00
arrival departure downstream Jaccard (7,772 records)	0.232	0.37 [0.24, 0.49]	4.5e-4	100.00 / 100.00
H17 Window competition size	0.185	7.39 [3.21, 11.57]	0.020	100.00 / 100.00
competitor trains 20 min (13,490 records)	-0.158	0.56 [0.18, 0.94]	0.072	100.00 / 100.00
H18 Co-window occupations	0.185	7.39 [3.21, 11.57]	0.020	100.00 / 100.00
competition set memberships 20 min (13,490 records)	-0.158	0.56 [0.18, 0.94]	0.072	100.00 / 100.00
H17 Window competition size	0.161	7.01 [2.85, 11.17]	0.028	100.00 / 100.00
competitor trains 10 min (13,490 records)	-0.150	0.55 [0.16, 0.93]	0.085	100.00 / 100.00
H18 Co-window occupations	0.161	7.01 [2.85, 11.17]	0.028	100.00 / 100.00
competition set memberships 10 min (13,490 records)	-0.150	0.55 [0.16, 0.93]	0.085	100.00 / 100.00
E15 Queue fast-train share	0.158	5.14 [2.24, 8.04]	0.020	100.00 / 100.00
queue fast fraction (4,061 records)	-0.004	-0.05 [-0.14, 0.04]	1.000	100.00 / 100.00
H17 Window competition size	0.149	5.54 [1.50, 9.59]	0.110	100.00 / 100.00
competitor trains 5 min (13,490 records)	-0.145	0.51 [0.15, 0.86]	0.086	100.00 / 100.00
H18 Co-window occupations	0.149	5.54 [1.50, 9.59]	0.110	100.00 / 100.00
competition set memberships 5 min (13,490 records)	-0.145	0.51 [0.15, 0.86]	0.086	100.00 / 100.00
H21 Shared-visit coupling	0.135	3.98 [1.07, 6.89]	0.110	100.00 / 100.00
same visit classes (13,490 records)	-0.124	0.47 [0.10, 0.85]	0.163	100.00 / 100.00
E14 Reference release rank	0.103	3.81 [-1.53, 9.14]	1.000	100.00 / 100.00
release queue rank (4,061 records)	-0.178	-0.21 [-0.39, -0.03]	0.269	100.00 / 100.00
E11 Co-located emergency trains	0.066	2.92 [-0.66, 6.50]	0.916	100.00 / 100.00
same stop count (13,490 records)	0.008	0.51 [0.22, 0.80]	0.025	100.00 / 100.00
H20 Support distance	-0.033	-1.30 [-2.94, 0.35]	0.981	100.00 / 100.00
nearest optional stop index (5,871 records)	0.026	0.66 [0.28, 1.05]	0.037	100.00 / 100.00
B17 Relation-layer overlap	0.033	1.90 [-0.90, 4.70]	1.000	100.00 / 100.00
entry headway station track Jaccard (13,490 records)	-0.047	-0.03 [-0.43, 0.36]	1.000	73.33 / 33.33
B17 Relation-layer overlap	0.033	1.90 [-0.90, 4.70]	1.000	100.00 / 100.00
exit headway station track Jaccard (13,490 records)	-0.047	-0.03 [-0.43, 0.36]	1.000	73.33 / 33.33
Infrastructure utilization				
R01 Mean occupation	0.308	10.50 [7.60, 13.39]	3.2e-6	100.00 / 100.00
mean occupancy 10 min next endpoint (13,490 records)	-0.264	-1.66 [-2.28, -1.03]	1.0e-3	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R13 Adjacent-station occupation	0.306	10.05 [7.62, 12.48]	2.6e-7	100.00 / 100.00
downstream occupancy (13,490 records)	-0.257	-0.98 [-1.51, -0.45]	0.018	100.00 / 100.00
R09 Reference headway buffer	-0.286	-5.19 [-7.54, -2.84]	0.003	100.00 / 100.00
0 previous buffer (12,890 records)	0.264	0.35 [0.22, 0.48]	1.0e-3	100.00 / 100.00
R09 Reference headway buffer	-0.244	-3.67 [-6.13, -1.20]	0.067	100.00 / 100.00
1 previous buffer (12,890 records)	0.209	-1.14 [-1.90, -0.37]	0.066	100.00 / 100.00
R07 Entry load	0.242	9.44 [4.86, 14.02]	0.006	100.00 / 100.00
entry load 20 min (13,490 records)	-0.187	0.49 [0.12, 0.86]	0.120	100.00 / 100.00
F18 Next stopping opportunity	0.232	7.51 [5.80, 9.22]	7.2e-8	100.00 / 100.00
next stop distance (10,415 records)	-0.169	-0.67 [-1.19, -0.15]	0.140	100.00 / 100.00
R07 Entry load	0.231	8.34 [3.63, 13.05]	0.020	100.00 / 100.00
entry load 10 min (13,490 records)	-0.193	0.46 [0.11, 0.82]	0.139	100.00 / 100.00
R04 Utilization	0.223	7.85 [3.92, 11.78]	0.008	100.00 / 100.00
utilization 20 min (13,490 records)	-0.179	0.34 [0.04, 0.65]	0.280	100.00 / 100.00
R12 Demand/capacity proxy	0.223	7.85 [3.92, 11.78]	0.008	100.00 / 100.00
demand capacity 20 min (13,490 records)	-0.179	0.34 [0.04, 0.65]	0.280	100.00 / 100.00
R07 Entry load	0.222	7.16 [2.28, 12.04]	0.071	100.00 / 100.00
entry load 5 min (13,490 records)	-0.191	0.42 [0.09, 0.75]	0.151	100.00 / 100.00
R04 Utilization	0.198	6.96 [3.03, 10.89]	0.020	100.00 / 100.00
utilization 10 min (13,490 records)	-0.177	0.25 [-0.03, 0.54]	0.711	100.00 / 100.00
R12 Demand/capacity proxy	0.198	6.96 [3.03, 10.89]	0.020	100.00 / 100.00
demand capacity 10 min (13,490 records)	-0.177	0.25 [-0.03, 0.54]	0.711	100.00 / 100.00
R01 Mean occupation	0.187	6.29 [2.53, 10.04]	0.030	100.00 / 100.00
mean occupancy 20 min (13,490 records)	-0.150	0.40 [0.10, 0.71]	0.118	100.00 / 100.00
F06 Directional pool size	0.187	5.91 [4.47, 7.35]	2.9e-7	100.00 / 100.00
direction pool next endpoint (13,490 records)	-0.152	-1.55 [-2.26, -0.83]	0.005	100.00 / 100.00
R10 Intersecting dwell distribution	0.185	7.39 [3.21, 11.57]	0.020	100.00 / 100.00
waiting duration 20 min count (13,490 records)	-0.158	0.56 [0.18, 0.94]	0.072	100.00 / 100.00
R14 Window competitors	0.185	7.39 [3.21, 11.57]	0.020	100.00 / 100.00
window competitors 20 min (13,490 records)	-0.158	0.56 [0.18, 0.94]	0.072	100.00 / 100.00
R08 Exit load	0.184	6.80 [3.67, 9.93]	0.003	100.00 / 100.00
exit load 5 min (13,490 records)	-0.147	0.39 [0.13, 0.65]	0.062	100.00 / 100.00
R10 Intersecting dwell distribution	0.184	1.39 [-0.17, 2.95]	0.711	100.00 / 100.00
waiting duration 20 min sum (13,490 records)	-0.139	0.25 [0.11, 0.40]	0.025	100.00 / 100.00
R15 Free windows	-0.183	-6.12 [-9.28, -2.95]	0.010	100.00 / 100.00
free window maximum 20 min (13,490 records)	0.153	-0.24 [-0.50, 0.02]	0.603	100.00 / 100.00
B07 Regional saturation	0.181	5.62 [2.70, 8.53]	0.010	100.00 / 100.00
saturated fraction 20 min (13,490 records)	-0.153	0.22 [0.02, 0.42]	0.280	100.00 / 100.00
R05 Saturated fraction	0.181	5.62 [2.70, 8.53]	0.010	100.00 / 100.00
saturated fraction 20 min (13,490 records)	-0.153	0.22 [0.02, 0.42]	0.280	100.00 / 100.00
R15 Free windows	-0.181	-5.62 [-8.53, -2.70]	0.010	100.00 / 100.00
free window total 20 min (13,490 records)	0.153	-0.22 [-0.42, -0.02]	0.280	100.00 / 100.00
R10 Intersecting dwell distribution	0.181	0.90 [-0.11, 1.90]	0.718	100.00 / 100.00
waiting duration 5 min interquartile range (12,290 records)	-0.155	0.13 [0.03, 0.23]	0.161	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
E10 Emergency pool capacity	-0.180	-5.71 [-8.84, -2.59]	0.016	100.00 / 100.00
emergency pool capacity (4,061 records)	0.088	0.11 [0.03, 0.19]	0.075	100.00 / 100.00
R06 Saturation persistence	0.180	3.29 [1.25, 5.33]	0.039	100.00 / 100.00
longest saturation 20 min (13,490 records)	-0.148	0.18 [0.02, 0.33]	0.272	100.00 / 100.00
R10 Intersecting dwell distribution	0.179	0.85 [-0.11, 1.81]	0.735	100.00 / 100.00
waiting duration 5 min standard deviation (12,290 records)	-0.153	0.13 [0.02, 0.24]	0.236	100.00 / 100.00
H19 Peak demand/capacity	0.178	5.43 [2.65, 8.20]	0.009	100.00 / 100.00
peak demand capacity 20 min (13,490 records)	-0.155	0.11 [-0.12, 0.34]	1.000	100.00 / 66.67
R15 Free windows	0.178	6.27 [3.22, 9.33]	0.006	100.00 / 100.00
free window count 20 min (13,490 records)	-0.150	0.20 [-0.01, 0.41]	0.518	100.00 / 100.00
R03 Minimum free capacity	-0.171	-5.34 [-7.45, -3.24]	6.0e-4	100.00 / 100.00
minimum free 20 min (13,490 records)	0.149	0.23 [0.08, 0.39]	0.053	100.00 / 100.00
R01 Mean occupation	0.166	5.46 [1.60, 9.33]	0.093	100.00 / 100.00
mean occupancy 10 min (13,490 records)	-0.150	0.39 [0.08, 0.70]	0.157	100.00 / 100.00
R15 Free windows	-0.165	-5.71 [-8.91, -2.51]	0.020	100.00 / 100.00
free window maximum 10 min (13,490 records)	0.139	-0.24 [-0.48, 0.00]	0.497	100.00 / 100.00
R10 Intersecting dwell distribution	0.164	0.55 [-0.66, 1.77]	1.000	100.00 / 100.00
waiting duration 10 min sum (13,490 records)	-0.134	0.23 [0.10, 0.36]	0.020	100.00 / 100.00
H19 Peak demand/capacity	0.164	5.01 [2.18, 7.83]	0.020	100.00 / 100.00
peak demand capacity 10 min (13,490 records)	-0.146	0.08 [-0.13, 0.30]	1.000	100.00 / 66.67
B07 Regional saturation	0.162	5.30 [2.33, 8.26]	0.020	100.00 / 100.00
saturated fraction 10 min (13,490 records)	-0.139	0.24 [0.03, 0.45]	0.266	100.00 / 100.00
R05 Saturated fraction	0.162	5.30 [2.33, 8.26]	0.020	100.00 / 100.00
saturated fraction 10 min (13,490 records)	-0.139	0.24 [0.03, 0.45]	0.266	100.00 / 100.00
R15 Free windows	-0.162	-5.30 [-8.26, -2.33]	0.020	100.00 / 100.00
free window total 10 min (13,490 records)	0.139	-0.24 [-0.45, -0.03]	0.266	100.00 / 100.00
R06 Saturation persistence	0.161	3.23 [1.02, 5.45]	0.073	100.00 / 100.00
longest saturation 10 min (13,490 records)	-0.135	0.20 [0.03, 0.37]	0.250	100.00 / 100.00
R10 Intersecting dwell distribution	0.161	7.01 [2.85, 11.17]	0.028	100.00 / 100.00
waiting duration 10 min count (13,490 records)	-0.150	0.55 [0.16, 0.93]	0.085	100.00 / 100.00
R14 Window competitors	0.161	7.01 [2.85, 11.17]	0.028	100.00 / 100.00
window competitors 10 min (13,490 records)	-0.150	0.55 [0.16, 0.93]	0.085	100.00 / 100.00
R15 Free windows	0.156	5.40 [2.37, 8.43]	0.020	100.00 / 100.00
free window count 10 min (13,490 records)	-0.134	0.24 [0.01, 0.46]	0.370	100.00 / 100.00
R08 Exit load	0.155	5.73 [3.01, 8.45]	0.004	100.00 / 100.00
exit load 10 min (13,490 records)	-0.099	0.76 [0.34, 1.17]	0.020	100.00 / 100.00
B07 Regional saturation	0.154	4.58 [1.74, 7.43]	0.039	100.00 / 100.00
saturated fraction 5 min (13,490 records)	-0.132	0.25 [0.03, 0.46]	0.275	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R05 Saturated fraction	0.154	4.58 [1.74, 7.43]	0.039	100.00 / 100.00
saturated fraction 5 min (13,490 records)	-0.132	0.25 [0.03, 0.46]	0.275	100.00 / 100.00
R15 Free windows	-0.154	-4.58 [-7.43, -1.74]	0.039	100.00 / 100.00
free window total 5 min (13,490 records)	0.132	-0.25 [-0.46, -0.03]	0.275	100.00 / 100.00
R03 Minimum free capacity	-0.153	-4.83 [-6.89, -2.77]	0.001	100.00 / 100.00
minimum free 10 min (13,490 records)	0.136	0.27 [0.12, 0.42]	0.023	100.00 / 100.00
R15 Free windows	-0.152	-4.58 [-7.53, -1.63]	0.049	100.00 / 100.00
free window maximum 5 min (13,490 records)	0.130	-0.26 [-0.49, -0.03]	0.272	100.00 / 100.00
R04 Utilization	0.152	5.13 [1.87, 8.39]	0.047	100.00 / 100.00
utilization 5 min (13,490 records)	-0.143	0.00 [-0.21, 0.21]	1.000	53.33 / 66.67
R12 Demand/capacity proxy	0.152	5.13 [1.87, 8.39]	0.047	100.00 / 100.00
demand capacity 5 min (13,490 records)	-0.143	0.00 [-0.21, 0.21]	1.000	53.33 / 66.67
H19 Peak demand/capacity	0.151	4.67 [1.92, 7.43]	0.027	100.00 / 100.00
peak demand capacity 5 min (13,490 records)	-0.138	0.06 [-0.15, 0.27]	1.000	100.00 / 66.67
R06 Saturation persistence	0.151	3.61 [1.30, 5.92]	0.047	100.00 / 100.00
longest saturation 5 min (13,490 records)	-0.127	0.22 [0.02, 0.42]	0.323	100.00 / 100.00
R15 Free windows	0.150	4.92 [1.69, 8.15]	0.057	100.00 / 100.00
free window count 5 min (13,490 records)	-0.131	0.20 [-0.02, 0.43]	0.711	100.00 / 100.00
R10 Intersecting dwell distribution	0.149	5.54 [1.50, 9.59]	0.110	100.00 / 100.00
waiting duration 5 min count (13,490 records)	-0.145	0.51 [0.15, 0.86]	0.086	100.00 / 100.00
R14 Window competitors	0.149	5.54 [1.50, 9.59]	0.110	100.00 / 100.00
window competitors 5 min (13,490 records)	-0.145	0.51 [0.15, 0.86]	0.086	100.00 / 100.00
R10 Intersecting dwell distribution	0.142	0.61 [-0.22, 1.44]	1.000	100.00 / 100.00
waiting duration 10 min interquartile range (12,290 records)	-0.100	0.17 [0.07, 0.28]	0.033	100.00 / 100.00
R03 Minimum free capacity	-0.139	-4.39 [-6.31, -2.48]	0.002	100.00 / 100.00
minimum free 5 min (13,490 records)	0.126	0.29 [0.14, 0.45]	0.016	100.00 / 100.00
R13 Adjacent-station occupation	0.137	4.77 [2.36, 7.19]	0.008	100.00 / 100.00
occupancy difference (13,490 records)	-0.119	-0.95 [-1.41, -0.49]	0.008	100.00 / 100.00
R16 Planned-arrival overlap	-0.137	-4.20 [-6.72, -1.69]	0.030	100.00 / 100.00
planned arrival release overlap 5 min (13,490 records)	0.082	-0.36 [-0.74, 0.01]	0.522	100.00 / 100.00
R10 Intersecting dwell distribution	0.136	-0.11 [-0.97, 0.74]	1.000	86.67 / 66.67
waiting duration 5 min sum (13,490 records)	-0.119	0.22 [0.10, 0.34]	0.015	100.00 / 100.00
R10 Intersecting dwell distribution	0.135	0.94 [0.04, 1.85]	0.413	100.00 / 100.00
waiting duration 10 min standard deviation (12,290 records)	-0.095	0.18 [0.06, 0.29]	0.047	100.00 / 100.00
R13 Adjacent-station occupation	0.135	3.82 [0.35, 7.29]	0.323	100.00 / 100.00
upstream occupancy (13,490 records)	-0.106	0.32 [0.10, 0.54]	0.068	100.00 / 100.00
F16 Downstream capacity gradient	0.129	4.14 [3.02, 5.26]	2.2e-6	100.00 / 100.00
next capacity delta (13,490 records)	-0.106	-1.30 [-1.90, -0.69]	0.005	100.00 / 100.00
R02 Peak occupation	0.128	4.28 [1.08, 7.47]	0.126	100.00 / 100.00
peak occupancy 20 min (13,490 records)	-0.110	0.43 [0.11, 0.75]	0.123	100.00 / 100.00
R10 Intersecting dwell distribution	0.128	0.31 [-0.52, 1.15]	1.000	100.00 / 66.67
waiting duration 10 min maximum (12,290 records)	-0.081	0.22 [0.10, 0.34]	0.015	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R08 Exit load	0.127	5.19 [1.07, 9.31]	0.176	100.00 / 100.00
exit load 20 min (13,490 records)	-0.087	0.96 [0.51, 1.42]	0.007	100.00 / 100.00
R01 Mean occupation	0.126	3.94 [0.49, 7.39]	0.282	100.00 / 100.00
mean occupancy 5 min (13,490 records)	-0.118	0.27 [0.00, 0.53]	0.432	100.00 / 100.00
R10 Intersecting dwell distribution	0.126	-0.22 [-0.94, 0.49]	1.000	100.00 / 66.67
waiting duration 5 min maximum (12,290 records)	-0.098	0.21 [0.09, 0.32]	0.020	100.00 / 100.00
R10 Intersecting dwell distribution	0.125	0.29 [-0.57, 1.16]	1.000	93.33 / 100.00
waiting duration 20 min 90th percentile (12,290 records)	-0.075	0.23 [0.11, 0.36]	0.015	100.00 / 100.00
R02 Peak occupation	0.119	3.92 [0.62, 7.23]	0.235	100.00 / 100.00
peak occupancy 10 min (13,490 records)	-0.105	0.41 [0.10, 0.72]	0.131	100.00 / 100.00
R10 Intersecting dwell distribution	0.116	0.95 [-0.08, 1.99]	0.628	100.00 / 100.00
waiting duration 20 min maximum (12,290 records)	-0.061	0.24 [0.11, 0.37]	0.018	100.00 / 100.00
R16 Planned-arrival overlap	-0.116	-3.58 [-5.86, -1.29]	0.047	100.00 / 100.00
planned arrival release overlap 10 min (13,490 records)	0.070	-0.16 [-0.40, 0.07]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution	0.114	-0.02 [-0.76, 0.71]	1.000	53.33 / 66.67
waiting duration 10 min 90th percentile (12,290 records)	-0.073	0.22 [0.11, 0.34]	0.015	100.00 / 100.00
R10 Intersecting dwell distribution	0.114	-0.74 [-1.41, -0.06]	0.328	100.00 / 66.67
waiting duration 20 min median (12,290 records)	-0.085	0.13 [0.05, 0.22]	0.053	100.00 / 100.00
R02 Peak occupation	0.110	3.62 [0.39, 6.85]	0.300	100.00 / 100.00
peak occupancy 5 min (13,490 records)	-0.100	0.40 [0.09, 0.71]	0.147	100.00 / 100.00
R11 Flow imbalance	-0.105	-2.77 [-5.68, 0.14]	0.582	100.00 / 100.00
arrivals minus departures 5 min (13,490 records)	0.090	-0.14 [-0.41, 0.13]	1.000	100.00 / 100.00
R16 Planned-arrival overlap	-0.104	-3.06 [-5.26, -0.86]	0.103	100.00 / 100.00
planned arrival release overlap 20 min (13,490 records)	0.065	-0.02 [-0.19, 0.15]	1.000	73.33 / 33.33
R10 Intersecting dwell distribution	0.103	-0.43 [-1.12, 0.27]	1.000	100.00 / 66.67
waiting duration 20 min mean (12,290 records)	-0.059	0.19 [0.09, 0.30]	0.018	100.00 / 100.00
R10 Intersecting dwell distribution	0.091	-0.36 [-1.06, 0.34]	1.000	100.00 / 66.67
waiting duration 5 min 90th percentile (12,290 records)	-0.067	0.21 [0.10, 0.32]	0.018	100.00 / 100.00
R10 Intersecting dwell distribution	0.090	-0.81 [-1.46, -0.17]	0.177	100.00 / 100.00
waiting duration 10 min median (12,290 records)	-0.062	0.17 [0.08, 0.25]	0.014	100.00 / 100.00
R10 Intersecting dwell distribution	0.083	1.23 [0.23, 2.24]	0.199	100.00 / 100.00
waiting duration 20 min standard deviation (12,290 records)	-0.037	0.21 [0.09, 0.34]	0.027	100.00 / 100.00
E18 Parking concentration	-0.082	-2.08 [-4.90, 0.75]	1.000	100.00 / 100.00
parking concentration (13,490 records)	0.062	0.12 [-0.27, 0.52]	1.000	100.00 / 66.67
R10 Intersecting dwell distribution	0.078	0.19 [-0.65, 1.04]	1.000	86.67 / 66.67
waiting duration 20 min interquartile range (12,290 records)	-0.032	0.21 [0.10, 0.33]	0.012	100.00 / 100.00
R10 Intersecting dwell distribution	0.077	-0.64 [-1.24, -0.04]	0.358	100.00 / 100.00
waiting duration 10 min mean (12,290 records)	-0.042	0.20 [0.10, 0.30]	0.011	100.00 / 100.00
B13 Adjacent-window difference	-0.074	-1.59 [-3.46, 0.29]	0.820	100.00 / 100.00
adjacent occupancy difference 5 min (13,490 records)	0.065	-0.11 [-0.31, 0.09]	1.000	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R17 Occupation change rate	-0.074	-1.59 [-3.46, 0.29]	0.820	100.00 / 100.00
occupancy rate 5 min (13,490 records)	0.065	-0.11 [-0.31, 0.09]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution	-0.074	-3.53 [-5.34, -1.72]	0.009	100.00 / 100.00
waiting duration 20 min zero fraction (12,290 records)	0.069	0.61 [0.33, 0.89]	0.004	100.00 / 100.00
B13 Adjacent-window difference	-0.059	-2.39 [-5.40, 0.63]	0.994	100.00 / 100.00
adjacent occupancy difference 20 min (13,490 records)	0.068	-0.12 [-0.24, 0.00]	0.471	100.00 / 100.00
R17 Occupation change rate	-0.059	-2.39 [-5.40, 0.63]	0.994	100.00 / 100.00
occupancy rate 20 min (13,490 records)	0.068	-0.12 [-0.24, 0.00]	0.471	100.00 / 100.00
R10 Intersecting dwell distribution	0.057	-0.85 [-1.43, -0.28]	0.063	100.00 / 100.00
waiting duration 5 min mean (12,290 records)	-0.037	0.20 [0.10, 0.30]	0.010	100.00 / 100.00
R10 Intersecting dwell distribution	0.057	-0.95 [-1.57, -0.33]	0.053	100.00 / 100.00
waiting duration 5 min median (12,290 records)	-0.037	0.18 [0.09, 0.27]	0.009	100.00 / 100.00
R18 Opposite-direction utilization	0.048	1.91 [0.25, 3.58]	0.275	100.00 / 100.00
opposite utilization 10 min (13,490 records)	-0.047	0.29 [0.14, 0.45]	0.015	100.00 / 100.00
R18 Opposite-direction utilization	0.044	2.52 [0.59, 4.45]	0.143	100.00 / 100.00
opposite utilization 20 min (13,490 records)	-0.048	0.28 [0.09, 0.48]	0.072	100.00 / 100.00
F17 Distance to a hub	-0.041	-1.35 [-2.79, 0.08]	0.594	100.00 / 100.00
hub distance (13,490 records)	0.039	0.00 [-0.13, 0.12]	1.000	66.67 / 66.67
R10 Intersecting dwell distribution	-0.037	-2.77 [-4.60, -0.94]	0.060	100.00 / 100.00
waiting duration 10 min zero fraction (12,290 records)	0.034	0.55 [0.30, 0.80]	0.005	100.00 / 100.00
R18 Opposite-direction utilization	0.037	1.33 [-0.19, 2.86]	0.755	100.00 / 100.00
opposite utilization 5 min (13,490 records)	-0.031	0.41 [0.19, 0.62]	0.014	100.00 / 100.00
F01 Facility role	-0.031	-0.97 [-2.08, 0.14]	0.760	100.00 / 100.00
hub (13,490 records)	0.027	0.60 [0.30, 0.89]	0.008	100.00 / 100.00
F06 Directional pool size	-0.031	-0.97 [-2.08, 0.14]	0.760	100.00 / 100.00
direction pool (13,490 records)	0.027	0.60 [0.30, 0.89]	0.008	100.00 / 100.00
F15 Parallel-capacity proxy	-0.031	-0.97 [-2.08, 0.14]	0.760	100.00 / 100.00
parallel capacity (13,490 records)	0.027	0.60 [0.30, 0.89]	0.008	100.00 / 100.00
R10 Intersecting dwell distribution	-0.030	-1.33 [-1.98, -0.68]	0.006	100.00 / 100.00
waiting duration 5 min minimum (12,290 records)	0.037	0.16 [0.10, 0.22]	0.001	100.00 / 100.00
F04 Physical betweenness	0.029	-0.08 [-1.80, 1.64]	1.000	66.67 / 66.67
betweenness (13,490 records)	-0.028	0.59 [0.29, 0.89]	0.011	100.00 / 100.00
F05 Physical articulation point	0.028	0.89 [0.00, 1.77]	0.468	100.00 / 100.00
articulation (13,490 records)	-0.038	-0.17 [-0.28, -0.05]	0.078	100.00 / 100.00
E23 Initial station occupation	-0.025	-0.80 [-1.69, 0.10]	0.718	100.00 / 100.00
initial station occupied (13,490 records)	0.035	0.37 [0.19, 0.56]	0.011	100.00 / 100.00
R10 Intersecting dwell distribution	0.023	-1.06 [-1.62, -0.49]	0.013	100.00 / 100.00
waiting duration 20 min minimum (12,290 records)	-0.020	0.09 [0.04, 0.15]	0.037	100.00 / 100.00
F01 Facility role	-0.021	-0.67 [-2.45, 1.11]	1.000	100.00 / 100.00
trunk (13,490 records)	0.018	0.81 [0.42, 1.21]	0.008	100.00 / 100.00
F02 Physical degree	-0.015	-0.65 [-1.84, 0.54]	1.000	100.00 / 100.00
undirected degree (13,490 records)	0.007	0.50 [0.25, 0.75]	0.011	100.00 / 100.00
F03 Branch-count proxy	-0.015	-0.65 [-1.84, 0.54]	1.000	100.00 / 100.00
branch count (13,490 records)	0.007	0.50 [0.25, 0.75]	0.011	100.00 / 100.00
F07 Compatible tracks	-0.015	-0.36 [-1.57, 0.86]	1.000	100.00 / 66.67
compatible tracks (13,490 records)	0.007	0.40 [0.19, 0.61]	0.016	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R10 Intersecting dwell distribution	-0.013	-1.99 [-3.81, -0.16]	0.340	100.00 / 100.00
waiting duration 5 min zero fraction	0.016	0.52 [0.27, 0.76]	0.006	100.00 / 100.00
(12,290 records)				
R11 Flow imbalance	-0.012	-0.04 [-1.63, 1.55]	1.000	60.00 / 33.33
arrivals minus departures 10 min	0.002	-0.21 [-0.39, -0.03]	0.239	100.00 / 100.00
(13,490 records)				
R10 Intersecting dwell distribution	-0.011	-1.16 [-1.76, -0.55]	0.011	100.00 / 100.00
waiting duration 10 min minimum	0.017	0.13 [0.07, 0.18]	0.003	100.00 / 100.00
(12,290 records)				
B13 Adjacent-window difference	-0.009	-0.81 [-3.22, 1.59]	1.000	100.00 / 100.00
adjacent occupancy difference 10 min	0.017	-0.10 [-0.26, 0.06]	1.000	100.00 / 100.00
(13,490 records)				
R17 Occupation change rate	-0.009	-0.81 [-3.22, 1.59]	1.000	100.00 / 100.00
occupancy rate 10 min (13,490 records)	0.017	-0.10 [-0.26, 0.06]	1.000	100.00 / 100.00
R11 Flow imbalance	0.000	-0.39 [-2.80, 2.03]	1.000	100.00 / 66.67
arrivals minus departures 20 min	-0.008	-0.12 [-0.27, 0.02]	0.879	100.00 / 100.00
(13,490 records)				
Delay propagation patterns				
G03 Activity delay increment	0.925	12.77 [10.40, 15.15]	7.8e-10	100.00 / 100.00
run delay increment (13,490 records)	-0.926	-8.84 [-8.85, -8.84]	1.0e-91	100.00 / 100.00
G04 Activity-duration increment	0.925	12.77 [10.40, 15.15]	7.8e-10	100.00 / 100.00
run increment (13,490 records)	-0.926	-8.84 [-8.85, -8.84]	1.0e-91	100.00 / 100.00
G19 Endpoint contrasts	0.782	10.50 [8.27, 12.73]	2.1e-8	100.00 / 100.00
run G02 departure delay difference	-0.747	-7.12 [-8.56, -5.69]	4.9e-9	100.00 / 100.00
(13,490 records)				
G10 Edge-slack distribution	0.744	10.94 [7.90, 13.99]	3.5e-6	100.00 / 100.00
outgoing slack minimum departure	-0.758	-7.64 [-8.27, -7.00]	1.7e-20	100.00 / 100.00
(13,490 records)				
G19 Endpoint contrasts	0.655	5.96 [3.33, 8.58]	0.002	100.00 / 100.00
run G01 arrival delay difference	-0.633	-4.39 [-6.42, -2.36]	0.005	100.00 / 100.00
(13,490 records)				
H25 Local assortativity	-0.450	-16.28 [-19.89, -12.67]	5.2e-8	100.00 / 100.00
local same type fraction departure	0.447	0.50 [0.40, 0.60]	4.9e-9	100.00 / 100.00
(7,802 records)				
H25 Local assortativity	0.427	14.42 [10.01, 18.84]	1.5e-5	100.00 / 100.00
local delay absolute difference	-0.424	-0.49 [-0.64, -0.34]	4.1e-5	100.00 / 100.00
departure (7,802 records)				
G10 Edge-slack distribution	-0.423	-13.83 [-17.32, -10.33]	5.7e-7	100.00 / 100.00
outgoing slack zero fraction departure	0.421	2.14 [1.36, 2.91]	9.0e-4	100.00 / 100.00
(13,490 records)				
B05 Regional delay distribution	0.362	11.76 [6.98, 16.54]	8.3e-4	100.00 / 100.00
region delay 10 min standard deviation	-0.301	0.11 [-0.19, 0.41]	1.000	100.00 / 66.67
(13,490 records)				
B06 Regional delay variance	0.362	9.83 [5.56, 14.10]	0.002	100.00 / 100.00
delay variance 10 min (13,490 records)	-0.301	0.00 [-0.22, 0.22]	1.000	46.67 / 66.67
B05 Regional delay distribution	0.360	12.45 [7.66, 17.24]	4.4e-4	100.00 / 100.00
region delay 20 min interquartile range	-0.292	0.23 [-0.12, 0.58]	1.000	100.00 / 66.67
(13,490 records)				
G15 Active-neighbor dispersion	0.356	8.94 [5.21, 12.68]	0.001	100.00 / 100.00
delay variance departure (7,802	-0.355	-0.34 [-0.49, -0.18]	0.005	100.00 / 100.00
records)				
B05 Regional delay distribution	0.353	11.75 [6.99, 16.51]	8.1e-4	100.00 / 100.00
region delay 20 min standard deviation	-0.278	0.26 [-0.10, 0.62]	1.000	100.00 / 66.67
(13,490 records)				
B06 Regional delay variance	0.353	11.19 [6.84, 15.53]	4.9e-4	100.00 / 100.00
delay variance 20 min (13,490 records)	-0.278	0.10 [-0.19, 0.38]	1.000	100.00 / 66.67

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H25 Local assortativity	-0.352	-11.75 [-15.01, -8.49]	3.4e-6	100.00 / 100.00
local same type fraction arrival (7,303 records)	0.358	0.35 [0.25, 0.46]	2.0e-5	100.00 / 100.00
B05 Regional delay distribution	0.352	10.27 [5.25, 15.30]	0.006	100.00 / 100.00
region delay 5 min standard deviation (13,490 records)	-0.314	0.02 [-0.24, 0.29]	1.000	80.00 / 66.67
B06 Regional delay variance	0.352	7.55 [3.62, 11.49]	0.011	100.00 / 100.00
delay variance 5 min (13,490 records)	-0.314	-0.03 [-0.24, 0.18]	1.000	93.33 / 66.67
H25 Local assortativity	0.352	10.36 [6.72, 14.00]	1.4e-4	100.00 / 100.00
local delay absolute difference arrival (7,303 records)	-0.353	-0.31 [-0.43, -0.18]	0.002	100.00 / 100.00
B05 Regional delay distribution	0.351	10.83 [5.84, 15.82]	0.003	100.00 / 100.00
region delay 10 min interquartile range (13,490 records)	-0.298	0.08 [-0.21, 0.37]	1.000	100.00 / 66.67
G15 Active-neighbor dispersion	0.348	10.71 [5.97, 15.45]	0.002	100.00 / 100.00
delay range departure (7,802 records)	-0.347	-0.37 [-0.54, -0.20]	0.005	100.00 / 100.00
G15 Active-neighbor dispersion	0.341	7.57 [3.02, 12.12]	0.031	100.00 / 100.00
delay coefficient of variation departure (7,802 records)	-0.340	-0.28 [-0.45, -0.11]	0.037	100.00 / 100.00
H16 Local motifs	0.325	10.30 [5.27, 15.33]	0.006	100.00 / 100.00
fan-in pairs departure (13,490 records)	-0.301	0.05 [-0.20, 0.31]	1.000	93.33 / 66.67
G19 Endpoint contrasts	-0.311	-8.86 [-12.11, -5.60]	2.2e-4	100.00 / 100.00
G08 active outgoing degree endpoint contrast (13,490 records)	0.341	0.42 [0.33, 0.52]	9.7e-8	100.00 / 100.00
B10 Affected-train share	0.308	9.70 [6.25, 13.14]	1.5e-4	100.00 / 100.00
affected fraction (13,490 records)	-0.260	-0.72 [-1.15, -0.28]	0.034	100.00 / 100.00
B16 Global reachability	0.306	9.70 [6.28, 13.12]	1.4e-4	100.00 / 100.00
reachable pair fraction (13,490 records)	-0.259	-0.75 [-1.19, -0.31]	0.028	100.00 / 100.00
B15 Global degree distribution	0.306	9.73 [6.31, 13.15]	1.4e-4	100.00 / 100.00
outgoing degree mean (13,490 records)	-0.252	-0.68 [-1.12, -0.24]	0.054	100.00 / 100.00
B15 Global degree distribution	0.306	9.73 [6.31, 13.15]	1.4e-4	100.00 / 100.00
outgoing degree sum (13,490 records)	-0.252	-0.68 [-1.12, -0.24]	0.054	100.00 / 100.00
B15 Global degree distribution	0.305	9.72 [6.30, 13.13]	1.4e-4	100.00 / 100.00
incoming degree mean (13,490 records)	-0.251	-0.67 [-1.11, -0.23]	0.057	100.00 / 100.00
B15 Global degree distribution	0.305	9.72 [6.30, 13.13]	1.4e-4	100.00 / 100.00
incoming degree sum (13,490 records)	-0.251	-0.67 [-1.11, -0.23]	0.057	100.00 / 100.00
B11 Global path distribution	0.304	9.81 [6.37, 13.25]	1.4e-4	100.00 / 100.00
root longest paths mean (13,490 records)	-0.256	-0.75 [-1.20, -0.31]	0.028	100.00 / 100.00
B11 Global path distribution	0.304	9.81 [6.37, 13.25]	1.4e-4	100.00 / 100.00
root longest paths sum (13,490 records)	-0.256	-0.75 [-1.20, -0.31]	0.028	100.00 / 100.00
B11 Global path distribution	-0.304	-9.66 [-13.07, -6.25]	1.4e-4	100.00 / 100.00
root longest paths zero fraction (13,490 records)	0.249	0.67 [0.23, 1.11]	0.054	100.00 / 100.00
B15 Global degree distribution	-0.304	-9.66 [-13.07, -6.25]	1.4e-4	100.00 / 100.00
incoming degree zero fraction (13,490 records)	0.249	0.67 [0.23, 1.11]	0.054	100.00 / 100.00
G15 Active-neighbor dispersion	0.304	8.19 [5.22, 11.16]	1.9e-4	100.00 / 100.00
delay variance arrival (7,303 records)	-0.309	-0.31 [-0.43, -0.18]	0.002	100.00 / 100.00
B11 Global path distribution	0.301	9.23 [5.89, 12.57]	1.9e-4	100.00 / 100.00
root longest paths 90th percentile (13,490 records)	-0.258	-0.73 [-1.14, -0.32]	0.023	100.00 / 100.00
B15 Global degree distribution	-0.301	-9.58 [-12.97, -6.19]	1.4e-4	100.00 / 100.00
outgoing degree zero fraction (13,490 records)	0.245	0.66 [0.23, 1.09]	0.054	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B11 Global path distribution root longest paths interquartile range (13,490 records)	0.299 -0.257	9.58 [6.09, 13.07] -0.78 [-1.23, -0.32]	2.0e-4 0.028	100.00 / 100.00 100.00 / 100.00
G15 Active-neighbor dispersion delay range arrival (7,303 records)	0.298 -0.303	9.77 [5.88, 13.66] -0.36 [-0.50, -0.22]	6.8e-4 0.002	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution outgoing degree standard deviation (13,490 records)	0.297 -0.258	8.60 [5.27, 11.93] -0.69 [-1.07, -0.31]	4.8e-4 0.020	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths standard deviation (13,490 records)	0.295 -0.254	9.17 [5.75, 12.58] -0.74 [-1.17, -0.31]	2.7e-4 0.027	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution outgoing degree 90th percentile (13,490 records)	0.293 -0.244	9.14 [5.83, 12.46] -0.85 [-1.17, -0.52]	1.9e-4 0.001	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths maximum (13,490 records)	0.292 -0.250	8.93 [5.46, 12.39] -0.71 [-1.14, -0.27]	4.9e-4 0.038	100.00 / 100.00 100.00 / 100.00
G15 Active-neighbor dispersion delay coefficient of variation arrival (7,303 records)	0.289 -0.298	6.07 [3.16, 8.99] -0.23 [-0.34, -0.13]	0.005 0.004	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min interquartile range (13,490 records)	0.289 -0.250	7.08 [2.84, 11.31] 0.06 [-0.15, 0.27]	0.030 1.000	100.00 / 100.00 100.00 / 66.67
G09 Active cross-train links active cross departure (13,490 records)	0.289 -0.241	9.06 [3.86, 14.26] 0.31 [0.00, 0.61]	0.021 0.445	100.00 / 100.00 100.00 / 100.00
G09 Active cross-train links active cross fraction departure (13,490 records)	0.286 -0.239	8.80 [3.50, 14.10] 0.31 [0.00, 0.61]	0.031 0.450	100.00 / 100.00 100.00 / 100.00
B11 Global path distribution root longest paths median (13,490 records)	0.282 -0.239	8.63 [5.57, 11.69] -0.66 [-1.09, -0.22]	1.5e-4 0.057	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution incoming degree 90th percentile (13,490 records)	0.277 -0.227	8.58 [5.54, 11.62] -0.74 [-1.06, -0.42]	1.4e-4 0.003	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration time20 concentration (13,490 records)	-0.271 0.232	-6.89 [-10.30, -3.48] 0.66 [0.34, 0.98]	0.007 0.008	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution incoming degree standard deviation (13,490 records)	0.270 -0.232	8.01 [4.85, 11.17] -0.66 [-1.01, -0.31]	6.0e-4 0.016	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution outgoing degree median (13,490 records)	0.267 -0.220	8.46 [5.40, 11.53] -0.55 [-1.00, -0.11]	1.9e-4 0.172	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution incoming degree median (13,490 records)	0.265 -0.221	8.41 [5.24, 11.57] -0.57 [-1.01, -0.14]	3.1e-4 0.130	100.00 / 100.00 100.00 / 100.00
B16 Global reachability mean reachable distance (13,490 records)	0.264 -0.231	8.16 [4.83, 11.50] -0.64 [-1.07, -0.21]	8.7e-4 0.062	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration time20 entropy (13,490 records)	0.264 -0.229	7.37 [4.24, 10.50] -0.72 [-1.04, -0.41]	0.001 0.004	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support minimum departure (13,490 records)	0.263 -0.215	6.94 [3.34, 10.55] 0.42 [0.10, 0.74]	0.010 0.131	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack interquartile range departure (13,490 records)	-0.263 0.239	-4.97 [-7.07, -2.86] 0.39 [0.25, 0.53]	0.001 9.0e-4	100.00 / 100.00 100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution incoming slack standard deviation departure (13,490 records)	-0.262 0.239	-4.97 [-7.07, -2.87] 0.39 [0.25, 0.53]	0.001 9.0e-4	100.00 / 100.00 100.00 / 100.00
B15 Global degree distribution outgoing degree interquartile range (13,490 records)	0.261 -0.226	8.32 [5.27, 11.37] -0.77 [-1.22, -0.32]	2.1e-4 0.027	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay zero fraction departure (12,890 records)	-0.261 0.213	-8.42 [-12.95, -3.90] -0.53 [-0.97, -0.10]	0.015 0.181	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack maximum departure (13,490 records)	-0.259 0.239	-4.86 [-6.96, -2.77] 0.39 [0.24, 0.53]	0.002 1.0e-3	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack 90th percentile departure (13,490 records)	-0.258 0.239	-4.83 [-6.92, -2.75] 0.38 [0.24, 0.53]	0.002 1.0e-3	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum departure (12,890 records)	0.257 -0.213	7.35 [3.19, 11.51] 0.46 [0.09, 0.83]	0.020 0.174	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay mean departure (12,890 records)	0.257 -0.213	7.35 [3.19, 11.51] 0.46 [0.09, 0.83]	0.020 0.174	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay median departure (12,890 records)	0.257 -0.213	7.35 [3.19, 11.51] 0.46 [0.09, 0.83]	0.020 0.174	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum departure (12,890 records)	0.257 -0.213	7.35 [3.19, 11.51] 0.46 [0.09, 0.83]	0.020 0.174	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay 90th percentile departure (12,890 records)	0.257 -0.213	7.35 [3.19, 11.51] 0.46 [0.09, 0.83]	0.020 0.174	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack sum departure (13,490 records)	-0.257 0.239	-4.50 [-6.50, -2.50] 0.38 [0.23, 0.52]	0.002 0.001	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum departure (13,490 records)	0.256 -0.214	7.20 [3.10, 11.30] 0.45 [0.09, 0.80]	0.020 0.163	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack mean departure (13,490 records)	-0.256 0.239	-4.66 [-6.70, -2.62] 0.38 [0.23, 0.52]	0.002 0.001	100.00 / 100.00 100.00 / 100.00
H13 Weak-component size component events arrival (13,490 records)	0.255 -0.183	8.31 [4.60, 12.03] 0.55 [0.12, 0.98]	0.002 0.156	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack median departure (13,490 records)	-0.253 0.236	-4.41 [-6.40, -2.42] 0.36 [0.22, 0.50]	0.003 0.001	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum arrival (13,490 records)	0.250 -0.229	7.52 [3.19, 11.86] 0.36 [0.03, 0.70]	0.022 0.305	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction departure (13,490 records)	-0.249 0.181	-7.88 [-12.08, -3.67] -0.76 [-1.30, -0.22]	0.013 0.085	100.00 / 100.00 100.00 / 100.00
H13 Weak-component size component trains arrival (13,490 records)	0.248 -0.182	8.31 [4.48, 12.14] 0.57 [0.13, 1.00]	0.003 0.133	100.00 / 100.00 100.00 / 100.00
G08 Active degree incoming active departure (13,490 records)	0.248 -0.191	8.69 [3.83, 13.54] 0.66 [0.15, 1.18]	0.020 0.148	100.00 / 100.00 100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G13 Upstream-neighbor delay	0.248	7.70 [3.42, 11.98]	0.019	100.00 / 100.00
neighbor delay maximum arrival (11,733 records)	-0.219	0.49 [0.10, 0.88]	0.166	100.00 / 100.00
G13 Upstream-neighbor delay	0.248	7.70 [3.42, 11.97]	0.019	100.00 / 100.00
neighbor delay 90th percentile arrival (11,733 records)	-0.219	0.49 [0.10, 0.88]	0.166	100.00 / 100.00
G13 Upstream-neighbor delay	0.247	7.48 [3.29, 11.68]	0.020	100.00 / 100.00
neighbor delay mean arrival (11,733 records)	-0.220	0.48 [0.10, 0.86]	0.160	100.00 / 100.00
G13 Upstream-neighbor delay	0.247	7.48 [3.29, 11.68]	0.020	100.00 / 100.00
neighbor delay median arrival (11,733 records)	-0.220	0.48 [0.10, 0.86]	0.160	100.00 / 100.00
G12 Delay-support distribution	0.244	7.00 [4.48, 9.53]	1.8e-4	100.00 / 100.00
outgoing support interquartile range arrival (13,490 records)	-0.185	0.32 [0.06, 0.57]	0.185	100.00 / 100.00
G12 Delay-support distribution	0.244	7.00 [4.48, 9.53]	1.8e-4	100.00 / 100.00
outgoing support standard deviation arrival (13,490 records)	-0.185	0.32 [0.06, 0.57]	0.185	100.00 / 100.00
G19 Endpoint contrasts	0.244	7.39 [4.34, 10.44]	9.9e-4	100.00 / 100.00
G08 active incoming degree endpoint contrast (13,490 records)	-0.215	-0.04 [-0.14, 0.07]	1.000	100.00 / 100.00
G01 Reference arrival delay	0.239	6.07 [3.16, 8.97]	0.005	100.00 / 100.00
arrival delay next endpoint (13,490 records)	-0.195	-1.22 [-1.81, -0.62]	0.008	100.00 / 100.00
G19 Endpoint contrasts	0.239	5.74 [3.15, 8.33]	0.003	100.00 / 100.00
H04 upstream reachability endpoint contrast (13,490 records)	-0.192	0.13 [-0.08, 0.34]	1.000	100.00 / 100.00
H13 Weak-component size	0.238	7.80 [4.20, 11.41]	0.004	100.00 / 100.00
component events departure (13,490 records)	-0.161	0.62 [0.16, 1.08]	0.118	100.00 / 100.00
G02 Reference departure delay	0.237	6.14 [3.27, 9.01]	0.004	100.00 / 100.00
departure delay next endpoint (13,490 records)	-0.193	-1.24 [-1.83, -0.65]	0.007	100.00 / 100.00
G12 Delay-support distribution	0.236	6.01 [3.14, 8.87]	0.005	100.00 / 100.00
outgoing support standard deviation departure (13,490 records)	-0.167	0.43 [0.14, 0.72]	0.064	100.00 / 100.00
G13 Upstream-neighbor delay	-0.236	-7.72 [-12.17, -3.27]	0.023	100.00 / 100.00
neighbor delay zero fraction arrival (11,733 records)	0.208	-0.60 [-1.05, -0.15]	0.126	100.00 / 100.00
G12 Delay-support distribution	0.234	5.70 [2.91, 8.49]	0.006	100.00 / 100.00
outgoing support interquartile range departure (13,490 records)	-0.164	0.43 [0.15, 0.71]	0.057	100.00 / 100.00
H13 Weak-component size	0.234	7.78 [4.08, 11.49]	0.005	100.00 / 100.00
component trains departure (13,490 records)	-0.163	0.64 [0.18, 1.10]	0.097	100.00 / 100.00
H15 Chain and crossing length	0.230	5.98 [0.72, 11.24]	0.286	100.00 / 100.00
longest crossings departure (13,490 records)	-0.225	0.35 [0.07, 0.64]	0.173	100.00 / 100.00
B05 Regional delay distribution	0.229	7.32 [3.43, 11.21]	0.013	100.00 / 100.00
region delay 20 min sum (13,490 records)	-0.172	0.48 [0.13, 0.84]	0.110	100.00 / 100.00
B15 Global degree distribution	0.228	6.98 [3.91, 10.05]	0.002	100.00 / 100.00
outgoing degree maximum (13,490 records)	-0.187	-0.50 [-0.86, -0.13]	0.111	100.00 / 100.00
B08 Community structure	0.227	7.10 [3.23, 10.96]	0.016	100.00 / 100.00
community events arrival (13,490 records)	-0.165	0.58 [0.17, 1.00]	0.090	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G19 Endpoint contrasts run H10 betweenness departure difference (13,490 records)	0.225 -0.196	3.82 [2.74, 4.90] -1.47 [-2.17, -0.76]	4.2e-6 0.007	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum arrival (11,733 records)	0.223 -0.200	6.76 [2.89, 10.63] 0.44 [0.10, 0.78]	0.021 0.145	100.00 / 100.00 100.00 / 100.00
B08 Community structure community events departure (13,490 records)	0.222 -0.150	7.33 [3.42, 11.24] 0.63 [0.19, 1.06]	0.013 0.077	100.00 / 100.00 100.00 / 100.00
H06 Upstream trains upstream departure (13,490 records)	0.222 -0.190	6.15 [0.93, 11.36] 0.47 [0.13, 0.80]	0.243 0.099	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction arrival (12,610 records)	-0.221 0.169	-7.33 [-11.45, -3.20] -0.71 [-1.20, -0.22]	0.020 0.073	100.00 / 100.00 100.00 / 100.00
B02 Local edges and density edges 20 min (13,490 records)	0.217 -0.180	8.66 [4.22, 13.09] 0.56 [0.16, 0.95]	0.009 0.086	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min sum (13,490 records)	0.215 -0.159	6.57 [2.51, 10.64] 0.53 [0.16, 0.90]	0.039 0.084	100.00 / 100.00 100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum departure (13,490 records)	0.212 -0.180	5.92 [1.14, 10.70] 0.42 [0.10, 0.73]	0.187 0.120	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support median arrival (12,610 records)	0.209 -0.172	5.42 [2.08, 8.77] 0.51 [0.17, 0.85]	0.038 0.064	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration station entropy (13,490 records)	0.209 -0.143	4.48 [1.66, 7.30] -0.42 [-0.65, -0.18]	0.044 0.024	100.00 / 100.00 100.00 / 100.00
H11 Harmonic closeness incoming harmonic departure (13,490 records)	0.203 -0.172	6.25 [1.32, 11.19] 0.66 [0.20, 1.11]	0.171 0.077	100.00 / 100.00 100.00 / 100.00
B01 Local region size region events 20 min (13,490 records)	0.201 -0.165	8.09 [3.71, 12.47] 0.63 [0.21, 1.05]	0.015 0.062	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min count (13,490 records)	0.201 -0.165	8.09 [3.71, 12.47] 0.63 [0.21, 1.05]	0.015 0.062	100.00 / 100.00 100.00 / 100.00
B01 Local region size region trains 20 min (13,490 records)	0.201 -0.160	7.85 [3.47, 12.24] 0.63 [0.22, 1.05]	0.019 0.056	100.00 / 100.00 100.00 / 100.00
G09 Active cross-train links active cross arrival (13,490 records)	0.200 -0.189	6.25 [2.01, 10.49] 0.32 [0.03, 0.61]	0.070 0.305	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support minimum arrival (12,610 records)	0.199 -0.170	5.11 [2.11, 8.11] 0.43 [0.14, 0.72]	0.026 0.066	100.00 / 100.00 100.00 / 100.00
G09 Active cross-train links active cross fraction arrival (13,490 records)	0.197 -0.187	6.14 [1.98, 10.30] 0.32 [0.03, 0.61]	0.068 0.297	100.00 / 100.00 100.00 / 100.00
G19 Endpoint contrasts run T18 remaining planned difference (13,490 records)	0.197 -0.161	6.28 [4.33, 8.24] 0.02 [-0.37, 0.42]	2.0e-5 1.000	100.00 / 100.00 93.33 / 66.67
G12 Delay-support distribution incoming support median departure (13,490 records)	0.191 -0.133	5.32 [1.93, 8.71] 0.64 [0.23, 1.05]	0.047 0.051	100.00 / 100.00 100.00 / 100.00
B12 Delay concentration station concentration (13,490 records)	-0.190 0.128	-3.07 [-5.58, -0.55] 0.27 [0.08, 0.46]	0.202 0.080	100.00 / 100.00 100.00 / 66.67
B05 Regional delay distribution region delay 5 min sum (13,490 records)	0.189 -0.135	5.17 [1.48, 8.86] 0.59 [0.21, 0.97]	0.097 0.050	100.00 / 100.00 100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support sum arrival (13,490 records)	0.188 -0.143	5.12 [1.68, 8.55] 0.55 [0.19, 0.92]	0.065 0.054	100.00 / 100.00 100.00 / 100.00
H09 Reachable delay totals upstream event delay departure (13,490 records)	0.188 -0.160	4.31 [0.11, 8.50] 0.43 [0.14, 0.73]	0.431 0.072	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support mean departure (13,490 records)	0.188 -0.129	5.33 [1.97, 8.69] 0.65 [0.24, 1.07]	0.044 0.047	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support sum departure (13,490 records)	0.187 -0.129	5.34 [1.86, 8.83] 0.66 [0.24, 1.07]	0.055 0.047	100.00 / 100.00 100.00 / 100.00
B02 Local edges and density edges 10 min (13,490 records)	0.187 -0.172	7.65 [3.09, 12.21] 0.58 [0.18, 0.98]	0.029 0.078	100.00 / 100.00 100.00 / 100.00
H04 Upstream reachability upstream departure (13,490 records)	0.185 -0.158	4.11 [-0.32, 8.53] 0.49 [0.17, 0.81]	0.627 0.053	100.00 / 100.00 100.00 / 100.00
B01 Local region size region trains 10 min (13,490 records)	0.185 -0.161	7.82 [3.33, 12.31] 0.63 [0.20, 1.06]	0.021 0.071	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support zero fraction arrival (13,490 records)	-0.184 0.115	-5.87 [-9.61, -2.13] -0.82 [-1.36, -0.29]	0.047 0.054	100.00 / 100.00 100.00 / 100.00
H24 Projection core number core departure (13,490 records)	0.183 -0.106	5.81 [2.09, 9.53] 0.82 [0.30, 1.34]	0.048 0.046	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min maximum (13,490 records)	0.183 -0.125	5.47 [1.88, 9.07] 0.72 [0.25, 1.18]	0.057 0.054	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min 90th percentile (13,490 records)	0.182 -0.125	5.40 [1.83, 8.98] 0.71 [0.25, 1.18]	0.060 0.054	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support mean arrival (12,610 records)	0.181 -0.135	4.91 [1.68, 8.13] 0.61 [0.23, 1.00]	0.057 0.044	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack zero fraction departure (13,490 records)	0.180 -0.185	5.42 [2.00, 8.84] 0.10 [-0.16, 0.36]	0.044 1.000	100.00 / 100.00 100.00 / 66.67
G10 Edge-slack distribution incoming slack minimum arrival (12,610 records)	0.179 -0.125	1.26 [-0.04, 2.55] 0.07 [0.00, 0.14]	0.541 0.544	100.00 / 100.00 100.00 / 66.67
H24 Projection core number core arrival (13,490 records)	0.179 -0.119	5.49 [1.89, 9.09] 0.73 [0.25, 1.21]	0.057 0.059	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min maximum (13,490 records)	0.178 -0.120	5.07 [1.55, 8.58] 0.70 [0.25, 1.15]	0.082 0.049	100.00 / 100.00 100.00 / 100.00
B01 Local region size region events 10 min (13,490 records)	0.178 -0.161	6.98 [2.53, 11.44] 0.65 [0.22, 1.07]	0.047 0.057	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min count (13,490 records)	0.178 -0.161	6.98 [2.53, 11.44] 0.65 [0.22, 1.07]	0.047 0.057	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min zero fraction (13,490 records)	-0.177 0.103	-5.72 [-9.32, -2.13] -0.90 [-1.48, -0.31]	0.043 0.054	100.00 / 100.00 100.00 / 100.00
G16 Incident-link imbalance pressure balance arrival (13,490 records)	-0.177 0.181	-3.97 [-7.13, -0.81] -0.15 [-0.33, 0.03]	0.177 0.818	100.00 / 100.00 100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay interquartile range arrival (11,733 records)	0.176 -0.158	3.22 [1.07, 5.36] 0.19 [0.02, 0.36]	0.063 0.289	100.00 / 100.00 100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G13 Upstream-neighbor delay	0.176	3.22 [1.07, 5.36]	0.063	100.00 / 100.00
neighbor delay standard deviation arrival (11,733 records)	-0.158	0.19 [0.02, 0.36]	0.289	100.00 / 100.00
H02 Longest active path	0.176	5.10 [0.59, 9.60]	0.291	100.00 / 100.00
longest path departure (13,490 records)	-0.145	0.64 [0.23, 1.04]	0.049	100.00 / 100.00
H15 Chain and crossing length	0.176	5.10 [0.59, 9.60]	0.291	100.00 / 100.00
longest chain departure (13,490 records)	-0.145	0.64 [0.23, 1.04]	0.049	100.00 / 100.00
B05 Regional delay distribution	-0.176	-5.77 [-9.41, -2.13]	0.044	100.00 / 100.00
region delay 10 min zero fraction (13,490 records)	0.101	-0.89 [-1.48, -0.31]	0.055	100.00 / 100.00
B05 Regional delay distribution	-0.174	-5.76 [-9.44, -2.07]	0.048	100.00 / 100.00
region delay 20 min zero fraction (13,490 records)	0.100	-0.90 [-1.49, -0.32]	0.054	100.00 / 100.00
G12 Delay-support distribution	0.173	3.92 [1.05, 6.79]	0.112	100.00 / 100.00
outgoing support sum arrival (13,490 records)	-0.119	0.62 [0.25, 1.00]	0.033	100.00 / 100.00
B15 Global degree distribution	0.173	5.47 [2.41, 8.53]	0.020	100.00 / 100.00
incoming degree interquartile range (13,490 records)	-0.146	-0.48 [-0.90, -0.06]	0.266	100.00 / 100.00
B05 Regional delay distribution	0.172	4.77 [1.32, 8.22]	0.106	100.00 / 100.00
region delay 10 min 90th percentile (13,490 records)	-0.114	0.71 [0.26, 1.16]	0.045	100.00 / 100.00
G12 Delay-support distribution	-0.168	-5.33 [-8.95, -1.71]	0.070	100.00 / 100.00
outgoing support zero fraction departure (13,490 records)	0.095	-0.88 [-1.45, -0.31]	0.051	100.00 / 100.00
G12 Delay-support distribution	0.166	3.64 [0.88, 6.40]	0.136	100.00 / 100.00
outgoing support sum departure (13,490 records)	-0.103	0.64 [0.27, 1.02]	0.026	100.00 / 100.00
B02 Local edges and density	-0.164	-5.00 [-8.72, -1.29]	0.122	100.00 / 100.00
density 20 min (13,426 records)	0.130	-0.97 [-1.52, -0.43]	0.023	100.00 / 100.00
B05 Regional delay distribution	0.164	4.45 [1.06, 7.85]	0.140	100.00 / 100.00
region delay 5 min maximum (13,490 records)	-0.108	0.71 [0.27, 1.15]	0.042	100.00 / 100.00
H15 Chain and crossing length	0.163	4.14 [0.05, 8.23]	0.456	100.00 / 100.00
longest crossings arrival (13,490 records)	-0.168	0.33 [0.08, 0.59]	0.121	100.00 / 100.00
B01 Local region size	0.160	5.93 [1.55, 10.31]	0.118	100.00 / 100.00
region trains 5 min (13,490 records)	-0.147	0.63 [0.21, 1.05]	0.057	100.00 / 100.00
H10 Active betweenness	0.157	4.15 [1.85, 6.45]	0.019	100.00 / 100.00
betweenness departure next endpoint (13,490 records)	-0.108	-0.76 [-1.32, -0.20]	0.114	100.00 / 100.00
G12 Delay-support distribution	0.155	3.63 [0.76, 6.51]	0.172	100.00 / 100.00
outgoing support maximum arrival (13,490 records)	-0.101	0.69 [0.29, 1.09]	0.028	100.00 / 100.00
B02 Local edges and density	-0.155	-4.68 [-8.43, -0.93]	0.180	100.00 / 100.00
density 10 min (13,341 records)	0.130	-0.81 [-1.29, -0.34]	0.028	100.00 / 100.00
B05 Regional delay distribution	0.155	4.11 [0.89, 7.34]	0.165	100.00 / 100.00
region delay 5 min 90th percentile (13,490 records)	-0.099	0.72 [0.28, 1.16]	0.036	100.00 / 100.00
G01 Reference arrival delay	0.154	3.60 [0.74, 6.45]	0.176	100.00 / 100.00
arrival delay (13,490 records)	-0.100	0.69 [0.29, 1.09]	0.028	100.00 / 100.00
G12 Delay-support distribution	0.154	3.52 [0.67, 6.38]	0.191	100.00 / 100.00
outgoing support 90th percentile arrival (13,490 records)	-0.100	0.69 [0.29, 1.09]	0.028	100.00 / 100.00
B04 Boundary connections	0.152	5.66 [1.88, 9.44]	0.063	100.00 / 100.00
entering 20 min (13,490 records)	-0.130	0.60 [0.23, 0.97]	0.038	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution region delay 20 min mean (13,490 records)	0.151 -0.095	3.72 [0.67, 6.77] 0.73 [0.29, 1.17]	0.202 0.033	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min mean (13,490 records)	0.151 -0.094	3.62 [0.59, 6.65] 0.73 [0.30, 1.16]	0.225 0.030	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support mean arrival (13,490 records)	0.149 -0.096	3.06 [0.29, 5.82] 0.69 [0.29, 1.08]	0.318 0.026	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support median arrival (13,490 records)	0.149 -0.096	3.06 [0.29, 5.82] 0.69 [0.29, 1.08]	0.318 0.026	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 20 min median (13,490 records)	0.149 -0.094	3.66 [0.64, 6.69] 0.74 [0.30, 1.18]	0.211 0.030	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support 90th percentile arrival (12,610 records)	0.149 -0.104	3.77 [0.79, 6.75] 0.70 [0.29, 1.11]	0.172 0.028	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction departure (10,811 records)	-0.147 0.091	-4.74 [-7.36, -2.12] -0.27 [-0.63, 0.10]	0.019 1.000	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 10 min median (13,490 records)	0.145 -0.089	3.47 [0.51, 6.42] 0.74 [0.31, 1.17]	0.249 0.028	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support 90th percentile departure (13,490 records)	0.144 -0.089	3.60 [0.64, 6.57] 0.74 [0.31, 1.18]	0.207 0.027	100.00 / 100.00 100.00 / 100.00
H08 Spatial reachability upstream stations departure (13,490 records)	0.144 -0.116	3.87 [0.14, 7.60] 0.71 [0.27, 1.15]	0.417 0.038	100.00 / 100.00 100.00 / 100.00
B04 Boundary connections leaving 10 min (13,490 records)	0.144 -0.126	4.64 [1.26, 8.02] 0.54 [0.11, 0.97]	0.110 0.163	100.00 / 100.00 100.00 / 100.00
B02 Local edges and density edges 5 min (13,490 records)	0.144 -0.138	4.96 [1.30, 8.63] 0.55 [0.20, 0.90]	0.118 0.049	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min mean (13,490 records)	0.143 -0.087	3.39 [0.44, 6.34] 0.74 [0.31, 1.17]	0.273 0.028	100.00 / 100.00 100.00 / 100.00
H07 Downstream trains downstream arrival (13,490 records)	0.140 -0.095	3.44 [0.22, 6.65] 0.48 [0.20, 0.76]	0.367 0.028	100.00 / 100.00 100.00 / 100.00
B04 Boundary connections leaving 5 min (13,490 records)	0.140 -0.126	4.46 [1.22, 7.69] 0.48 [0.04, 0.93]	0.108 0.325	100.00 / 100.00 100.00 / 100.00
G08 Active degree outgoing active arrival (13,490 records)	0.139 -0.096	4.19 [1.01, 7.36] 0.70 [0.26, 1.14]	0.136 0.044	100.00 / 100.00 100.00 / 100.00
B01 Local region size region events 5 min (13,490 records)	0.138 -0.134	4.70 [1.02, 8.39] 0.61 [0.22, 0.99]	0.165 0.044	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min count (13,490 records)	0.138 -0.134	4.70 [1.02, 8.39] 0.61 [0.22, 0.99]	0.165 0.044	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support maximum departure (13,490 records)	0.138 -0.082	3.25 [0.43, 6.07] 0.75 [0.32, 1.18]	0.273 0.026	100.00 / 100.00 100.00 / 100.00
H08 Spatial reachability upstream branches departure (13,490 records)	0.138 -0.109	3.90 [0.18, 7.63] 0.78 [0.31, 1.25]	0.402 0.032	100.00 / 100.00 100.00 / 100.00
G02 Reference departure delay departure delay (13,490 records)	0.138 -0.081	3.25 [0.43, 6.08] 0.75 [0.32, 1.18]	0.273 0.026	100.00 / 100.00 100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support maximum arrival (12,610 records)	0.138 -0.093	3.26 [0.43, 6.09] 0.72 [0.31, 1.13]	0.273 0.025	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction arrival (9,875 records)	-0.137 0.087	-4.47 [-7.16, -1.77] -0.37 [-0.79, 0.05]	0.033 0.759	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min median (13,490 records)	0.137 -0.081	3.15 [0.31, 5.98] 0.75 [0.32, 1.18]	0.313 0.025	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution outgoing support 90th percentile departure (13,490 records)	0.136 -0.079	3.10 [0.28, 5.92] 0.75 [0.32, 1.18]	0.324 0.025	100.00 / 100.00 100.00 / 100.00
G12 Delay-support distribution incoming support maximum departure (13,490 records)	0.135 -0.080	3.18 [0.34, 6.02] 0.76 [0.32, 1.19]	0.303 0.025	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution outgoing slack median departure (13,490 records)	0.135 -0.154	-0.37 [-3.32, 2.59] -1.90 [-3.26, -0.55]	1.000 0.090	100.00 / 66.67 100.00 / 100.00
G12 Delay-support distribution outgoing support mean departure (13,490 records)	0.132 -0.077	2.54 [-0.23, 5.32] 0.74 [0.32, 1.15]	0.644 0.024	100.00 / 100.00 100.00 / 100.00
H06 Upstream trains upstream arrival (13,490 records)	0.131 -0.110	3.29 [-0.38, 6.96] 0.47 [0.17, 0.76]	0.704 0.044	100.00 / 100.00 100.00 / 100.00
B04 Boundary connections entering 10 min (13,490 records)	0.131 -0.127	4.53 [0.91, 8.15] 0.56 [0.22, 0.89]	0.178 0.035	100.00 / 100.00 100.00 / 100.00
H01 Shortest source distance source edges departure (6,850 records)	0.130 -0.167	4.02 [0.26, 7.79] -0.08 [-0.16, 0.00]	0.367 0.412	100.00 / 100.00 100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay sum departure (13,490 records)	0.130 -0.086	2.38 [0.23, 4.52] 0.32 [0.01, 0.63]	0.313 0.432	100.00 / 100.00 100.00 / 100.00
G19 Endpoint contrasts run F06 direction pool difference (13,490 records)	0.129 -0.106	4.14 [3.02, 5.26] -1.30 [-1.90, -0.69]	2.2e-6 0.005	100.00 / 100.00 100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum arrival (13,490 records)	0.129 -0.108	3.27 [-0.17, 6.72] 0.42 [0.15, 0.68]	0.583 0.054	100.00 / 100.00 100.00 / 100.00
G03 Activity delay increment dwell delay increment (13,490 records)	0.127 -0.087	-0.73 [-1.28, -0.18] 0.20 [0.11, 0.29]	0.133 0.004	100.00 / 100.00 100.00 / 100.00
G04 Activity-duration increment dwell increment (13,490 records)	0.127 -0.087	-0.73 [-1.28, -0.18] 0.20 [0.11, 0.29]	0.133 0.004	100.00 / 100.00 100.00 / 100.00
G11 Near-tight links slack at most 1 minute departure (13,490 records)	-0.126 0.145	-3.85 [-5.45, -2.24] 1.80 [1.02, 2.58]	0.001 0.003	100.00 / 100.00 100.00 / 100.00
B05 Regional delay distribution region delay 5 min minimum (13,490 records)	0.126 -0.070	2.63 [-0.05, 5.31] 0.75 [0.33, 1.16]	0.512 0.023	100.00 / 100.00 100.00 / 100.00
H09 Reachable delay totals downstream train maximum delay sum arrival (13,490 records)	0.126 -0.085	2.11 [-0.36, 4.57] 0.43 [0.19, 0.68]	0.800 0.024	100.00 / 100.00 100.00 / 100.00
H12 PageRank pagerank arrival (13,490 records)	-0.125 0.122	-3.27 [-4.65, -1.88] 0.74 [0.39, 1.10]	0.001 0.007	100.00 / 100.00 100.00 / 100.00
G10 Edge-slack distribution incoming slack zero fraction arrival (12,610 records)	-0.124 0.103	-4.07 [-5.89, -2.24] 0.27 [0.03, 0.51]	0.003 0.270	100.00 / 100.00 100.00 / 100.00
G19 Endpoint contrasts H05 downstream reachability endpoint contrast (13,490 records)	-0.123 0.098	-0.09 [-0.94, 0.77] 0.00 [-0.02, 0.03]	1.000 1.000	100.00 / 100.00 86.67 / 66.67

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution	0.123	2.40 [-0.21, 5.02]	0.639	100.00 / 100.00
outgoing support minimum arrival (13,490 records)	-0.073	0.68 [0.29, 1.06]	0.024	100.00 / 100.00
G12 Delay-support distribution	0.121	2.42 [-0.34, 5.18]	0.749	100.00 / 100.00
outgoing support median departure (13,490 records)	-0.068	0.73 [0.32, 1.14]	0.023	100.00 / 100.00
B09 Boundary pressure imbalance	-0.121	-2.54 [-4.98, -0.10]	0.410	100.00 / 100.00
cross boundary pressure balance 10 min (13,490 records)	0.143	0.09 [-0.06, 0.25]	1.000	100.00 / 100.00
H09 Reachable delay totals	0.119	2.52 [-0.64, 5.67]	0.981	100.00 / 100.00
upstream event delay arrival (13,490 records)	-0.100	0.42 [0.15, 0.68]	0.044	100.00 / 100.00
H08 Spatial reachability	0.118	2.79 [0.02, 5.55]	0.465	100.00 / 100.00
downstream branches arrival (13,490 records)	-0.066	0.67 [0.29, 1.05]	0.023	100.00 / 100.00
B04 Boundary connections	0.117	4.12 [1.13, 7.11]	0.108	100.00 / 100.00
leaving 20 min (13,490 records)	-0.095	0.61 [0.18, 1.04]	0.086	100.00 / 100.00
G14 Downstream-neighbor delay	0.117	2.19 [0.08, 4.30]	0.417	100.00 / 100.00
neighbor delay sum arrival (13,490 records)	-0.088	0.31 [-0.01, 0.63]	0.500	100.00 / 100.00
G10 Edge-slack distribution	-0.116	-3.33 [-5.02, -1.64]	0.009	100.00 / 100.00
incoming slack median arrival (12,610 records)	0.113	0.21 [0.09, 0.34]	0.026	100.00 / 100.00
B09 Boundary pressure imbalance	-0.116	-1.84 [-3.83, 0.15]	0.628	100.00 / 100.00
cross boundary pressure balance 5 min (13,490 records)	0.132	0.08 [-0.09, 0.24]	1.000	100.00 / 100.00
B05 Regional delay distribution	0.115	2.10 [-0.33, 4.53]	0.778	100.00 / 100.00
region delay 10 min minimum (13,490 records)	-0.060	0.74 [0.33, 1.15]	0.022	100.00 / 100.00
H04 Upstream reachability	0.115	2.27 [-0.99, 5.54]	1.000	100.00 / 100.00
upstream arrival (13,490 records)	-0.096	0.47 [0.19, 0.76]	0.031	100.00 / 100.00
G11 Near-tight links	0.115	4.05 [0.99, 7.11]	0.133	100.00 / 100.00
slack at most 2 minutes arrival (13,490 records)	-0.124	0.39 [0.11, 0.67]	0.091	100.00 / 100.00
G10 Edge-slack distribution	0.114	1.09 [-2.42, 4.60]	1.000	100.00 / 66.67
outgoing slack mean departure (13,490 records)	-0.133	-3.06 [-4.77, -1.34]	0.023	100.00 / 100.00
H02 Longest active path	0.114	3.21 [-0.18, 6.60]	0.583	100.00 / 100.00
longest path arrival (13,490 records)	-0.090	0.59 [0.24, 0.95]	0.032	100.00 / 100.00
H15 Chain and crossing length	0.114	3.21 [-0.18, 6.60]	0.583	100.00 / 100.00
longest chain arrival (13,490 records)	-0.090	0.59 [0.24, 0.95]	0.032	100.00 / 100.00
G10 Edge-slack distribution	-0.113	-2.03 [-3.15, -0.91]	0.018	100.00 / 100.00
incoming slack standard deviation arrival (12,610 records)	0.101	0.16 [0.04, 0.28]	0.131	100.00 / 100.00
G10 Edge-slack distribution	-0.111	-2.03 [-3.19, -0.87]	0.021	100.00 / 100.00
incoming slack interquartile range arrival (12,610 records)	0.100	0.12 [0.00, 0.25]	0.527	100.00 / 100.00
B02 Local edges and density	-0.111	-3.66 [-7.07, -0.26]	0.356	100.00 / 100.00
density 5 min (13,148 records)	0.093	-0.74 [-1.17, -0.32]	0.026	100.00 / 100.00
H01 Shortest source distance	0.110	3.43 [-0.52, 7.39]	0.769	100.00 / 100.00
source minimum time lag departure (6,850 records)	-0.166	-0.08 [-0.17, 0.01]	0.819	100.00 / 100.00
H16 Local motifs	0.109	3.46 [0.88, 6.03]	0.124	100.00 / 100.00
diamonds at source arrival (13,490 records)	-0.089	0.20 [0.05, 0.35]	0.129	100.00 / 100.00
B03 Active local density	0.108	2.49 [0.54, 4.45]	0.165	100.00 / 100.00
active density 5 min (13,148 records)	-0.041	0.80 [0.33, 1.28]	0.029	100.00 / 100.00

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H08 Spatial reachability	0.108	3.08 [-0.14, 6.29]	0.570	100.00 / 100.00
upstream stations arrival (13,490 records)	-0.088	0.63 [0.24, 1.03]	0.041	100.00 / 100.00
B05 Regional delay distribution	0.107	1.71 [-0.68, 4.11]	1.000	100.00 / 100.00
region delay 20 min minimum (13,490 records)	-0.056	0.72 [0.32, 1.12]	0.022	100.00 / 100.00
H11 Harmonic closeness	0.106	2.67 [-0.55, 5.89]	0.882	100.00 / 100.00
incoming harmonic arrival (13,490 records)	-0.085	0.63 [0.25, 1.01]	0.034	100.00 / 100.00
G10 Edge-slack distribution	-0.106	-2.44 [-3.83, -1.05]	0.020	100.00 / 100.00
incoming slack mean arrival (12,610 records)	0.107	0.18 [0.07, 0.30]	0.037	100.00 / 100.00
G10 Edge-slack distribution	-0.105	-2.11 [-3.34, -0.89]	0.023	100.00 / 100.00
incoming slack 90th percentile arrival (12,610 records)	0.103	0.15 [0.03, 0.27]	0.154	100.00 / 100.00
H08 Spatial reachability	0.104	3.05 [-0.05, 6.14]	0.512	100.00 / 100.00
upstream branches arrival (13,490 records)	-0.083	0.68 [0.27, 1.09]	0.032	100.00 / 100.00
H23 Local clustering	0.104	3.21 [2.20, 4.22]	2.5e-5	100.00 / 100.00
undirected clustering arrival (13,490 records)	-0.101	-0.02 [-0.05, 0.02]	1.000	100.00 / 66.67
B15 Global degree distribution	0.103	3.30 [1.70, 4.91]	0.006	100.00 / 100.00
incoming degree maximum (13,490 records)	-0.101	-0.38 [-0.63, -0.13]	0.057	100.00 / 100.00
G12 Delay-support distribution	0.102	1.81 [-0.83, 4.45]	1.000	100.00 / 100.00
outgoing support minimum departure (13,490 records)	-0.055	0.71 [0.32, 1.10]	0.022	100.00 / 100.00
G12 Delay-support distribution	0.101	-1.03 [-2.02, -0.04]	0.417	100.00 / 100.00
incoming support interquartile range arrival (12,610 records)	-0.059	0.51 [0.28, 0.74]	0.004	100.00 / 100.00
G08 Active degree	-0.101	-3.36 [-5.07, -1.64]	0.009	100.00 / 100.00
outgoing active departure (13,490 records)	0.165	1.05 [0.59, 1.51]	0.003	100.00 / 100.00
G12 Delay-support distribution	0.101	-1.09 [-2.07, -0.11]	0.313	100.00 / 100.00
incoming support standard deviation arrival (12,610 records)	-0.058	0.51 [0.28, 0.73]	0.004	100.00 / 100.00
G10 Edge-slack distribution	-0.100	-1.78 [-2.94, -0.61]	0.056	100.00 / 100.00
incoming slack maximum arrival (12,610 records)	0.098	0.13 [0.01, 0.26]	0.384	100.00 / 100.00
H16 Local motifs	-0.097	-2.92 [-4.16, -1.67]	0.001	100.00 / 100.00
fan-out pairs departure (13,490 records)	0.127	0.47 [0.29, 0.65]	0.001	100.00 / 100.00
G14 Downstream-neighbor delay	0.096	3.05 [0.84, 5.27]	0.108	100.00 / 100.00
neighbor delay count departure (13,490 records)	-0.069	0.31 [-0.17, 0.78]	1.000	100.00 / 100.00
G10 Edge-slack distribution	0.093	-0.55 [-3.00, 1.89]	1.000	100.00 / 66.67
outgoing slack sum departure (13,490 records)	-0.110	-1.91 [-2.92, -0.90]	0.015	100.00 / 100.00
G07 Degree by relation	0.091	2.90 [0.82, 4.98]	0.102	100.00 / 100.00
outgoing entry headway departure (13,490 records)	-0.064	0.25 [-0.32, 0.83]	1.000	93.33 / 66.67
H01 Shortest source distance	0.090	2.77 [-0.52, 6.05]	0.834	100.00 / 100.00
source edges arrival (5,927 records)	-0.152	-0.06 [-0.14, 0.01]	0.863	100.00 / 100.00
G13 Upstream-neighbor delay	0.090	2.76 [1.08, 4.45]	0.033	100.00 / 100.00
neighbor delay count arrival (13,490 records)	-0.093	-0.14 [-0.42, 0.15]	1.000	93.33 / 66.67

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Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G08 Active degree	0.089	2.32 [-0.18, 4.82]	0.622	100.00 / 100.00
incoming active arrival (13,490 records)	-0.053	0.69 [0.27, 1.10]	0.032	100.00 / 100.00
B03 Active local density	0.089	1.01 [-0.64, 2.66]	1.000	100.00 / 100.00
active density 10 min (13,341 records)	-0.031	0.72 [0.31, 1.14]	0.026	100.00 / 100.00
H16 Local motifs	-0.088	-2.79 [-3.91, -1.66]	7.7e-4	100.00 / 100.00
diamonds at source departure (13,490 records)	0.101	0.39 [0.24, 0.53]	1.0e-3	100.00 / 100.00
B09 Boundary pressure imbalance	-0.087	-3.16 [-6.27, -0.04]	0.455	100.00 / 100.00
cross boundary pressure balance 20 min (13,490 records)	0.120	0.08 [-0.05, 0.21]	1.000	100.00 / 100.00
G19 Endpoint contrasts	0.087	4.64 [2.83, 6.46]	5.2e-4	100.00 / 100.00
run R01 mean occupancy 10 min difference (13,490 records)	-0.055	-1.91 [-2.71, -1.12]	0.002	100.00 / 100.00
G05 Event degree	0.086	2.74 [0.55, 4.92]	0.177	100.00 / 100.00
outgoing arrival (13,490 records)	-0.077	0.18 [-0.35, 0.71]	1.000	93.33 / 66.67
G06 Cross-train degree	0.086	2.74 [0.55, 4.92]	0.177	100.00 / 100.00
outgoing cross arrival (13,490 records)	-0.077	0.18 [-0.35, 0.71]	1.000	93.33 / 66.67
G07 Degree by relation	0.086	2.74 [0.55, 4.92]	0.177	100.00 / 100.00
outgoing exit headway arrival (13,490 records)	-0.077	0.18 [-0.35, 0.71]	1.000	93.33 / 66.67
G10 Edge-slack distribution	0.086	2.74 [0.55, 4.92]	0.177	100.00 / 100.00
outgoing slack count arrival (13,490 records)	-0.077	0.18 [-0.35, 0.71]	1.000	93.33 / 66.67
G12 Delay-support distribution	0.086	2.74 [0.55, 4.92]	0.177	100.00 / 100.00
outgoing support count arrival (13,490 records)	-0.077	0.18 [-0.35, 0.71]	1.000	93.33 / 66.67
G14 Downstream-neighbor delay	0.086	2.74 [0.55, 4.92]	0.177	100.00 / 100.00
neighbor delay count arrival (13,490 records)	-0.077	0.18 [-0.35, 0.71]	1.000	93.33 / 66.67
G12 Delay-support distribution	-0.086	-3.48 [-4.91, -2.06]	8.7e-4	100.00 / 100.00
incoming support interquartile range departure (13,490 records)	0.126	0.55 [0.34, 0.76]	0.001	100.00 / 100.00
G12 Delay-support distribution	-0.086	-3.48 [-4.90, -2.06]	8.7e-4	100.00 / 100.00
incoming support standard deviation departure (13,490 records)	0.126	0.55 [0.34, 0.76]	0.001	100.00 / 100.00
B03 Active local density	0.086	-0.53 [-2.02, 0.95]	1.000	86.67 / 33.33
active density 20 min (13,426 records)	-0.030	0.68 [0.32, 1.05]	0.016	100.00 / 100.00
B04 Boundary connections	0.085	2.99 [0.00, 5.98]	0.476	100.00 / 100.00
entering 5 min (13,490 records)	-0.096	0.46 [0.18, 0.73]	0.033	100.00 / 100.00
G10 Edge-slack distribution	0.084	-0.35 [-3.35, 2.65]	1.000	100.00 / 66.67
outgoing slack 90th percentile departure (13,490 records)	-0.101	-2.22 [-3.47, -0.98]	0.023	100.00 / 100.00
H11 Harmonic closeness	0.081	1.38 [-0.93, 3.68]	1.000	100.00 / 100.00
outgoing harmonic arrival (13,490 records)	-0.038	0.63 [0.31, 0.96]	0.012	100.00 / 100.00
G14 Downstream-neighbor delay	0.080	1.17 [-0.88, 3.22]	1.000	100.00 / 66.67
neighbor delay minimum departure (10,811 records)	-0.048	0.19 [-0.10, 0.48]	1.000	100.00 / 100.00
G10 Edge-slack distribution	0.079	-0.34 [-3.31, 2.62]	1.000	100.00 / 66.67
outgoing slack maximum departure (13,490 records)	-0.096	-2.26 [-3.47, -1.05]	0.017	100.00 / 100.00
G14 Downstream-neighbor delay	0.078	1.21 [-0.83, 3.25]	1.000	100.00 / 100.00
neighbor delay mean departure (10,811 records)	-0.047	0.19 [-0.10, 0.49]	1.000	100.00 / 100.00

Continued on the next page.

Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G14 Downstream-neighbor delay	0.078	1.21 [-0.83, 3.25]	1.000	100.00 / 100.00
neighbor delay median departure (10,811 records)	-0.047	0.19 [-0.10, 0.49]	1.000	100.00 / 100.00
H05 Downstream reachability	0.078	1.14 [-0.92, 3.20]	1.000	100.00 / 100.00
downstream arrival (13,490 records)	-0.035	0.44 [0.22, 0.67]	0.012	100.00 / 100.00
G10 Edge-slack distribution	-0.078	-2.14 [-3.41, -0.86]	0.029	100.00 / 100.00
incoming slack sum arrival (13,490 records)	0.082	0.16 [0.05, 0.28]	0.083	100.00 / 100.00
H09 Reachable delay totals	0.077	0.23 [-1.27, 1.73]	1.000	93.33 / 66.67
downstream event delay arrival (13,490 records)	-0.034	0.40 [0.19, 0.60]	0.012	100.00 / 100.00
G14 Downstream-neighbor delay	0.076	1.22 [-0.79, 3.23]	1.000	100.00 / 100.00
neighbor delay 90th percentile departure (10,811 records)	-0.045	0.19 [-0.10, 0.49]	1.000	100.00 / 100.00
G14 Downstream-neighbor delay	0.076	1.22 [-0.78, 3.23]	1.000	100.00 / 100.00
neighbor delay maximum departure (10,811 records)	-0.044	0.19 [-0.10, 0.49]	1.000	100.00 / 100.00
H23 Local clustering	0.075	3.50 [3.09, 3.92]	5.0e-15	100.00 / 100.00
undirected clustering departure (13,490 records)	-0.075	-0.02 [-0.05, 0.01]	1.000	100.00 / 66.67
H08 Spatial reachability	0.072	1.54 [-1.00, 4.07]	1.000	100.00 / 100.00
downstream stations arrival (13,490 records)	-0.026	0.67 [0.32, 1.03]	0.014	100.00 / 100.00
G14 Downstream-neighbor delay	0.071	1.09 [-1.05, 3.22]	1.000	100.00 / 66.67
neighbor delay maximum arrival (9,875 records)	-0.048	0.28 [-0.05, 0.60]	0.818	100.00 / 100.00
G14 Downstream-neighbor delay	0.071	1.09 [-1.05, 3.22]	1.000	100.00 / 66.67
neighbor delay mean arrival (9,875 records)	-0.048	0.28 [-0.05, 0.60]	0.818	100.00 / 100.00
G14 Downstream-neighbor delay	0.071	1.09 [-1.05, 3.22]	1.000	100.00 / 66.67
neighbor delay median arrival (9,875 records)	-0.048	0.28 [-0.05, 0.60]	0.818	100.00 / 100.00
G14 Downstream-neighbor delay	0.071	1.09 [-1.05, 3.22]	1.000	100.00 / 66.67
neighbor delay minimum arrival (9,875 records)	-0.048	0.28 [-0.05, 0.60]	0.818	100.00 / 100.00
G14 Downstream-neighbor delay	0.071	1.09 [-1.05, 3.22]	1.000	100.00 / 66.67
neighbor delay 90th percentile arrival (9,875 records)	-0.048	0.28 [-0.05, 0.60]	0.818	100.00 / 100.00
H16 Local motifs	0.069	2.20 [-0.33, 4.73]	0.767	100.00 / 100.00
fan-out pairs arrival (13,490 records)	-0.061	0.37 [0.16, 0.58]	0.025	100.00 / 100.00
G10 Edge-slack distribution	-0.068	-2.34 [-3.64, -1.04]	0.019	100.00 / 100.00
outgoing slack zero fraction arrival (13,490 records)	0.041	0.04 [-0.32, 0.40]	1.000	80.00 / 66.67
G10 Edge-slack distribution	0.065	-0.95 [-1.52, -0.39]	0.029	100.00 / 100.00
outgoing slack minimum arrival (13,490 records)	-0.023	0.19 [0.10, 0.27]	0.004	100.00 / 100.00
G10 Edge-slack distribution	0.063	0.93 [0.11, 1.76]	0.285	100.00 / 100.00
incoming slack minimum departure (13,490 records)	-0.014	-0.05 [-0.12, 0.02]	1.000	93.33 / 66.67
G11 Near-tight links	0.062	2.42 [-1.49, 6.34]	1.000	100.00 / 100.00
slack at most 2 minutes departure (13,490 records)	-0.023	1.72 [0.91, 2.52]	0.006	100.00 / 100.00
G14 Downstream-neighbor delay	0.062	0.31 [-0.85, 1.47]	1.000	100.00 / 66.67
neighbor delay interquartile range departure (10,811 records)	-0.034	0.04 [-0.15, 0.22]	1.000	100.00 / 66.67

Continued on the next page.

Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G14 Downstream-neighbor delay	0.062	0.31 [-0.85, 1.47]	1.000	100.00 / 66.67
neighbor delay standard deviation departure (10,811 records)	-0.034	0.04 [-0.15, 0.22]	1.000	100.00 / 66.67
H01 Shortest source distance	0.060	1.81 [-1.49, 5.11]	1.000	100.00 / 100.00
source minimum time lag arrival (5,927 records)	-0.145	-0.05 [-0.14, 0.04]	1.000	100.00 / 100.00
H09 Reachable delay totals	-0.058	0.09 [-1.38, 1.56]	1.000	80.00 / 66.67
downstream event delay departure (13,490 records)	0.115	0.41 [0.20, 0.61]	0.011	100.00 / 100.00
H09 Reachable delay totals	-0.056	0.03 [-1.69, 1.76]	1.000	60.00 / 66.67
downstream train maximum delay sum departure (13,490 records)	0.113	0.53 [0.26, 0.80]	0.011	100.00 / 100.00
H10 Active betweenness	-0.051	0.95 [-1.41, 3.31]	1.000	100.00 / 100.00
betweenness departure (13,490 records)	0.086	0.47 [0.23, 0.71]	0.012	100.00 / 100.00
G07 Degree by relation	-0.049	-1.57 [-2.29, -0.84]	0.004	100.00 / 100.00
incoming emergency departure (13,490 records)	0.057	0.22 [0.14, 0.31]	1.0e-3	100.00 / 100.00
G18 External-source share	-0.049	-1.50 [-2.23, -0.77]	0.006	100.00 / 100.00
source fraction departure (13,490 records)	0.057	0.22 [0.14, 0.31]	1.0e-3	100.00 / 100.00
G05 Event degree	0.049	1.91 [0.22, 3.60]	0.294	100.00 / 100.00
outgoing departure (13,490 records)	-0.032	0.28 [-0.23, 0.79]	1.000	93.33 / 66.67
G06 Cross-train degree	0.049	1.91 [0.22, 3.60]	0.294	100.00 / 100.00
outgoing cross departure (13,490 records)	-0.032	0.28 [-0.23, 0.79]	1.000	93.33 / 66.67
G10 Edge-slack distribution	0.049	1.91 [0.22, 3.60]	0.294	100.00 / 100.00
outgoing slack count departure (13,490 records)	-0.032	0.28 [-0.23, 0.79]	1.000	93.33 / 66.67
G12 Delay-support distribution	0.049	1.91 [0.22, 3.60]	0.294	100.00 / 100.00
outgoing support count departure (13,490 records)	-0.032	0.28 [-0.23, 0.79]	1.000	93.33 / 66.67
H11 Harmonic closeness	-0.049	-0.28 [-2.12, 1.57]	1.000	93.33 / 33.33
outgoing harmonic departure (13,490 records)	0.109	0.73 [0.39, 1.07]	0.005	100.00 / 100.00
B08 Community structure	-0.048	-1.56 [-2.67, -0.45]	0.094	100.00 / 100.00
cross community edges departure (13,490 records)	0.045	0.38 [0.22, 0.54]	0.002	100.00 / 100.00
G17 Relation-type entropy	-0.048	0.82 [-0.14, 1.78]	0.800	100.00 / 100.00
type entropy arrival (13,490 records)	0.028	-0.14 [-0.30, 0.02]	0.759	100.00 / 100.00
G16 Incident-link imbalance	0.048	1.85 [0.18, 3.52]	0.313	100.00 / 100.00
degree balance departure (13,490 records)	-0.033	0.22 [-0.28, 0.71]	1.000	93.33 / 66.67
G10 Edge-slack distribution	-0.045	-3.18 [-5.09, -1.27]	0.031	100.00 / 100.00
outgoing slack standard deviation departure (13,490 records)	0.039	-0.34 [-0.85, 0.16]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.045	-3.18 [-5.11, -1.25]	0.033	100.00 / 100.00
outgoing slack interquartile range departure (13,490 records)	0.038	-0.34 [-0.84, 0.16]	1.000	100.00 / 100.00
G07 Degree by relation	0.044	1.39 [0.33, 2.46]	0.143	100.00 / 100.00
incoming exit headway arrival (13,490 records)	-0.059	-0.13 [-0.29, 0.03]	0.968	100.00 / 100.00
G16 Incident-link imbalance	0.043	0.56 [-0.86, 1.98]	1.000	100.00 / 66.67
degree balance arrival (13,490 records)	-0.024	0.21 [-0.21, 0.63]	1.000	93.33 / 100.00
H05 Downstream reachability	-0.042	1.11 [-0.95, 3.17]	1.000	100.00 / 100.00
downstream departure (13,490 records)	0.103	0.45 [0.22, 0.68]	0.010	100.00 / 100.00

Continued on the next page.

Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B08 Community structure	-0.041	-1.43 [-2.50, -0.36]	0.124	100.00 / 100.00
cross community edges arrival (13,490 records)	0.041	0.35 [0.20, 0.49]	0.002	100.00 / 100.00
G06 Cross-train degree	0.039	1.24 [0.37, 2.11]	0.086	100.00 / 100.00
incoming cross departure (13,490 records)	-0.046	0.02 [-0.23, 0.27]	1.000	86.67 / 66.67
G07 Degree by relation	0.039	1.24 [0.37, 2.11]	0.086	100.00 / 100.00
incoming entry headway departure (13,490 records)	-0.046	0.02 [-0.23, 0.27]	1.000	86.67 / 66.67
G13 Upstream-neighbor delay	0.039	1.24 [0.37, 2.11]	0.086	100.00 / 100.00
neighbor delay count departure (13,490 records)	-0.046	0.02 [-0.23, 0.27]	1.000	86.67 / 66.67
H08 Spatial reachability	-0.037	1.65 [-1.16, 4.47]	1.000	100.00 / 100.00
downstream stations departure (13,490 records)	0.101	0.68 [0.34, 1.02]	0.010	100.00 / 100.00
H12 PageRank	0.035	-0.32 [-2.53, 1.89]	1.000	86.67 / 66.67
pagerank departure (13,490 records)	-0.020	0.74 [0.37, 1.12]	0.011	100.00 / 100.00
H03 Event time from release	0.033	1.13 [-1.94, 4.21]	1.000	100.00 / 66.67
time from release departure (13,490 records)	-0.045	1.80 [0.92, 2.68]	0.008	100.00 / 100.00
G17 Relation-type entropy	-0.032	-0.42 [-2.09, 1.24]	1.000	100.00 / 66.67
type entropy departure (13,490 records)	0.034	0.29 [-0.04, 0.62]	0.711	100.00 / 100.00
H03 Event time from release	0.031	1.34 [-1.62, 4.31]	1.000	100.00 / 100.00
time from release arrival (13,490 records)	-0.045	1.64 [0.82, 2.45]	0.010	100.00 / 100.00
H08 Spatial reachability	-0.030	1.02 [-1.52, 3.57]	1.000	100.00 / 66.67
downstream branches departure (13,490 records)	0.102	0.80 [0.41, 1.20]	0.009	100.00 / 100.00
G07 Degree by relation	0.028	0.89 [0.00, 1.77]	0.468	100.00 / 100.00
incoming run arrival (13,490 records)	-0.038	-0.17 [-0.28, -0.05]	0.078	100.00 / 100.00
G19 Endpoint contrasts	-0.027	-1.14 [-2.15, -0.14]	0.289	100.00 / 100.00
G05 incoming degree endpoint contrast (13,490 records)	0.042	0.16 [0.02, 0.30]	0.246	100.00 / 100.00
H07 Downstream trains	-0.026	1.12 [-1.16, 3.41]	1.000	100.00 / 100.00
downstream departure (13,490 records)	0.090	0.61 [0.30, 0.92]	0.011	100.00 / 100.00
G10 Edge-slack distribution	-0.023	-2.76 [-4.51, -1.01]	0.046	100.00 / 100.00
outgoing slack maximum arrival (13,490 records)	0.022	-0.20 [-0.62, 0.23]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.023	-2.74 [-4.49, -0.99]	0.047	100.00 / 100.00
outgoing slack 90th percentile arrival (13,490 records)	0.022	-0.18 [-0.60, 0.23]	1.000	100.00 / 100.00
H10 Active betweenness	0.022	0.25 [-1.59, 2.09]	1.000	93.33 / 66.67
betweenness arrival (13,490 records)	-0.003	0.43 [0.21, 0.65]	0.010	100.00 / 100.00
G07 Degree by relation	0.022	0.69 [-1.02, 2.40]	1.000	100.00 / 100.00
outgoing station track departure (13,490 records)	-0.012	0.25 [-0.13, 0.64]	1.000	100.00 / 66.67
G07 Degree by relation	0.020	0.65 [-0.16, 1.46]	0.976	100.00 / 66.67
incoming emergency arrival (13,490 records)	0.022	0.16 [0.09, 0.23]	0.003	100.00 / 100.00
G11 Near-tight links	0.019	0.40 [-1.09, 1.89]	1.000	100.00 / 66.67
slack at most 1 minute arrival (13,490 records)	-0.047	0.38 [0.16, 0.59]	0.027	100.00 / 100.00
G16 Incident-link imbalance	-0.019	-0.32 [-1.73, 1.09]	1.000	100.00 / 66.67
pressure balance departure (13,490 records)	0.038	0.33 [0.13, 0.53]	0.038	100.00 / 100.00

Continued on the next page.

Table S9 continued: Strategy-3 running durations

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution	-0.019	-2.48 [-4.14, -0.82]	0.064	100.00 / 100.00
outgoing slack mean arrival (13,490 records)	0.020	-0.08 [-0.41, 0.25]	1.000	100.00 / 66.67
G10 Edge-slack distribution	-0.019	-2.48 [-4.14, -0.82]	0.064	100.00 / 100.00
outgoing slack median arrival (13,490 records)	0.020	-0.08 [-0.41, 0.25]	1.000	100.00 / 66.67
G10 Edge-slack distribution	-0.019	-2.58 [-4.30, -0.87]	0.061	100.00 / 100.00
outgoing slack interquartile range arrival (13,490 records)	0.009	-0.33 [-0.81, 0.16]	1.000	100.00 / 100.00
G10 Edge-slack distribution	-0.019	-2.58 [-4.30, -0.87]	0.061	100.00 / 100.00
outgoing slack standard deviation arrival (13,490 records)	0.009	-0.33 [-0.81, 0.16]	1.000	100.00 / 100.00
G18 External-source share	0.018	0.59 [-0.23, 1.41]	1.000	100.00 / 66.67
source fraction arrival (12,610 records)	0.023	0.17 [0.09, 0.24]	0.003	100.00 / 100.00
G10 Edge-slack distribution	-0.018	-2.42 [-4.05, -0.78]	0.068	100.00 / 100.00
outgoing slack sum arrival (13,490 records)	0.018	-0.12 [-0.46, 0.22]	1.000	100.00 / 66.67
G05 Event degree	0.015	1.03 [-0.06, 2.13]	0.583	100.00 / 100.00
incoming arrival (13,490 records)	-0.029	-0.11 [-0.26, 0.05]	1.000	100.00 / 100.00
G10 Edge-slack distribution	0.015	1.03 [-0.06, 2.13]	0.583	100.00 / 100.00
incoming slack count arrival (13,490 records)	-0.029	-0.11 [-0.26, 0.05]	1.000	100.00 / 100.00
G12 Delay-support distribution	0.015	1.03 [-0.06, 2.13]	0.583	100.00 / 100.00
incoming support count arrival (13,490 records)	-0.029	-0.11 [-0.26, 0.05]	1.000	100.00 / 100.00
G06 Cross-train degree	0.013	0.79 [-0.34, 1.92]	1.000	100.00 / 100.00
incoming cross arrival (13,490 records)	-0.030	-0.10 [-0.28, 0.07]	1.000	100.00 / 100.00
G19 Endpoint contrasts	0.013	0.65 [-1.08, 2.37]	1.000	100.00 / 100.00
G05 outgoing degree endpoint contrast (13,490 records)	0.010	0.29 [-0.10, 0.67]	1.000	100.00 / 100.00
H16 Local motifs	-0.013	-0.34 [-1.59, 0.91]	1.000	100.00 / 100.00
fan-in pairs arrival (13,490 records)	0.007	0.30 [0.10, 0.50]	0.064	100.00 / 100.00
G05 Event degree	-0.007	-0.23 [-1.39, 0.93]	1.000	93.33 / 66.67
incoming departure (13,490 records)	0.007	0.18 [-0.01, 0.37]	0.624	100.00 / 100.00
G10 Edge-slack distribution	-0.007	-0.23 [-1.39, 0.93]	1.000	93.33 / 66.67
incoming slack count departure (13,490 records)	0.007	0.18 [-0.01, 0.37]	0.624	100.00 / 100.00
G12 Delay-support distribution	-0.007	-0.23 [-1.39, 0.93]	1.000	93.33 / 66.67
incoming support count departure (13,490 records)	0.007	0.18 [-0.01, 0.37]	0.624	100.00 / 100.00
G07 Degree by relation	0.006	0.20 [-1.11, 1.51]	1.000	93.33 / 66.67
incoming station track arrival (13,490 records)	-0.023	-0.07 [-0.26, 0.12]	1.000	100.00 / 100.00
G19 Endpoint contrasts	0.004	0.86 [-1.55, 3.27]	1.000	100.00 / 100.00
H10 active betweenness endpoint contrast (13,490 records)	0.026	0.00 [-0.05, 0.04]	1.000	86.67 / 33.33

Notes: Codes and definitions follow Table S1. Rank corr.: scenario-weighted Spearman correlation. Upper/lower lines: change probability/signed duration change. Effects per indicator standard deviation are in percentage points/minutes, respectively. Brackets contain 95% confidence intervals. Adjusted *p*: Benjamini–Yekutieli p-value; 1.1e-7 denotes 1.1×10^{-7}. Direction stable: unchanged coefficient sign after omitting one start-time group/one blockage location (percent). Pair differences are the following train value minus the leading train value. Record counts include only finite values of the indicated form.

S3.7 Strategy-3 train orders

Table S10: Complete indicator associations: Strategy-3 train orders.

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
Disruption conditions				
X04 Planned blockage overlap planned blockage overlap difference (40,130 records)	-0.544	-16.34 [-19.74, -12.95]	1.6e-8	100.00 / 100.00
X13 Arrival before reopening reaches stop before reopen difference (40,130 records)	-0.517	-15.69 [-18.79, -12.60]	6.1e-9	100.00 / 100.00
X05 Planned overlap fraction planned overlap fraction difference (40,130 records)	-0.516	-15.62 [-18.68, -12.57]	5.7e-9	100.00 / 100.00
E08 Minimum emergency hold minimum emergency hold leading train (40,130 records)	0.494	14.63 [9.94, 19.32]	1.4e-5	100.00 / 100.00
X04 Planned blockage overlap planned blockage overlap leading train (40,130 records)	0.492	15.32 [10.92, 19.72]	3.1e-6	100.00 / 100.00
X16 Blockage-only waiting blockage only wait difference (40,130 records)	-0.491	-13.73 [-17.16, -10.31]	4.5e-7	100.00 / 100.00
X13 Arrival before reopening reaches stop before reopen leading train (40,130 records)	0.482	15.12 [11.25, 18.99]	5.9e-7	100.00 / 100.00
X05 Planned overlap fraction planned overlap fraction leading train (40,130 records)	0.480	14.93 [11.11, 18.75]	5.9e-7	100.00 / 100.00
X16 Blockage-only waiting blockage only wait leading train (40,130 records)	0.454	10.72 [6.21, 15.22]	5.6e-4	100.00 / 100.00
E08 Minimum emergency hold minimum emergency hold difference (40,130 records)	-0.447	-16.57 [-20.32, -12.83]	7.8e-8	100.00 / 100.00
X18 Emergency origin hold origin emergency leading train (40,130 records)	0.447	14.04 [10.98, 17.10]	4.3e-8	100.00 / 100.00
X19 Emergency restriction type direct stop type difference (40,130 records)	-0.446	-12.45 [-15.07, -9.82]	2.2e-8	100.00 / 100.00
X19 Emergency restriction type direct stop type leading train (40,130 records)	0.423	10.79 [7.50, 14.08]	7.3e-6	100.00 / 100.00
X04 Planned blockage overlap planned blockage overlap absolute pair difference (40,130 records)	0.414	12.68 [9.23, 16.13]	1.4e-6	100.00 / 100.00
X16 Blockage-only waiting blockage only wait absolute pair difference (40,130 records)	0.402	11.09 [7.38, 14.79]	2.6e-5	100.00 / 100.00
X13 Arrival before reopening reaches stop before reopen absolute pair difference (40,130 records)	0.401	12.59 [9.45, 15.73]	4.5e-7	100.00 / 100.00
X05 Planned overlap fraction planned overlap fraction absolute pair difference (40,130 records)	0.400	12.49 [9.37, 15.60]	4.5e-7	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
E08 Minimum emergency hold minimum emergency hold absolute pair difference (40,130 records)	0.399	12.99 [8.65, 17.33]	2.6e-5	100.00 / 100.00
X01 Blocked-section exposure crosses closed difference (40,130 records)	-0.363	-11.37 [-14.06, -8.68]	1.8e-7	100.00 / 100.00
X19 Emergency restriction type direct stop type absolute pair difference (40,130 records)	0.348	9.88 [7.21, 12.55]	1.4e-6	100.00 / 100.00
E05 Emergency stopping position first stop index leading train (14,272 records)	-0.342	-16.39 [-19.03, -13.76]	6.0e-11	100.00 / 100.00
X18 Emergency origin hold origin emergency difference (40,130 records)	-0.317	-10.25 [-12.80, -7.70]	4.5e-7	100.00 / 100.00
X01 Blocked-section exposure crosses closed leading train (40,130 records)	0.315	9.90 [7.10, 12.69]	2.6e-6	100.00 / 100.00
D09 Time from start time reference from start leading train (40,130 records)	0.304	9.52 [6.68, 12.37]	5.4e-6	100.00 / 100.00
D09 Time from start time reference from start absolute pair difference (40,130 records)	-0.281	-6.89 [-9.30, -4.48]	5.0e-5	100.00 / 100.00
D09 Time from start time reference from start difference (40,130 records)	-0.281	-6.89 [-9.30, -4.48]	5.0e-5	100.00 / 100.00
D10 Time from reopening reference from release absolute pair difference (40,130 records)	-0.281	-6.89 [-9.30, -4.48]	5.0e-5	100.00 / 100.00
D10 Time from reopening reference from release difference (40,130 records)	-0.281	-6.89 [-9.30, -4.48]	5.0e-5	100.00 / 100.00
X02 Already past blocked section passed at start time leading train (21,930 records)	-0.270	-11.64 [-14.80, -8.47]	1.4e-6	100.00 / 100.00
X03 Planned clearance before start time planned clear before leading train (21,930 records)	-0.270	-11.64 [-14.80, -8.47]	1.4e-6	100.00 / 100.00
X12 Can clear before start time can clear without emergency leading train (21,930 records)	-0.270	-11.64 [-14.80, -8.47]	1.4e-6	100.00 / 100.00
X18 Emergency origin hold origin emergency absolute pair difference (40,130 records)	0.268	8.42 [5.35, 11.49]	8.8e-5	100.00 / 100.00
E20 Planned starts before release planned starts before reopen (40,130 records)	0.252	7.86 [4.59, 11.13]	5.0e-4	100.00 / 100.00
E12 Origin waiting count origin waiting (40,130 records)	0.249	7.31 [4.50, 10.12]	1.7e-4	100.00 / 100.00
E21 Train-state counts at blockage start origin hold (40,130 records)	0.249	7.31 [4.50, 10.12]	1.7e-4	100.00 / 100.00
E16 Emergency backlog backlog (40,130 records)	0.247	7.32 [4.57, 10.07]	1.3e-4	100.00 / 100.00
X01 Blocked-section exposure crosses closed absolute pair difference (40,130 records)	0.224	7.04 [2.88, 11.21]	0.023	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
E05 Emergency stopping position first stop index absolute pair difference (6,911 records)	-0.204	-6.12 [-9.07, -3.17]	0.003	100.00 / 100.00
E05 Emergency stopping position first stop index difference (6,911 records)	0.204	6.12 [3.17, 9.07]	0.003	100.00 / 100.00
E07 Emergency arrival time emergency arrival absolute pair difference (6,911 records)	-0.201	-4.59 [-5.63, -3.56]	7.8e-8	100.00 / 100.00
D10 Time from reopening reference from release leading train (40,130 records)	0.192	6.05 [3.84, 8.25]	9.0e-5	100.00 / 100.00
X07 Planned exit from start time planned exit from start time leading train (21,930 records)	0.188	8.07 [4.40, 11.74]	0.002	100.00 / 100.00
X06 Planned entry from start time planned entry from start time leading train (21,930 records)	0.179	7.85 [4.23, 11.47]	0.002	100.00 / 100.00
X09 Minimum time to entry minrun to entry difference (16,745 records)	-0.178	-6.32 [-10.00, -2.65]	0.019	100.00 / 100.00
D03 Blockage duration duration (40,130 records)	0.174	5.34 [3.11, 7.58]	5.4e-4	100.00 / 100.00
D08 Remaining blockage remaining (40,130 records)	0.174	5.34 [3.11, 7.58]	5.4e-4	100.00 / 100.00
D20 Lost resource time lost resource minutes (40,130 records)	0.174	5.34 [3.11, 7.58]	5.4e-4	100.00 / 100.00
I05 Predicted remaining duration predicted remaining (40,130 records)	0.174	5.34 [3.11, 7.58]	5.4e-4	100.00 / 100.00
I06 Remaining-duration bounds lower (40,130 records)	0.174	5.34 [3.11, 7.58]	5.4e-4	100.00 / 100.00
I06 Remaining-duration bounds upper (40,130 records)	0.174	5.34 [3.11, 7.58]	5.4e-4	100.00 / 100.00
I09 Remaining-duration quantiles 10th percentile (40,130 records)	0.174	5.34 [3.11, 7.58]	5.4e-4	100.00 / 100.00
I09 Remaining-duration quantiles median (40,130 records)	0.174	5.34 [3.11, 7.58]	5.4e-4	100.00 / 100.00
I09 Remaining-duration quantiles 90th percentile (40,130 records)	0.174	5.34 [3.11, 7.58]	5.4e-4	100.00 / 100.00
E02 Executed route fraction executed fraction leading train (40,130 records)	-0.171	-6.06 [-8.39, -3.74]	1.7e-4	100.00 / 100.00
E03 Current route position current route index leading train (40,130 records)	-0.171	-6.06 [-8.39, -3.74]	1.7e-4	100.00 / 100.00
X09 Minimum time to entry minrun to entry leading train (21,930 records)	0.164	6.96 [4.45, 9.47]	7.8e-5	100.00 / 100.00
E17 Early release concentration release first5 fraction (40,130 records)	-0.160	-4.14 [-6.62, -1.66]	0.025	100.00 / 100.00
X15 Earliest entry from release earliest entry minus release leading train (21,930 records)	-0.159	-5.77 [-9.65, -1.88]	0.063	100.00 / 100.00
D09 Time from start time reference from start following train (40,130 records)	0.157	4.93 [2.85, 7.00]	5.9e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
E05 Emergency stopping position first stop index following train (9,770 records)	-0.156	-3.27 [-4.43, -2.11]	6.0e-5	100.00 / 100.00
E02 Executed route fraction executed fraction absolute pair difference (40,130 records)	-0.156	-5.24 [-7.61, -2.88]	0.001	100.00 / 100.00
E02 Executed route fraction executed fraction difference (40,130 records)	0.156	5.24 [2.88, 7.61]	0.001	100.00 / 100.00
E03 Current route position current route index absolute pair difference (40,130 records)	-0.156	-5.24 [-7.61, -2.88]	0.001	100.00 / 100.00
E03 Current route position current route index difference (40,130 records)	0.156	5.24 [2.88, 7.61]	0.001	100.00 / 100.00
E06 Time to emergency station time to stop absolute pair difference (6,911 records)	-0.152	-3.41 [-5.49, -1.34]	0.028	100.00 / 100.00
E07 Emergency arrival time emergency arrival following train (9,770 records)	-0.151	-3.53 [-5.75, -1.30]	0.039	100.00 / 100.00
D09 Time from start time planned from start leading train (40,130 records)	0.145	3.84 [1.43, 6.24]	0.037	100.00 / 100.00
X02 Already past blocked section passed at start time difference (16,745 records)	0.142	4.69 [3.25, 6.13]	7.7e-6	100.00 / 100.00
X03 Planned clearance before start time planned clear before difference (16,745 records)	0.142	4.69 [3.25, 6.13]	7.7e-6	100.00 / 100.00
X12 Can clear before start time can clear without emergency difference (16,745 records)	0.142	4.69 [3.25, 6.13]	7.7e-6	100.00 / 100.00
X02 Already past blocked section passed at start time absolute pair difference (16,745 records)	-0.142	-4.69 [-6.13, -3.25]	7.7e-6	100.00 / 100.00
X03 Planned clearance before start time planned clear before absolute pair difference (16,745 records)	-0.142	-4.69 [-6.13, -3.25]	7.7e-6	100.00 / 100.00
X12 Can clear before start time can clear without emergency absolute pair difference (16,745 records)	-0.142	-4.69 [-6.13, -3.25]	7.7e-6	100.00 / 100.00
X08 Sections to blockage entry intervals to entry leading train (21,930 records)	0.138	5.94 [3.03, 8.84]	0.004	100.00 / 100.00
X06 Planned entry from start time planned entry from start time following train (21,675 records)	0.135	3.81 [1.42, 6.20]	0.037	100.00 / 100.00
E07 Emergency arrival time emergency arrival difference (6,911 records)	-0.134	-4.81 [-6.85, -2.78]	6.1e-4	100.00 / 100.00
D10 Time from reopening planned from release following train (40,130 records)	-0.133	-3.62 [-5.77, -1.46]	0.024	100.00 / 100.00
X07 Planned exit from start time planned exit from start time following train (21,675 records)	0.129	3.69 [1.34, 6.04]	0.042	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
E21 Train-state counts at blockage start finish committed run then hold (40,130 records)	0.126	4.72 [1.49, 7.96]	0.071	100.00 / 100.00
E06 Time to emergency station time to stop leading train (14,272 records)	-0.125	-3.79 [-8.18, 0.59]	0.954	100.00 / 100.00
E13 Station waiting count station waiting (40,130 records)	0.125	4.79 [1.61, 7.98]	0.058	100.00 / 100.00
X08 Sections to blockage entry intervals to entry absolute pair difference (16,745 records)	-0.119	-4.21 [-7.07, -1.34]	0.068	100.00 / 100.00
X08 Sections to blockage entry intervals to entry difference (16,745 records)	-0.119	-4.21 [-7.07, -1.34]	0.068	100.00 / 100.00
E01 Train state at start time state 0 leading train (40,130 records)	0.112	3.52 [1.49, 5.56]	0.019	100.00 / 100.00
E02 Executed route fraction executed fraction following train (40,130 records)	-0.110	-3.78 [-5.79, -1.77]	0.009	100.00 / 100.00
E03 Current route position current route index following train (40,130 records)	-0.110	-3.78 [-5.79, -1.77]	0.009	100.00 / 100.00
E01 Train state at start time state 0 following train (40,130 records)	0.108	3.40 [1.05, 5.76]	0.076	100.00 / 100.00
E06 Time to emergency station time to stop following train (9,770 records)	-0.107	-1.78 [-3.20, -0.37]	0.183	100.00 / 100.00
E06 Time to emergency station time to stop difference (6,911 records)	-0.106	-4.46 [-6.49, -2.44]	0.001	100.00 / 100.00
X17 Remaining route fraction remaining route fraction leading train (40,130 records)	0.105	3.36 [2.05, 4.67]	2.1e-4	100.00 / 100.00
X17 Remaining route fraction remaining route fraction following train (40,130 records)	0.105	3.36 [2.05, 4.67]	2.1e-4	100.00 / 100.00
E07 Emergency arrival time emergency arrival leading train (14,272 records)	-0.105	-7.13 [-16.23, 1.97]	1.000	100.00 / 100.00
E21 Train-state counts at blockage start completed (40,130 records)	-0.105	-3.97 [-6.63, -1.31]	0.061	100.00 / 100.00
X08 Sections to blockage entry intervals to entry following train (21,675 records)	0.101	2.78 [0.67, 4.89]	0.143	100.00 / 100.00
X10 Signed blockage distance signed distance to blockage leading train (21,930 records)	0.100	4.97 [1.58, 8.36]	0.070	100.00 / 100.00
E01 Train state at start time state 2 leading train (40,130 records)	-0.100	-3.14 [-5.10, -1.18]	0.036	100.00 / 100.00
D10 Time from reopening planned from release leading train (40,130 records)	-0.097	-2.76 [-4.70, -0.83]	0.084	100.00 / 100.00
E21 Train-state counts at blockage start hold current station (40,130 records)	0.087	2.79 [0.37, 5.21]	0.300	100.00 / 100.00
E04 Committed running remainder committed remaining absolute pair difference (40,130 records)	-0.086	-1.54 [-3.36, 0.28]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
D01 Blockage start time start (40,130 records)	-0.086	-3.93 [-6.70, -1.16]	0.087	100.00 / 100.00
E19 Committed event count fixed events (40,130 records)	-0.086	-3.99 [-6.73, -1.25]	0.072	100.00 / 100.00
E04 Committed running remainder committed remaining leading train (40,130 records)	-0.085	-1.40 [-3.13, 0.32]	1.000	100.00 / 100.00
D02 Blockage end end (40,130 records)	0.084	1.53 [-0.51, 3.58]	1.000	100.00 / 100.00
E04 Committed running remainder committed remaining following train (40,130 records)	-0.081	-2.63 [-4.19, -1.07]	0.023	100.00 / 100.00
E01 Train state at start time state 2 following train (40,130 records)	-0.080	-2.53 [-4.95, -0.11]	0.476	100.00 / 100.00
X04 Planned blockage overlap planned blockage overlap following train (40,130 records)	-0.078	-2.81 [-5.42, -0.20]	0.413	100.00 / 100.00
D09 Time from start time planned from start absolute pair difference (40,130 records)	-0.071	-2.38 [-5.04, 0.28]	0.858	100.00 / 100.00
D09 Time from start time planned from start difference (40,130 records)	-0.071	-2.38 [-5.04, 0.28]	0.858	100.00 / 100.00
D10 Time from reopening planned from release absolute pair difference (40,130 records)	-0.071	-2.38 [-5.04, 0.28]	0.858	100.00 / 100.00
D10 Time from reopening planned from release difference (40,130 records)	-0.071	-2.38 [-5.04, 0.28]	0.858	100.00 / 100.00
E01 Train state at start time state 1 following train (40,130 records)	-0.071	-2.24 [-3.03, -1.46]	5.2e-5	100.00 / 100.00
E09 Already stopped at start time already stopped following train (40,130 records)	-0.071	-2.24 [-3.03, -1.46]	5.2e-5	100.00 / 100.00
X19 Emergency restriction type direct stop type following train (40,130 records)	-0.070	-2.93 [-5.22, -0.65]	0.165	100.00 / 100.00
D11 Blocked section U11 U12 (40,130 records)	-0.069	-2.15 [-4.18, -0.13]	0.441	100.00 / 100.00
E26 Committed movement count committed runs (40,130 records)	-0.067	-2.25 [-5.16, 0.65]	1.000	100.00 / 100.00
X10 Signed blockage distance signed distance to blockage following train (21,675 records)	0.067	1.87 [-0.81, 4.56]	1.000	100.00 / 100.00
X09 Minimum time to entry minrun to entry absolute pair difference (16,745 records)	-0.066	-2.93 [-5.70, -0.16]	0.452	100.00 / 100.00
X16 Blockage-only waiting blockage only wait following train (40,130 records)	-0.063	-2.34 [-4.57, -0.11]	0.464	100.00 / 100.00
X13 Arrival before reopening reaches stop before reopen following train (40,130 records)	-0.058	-1.81 [-4.78, 1.15]	1.000	100.00 / 100.00
X05 Planned overlap fraction planned overlap fraction following train (40,130 records)	-0.057	-1.78 [-4.74, 1.17]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
X09 Minimum time to entry minrun to entry following train (21,675 records)	0.054	1.81 [-0.52, 4.15]	1.000	100.00 / 100.00
D04 Start-time wave phase start phase40 (40,130 records)	-0.049	-2.24 [-4.67, 0.18]	0.765	100.00 / 100.00
E08 Minimum emergency hold minimum emergency hold following train (40,130 records)	-0.047	-1.02 [-4.01, 1.96]	1.000	100.00 / 66.67
D10 Time from reopening reference from release following train (40,130 records)	-0.047	0.21 [-1.43, 1.85]	1.000	86.67 / 66.67
D11 Blocked section H1 C (40,130 records)	0.044	1.39 [-2.26, 5.04]	1.000	100.00 / 100.00
E01 Train state at start time state 1 absolute pair difference (40,130 records)	-0.044	-1.39 [-3.63, 0.85]	1.000	100.00 / 66.67
E09 Already stopped at start time already stopped absolute pair difference (40,130 records)	-0.044	-1.39 [-3.63, 0.85]	1.000	100.00 / 66.67
X06 Planned entry from start time planned entry from start time absolute pair difference (16,745 records)	0.042	0.79 [-1.05, 2.63]	1.000	93.33 / 66.67
X06 Planned entry from start time planned entry from start time difference (16,745 records)	0.042	0.79 [-1.05, 2.63]	1.000	93.33 / 66.67
X14 Earliest-entry estimate earliest entry no other conflicts absolute pair difference (16,745 records)	0.042	0.79 [-1.05, 2.63]	1.000	93.33 / 66.67
X14 Earliest-entry estimate earliest entry no other conflicts difference (16,745 records)	0.042	0.79 [-1.05, 2.63]	1.000	93.33 / 66.67
X15 Earliest entry from release earliest entry minus release absolute pair difference (16,745 records)	0.042	0.79 [-1.05, 2.63]	1.000	93.33 / 66.67
X15 Earliest entry from release earliest entry minus release difference (16,745 records)	0.042	0.79 [-1.05, 2.63]	1.000	93.33 / 66.67
X02 Already past blocked section passed at start time following train (21,675 records)	-0.040	-1.10 [-1.57, -0.62]	9.1e-4	100.00 / 100.00
X03 Planned clearance before start time planned clear before following train (21,675 records)	-0.040	-1.10 [-1.57, -0.62]	9.1e-4	100.00 / 100.00
X12 Can clear before start time can clear without emergency following train (21,675 records)	-0.040	-1.10 [-1.57, -0.62]	9.1e-4	100.00 / 100.00
E01 Train state at start time state 2 absolute pair difference (40,130 records)	-0.038	-1.18 [-2.74, 0.37]	1.000	100.00 / 100.00
X01 Blocked-section exposure crosses closed following train (40,130 records)	-0.038	-1.18 [-5.22, 2.86]	1.000	100.00 / 66.67
D09 Time from start time planned from start following train (40,130 records)	0.033	1.12 [-1.25, 3.50]	1.000	100.00 / 66.67
E01 Train state at start time state 2 difference (40,130 records)	0.032	0.91 [-1.60, 3.41]	1.000	100.00 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
X14 Earliest-entry estimate earliest entry no other conflicts following train (21,675 records)	-0.028	-0.64 [-2.20, 0.93]	1.000	100.00 / 66.67
X18 Emergency origin hold origin emergency following train (40,130 records)	0.026	0.81 [-1.67, 3.30]	1.000	93.33 / 66.67
D11 Blocked section U13 H1 (40,130 records)	0.024	0.76 [-1.96, 3.48]	1.000	100.00 / 50.00
X11 At blockage entrance at closed entry leading train (40,130 records)	0.024	0.74 [-1.00, 2.49]	1.000	100.00 / 66.67
X11 At blockage entrance at closed entry following train (40,130 records)	0.024	0.74 [-1.00, 2.49]	1.000	100.00 / 66.67
E01 Train state at start time state 1 difference (40,130 records)	-0.020	-0.69 [-2.90, 1.52]	1.000	93.33 / 66.67
E09 Already stopped at start time already stopped difference (40,130 records)	-0.020	-0.69 [-2.90, 1.52]	1.000	93.33 / 66.67
X15 Earliest entry from release earliest entry minus release following train (21,675 records)	-0.017	-0.06 [-1.51, 1.40]	1.000	46.67 / 66.67
E01 Train state at start time state 0 absolute pair difference (40,130 records)	-0.016	-0.50 [-2.20, 1.19]	1.000	93.33 / 66.67
E01 Train state at start time state 0 difference (40,130 records)	-0.016	-0.50 [-2.20, 1.19]	1.000	93.33 / 66.67
X07 Planned exit from start time planned exit from start time absolute pair difference (16,745 records)	0.015	0.19 [-1.52, 1.90]	1.000	60.00 / 66.67
X07 Planned exit from start time planned exit from start time difference (16,745 records)	0.015	0.19 [-1.52, 1.90]	1.000	60.00 / 66.67
E01 Train state at start time state 1 leading train (40,130 records)	-0.015	-0.46 [-2.82, 1.90]	1.000	100.00 / 33.33
E09 Already stopped at start time already stopped leading train (40,130 records)	-0.015	-0.46 [-2.82, 1.90]	1.000	100.00 / 33.33
D05 Reopening wave phase end phase40 (40,130 records)	-0.011	-0.39 [-2.53, 1.75]	1.000	93.33 / 33.33
E04 Committed running remainder committed remaining difference (40,130 records)	0.006	-0.81 [-2.56, 0.94]	1.000	93.33 / 66.67
X14 Earliest-entry estimate earliest entry no other conflicts leading train (21,930 records)	-0.002	-0.41 [-4.74, 3.91]	1.000	73.33 / 66.67
Timetable characteristics				
T12 Planned local flow planned flow 5 min leading train (40,130 records)	-0.267	-7.47 [-9.66, -5.29]	3.8e-6	100.00 / 100.00
T12 Planned local flow planned flow 5 min following train (40,130 records)	-0.242	-6.91 [-9.16, -4.66]	1.9e-5	100.00 / 100.00
T12 Planned local flow planned flow 10 min following train (40,130 records)	-0.220	-6.11 [-8.02, -4.20]	1.0e-5	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
T12 Planned local flow planned flow 10 min leading train (40,130 records)	-0.218	-5.75 [-7.71, -3.78]	3.7e-5	100.00 / 100.00
T12 Planned local flow planned flow 20 min leading train (40,130 records)	-0.195	-5.48 [-7.88, -3.09]	9.2e-4	100.00 / 100.00
T12 Planned local flow planned flow 5 min absolute pair difference (40,130 records)	-0.180	-5.10 [-6.96, -3.25]	8.6e-5	100.00 / 100.00
T12 Planned local flow planned flow 20 min following train (40,130 records)	-0.163	-4.37 [-6.36, -2.37]	0.002	100.00 / 100.00
T13 Planned train-type pattern fast fraction 10 min absolute pair difference (15,484 records)	-0.157	-3.01 [-5.71, -0.32]	0.346	100.00 / 100.00
F13 Minimum running time minimum run difference (40,130 records)	-0.149	-4.70 [-7.64, -1.76]	0.036	100.00 / 100.00
T04 Planned running duration planned run next difference (40,130 records)	-0.149	-4.70 [-7.64, -1.76]	0.036	100.00 / 100.00
T15 Fast-train indicator fast difference (40,130 records)	0.149	4.70 [1.76, 7.64]	0.036	100.00 / 100.00
T17 Total planned journey time planned total difference (40,130 records)	-0.149	-4.70 [-7.64, -1.76]	0.036	100.00 / 100.00
T13 Planned train-type pattern alternation 20 min leading train (16,758 records)	-0.148	-3.51 [-5.16, -1.86]	0.002	100.00 / 100.00
T13 Planned train-type pattern fast fraction 5 min absolute pair difference (12,069 records)	-0.144	-2.04 [-4.37, 0.29]	0.917	100.00 / 100.00
T10 Planned exit buffer previous buffer difference (36,410 records)	-0.137	-4.25 [-7.63, -0.87]	0.188	100.00 / 100.00
M06 Remaining candidate decisions remaining stopping decisions following train (40,130 records)	0.136	4.34 [2.74, 5.94]	1.0e-4	100.00 / 100.00
M06 Remaining candidate decisions remaining order decisions leading train (40,130 records)	0.136	4.16 [2.08, 6.24]	0.004	100.00 / 100.00
M06 Remaining candidate decisions remaining order decisions following train (40,130 records)	0.135	4.14 [2.06, 6.22]	0.005	100.00 / 100.00
T20 Remaining scheduled stops remaining scheduled stops leading train (40,130 records)	0.134	4.08 [1.66, 6.50]	0.023	100.00 / 100.00
T18 Remaining planned journey remaining planned difference (40,130 records)	-0.132	-3.64 [-6.70, -0.59]	0.252	100.00 / 100.00
T19 Remaining minimum journey remaining minimum difference (40,130 records)	-0.132	-3.64 [-6.70, -0.59]	0.252	100.00 / 100.00
M06 Remaining candidate decisions remaining stopping decisions difference (40,130 records)	0.132	3.89 [0.88, 6.90]	0.160	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
T18 Remaining planned journey remaining planned leading train (40,130 records)	0.132	4.13 [1.96, 6.30]	0.008	100.00 / 100.00
T19 Remaining minimum journey remaining minimum leading train (40,130 records)	0.132	4.13 [1.96, 6.30]	0.008	100.00 / 100.00
T10 Planned exit buffer previous buffer following train (40,130 records)	-0.128	-4.07 [-6.01, -2.13]	0.003	100.00 / 100.00
T02 Planned departure time planned departure following train (40,130 records)	-0.126	-4.01 [-5.85, -2.17]	0.002	100.00 / 100.00
F13 Minimum running time minimum run following train (40,130 records)	-0.125	-3.94 [-5.32, -2.56]	5.0e-5	100.00 / 100.00
T04 Planned running duration planned run next following train (40,130 records)	-0.125	-3.94 [-5.32, -2.56]	5.0e-5	100.00 / 100.00
T15 Fast-train indicator fast following train (40,130 records)	0.125	3.94 [2.56, 5.32]	5.0e-5	100.00 / 100.00
T17 Total planned journey time planned total following train (40,130 records)	-0.125	-3.94 [-5.32, -2.56]	5.0e-5	100.00 / 100.00
T01 Planned arrival time planned arrival following train (40,130 records)	-0.124	-3.97 [-5.81, -2.12]	0.002	100.00 / 100.00
T24 Planned terminal position terminal from release following train (40,130 records)	-0.119	-3.11 [-5.18, -1.04]	0.058	100.00 / 100.00
T09 Planned entry buffer previous buffer difference (36,410 records)	-0.118	-3.62 [-6.87, -0.38]	0.350	100.00 / 100.00
T09 Planned entry buffer previous buffer following train (40,130 records)	-0.115	-3.77 [-5.51, -2.03]	0.002	100.00 / 100.00
T20 Remaining scheduled stops remaining scheduled stops difference (40,130 records)	-0.115	-3.41 [-6.45, -0.37]	0.340	100.00 / 100.00
M06 Remaining candidate decisions remaining track-assignment decisions leading train (40,130 records)	0.115	3.69 [2.33, 5.05]	1.1e-4	100.00 / 100.00
T13 Planned train-type pattern alternation 20 min following train (13,282 records)	-0.113	-3.59 [-6.12, -1.05]	0.088	100.00 / 100.00
F12 Minimum dwell minimum dwell difference (40,130 records)	-0.113	-3.53 [-5.42, -1.64]	0.009	100.00 / 100.00
F14 Allowed stopping mode required difference (40,130 records)	-0.113	-3.53 [-5.42, -1.64]	0.009	100.00 / 100.00
T03 Planned dwell planned dwell difference (40,130 records)	-0.113	-3.53 [-5.42, -1.64]	0.009	100.00 / 100.00
T05 Planned stopping mode planned stop difference (40,130 records)	-0.113	-3.53 [-5.42, -1.64]	0.009	100.00 / 100.00
F14 Allowed stopping mode optional difference (40,130 records)	0.108	3.40 [1.56, 5.23]	0.010	100.00 / 100.00

Continued on the next page.

Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
M06 Remaining candidate decisions remaining duration decisions leading train (40,130 records)	0.105	3.41 [2.10, 4.71]	1.7e-4	100.00 / 100.00
M06 Remaining candidate decisions remaining duration decisions following train (40,130 records)	0.105	3.41 [2.10, 4.71]	1.7e-4	100.00 / 100.00
T22 Route extent remaining edges leading train (40,130 records)	0.105	3.36 [2.05, 4.67]	2.1e-4	100.00 / 100.00
T22 Route extent remaining edges following train (40,130 records)	0.105	3.36 [2.05, 4.67]	2.1e-4	100.00 / 100.00
M06 Remaining candidate decisions remaining track-assignment decisions following train (40,130 records)	0.105	3.45 [2.15, 4.75]	1.4e-4	100.00 / 100.00
T13 Planned train-type pattern alternation 20 min absolute pair difference (12,732 records)	0.104	2.24 [0.62, 3.87]	0.106	100.00 / 100.00
T06 Planned track label planned track index leading train (38,510 records)	0.103	3.00 [0.91, 5.10]	0.081	100.00 / 100.00
F13 Minimum running time minimum run leading train (40,130 records)	0.101	3.18 [-0.07, 6.44]	0.621	100.00 / 100.00
T04 Planned running duration planned run next leading train (40,130 records)	0.101	3.18 [-0.07, 6.44]	0.621	100.00 / 100.00
T15 Fast-train indicator fast leading train (40,130 records)	-0.101	-3.18 [-6.44, 0.07]	0.621	100.00 / 100.00
T17 Total planned journey time planned total leading train (40,130 records)	0.101	3.18 [-0.07, 6.44]	0.621	100.00 / 100.00
T06 Planned track label planned track index difference (38,510 records)	-0.100	-3.00 [-5.28, -0.71]	0.146	100.00 / 100.00
T21 Nearby scheduled-stop distance next scheduled-stop distance difference (40,130 records)	0.093	2.97 [0.58, 5.35]	0.199	100.00 / 100.00
T21 Nearby scheduled-stop distance previous scheduled-stop distance difference (40,130 records)	0.090	2.88 [0.55, 5.21]	0.208	100.00 / 100.00
T21 Nearby scheduled-stop distance next scheduled-stop distance following train (40,130 records)	0.088	2.78 [1.75, 3.81]	1.1e-4	100.00 / 100.00
T24 Planned terminal position planned terminal following train (40,130 records)	-0.086	-3.27 [-5.12, -1.41]	0.016	100.00 / 100.00
T24 Planned terminal position planned terminal absolute pair difference (40,130 records)	-0.085	-3.52 [-6.81, -0.24]	0.424	100.00 / 100.00
T24 Planned terminal position planned terminal difference (40,130 records)	-0.085	-3.52 [-6.81, -0.24]	0.424	100.00 / 100.00
T24 Planned terminal position terminal from release absolute pair difference (40,130 records)	-0.085	-3.52 [-6.81, -0.24]	0.424	100.00 / 100.00

Continued on the next page.

Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
T24 Planned terminal position terminal from release difference (40,130 records)	-0.085	-3.52 [-6.81, -0.24]	0.424	100.00 / 100.00
T11 Planned track buffer planned track buffer following train (37,840 records)	-0.084	-2.79 [-4.42, -1.16]	0.020	100.00 / 100.00
T13 Planned train-type pattern alternation 20 min difference (12,732 records)	0.082	1.89 [0.23, 3.55]	0.318	100.00 / 100.00
T13 Planned train-type pattern fast fraction 20 min absolute pair difference (21,152 records)	-0.082	-1.80 [-3.38, -0.23]	0.305	100.00 / 100.00
T12 Planned local flow planned flow 20 min absolute pair difference (40,130 records)	-0.075	-1.82 [-3.58, -0.05]	0.510	100.00 / 100.00
T12 Planned local flow planned flow 10 min absolute pair difference (40,130 records)	-0.074	-2.06 [-3.79, -0.33]	0.251	100.00 / 100.00
F12 Minimum dwell minimum dwell following train (40,130 records)	-0.073	-2.28 [-3.35, -1.21]	0.002	100.00 / 100.00
F14 Allowed stopping mode required following train (40,130 records)	-0.073	-2.28 [-3.35, -1.21]	0.002	100.00 / 100.00
T03 Planned dwell planned dwell following train (40,130 records)	-0.073	-2.28 [-3.35, -1.21]	0.002	100.00 / 100.00
T05 Planned stopping mode planned stop following train (40,130 records)	-0.073	-2.28 [-3.35, -1.21]	0.002	100.00 / 100.00
F14 Allowed stopping mode optional leading train (40,130 records)	-0.072	-2.27 [-4.06, -0.48]	0.179	100.00 / 100.00
F14 Allowed stopping mode optional following train (40,130 records)	0.072	2.27 [1.27, 3.26]	0.001	100.00 / 100.00
T02 Planned departure time planned departure absolute pair difference (40,130 records)	-0.071	-2.38 [-5.04, 0.28]	0.858	100.00 / 100.00
T02 Planned departure time planned departure difference (40,130 records)	-0.071	-2.38 [-5.04, 0.28]	0.858	100.00 / 100.00
T21 Nearby scheduled-stop distance previous scheduled-stop distance following train (40,130 records)	0.071	2.15 [1.56, 2.74]	1.6e-6	100.00 / 100.00
F12 Minimum dwell minimum dwell leading train (40,130 records)	0.070	2.20 [0.51, 3.90]	0.153	100.00 / 100.00
F14 Allowed stopping mode required leading train (40,130 records)	0.070	2.20 [0.51, 3.90]	0.153	100.00 / 100.00
T03 Planned dwell planned dwell leading train (40,130 records)	0.070	2.20 [0.51, 3.90]	0.153	100.00 / 100.00
T05 Planned stopping mode planned stop leading train (40,130 records)	0.070	2.20 [0.51, 3.90]	0.153	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
M06 Remaining candidate decisions remaining track-assignment decisions absolute pair difference (40,130 records)	-0.070	-2.19 [-3.29, -1.10]	0.004	100.00 / 100.00
M06 Remaining candidate decisions remaining track-assignment decisions difference (40,130 records)	-0.070	-2.19 [-3.29, -1.10]	0.004	100.00 / 100.00
T12 Planned local flow planned flow 20 min difference (40,130 records)	0.068	1.64 [-0.10, 3.38]	0.716	100.00 / 100.00
T13 Planned train-type pattern fast fraction 20 min leading train (27,329 records)	-0.067	-1.42 [-2.99, 0.15]	0.831	100.00 / 100.00
T11 Planned track buffer planned track buffer leading train (29,180 records)	-0.067	-1.35 [-3.19, 0.49]	1.000	100.00 / 100.00
T13 Planned train-type pattern fast fraction 5 min leading train (17,642 records)	-0.067	-1.20 [-3.14, 0.74]	1.000	100.00 / 100.00
T01 Planned arrival time planned arrival absolute pair difference (40,130 records)	-0.066	-2.21 [-4.83, 0.40]	1.000	100.00 / 100.00
T01 Planned arrival time planned arrival difference (40,130 records)	-0.066	-2.21 [-4.83, 0.40]	1.000	100.00 / 100.00
T24 Planned terminal position terminal from release leading train (40,130 records)	-0.063	-1.53 [-3.40, 0.34]	1.000	100.00 / 100.00
T13 Planned train-type pattern fast fraction 10 min leading train (21,932 records)	-0.061	-1.31 [-3.15, 0.54]	1.000	100.00 / 100.00
T18 Remaining planned journey remaining planned following train (40,130 records)	0.057	1.61 [0.79, 2.43]	0.005	100.00 / 100.00
T19 Remaining minimum journey remaining minimum following train (40,130 records)	0.057	1.61 [0.79, 2.43]	0.005	100.00 / 100.00
M06 Remaining candidate decisions remaining stopping decisions leading train (40,130 records)	-0.056	-1.31 [-4.46, 1.84]	1.000	100.00 / 100.00
T11 Planned track buffer planned track buffer absolute pair difference (29,180 records)	0.053	2.25 [-0.05, 4.54]	0.618	100.00 / 100.00
M06 Remaining candidate decisions remaining order decisions difference (40,130 records)	-0.050	-1.57 [-2.74, -0.39]	0.131	100.00 / 100.00
T06 Planned track label planned track index following train (38,510 records)	-0.046	-0.91 [-1.99, 0.17]	1.000	100.00 / 100.00
T10 Planned exit buffer previous buffer leading train (36,410 records)	0.044	1.12 [-1.60, 3.85]	1.000	100.00 / 100.00
T20 Remaining scheduled stops remaining scheduled stops following train (40,130 records)	0.043	1.14 [0.35, 1.93]	0.077	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
T01 Planned arrival time planned arrival leading train (40,130 records)	-0.043	-1.64 [-4.39, 1.10]	1.000	100.00 / 100.00
T02 Planned departure time planned departure leading train (40,130 records)	-0.040	-1.57 [-4.33, 1.19]	1.000	100.00 / 100.00
T13 Planned train-type pattern fast fraction 20 min following train (21,928 records)	-0.039	-0.84 [-2.65, 0.97]	1.000	100.00 / 100.00
T21 Nearby scheduled-stop distance next scheduled-stop distance leading train (40,130 records)	-0.039	-1.22 [-3.38, 0.95]	1.000	100.00 / 100.00
T21 Nearby scheduled-stop distance previous scheduled-stop distance leading train (40,130 records)	-0.039	-1.20 [-3.15, 0.75]	1.000	100.00 / 100.00
T11 Planned track buffer planned track buffer difference (29,180 records)	-0.038	-2.09 [-5.98, 1.80]	1.000	100.00 / 100.00
F12 Minimum dwell minimum dwell absolute pair difference (40,130 records)	-0.036	-1.14 [-1.81, -0.48]	0.020	100.00 / 100.00
F14 Allowed stopping mode required absolute pair difference (40,130 records)	-0.036	-1.14 [-1.81, -0.48]	0.020	100.00 / 100.00
T03 Planned dwell planned dwell absolute pair difference (40,130 records)	-0.036	-1.14 [-1.81, -0.48]	0.020	100.00 / 100.00
T05 Planned stopping mode planned stop absolute pair difference (40,130 records)	-0.036	-1.14 [-1.81, -0.48]	0.020	100.00 / 100.00
F14 Allowed stopping mode optional absolute pair difference (40,130 records)	-0.034	-1.08 [-1.27, -0.88]	1.4e-9	100.00 / 100.00
T20 Remaining scheduled stops remaining scheduled stops absolute pair difference (40,130 records)	0.034	1.15 [-0.21, 2.52]	1.000	100.00 / 100.00
T14 Departure wave origin wave40 following train (40,130 records)	-0.033	-1.07 [-2.85, 0.71]	1.000	100.00 / 100.00
F13 Minimum running time minimum run absolute pair difference (40,130 records)	-0.032	-1.02 [-1.88, -0.16]	0.266	100.00 / 66.67
T04 Planned running duration planned run next absolute pair difference (40,130 records)	-0.032	-1.02 [-1.88, -0.16]	0.266	100.00 / 66.67
T15 Fast-train indicator fast absolute pair difference (40,130 records)	-0.032	-1.02 [-1.88, -0.16]	0.266	100.00 / 66.67
T17 Total planned journey time planned total absolute pair difference (40,130 records)	-0.032	-1.02 [-1.88, -0.16]	0.266	100.00 / 66.67
T14 Departure wave origin wave40 leading train (40,130 records)	-0.032	-1.23 [-3.83, 1.36]	1.000	100.00 / 100.00
T13 Planned train-type pattern fast fraction 10 min difference (15,484 records)	0.028	0.29 [-0.74, 1.32]	1.000	100.00 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
T13 Planned train-type pattern fast fraction 5 min difference (12,069 records)	0.027	0.23 [-0.40, 0.86]	1.000	100.00 / 66.67
T09 Planned entry buffer previous buffer absolute pair difference (36,410 records)	0.026	0.74 [-1.13, 2.61]	1.000	100.00 / 100.00
T12 Planned local flow planned flow 5 min difference (40,130 records)	0.026	0.41 [-0.88, 1.70]	1.000	93.33 / 33.33
T14 Departure wave origin wave40 absolute pair difference (40,130 records)	0.023	0.61 [-1.74, 2.95]	1.000	93.33 / 66.67
T14 Departure wave origin wave40 difference (40,130 records)	0.023	0.61 [-1.74, 2.95]	1.000	93.33 / 66.67
T09 Planned entry buffer previous buffer leading train (36,410 records)	0.021	0.53 [-1.96, 3.02]	1.000	100.00 / 100.00
T24 Planned terminal position planned terminal leading train (40,130 records)	-0.019	0.26 [-2.97, 3.49]	1.000	73.33 / 66.67
T13 Planned train-type pattern alternation 10 min following train (9,509 records)	0.015	0.39 [-0.50, 1.28]	1.000	93.33 / 66.67
T13 Planned train-type pattern alternation 10 min leading train (9,569 records)	0.014	0.37 [0.11, 0.63]	0.084	100.00 / 100.00
T13 Planned train-type pattern fast fraction 20 min difference (21,152 records)	0.014	0.11 [-1.01, 1.23]	1.000	93.33 / 66.67
T13 Planned train-type pattern fast fraction 5 min following train (17,064 records)	-0.013	-0.20 [-1.48, 1.09]	1.000	86.67 / 66.67
T10 Planned exit buffer previous buffer absolute pair difference (36,410 records)	0.013	0.12 [-1.82, 2.05]	1.000	73.33 / 66.67
T18 Remaining planned journey remaining planned absolute pair difference (40,130 records)	0.012	0.96 [-0.36, 2.28]	1.000	100.00 / 66.67
T19 Remaining minimum journey remaining minimum absolute pair difference (40,130 records)	0.012	0.96 [-0.36, 2.28]	1.000	100.00 / 66.67
T13 Planned train-type pattern fast fraction 10 min following train (19,188 records)	-0.012	-0.22 [-1.81, 1.38]	1.000	93.33 / 66.67
M06 Remaining candidate decisions remaining stopping decisions absolute pair difference (40,130 records)	0.010	0.61 [-0.60, 1.82]	1.000	100.00 / 66.67
T06 Planned track label planned track index absolute pair difference (38,510 records)	-0.008	-0.03 [-0.93, 0.86]	1.000	66.67 / 33.33
T12 Planned local flow planned flow 10 min difference (40,130 records)	-0.007	-0.55 [-2.01, 0.91]	1.000	100.00 / 66.67
T21 Nearby scheduled-stop distance next scheduled-stop distance absolute pair difference (40,130 records)	0.002	0.05 [-0.98, 1.07]	1.000	73.33 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
M06 Remaining candidate decisions remaining order decisions absolute pair difference (40,130 records)	-0.001	-0.04 [-1.71, 1.62]	1.000	73.33 / 66.67
T21 Nearby scheduled-stop distance previous scheduled-stop distance absolute pair difference (40,130 records)	0.000	-0.01 [-1.05, 1.03]	1.000	33.33 / 66.67
Train interactions				
X22 Pair waiting contrast blockage wait difference (40,130 records)	-0.491	-13.73 [-17.16, -10.31]	4.5e-7	100.00 / 100.00
E11 Co-located emergency trains same stop count leading train (40,130 records)	0.446	11.30 [6.87, 15.73]	2.3e-4	100.00 / 100.00
E11 Co-located emergency trains same stop count difference (40,130 records)	-0.444	-11.91 [-14.96, -8.86]	5.9e-7	100.00 / 100.00
E11 Co-located emergency trains same stop count absolute pair difference (40,130 records)	0.368	9.23 [6.25, 12.21]	1.6e-5	100.00 / 100.00
X20 Same exposure category same exposure (40,130 records)	-0.359	-11.27 [-14.09, -8.45]	4.7e-7	100.00 / 100.00
H22 Shared downstream support pair downstream Jaccard (28,988 records)	0.314	8.91 [6.58, 11.23]	8.3e-7	100.00 / 100.00
H17 Window competition size competitor trains 20 min following train (40,130 records)	0.259	8.26 [5.84, 10.68]	4.0e-6	100.00 / 100.00
H18 Co-window occupations competition set memberships 20 min following train (40,130 records)	0.259	8.26 [5.84, 10.68]	4.0e-6	100.00 / 100.00
P16 Reference-delay contrast reference departure delay difference (40,130 records)	-0.250	-6.11 [-8.43, -3.80]	1.5e-4	100.00 / 100.00
H17 Window competition size competitor trains 20 min leading train (40,130 records)	0.247	7.62 [5.05, 10.20]	3.0e-5	100.00 / 100.00
H18 Co-window occupations competition set memberships 20 min leading train (40,130 records)	0.247	7.62 [5.05, 10.20]	3.0e-5	100.00 / 100.00
H22 Shared downstream support arrival departure downstream Jaccard difference (20,829 records)	-0.239	-9.03 [-12.31, -5.75]	8.4e-5	100.00 / 100.00
H17 Window competition size competitor trains 10 min following train (40,130 records)	0.227	7.42 [5.39, 9.46]	1.6e-6	100.00 / 100.00
H18 Co-window occupations competition set memberships 10 min following train (40,130 records)	0.227	7.42 [5.39, 9.46]	1.6e-6	100.00 / 100.00
H22 Shared downstream support arrival departure downstream Jaccard leading train (25,578 records)	0.224	6.10 [3.39, 8.81]	0.001	100.00 / 100.00
H17 Window competition size competitor trains 10 min leading train (40,130 records)	0.215	6.55 [4.50, 8.61]	1.1e-5	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H18 Co-window occupations competition set memberships 10 min leading train (40,130 records)	0.215	6.55 [4.50, 8.61]	1.1e-5	100.00 / 100.00
E15 Queue fast-train share queue fast fraction difference (6,911 records)	0.213	6.17 [3.81, 8.53]	1.7e-4	100.00 / 100.00
H17 Window competition size competitor trains 5 min following train (40,130 records)	0.197	6.31 [4.51, 8.12]	3.0e-6	100.00 / 100.00
H18 Co-window occupations competition set memberships 5 min following train (40,130 records)	0.197	6.31 [4.51, 8.12]	3.0e-6	100.00 / 100.00
H17 Window competition size competitor trains 5 min leading train (40,130 records)	0.196	5.98 [4.26, 7.70]	3.1e-6	100.00 / 100.00
H18 Co-window occupations competition set memberships 5 min leading train (40,130 records)	0.196	5.98 [4.26, 7.70]	3.1e-6	100.00 / 100.00
P18 Reference interval overlap reference interval overlap (40,130 records)	0.164	1.40 [-0.06, 2.86]	0.665	100.00 / 100.00
P10 Shared route extent common sections (40,130 records)	-0.153	-4.85 [-7.09, -2.61]	0.002	100.00 / 100.00
P11 Shared station pools common station pools (40,130 records)	-0.153	-4.85 [-7.09, -2.61]	0.002	100.00 / 100.00
P15 Parallel common stations parallel common stations (40,130 records)	-0.153	-4.85 [-7.09, -2.61]	0.002	100.00 / 100.00
P05 Running-duration contrast planned run difference (40,130 records)	-0.149	-4.70 [-7.64, -1.76]	0.036	100.00 / 100.00
H21 Shared-visit coupling same visit classes following train (40,130 records)	0.148	4.48 [2.51, 6.45]	1.0e-3	100.00 / 100.00
P12 Merge and split positions merge distance (40,130 records)	0.141	4.76 [2.94, 6.58]	1.6e-4	100.00 / 100.00
H17 Window competition size competitor trains 10 min absolute pair difference (40,130 records)	0.137	3.56 [1.74, 5.38]	0.006	100.00 / 100.00
H18 Co-window occupations competition set memberships 10 min absolute pair difference (40,130 records)	0.137	3.56 [1.74, 5.38]	0.006	100.00 / 100.00
P07 Remaining-journey contrast remaining minimum difference (40,130 records)	-0.132	-3.64 [-6.70, -0.59]	0.252	100.00 / 100.00
P07 Remaining-journey contrast remaining planned difference (40,130 records)	-0.132	-3.64 [-6.70, -0.59]	0.252	100.00 / 100.00
H21 Shared-visit coupling same visit classes leading train (40,130 records)	0.131	4.17 [2.24, 6.09]	0.002	100.00 / 100.00
P06 Train-type pair fast slow (40,130 records)	-0.128	-4.02 [-6.85, -1.19]	0.086	100.00 / 100.00
P10 Shared route extent common minimum time (40,130 records)	-0.123	-3.30 [-5.23, -1.36]	0.021	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
P06 Train-type pair slow fast (40,130 records)	0.120	3.77 [1.47, 6.07]	0.030	100.00 / 100.00
H22 Shared downstream support arrival departure downstream Jaccard absolute pair difference (20,829 records)	0.113	3.16 [0.03, 6.29]	0.548	100.00 / 100.00
H21 Shared-visit coupling same visit classes difference (40,130 records)	0.108	3.40 [1.56, 5.23]	0.010	100.00 / 100.00
E15 Queue fast-train share queue fast fraction following train (9,770 records)	0.108	2.38 [0.71, 4.06]	0.086	100.00 / 100.00
H17 Window competition size competitor trains 5 min absolute pair difference (40,130 records)	0.089	2.35 [1.00, 3.71]	0.018	100.00 / 100.00
H18 Co-window occupations competition set memberships 5 min absolute pair difference (40,130 records)	0.089	2.35 [1.00, 3.71]	0.018	100.00 / 100.00
P09 Stop-pattern disagreement stop disagreements (40,130 records)	-0.088	-2.43 [-3.08, -1.78]	1.1e-6	100.00 / 100.00
P08 Terminal-time contrast terminal difference (40,130 records)	-0.085	-3.52 [-6.81, -0.24]	0.424	100.00 / 100.00
P04 Planned exit gap planned exit gap (40,130 records)	-0.082	-2.66 [-5.42, 0.10]	0.657	100.00 / 100.00
H17 Window competition size competitor trains 20 min absolute pair difference (40,130 records)	0.082	1.69 [-0.01, 3.40]	0.586	100.00 / 100.00
H18 Co-window occupations competition set memberships 20 min absolute pair difference (40,130 records)	0.082	1.69 [-0.01, 3.40]	0.586	100.00 / 100.00
H17 Window competition size competitor trains 20 min difference (40,130 records)	0.081	1.84 [-0.44, 4.12]	1.000	100.00 / 100.00
H18 Co-window occupations competition set memberships 20 min difference (40,130 records)	0.081	1.84 [-0.44, 4.12]	1.000	100.00 / 100.00
P17 Same reference track reference same track (40,130 records)	-0.078	-2.46 [-3.45, -1.47]	3.3e-4	100.00 / 100.00
E15 Queue fast-train share queue fast fraction leading train (14,272 records)	-0.076	-3.61 [-8.05, 0.83]	1.000	100.00 / 66.67
P03 Planned entry gap planned entry gap (40,130 records)	-0.071	-2.38 [-5.04, 0.28]	0.858	100.00 / 100.00
E14 Reference release rank release queue rank following train (9,770 records)	0.061	1.41 [-1.24, 4.06]	1.000	100.00 / 100.00
E15 Queue fast-train share queue fast fraction absolute pair difference (6,911 records)	-0.056	-2.18 [-4.83, 0.47]	1.000	100.00 / 100.00
H17 Window competition size competitor trains 10 min difference (40,130 records)	0.051	1.54 [-0.54, 3.63]	1.000	100.00 / 100.00
H18 Co-window occupations competition set memberships 10 min difference (40,130 records)	0.051	1.54 [-0.54, 3.63]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H20 Support distance pair graph distance (8,391 records)	0.051	0.45 [-2.17, 3.06]	1.000	73.33 / 66.67
E11 Co-located emergency trains same stop count following train (40,130 records)	-0.049	-0.83 [-4.02, 2.36]	1.000	93.33 / 100.00
X21 Pair entry-time contrast entry relative release difference (16,745 records)	0.042	0.79 [-1.05, 2.63]	1.000	93.33 / 66.67
P06 Train-type pair fast fast (40,130 records)	0.035	1.10 [-0.60, 2.80]	1.000	100.00 / 100.00
H20 Support distance nearest optional stop index following train (15,301 records)	-0.035	-1.28 [-2.27, -0.28]	0.174	100.00 / 100.00
H21 Shared-visit coupling same visit classes absolute pair difference (40,130 records)	-0.034	-1.08 [-1.27, -0.88]	1.4e-9	100.00 / 100.00
P02 Planned rank distance planned rank distance (40,130 records)	0.032	0.53 [-1.04, 2.11]	1.000	93.33 / 100.00
B17 Relation-layer overlap entry headway station track Jaccard (40,130 records)	-0.028	0.84 [-1.99, 3.67]	1.000	86.67 / 100.00
B17 Relation-layer overlap exit headway station track Jaccard (40,130 records)	-0.028	0.84 [-1.99, 3.67]	1.000	86.67 / 100.00
E14 Reference release rank release queue rank absolute pair difference (6,911 records)	-0.024	-0.64 [-3.69, 2.41]	1.000	93.33 / 66.67
E14 Reference release rank release queue rank leading train (14,272 records)	0.020	0.79 [-3.82, 5.40]	1.000	86.67 / 66.67
E14 Reference release rank release queue rank difference (6,911 records)	-0.018	-0.30 [-3.55, 2.96]	1.000	86.67 / 100.00
H17 Window competition size competitor trains 5 min difference (40,130 records)	0.014	0.27 [-1.38, 1.93]	1.000	100.00 / 100.00
H18 Co-window occupations competition set memberships 5 min difference (40,130 records)	0.014	0.27 [-1.38, 1.93]	1.000	100.00 / 100.00
P13 Catch-upstream proxy potential catchup (40,130 records)	-0.012	-0.41 [-0.95, 0.13]	1.000	100.00 / 33.33
P14 Potential order positions free order positions (40,130 records)	-0.011	-0.62 [-2.09, 0.85]	1.000	100.00 / 66.67
P06 Train-type pair slow slow (40,130 records)	0.009	0.30 [-1.23, 1.82]	1.000	93.33 / 66.67
H22 Shared downstream support arrival departure downstream Jaccard following train (25,009 records)	0.006	-2.03 [-4.20, 0.13]	0.723	100.00 / 100.00
P12 Merge and split positions split remaining (40,130 records)	-0.006	-0.36 [-1.41, 0.69]	1.000	100.00 / 66.67
H20 Support distance nearest optional stop index leading train (21,036 records)	0.002	0.06 [-0.65, 0.76]	1.000	86.67 / 66.67
X23 Pair blockage-rank gap blockage planned rank difference (16,745 records)	0.001	-0.23 [-2.01, 1.56]	1.000	80.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
Infrastructure utilization				
E10 Emergency pool capacity emergency pool capacity leading train (14,272 records)	-0.334	-15.74 [-19.86, -11.61]	9.2e-7	100.00 / 100.00
R07 Entry load entry load 20 min following train (40,130 records)	0.273	8.62 [6.20, 11.03]	2.2e-6	100.00 / 100.00
R16 Planned-arrival overlap planned arrival release overlap 5 min leading train (40,130 records)	-0.267	-7.34 [-9.67, -5.00]	1.3e-5	100.00 / 100.00
R04 Utilization utilization 20 min following train (40,130 records)	0.259	8.22 [5.82, 10.61]	3.8e-6	100.00 / 100.00
R12 Demand/capacity proxy demand capacity 20 min following train (40,130 records)	0.259	8.22 [5.82, 10.61]	3.8e-6	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min count following train (40,130 records)	0.259	8.26 [5.84, 10.68]	4.0e-6	100.00 / 100.00
R14 Window competitors window competitors 20 min following train (40,130 records)	0.259	8.26 [5.84, 10.68]	4.0e-6	100.00 / 100.00
R16 Planned-arrival overlap planned arrival release overlap 10 min leading train (40,130 records)	-0.254	-6.93 [-9.02, -4.84]	6.2e-6	100.00 / 100.00
R16 Planned-arrival overlap planned arrival release overlap 20 min leading train (40,130 records)	-0.250	-6.98 [-9.21, -4.76]	1.3e-5	100.00 / 100.00
R07 Entry load entry load 20 min leading train (40,130 records)	0.249	7.23 [4.85, 9.62]	2.2e-5	100.00 / 100.00
R01 Mean occupation mean occupancy 20 min following train (40,130 records)	0.249	7.50 [5.14, 9.86]	1.1e-5	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min count leading train (40,130 records)	0.247	7.62 [5.05, 10.20]	3.0e-5	100.00 / 100.00
R14 Window competitors window competitors 20 min leading train (40,130 records)	0.247	7.62 [5.05, 10.20]	3.0e-5	100.00 / 100.00
R09 Reference headway buffer 0 previous buffer following train (40,130 records)	-0.242	-5.00 [-7.00, -3.00]	2.9e-4	100.00 / 100.00
R08 Exit load exit load 10 min difference (40,130 records)	0.239	7.77 [5.28, 10.25]	1.3e-5	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min sum following train (40,130 records)	0.238	2.83 [0.23, 5.42]	0.391	100.00 / 100.00
R08 Exit load exit load 20 min following train (40,130 records)	0.232	6.96 [4.88, 9.04]	5.3e-6	100.00 / 100.00
R08 Exit load exit load 10 min following train (40,130 records)	0.232	7.12 [4.92, 9.32]	8.6e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R09 Reference headway buffer 1 previous buffer following train (40,130 records)	-0.232	-5.01 [-7.02, -3.01]	3.0e-4	100.00 / 100.00
R08 Exit load exit load 5 min following train (40,130 records)	0.231	7.21 [4.94, 9.47]	1.1e-5	100.00 / 100.00
R04 Utilization utilization 20 min leading train (40,130 records)	0.230	7.10 [4.72, 9.47]	2.7e-5	100.00 / 100.00
R12 Demand/capacity proxy demand capacity 20 min leading train (40,130 records)	0.230	7.10 [4.72, 9.47]	2.7e-5	100.00 / 100.00
R07 Entry load entry load 10 min following train (40,130 records)	0.230	7.35 [5.46, 9.24]	6.6e-7	100.00 / 100.00
R09 Reference headway buffer 0 previous buffer difference (36,410 records)	-0.228	-6.28 [-8.65, -3.92]	1.5e-4	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min count following train (40,130 records)	0.227	7.42 [5.39, 9.46]	1.6e-6	100.00 / 100.00
R14 Window competitors window competitors 10 min following train (40,130 records)	0.227	7.42 [5.39, 9.46]	1.6e-6	100.00 / 100.00
R16 Planned-arrival overlap planned arrival release overlap 5 min following train (40,130 records)	-0.227	-6.38 [-8.35, -4.40]	8.8e-6	100.00 / 100.00
R09 Reference headway buffer 1 previous buffer difference (36,410 records)	-0.226	-6.18 [-8.50, -3.85]	1.5e-4	100.00 / 100.00
R01 Mean occupation mean occupancy 20 min leading train (40,130 records)	0.222	6.12 [3.90, 8.34]	8.3e-5	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min sum following train (40,130 records)	0.221	1.94 [-0.85, 4.73]	1.000	100.00 / 100.00
R01 Mean occupation mean occupancy 10 min following train (40,130 records)	0.221	6.84 [4.82, 8.86]	4.4e-6	100.00 / 100.00
R13 Adjacent-station occupation upstream occupancy leading train (40,130 records)	0.220	6.95 [4.13, 9.76]	3.6e-4	100.00 / 100.00
R04 Utilization utilization 10 min following train (40,130 records)	0.219	7.14 [5.20, 9.09]	1.5e-6	100.00 / 100.00
R12 Demand/capacity proxy demand capacity 10 min following train (40,130 records)	0.219	7.14 [5.20, 9.09]	1.5e-6	100.00 / 100.00
B13 Adjacent-window difference adjacent occupancy difference 5 min difference (40,130 records)	-0.218	-6.63 [-9.05, -4.20]	9.4e-5	100.00 / 100.00
R17 Occupation change rate occupancy rate 5 min difference (40,130 records)	-0.218	-6.63 [-9.05, -4.20]	9.4e-5	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R16 Planned-arrival overlap planned arrival release overlap 5 min absolute pair difference (40,130 records)	-0.216	-6.13 [-8.08, -4.17]	1.3e-5	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min count leading train (40,130 records)	0.215	6.55 [4.50, 8.61]	1.1e-5	100.00 / 100.00
R14 Window competitors window competitors 10 min leading train (40,130 records)	0.215	6.55 [4.50, 8.61]	1.1e-5	100.00 / 100.00
R01 Mean occupation mean occupancy 10 min leading train (40,130 records)	0.210	5.92 [3.99, 7.85]	1.9e-5	100.00 / 100.00
R04 Utilization utilization 10 min leading train (40,130 records)	0.208	6.50 [4.55, 8.46]	5.8e-6	100.00 / 100.00
R12 Demand/capacity proxy demand capacity 10 min leading train (40,130 records)	0.208	6.50 [4.55, 8.46]	5.8e-6	100.00 / 100.00
R16 Planned-arrival overlap planned arrival release overlap 10 min following train (40,130 records)	-0.207	-5.95 [-7.74, -4.15]	6.2e-6	100.00 / 100.00
R08 Exit load exit load 5 min difference (40,130 records)	0.207	6.73 [4.22, 9.23]	1.2e-4	100.00 / 100.00
R07 Entry load entry load 5 min following train (40,130 records)	0.206	6.49 [4.64, 8.35]	3.0e-6	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min sum leading train (40,130 records)	0.204	2.45 [0.54, 4.37]	0.169	100.00 / 100.00
R13 Adjacent-station occupation downstream occupancy following train (40,130 records)	0.203	6.04 [3.72, 8.35]	1.7e-4	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min interquartile range following train (38,510 records)	0.202	3.29 [1.52, 5.05]	0.009	100.00 / 100.00
B13 Adjacent-window difference adjacent occupancy difference 5 min leading train (40,130 records)	0.201	6.11 [2.96, 9.25]	0.006	100.00 / 100.00
R17 Occupation change rate occupancy rate 5 min leading train (40,130 records)	0.201	6.11 [2.96, 9.25]	0.006	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min standard deviation following train (38,510 records)	0.201	3.39 [1.58, 5.21]	0.009	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min interquartile range following train (38,510 records)	0.200	3.27 [1.84, 4.70]	9.3e-4	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min standard deviation following train (38,510 records)	0.198	3.56 [1.98, 5.15]	0.001	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min count following train (40,130 records)	0.197	6.31 [4.51, 8.12]	3.0e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R14 Window competitors window competitors 5 min following train (40,130 records)	0.197	6.31 [4.51, 8.12]	3.0e-6	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min count leading train (40,130 records)	0.196	5.98 [4.26, 7.70]	3.1e-6	100.00 / 100.00
R14 Window competitors window competitors 5 min leading train (40,130 records)	0.196	5.98 [4.26, 7.70]	3.1e-6	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min sum following train (40,130 records)	0.192	1.86 [-0.75, 4.47]	1.000	100.00 / 100.00
R01 Mean occupation mean occupancy 5 min leading train (40,130 records)	0.191	5.74 [3.85, 7.64]	2.2e-5	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 5 min difference (40,130 records)	-0.191	-5.85 [-7.92, -3.77]	6.0e-5	100.00 / 100.00
R04 Utilization utilization 5 min leading train (40,130 records)	0.185	5.82 [3.97, 7.66]	1.3e-5	100.00 / 100.00
R12 Demand/capacity proxy demand capacity 5 min leading train (40,130 records)	0.185	5.82 [3.97, 7.66]	1.3e-5	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min sum leading train (40,130 records)	0.184	1.71 [-0.08, 3.50]	0.676	100.00 / 100.00
R08 Exit load exit load 20 min difference (40,130 records)	0.184	5.21 [2.98, 7.45]	7.2e-4	100.00 / 100.00
R01 Mean occupation mean occupancy 5 min following train (40,130 records)	0.184	5.56 [4.00, 7.12]	2.3e-6	100.00 / 100.00
R02 Peak occupation peak occupancy 20 min leading train (40,130 records)	0.178	5.50 [3.74, 7.25]	1.3e-5	100.00 / 100.00
E10 Emergency pool capacity emergency pool capacity absolute pair difference (6,911 records)	-0.178	-5.40 [-7.19, -3.60]	2.5e-5	100.00 / 100.00
R07 Entry load entry load 10 min leading train (40,130 records)	0.178	5.64 [4.00, 7.27]	3.6e-6	100.00 / 100.00
R02 Peak occupation peak occupancy 20 min following train (40,130 records)	0.178	5.41 [3.85, 6.97]	3.2e-6	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min interquartile range leading train (38,510 records)	0.177	3.27 [1.55, 4.99]	0.007	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min standard deviation leading train (38,510 records)	0.177	3.21 [1.34, 5.08]	0.020	100.00 / 100.00
R16 Planned-arrival overlap planned arrival release overlap 10 min absolute pair difference (40,130 records)	-0.176	-4.77 [-6.53, -3.01]	1.0e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R02 Peak occupation peak occupancy 5 min leading train (40,130 records)	0.176	5.50 [3.75, 7.25]	1.3e-5	100.00 / 100.00
B07 Regional saturation saturated fraction 20 min leading train (40,130 records)	0.175	5.55 [3.79, 7.31]	1.3e-5	100.00 / 100.00
R05 Saturated fraction saturated fraction 20 min leading train (40,130 records)	0.175	5.55 [3.79, 7.31]	1.3e-5	100.00 / 100.00
R15 Free windows free window total 20 min leading train (40,130 records)	-0.175	-5.55 [-7.31, -3.79]	1.3e-5	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min standard deviation leading train (38,510 records)	0.174	3.51 [1.50, 5.51]	0.017	100.00 / 100.00
R02 Peak occupation peak occupancy 10 min leading train (40,130 records)	0.174	5.36 [3.71, 7.01]	7.9e-6	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min sum leading train (40,130 records)	0.173	1.48 [-0.27, 3.23]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min maximum following train (38,510 records)	0.173	2.33 [0.43, 4.23]	0.215	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min interquartile range leading train (38,510 records)	0.173	3.32 [1.49, 5.14]	0.012	100.00 / 100.00
R16 Planned-arrival overlap planned arrival release overlap 20 min following train (40,130 records)	-0.171	-4.68 [-6.54, -2.82]	2.8e-4	100.00 / 100.00
B07 Regional saturation saturated fraction 20 min following train (40,130 records)	0.170	5.94 [4.20, 7.67]	3.8e-6	100.00 / 100.00
R05 Saturated fraction saturated fraction 20 min following train (40,130 records)	0.170	5.94 [4.20, 7.67]	3.8e-6	100.00 / 100.00
R15 Free windows free window total 20 min following train (40,130 records)	-0.170	-5.94 [-7.67, -4.20]	3.8e-6	100.00 / 100.00
R02 Peak occupation peak occupancy 10 min following train (40,130 records)	0.170	5.27 [3.67, 6.86]	6.7e-6	100.00 / 100.00
H19 Peak demand/capacity peak demand capacity 20 min leading train (40,130 records)	0.167	5.00 [3.41, 6.58]	1.2e-5	100.00 / 100.00
R06 Saturation persistence longest saturation 20 min leading train (40,130 records)	0.167	4.15 [2.96, 5.34]	3.1e-6	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min standard deviation following train (38,510 records)	0.166	3.35 [2.50, 4.19]	4.9e-7	100.00 / 100.00
H19 Peak demand/capacity peak demand capacity 20 min following train (40,130 records)	0.166	4.90 [3.57, 6.23]	1.4e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R10 Intersecting dwell distribution waiting duration 5 min maximum following train (38,510 records)	0.164	2.01 [-0.08, 4.11]	0.664	100.00 / 100.00
R17 Occupation change rate occupancy rate 5 min absolute pair difference (40,130 records)	0.163	4.91 [2.92, 6.90]	3.6e-4	100.00 / 100.00
B13 Adjacent-window difference adjacent occupancy difference 5 min absolute pair difference (40,130 records)	0.163	4.91 [2.92, 6.90]	3.6e-4	100.00 / 100.00
B07 Regional saturation saturated fraction 10 min leading train (40,130 records)	0.163	5.23 [3.74, 6.71]	2.8e-6	100.00 / 100.00
R05 Saturated fraction saturated fraction 10 min leading train (40,130 records)	0.163	5.23 [3.74, 6.71]	2.8e-6	100.00 / 100.00
R15 Free windows free window total 10 min leading train (40,130 records)	-0.163	-5.23 [-6.71, -3.74]	2.8e-6	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min interquartile range following train (38,510 records)	0.162	3.13 [2.16, 4.11]	9.4e-6	100.00 / 100.00
R04 Utilization utilization 5 min following train (40,130 records)	0.162	5.27 [3.91, 6.64]	6.9e-7	100.00 / 100.00
R12 Demand/capacity proxy demand capacity 5 min following train (40,130 records)	0.162	5.27 [3.91, 6.64]	6.9e-7	100.00 / 100.00
R02 Peak occupation peak occupancy 5 min following train (40,130 records)	0.162	5.03 [3.70, 6.37]	1.1e-6	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min 90th percentile following train (38,510 records)	0.160	1.99 [-0.13, 4.11]	0.725	100.00 / 100.00
R13 Adjacent-station occupation occupancy difference absolute pair difference (40,130 records)	0.160	5.13 [2.40, 7.87]	0.009	100.00 / 100.00
R15 Free windows free window maximum 20 min leading train (40,130 records)	-0.160	-4.96 [-6.43, -3.50]	4.4e-6	100.00 / 100.00
R15 Free windows free window maximum 20 min following train (40,130 records)	-0.160	-5.04 [-6.29, -3.80]	4.2e-7	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min maximum following train (38,510 records)	0.158	2.50 [1.17, 3.84]	0.008	100.00 / 100.00
R13 Adjacent-station occupation occupancy difference difference (40,130 records)	0.158	5.17 [2.49, 7.84]	0.006	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 5 min leading train (40,130 records)	0.158	4.93 [2.86, 7.01]	5.7e-4	100.00 / 100.00
H19 Peak demand/capacity peak demand capacity 10 min leading train (40,130 records)	0.157	4.71 [3.24, 6.18]	9.7e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R07 Entry load entry load 10 min absolute pair difference (40,130 records)	0.157	4.87 [2.49, 7.25]	0.003	100.00 / 100.00
H19 Peak demand/capacity peak demand capacity 5 min leading train (40,130 records)	0.156	4.69 [3.17, 6.21]	1.6e-5	100.00 / 100.00
R06 Saturation persistence longest saturation 5 min leading train (40,130 records)	0.155	4.99 [3.66, 6.31]	1.1e-6	100.00 / 100.00
R06 Saturation persistence longest saturation 20 min following train (40,130 records)	0.155	4.15 [2.86, 5.43]	8.6e-6	100.00 / 100.00
R06 Saturation persistence longest saturation 10 min leading train (40,130 records)	0.155	4.39 [3.14, 5.64]	2.8e-6	100.00 / 100.00
R15 Free windows free window maximum 5 min leading train (40,130 records)	-0.155	-4.78 [-6.22, -3.33]	6.5e-6	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min 90th percentile following train (38,510 records)	0.155	2.05 [0.53, 3.58]	0.123	100.00 / 100.00
H19 Peak demand/capacity peak demand capacity 10 min following train (40,130 records)	0.154	4.53 [3.26, 5.80]	2.2e-6	100.00 / 100.00
R15 Free windows free window maximum 10 min leading train (40,130 records)	-0.154	-4.90 [-6.31, -3.48]	3.4e-6	100.00 / 100.00
R15 Free windows free window count 20 min leading train (40,130 records)	0.154	4.98 [3.38, 6.57]	1.4e-5	100.00 / 100.00
B07 Regional saturation saturated fraction 5 min leading train (40,130 records)	0.153	4.82 [3.54, 6.09]	1.0e-6	100.00 / 100.00
R05 Saturated fraction saturated fraction 5 min leading train (40,130 records)	0.153	4.82 [3.54, 6.09]	1.0e-6	100.00 / 100.00
R15 Free windows free window total 5 min leading train (40,130 records)	-0.153	-4.82 [-6.09, -3.54]	1.0e-6	100.00 / 100.00
B07 Regional saturation saturated fraction 10 min following train (40,130 records)	0.153	5.42 [4.06, 6.78]	4.8e-7	100.00 / 100.00
R05 Saturated fraction saturated fraction 10 min following train (40,130 records)	0.153	5.42 [4.06, 6.78]	4.8e-7	100.00 / 100.00
R15 Free windows free window total 10 min following train (40,130 records)	-0.153	-5.42 [-6.78, -4.06]	4.8e-7	100.00 / 100.00
R17 Occupation change rate occupancy rate 10 min difference (40,130 records)	-0.153	-4.68 [-7.10, -2.25]	0.007	100.00 / 100.00
B13 Adjacent-window difference adjacent occupancy difference 10 min difference (40,130 records)	-0.153	-4.68 [-7.10, -2.25]	0.007	100.00 / 100.00
R08 Exit load exit load 5 min absolute pair difference (40,130 records)	0.153	5.36 [3.03, 7.68]	8.6e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R10 Intersecting dwell distribution waiting duration 20 min 90th percentile leading train (38,510 records)	0.151	1.93 [0.32, 3.54]	0.248	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min 90th percentile following train (38,510 records)	0.151	1.78 [-0.48, 4.04]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min standard deviation leading train (38,510 records)	0.149	3.31 [1.54, 5.08]	0.009	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min maximum leading train (38,510 records)	0.149	2.34 [0.73, 3.95]	0.073	100.00 / 100.00
R15 Free windows free window count 20 min following train (40,130 records)	0.149	5.17 [3.66, 6.67]	3.8e-6	100.00 / 100.00
R15 Free windows free window maximum 10 min following train (40,130 records)	-0.148	-4.88 [-6.12, -3.64]	5.7e-7	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min interquartile range leading train (38,510 records)	0.147	3.12 [1.50, 4.73]	0.006	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 10 min difference (40,130 records)	-0.147	-4.51 [-6.58, -2.44]	0.002	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 20 min absolute pair difference (40,130 records)	0.146	4.08 [2.06, 6.10]	0.004	100.00 / 100.00
R13 Adjacent-station occupation downstream occupancy absolute pair difference (40,130 records)	0.146	4.50 [2.32, 6.69]	0.003	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min maximum leading train (38,510 records)	0.145	1.96 [0.39, 3.53]	0.194	100.00 / 100.00
R15 Free windows free window maximum 20 min absolute pair difference (40,130 records)	0.143	2.65 [1.14, 4.16]	0.016	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min maximum leading train (38,510 records)	0.142	1.87 [0.34, 3.40]	0.222	100.00 / 100.00
R07 Entry load entry load 5 min difference (40,130 records)	0.142	4.07 [2.03, 6.10]	0.004	100.00 / 100.00
R15 Free windows free window maximum 10 min absolute pair difference (40,130 records)	0.141	3.48 [1.91, 5.04]	0.001	100.00 / 100.00
R13 Adjacent-station occupation downstream occupancy difference (40,130 records)	0.140	4.21 [1.85, 6.57]	0.014	100.00 / 100.00
R06 Saturation persistence longest saturation 10 min following train (40,130 records)	0.140	4.19 [3.15, 5.23]	4.5e-7	100.00 / 100.00
B07 Regional saturation saturated fraction 5 min following train (40,130 records)	0.139	4.72 [3.62, 5.82]	1.3e-7	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R05 Saturated fraction saturated fraction 5 min following train (40,130 records)	0.139	4.72 [3.62, 5.82]	1.3e-7	100.00 / 100.00
R15 Free windows free window total 5 min following train (40,130 records)	-0.139	-4.72 [-5.82, -3.62]	1.3e-7	100.00 / 100.00
R03 Minimum free capacity minimum free 20 min leading train (40,130 records)	-0.139	-3.99 [-5.51, -2.47]	1.5e-4	100.00 / 100.00
H19 Peak demand/capacity peak demand capacity 5 min following train (40,130 records)	0.139	4.08 [2.89, 5.27]	3.8e-6	100.00 / 100.00
B13 Adjacent-window difference adjacent occupancy difference 10 min leading train (40,130 records)	0.138	3.45 [0.74, 6.16]	0.174	100.00 / 100.00
R17 Occupation change rate occupancy rate 10 min leading train (40,130 records)	0.138	3.45 [0.74, 6.16]	0.174	100.00 / 100.00
R15 Free windows free window maximum 5 min following train (40,130 records)	-0.137	-4.48 [-5.65, -3.32]	7.4e-7	100.00 / 100.00
R15 Free windows free window count 10 min absolute pair difference (40,130 records)	0.137	3.94 [1.98, 5.89]	0.004	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min count absolute pair difference (40,130 records)	0.137	3.56 [1.74, 5.38]	0.006	100.00 / 100.00
R14 Window competitors window competitors 10 min absolute pair difference (40,130 records)	0.137	3.56 [1.74, 5.38]	0.006	100.00 / 100.00
R08 Exit load exit load 20 min leading train (40,130 records)	0.136	4.45 [2.72, 6.18]	2.1e-4	100.00 / 100.00
B07 Regional saturation saturated fraction 10 min absolute pair difference (40,130 records)	0.135	3.28 [1.74, 4.83]	0.002	100.00 / 100.00
R05 Saturated fraction saturated fraction 10 min absolute pair difference (40,130 records)	0.135	3.28 [1.74, 4.83]	0.002	100.00 / 100.00
R15 Free windows free window total 10 min absolute pair difference (40,130 records)	0.135	3.28 [1.74, 4.83]	0.002	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min 90th percentile leading train (38,510 records)	0.133	1.62 [0.09, 3.14]	0.450	100.00 / 100.00
E10 Emergency pool capacity emergency pool capacity difference (6,911 records)	0.133	3.72 [0.97, 6.47]	0.121	100.00 / 100.00
R03 Minimum free capacity minimum free 20 min following train (40,130 records)	-0.133	-3.99 [-5.35, -2.64]	3.1e-5	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 5 min absolute pair difference (40,130 records)	0.132	4.06 [2.38, 5.73]	4.6e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B07 Regional saturation saturated fraction 5 min absolute pair difference (40,130 records)	0.131	3.44 [2.06, 4.82]	3.1e-4	100.00 / 100.00
R05 Saturated fraction saturated fraction 5 min absolute pair difference (40,130 records)	0.131	3.44 [2.06, 4.82]	3.1e-4	100.00 / 100.00
R15 Free windows free window total 5 min absolute pair difference (40,130 records)	0.131	3.44 [2.06, 4.82]	3.1e-4	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min 90th percentile leading train (38,510 records)	0.130	1.69 [0.14, 3.25]	0.391	100.00 / 100.00
R15 Free windows free window maximum 5 min absolute pair difference (40,130 records)	0.130	3.19 [1.92, 4.46]	2.8e-4	100.00 / 100.00
E18 Parking concentration parking concentration (40,130 records)	-0.128	-3.82 [-6.19, -1.45]	0.034	100.00 / 100.00
R07 Entry load entry load 20 min difference (40,130 records)	0.128	3.49 [0.98, 6.00]	0.099	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min minimum absolute pair difference (38,510 records)	-0.128	-1.25 [-1.99, -0.52]	0.021	100.00 / 100.00
R13 Adjacent-station occupation occupancy difference leading train (40,130 records)	-0.128	-4.18 [-6.82, -1.54]	0.039	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 10 min leading train (40,130 records)	0.127	3.86 [1.23, 6.49]	0.069	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min mean leading train (38,510 records)	0.127	0.41 [-1.00, 1.82]	1.000	100.00 / 33.33
R10 Intersecting dwell distribution waiting duration 10 min mean following train (38,510 records)	0.127	0.63 [-1.41, 2.68]	1.000	100.00 / 33.33
R13 Adjacent-station occupation upstream occupancy following train (40,130 records)	0.126	3.82 [2.38, 5.25]	1.3e-4	100.00 / 100.00
R03 Minimum free capacity minimum free 10 min leading train (40,130 records)	-0.126	-3.63 [-5.01, -2.26]	1.4e-4	100.00 / 100.00
R16 Planned-arrival overlap planned arrival release overlap 20 min absolute pair difference (40,130 records)	-0.126	-3.55 [-5.22, -1.88]	0.002	100.00 / 100.00
R06 Saturation persistence longest saturation 5 min following train (40,130 records)	0.126	3.95 [3.15, 4.75]	9.6e-9	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min mean leading train (38,510 records)	0.126	0.44 [-0.87, 1.75]	1.000	100.00 / 33.33
R10 Intersecting dwell distribution waiting duration 5 min mean following train (38,510 records)	0.125	0.64 [-1.44, 2.72]	1.000	100.00 / 33.33

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R15 Free windows free window count 10 min following train (40,130 records)	0.124	4.28 [3.14, 5.43]	1.1e-6	100.00 / 100.00
R07 Entry load entry load 5 min leading train (40,130 records)	0.124	3.59 [2.14, 5.04]	3.4e-4	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 10 min absolute pair difference (40,130 records)	0.123	3.73 [2.21, 5.26]	3.9e-4	100.00 / 100.00
R13 Adjacent-station occupation upstream occupancy absolute pair difference (40,130 records)	0.123	3.57 [1.54, 5.59]	0.016	100.00 / 100.00
R03 Minimum free capacity minimum free 5 min leading train (40,130 records)	-0.123	-3.57 [-4.94, -2.20]	1.7e-4	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min median leading train (38,510 records)	0.120	-0.36 [-1.73, 1.02]	1.000	93.33 / 66.67
R10 Intersecting dwell distribution waiting duration 5 min standard deviation absolute pair difference (38,510 records)	0.118	1.44 [-0.03, 2.92]	0.615	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min interquartile range absolute pair difference (38,510 records)	0.118	1.75 [0.28, 3.22]	0.253	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min mean leading train (38,510 records)	0.118	0.46 [-0.93, 1.85]	1.000	100.00 / 33.33
R18 Opposite-direction utilization opposite utilization 20 min following train (40,130 records)	0.116	4.22 [2.79, 5.66]	3.4e-5	100.00 / 100.00
R15 Free windows free window count 10 min leading train (40,130 records)	0.116	3.44 [2.41, 4.47]	5.3e-6	100.00 / 100.00
R03 Minimum free capacity minimum free 10 min following train (40,130 records)	-0.115	-3.49 [-4.71, -2.27]	5.0e-5	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min median following train (38,510 records)	0.115	0.11 [-1.62, 1.83]	1.000	73.33 / 33.33
R10 Intersecting dwell distribution waiting duration 20 min mean following train (38,510 records)	0.115	0.60 [-1.23, 2.43]	1.000	100.00 / 33.33
R07 Entry load entry load 20 min absolute pair difference (40,130 records)	0.115	3.01 [1.03, 4.99]	0.053	100.00 / 100.00
R13 Adjacent-station occupation upstream occupancy difference (40,130 records)	-0.115	-3.62 [-5.69, -1.55]	0.017	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min median leading train (38,510 records)	0.113	0.27 [-1.23, 1.77]	1.000	93.33 / 33.33
R17 Occupation change rate occupancy rate 10 min absolute pair difference (40,130 records)	0.112	3.42 [1.29, 5.55]	0.035	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B13 Adjacent-window difference adjacent occupancy difference 10 min absolute pair difference (40,130 records)	0.112	3.42 [1.29, 5.55]	0.035	100.00 / 100.00
R07 Entry load entry load 10 min difference (40,130 records)	0.111	3.59 [0.73, 6.45]	0.189	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min median leading train (38,510 records)	0.110	0.08 [-1.45, 1.60]	1.000	60.00 / 33.33
R10 Intersecting dwell distribution waiting duration 5 min median following train (38,510 records)	0.109	0.30 [-1.57, 2.18]	1.000	93.33 / 33.33
R15 Free windows free window count 5 min following train (40,130 records)	0.109	3.74 [2.64, 4.85]	4.5e-6	100.00 / 100.00
B07 Regional saturation saturated fraction 20 min absolute pair difference (40,130 records)	0.109	2.24 [0.86, 3.62]	0.031	100.00 / 100.00
R05 Saturated fraction saturated fraction 20 min absolute pair difference (40,130 records)	0.109	2.24 [0.86, 3.62]	0.031	100.00 / 100.00
R15 Free windows free window total 20 min absolute pair difference (40,130 records)	0.109	2.24 [0.86, 3.62]	0.031	100.00 / 100.00
B13 Adjacent-window difference adjacent occupancy difference 20 min difference (40,130 records)	-0.108	-2.62 [-4.70, -0.54]	0.186	100.00 / 100.00
R17 Occupation change rate occupancy rate 20 min difference (40,130 records)	-0.108	-2.62 [-4.70, -0.54]	0.186	100.00 / 100.00
R08 Exit load exit load 20 min absolute pair difference (40,130 records)	0.108	3.35 [1.24, 5.46]	0.038	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min median following train (38,510 records)	0.107	-0.21 [-1.53, 1.12]	1.000	93.33 / 66.67
R07 Entry load entry load 5 min absolute pair difference (40,130 records)	0.107	3.08 [1.59, 4.57]	0.003	100.00 / 100.00
B13 Adjacent-window difference adjacent occupancy difference 20 min following train (40,130 records)	-0.106	-2.72 [-4.63, -0.80]	0.088	100.00 / 100.00
R17 Occupation change rate occupancy rate 20 min following train (40,130 records)	-0.106	-2.72 [-4.63, -0.80]	0.088	100.00 / 100.00
R16 Planned-arrival overlap planned arrival release overlap 20 min difference (40,130 records)	0.105	2.97 [1.69, 4.26]	8.2e-4	100.00 / 100.00
R08 Exit load exit load 10 min absolute pair difference (40,130 records)	0.103	4.40 [2.06, 6.73]	0.008	100.00 / 100.00
R18 Opposite-direction utilization opposite utilization 20 min difference (40,130 records)	0.102	1.81 [0.54, 3.09]	0.084	100.00 / 100.00
F01 Facility role trunk leading train (40,130 records)	0.101	3.18 [2.12, 4.24]	2.6e-5	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
F01 Facility role trunk following train (40,130 records)	0.101	3.18 [2.12, 4.24]	2.6e-5	100.00 / 100.00
R15 Free windows free window count 20 min absolute pair difference (40,130 records)	0.100	2.39 [0.83, 3.95]	0.049	100.00 / 100.00
R15 Free windows free window count 5 min absolute pair difference (40,130 records)	0.099	2.81 [1.64, 3.98]	5.1e-4	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 20 min difference (40,130 records)	-0.099	-3.28 [-5.21, -1.35]	0.022	100.00 / 100.00
R03 Minimum free capacity minimum free 5 min following train (40,130 records)	-0.096	-2.89 [-4.03, -1.75]	2.5e-4	100.00 / 100.00
R18 Opposite-direction utilization opposite utilization 10 min following train (40,130 records)	0.094	3.04 [2.02, 4.06]	2.8e-5	100.00 / 100.00
R13 Adjacent-station occupation occupancy difference following train (40,130 records)	0.092	2.26 [0.31, 4.21]	0.288	100.00 / 100.00
R09 Reference headway buffer 1 previous buffer leading train (36,410 records)	0.092	4.06 [1.55, 6.56]	0.033	100.00 / 100.00
B13 Adjacent-window difference adjacent occupancy difference 5 min following train (40,130 records)	-0.092	-2.30 [-3.75, -0.84]	0.040	100.00 / 100.00
R17 Occupation change rate occupancy rate 5 min following train (40,130 records)	-0.092	-2.30 [-3.75, -0.84]	0.040	100.00 / 100.00
R15 Free windows free window count 5 min leading train (40,130 records)	0.090	2.15 [1.02, 3.28]	0.007	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min count absolute pair difference (40,130 records)	0.089	2.35 [1.00, 3.71]	0.018	100.00 / 100.00
R14 Window competitors window competitors 5 min absolute pair difference (40,130 records)	0.089	2.35 [1.00, 3.71]	0.018	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min zero fraction absolute pair difference (38,510 records)	-0.089	-4.26 [-5.94, -2.57]	2.6e-4	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min maximum absolute pair difference (38,510 records)	-0.089	-0.06 [-1.58, 1.47]	1.000	66.67 / 66.67
R10 Intersecting dwell distribution waiting duration 20 min maximum absolute pair difference (38,510 records)	-0.088	-0.14 [-1.48, 1.20]	1.000	66.67 / 66.67
R06 Saturation persistence longest saturation 5 min absolute pair difference (40,130 records)	0.088	2.58 [1.31, 3.86]	0.004	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 5 min following train (40,130 records)	-0.087	-2.76 [-4.38, -1.15]	0.021	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
F18 Next stopping opportunity next stop distance following train (34,175 records)	0.087	2.83 [1.49, 4.18]	0.002	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min zero fraction absolute pair difference (38,510 records)	-0.087	-4.58 [-6.26, -2.90]	9.5e-5	100.00 / 100.00
R18 Opposite-direction utilization opposite utilization 5 min leading train (40,130 records)	0.086	2.24 [1.43, 3.04]	8.0e-5	100.00 / 100.00
R18 Opposite-direction utilization opposite utilization 5 min following train (40,130 records)	0.085	2.33 [1.50, 3.16]	6.6e-5	100.00 / 100.00
R09 Reference headway buffer 0 previous buffer leading train (36,410 records)	0.084	4.17 [1.64, 6.70]	0.028	100.00 / 100.00
R02 Peak occupation peak occupancy 20 min absolute pair difference (40,130 records)	-0.084	-2.49 [-3.52, -1.45]	4.9e-4	100.00 / 100.00
R03 Minimum free capacity minimum free 20 min absolute pair difference (40,130 records)	-0.084	-2.49 [-3.52, -1.45]	4.9e-4	100.00 / 100.00
H19 Peak demand/capacity peak demand capacity 20 min absolute pair difference (40,130 records)	-0.084	-2.46 [-3.52, -1.40]	7.7e-4	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 20 min leading train (40,130 records)	0.084	2.49 [0.42, 4.56]	0.239	100.00 / 66.67
R10 Intersecting dwell distribution waiting duration 20 min count absolute pair difference (40,130 records)	0.082	1.69 [-0.01, 3.40]	0.586	100.00 / 100.00
R14 Window competitors window competitors 20 min absolute pair difference (40,130 records)	0.082	1.69 [-0.01, 3.40]	0.586	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min count difference (40,130 records)	0.081	1.84 [-0.44, 4.12]	1.000	100.00 / 100.00
R14 Window competitors window competitors 20 min difference (40,130 records)	0.081	1.84 [-0.44, 4.12]	1.000	100.00 / 100.00
B13 Adjacent-window difference adjacent occupancy difference 10 min following train (40,130 records)	-0.079	-2.07 [-3.82, -0.33]	0.255	100.00 / 100.00
R17 Occupation change rate occupancy rate 10 min following train (40,130 records)	-0.079	-2.07 [-3.82, -0.33]	0.255	100.00 / 100.00
F04 Physical betweenness betweenness leading train (40,130 records)	0.079	2.88 [1.83, 3.93]	9.2e-5	100.00 / 100.00
F04 Physical betweenness betweenness following train (40,130 records)	0.079	2.88 [1.83, 3.93]	9.2e-5	100.00 / 100.00
E10 Emergency pool capacity emergency pool capacity following train (9,770 records)	-0.077	-1.73 [-2.64, -0.81]	0.008	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R10 Intersecting dwell distribution waiting duration 10 min zero fraction difference (38,510 records)	0.077	1.87 [0.92, 2.82]	0.005	100.00 / 100.00
R15 Free windows free window count 5 min difference (40,130 records)	0.075	2.25 [1.02, 3.48]	0.011	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min minimum following train (38,510 records)	-0.075	-1.42 [-2.02, -0.82]	5.9e-4	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min zero fraction leading train (38,510 records)	-0.075	-3.18 [-4.52, -1.84]	5.7e-4	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min mean difference (38,510 records)	-0.073	-0.04 [-1.53, 1.45]	1.000	40.00 / 33.33
R10 Intersecting dwell distribution waiting duration 10 min standard deviation absolute pair difference (38,510 records)	0.073	0.82 [-0.68, 2.32]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min zero fraction absolute pair difference (38,510 records)	-0.072	-3.45 [-4.73, -2.16]	1.1e-4	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min interquartile range absolute pair difference (38,510 records)	0.072	1.43 [-0.05, 2.92]	0.653	100.00 / 100.00
R18 Opposite-direction utilization opposite utilization 10 min leading train (40,130 records)	0.072	2.36 [1.53, 3.19]	5.5e-5	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min minimum absolute pair difference (38,510 records)	-0.071	-1.15 [-1.97, -0.32]	0.098	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min zero fraction difference (38,510 records)	0.071	2.09 [1.11, 3.07]	0.002	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min 90th percentile difference (38,510 records)	-0.071	-0.72 [-2.35, 0.92]	1.000	100.00 / 66.67
R10 Intersecting dwell distribution waiting duration 5 min maximum absolute pair difference (38,510 records)	-0.068	0.20 [-1.19, 1.59]	1.000	86.67 / 33.33
R10 Intersecting dwell distribution waiting duration 20 min median absolute pair difference (38,510 records)	-0.068	-0.70 [-2.05, 0.64]	1.000	100.00 / 66.67
R16 Planned-arrival overlap planned arrival release overlap 5 min difference (40,130 records)	0.066	1.46 [-0.12, 3.03]	0.768	100.00 / 100.00
R13 Adjacent-station occupation downstream occupancy leading train (40,130 records)	0.065	2.53 [0.68, 4.38]	0.113	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min median difference (38,510 records)	-0.065	0.29 [-1.08, 1.67]	1.000	93.33 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R11 Flow imbalance arrivals minus departures 10 min following train (40,130 records)	-0.063	-1.57 [-2.72, -0.42]	0.112	100.00 / 100.00
F17 Distance to a hub hub distance leading train (40,130 records)	-0.062	-1.88 [-2.76, -1.00]	0.002	100.00 / 100.00
F17 Distance to a hub hub distance following train (40,130 records)	-0.062	-1.88 [-2.76, -1.00]	0.002	100.00 / 100.00
R16 Planned-arrival overlap planned arrival release overlap 10 min difference (40,130 records)	0.061	1.23 [-0.12, 2.57]	0.802	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min minimum difference (38,510 records)	-0.060	0.75 [-0.02, 1.52]	0.623	100.00 / 100.00
F18 Next stopping opportunity next stop distance difference (32,600 records)	0.060	2.07 [-1.13, 5.28]	1.000	100.00 / 100.00
R01 Mean occupation mean occupancy 20 min absolute pair difference (40,130 records)	0.059	0.55 [-0.68, 1.78]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min median absolute pair difference (38,510 records)	-0.059	-0.36 [-1.83, 1.11]	1.000	100.00 / 66.67
R10 Intersecting dwell distribution waiting duration 10 min zero fraction leading train (38,510 records)	-0.058	-3.04 [-4.23, -1.85]	2.3e-4	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min minimum difference (38,510 records)	-0.058	0.77 [-0.07, 1.61]	0.780	100.00 / 100.00
R04 Utilization utilization 20 min absolute pair difference (40,130 records)	0.058	0.81 [-0.62, 2.24]	1.000	100.00 / 100.00
R12 Demand/capacity proxy demand capacity 20 min absolute pair difference (40,130 records)	0.058	0.81 [-0.62, 2.24]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min mean difference (38,510 records)	-0.057	-0.09 [-1.53, 1.35]	1.000	53.33 / 33.33
R06 Saturation persistence longest saturation 10 min absolute pair difference (40,130 records)	0.057	1.79 [0.74, 2.85]	0.022	100.00 / 100.00
R04 Utilization utilization 10 min absolute pair difference (40,130 records)	0.056	1.44 [0.04, 2.85]	0.512	100.00 / 100.00
R12 Demand/capacity proxy demand capacity 10 min absolute pair difference (40,130 records)	0.056	1.44 [0.04, 2.85]	0.512	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min standard deviation difference (38,510 records)	-0.056	-1.22 [-2.70, 0.26]	1.000	100.00 / 100.00
R11 Flow imbalance arrivals minus departures 20 min following train (40,130 records)	-0.056	-1.50 [-2.96, -0.04]	0.510	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B13 Adjacent-window difference adjacent occupancy difference 20 min absolute pair difference (40,130 records)	0.055	1.31 [-0.68, 3.30]	1.000	100.00 / 100.00
R17 Occupation change rate occupancy rate 20 min absolute pair difference (40,130 records)	0.055	1.31 [-0.68, 3.30]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min 90th percentile absolute pair difference (38,510 records)	-0.054	0.02 [-1.49, 1.52]	1.000	33.33 / 33.33
R01 Mean occupation mean occupancy 10 min absolute pair difference (40,130 records)	0.054	0.94 [-0.13, 2.01]	0.898	100.00 / 100.00
R15 Free windows free window count 10 min difference (40,130 records)	0.053	1.91 [0.22, 3.59]	0.326	100.00 / 100.00
R06 Saturation persistence longest saturation 5 min difference (40,130 records)	-0.053	-2.01 [-3.01, -1.02]	0.004	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min 90th percentile absolute pair difference (38,510 records)	-0.052	0.00 [-1.55, 1.55]	1.000	66.67 / 66.67
R10 Intersecting dwell distribution waiting duration 10 min minimum following train (38,510 records)	-0.052	-1.34 [-1.86, -0.81]	2.4e-4	100.00 / 100.00
R02 Peak occupation peak occupancy 10 min absolute pair difference (40,130 records)	-0.051	-1.51 [-2.30, -0.72]	0.007	100.00 / 100.00
R03 Minimum free capacity minimum free 10 min absolute pair difference (40,130 records)	-0.051	-1.51 [-2.30, -0.72]	0.007	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min count difference (40,130 records)	0.051	1.54 [-0.54, 3.63]	1.000	100.00 / 100.00
R14 Window competitors window competitors 10 min difference (40,130 records)	0.051	1.54 [-0.54, 3.63]	1.000	100.00 / 100.00
H19 Peak demand/capacity peak demand capacity 10 min absolute pair difference (40,130 records)	-0.051	-1.46 [-2.28, -0.63]	0.016	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min mean difference (38,510 records)	-0.050	-0.09 [-1.36, 1.18]	1.000	60.00 / 33.33
R08 Exit load exit load 10 min leading train (40,130 records)	-0.050	-0.56 [-2.64, 1.53]	1.000	100.00 / 66.67
R10 Intersecting dwell distribution waiting duration 20 min zero fraction difference (38,510 records)	0.049	1.95 [0.96, 2.94]	0.005	100.00 / 100.00
R09 Reference headway buffer 0 previous buffer absolute pair difference (36,410 records)	0.049	2.43 [0.07, 4.78]	0.503	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min median difference (38,510 records)	-0.048	-0.02 [-1.50, 1.45]	1.000	46.67 / 33.33

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R18 Opposite-direction utilization opposite utilization 20 min leading train (40,130 records)	0.047	2.81 [1.59, 4.03]	8.5e-4	100.00 / 100.00
R01 Mean occupation mean occupancy 5 min difference (40,130 records)	-0.046	-1.41 [-2.44, -0.38]	0.112	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min 90th percentile difference (38,510 records)	-0.046	-0.58 [-2.17, 1.01]	1.000	100.00 / 33.33
R10 Intersecting dwell distribution waiting duration 5 min minimum absolute pair difference (38,510 records)	-0.046	-1.07 [-1.93, -0.21]	0.195	100.00 / 100.00
R04 Utilization utilization 5 min difference (40,130 records)	-0.045	-1.42 [-2.40, -0.44]	0.074	100.00 / 100.00
R12 Demand/capacity proxy demand capacity 5 min difference (40,130 records)	-0.045	-1.42 [-2.40, -0.44]	0.074	100.00 / 100.00
R09 Reference headway buffer 1 previous buffer absolute pair difference (36,410 records)	0.044	2.16 [-0.15, 4.47]	0.729	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min maximum difference (38,510 records)	-0.042	-0.67 [-2.15, 0.81]	1.000	100.00 / 66.67
R10 Intersecting dwell distribution waiting duration 10 min mean absolute pair difference (38,510 records)	-0.042	-0.44 [-1.80, 0.93]	1.000	100.00 / 66.67
R10 Intersecting dwell distribution waiting duration 20 min zero fraction leading train (38,510 records)	-0.042	-2.90 [-3.92, -1.88]	5.6e-5	100.00 / 100.00
R02 Peak occupation peak occupancy 5 min difference (40,130 records)	-0.042	-1.25 [-2.30, -0.20]	0.251	100.00 / 100.00
R03 Minimum free capacity minimum free 5 min difference (40,130 records)	0.042	1.25 [0.20, 2.30]	0.251	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min 90th percentile difference (38,510 records)	-0.041	-0.67 [-2.06, 0.71]	1.000	100.00 / 66.67
H19 Peak demand/capacity peak demand capacity 5 min difference (40,130 records)	-0.041	-1.19 [-2.30, -0.07]	0.440	100.00 / 100.00
F01 Facility role hub leading train (40,130 records)	0.041	1.28 [0.52, 2.05]	0.024	100.00 / 100.00
F01 Facility role hub following train (40,130 records)	0.041	1.28 [0.52, 2.05]	0.024	100.00 / 100.00
F06 Directional pool size direction pool leading train (40,130 records)	0.041	1.28 [0.52, 2.05]	0.024	100.00 / 100.00
F06 Directional pool size direction pool following train (40,130 records)	0.041	1.28 [0.52, 2.05]	0.024	100.00 / 100.00
F15 Parallel-capacity proxy parallel capacity leading train (40,130 records)	0.041	1.28 [0.52, 2.05]	0.024	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
F15 Parallel-capacity proxy parallel capacity following train (40,130 records)	0.041	1.28 [0.52, 2.05]	0.024	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min maximum difference (38,510 records)	-0.041	-0.77 [-2.13, 0.59]	1.000	100.00 / 66.67
R10 Intersecting dwell distribution waiting duration 20 min sum absolute pair difference (40,130 records)	0.040	0.30 [-1.21, 1.80]	1.000	100.00 / 33.33
R10 Intersecting dwell distribution waiting duration 20 min mean absolute pair difference (38,510 records)	-0.040	-0.56 [-1.90, 0.77]	1.000	100.00 / 66.67
R10 Intersecting dwell distribution waiting duration 5 min median difference (38,510 records)	-0.038	-0.13 [-1.55, 1.30]	1.000	66.67 / 33.33
R06 Saturation persistence longest saturation 10 min difference (40,130 records)	-0.038	-1.28 [-2.11, -0.46]	0.044	100.00 / 100.00
F02 Physical degree undirected degree leading train (40,130 records)	0.036	1.18 [0.42, 1.94]	0.047	100.00 / 100.00
F02 Physical degree undirected degree following train (40,130 records)	0.036	1.18 [0.42, 1.94]	0.047	100.00 / 100.00
F03 Branch-count proxy branch count leading train (40,130 records)	0.036	1.18 [0.42, 1.94]	0.047	100.00 / 100.00
F03 Branch-count proxy branch count following train (40,130 records)	0.036	1.18 [0.42, 1.94]	0.047	100.00 / 100.00
F07 Compatible tracks compatible tracks leading train (40,130 records)	0.036	1.06 [0.30, 1.81]	0.094	100.00 / 100.00
F07 Compatible tracks compatible tracks following train (40,130 records)	0.036	1.06 [0.30, 1.81]	0.094	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min interquartile range difference (38,510 records)	-0.036	-1.45 [-3.12, 0.22]	0.936	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min maximum difference (38,510 records)	-0.036	-0.56 [-2.12, 1.01]	1.000	100.00 / 66.67
R18 Opposite-direction utilization opposite utilization 10 min difference (40,130 records)	0.036	0.83 [-0.12, 1.79]	0.930	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min minimum difference (38,510 records)	-0.035	0.99 [0.14, 1.84]	0.286	100.00 / 100.00
R18 Opposite-direction utilization opposite utilization 5 min absolute pair difference (40,130 records)	0.035	0.14 [-0.96, 1.24]	1.000	86.67 / 66.67
R10 Intersecting dwell distribution waiting duration 20 min zero fraction following train (38,510 records)	0.035	-1.03 [-2.45, 0.39]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min sum absolute pair difference (40,130 records)	0.034	0.38 [-1.08, 1.84]	1.000	100.00 / 33.33

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R04 Utilization utilization 20 min difference (40,130 records)	0.033	0.83 [-0.75, 2.41]	1.000	100.00 / 66.67
R12 Demand/capacity proxy demand capacity 20 min difference (40,130 records)	0.033	0.83 [-0.75, 2.41]	1.000	100.00 / 66.67
R01 Mean occupation mean occupancy 20 min difference (40,130 records)	0.033	0.54 [-0.88, 1.95]	1.000	100.00 / 66.67
R15 Free windows free window count 20 min difference (40,130 records)	0.031	1.16 [-0.61, 2.93]	1.000	100.00 / 100.00
B07 Regional saturation saturated fraction 20 min difference (40,130 records)	0.031	0.49 [-0.84, 1.82]	1.000	100.00 / 100.00
R05 Saturated fraction saturated fraction 20 min difference (40,130 records)	0.031	0.49 [-0.84, 1.82]	1.000	100.00 / 100.00
R15 Free windows free window total 20 min difference (40,130 records)	-0.031	-0.49 [-1.82, 0.84]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min minimum leading train (38,510 records)	0.031	-1.47 [-2.17, -0.76]	0.003	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min 90th percentile absolute pair difference (38,510 records)	-0.031	0.16 [-1.26, 1.58]	1.000	80.00 / 33.33
R10 Intersecting dwell distribution waiting duration 10 min zero fraction following train (38,510 records)	0.030	-1.14 [-2.25, -0.03]	0.503	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min interquartile range difference (38,510 records)	-0.029	-1.60 [-3.07, -0.13]	0.393	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 20 min sum difference (40,130 records)	0.029	-0.62 [-2.13, 0.88]	1.000	100.00 / 33.33
R06 Saturation persistence longest saturation 20 min absolute pair difference (40,130 records)	0.029	0.94 [0.02, 1.86]	0.520	100.00 / 100.00
R18 Opposite-direction utilization opposite utilization 10 min absolute pair difference (40,130 records)	0.029	0.53 [-0.73, 1.80]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min standard deviation difference (38,510 records)	-0.029	-1.24 [-2.76, 0.29]	1.000	100.00 / 100.00
R06 Saturation persistence longest saturation 20 min difference (40,130 records)	-0.027	-1.01 [-1.71, -0.31]	0.077	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min median absolute pair difference (38,510 records)	-0.025	-0.14 [-1.61, 1.33]	1.000	73.33 / 66.67
R10 Intersecting dwell distribution waiting duration 20 min interquartile range absolute pair difference (38,510 records)	0.024	1.15 [-0.53, 2.83]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
F18 Next stopping opportunity next stop distance absolute pair difference (32,600 records)	0.024	0.81 [-1.28, 2.90]	1.000	100.00 / 100.00
R18 Opposite-direction utilization opposite utilization 20 min absolute pair difference (40,130 records)	0.024	0.72 [-0.59, 2.03]	1.000	100.00 / 100.00
R02 Peak occupation peak occupancy 10 min difference (40,130 records)	-0.023	-0.57 [-1.39, 0.25]	1.000	100.00 / 100.00
R03 Minimum free capacity minimum free 10 min difference (40,130 records)	0.023	0.57 [-0.25, 1.39]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min sum absolute pair difference (40,130 records)	0.023	0.33 [-1.00, 1.66]	1.000	100.00 / 33.33
R10 Intersecting dwell distribution waiting duration 5 min sum difference (40,130 records)	-0.023	-0.55 [-1.82, 0.72]	1.000	100.00 / 33.33
R10 Intersecting dwell distribution waiting duration 20 min standard deviation absolute pair difference (38,510 records)	0.023	0.30 [-1.13, 1.72]	1.000	100.00 / 66.67
H19 Peak demand/capacity peak demand capacity 10 min difference (40,130 records)	-0.022	-0.57 [-1.46, 0.33]	1.000	100.00 / 100.00
B13 Adjacent-window difference adjacent occupancy difference 20 min leading train (40,130 records)	0.022	-0.28 [-2.35, 1.80]	1.000	86.67 / 66.67
R17 Occupation change rate occupancy rate 20 min leading train (40,130 records)	0.022	-0.28 [-2.35, 1.80]	1.000	86.67 / 66.67
R10 Intersecting dwell distribution waiting duration 5 min standard deviation difference (38,510 records)	-0.021	-1.71 [-3.16, -0.26]	0.268	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min interquartile range difference (38,510 records)	-0.020	-1.59 [-2.96, -0.22]	0.292	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min zero fraction following train (38,510 records)	0.020	-0.88 [-1.72, -0.05]	0.456	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 10 min minimum leading train (38,510 records)	0.017	-1.45 [-2.13, -0.77]	0.002	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min mean absolute pair difference (38,510 records)	-0.015	-0.29 [-1.57, 0.99]	1.000	86.67 / 66.67
F16 Downstream capacity gradient next capacity delta leading train (40,130 records)	0.014	0.47 [-0.08, 1.01]	1.000	100.00 / 100.00
F16 Downstream capacity gradient next capacity delta following train (40,130 records)	0.014	0.47 [-0.08, 1.01]	1.000	100.00 / 100.00
R02 Peak occupation peak occupancy 20 min difference (40,130 records)	-0.014	-0.38 [-1.31, 0.55]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R03 Minimum free capacity minimum free 20 min difference (40,130 records)	0.014	0.38 [-0.55, 1.31]	1.000	100.00 / 100.00
R10 Intersecting dwell distribution waiting duration 5 min count difference (40,130 records)	0.014	0.27 [-1.38, 1.93]	1.000	100.00 / 100.00
R14 Window competitors window competitors 5 min difference (40,130 records)	0.014	0.27 [-1.38, 1.93]	1.000	100.00 / 100.00
H19 Peak demand/capacity peak demand capacity 20 min difference (40,130 records)	-0.014	-0.39 [-1.41, 0.62]	1.000	100.00 / 100.00
R18 Opposite-direction utilization opposite utilization 5 min difference (40,130 records)	0.014	0.08 [-0.62, 0.79]	1.000	80.00 / 66.67
R10 Intersecting dwell distribution waiting duration 5 min minimum following train (38,510 records)	-0.014	-1.08 [-1.47, -0.69]	7.7e-5	93.33 / 66.67
R01 Mean occupation mean occupancy 10 min difference (40,130 records)	-0.013	-0.15 [-1.42, 1.11]	1.000	73.33 / 33.33
R15 Free windows free window maximum 20 min difference (40,130 records)	-0.012	-0.09 [-1.38, 1.21]	1.000	80.00 / 66.67
R02 Peak occupation peak occupancy 5 min absolute pair difference (40,130 records)	-0.012	-0.28 [-0.81, 0.26]	1.000	100.00 / 66.67
R03 Minimum free capacity minimum free 5 min absolute pair difference (40,130 records)	-0.012	-0.28 [-0.81, 0.26]	1.000	100.00 / 66.67
R10 Intersecting dwell distribution waiting duration 20 min minimum leading train (38,510 records)	-0.011	-1.51 [-2.15, -0.87]	6.0e-4	100.00 / 100.00
F05 Physical articulation point articulation leading train (40,130 records)	-0.011	-0.34 [-0.96, 0.27]	1.000	100.00 / 66.67
F05 Physical articulation point articulation following train (40,130 records)	-0.011	-0.34 [-0.96, 0.27]	1.000	100.00 / 66.67
H19 Peak demand/capacity peak demand capacity 5 min absolute pair difference (40,130 records)	-0.011	-0.19 [-0.84, 0.46]	1.000	93.33 / 66.67
E23 Initial station occupation initial station occupied leading train (40,130 records)	0.010	0.32 [-1.00, 1.65]	1.000	93.33 / 66.67
E23 Initial station occupation initial station occupied following train (40,130 records)	0.010	0.32 [-1.00, 1.65]	1.000	93.33 / 66.67
R04 Utilization utilization 10 min difference (40,130 records)	-0.010	0.04 [-1.51, 1.59]	1.000	46.67 / 66.67
R12 Demand/capacity proxy demand capacity 10 min difference (40,130 records)	-0.010	0.04 [-1.51, 1.59]	1.000	46.67 / 66.67
R01 Mean occupation mean occupancy 5 min absolute pair difference (40,130 records)	0.009	0.32 [-0.14, 0.78]	1.000	100.00 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
R15 Free windows free window maximum 10 min difference (40,130 records)	-0.008	0.04 [-1.11, 1.19]	1.000	60.00 / 66.67
R04 Utilization utilization 5 min absolute pair difference (40,130 records)	0.007	0.13 [-0.36, 0.62]	1.000	93.33 / 33.33
R12 Demand/capacity proxy demand capacity 5 min absolute pair difference (40,130 records)	0.007	0.13 [-0.36, 0.62]	1.000	93.33 / 33.33
B07 Regional saturation saturated fraction 5 min difference (40,130 records)	0.007	-0.60 [-1.52, 0.32]	1.000	100.00 / 100.00
R05 Saturated fraction saturated fraction 5 min difference (40,130 records)	0.007	-0.60 [-1.52, 0.32]	1.000	100.00 / 100.00
R15 Free windows free window total 5 min difference (40,130 records)	-0.007	0.60 [-0.32, 1.52]	1.000	100.00 / 100.00
F18 Next stopping opportunity next stop distance leading train (33,500 records)	-0.006	-0.21 [-3.20, 2.78]	1.000	86.67 / 33.33
R15 Free windows free window maximum 5 min difference (40,130 records)	0.005	0.58 [-0.39, 1.55]	1.000	100.00 / 100.00
R08 Exit load exit load 5 min leading train (40,130 records)	0.005	0.86 [-1.02, 2.74]	1.000	100.00 / 66.67
R15 Free windows free window total 10 min difference (40,130 records)	-0.004	0.12 [-1.09, 1.33]	1.000	80.00 / 66.67
B07 Regional saturation saturated fraction 10 min difference (40,130 records)	0.004	-0.12 [-1.33, 1.09]	1.000	80.00 / 66.67
R05 Saturated fraction saturated fraction 10 min difference (40,130 records)	0.004	-0.12 [-1.33, 1.09]	1.000	80.00 / 66.67
R10 Intersecting dwell distribution waiting duration 10 min sum difference (40,130 records)	-0.002	-0.58 [-2.02, 0.86]	1.000	100.00 / 33.33
Delay propagation patterns				
H07 Downstream trains downstream departure leading train (40,130 records)	0.388	11.80 [8.51, 15.09]	2.0e-6	100.00 / 100.00
H07 Downstream trains downstream arrival leading train (40,130 records)	0.386	12.18 [9.21, 15.14]	3.2e-7	100.00 / 100.00
H09 Reachable delay totals downstream train maximum delay sum arrival leading train (40,130 records)	0.385	11.74 [8.71, 14.76]	6.7e-7	100.00 / 100.00
H09 Reachable delay totals downstream train maximum delay sum departure leading train (40,130 records)	0.384	11.23 [7.84, 14.62]	6.2e-6	100.00 / 100.00
H10 Active betweenness betweenness departure leading train (40,130 records)	0.383	9.95 [6.75, 13.15]	1.5e-5	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H09 Reachable delay totals downstream event delay arrival leading train (40,130 records)	0.380	11.66 [8.78, 14.55]	4.2e-7	100.00 / 100.00
H11 Harmonic closeness outgoing harmonic departure leading train (40,130 records)	0.380	12.39 [9.55, 15.23]	1.0e-7	100.00 / 100.00
H11 Harmonic closeness outgoing harmonic arrival leading train (40,130 records)	0.379	12.61 [10.10, 15.12]	6.9e-9	100.00 / 100.00
H05 Downstream reachability downstream arrival leading train (40,130 records)	0.378	12.11 [9.41, 14.82]	6.6e-8	100.00 / 100.00
H08 Spatial reachability downstream stations arrival leading train (40,130 records)	0.377	11.95 [9.86, 14.05]	5.0e-10	100.00 / 100.00
H09 Reachable delay totals downstream event delay departure leading train (40,130 records)	0.376	10.76 [7.47, 14.05]	7.5e-6	100.00 / 100.00
H05 Downstream reachability downstream departure leading train (40,130 records)	0.372	11.24 [8.21, 14.27]	1.4e-6	100.00 / 100.00
H08 Spatial reachability downstream branches arrival leading train (40,130 records)	0.365	11.60 [9.42, 13.77]	2.1e-9	100.00 / 100.00
H13 Weak-component size component events arrival leading train (40,130 records)	0.365	11.48 [8.69, 14.27]	3.1e-7	100.00 / 100.00
H08 Spatial reachability downstream stations departure leading train (40,130 records)	0.364	11.29 [8.74, 13.84]	7.8e-8	100.00 / 100.00
H08 Spatial reachability downstream branches departure leading train (40,130 records)	0.361	11.27 [8.71, 13.84]	8.9e-8	100.00 / 100.00
H13 Weak-component size component trains arrival leading train (40,130 records)	0.358	11.27 [8.24, 14.30]	1.3e-6	100.00 / 100.00
H10 Active betweenness betweenness arrival leading train (40,130 records)	0.356	9.77 [6.61, 12.92]	1.6e-5	100.00 / 100.00
H13 Weak-component size component events departure leading train (40,130 records)	0.356	11.15 [8.33, 13.96]	5.0e-7	100.00 / 100.00
B05 Regional delay distribution region delay 20 min sum leading train (40,130 records)	0.354	9.77 [6.82, 12.72]	6.3e-6	100.00 / 100.00
B05 Regional delay distribution region delay 5 min sum leading train (40,130 records)	0.351	9.96 [7.06, 12.85]	3.7e-6	100.00 / 100.00
G08 Active degree outgoing active departure leading train (40,130 records)	0.349	11.11 [8.04, 14.19]	1.8e-6	100.00 / 100.00
H13 Weak-component size component trains departure leading train (40,130 records)	0.349	10.96 [7.91, 14.01]	2.0e-6	100.00 / 100.00
B05 Regional delay distribution region delay 10 min sum leading train (40,130 records)	0.348	9.68 [6.83, 12.52]	4.2e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H24 Projection core number core arrival leading train (40,130 records)	0.348	10.93 [8.15, 13.70]	5.4e-7	100.00 / 100.00
G12 Delay-support distribution incoming support sum arrival leading train (40,130 records)	0.347	9.27 [6.34, 12.21]	1.2e-5	100.00 / 100.00
B08 Community structure community events departure leading train (40,130 records)	0.346	10.64 [7.88, 13.40]	7.3e-7	100.00 / 100.00
B08 Community structure community events arrival leading train (40,130 records)	0.345	10.50 [7.71, 13.30]	1.1e-6	100.00 / 100.00
G12 Delay-support distribution incoming support mean arrival leading train (38,990 records)	0.344	9.53 [6.87, 12.19]	2.1e-6	100.00 / 100.00
G12 Delay-support distribution incoming support maximum arrival leading train (38,990 records)	0.343	10.28 [7.31, 13.25]	3.3e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support minimum arrival leading train (40,130 records)	0.342	9.64 [6.79, 12.49]	4.6e-6	100.00 / 100.00
H24 Projection core number core departure leading train (40,130 records)	0.341	10.70 [7.67, 13.72]	2.6e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support zero fraction arrival leading train (40,130 records)	-0.341	-10.64 [-13.50, -7.78]	1.3e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support zero fraction departure leading train (40,130 records)	-0.341	-10.62 [-13.51, -7.73]	1.4e-6	100.00 / 100.00
G12 Delay-support distribution incoming support 90th percentile arrival leading train (38,990 records)	0.340	10.13 [7.23, 13.03]	3.0e-6	100.00 / 100.00
H09 Reachable delay totals downstream train maximum delay sum departure difference (40,130 records)	-0.339	-9.08 [-12.44, -5.72]	1.1e-4	100.00 / 100.00
B05 Regional delay distribution region delay 20 min standard deviation leading train (40,130 records)	0.336	10.28 [6.70, 13.86]	4.8e-5	100.00 / 100.00
B06 Regional delay variance delay variance 20 min leading train (40,130 records)	0.336	8.75 [5.44, 12.06]	1.5e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support mean arrival leading train (40,130 records)	0.336	9.85 [6.92, 12.78]	5.0e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support median arrival leading train (40,130 records)	0.336	9.85 [6.92, 12.78]	5.0e-6	100.00 / 100.00
G01 Reference arrival delay arrival delay leading train (40,130 records)	0.336	9.96 [6.99, 12.93]	5.1e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support 90th percentile arrival leading train (40,130 records)	0.334	9.95 [6.98, 12.91]	5.1e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support maximum arrival leading train (40,130 records)	0.334	9.96 [6.99, 12.93]	5.1e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H09 Reachable delay totals downstream event delay departure difference (40,130 records)	-0.333	-8.84 [-12.07, -5.61]	9.1e-5	100.00 / 100.00
H09 Reachable delay totals downstream train maximum delay sum arrival difference (40,130 records)	-0.332	-9.52 [-12.39, -6.66]	6.0e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support sum arrival leading train (40,130 records)	0.331	9.52 [6.54, 12.50]	1.0e-5	100.00 / 100.00
B05 Regional delay distribution region delay 10 min standard deviation leading train (40,130 records)	0.331	9.89 [6.39, 13.40]	6.0e-5	100.00 / 100.00
B06 Regional delay variance delay variance 10 min leading train (40,130 records)	0.331	7.72 [4.39, 11.06]	8.0e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support median departure leading train (40,130 records)	0.328	9.49 [6.45, 12.52]	1.4e-5	100.00 / 100.00
B05 Regional delay distribution region delay 20 min interquartile range leading train (40,130 records)	0.328	9.84 [6.16, 13.52]	1.3e-4	100.00 / 100.00
H09 Reachable delay totals downstream event delay arrival difference (40,130 records)	-0.328	-9.81 [-12.52, -7.11]	1.7e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support sum departure leading train (40,130 records)	0.327	9.38 [6.52, 12.24]	7.1e-6	100.00 / 100.00
B05 Regional delay distribution region delay 20 min zero fraction leading train (40,130 records)	-0.327	-10.14 [-12.88, -7.40]	1.4e-6	100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction arrival leading train (38,990 records)	-0.323	-9.87 [-12.63, -7.12]	2.1e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support mean departure leading train (40,130 records)	0.322	9.24 [6.26, 12.22]	1.5e-5	100.00 / 100.00
G12 Delay-support distribution incoming support mean departure leading train (40,130 records)	0.321	8.87 [6.12, 11.61]	8.7e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support 90th percentile departure leading train (40,130 records)	0.319	9.37 [6.43, 12.31]	1.1e-5	100.00 / 100.00
G12 Delay-support distribution incoming support sum departure leading train (40,130 records)	0.318	8.71 [5.73, 11.69]	3.7e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support minimum departure leading train (40,130 records)	0.318	8.63 [5.72, 11.54]	3.0e-5	100.00 / 100.00
B05 Regional delay distribution region delay 20 min standard deviation following train (40,130 records)	0.317	10.00 [6.54, 13.45]	4.3e-5	100.00 / 100.00
B06 Regional delay variance delay variance 20 min following train (40,130 records)	0.317	8.95 [5.81, 12.10]	5.3e-5	100.00 / 100.00
G12 Delay-support distribution incoming support maximum departure leading train (40,130 records)	0.317	9.37 [6.45, 12.29]	9.6e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support 90th percentile departure leading train (40,130 records)	0.316	9.36 [6.45, 12.27]	9.3e-6	100.00 / 100.00
H13 Weak-component size component events departure following train (40,130 records)	0.316	9.91 [7.10, 12.73]	2.8e-6	100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction departure following train (40,130 records)	-0.315	-9.69 [-12.25, -7.13]	1.0e-6	100.00 / 100.00
G02 Reference departure delay departure delay leading train (40,130 records)	0.315	9.32 [6.41, 12.23]	9.9e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support maximum departure leading train (40,130 records)	0.315	9.32 [6.41, 12.23]	9.9e-6	100.00 / 100.00
B05 Regional delay distribution region delay 10 min zero fraction leading train (40,130 records)	-0.314	-9.81 [-12.45, -7.17]	1.3e-6	100.00 / 100.00
G08 Active degree outgoing active arrival leading train (40,130 records)	0.314	9.64 [7.09, 12.20]	1.1e-6	100.00 / 100.00
B05 Regional delay distribution region delay 5 min maximum leading train (40,130 records)	0.314	9.39 [6.47, 12.31]	9.4e-6	100.00 / 100.00
B05 Regional delay distribution region delay 5 min 90th percentile leading train (40,130 records)	0.314	9.35 [6.44, 12.27]	9.8e-6	100.00 / 100.00
B05 Regional delay distribution region delay 20 min maximum leading train (40,130 records)	0.313	9.54 [6.62, 12.47]	7.7e-6	100.00 / 100.00
B05 Regional delay distribution region delay 10 min 90th percentile leading train (40,130 records)	0.313	9.36 [6.44, 12.28]	9.7e-6	100.00 / 100.00
B05 Regional delay distribution region delay 10 min maximum leading train (40,130 records)	0.312	9.38 [6.48, 12.28]	8.7e-6	100.00 / 100.00
B05 Regional delay distribution region delay 5 min median leading train (40,130 records)	0.311	9.10 [6.20, 12.00]	1.3e-5	100.00 / 100.00
G12 Delay-support distribution incoming support interquartile range arrival leading train (38,990 records)	0.311	5.54 [3.14, 7.93]	8.1e-4	100.00 / 100.00
H16 Local motifs fan-out pairs departure leading train (40,130 records)	0.311	9.67 [7.42, 11.92]	1.3e-7	100.00 / 100.00
B05 Regional delay distribution region delay 5 min zero fraction leading train (40,130 records)	-0.311	-9.75 [-12.37, -7.13]	1.3e-6	100.00 / 100.00
B05 Regional delay distribution region delay 20 min 90th percentile leading train (40,130 records)	0.310	9.45 [6.53, 12.36]	8.4e-6	100.00 / 100.00
G12 Delay-support distribution incoming support standard deviation arrival leading train (38,990 records)	0.310	5.45 [3.07, 7.84]	9.4e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution region delay 10 min standard deviation following train (40,130 records)	0.310	9.48 [6.47, 12.50]	1.3e-5	100.00 / 100.00
B06 Regional delay variance delay variance 10 min following train (40,130 records)	0.310	7.96 [5.23, 10.70]	3.8e-5	100.00 / 100.00
B05 Regional delay distribution region delay 5 min mean leading train (40,130 records)	0.309	9.02 [6.16, 11.89]	1.3e-5	100.00 / 100.00
B05 Regional delay distribution region delay 10 min median leading train (40,130 records)	0.308	8.91 [6.07, 11.75]	1.3e-5	100.00 / 100.00
B05 Regional delay distribution region delay 20 min interquartile range following train (40,130 records)	0.306	9.62 [6.48, 12.76]	1.9e-5	100.00 / 100.00
G09 Active cross-train links active cross departure following train (40,130 records)	0.306	9.55 [7.35, 11.76]	1.2e-7	100.00 / 100.00
H13 Weak-component size component trains departure following train (40,130 records)	0.305	9.74 [6.70, 12.78]	9.7e-6	100.00 / 100.00
G09 Active cross-train links active cross fraction departure following train (40,130 records)	0.304	9.52 [7.20, 11.85]	3.4e-7	100.00 / 100.00
B05 Regional delay distribution region delay 10 min mean leading train (40,130 records)	0.304	8.79 [6.00, 11.59]	1.3e-5	100.00 / 100.00
H02 Longest active path longest path departure leading train (40,130 records)	0.304	8.30 [5.76, 10.85]	7.7e-6	100.00 / 100.00
H15 Chain and crossing length longest chain departure leading train (40,130 records)	0.304	8.30 [5.76, 10.85]	7.7e-6	100.00 / 100.00
B05 Regional delay distribution region delay 20 min median leading train (40,130 records)	0.304	8.68 [5.91, 11.45]	1.3e-5	100.00 / 100.00
H08 Spatial reachability upstream stations departure leading train (40,130 records)	0.303	9.08 [6.70, 11.47]	9.5e-7	100.00 / 100.00
G12 Delay-support distribution incoming support median departure leading train (40,130 records)	0.303	8.41 [5.72, 11.09]	1.3e-5	100.00 / 100.00
B05 Regional delay distribution region delay 5 min minimum leading train (40,130 records)	0.301	8.31 [5.61, 11.02]	1.8e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support median departure difference (40,130 records)	-0.301	-6.81 [-9.47, -4.16]	2.1e-4	100.00 / 100.00
B05 Regional delay distribution region delay 20 min mean leading train (40,130 records)	0.301	8.62 [5.87, 11.36]	1.3e-5	100.00 / 100.00
H08 Spatial reachability upstream branches departure leading train (40,130 records)	0.300	9.26 [6.72, 11.79]	1.6e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support mean arrival difference (40,130 records)	-0.300	-6.49 [-9.07, -3.90]	2.9e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution outgoing support median arrival difference (40,130 records)	-0.300	-6.49 [-9.07, -3.90]	2.9e-4	100.00 / 100.00
H09 Reachable delay totals downstream event delay arrival absolute pair difference (40,130 records)	0.300	8.96 [6.41, 11.51]	2.9e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support minimum arrival difference (40,130 records)	-0.299	-6.59 [-9.31, -3.88]	4.4e-4	100.00 / 100.00
H13 Weak-component size component events arrival following train (40,130 records)	0.298	9.21 [6.53, 11.89]	3.7e-6	100.00 / 100.00
G12 Delay-support distribution incoming support maximum arrival difference (38,773 records)	-0.298	-6.96 [-9.29, -4.62]	2.8e-5	100.00 / 100.00
H09 Reachable delay totals upstream event delay departure leading train (40,130 records)	0.298	6.42 [3.89, 8.94]	2.4e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support mean departure difference (40,130 records)	-0.297	-6.46 [-8.93, -3.99]	1.7e-4	100.00 / 100.00
B08 Community structure community events departure following train (40,130 records)	0.297	9.16 [6.57, 11.75]	2.6e-6	100.00 / 100.00
G16 Incident-link imbalance pressure balance departure leading train (40,130 records)	0.296	7.84 [5.26, 10.42]	2.2e-5	100.00 / 100.00
H24 Projection core number core departure following train (40,130 records)	0.295	9.26 [6.98, 11.55]	4.0e-7	100.00 / 100.00
H09 Reachable delay totals downstream event delay departure absolute pair difference (40,130 records)	0.295	7.92 [5.02, 10.83]	9.6e-5	100.00 / 100.00
H09 Reachable delay totals upstream event delay arrival leading train (40,130 records)	0.294	5.71 [3.36, 8.06]	4.3e-4	100.00 / 100.00
H02 Longest active path longest path arrival leading train (40,130 records)	0.294	7.58 [5.26, 9.91]	7.6e-6	100.00 / 100.00
H15 Chain and crossing length longest chain arrival leading train (40,130 records)	0.294	7.58 [5.26, 9.91]	7.6e-6	100.00 / 100.00
G08 Active degree incoming active departure following train (40,130 records)	0.293	9.25 [6.85, 11.65]	7.4e-7	100.00 / 100.00
B05 Regional delay distribution region delay 20 min sum following train (40,130 records)	0.292	7.94 [5.49, 10.39]	8.4e-6	100.00 / 100.00
H05 Downstream reachability downstream arrival difference (40,130 records)	-0.292	-8.99 [-11.63, -6.36]	4.0e-6	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay zero fraction departure following train (40,130 records)	-0.291	-9.13 [-11.63, -6.63]	1.6e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H09 Reachable delay totals downstream train maximum delay sum arrival absolute pair difference (40,130 records)	0.290	8.32 [5.67, 10.97]	1.3e-5	100.00 / 100.00
B05 Regional delay distribution region delay 10 min minimum leading train (40,130 records)	0.290	7.59 [5.15, 10.02]	1.4e-5	100.00 / 100.00
H05 Downstream reachability downstream arrival absolute pair difference (40,130 records)	0.288	8.76 [6.38, 11.15]	1.5e-6	100.00 / 100.00
H05 Downstream reachability downstream departure difference (40,130 records)	-0.288	-7.72 [-10.74, -4.70]	2.3e-4	100.00 / 100.00
H07 Downstream trains downstream arrival difference (40,130 records)	-0.288	-8.34 [-11.12, -5.56]	2.6e-5	100.00 / 100.00
H07 Downstream trains downstream departure difference (40,130 records)	-0.286	-7.66 [-10.89, -4.42]	6.0e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support 90th percentile arrival difference (40,130 records)	-0.285	-6.33 [-8.79, -3.87]	2.0e-4	100.00 / 100.00
H13 Weak-component size component trains arrival following train (40,130 records)	0.284	9.01 [6.14, 11.88]	1.3e-5	100.00 / 100.00
G19 Endpoint contrasts H04 upstream reachability endpoint contrast following train (40,130 records)	0.283	5.28 [3.10, 7.45]	4.4e-4	100.00 / 100.00
H09 Reachable delay totals downstream train maximum delay sum departure absolute pair difference (40,130 records)	0.283	7.70 [4.74, 10.65]	1.7e-4	100.00 / 100.00
B05 Regional delay distribution region delay 10 min interquartile range following train (40,130 records)	0.283	8.42 [5.72, 11.13]	1.5e-5	100.00 / 100.00
B05 Regional delay distribution region delay 20 min minimum leading train (40,130 records)	0.282	7.08 [4.63, 9.53]	4.3e-5	100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction departure leading train (40,130 records)	-0.281	-8.93 [-11.64, -6.21]	6.8e-6	100.00 / 100.00
H03 Event time from release time from release departure absolute pair difference (40,130 records)	-0.281	-6.89 [-9.30, -4.48]	5.0e-5	100.00 / 100.00
H03 Event time from release time from release departure difference (40,130 records)	-0.281	-6.89 [-9.30, -4.48]	5.0e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack mean departure leading train (40,130 records)	-0.279	-5.71 [-7.73, -3.68]	6.0e-5	100.00 / 100.00
B11 Global path distribution root longest paths zero fraction (40,130 records)	-0.278	-8.76 [-11.69, -5.83]	2.7e-5	100.00 / 100.00
B15 Global degree distribution incoming degree zero fraction (40,130 records)	-0.278	-8.76 [-11.69, -5.83]	2.7e-5	100.00 / 100.00

Continued on the next page.

Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution outgoing support 90th percentile departure difference (40,130 records)	-0.278	-6.28 [-8.67, -3.89]	1.6e-4	100.00 / 100.00
B15 Global degree distribution outgoing degree zero fraction (40,130 records)	-0.278	-8.76 [-11.64, -5.88]	2.2e-5	100.00 / 100.00
H08 Spatial reachability upstream branches arrival leading train (40,130 records)	0.278	8.03 [5.79, 10.28]	2.2e-6	100.00 / 100.00
B01 Local region size region trains 20 min following train (40,130 records)	0.276	8.64 [6.09, 11.19]	4.5e-6	100.00 / 100.00
B15 Global degree distribution incoming degree mean (40,130 records)	0.276	8.72 [5.80, 11.64]	2.8e-5	100.00 / 100.00
B15 Global degree distribution incoming degree sum (40,130 records)	0.276	8.72 [5.80, 11.64]	2.8e-5	100.00 / 100.00
H11 Harmonic closeness outgoing harmonic arrival difference (40,130 records)	-0.276	-8.64 [-10.98, -6.30]	1.4e-6	100.00 / 100.00
B05 Regional delay distribution region delay 10 min sum following train (40,130 records)	0.276	7.38 [5.24, 9.52]	3.7e-6	100.00 / 100.00
G10 Edge-slack distribution outgoing slack sum departure leading train (40,130 records)	-0.276	-5.62 [-7.59, -3.64]	5.4e-5	100.00 / 100.00
H08 Spatial reachability upstream stations arrival leading train (40,130 records)	0.275	7.51 [5.25, 9.77]	6.2e-6	100.00 / 100.00
H15 Chain and crossing length longest crossings departure following train (40,130 records)	0.275	6.98 [4.03, 9.93]	6.1e-4	100.00 / 100.00
B15 Global degree distribution outgoing degree mean (40,130 records)	0.274	8.69 [5.76, 11.63]	3.1e-5	100.00 / 100.00
B15 Global degree distribution outgoing degree sum (40,130 records)	0.274	8.69 [5.76, 11.63]	3.1e-5	100.00 / 100.00
G01 Reference arrival delay arrival delay difference (40,130 records)	-0.274	-6.31 [-8.74, -3.88]	1.8e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack 90th percentile departure leading train (40,130 records)	-0.274	-5.88 [-8.00, -3.76]	7.6e-5	100.00 / 100.00
B08 Community structure community events arrival following train (40,130 records)	0.274	8.40 [5.96, 10.85]	3.8e-6	100.00 / 100.00
G19 Endpoint contrasts H10 active betweenness endpoint contrast absolute pair difference (40,130 records)	0.273	5.80 [3.33, 8.27]	6.8e-4	100.00 / 100.00
G12 Delay-support distribution incoming support minimum departure following train (40,130 records)	0.272	6.91 [4.77, 9.06]	9.3e-6	100.00 / 100.00
H03 Event time from release time from release arrival difference (40,130 records)	-0.272	-6.82 [-9.53, -4.11]	2.8e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support maximum arrival difference (40,130 records)	-0.272	-6.29 [-8.71, -3.86]	1.8e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution outgoing support sum arrival difference (40,130 records)	-0.272	-6.28 [-8.70, -3.86]	1.8e-4	100.00 / 100.00
B05 Regional delay distribution region delay 5 min standard deviation following train (40,130 records)	0.272	8.05 [5.53, 10.56]	9.9e-6	100.00 / 100.00
B06 Regional delay variance delay variance 5 min following train (40,130 records)	0.272	6.20 [3.81, 8.59]	1.8e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support sum departure difference (40,130 records)	-0.272	-6.40 [-8.84, -3.96]	1.6e-4	100.00 / 100.00
G12 Delay-support distribution incoming support standard deviation departure leading train (40,130 records)	0.272	4.81 [2.25, 7.37]	0.008	100.00 / 100.00
G10 Edge-slack distribution outgoing slack maximum departure leading train (40,130 records)	-0.271	-5.92 [-8.06, -3.78]	8.2e-5	100.00 / 100.00
H05 Downstream reachability downstream departure absolute pair difference (40,130 records)	0.271	7.36 [4.74, 9.97]	6.2e-5	100.00 / 100.00
G12 Delay-support distribution incoming support interquartile range departure leading train (40,130 records)	0.270	4.74 [2.17, 7.31]	0.010	100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum arrival leading train (40,130 records)	0.270	5.08 [2.96, 7.20]	5.1e-4	100.00 / 100.00
H06 Upstream trains upstream departure following train (40,130 records)	0.270	7.27 [4.37, 10.17]	2.9e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack standard deviation departure leading train (40,130 records)	-0.269	-5.83 [-7.97, -3.70]	9.5e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack interquartile range departure leading train (40,130 records)	-0.268	-5.86 [-7.99, -3.74]	8.4e-5	100.00 / 100.00
H11 Harmonic closeness outgoing harmonic departure difference (40,130 records)	-0.268	-7.99 [-10.89, -5.09]	8.5e-5	100.00 / 100.00
H15 Chain and crossing length longest crossings departure difference (40,130 records)	0.267	6.92 [3.97, 9.88]	7.0e-4	100.00 / 100.00
B05 Regional delay distribution region delay 5 min standard deviation leading train (40,130 records)	0.267	7.96 [4.70, 11.22]	4.1e-4	100.00 / 100.00
B06 Regional delay variance delay variance 5 min leading train (40,130 records)	0.267	6.10 [3.06, 9.14]	0.004	100.00 / 100.00
B01 Local region size region events 20 min following train (40,130 records)	0.267	8.56 [6.06, 11.05]	3.8e-6	100.00 / 100.00
B05 Regional delay distribution region delay 20 min count following train (40,130 records)	0.267	8.56 [6.06, 11.05]	3.8e-6	100.00 / 100.00
B02 Local edges and density edges 20 min following train (40,130 records)	0.267	8.48 [5.91, 11.05]	6.5e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B15 Global degree distribution outgoing degree 90th percentile (40,130 records)	0.266	8.37 [5.52, 11.23]	3.5e-5	100.00 / 100.00
H07 Downstream trains downstream arrival absolute pair difference (40,130 records)	0.266	7.85 [5.37, 10.32]	1.2e-5	100.00 / 100.00
H04 Upstream reachability upstream arrival leading train (40,130 records)	0.266	5.23 [3.10, 7.36]	3.7e-4	100.00 / 100.00
B10 Affected-train share affected fraction (40,130 records)	0.266	8.32 [5.19, 11.45]	1.3e-4	100.00 / 100.00
G12 Delay-support distribution incoming support minimum arrival leading train (38,990 records)	0.265	7.90 [5.65, 10.15]	2.9e-6	100.00 / 100.00
G12 Delay-support distribution incoming support median arrival leading train (38,990 records)	0.265	7.64 [5.11, 10.16]	2.3e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support standard deviation departure leading train (40,130 records)	0.264	6.52 [3.58, 9.47]	0.001	100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum departure leading train (40,130 records)	0.264	5.55 [3.28, 7.83]	4.1e-4	100.00 / 100.00
G19 Endpoint contrasts H05 downstream reachability endpoint contrast leading train (40,130 records)	-0.264	-1.13 [-2.41, 0.14]	0.864	100.00 / 100.00
H04 Upstream reachability upstream departure leading train (40,130 records)	0.263	5.94 [3.60, 8.28]	2.5e-4	100.00 / 100.00
B01 Local region size region trains 20 min leading train (40,130 records)	0.263	7.94 [5.21, 10.68]	4.0e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support interquartile range departure leading train (40,130 records)	0.262	6.40 [3.53, 9.26]	0.001	100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum departure following train (40,130 records)	0.262	6.89 [4.29, 9.48]	1.4e-4	100.00 / 100.00
H01 Shortest source distance source minimum time lag arrival difference (12,797 records)	-0.261	-11.20 [-14.86, -7.54]	1.9e-5	100.00 / 100.00
B11 Global path distribution root longest paths mean (40,130 records)	0.260	8.09 [4.84, 11.33]	3.2e-4	100.00 / 100.00
B11 Global path distribution root longest paths sum (40,130 records)	0.260	8.09 [4.84, 11.33]	3.2e-4	100.00 / 100.00
B06 Regional delay variance delay variance 5 min absolute pair difference (40,130 records)	0.260	4.78 [2.54, 7.03]	0.002	100.00 / 100.00
H15 Chain and crossing length longest crossings departure absolute pair difference (40,130 records)	0.260	6.58 [3.59, 9.57]	0.001	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum departure following train (40,130 records)	0.259	7.09 [4.69, 9.49]	3.1e-5	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G13 Upstream-neighbor delay neighbor delay mean departure following train (40,130 records)	0.259	7.09 [4.69, 9.49]	3.1e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay median departure following train (40,130 records)	0.259	7.09 [4.69, 9.49]	3.1e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum departure following train (40,130 records)	0.259	7.09 [4.69, 9.49]	3.1e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay 90th percentile departure following train (40,130 records)	0.259	7.09 [4.69, 9.49]	3.1e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum departure following train (40,130 records)	0.259	7.09 [4.69, 9.49]	3.1e-5	100.00 / 100.00
G12 Delay-support distribution incoming support 90th percentile arrival difference (38,773 records)	-0.258	-6.48 [-8.66, -4.31]	2.8e-5	100.00 / 100.00
B16 Global reachability reachable pair fraction (40,130 records)	0.257	7.78 [4.48, 11.07]	6.2e-4	100.00 / 100.00
B05 Regional delay distribution region delay 5 min zero fraction following train (40,130 records)	-0.257	-8.12 [-10.44, -5.79]	3.0e-6	100.00 / 100.00
G09 Active cross-train links active cross departure leading train (40,130 records)	0.257	7.18 [5.01, 9.35]	6.3e-6	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay sum departure leading train (40,130 records)	0.256	7.32 [4.84, 9.79]	3.1e-5	100.00 / 100.00
H16 Local motifs diamonds at source departure leading train (40,130 records)	0.256	8.07 [5.15, 10.99]	8.0e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum arrival leading train (34,618 records)	0.256	8.51 [5.78, 11.23]	1.7e-5	100.00 / 100.00
G09 Active cross-train links active cross fraction departure leading train (40,130 records)	0.255	7.16 [5.00, 9.32]	6.2e-6	100.00 / 100.00
H11 Harmonic closeness incoming harmonic departure following train (40,130 records)	0.255	7.67 [5.46, 9.88]	3.3e-6	100.00 / 100.00
H01 Shortest source distance source minimum time lag arrival absolute pair difference (12,797 records)	-0.255	-10.88 [-14.51, -7.26]	2.6e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack mean departure absolute pair difference (40,130 records)	-0.255	-5.62 [-7.61, -3.63]	6.0e-5	100.00 / 100.00
B15 Global degree distribution incoming degree 90th percentile (40,130 records)	0.255	7.88 [5.26, 10.50]	2.6e-5	100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction arrival following train (38,806 records)	-0.255	-8.23 [-10.78, -5.67]	9.3e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution outgoing slack sum departure absolute pair difference (40,130 records)	-0.254	-5.54 [-7.51, -3.58]	6.0e-5	100.00 / 100.00
H16 Local motifs fan-in pairs departure following train (40,130 records)	0.254	7.97 [5.95, 9.99]	5.2e-7	100.00 / 100.00
G12 Delay-support distribution incoming support maximum departure difference (40,130 records)	-0.253	-6.16 [-8.49, -3.83]	1.4e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack 90th percentile departure absolute pair difference (40,130 records)	-0.253	-5.78 [-7.86, -3.69]	7.7e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction departure leading train (40,130 records)	-0.253	-7.87 [-10.13, -5.60]	3.2e-6	100.00 / 100.00
G10 Edge-slack distribution outgoing slack maximum departure absolute pair difference (40,130 records)	-0.252	-5.82 [-7.93, -3.71]	8.3e-5	100.00 / 100.00
B05 Regional delay distribution region delay 5 min sum following train (40,130 records)	0.252	6.44 [4.60, 8.28]	3.0e-6	100.00 / 100.00
B01 Local region size region events 20 min leading train (40,130 records)	0.252	7.62 [5.02, 10.21]	3.4e-5	100.00 / 100.00
B05 Regional delay distribution region delay 20 min count leading train (40,130 records)	0.252	7.62 [5.02, 10.21]	3.4e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack standard deviation departure absolute pair difference (40,130 records)	-0.252	-5.74 [-7.85, -3.63]	9.8e-5	100.00 / 100.00
B05 Regional delay distribution region delay 10 min zero fraction following train (40,130 records)	-0.251	-8.07 [-10.38, -5.76]	3.0e-6	100.00 / 100.00
G02 Reference departure delay departure delay difference (40,130 records)	-0.250	-6.11 [-8.43, -3.80]	1.5e-4	100.00 / 100.00
B02 Local edges and density density 20 min following train (40,112 records)	-0.250	-7.29 [-9.25, -5.34]	1.3e-6	100.00 / 100.00
G12 Delay-support distribution outgoing support maximum departure difference (40,130 records)	-0.250	-6.11 [-8.42, -3.79]	1.5e-4	100.00 / 100.00
H08 Spatial reachability downstream branches departure following train (40,130 records)	0.250	7.82 [6.22, 9.42]	1.2e-8	100.00 / 100.00
G08 Active degree incoming active departure leading train (40,130 records)	0.250	7.32 [5.08, 9.56]	7.7e-6	100.00 / 100.00
H03 Event time from release time from release arrival absolute pair difference (40,130 records)	-0.249	-6.08 [-8.53, -3.63]	3.3e-4	100.00 / 100.00
G08 Active degree incoming active arrival leading train (40,130 records)	0.249	6.59 [4.45, 8.72]	1.7e-5	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution region delay 10 min interquartile range leading train (40,130 records)	0.249	7.82 [4.75, 10.90]	2.4e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack interquartile range departure absolute pair difference (40,130 records)	-0.249	-5.76 [-7.86, -3.67]	8.7e-5	100.00 / 100.00
H10 Active betweenness betweenness departure absolute pair difference (40,130 records)	0.248	6.41 [4.11, 8.71]	7.2e-5	100.00 / 100.00
B05 Regional delay distribution region delay 5 min mean difference (40,130 records)	-0.248	-5.86 [-8.02, -3.71]	1.0e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack median departure leading train (40,130 records)	-0.248	-5.18 [-6.97, -3.38]	4.4e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay mean arrival leading train (34,618 records)	0.248	8.41 [5.62, 11.20]	2.8e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay median arrival leading train (34,618 records)	0.248	8.41 [5.62, 11.20]	2.8e-5	100.00 / 100.00
B03 Active local density active density 10 min leading train (39,724 records)	0.248	5.35 [3.74, 6.97]	6.2e-6	100.00 / 100.00
H01 Shortest source distance source minimum time lag departure absolute pair difference (18,242 records)	-0.248	-10.70 [-14.31, -7.09]	3.0e-5	100.00 / 100.00
H01 Shortest source distance source minimum time lag departure difference (18,242 records)	-0.248	-10.70 [-14.31, -7.09]	3.0e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support zero fraction departure following train (40,130 records)	-0.247	-7.86 [-10.10, -5.61]	3.0e-6	100.00 / 100.00
H07 Downstream trains downstream departure absolute pair difference (40,130 records)	0.247	6.78 [4.04, 9.53]	3.5e-4	100.00 / 100.00
B02 Local edges and density edges 20 min leading train (40,130 records)	0.247	7.43 [4.79, 10.07]	6.2e-5	100.00 / 100.00
H11 Harmonic closeness incoming harmonic departure leading train (40,130 records)	0.247	7.43 [4.95, 9.90]	2.6e-5	100.00 / 100.00
H11 Harmonic closeness incoming harmonic arrival leading train (40,130 records)	0.246	6.90 [4.55, 9.25]	3.4e-5	100.00 / 100.00
B03 Active local density active density 20 min leading train (39,956 records)	0.246	3.79 [2.33, 5.26]	1.8e-4	100.00 / 100.00
B11 Global path distribution root longest paths 90th percentile (40,130 records)	0.246	7.77 [4.65, 10.89]	3.2e-4	100.00 / 100.00
H11 Harmonic closeness outgoing harmonic arrival absolute pair difference (40,130 records)	0.246	7.46 [5.42, 9.50]	1.5e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution outgoing support zero fraction arrival following train (40,130 records)	-0.246	-7.80 [-9.76, -5.84]	4.8e-7	100.00 / 100.00
H08 Spatial reachability downstream stations departure following train (40,130 records)	0.245	7.47 [5.84, 9.10]	4.3e-8	100.00 / 100.00
H05 Downstream reachability downstream departure following train (40,130 records)	0.244	5.57 [3.28, 7.86]	4.3e-4	100.00 / 100.00
H07 Downstream trains downstream departure following train (40,130 records)	0.244	6.17 [3.79, 8.54]	1.8e-4	100.00 / 100.00
B01 Local region size region trains 10 min following train (40,130 records)	0.244	7.94 [5.76, 10.13]	1.7e-6	100.00 / 100.00
B05 Regional delay distribution region delay 5 min median difference (40,130 records)	-0.243	-5.98 [-8.26, -3.70]	1.6e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support sum departure following train (40,130 records)	0.243	5.75 [3.94, 7.56]	1.1e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum arrival following train (40,130 records)	0.241	6.79 [4.48, 9.10]	3.4e-5	100.00 / 100.00
G16 Incident-link imbalance pressure balance departure absolute pair difference (40,130 records)	0.241	4.45 [1.74, 7.16]	0.029	100.00 / 100.00
B08 Community structure cross community edges arrival leading train (40,130 records)	0.241	6.95 [5.83, 8.08]	6.0e-11	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction arrival leading train (38,060 records)	-0.240	-7.46 [-9.71, -5.21]	6.2e-6	100.00 / 100.00
B05 Regional delay distribution region delay 20 min zero fraction following train (40,130 records)	-0.240	-7.85 [-10.10, -5.61]	3.0e-6	100.00 / 100.00
B11 Global path distribution root longest paths interquartile range (40,130 records)	0.239	7.73 [4.40, 11.05]	7.6e-4	100.00 / 100.00
H11 Harmonic closeness outgoing harmonic departure following train (40,130 records)	0.238	6.44 [4.47, 8.41]	7.3e-6	100.00 / 100.00
B03 Active local density active density 5 min leading train (39,199 records)	0.238	6.29 [4.24, 8.35]	1.9e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay 90th percentile arrival leading train (34,618 records)	0.238	7.90 [5.07, 10.73]	7.9e-5	100.00 / 100.00
B05 Regional delay distribution region delay 5 min standard deviation absolute pair difference (40,130 records)	0.237	5.59 [3.35, 7.82]	3.0e-4	100.00 / 100.00
G16 Incident-link imbalance pressure balance departure difference (40,130 records)	-0.237	-5.28 [-8.20, -2.35]	0.013	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B11 Global path distribution root longest paths standard deviation (40,130 records)	0.237	7.49 [4.25, 10.73]	8.1e-4	100.00 / 100.00
H24 Projection core number core arrival following train (40,130 records)	0.235	7.31 [5.47, 9.15]	4.8e-7	100.00 / 100.00
B15 Global degree distribution incoming degree median (40,130 records)	0.235	7.39 [4.59, 10.19]	1.5e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack mean departure difference (40,130 records)	0.235	4.51 [3.02, 6.01]	2.4e-5	100.00 / 100.00
B05 Regional delay distribution region delay 5 min 90th percentile difference (40,130 records)	-0.234	-5.76 [-7.86, -3.66]	8.9e-5	100.00 / 100.00
B15 Global degree distribution outgoing degree standard deviation (40,130 records)	0.234	7.43 [4.36, 10.50]	4.5e-4	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum arrival leading train (34,618 records)	0.233	7.74 [4.90, 10.58]	1.1e-4	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum arrival following train (38,453 records)	0.233	6.81 [4.48, 9.14]	3.6e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack sum departure difference (40,130 records)	0.233	4.51 [3.05, 5.97]	1.7e-5	100.00 / 100.00
B02 Local edges and density density 20 min leading train (39,956 records)	-0.233	-6.98 [-9.07, -4.90]	5.2e-6	100.00 / 100.00
B11 Global path distribution root longest paths median (40,130 records)	0.233	6.66 [3.39, 9.93]	0.004	100.00 / 100.00
B11 Global path distribution root longest paths maximum (40,130 records)	0.232	7.26 [4.08, 10.44]	9.5e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack 90th percentile departure difference (40,130 records)	0.232	4.68 [3.13, 6.24]	2.4e-5	100.00 / 100.00
B12 Delay concentration station entropy (40,130 records)	0.232	4.58 [1.84, 7.31]	0.025	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay 90th percentile arrival following train (38,453 records)	0.232	6.76 [4.43, 9.08]	4.0e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay sum departure following train (40,130 records)	0.232	5.45 [3.36, 7.53]	1.7e-4	100.00 / 100.00
G12 Delay-support distribution incoming support mean departure following train (40,130 records)	0.232	6.07 [3.99, 8.16]	3.8e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack maximum departure difference (40,130 records)	0.231	4.73 [3.16, 6.31]	2.6e-5	100.00 / 100.00
H10 Active betweenness betweenness arrival difference (40,130 records)	-0.230	-6.49 [-9.08, -3.91]	2.9e-4	100.00 / 100.00

Continued on the next page.

Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B01 Local region size region trains 10 min leading train (40,130 records)	0.230	7.02 [4.79, 9.25]	1.3e-5	100.00 / 100.00
G12 Delay-support distribution incoming support minimum departure leading train (40,130 records)	0.230	6.88 [4.51, 9.24]	4.0e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay zero fraction arrival following train (38,453 records)	-0.229	-7.29 [-9.64, -4.94]	1.6e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay mean arrival following train (38,453 records)	0.229	6.40 [4.12, 8.69]	6.7e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay median arrival following train (38,453 records)	0.229	6.40 [4.12, 8.69]	6.7e-5	100.00 / 100.00
H06 Upstream trains upstream arrival leading train (40,130 records)	0.229	4.39 [2.47, 6.30]	9.1e-4	100.00 / 100.00
H10 Active betweenness betweenness arrival absolute pair difference (40,130 records)	0.229	6.33 [4.14, 8.51]	4.2e-5	100.00 / 100.00
H09 Reachable delay totals downstream event delay departure following train (40,130 records)	0.229	4.09 [2.38, 5.81]	5.3e-4	100.00 / 100.00
G12 Delay-support distribution incoming support median departure following train (40,130 records)	0.229	5.91 [3.72, 8.10]	1.1e-4	100.00 / 100.00
H09 Reachable delay totals upstream event delay departure following train (40,130 records)	0.228	5.30 [3.21, 7.39]	2.5e-4	100.00 / 100.00
G12 Delay-support distribution incoming support sum departure following train (40,130 records)	0.228	5.89 [3.69, 8.08]	1.2e-4	100.00 / 100.00
H11 Harmonic closeness outgoing harmonic departure absolute pair difference (40,130 records)	0.227	6.43 [4.01, 8.85]	1.4e-4	100.00 / 100.00
G09 Active cross-train links active cross arrival following train (40,130 records)	0.227	6.97 [4.69, 9.25]	2.0e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack interquartile range departure difference (40,130 records)	0.226	4.70 [3.15, 6.24]	2.3e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction departure following train (29,690 records)	-0.226	-7.64 [-9.40, -5.88]	1.5e-7	100.00 / 100.00
G12 Delay-support distribution incoming support median arrival following train (38,806 records)	0.225	5.73 [3.71, 7.75]	5.6e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack standard deviation departure difference (40,130 records)	0.225	4.65 [3.10, 6.21]	2.6e-5	100.00 / 100.00
G09 Active cross-train links active cross fraction arrival following train (40,130 records)	0.225	6.98 [4.73, 9.23]	1.5e-5	100.00 / 100.00
H04 Upstream reachability upstream departure following train (40,130 records)	0.225	5.45 [3.24, 7.66]	3.6e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support interquartile range departure difference (40,130 records)	-0.225	-5.41 [-7.36, -3.46]	7.5e-5	100.00 / 100.00
B15 Global degree distribution outgoing degree median (40,130 records)	0.225	7.00 [4.26, 9.74]	2.3e-4	100.00 / 100.00
H08 Spatial reachability downstream stations arrival difference (40,130 records)	-0.224	-7.02 [-8.55, -5.50]	4.3e-8	100.00 / 100.00
B15 Global degree distribution outgoing degree interquartile range (40,130 records)	0.224	6.96 [3.95, 9.98]	8.2e-4	100.00 / 100.00
B12 Delay concentration time20 concentration (40,130 records)	-0.223	-6.27 [-9.07, -3.47]	0.001	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum arrival leading train (40,130 records)	0.222	7.09 [4.43, 9.74]	1.3e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support standard deviation departure following train (40,130 records)	0.222	5.56 [3.52, 7.60]	9.6e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support interquartile range departure following train (40,130 records)	0.222	5.42 [3.38, 7.46]	1.4e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support minimum departure difference (40,130 records)	-0.222	-6.11 [-8.41, -3.80]	1.4e-4	100.00 / 100.00
B04 Boundary connections entering 20 min following train (40,130 records)	0.222	7.14 [5.26, 9.03]	1.0e-6	100.00 / 100.00
G19 Endpoint contrasts H04 upstream reachability endpoint contrast leading train (40,130 records)	0.221	2.89 [1.70, 4.07]	4.3e-4	100.00 / 100.00
B05 Regional delay distribution region delay 10 min median difference (40,130 records)	-0.221	-5.65 [-7.74, -3.56]	1.1e-4	100.00 / 100.00
B12 Delay concentration station concentration (40,130 records)	-0.221	-3.16 [-5.58, -0.73]	0.154	100.00 / 100.00
B15 Global degree distribution outgoing degree maximum (40,130 records)	0.220	6.70 [3.88, 9.53]	5.8e-4	100.00 / 100.00
B05 Regional delay distribution region delay 20 min maximum following train (40,130 records)	0.220	6.49 [4.22, 8.76]	5.0e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support sum arrival following train (40,130 records)	0.220	5.03 [3.43, 6.62]	1.3e-5	100.00 / 100.00
B05 Regional delay distribution region delay 10 min mean difference (40,130 records)	-0.220	-5.44 [-7.39, -3.49]	6.8e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack median departure difference (40,130 records)	0.220	4.10 [2.74, 5.45]	2.3e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay zero fraction arrival leading train (34,618 records)	-0.220	-7.05 [-9.69, -4.41]	1.4e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution region delay 5 min interquartile range absolute pair difference (40,130 records)	0.219	5.72 [3.66, 7.78]	7.6e-5	100.00 / 100.00
G12 Delay-support distribution incoming support standard deviation departure difference (40,130 records)	-0.219	-5.42 [-7.36, -3.48]	6.8e-5	100.00 / 100.00
H06 Upstream trains upstream departure leading train (40,130 records)	0.218	4.80 [2.65, 6.95]	0.001	100.00 / 100.00
B01 Local region size region events 10 min following train (40,130 records)	0.218	7.13 [5.17, 9.09]	1.6e-6	100.00 / 100.00
B05 Regional delay distribution region delay 10 min count following train (40,130 records)	0.218	7.13 [5.17, 9.09]	1.6e-6	100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum departure absolute pair difference (40,130 records)	0.218	5.33 [2.83, 7.83]	0.002	100.00 / 100.00
H16 Local motifs fan-out pairs arrival leading train (40,130 records)	0.218	6.84 [4.96, 8.73]	1.7e-6	100.00 / 100.00
H10 Active betweenness betweenness departure following train (40,130 records)	0.217	4.93 [2.68, 7.18]	0.002	100.00 / 100.00
H02 Longest active path longest path departure following train (40,130 records)	0.217	6.13 [3.97, 8.30]	5.8e-5	100.00 / 100.00
H15 Chain and crossing length longest chain departure following train (40,130 records)	0.217	6.13 [3.97, 8.30]	5.8e-5	100.00 / 100.00
B05 Regional delay distribution region delay 20 min 90th percentile following train (40,130 records)	0.217	6.30 [4.00, 8.59]	8.9e-5	100.00 / 100.00
B01 Local region size region events 10 min leading train (40,130 records)	0.216	6.74 [4.70, 8.78]	6.5e-6	100.00 / 100.00
B05 Regional delay distribution region delay 10 min count leading train (40,130 records)	0.216	6.74 [4.70, 8.78]	6.5e-6	100.00 / 100.00
B05 Regional delay distribution region delay 10 min maximum following train (40,130 records)	0.215	5.89 [3.60, 8.18]	2.1e-4	100.00 / 100.00
B06 Regional delay variance delay variance 10 min absolute pair difference (40,130 records)	0.215	4.27 [2.38, 6.15]	0.001	100.00 / 100.00
H16 Local motifs fan-in pairs departure difference (40,130 records)	0.215	6.62 [4.58, 8.65]	8.0e-6	100.00 / 100.00
G12 Delay-support distribution incoming support 90th percentile departure difference (40,130 records)	-0.215	-5.84 [-8.18, -3.50]	3.0e-4	100.00 / 100.00
H09 Reachable delay totals downstream train maximum delay sum departure following train (40,130 records)	0.213	4.36 [2.62, 6.10]	2.9e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G14 Downstream-neighbor delay neighbor delay maximum arrival leading train (38,060 records)	0.213	5.82 [3.61, 8.02]	1.5e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay mean arrival leading train (38,060 records)	0.213	5.82 [3.61, 8.02]	1.5e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay median arrival leading train (38,060 records)	0.213	5.82 [3.61, 8.02]	1.5e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay minimum arrival leading train (38,060 records)	0.213	5.82 [3.61, 8.02]	1.5e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay 90th percentile arrival leading train (38,060 records)	0.213	5.82 [3.61, 8.02]	1.5e-4	100.00 / 100.00
G16 Incident-link imbalance pressure balance arrival absolute pair difference (40,130 records)	0.213	3.36 [1.10, 5.62]	0.063	100.00 / 100.00
B05 Regional delay distribution region delay 5 min interquartile range following train (40,130 records)	0.213	6.02 [4.27, 7.78]	3.8e-6	100.00 / 100.00
B12 Delay concentration time20 entropy (40,130 records)	0.213	6.59 [3.93, 9.24]	3.3e-4	100.00 / 100.00
G08 Active degree outgoing active departure following train (40,130 records)	0.212	6.40 [5.08, 7.73]	1.5e-8	100.00 / 100.00
B05 Regional delay distribution region delay 5 min maximum difference (40,130 records)	-0.212	-5.63 [-7.65, -3.61]	7.0e-5	100.00 / 100.00
H06 Upstream trains upstream departure difference (40,130 records)	0.211	5.71 [3.15, 8.27]	0.001	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay maximum departure leading train (40,130 records)	0.211	5.98 [3.65, 8.30]	2.1e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay mean departure leading train (40,130 records)	0.211	5.81 [3.53, 8.08]	2.2e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay median departure leading train (40,130 records)	0.211	5.81 [3.53, 8.08]	2.2e-4	100.00 / 100.00
B04 Boundary connections leaving 20 min following train (40,130 records)	0.211	6.47 [4.32, 8.61]	2.4e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay 90th percentile departure leading train (40,130 records)	0.210	5.95 [3.64, 8.27]	2.1e-4	100.00 / 100.00
B08 Community structure cross community edges departure leading train (40,130 records)	0.210	6.69 [5.37, 8.01]	6.1e-9	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum departure absolute pair difference (36,410 records)	0.210	4.00 [1.98, 6.01]	0.005	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay mean departure absolute pair difference (36,410 records)	0.210	4.00 [1.98, 6.01]	0.005	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G13 Upstream-neighbor delay neighbor delay median departure absolute pair difference (36,410 records)	0.210	4.00 [1.98, 6.01]	0.005	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum departure absolute pair difference (36,410 records)	0.210	4.00 [1.98, 6.01]	0.005	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay 90th percentile departure absolute pair difference (36,410 records)	0.210	4.00 [1.98, 6.01]	0.005	100.00 / 100.00
B04 Boundary connections entering 20 min leading train (40,130 records)	0.210	6.80 [4.48, 9.11]	3.4e-5	100.00 / 100.00
B02 Local edges and density edges 10 min following train (40,130 records)	0.210	6.95 [4.93, 8.97]	3.6e-6	100.00 / 100.00
B05 Regional delay distribution region delay 10 min 90th percentile following train (40,130 records)	0.209	5.57 [3.34, 7.81]	3.1e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay minimum departure leading train (40,130 records)	0.209	5.49 [3.31, 7.67]	2.8e-4	100.00 / 100.00
B15 Global degree distribution incoming degree standard deviation (40,130 records)	0.208	7.06 [4.15, 9.97]	4.4e-4	100.00 / 100.00
H16 Local motifs fan-in pairs departure absolute pair difference (40,130 records)	0.208	6.53 [4.75, 8.31]	1.5e-6	100.00 / 100.00
B02 Local edges and density density 10 min following train (40,027 records)	-0.207	-6.22 [-7.82, -4.63]	5.9e-7	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay maximum arrival difference (26,950 records)	-0.207	-5.19 [-7.29, -3.08]	3.9e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay mean arrival difference (26,950 records)	-0.207	-5.19 [-7.29, -3.08]	3.9e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay median arrival difference (26,950 records)	-0.207	-5.19 [-7.29, -3.08]	3.9e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay minimum arrival difference (26,950 records)	-0.207	-5.19 [-7.29, -3.08]	3.9e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay 90th percentile arrival difference (26,950 records)	-0.207	-5.19 [-7.29, -3.08]	3.9e-4	100.00 / 100.00
H08 Spatial reachability downstream branches arrival difference (40,130 records)	-0.207	-6.54 [-8.18, -4.90]	4.8e-7	100.00 / 100.00
B01 Local region size region trains 5 min following train (40,130 records)	0.206	6.67 [4.79, 8.54]	2.3e-6	100.00 / 100.00
B05 Regional delay distribution region delay 5 min maximum following train (40,130 records)	0.206	5.30 [3.10, 7.51]	4.9e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution region delay 5 min minimum difference (40,130 records)	-0.206	-5.68 [-7.70, -3.66]	6.3e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack interquartile range arrival leading train (40,130 records)	-0.206	-4.83 [-6.65, -3.00]	1.4e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack standard deviation arrival leading train (40,130 records)	-0.206	-4.83 [-6.65, -3.00]	1.4e-4	100.00 / 100.00
B05 Regional delay distribution region delay 5 min interquartile range leading train (40,130 records)	0.205	5.71 [3.17, 8.24]	0.001	100.00 / 100.00
B05 Regional delay distribution region delay 5 min sum difference (40,130 records)	-0.205	-5.45 [-7.45, -3.44]	1.0e-4	100.00 / 100.00
B02 Local edges and density density 10 min leading train (39,724 records)	-0.205	-6.42 [-8.20, -4.63]	2.0e-6	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum departure absolute pair difference (40,130 records)	0.205	3.89 [2.04, 5.73]	0.002	100.00 / 100.00
G10 Edge-slack distribution outgoing slack zero fraction departure difference (40,130 records)	-0.204	-6.59 [-8.83, -4.36]	3.1e-5	100.00 / 100.00
G12 Delay-support distribution incoming support mean arrival following train (38,806 records)	0.204	5.15 [3.15, 7.16]	2.1e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack maximum arrival leading train (40,130 records)	-0.204	-4.65 [-6.72, -2.57]	0.001	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum departure leading train (36,410 records)	0.204	6.64 [3.89, 9.39]	5.1e-4	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay mean departure leading train (36,410 records)	0.204	6.64 [3.89, 9.39]	5.1e-4	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay median departure leading train (36,410 records)	0.204	6.64 [3.89, 9.39]	5.1e-4	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum departure leading train (36,410 records)	0.204	6.64 [3.89, 9.39]	5.1e-4	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay 90th percentile departure leading train (36,410 records)	0.204	6.64 [3.89, 9.39]	5.1e-4	100.00 / 100.00
G12 Delay-support distribution incoming support 90th percentile departure following train (40,130 records)	0.204	5.04 [2.92, 7.17]	5.7e-4	100.00 / 100.00
B05 Regional delay distribution region delay 5 min 90th percentile following train (40,130 records)	0.203	5.16 [2.98, 7.34]	6.0e-4	100.00 / 100.00
G12 Delay-support distribution incoming support minimum departure absolute pair difference (40,130 records)	0.203	3.62 [1.62, 5.63]	0.013	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution outgoing slack 90th percentile arrival leading train (40,130 records)	-0.203	-4.55 [-6.64, -2.46]	0.002	100.00 / 100.00
B05 Regional delay distribution region delay 10 min interquartile range absolute pair difference (40,130 records)	0.202	5.20 [3.09, 7.31]	3.7e-4	100.00 / 100.00
H08 Spatial reachability downstream stations departure difference (40,130 records)	-0.202	-5.52 [-7.37, -3.66]	3.0e-5	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum arrival following train (38,453 records)	0.201	5.62 [3.44, 7.80]	2.0e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack mean arrival leading train (40,130 records)	-0.201	-3.72 [-5.83, -1.60]	0.016	100.00 / 100.00
G10 Edge-slack distribution outgoing slack median arrival leading train (40,130 records)	-0.201	-3.72 [-5.83, -1.60]	0.016	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay sum arrival leading train (40,130 records)	0.201	5.55 [3.26, 7.85]	4.6e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack sum arrival leading train (40,130 records)	-0.201	-3.65 [-5.76, -1.54]	0.019	100.00 / 100.00
B02 Local edges and density edges 10 min leading train (40,130 records)	0.201	6.27 [4.25, 8.29]	1.5e-5	100.00 / 100.00
G12 Delay-support distribution incoming support sum arrival following train (40,130 records)	0.200	5.02 [3.01, 7.02]	3.1e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack median departure absolute pair difference (40,130 records)	-0.200	-5.15 [-6.96, -3.35]	5.0e-5	100.00 / 100.00
B16 Global reachability mean reachable distance (40,130 records)	0.199	6.34 [3.32, 9.36]	0.003	100.00 / 100.00
G11 Near-tight links slack at most 2 minutes departure leading train (40,130 records)	0.199	6.12 [4.27, 7.96]	6.2e-6	100.00 / 100.00
G12 Delay-support distribution incoming support maximum departure following train (40,130 records)	0.199	4.80 [2.70, 6.91]	9.5e-4	100.00 / 100.00
B05 Regional delay distribution region delay 5 min zero fraction absolute pair difference (40,130 records)	-0.199	-6.39 [-8.60, -4.18]	4.3e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay 90th percentile departure difference (29,690 records)	-0.199	-5.20 [-7.75, -2.64]	0.004	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction arrival following train (26,950 records)	-0.199	-6.54 [-8.50, -4.58]	6.5e-6	100.00 / 100.00
B05 Regional delay distribution region delay 20 min mean following train (40,130 records)	0.199	5.03 [2.91, 7.15]	5.9e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution outgoing support maximum departure following train (40,130 records)	0.199	4.80 [2.70, 6.91]	9.5e-4	100.00 / 100.00
G02 Reference departure delay departure delay following train (40,130 records)	0.198	4.80 [2.70, 6.91]	9.6e-4	100.00 / 100.00
G19 Endpoint contrasts H04 upstream reachability endpoint contrast absolute pair difference (40,130 records)	0.198	4.38 [2.40, 6.36]	0.001	100.00 / 100.00
B05 Regional delay distribution region delay 10 min mean following train (40,130 records)	0.198	4.79 [2.66, 6.92]	0.001	100.00 / 100.00
B05 Regional delay distribution region delay 10 min median following train (40,130 records)	0.198	4.73 [2.63, 6.84]	0.001	100.00 / 100.00
G12 Delay-support distribution incoming support minimum arrival following train (38,806 records)	0.198	4.80 [2.76, 6.84]	6.5e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support 90th percentile departure following train (40,130 records)	0.197	4.72 [2.61, 6.83]	0.001	100.00 / 100.00
B05 Regional delay distribution region delay 20 min median following train (40,130 records)	0.197	4.95 [2.86, 7.03]	5.9e-4	100.00 / 100.00
B05 Regional delay distribution region delay 5 min mean following train (40,130 records)	0.197	4.67 [2.58, 6.76]	0.001	100.00 / 100.00
B05 Regional delay distribution region delay 5 min median following train (40,130 records)	0.197	4.66 [2.58, 6.74]	0.001	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum departure leading train (40,130 records)	0.196	6.31 [3.59, 9.04]	7.9e-4	100.00 / 100.00
G10 Edge-slack distribution incoming slack interquartile range departure following train (40,130 records)	-0.196	-4.27 [-5.76, -2.79]	4.5e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support maximum arrival following train (40,130 records)	0.196	4.56 [2.84, 6.28]	1.4e-4	100.00 / 100.00
G10 Edge-slack distribution incoming slack standard deviation departure following train (40,130 records)	-0.196	-4.27 [-5.74, -2.80]	4.0e-5	100.00 / 100.00
H06 Upstream trains upstream departure absolute pair difference (40,130 records)	0.196	5.18 [2.55, 7.81]	0.005	100.00 / 100.00
B05 Regional delay distribution region delay 10 min zero fraction absolute pair difference (40,130 records)	-0.196	-6.37 [-8.58, -4.15]	4.5e-5	100.00 / 100.00
H15 Chain and crossing length longest crossings arrival following train (40,130 records)	0.195	5.20 [2.78, 7.62]	0.002	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution incoming slack maximum departure following train (40,130 records)	-0.195	-4.27 [-5.75, -2.80]	4.3e-5	100.00 / 100.00
G10 Edge-slack distribution incoming slack 90th percentile departure following train (40,130 records)	-0.195	-4.28 [-5.73, -2.82]	3.4e-5	100.00 / 100.00
H16 Local motifs fan-out pairs departure difference (40,130 records)	-0.195	-6.16 [-8.51, -3.81]	1.6e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support mean departure following train (40,130 records)	0.195	4.32 [2.21, 6.43]	0.003	100.00 / 100.00
G10 Edge-slack distribution incoming slack sum departure following train (40,130 records)	-0.195	-4.25 [-5.68, -2.82]	2.8e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support 90th percentile arrival following train (40,130 records)	0.195	4.49 [2.78, 6.20]	1.6e-4	100.00 / 100.00
G10 Edge-slack distribution incoming slack mean departure following train (40,130 records)	-0.195	-4.25 [-5.66, -2.84]	2.5e-5	100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum departure difference (40,130 records)	0.194	5.10 [2.76, 7.45]	0.002	100.00 / 100.00
B01 Local region size region trains 5 min leading train (40,130 records)	0.194	6.01 [4.21, 7.82]	5.8e-6	100.00 / 100.00
G10 Edge-slack distribution outgoing slack mean arrival difference (40,130 records)	0.194	3.54 [1.36, 5.72]	0.032	100.00 / 100.00
G10 Edge-slack distribution outgoing slack median arrival difference (40,130 records)	0.194	3.54 [1.36, 5.72]	0.032	100.00 / 100.00
B05 Regional delay distribution region delay 10 min 90th percentile difference (40,130 records)	-0.194	-5.30 [-7.19, -3.41]	6.4e-5	100.00 / 100.00
B04 Boundary connections entering 10 min leading train (40,130 records)	0.193	6.18 [4.10, 8.26]	2.9e-5	100.00 / 100.00
G19 Endpoint contrasts H04 upstream reachability endpoint contrast difference (40,130 records)	0.193	3.58 [1.89, 5.27]	0.002	100.00 / 100.00
B05 Regional delay distribution region delay 20 min mean difference (40,130 records)	-0.192	-4.95 [-6.70, -3.21]	5.3e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay maximum departure difference (29,690 records)	-0.192	-5.18 [-7.80, -2.55]	0.005	100.00 / 100.00
G12 Delay-support distribution outgoing support mean arrival following train (40,130 records)	0.192	4.19 [2.50, 5.88]	3.3e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support median arrival following train (40,130 records)	0.192	4.19 [2.50, 5.88]	3.3e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G01 Reference arrival delay arrival delay following train (40,130 records)	0.192	4.54 [2.82, 6.25]	1.4e-4	100.00 / 100.00
B09 Boundary pressure imbalance cross boundary pressure balance 5 min absolute pair difference (40,130 records)	0.192	2.71 [0.35, 5.08]	0.305	100.00 / 100.00
G10 Edge-slack distribution outgoing slack sum arrival difference (40,130 records)	0.192	3.48 [1.28, 5.68]	0.039	100.00 / 100.00
G10 Edge-slack distribution incoming slack median departure following train (40,130 records)	-0.192	-4.04 [-5.42, -2.67]	3.3e-5	100.00 / 100.00
H16 Local motifs fan-out pairs departure absolute pair difference (40,130 records)	0.192	6.10 [4.10, 8.10]	2.0e-5	100.00 / 100.00
H03 Event time from release time from release departure leading train (40,130 records)	0.192	6.05 [3.84, 8.25]	9.0e-5	100.00 / 100.00
H13 Weak-component size component events departure absolute pair difference (40,130 records)	-0.191	-4.30 [-5.95, -2.65]	1.7e-4	100.00 / 100.00
H03 Event time from release time from release arrival leading train (40,130 records)	0.190	5.70 [3.34, 8.05]	4.6e-4	100.00 / 100.00
G08 Active degree outgoing active departure difference (40,130 records)	-0.190	-5.65 [-8.02, -3.28]	5.6e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack 90th percentile arrival difference (40,130 records)	0.189	3.96 [1.95, 5.97]	0.005	100.00 / 100.00
B01 Local region size region events 5 min leading train (40,130 records)	0.189	5.89 [4.01, 7.77]	1.3e-5	100.00 / 100.00
B05 Regional delay distribution region delay 5 min count leading train (40,130 records)	0.189	5.89 [4.01, 7.77]	1.3e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support median departure following train (40,130 records)	0.189	4.32 [2.22, 6.42]	0.003	100.00 / 100.00
H13 Weak-component size component trains departure absolute pair difference (40,130 records)	-0.189	-4.46 [-6.16, -2.76]	1.5e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack interquartile range arrival absolute pair difference (40,130 records)	-0.188	-4.70 [-6.45, -2.94]	1.3e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack standard deviation arrival absolute pair difference (40,130 records)	-0.188	-4.70 [-6.45, -2.94]	1.3e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack maximum arrival difference (40,130 records)	0.187	3.99 [2.02, 5.97]	0.004	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution region delay 20 min zero fraction absolute pair difference (40,130 records)	-0.187	-6.18 [-8.32, -4.04]	4.4e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay sum arrival following train (40,130 records)	0.187	4.49 [2.82, 6.15]	1.1e-4	100.00 / 100.00
H16 Local motifs diamonds at source arrival leading train (40,130 records)	0.187	5.87 [4.29, 7.46]	1.4e-6	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay mean departure difference (29,690 records)	-0.187	-5.07 [-7.31, -2.82]	0.001	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay median departure difference (29,690 records)	-0.187	-5.07 [-7.31, -2.82]	0.001	100.00 / 100.00
H08 Spatial reachability downstream branches departure difference (40,130 records)	-0.187	-5.21 [-7.12, -3.30]	9.6e-5	100.00 / 100.00
H07 Downstream trains downstream arrival following train (40,130 records)	0.186	4.84 [2.72, 6.95]	9.3e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack zero fraction departure leading train (40,130 records)	0.186	5.60 [2.52, 8.67]	0.012	100.00 / 100.00
B04 Boundary connections leaving 10 min following train (40,130 records)	0.186	5.81 [3.97, 7.66]	1.3e-5	100.00 / 100.00
H08 Spatial reachability upstream branches departure following train (40,130 records)	0.185	5.85 [4.54, 7.17]	7.8e-8	100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum arrival absolute pair difference (40,130 records)	0.185	4.18 [2.15, 6.21]	0.003	100.00 / 100.00
B05 Regional delay distribution region delay 5 min minimum following train (40,130 records)	0.185	3.95 [2.02, 5.89]	0.004	100.00 / 100.00
B05 Regional delay distribution region delay 20 min median difference (40,130 records)	-0.185	-5.11 [-6.95, -3.28]	7.1e-5	100.00 / 100.00
H08 Spatial reachability downstream branches arrival following train (40,130 records)	0.184	5.65 [4.12, 7.19]	1.4e-6	100.00 / 100.00
G10 Edge-slack distribution incoming slack median departure difference (40,130 records)	-0.184	-4.91 [-6.92, -2.91]	3.9e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support interquartile range arrival following train (40,130 records)	0.183	5.14 [3.32, 6.95]	5.7e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support standard deviation arrival following train (40,130 records)	0.183	5.14 [3.32, 6.95]	5.7e-5	100.00 / 100.00
B08 Community structure cross community edges arrival difference (40,130 records)	-0.183	-5.34 [-6.11, -4.57]	4.5e-12	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H15 Chain and crossing length longest crossings arrival absolute pair difference (40,130 records)	0.183	4.93 [2.53, 7.32]	0.003	100.00 / 100.00
G10 Edge-slack distribution outgoing slack zero fraction arrival difference (40,130 records)	-0.183	-5.72 [-7.79, -3.65]	8.1e-5	100.00 / 100.00
G12 Delay-support distribution incoming support mean arrival difference (38,773 records)	-0.183	-4.64 [-6.51, -2.76]	3.4e-4	100.00 / 100.00
G19 Endpoint contrasts H05 downstream reachability endpoint contrast difference (40,130 records)	0.183	2.89 [1.26, 4.51]	0.014	100.00 / 100.00
B02 Local edges and density density 5 min leading train (39,199 records)	-0.183	-5.98 [-7.79, -4.17]	6.4e-6	100.00 / 100.00
G10 Edge-slack distribution incoming slack interquartile range departure difference (40,130 records)	-0.182	-5.27 [-7.18, -3.35]	8.5e-5	100.00 / 100.00
G08 Active degree outgoing active arrival following train (40,130 records)	0.182	5.71 [4.09, 7.33]	2.8e-6	100.00 / 100.00
G11 Near-tight links slack at most 2 minutes departure following train (40,130 records)	0.181	5.74 [4.29, 7.19]	4.9e-7	100.00 / 100.00
G10 Edge-slack distribution incoming slack standard deviation departure difference (40,130 records)	-0.181	-5.32 [-7.22, -3.42]	6.7e-5	100.00 / 100.00
G10 Edge-slack distribution incoming slack 90th percentile departure difference (40,130 records)	-0.181	-5.22 [-7.13, -3.32]	8.8e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support interquartile range arrival leading train (40,130 records)	0.181	6.04 [3.62, 8.46]	3.1e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support standard deviation arrival leading train (40,130 records)	0.181	6.04 [3.62, 8.46]	3.1e-4	100.00 / 100.00
G10 Edge-slack distribution incoming slack maximum departure difference (40,130 records)	-0.181	-5.21 [-7.13, -3.30]	9.6e-5	100.00 / 100.00
H15 Chain and crossing length longest crossings arrival difference (40,130 records)	0.180	5.02 [2.66, 7.38]	0.002	100.00 / 100.00
G10 Edge-slack distribution incoming slack mean departure difference (40,130 records)	-0.180	-5.15 [-7.04, -3.25]	1.0e-4	100.00 / 100.00
G11 Near-tight links slack at most 1 minute departure leading train (40,130 records)	0.180	5.49 [3.74, 7.23]	1.3e-5	100.00 / 100.00
H06 Upstream trains upstream arrival following train (40,130 records)	0.180	5.08 [2.79, 7.37]	0.001	100.00 / 100.00
B04 Boundary connections entering 5 min leading train (40,130 records)	0.180	5.63 [3.53, 7.73]	1.2e-4	100.00 / 100.00
B02 Local edges and density edges 5 min leading train (40,130 records)	0.179	5.27 [3.48, 7.05]	3.1e-5	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution incoming slack sum departure difference (40,130 records)	-0.179	-4.92 [-6.89, -2.94]	3.2e-4	100.00 / 100.00
G12 Delay-support distribution incoming support 90th percentile arrival following train (38,806 records)	0.179	4.43 [2.58, 6.27]	5.1e-4	100.00 / 100.00
H05 Downstream reachability downstream arrival following train (40,130 records)	0.178	4.03 [2.04, 6.02]	0.004	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay sum departure difference (40,130 records)	-0.178	-3.92 [-6.18, -1.65]	0.019	100.00 / 100.00
G12 Delay-support distribution incoming support interquartile range arrival difference (38,773 records)	-0.178	-4.48 [-6.29, -2.68]	3.3e-4	100.00 / 100.00
B05 Regional delay distribution region delay 10 min minimum following train (40,130 records)	0.178	3.68 [1.74, 5.63]	0.008	100.00 / 100.00
G12 Delay-support distribution incoming support standard deviation arrival difference (38,773 records)	-0.178	-4.46 [-6.27, -2.65]	3.7e-4	100.00 / 100.00
H11 Harmonic closeness outgoing harmonic arrival following train (40,130 records)	0.177	4.82 [3.04, 6.61]	1.1e-4	100.00 / 100.00
G19 Endpoint contrasts G08 active incoming degree endpoint contrast absolute pair difference (40,130 records)	0.176	5.41 [4.20, 6.61]	6.6e-8	100.00 / 100.00
H08 Spatial reachability downstream stations arrival following train (40,130 records)	0.176	5.37 [3.95, 6.79]	1.1e-6	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay interquartile range arrival following train (38,453 records)	0.176	3.47 [2.06, 4.87]	3.7e-4	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay standard deviation arrival following train (38,453 records)	0.176	3.47 [2.06, 4.87]	3.7e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support standard deviation departure absolute pair difference (40,130 records)	0.175	3.94 [1.48, 6.41]	0.036	100.00 / 100.00
G16 Incident-link imbalance pressure balance arrival following train (40,130 records)	-0.174	-3.20 [-5.35, -1.04]	0.063	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay zero fraction departure leading train (36,410 records)	-0.174	-5.55 [-8.08, -3.02]	0.002	100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum arrival following train (40,130 records)	0.174	4.91 [2.77, 7.04]	9.0e-4	100.00 / 100.00
H09 Reachable delay totals upstream event delay departure absolute pair difference (40,130 records)	0.174	2.73 [0.91, 4.56]	0.058	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support minimum arrival absolute pair difference (38,773 records)	0.174	3.07 [1.20, 4.94]	0.029	100.00 / 100.00
B05 Regional delay distribution region delay 10 min minimum difference (40,130 records)	-0.174	-5.06 [-6.73, -3.39]	2.2e-5	100.00 / 100.00
B04 Boundary connections leaving 20 min leading train (40,130 records)	0.173	5.43 [2.77, 8.09]	0.004	100.00 / 100.00
G12 Delay-support distribution outgoing support zero fraction departure absolute pair difference (40,130 records)	-0.173	-5.59 [-7.56, -3.62]	5.5e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay maximum departure following train (29,690 records)	0.173	4.34 [2.92, 5.76]	2.4e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay 90th percentile departure following train (29,690 records)	0.173	4.31 [2.88, 5.73]	2.6e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support minimum arrival following train (40,130 records)	0.172	3.77 [2.12, 5.41]	9.4e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay mean departure following train (29,690 records)	0.172	4.15 [2.72, 5.58]	4.5e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay median departure following train (29,690 records)	0.172	4.15 [2.72, 5.58]	4.5e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support minimum departure following train (40,130 records)	0.172	3.72 [1.67, 5.76]	0.012	100.00 / 100.00
G12 Delay-support distribution incoming support maximum arrival following train (38,806 records)	0.170	3.99 [2.18, 5.79]	0.001	100.00 / 100.00
G08 Active degree incoming active arrival following train (40,130 records)	0.170	5.30 [3.56, 7.05]	2.2e-5	100.00 / 100.00
B05 Regional delay distribution region delay 20 min minimum difference (40,130 records)	-0.169	-4.80 [-6.46, -3.13]	4.4e-5	100.00 / 100.00
H16 Local motifs diamonds at source departure difference (40,130 records)	-0.169	-5.23 [-8.76, -1.70]	0.065	100.00 / 100.00
G12 Delay-support distribution incoming support sum arrival difference (40,130 records)	-0.169	-4.22 [-6.20, -2.24]	0.002	100.00 / 100.00
G10 Edge-slack distribution outgoing slack interquartile range arrival difference (40,130 records)	0.169	3.79 [2.41, 5.17]	8.5e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack standard deviation arrival difference (40,130 records)	0.169	3.79 [2.41, 5.17]	8.5e-5	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support median arrival absolute pair difference (38,773 records)	0.169	2.67 [1.05, 4.30]	0.029	100.00 / 100.00
G12 Delay-support distribution outgoing support interquartile range departure absolute pair difference (40,130 records)	0.168	3.98 [1.55, 6.40]	0.029	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum arrival absolute pair difference (40,130 records)	0.168	3.63 [1.93, 5.34]	0.002	100.00 / 100.00
G19 Endpoint contrasts G08 active outgoing degree endpoint contrast absolute pair difference (40,130 records)	0.168	5.07 [3.11, 7.03]	1.9e-4	100.00 / 100.00
G09 Active cross-train links active cross arrival leading train (40,130 records)	0.168	4.38 [2.82, 5.93]	6.1e-5	100.00 / 100.00
B08 Community structure cross community edges arrival absolute pair difference (40,130 records)	0.167	4.81 [4.21, 5.41]	1.3e-13	100.00 / 100.00
H09 Reachable delay totals downstream event delay arrival following train (40,130 records)	0.167	2.78 [1.23, 4.33]	0.013	100.00 / 100.00
B01 Local region size region events 5 min following train (40,130 records)	0.167	5.14 [3.58, 6.70]	6.7e-6	100.00 / 100.00
B05 Regional delay distribution region delay 5 min count following train (40,130 records)	0.167	5.14 [3.58, 6.70]	6.7e-6	100.00 / 100.00
B04 Boundary connections entering 10 min following train (40,130 records)	0.166	5.29 [3.86, 6.72]	1.4e-6	100.00 / 100.00
B04 Boundary connections leaving 5 min following train (40,130 records)	0.165	5.17 [3.40, 6.94]	3.6e-5	100.00 / 100.00
G09 Active cross-train links active cross fraction arrival leading train (40,130 records)	0.165	4.38 [2.87, 5.89]	4.0e-5	100.00 / 100.00
B05 Regional delay distribution region delay 20 min minimum following train (40,130 records)	0.165	3.34 [1.43, 5.24]	0.017	100.00 / 100.00
H08 Spatial reachability upstream branches departure absolute pair difference (40,130 records)	-0.165	-5.18 [-7.22, -3.14]	2.4e-4	100.00 / 100.00
H09 Reachable delay totals downstream train maximum delay sum arrival following train (40,130 records)	0.165	3.32 [1.72, 4.93]	0.003	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay minimum departure following train (29,690 records)	0.164	3.92 [2.49, 5.36]	1.0e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack maximum arrival absolute pair difference (40,130 records)	-0.164	-4.08 [-5.98, -2.18]	0.002	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G14 Downstream-neighbor delay neighbor delay sum departure absolute pair difference (40,130 records)	0.163	3.19 [1.10, 5.28]	0.051	100.00 / 100.00
G11 Near-tight links slack at most 2 minutes arrival absolute pair difference (40,130 records)	0.163	5.07 [3.37, 6.76]	2.6e-5	100.00 / 100.00
H08 Spatial reachability upstream stations departure following train (40,130 records)	0.162	4.74 [3.16, 6.32]	2.7e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack 90th percentile arrival absolute pair difference (40,130 records)	-0.160	-3.94 [-5.83, -2.05]	0.003	100.00 / 100.00
H11 Harmonic closeness incoming harmonic arrival following train (40,130 records)	0.160	4.60 [2.90, 6.29]	1.1e-4	100.00 / 100.00
B05 Regional delay distribution region delay 10 min standard deviation absolute pair difference (40,130 records)	0.160	2.99 [1.10, 4.88]	0.039	100.00 / 100.00
H24 Projection core number core arrival difference (40,130 records)	-0.160	-4.26 [-6.34, -2.17]	0.004	100.00 / 100.00
G12 Delay-support distribution incoming support interquartile range departure absolute pair difference (40,130 records)	0.160	2.33 [0.44, 4.22]	0.208	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay count departure leading train (40,130 records)	0.160	5.01 [3.34, 6.68]	2.6e-5	100.00 / 100.00
H16 Local motifs diamonds at source departure absolute pair difference (40,130 records)	0.159	5.04 [2.92, 7.16]	5.7e-4	100.00 / 100.00
G11 Near-tight links slack at most 1 minute arrival absolute pair difference (40,130 records)	0.159	4.79 [3.16, 6.42]	3.4e-5	100.00 / 100.00
B03 Active local density active density 5 min following train (39,648 records)	0.159	3.64 [2.07, 5.21]	7.9e-4	100.00 / 100.00
H09 Reachable delay totals upstream event delay arrival absolute pair difference (40,130 records)	0.159	2.20 [0.58, 3.82]	0.118	100.00 / 100.00
G09 Active cross-train links active cross fraction departure difference (40,130 records)	0.158	3.89 [2.34, 5.44]	2.9e-4	100.00 / 100.00
B02 Local edges and density edges 5 min following train (40,130 records)	0.158	4.88 [3.35, 6.41]	1.0e-5	100.00 / 100.00
G12 Delay-support distribution incoming support standard deviation departure absolute pair difference (40,130 records)	0.158	2.33 [0.44, 4.21]	0.208	100.00 / 100.00
G19 Endpoint contrasts G08 active incoming degree endpoint contrast following train (40,130 records)	0.156	4.78 [3.35, 6.21]	5.5e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H10 Active betweenness betweenness departure difference (40,130 records)	-0.156	-4.70 [-7.65, -1.75]	0.037	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay interquartile range arrival absolute pair difference (34,608 records)	0.155	2.32 [1.02, 3.62]	0.014	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay standard deviation arrival absolute pair difference (34,608 records)	0.155	2.32 [1.02, 3.62]	0.014	100.00 / 100.00
G09 Active cross-train links active cross fraction departure absolute pair difference (40,130 records)	0.155	3.14 [1.57, 4.70]	0.004	100.00 / 100.00
G19 Endpoint contrasts H10 active betweenness endpoint contrast following train (40,130 records)	0.154	3.40 [2.08, 4.72]	2.0e-4	100.00 / 100.00
G10 Edge-slack distribution incoming slack median arrival difference (38,773 records)	-0.154	-4.55 [-6.90, -2.20]	0.006	100.00 / 100.00
B05 Regional delay distribution region delay 10 min maximum difference (40,130 records)	-0.154	-4.94 [-6.70, -3.18]	6.3e-5	100.00 / 100.00
H09 Reachable delay totals upstream event delay arrival following train (40,130 records)	0.154	3.69 [1.93, 5.45]	0.003	100.00 / 100.00
G12 Delay-support distribution outgoing support sum departure absolute pair difference (40,130 records)	0.153	2.26 [0.02, 4.51]	0.550	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay interquartile range departure leading train (40,130 records)	0.152	2.09 [-0.07, 4.26]	0.649	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay standard deviation departure leading train (40,130 records)	0.152	2.09 [-0.07, 4.26]	0.649	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay mean arrival absolute pair difference (34,608 records)	0.152	2.40 [0.42, 4.38]	0.233	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay median arrival absolute pair difference (34,608 records)	0.152	2.40 [0.42, 4.38]	0.233	100.00 / 100.00
H06 Upstream trains upstream arrival absolute pair difference (40,130 records)	0.152	3.81 [1.79, 5.83]	0.008	100.00 / 100.00
H04 Upstream reachability upstream arrival following train (40,130 records)	0.152	3.65 [1.82, 5.49]	0.005	100.00 / 100.00
G10 Edge-slack distribution outgoing slack sum arrival absolute pair difference (40,130 records)	-0.151	-2.92 [-4.74, -1.11]	0.034	100.00 / 100.00
B02 Local edges and density density 5 min following train (39,648 records)	-0.151	-4.81 [-6.23, -3.38]	4.9e-6	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support standard deviation arrival absolute pair difference (38,773 records)	0.150	1.88 [0.17, 3.58]	0.374	100.00 / 100.00
G19 Endpoint contrasts H05 downstream reachability endpoint contrast absolute pair difference (40,130 records)	0.150	3.91 [1.92, 5.90]	0.005	100.00 / 100.00
H02 Longest active path longest path arrival following train (40,130 records)	0.150	3.97 [2.13, 5.82]	0.002	100.00 / 100.00
H15 Chain and crossing length longest chain arrival following train (40,130 records)	0.150	3.97 [2.13, 5.82]	0.002	100.00 / 100.00
B06 Regional delay variance delay variance 20 min absolute pair difference (40,130 records)	0.149	2.88 [0.91, 4.86]	0.071	100.00 / 100.00
G10 Edge-slack distribution outgoing slack mean arrival absolute pair difference (40,130 records)	-0.148	-2.86 [-4.70, -1.02]	0.044	100.00 / 100.00
G10 Edge-slack distribution outgoing slack median arrival absolute pair difference (40,130 records)	-0.148	-2.86 [-4.70, -1.02]	0.044	100.00 / 100.00
H24 Projection core number core departure absolute pair difference (40,130 records)	-0.148	-4.74 [-6.46, -3.02]	8.4e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay 90th percentile departure absolute pair difference (29,690 records)	0.148	1.64 [-0.94, 4.23]	1.000	100.00 / 66.67
G12 Delay-support distribution incoming support interquartile range arrival absolute pair difference (38,773 records)	0.147	1.91 [0.23, 3.58]	0.322	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay minimum departure difference (29,690 records)	-0.147	-4.50 [-6.43, -2.57]	7.9e-4	100.00 / 100.00
G12 Delay-support distribution incoming support sum arrival absolute pair difference (40,130 records)	0.147	1.62 [0.05, 3.20]	0.497	100.00 / 100.00
B08 Community structure cross community edges departure absolute pair difference (40,130 records)	0.147	4.56 [3.55, 5.56]	5.6e-8	100.00 / 100.00
B15 Global degree distribution incoming degree interquartile range (40,130 records)	0.147	4.61 [1.33, 7.88]	0.092	100.00 / 100.00
H09 Reachable delay totals upstream event delay departure difference (40,130 records)	0.147	2.21 [0.31, 4.11]	0.285	100.00 / 100.00
G08 Active degree outgoing active arrival difference (40,130 records)	-0.146	-4.17 [-5.87, -2.48]	3.7e-4	100.00 / 100.00
B09 Boundary pressure imbalance cross boundary pressure balance 10 min absolute pair difference (40,130 records)	0.146	1.76 [-0.54, 4.07]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution outgoing support zero fraction arrival absolute pair difference (40,130 records)	-0.146	-4.67 [-6.99, -2.35]	0.004	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay 90th percentile arrival absolute pair difference (34,608 records)	0.146	2.19 [0.33, 4.05]	0.270	100.00 / 100.00
H04 Upstream reachability upstream departure difference (40,130 records)	0.146	2.72 [0.90, 4.54]	0.060	100.00 / 100.00
B01 Local region size region trains 10 min absolute pair difference (40,130 records)	0.145	3.82 [1.97, 5.67]	0.003	100.00 / 100.00
H25 Local assortativity local delay absolute difference departure following train (25,167 records)	0.145	5.16 [2.92, 7.40]	8.7e-4	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum arrival absolute pair difference (34,608 records)	0.145	2.32 [0.36, 4.28]	0.260	100.00 / 100.00
G15 Active-neighbor dispersion delay variance departure leading train (25,645 records)	0.145	4.63 [2.16, 7.10]	0.009	100.00 / 100.00
H01 Shortest source distance source edges departure absolute pair difference (18,242 records)	-0.143	-6.43 [-9.45, -3.41]	0.002	100.00 / 100.00
G15 Active-neighbor dispersion delay range departure leading train (25,645 records)	0.143	5.39 [2.59, 8.19]	0.006	100.00 / 100.00
H25 Local assortativity local delay absolute difference departure leading train (25,645 records)	0.143	5.19 [2.18, 8.21]	0.019	100.00 / 100.00
B02 Local edges and density edges 10 min absolute pair difference (40,130 records)	0.143	3.99 [2.14, 5.85]	0.002	100.00 / 100.00
H01 Shortest source distance source edges departure difference (18,242 records)	-0.142	-6.33 [-9.32, -3.35]	0.002	100.00 / 100.00
G12 Delay-support distribution incoming support interquartile range arrival following train (38,806 records)	0.142	-0.12 [-0.95, 0.71]	1.000	80.00 / 66.67
B04 Boundary connections leaving 10 min leading train (40,130 records)	0.142	4.50 [2.00, 6.99]	0.013	100.00 / 100.00
G15 Active-neighbor dispersion delay variance departure following train (25,167 records)	0.142	4.39 [2.89, 5.89]	3.5e-5	100.00 / 100.00
G10 Edge-slack distribution incoming slack minimum arrival following train (38,806 records)	0.141	3.02 [1.27, 4.77]	0.019	100.00 / 100.00
G12 Delay-support distribution incoming support standard deviation arrival following train (38,806 records)	0.141	-0.18 [-1.01, 0.65]	1.000	86.67 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G15 Active-neighbor dispersion delay range departure following train (25,167 records)	0.141	4.86 [2.66, 7.05]	0.001	100.00 / 100.00
B09 Boundary pressure imbalance cross boundary pressure balance 10 min following train (40,130 records)	-0.141	-1.55 [-2.90, -0.21]	0.299	100.00 / 100.00
G10 Edge-slack distribution incoming slack zero fraction departure difference (40,130 records)	0.140	4.35 [2.69, 6.01]	1.6e-4	100.00 / 100.00
B03 Active local density active density 10 min following train (40,027 records)	0.140	1.38 [0.04, 2.73]	0.509	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay maximum arrival following train (26,950 records)	0.139	3.17 [1.57, 4.77]	0.005	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay mean arrival following train (26,950 records)	0.139	3.17 [1.57, 4.77]	0.005	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay median arrival following train (26,950 records)	0.139	3.17 [1.57, 4.77]	0.005	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay minimum arrival following train (26,950 records)	0.139	3.17 [1.57, 4.77]	0.005	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay 90th percentile arrival following train (26,950 records)	0.139	3.17 [1.57, 4.77]	0.005	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay maximum departure absolute pair difference (29,690 records)	0.139	1.75 [-0.91, 4.40]	1.000	100.00 / 100.00
H08 Spatial reachability downstream stations arrival absolute pair difference (40,130 records)	0.139	4.97 [3.35, 6.58]	1.8e-5	100.00 / 100.00
G03 Activity delay increment dwell delay increment following train (40,130 records)	0.138	0.95 [-1.22, 3.11]	1.000	100.00 / 100.00
G04 Activity-duration increment dwell increment following train (40,130 records)	0.138	0.95 [-1.22, 3.11]	1.000	100.00 / 100.00
H16 Local motifs fan-out pairs arrival following train (40,130 records)	0.138	4.35 [2.83, 5.87]	5.0e-5	100.00 / 100.00
B03 Active local density active density 20 min following train (40,112 records)	0.138	-0.33 [-1.71, 1.04]	1.000	86.67 / 66.67
H16 Local motifs diamonds at source arrival following train (40,130 records)	0.138	4.32 [3.22, 5.42]	5.5e-7	100.00 / 100.00
H08 Spatial reachability upstream branches arrival following train (40,130 records)	0.137	4.01 [2.66, 5.36]	3.0e-5	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay sum arrival difference (40,130 records)	-0.136	-2.23 [-3.99, -0.46]	0.183	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G15 Active-neighbor dispersion delay coefficient of variation departure following train (25,167 records)	0.136	2.12 [-0.15, 4.39]	0.732	100.00 / 100.00
B01 Local region size region events 10 min absolute pair difference (40,130 records)	0.136	3.54 [1.80, 5.28]	0.004	100.00 / 100.00
B05 Regional delay distribution region delay 10 min count absolute pair difference (40,130 records)	0.136	3.54 [1.80, 5.28]	0.004	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay count departure following train (40,130 records)	0.136	4.28 [2.29, 6.28]	0.002	100.00 / 100.00
B09 Boundary pressure imbalance cross boundary pressure balance 20 min absolute pair difference (40,130 records)	0.135	2.04 [-0.29, 4.37]	0.911	100.00 / 100.00
H04 Upstream reachability upstream departure absolute pair difference (40,130 records)	0.135	2.37 [0.44, 4.30]	0.215	100.00 / 100.00
B05 Regional delay distribution region delay 10 min sum difference (40,130 records)	-0.135	-3.59 [-5.54, -1.65]	0.010	100.00 / 100.00
H25 Local assortativity local delay absolute difference arrival following train (24,085 records)	0.134	4.33 [1.82, 6.83]	0.019	100.00 / 100.00
H08 Spatial reachability upstream stations departure absolute pair difference (40,130 records)	-0.134	-4.50 [-6.57, -2.44]	0.002	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay interquartile range departure absolute pair difference (29,690 records)	0.133	1.24 [-1.03, 3.51]	1.000	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay standard deviation departure absolute pair difference (29,690 records)	0.133	1.24 [-1.03, 3.51]	1.000	100.00 / 100.00
G12 Delay-support distribution incoming support mean arrival absolute pair difference (38,773 records)	0.133	1.57 [-0.08, 3.21]	0.688	100.00 / 100.00
G10 Edge-slack distribution incoming slack median arrival following train (38,806 records)	-0.133	-2.83 [-4.47, -1.19]	0.019	100.00 / 100.00
H01 Shortest source distance source edges arrival difference (12,797 records)	-0.133	-5.71 [-8.48, -2.94]	0.003	100.00 / 100.00
H08 Spatial reachability upstream stations arrival following train (40,130 records)	0.132	3.64 [2.01, 5.28]	0.001	100.00 / 100.00
B08 Community structure cross community edges departure difference (40,130 records)	-0.132	-4.15 [-5.92, -2.38]	6.8e-4	100.00 / 100.00
G07 Degree by relation outgoing entry headway departure absolute pair difference (40,130 records)	-0.132	-4.14 [-5.78, -2.50]	2.6e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G07 Degree by relation outgoing entry headway departure difference (40,130 records)	0.132	4.14 [2.50, 5.78]	2.6e-4	100.00 / 100.00
G07 Degree by relation outgoing entry headway departure following train (40,130 records)	0.132	4.14 [2.50, 5.78]	2.6e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support sum arrival absolute pair difference (40,130 records)	0.130	1.67 [-0.72, 4.06]	1.000	100.00 / 100.00
G15 Active-neighbor dispersion delay coefficient of variation departure leading train (25,645 records)	0.130	2.23 [0.28, 4.17]	0.307	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay minimum arrival difference (34,608 records)	-0.129	-1.98 [-3.57, -0.39]	0.197	100.00 / 100.00
H01 Shortest source distance source edges arrival absolute pair difference (12,797 records)	-0.129	-5.51 [-8.33, -2.70]	0.005	100.00 / 100.00
G09 Active cross-train links active cross fraction arrival absolute pair difference (40,130 records)	0.129	2.87 [0.98, 4.77]	0.055	100.00 / 100.00
B05 Regional delay distribution region delay 20 min 90th percentile difference (40,130 records)	-0.128	-4.53 [-6.15, -2.90]	7.1e-5	100.00 / 100.00
H25 Local assortativity local same type fraction departure following train (25,167 records)	-0.127	-4.74 [-7.57, -1.91]	0.025	100.00 / 100.00
B09 Boundary pressure imbalance cross boundary pressure balance 20 min difference (40,130 records)	-0.127	-2.61 [-4.89, -0.32]	0.316	100.00 / 100.00
G09 Active cross-train links active cross fraction arrival difference (40,130 records)	0.126	3.29 [1.47, 5.11]	0.012	100.00 / 100.00
B04 Boundary connections leaving 5 min leading train (40,130 records)	0.126	3.94 [1.91, 5.98]	0.006	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum arrival absolute pair difference (34,608 records)	0.126	2.10 [0.26, 3.94]	0.311	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay mean departure absolute pair difference (29,690 records)	0.126	0.96 [-1.37, 3.29]	1.000	100.00 / 66.67
G14 Downstream-neighbor delay neighbor delay median departure absolute pair difference (29,690 records)	0.126	0.96 [-1.37, 3.29]	1.000	100.00 / 66.67
B09 Boundary pressure imbalance cross boundary pressure balance 20 min following train (40,130 records)	-0.126	-2.63 [-4.31, -0.95]	0.042	100.00 / 100.00
G10 Edge-slack distribution incoming slack standard deviation arrival following train (38,806 records)	-0.125	-1.77 [-2.70, -0.84]	0.008	100.00 / 100.00
H16 Local motifs fan-in pairs arrival absolute pair difference (40,130 records)	0.125	3.82 [2.52, 5.12]	3.4e-5	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B09 Boundary pressure imbalance cross boundary pressure balance 5 min following train (40,130 records)	-0.125	-0.77 [-1.69, 0.15]	1.000	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay interquartile range departure following train (29,690 records)	0.124	3.17 [1.32, 5.02]	0.021	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay standard deviation departure following train (29,690 records)	0.124	3.17 [1.32, 5.02]	0.021	100.00 / 100.00
G10 Edge-slack distribution incoming slack interquartile range arrival following train (38,806 records)	-0.124	-1.77 [-2.70, -0.84]	0.007	100.00 / 100.00
H16 Local motifs fan-in pairs arrival difference (40,130 records)	0.124	3.78 [2.36, 5.19]	1.3e-4	100.00 / 100.00
G10 Edge-slack distribution incoming slack zero fraction departure following train (40,130 records)	0.124	3.59 [1.98, 5.19]	0.001	100.00 / 100.00
G09 Active cross-train links active cross arrival difference (40,130 records)	0.123	3.29 [1.47, 5.11]	0.012	100.00 / 100.00
G10 Edge-slack distribution outgoing slack minimum arrival following train (40,130 records)	0.123	0.53 [-1.31, 2.37]	1.000	100.00 / 33.33
G12 Delay-support distribution incoming support mean departure difference (40,130 records)	-0.123	-3.55 [-5.73, -1.37]	0.031	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay maximum arrival absolute pair difference (26,950 records)	0.122	0.77 [-1.86, 3.40]	1.000	100.00 / 66.67
G14 Downstream-neighbor delay neighbor delay mean arrival absolute pair difference (26,950 records)	0.122	0.77 [-1.86, 3.40]	1.000	100.00 / 66.67
G14 Downstream-neighbor delay neighbor delay median arrival absolute pair difference (26,950 records)	0.122	0.77 [-1.86, 3.40]	1.000	100.00 / 66.67
G14 Downstream-neighbor delay neighbor delay minimum arrival absolute pair difference (26,950 records)	0.122	0.77 [-1.86, 3.40]	1.000	100.00 / 66.67
G14 Downstream-neighbor delay neighbor delay 90th percentile arrival absolute pair difference (26,950 records)	0.122	0.77 [-1.86, 3.40]	1.000	100.00 / 66.67
H16 Local motifs fan-out pairs departure following train (40,130 records)	0.122	3.52 [2.09, 4.95]	3.7e-4	100.00 / 100.00
H13 Weak-component size component events arrival absolute pair difference (40,130 records)	-0.122	-2.38 [-3.98, -0.79]	0.061	100.00 / 100.00
G03 Activity delay increment dwell delay increment difference (40,130 records)	0.122	1.18 [-1.08, 3.43]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G04 Activity-duration increment dwell increment difference (40,130 records)	0.122	1.18 [-1.08, 3.43]	1.000	100.00 / 100.00
H04 Upstream reachability upstream arrival absolute pair difference (40,130 records)	0.122	1.73 [0.04, 3.42]	0.512	100.00 / 100.00
H16 Local motifs fan-in pairs arrival following train (40,130 records)	0.121	3.77 [2.43, 5.10]	5.8e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack minimum arrival difference (40,130 records)	0.121	1.25 [-0.90, 3.40]	1.000	100.00 / 100.00
H11 Harmonic closeness incoming harmonic departure difference (40,130 records)	0.120	2.43 [0.80, 4.06]	0.063	100.00 / 100.00
H13 Weak-component size component trains arrival absolute pair difference (40,130 records)	-0.119	-2.50 [-4.13, -0.87]	0.050	100.00 / 100.00
G12 Delay-support distribution outgoing support minimum arrival absolute pair difference (40,130 records)	0.119	0.07 [-1.83, 1.97]	1.000	53.33 / 33.33
G09 Active cross-train links active cross departure difference (40,130 records)	0.119	3.01 [1.50, 4.52]	0.005	100.00 / 100.00
H25 Local assortativity local same type fraction departure leading train (25,645 records)	-0.118	-4.20 [-8.07, -0.32]	0.405	100.00 / 100.00
G12 Delay-support distribution incoming support sum departure difference (40,130 records)	-0.117	-3.51 [-5.84, -1.17]	0.058	100.00 / 100.00
G12 Delay-support distribution outgoing support mean arrival absolute pair difference (40,130 records)	0.117	-0.04 [-1.92, 1.85]	1.000	60.00 / 66.67
G12 Delay-support distribution outgoing support median arrival absolute pair difference (40,130 records)	0.117	-0.04 [-1.92, 1.85]	1.000	60.00 / 66.67
G13 Upstream-neighbor delay neighbor delay interquartile range arrival difference (34,608 records)	0.117	2.35 [1.11, 3.60]	0.008	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay standard deviation arrival difference (34,608 records)	0.117	2.35 [1.11, 3.60]	0.008	100.00 / 100.00
G09 Active cross-train links active cross arrival absolute pair difference (40,130 records)	0.117	2.84 [0.97, 4.71]	0.053	100.00 / 100.00
G12 Delay-support distribution outgoing support minimum departure absolute pair difference (40,130 records)	0.116	-0.85 [-2.46, 0.77]	1.000	93.33 / 100.00
G10 Edge-slack distribution incoming slack minimum arrival absolute pair difference (38,773 records)	0.115	1.61 [-0.89, 4.11]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G12 Delay-support distribution incoming support median departure difference (40,130 records)	-0.115	-3.09 [-5.03, -1.15]	0.037	100.00 / 100.00
G12 Delay-support distribution outgoing support interquartile range arrival absolute pair difference (40,130 records)	0.115	2.91 [1.07, 4.75]	0.039	100.00 / 100.00
G12 Delay-support distribution outgoing support standard deviation arrival absolute pair difference (40,130 records)	0.115	2.91 [1.07, 4.75]	0.039	100.00 / 100.00
G10 Edge-slack distribution incoming slack mean arrival following train (38,806 records)	-0.115	-1.60 [-3.25, 0.04]	0.631	100.00 / 100.00
G10 Edge-slack distribution outgoing slack zero fraction departure following train (40,130 records)	-0.115	-3.67 [-5.68, -1.66]	0.011	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay count arrival following train (40,130 records)	0.114	3.21 [2.10, 4.32]	4.3e-5	100.00 / 100.00
G12 Delay-support distribution outgoing support mean departure absolute pair difference (40,130 records)	0.114	-0.89 [-2.50, 0.72]	1.000	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay zero fraction departure difference (36,410 records)	-0.114	-3.69 [-5.02, -2.37]	8.3e-5	100.00 / 100.00
G10 Edge-slack distribution incoming slack minimum arrival difference (38,773 records)	0.114	3.09 [1.05, 5.14]	0.055	100.00 / 100.00
G12 Delay-support distribution incoming support 90th percentile arrival absolute pair difference (38,773 records)	0.114	0.26 [-1.51, 2.02]	1.000	80.00 / 33.33
G13 Upstream-neighbor delay neighbor delay count arrival difference (40,130 records)	0.114	3.35 [1.74, 4.96]	0.003	100.00 / 100.00
G10 Edge-slack distribution incoming slack 90th percentile arrival following train (38,806 records)	-0.114	-1.47 [-2.76, -0.18]	0.316	100.00 / 100.00
G10 Edge-slack distribution incoming slack zero fraction departure absolute pair difference (40,130 records)	0.113	3.44 [1.97, 4.91]	6.9e-4	100.00 / 100.00
G12 Delay-support distribution incoming support median arrival difference (38,773 records)	-0.113	-1.72 [-3.53, 0.08]	0.682	100.00 / 66.67
H12 PageRank pagerank departure following train (40,130 records)	0.113	1.41 [0.20, 2.63]	0.289	100.00 / 100.00
H25 Local assortativity local same type fraction arrival following train (24,085 records)	-0.113	-4.09 [-6.74, -1.44]	0.048	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay sum arrival absolute pair difference (40,130 records)	0.113	1.00 [-0.83, 2.83]	1.000	100.00 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution region delay 20 min maximum difference (40,130 records)	-0.113	-4.40 [-6.05, -2.75]	1.3e-4	100.00 / 100.00
G16 Incident-link imbalance degree balance departure difference (40,130 records)	0.113	3.57 [2.27, 4.87]	8.8e-5	100.00 / 100.00
G10 Edge-slack distribution outgoing slack zero fraction arrival leading train (40,130 records)	0.112	3.27 [1.23, 5.32]	0.035	100.00 / 100.00
H12 PageRank pagerank departure leading train (40,130 records)	0.112	1.57 [0.17, 2.96]	0.336	100.00 / 100.00
B05 Regional delay distribution region delay 5 min sum absolute pair difference (40,130 records)	0.111	1.08 [-0.96, 3.12]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack sum arrival following train (40,130 records)	-0.111	-1.62 [-3.28, 0.05]	0.632	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction departure absolute pair difference (29,690 records)	-0.111	-3.67 [-5.17, -2.18]	4.3e-4	100.00 / 100.00
G07 Degree by relation incoming emergency arrival difference (40,130 records)	-0.111	-3.48 [-4.73, -2.23]	7.4e-5	100.00 / 100.00
G12 Delay-support distribution incoming support maximum arrival absolute pair difference (38,773 records)	0.111	0.43 [-1.25, 2.11]	1.000	100.00 / 100.00
G03 Activity delay increment dwell delay increment absolute pair difference (40,130 records)	0.110	0.89 [-0.86, 2.65]	1.000	100.00 / 66.67
G04 Activity-duration increment dwell increment absolute pair difference (40,130 records)	0.110	0.89 [-0.86, 2.65]	1.000	100.00 / 66.67
G10 Edge-slack distribution incoming slack maximum arrival following train (38,806 records)	-0.110	-1.13 [-2.38, 0.11]	0.806	100.00 / 66.67
G10 Edge-slack distribution incoming slack zero fraction arrival absolute pair difference (38,773 records)	0.110	2.67 [0.68, 4.65]	0.124	100.00 / 100.00
G05 Event degree outgoing arrival absolute pair difference (40,130 records)	-0.109	-3.44 [-4.72, -2.16]	1.1e-4	100.00 / 100.00
G05 Event degree outgoing arrival difference (40,130 records)	0.109	3.44 [2.16, 4.72]	1.1e-4	100.00 / 100.00
G06 Cross-train degree outgoing cross arrival absolute pair difference (40,130 records)	-0.109	-3.44 [-4.72, -2.16]	1.1e-4	100.00 / 100.00
G06 Cross-train degree outgoing cross arrival difference (40,130 records)	0.109	3.44 [2.16, 4.72]	1.1e-4	100.00 / 100.00
G07 Degree by relation outgoing exit headway arrival absolute pair difference (40,130 records)	-0.109	-3.44 [-4.72, -2.16]	1.1e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G07 Degree by relation outgoing exit headway arrival difference (40,130 records)	0.109	3.44 [2.16, 4.72]	1.1e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack count arrival absolute pair difference (40,130 records)	-0.109	-3.44 [-4.72, -2.16]	1.1e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack count arrival difference (40,130 records)	0.109	3.44 [2.16, 4.72]	1.1e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support count arrival absolute pair difference (40,130 records)	-0.109	-3.44 [-4.72, -2.16]	1.1e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support count arrival difference (40,130 records)	0.109	3.44 [2.16, 4.72]	1.1e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay count arrival absolute pair difference (40,130 records)	-0.109	-3.44 [-4.72, -2.16]	1.1e-4	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay count arrival difference (40,130 records)	0.109	3.44 [2.16, 4.72]	1.1e-4	100.00 / 100.00
G16 Incident-link imbalance degree balance arrival difference (40,130 records)	0.109	3.27 [1.64, 4.90]	0.004	100.00 / 100.00
G19 Endpoint contrasts H05 downstream reachability endpoint contrast following train (40,130 records)	-0.109	2.34 [1.35, 3.33]	6.0e-4	100.00 / 100.00
B02 Local edges and density edges 20 min absolute pair difference (40,130 records)	0.108	2.54 [0.68, 4.39]	0.111	100.00 / 100.00
G12 Delay-support distribution incoming support minimum arrival difference (38,773 records)	-0.107	-2.36 [-3.88, -0.84]	0.044	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay minimum departure absolute pair difference (29,690 records)	0.107	0.27 [-1.82, 2.36]	1.000	73.33 / 66.67
H11 Harmonic closeness incoming harmonic arrival absolute pair difference (40,130 records)	0.107	1.19 [-0.40, 2.78]	1.000	100.00 / 100.00
G05 Event degree outgoing arrival following train (40,130 records)	0.107	3.35 [1.83, 4.86]	0.001	100.00 / 100.00
G06 Cross-train degree outgoing cross arrival following train (40,130 records)	0.107	3.35 [1.83, 4.86]	0.001	100.00 / 100.00
G07 Degree by relation outgoing exit headway arrival following train (40,130 records)	0.107	3.35 [1.83, 4.86]	0.001	100.00 / 100.00
G10 Edge-slack distribution outgoing slack count arrival following train (40,130 records)	0.107	3.35 [1.83, 4.86]	0.001	100.00 / 100.00
G12 Delay-support distribution outgoing support count arrival following train (40,130 records)	0.107	3.35 [1.83, 4.86]	0.001	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G14 Downstream-neighbor delay neighbor delay count arrival following train (40,130 records)	0.107	3.35 [1.83, 4.86]	0.001	100.00 / 100.00
G10 Edge-slack distribution incoming slack mean arrival difference (38,773 records)	-0.106	-3.41 [-5.42, -1.40]	0.022	100.00 / 100.00
G19 Endpoint contrasts G08 active incoming degree endpoint contrast difference (40,130 records)	0.106	3.20 [1.74, 4.67]	0.002	100.00 / 100.00
G16 Incident-link imbalance degree balance departure following train (40,130 records)	0.106	3.59 [1.83, 5.34]	0.003	100.00 / 100.00
B01 Local region size region trains 5 min absolute pair difference (40,130 records)	0.106	3.03 [1.71, 4.35]	9.0e-4	100.00 / 100.00
G16 Incident-link imbalance degree balance departure absolute pair difference (40,130 records)	-0.106	-3.46 [-4.86, -2.05]	3.7e-4	100.00 / 100.00
H08 Spatial reachability downstream branches arrival absolute pair difference (40,130 records)	0.105	3.89 [2.31, 5.48]	3.7e-4	100.00 / 100.00
G15 Active-neighbor dispersion delay variance arrival following train (24,085 records)	0.105	3.51 [1.48, 5.54]	0.019	100.00 / 100.00
G16 Incident-link imbalance degree balance arrival following train (40,130 records)	0.105	2.89 [1.50, 4.27]	0.003	100.00 / 100.00
G07 Degree by relation incoming emergency arrival absolute pair difference (40,130 records)	0.105	3.29 [1.99, 4.59]	2.6e-4	100.00 / 100.00
G11 Near-tight links slack at most 1 minute departure following train (40,130 records)	0.105	3.18 [2.41, 3.95]	2.7e-7	100.00 / 100.00
G17 Relation-type entropy type entropy arrival following train (40,130 records)	-0.104	-0.70 [-1.50, 0.09]	0.880	100.00 / 66.67
H25 Local assortativity local delay absolute difference arrival difference (18,212 records)	0.104	4.59 [2.37, 6.82]	0.003	100.00 / 100.00
G12 Delay-support distribution outgoing support median departure absolute pair difference (40,130 records)	0.104	-0.44 [-2.24, 1.36]	1.000	93.33 / 66.67
B04 Boundary connections entering 5 min following train (40,130 records)	0.104	3.16 [1.73, 4.58]	0.001	100.00 / 100.00
G10 Edge-slack distribution incoming slack standard deviation arrival difference (38,773 records)	-0.104	-3.46 [-5.02, -1.89]	0.001	100.00 / 100.00
G15 Active-neighbor dispersion delay range arrival following train (24,085 records)	0.103	3.93 [1.35, 6.50]	0.052	100.00 / 100.00
G12 Delay-support distribution outgoing support 90th percentile arrival absolute pair difference (40,130 records)	0.103	-0.15 [-2.09, 1.78]	1.000	73.33 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B02 Local edges and density edges 20 min difference (40,130 records)	0.103	2.45 [0.12, 4.78]	0.464	100.00 / 100.00
B01 Local region size region events 20 min absolute pair difference (40,130 records)	0.103	2.20 [0.40, 3.99]	0.221	100.00 / 100.00
B05 Regional delay distribution region delay 20 min count absolute pair difference (40,130 records)	0.103	2.20 [0.40, 3.99]	0.221	100.00 / 100.00
G05 Event degree outgoing departure following train (40,130 records)	0.102	3.45 [1.71, 5.19]	0.005	100.00 / 100.00
G06 Cross-train degree outgoing cross departure following train (40,130 records)	0.102	3.45 [1.71, 5.19]	0.005	100.00 / 100.00
G10 Edge-slack distribution outgoing slack count departure following train (40,130 records)	0.102	3.45 [1.71, 5.19]	0.005	100.00 / 100.00
G12 Delay-support distribution outgoing support count departure following train (40,130 records)	0.102	3.45 [1.71, 5.19]	0.005	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay mean arrival difference (34,608 records)	-0.102	-1.22 [-2.90, 0.46]	1.000	100.00 / 66.67
G13 Upstream-neighbor delay neighbor delay median arrival difference (34,608 records)	-0.102	-1.22 [-2.90, 0.46]	1.000	100.00 / 66.67
G10 Edge-slack distribution incoming slack sum arrival difference (40,130 records)	-0.102	-3.31 [-5.29, -1.33]	0.025	100.00 / 100.00
B01 Local region size region events 20 min difference (40,130 records)	0.102	2.33 [0.08, 4.59]	0.495	100.00 / 100.00
B05 Regional delay distribution region delay 20 min count difference (40,130 records)	0.102	2.33 [0.08, 4.59]	0.495	100.00 / 100.00
G05 Event degree outgoing departure absolute pair difference (40,130 records)	-0.102	-3.34 [-4.63, -2.05]	1.9e-4	100.00 / 100.00
G06 Cross-train degree outgoing cross departure absolute pair difference (40,130 records)	-0.102	-3.34 [-4.63, -2.05]	1.9e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack count departure absolute pair difference (40,130 records)	-0.102	-3.34 [-4.63, -2.05]	1.9e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support count departure absolute pair difference (40,130 records)	-0.102	-3.34 [-4.63, -2.05]	1.9e-4	100.00 / 100.00
B09 Boundary pressure imbalance cross boundary pressure balance 5 min difference (40,130 records)	-0.102	-2.19 [-4.40, 0.01]	0.582	100.00 / 100.00
G10 Edge-slack distribution incoming slack 90th percentile arrival difference (38,773 records)	-0.102	-3.20 [-4.96, -1.45]	0.011	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H08 Spatial reachability upstream branches arrival difference (40,130 records)	-0.101	-3.03 [-4.60, -1.47]	0.006	100.00 / 100.00
G16 Incident-link imbalance pressure balance arrival leading train (40,130 records)	-0.101	-2.41 [-4.71, -0.12]	0.458	100.00 / 66.67
B05 Regional delay distribution region delay 10 min sum absolute pair difference (40,130 records)	0.101	0.84 [-1.14, 2.83]	1.000	100.00 / 100.00
G09 Active cross-train links active cross departure absolute pair difference (40,130 records)	0.101	2.23 [0.95, 3.50]	0.017	100.00 / 100.00
G12 Delay-support distribution incoming support median departure absolute pair difference (40,130 records)	0.101	0.49 [-1.14, 2.12]	1.000	100.00 / 33.33
H11 Harmonic closeness incoming harmonic departure absolute pair difference (40,130 records)	0.100	1.23 [-0.53, 2.99]	1.000	100.00 / 100.00
G05 Event degree outgoing departure difference (40,130 records)	0.099	3.21 [2.02, 4.41]	1.1e-4	100.00 / 100.00
G06 Cross-train degree outgoing cross departure difference (40,130 records)	0.099	3.21 [2.02, 4.41]	1.1e-4	100.00 / 100.00
G10 Edge-slack distribution outgoing slack count departure difference (40,130 records)	0.099	3.21 [2.02, 4.41]	1.1e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support count departure difference (40,130 records)	0.099	3.21 [2.02, 4.41]	1.1e-4	100.00 / 100.00
G10 Edge-slack distribution incoming slack interquartile range arrival difference (38,773 records)	-0.098	-3.40 [-4.95, -1.84]	0.002	100.00 / 100.00
H10 Active betweenness betweenness arrival following train (40,130 records)	0.098	2.67 [1.07, 4.28]	0.025	100.00 / 100.00
G10 Edge-slack distribution outgoing slack minimum arrival absolute pair difference (40,130 records)	0.098	0.61 [-0.99, 2.21]	1.000	100.00 / 66.67
B04 Boundary connections leaving 5 min difference (40,130 records)	0.098	2.47 [1.29, 3.64]	0.003	100.00 / 100.00
B03 Active local density active density 20 min difference (39,938 records)	-0.097	-2.83 [-3.99, -1.66]	4.5e-4	100.00 / 100.00
B05 Regional delay distribution region delay 20 min interquartile range absolute pair difference (40,130 records)	0.097	1.92 [0.01, 3.82]	0.558	100.00 / 100.00
H13 Weak-component size component trains arrival difference (40,130 records)	-0.096	-2.96 [-4.80, -1.12]	0.035	100.00 / 100.00
B05 Regional delay distribution region delay 20 min maximum absolute pair difference (40,130 records)	-0.096	-3.78 [-5.35, -2.21]	4.9e-4	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H13 Weak-component size component events arrival difference (40,130 records)	-0.096	-3.05 [-4.88, -1.23]	0.025	100.00 / 100.00
G11 Near-tight links slack at most 1 minute departure absolute pair difference (40,130 records)	0.096	2.89 [1.62, 4.17]	0.001	100.00 / 100.00
G12 Delay-support distribution incoming support mean departure absolute pair difference (40,130 records)	0.094	0.32 [-1.32, 1.97]	1.000	86.67 / 33.33
G07 Degree by relation incoming emergency arrival leading train (40,130 records)	0.093	2.93 [1.63, 4.23]	0.001	100.00 / 100.00
G01 Reference arrival delay arrival delay absolute pair difference (40,130 records)	0.093	-0.16 [-2.11, 1.79]	1.000	73.33 / 66.67
B09 Boundary pressure imbalance cross boundary pressure balance 10 min difference (40,130 records)	-0.093	-1.92 [-4.15, 0.32]	0.975	100.00 / 100.00
G12 Delay-support distribution incoming support sum departure absolute pair difference (40,130 records)	0.093	0.15 [-1.39, 1.69]	1.000	80.00 / 33.33
G18 External-source share source fraction arrival leading train (38,990 records)	0.093	2.94 [1.61, 4.26]	0.001	100.00 / 100.00
H06 Upstream trains upstream arrival difference (40,130 records)	0.093	3.30 [1.32, 5.28]	0.025	100.00 / 100.00
H24 Projection core number core departure difference (40,130 records)	-0.092	-2.33 [-3.92, -0.74]	0.068	100.00 / 100.00
H02 Longest active path longest path arrival absolute pair difference (40,130 records)	0.091	0.26 [-1.40, 1.92]	1.000	86.67 / 66.67
H15 Chain and crossing length longest chain arrival absolute pair difference (40,130 records)	0.091	0.26 [-1.40, 1.92]	1.000	86.67 / 66.67
G12 Delay-support distribution outgoing support maximum arrival absolute pair difference (40,130 records)	0.091	-0.19 [-2.14, 1.76]	1.000	73.33 / 66.67
G15 Active-neighbor dispersion delay coefficient of variation arrival following train (24,085 records)	0.091	1.66 [-0.36, 3.69]	1.000	100.00 / 66.67
G17 Relation-type entropy type entropy arrival difference (40,130 records)	-0.090	-2.31 [-3.62, -1.00]	0.016	100.00 / 100.00
B03 Active local density active density 10 min difference (39,628 records)	-0.090	-2.54 [-3.96, -1.12]	0.014	100.00 / 100.00
G19 Endpoint contrasts G08 active outgoing degree endpoint contrast leading train (40,130 records)	0.090	2.88 [1.53, 4.23]	0.002	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B04 Boundary connections entering 5 min difference (40,130 records)	-0.089	-2.74 [-4.88, -0.61]	0.165	100.00 / 100.00
H08 Spatial reachability downstream stations departure absolute pair difference (40,130 records)	0.088	2.70 [0.36, 5.04]	0.300	100.00 / 100.00
G12 Delay-support distribution outgoing support zero fraction arrival difference (40,130 records)	0.087	2.89 [1.07, 4.71]	0.038	100.00 / 100.00
H25 Local assortativity local same type fraction arrival difference (18,212 records)	-0.087	-3.93 [-6.42, -1.43]	0.040	100.00 / 100.00
G11 Near-tight links slack at most 2 minutes arrival following train (40,130 records)	0.087	2.93 [1.62, 4.23]	0.001	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction arrival absolute pair difference (26,950 records)	-0.087	-2.86 [-4.73, -1.00]	0.050	100.00 / 100.00
G08 Active degree incoming active departure difference (40,130 records)	0.087	2.37 [1.00, 3.75]	0.019	100.00 / 100.00
B04 Boundary connections leaving 10 min difference (40,130 records)	0.086	2.38 [0.87, 3.89]	0.040	100.00 / 100.00
G10 Edge-slack distribution outgoing slack zero fraction arrival following train (40,130 records)	-0.086	-3.18 [-4.10, -2.25]	3.7e-6	100.00 / 100.00
B01 Local region size region trains 20 min difference (40,130 records)	0.086	1.96 [-0.36, 4.28]	1.000	100.00 / 100.00
B02 Local edges and density edges 5 min absolute pair difference (40,130 records)	0.085	1.92 [0.99, 2.86]	0.003	100.00 / 100.00
G10 Edge-slack distribution incoming slack maximum arrival difference (38,773 records)	-0.085	-2.76 [-4.37, -1.16]	0.019	100.00 / 100.00
G19 Endpoint contrasts H10 active betweenness endpoint contrast difference (40,130 records)	0.085	2.06 [0.85, 3.26]	0.021	100.00 / 100.00
B01 Local region size region trains 20 min absolute pair difference (40,130 records)	0.085	1.79 [0.05, 3.54]	0.509	100.00 / 100.00
H16 Local motifs diamonds at source departure following train (40,130 records)	0.085	2.67 [1.33, 4.00]	0.004	100.00 / 100.00
G16 Incident-link imbalance degree balance arrival absolute pair difference (40,130 records)	-0.085	-2.53 [-4.18, -0.89]	0.048	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay 90th percentile arrival difference (34,608 records)	-0.085	-0.47 [-2.11, 1.17]	1.000	93.33 / 66.67
G12 Delay-support distribution outgoing support 90th percentile departure absolute pair difference (40,130 records)	0.084	-0.93 [-2.73, 0.87]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H12 PageRank pagerank arrival absolute pair difference (40,130 records)	0.084	-1.05 [-2.36, 0.27]	1.000	100.00 / 66.67
G19 Endpoint contrasts G05 outgoing degree endpoint contrast following train (40,130 records)	0.084	2.56 [1.04, 4.09]	0.024	100.00 / 100.00
H12 PageRank pagerank departure absolute pair difference (40,130 records)	0.083	-0.43 [-1.50, 0.64]	1.000	93.33 / 66.67
H01 Shortest source distance source minimum time lag departure leading train (23,914 records)	0.082	2.96 [-0.87, 6.79]	1.000	100.00 / 66.67
H02 Longest active path longest path departure difference (40,130 records)	0.082	0.30 [-1.42, 2.02]	1.000	86.67 / 66.67
H15 Chain and crossing length longest chain departure difference (40,130 records)	0.082	0.30 [-1.42, 2.02]	1.000	86.67 / 66.67
G12 Delay-support distribution outgoing support zero fraction departure difference (40,130 records)	0.082	3.12 [1.69, 4.56]	0.002	100.00 / 100.00
B02 Local edges and density density 20 min difference (39,938 records)	-0.081	-1.19 [-2.77, 0.40]	1.000	100.00 / 100.00
G12 Delay-support distribution outgoing support standard deviation departure difference (40,130 records)	-0.081	-1.87 [-4.25, 0.52]	1.000	100.00 / 100.00
G11 Near-tight links slack at most 2 minutes arrival leading train (40,130 records)	0.081	2.13 [0.95, 3.31]	0.013	100.00 / 100.00
H08 Spatial reachability upstream branches departure difference (40,130 records)	-0.080	-2.70 [-4.39, -1.01]	0.036	100.00 / 100.00
B15 Global degree distribution incoming degree maximum (40,130 records)	0.080	2.58 [1.01, 4.14]	0.028	100.00 / 100.00
H16 Local motifs fan-out pairs arrival difference (40,130 records)	-0.080	-2.50 [-3.56, -1.43]	7.3e-4	100.00 / 100.00
G12 Delay-support distribution outgoing support interquartile range departure difference (40,130 records)	-0.079	-1.94 [-4.28, 0.40]	1.000	100.00 / 100.00
G11 Near-tight links slack at most 2 minutes departure absolute pair difference (40,130 records)	0.077	2.17 [1.32, 3.03]	2.5e-4	100.00 / 100.00
G11 Near-tight links slack at most 1 minute departure difference (40,130 records)	-0.077	-2.65 [-4.53, -0.77]	0.092	100.00 / 100.00
B05 Regional delay distribution region delay 20 min 90th percentile absolute pair difference (40,130 records)	-0.076	-3.58 [-5.21, -1.95]	0.002	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay count departure absolute pair difference (40,130 records)	-0.075	-2.08 [-3.40, -0.76]	0.040	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H08 Spatial reachability upstream stations departure difference (40,130 records)	-0.075	-3.03 [-4.79, -1.27]	0.019	100.00 / 100.00
G12 Delay-support distribution incoming support 90th percentile departure absolute pair difference (40,130 records)	0.075	-1.28 [-2.99, 0.44]	1.000	100.00 / 100.00
B05 Regional delay distribution region delay 20 min sum absolute pair difference (40,130 records)	0.073	0.75 [-1.13, 2.63]	1.000	100.00 / 100.00
B01 Local region size region events 5 min absolute pair difference (40,130 records)	0.073	1.86 [0.89, 2.84]	0.007	100.00 / 100.00
B05 Regional delay distribution region delay 5 min count absolute pair difference (40,130 records)	0.073	1.86 [0.89, 2.84]	0.007	100.00 / 100.00
G10 Edge-slack distribution outgoing slack minimum departure following train (40,130 records)	0.073	-0.22 [-0.63, 0.18]	1.000	100.00 / 33.33
G12 Delay-support distribution incoming support zero fraction departure absolute pair difference (40,130 records)	-0.073	-2.94 [-4.97, -0.91]	0.076	100.00 / 100.00
B05 Regional delay distribution region delay 20 min standard deviation absolute pair difference (40,130 records)	0.071	0.08 [-1.75, 1.91]	1.000	73.33 / 66.67
B04 Boundary connections leaving 20 min difference (40,130 records)	0.071	2.00 [0.12, 3.88]	0.437	100.00 / 100.00
H01 Shortest source distance source minimum time lag departure following train (22,528 records)	-0.070	-2.95 [-6.82, 0.92]	1.000	100.00 / 66.67
G10 Edge-slack distribution outgoing slack minimum departure difference (40,130 records)	0.070	0.45 [-0.06, 0.97]	0.921	100.00 / 100.00
G19 Endpoint contrasts G08 active outgoing degree endpoint contrast difference (40,130 records)	-0.069	-2.09 [-3.89, -0.28]	0.295	100.00 / 100.00
G07 Degree by relation outgoing station track departure following train (40,130 records)	0.069	2.17 [0.46, 3.88]	0.177	100.00 / 100.00
G10 Edge-slack distribution incoming slack median departure leading train (40,130 records)	0.069	3.36 [1.14, 5.57]	0.054	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum departure difference (36,410 records)	-0.068	0.79 [-1.02, 2.59]	1.000	93.33 / 33.33
G13 Upstream-neighbor delay neighbor delay mean departure difference (36,410 records)	-0.068	0.79 [-1.02, 2.59]	1.000	93.33 / 33.33
G13 Upstream-neighbor delay neighbor delay median departure difference (36,410 records)	-0.068	0.79 [-1.02, 2.59]	1.000	93.33 / 33.33
G13 Upstream-neighbor delay neighbor delay minimum departure difference (36,410 records)	-0.068	0.79 [-1.02, 2.59]	1.000	93.33 / 33.33

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G13 Upstream-neighbor delay neighbor delay 90th percentile departure difference (36,410 records)	-0.068	0.79 [-1.02, 2.59]	1.000	93.33 / 33.33
G15 Active-neighbor dispersion delay variance departure absolute pair difference (21,016 records)	0.067	2.18 [-1.26, 5.61]	1.000	100.00 / 100.00
B02 Local edges and density density 5 min absolute pair difference (38,748 records)	0.066	0.79 [-0.22, 1.80]	1.000	100.00 / 100.00
H12 PageRank pagerank arrival leading train (40,130 records)	0.066	1.01 [-0.37, 2.39]	1.000	100.00 / 100.00
H09 Reachable delay totals upstream train maximum delay sum arrival difference (40,130 records)	0.066	2.79 [0.89, 4.68]	0.068	100.00 / 100.00
G15 Active-neighbor dispersion delay variance arrival difference (18,212 records)	0.066	2.38 [0.82, 3.95]	0.052	100.00 / 100.00
G19 Endpoint contrasts G05 outgoing degree endpoint contrast absolute pair difference (40,130 records)	-0.065	-2.11 [-3.11, -1.11]	0.003	100.00 / 100.00
H15 Chain and crossing length longest crossings departure leading train (40,130 records)	0.064	2.45 [0.46, 4.43]	0.211	100.00 / 100.00
G10 Edge-slack distribution incoming slack standard deviation departure leading train (40,130 records)	0.064	3.33 [0.98, 5.69]	0.088	100.00 / 100.00
G10 Edge-slack distribution incoming slack interquartile range departure leading train (40,130 records)	0.064	3.24 [0.87, 5.61]	0.111	100.00 / 100.00
H08 Spatial reachability upstream branches arrival absolute pair difference (40,130 records)	-0.064	-2.65 [-4.28, -1.03]	0.030	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay maximum arrival difference (34,608 records)	-0.064	-0.29 [-1.91, 1.33]	1.000	80.00 / 66.67
G10 Edge-slack distribution incoming slack mean departure leading train (40,130 records)	0.064	3.24 [0.95, 5.53]	0.089	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay interquartile range departure difference (29,690 records)	-0.063	-0.41 [-2.76, 1.93]	1.000	100.00 / 66.67
G14 Downstream-neighbor delay neighbor delay standard deviation departure difference (29,690 records)	-0.063	-0.41 [-2.76, 1.93]	1.000	100.00 / 66.67
H13 Weak-component size component events departure difference (40,130 records)	-0.063	-2.17 [-3.55, -0.78]	0.043	100.00 / 100.00
H13 Weak-component size component trains departure difference (40,130 records)	-0.063	-2.02 [-3.42, -0.62]	0.077	100.00 / 100.00
G10 Edge-slack distribution incoming slack 90th percentile departure leading train (40,130 records)	0.063	3.23 [0.88, 5.57]	0.107	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H25 Local assortativity local delay absolute difference departure absolute pair difference (21,016 records)	0.063	2.42 [-2.20, 7.03]	1.000	100.00 / 66.67
G10 Edge-slack distribution incoming slack maximum departure leading train (40,130 records)	0.063	3.18 [0.81, 5.54]	0.124	100.00 / 100.00
G10 Edge-slack distribution incoming slack sum departure leading train (40,130 records)	0.063	2.93 [0.64, 5.21]	0.169	100.00 / 100.00
B02 Local edges and density density 10 min absolute pair difference (39,628 records)	0.062	0.12 [-1.11, 1.36]	1.000	86.67 / 66.67
G19 Endpoint contrasts G05 outgoing degree endpoint contrast difference (40,130 records)	0.062	1.89 [0.67, 3.11]	0.045	100.00 / 100.00
B05 Regional delay distribution region delay 20 min zero fraction difference (40,130 records)	0.062	2.73 [1.25, 4.21]	0.010	100.00 / 100.00
H16 Local motifs diamonds at source arrival difference (40,130 records)	-0.062	-1.93 [-3.26, -0.59]	0.076	100.00 / 100.00
H15 Chain and crossing length longest crossings arrival leading train (40,130 records)	0.061	1.92 [0.25, 3.58]	0.305	100.00 / 100.00
G12 Delay-support distribution incoming support maximum departure absolute pair difference (40,130 records)	0.061	-1.06 [-2.91, 0.79]	1.000	100.00 / 100.00
B05 Regional delay distribution region delay 5 min mean absolute pair difference (40,130 records)	0.060	-1.44 [-3.06, 0.17]	0.861	100.00 / 100.00
H16 Local motifs fan-out pairs arrival absolute pair difference (40,130 records)	0.060	1.87 [0.55, 3.20]	0.090	100.00 / 100.00
G02 Reference departure delay departure delay absolute pair difference (40,130 records)	0.060	-1.09 [-2.94, 0.77]	1.000	100.00 / 100.00
B04 Boundary connections leaving 5 min absolute pair difference (40,130 records)	-0.060	-1.73 [-2.74, -0.73]	0.019	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction arrival difference (26,950 records)	0.059	2.04 [0.71, 3.36]	0.050	100.00 / 100.00
H02 Longest active path longest path departure absolute pair difference (40,130 records)	0.059	-0.43 [-2.38, 1.52]	1.000	93.33 / 100.00
H15 Chain and crossing length longest chain departure absolute pair difference (40,130 records)	0.059	-0.43 [-2.38, 1.52]	1.000	93.33 / 100.00
G12 Delay-support distribution outgoing support maximum departure absolute pair difference (40,130 records)	0.059	-1.09 [-2.94, 0.76]	1.000	100.00 / 100.00
G08 Active degree incoming active arrival absolute pair difference (40,130 records)	0.059	1.27 [0.03, 2.52]	0.512	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B01 Local region size region trains 10 min difference (40,130 records)	0.058	1.76 [-0.42, 3.93]	1.000	100.00 / 100.00
B05 Regional delay distribution region delay 5 min zero fraction difference (40,130 records)	0.058	1.94 [0.52, 3.36]	0.112	100.00 / 100.00
B04 Boundary connections leaving 10 min absolute pair difference (40,130 records)	-0.058	-1.68 [-3.12, -0.25]	0.276	100.00 / 100.00
G10 Edge-slack distribution incoming slack zero fraction arrival leading train (38,990 records)	-0.058	-1.77 [-3.51, -0.04]	0.520	100.00 / 100.00
B04 Boundary connections leaving 20 min absolute pair difference (40,130 records)	-0.057	-1.77 [-2.89, -0.65]	0.039	100.00 / 100.00
G17 Relation-type entropy type entropy arrival absolute pair difference (40,130 records)	-0.057	0.96 [-0.73, 2.65]	1.000	100.00 / 100.00
B04 Boundary connections entering 5 min absolute pair difference (40,130 records)	0.056	1.62 [-0.15, 3.38]	0.789	100.00 / 100.00
G15 Active-neighbor dispersion delay range arrival difference (18,212 records)	0.056	2.47 [0.20, 4.73]	0.392	100.00 / 100.00
B05 Regional delay distribution region delay 10 min zero fraction difference (40,130 records)	0.056	2.09 [0.66, 3.52]	0.071	100.00 / 100.00
B05 Regional delay distribution region delay 20 min sum difference (40,130 records)	-0.056	-1.99 [-3.89, -0.10]	0.461	100.00 / 100.00
B08 Community structure community events departure absolute pair difference (40,130 records)	-0.055	-1.43 [-3.68, 0.81]	1.000	100.00 / 100.00
H03 Event time from release time from release arrival following train (40,130 records)	-0.055	-0.10 [-1.88, 1.68]	1.000	53.33 / 66.67
G15 Active-neighbor dispersion delay coefficient of variation arrival difference (18,212 records)	0.055	1.15 [-0.89, 3.19]	1.000	100.00 / 100.00
B04 Boundary connections entering 10 min absolute pair difference (40,130 records)	0.055	0.89 [-0.79, 2.56]	1.000	100.00 / 100.00
G16 Incident-link imbalance pressure balance departure following train (40,130 records)	0.054	1.93 [-0.02, 3.88]	0.586	100.00 / 100.00
G15 Active-neighbor dispersion delay range departure absolute pair difference (21,016 records)	0.054	1.95 [-1.98, 5.89]	1.000	100.00 / 66.67
G08 Active degree outgoing active departure absolute pair difference (40,130 records)	0.054	1.49 [-0.36, 3.35]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack median arrival leading train (38,990 records)	0.053	3.37 [0.50, 6.23]	0.270	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay sum departure difference (40,130 records)	-0.052	0.98 [-0.80, 2.75]	1.000	100.00 / 33.33

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution region delay 5 min median absolute pair difference (40,130 records)	0.052	-1.25 [-2.93, 0.43]	1.000	100.00 / 100.00
G15 Active-neighbor dispersion delay coefficient of variation departure absolute pair difference (21,016 records)	0.051	-0.02 [-3.39, 3.35]	1.000	40.00 / 33.33
G13 Upstream-neighbor delay neighbor delay interquartile range arrival leading train (34,618 records)	0.051	0.15 [-1.21, 1.51]	1.000	73.33 / 33.33
G13 Upstream-neighbor delay neighbor delay standard deviation arrival leading train (34,618 records)	0.051	0.15 [-1.21, 1.51]	1.000	73.33 / 33.33
G11 Near-tight links slack at most 1 minute arrival leading train (40,130 records)	0.051	1.20 [0.33, 2.07]	0.102	100.00 / 100.00
H25 Local assortativity local same type fraction departure difference (21,016 records)	-0.051	-1.80 [-4.97, 1.37]	1.000	100.00 / 100.00
B03 Active local density active density 5 min difference (38,748 records)	-0.050	-1.63 [-3.14, -0.13]	0.401	100.00 / 100.00
B08 Community structure community events arrival difference (40,130 records)	-0.050	-1.53 [-3.95, 0.90]	1.000	100.00 / 100.00
B08 Community structure cross community edges departure following train (40,130 records)	0.049	1.47 [0.54, 2.39]	0.039	100.00 / 100.00
G10 Edge-slack distribution outgoing slack median departure following train (40,130 records)	0.049	-1.96 [-2.96, -0.95]	0.006	100.00 / 100.00
G07 Degree by relation outgoing station track departure difference (40,130 records)	0.049	1.49 [1.08, 1.91]	2.0e-6	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay count departure difference (40,130 records)	0.049	1.20 [-0.18, 2.58]	0.930	100.00 / 100.00
B04 Boundary connections entering 20 min difference (40,130 records)	0.049	0.61 [-1.35, 2.56]	1.000	100.00 / 100.00
H25 Local assortativity local same type fraction departure absolute pair difference (21,016 records)	0.048	2.08 [-2.58, 6.74]	1.000	100.00 / 66.67
G05 Event degree incoming departure difference (40,130 records)	-0.048	-1.52 [-3.39, 0.35]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack count departure difference (40,130 records)	-0.048	-1.52 [-3.39, 0.35]	1.000	100.00 / 100.00
G12 Delay-support distribution incoming support count departure difference (40,130 records)	-0.048	-1.52 [-3.39, 0.35]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack zero fraction departure leading train (40,130 records)	-0.048	-1.44 [-3.02, 0.13]	0.780	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G07 Degree by relation outgoing station track departure absolute pair difference (40,130 records)	-0.048	-1.50 [-1.85, -1.16]	1.1e-7	100.00 / 100.00
H03 Event time from release time from release departure following train (40,130 records)	-0.047	0.21 [-1.43, 1.85]	1.000	86.67 / 66.67
H12 PageRank pagerank arrival difference (40,130 records)	-0.046	-1.53 [-2.81, -0.26]	0.242	100.00 / 100.00
G10 Edge-slack distribution outgoing slack minimum departure absolute pair difference (40,130 records)	0.046	-0.63 [-0.90, -0.35]	9.1e-4	100.00 / 100.00
B05 Regional delay distribution region delay 5 min minimum absolute pair difference (40,130 records)	0.046	-1.26 [-2.85, 0.33]	1.000	100.00 / 100.00
G10 Edge-slack distribution outgoing slack mean departure following train (40,130 records)	0.046	-1.96 [-3.07, -0.84]	0.016	100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction departure difference (40,130 records)	-0.046	-1.12 [-2.38, 0.15]	0.890	100.00 / 100.00
H04 Upstream reachability upstream arrival difference (40,130 records)	0.046	1.10 [-0.58, 2.78]	1.000	100.00 / 100.00
G10 Edge-slack distribution outgoing slack sum departure following train (40,130 records)	0.045	-2.01 [-3.11, -0.90]	0.012	100.00 / 100.00
G12 Delay-support distribution incoming support minimum departure difference (40,130 records)	-0.045	0.15 [-1.94, 2.24]	1.000	66.67 / 33.33
G05 Event degree incoming departure following train (40,130 records)	-0.044	-1.40 [-1.81, -0.98]	4.4e-6	100.00 / 100.00
G07 Degree by relation incoming emergency departure following train (40,130 records)	-0.044	-1.40 [-1.81, -0.98]	4.4e-6	100.00 / 100.00
G10 Edge-slack distribution incoming slack count departure following train (40,130 records)	-0.044	-1.40 [-1.81, -0.98]	4.4e-6	100.00 / 100.00
G12 Delay-support distribution incoming support count departure following train (40,130 records)	-0.044	-1.40 [-1.81, -0.98]	4.4e-6	100.00 / 100.00
G18 External-source share source fraction departure following train (40,130 records)	-0.044	-1.40 [-1.81, -0.98]	4.4e-6	100.00 / 100.00
G10 Edge-slack distribution outgoing slack 90th percentile departure following train (40,130 records)	0.044	-2.02 [-3.21, -0.83]	0.022	100.00 / 100.00
G10 Edge-slack distribution outgoing slack maximum departure following train (40,130 records)	0.044	-2.00 [-3.21, -0.80]	0.026	100.00 / 100.00
H01 Shortest source distance source edges departure leading train (23,914 records)	0.044	1.47 [-2.37, 5.32]	1.000	100.00 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H08 Spatial reachability downstream branches departure absolute pair difference (40,130 records)	0.043	1.25 [-0.97, 3.47]	1.000	100.00 / 100.00
G19 Endpoint contrasts G05 incoming degree endpoint contrast absolute pair difference (40,130 records)	0.043	1.02 [-1.28, 3.32]	1.000	100.00 / 100.00
B08 Community structure community events departure difference (40,130 records)	-0.043	-0.97 [-2.70, 0.75]	1.000	100.00 / 100.00
B03 Active local density active density 20 min absolute pair difference (39,938 records)	-0.042	-4.07 [-5.89, -2.26]	0.001	100.00 / 100.00
B05 Regional delay distribution region delay 5 min 90th percentile absolute pair difference (40,130 records)	0.042	-1.59 [-3.27, 0.10]	0.717	100.00 / 100.00
G10 Edge-slack distribution outgoing slack zero fraction departure absolute pair difference (40,130 records)	-0.041	-1.40 [-2.54, -0.26]	0.212	100.00 / 100.00
G07 Degree by relation incoming emergency departure difference (40,130 records)	-0.040	-1.27 [-3.61, 1.07]	1.000	100.00 / 66.67
G10 Edge-slack distribution outgoing slack interquartile range departure following train (40,130 records)	0.040	-2.02 [-3.26, -0.79]	0.029	100.00 / 100.00
G17 Relation-type entropy type entropy departure absolute pair difference (40,130 records)	-0.040	-0.84 [-1.92, 0.25]	1.000	100.00 / 100.00
G10 Edge-slack distribution outgoing slack standard deviation departure following train (40,130 records)	0.040	-2.04 [-3.28, -0.79]	0.030	100.00 / 100.00
G18 External-source share source fraction departure difference (40,130 records)	-0.040	-1.17 [-3.45, 1.11]	1.000	100.00 / 66.67
H25 Local assortativity local delay absolute difference departure difference (21,016 records)	0.040	0.90 [-1.82, 3.61]	1.000	100.00 / 100.00
B05 Regional delay distribution region delay 10 min maximum absolute pair difference (40,130 records)	-0.040	-2.71 [-4.41, -1.01]	0.037	100.00 / 100.00
G10 Edge-slack distribution outgoing slack zero fraction arrival absolute pair difference (40,130 records)	-0.039	-0.93 [-1.49, -0.36]	0.028	100.00 / 100.00
G10 Edge-slack distribution incoming slack minimum departure difference (40,130 records)	0.039	0.49 [0.18, 0.79]	0.035	100.00 / 100.00
B04 Boundary connections entering 10 min difference (40,130 records)	-0.038	-1.30 [-3.18, 0.58]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H08 Spatial reachability upstream stations arrival difference (40,130 records)	-0.038	-2.65 [-4.24, -1.06]	0.025	100.00 / 100.00
H25 Local assortativity local delay absolute difference arrival absolute pair difference (18,212 records)	0.037	2.21 [-0.63, 5.05]	1.000	100.00 / 100.00
H12 PageRank pagerank departure difference (40,130 records)	0.037	0.34 [-0.70, 1.38]	1.000	100.00 / 66.67
H16 Local motifs fan-in pairs departure leading train (40,130 records)	0.037	1.17 [-0.45, 2.78]	1.000	100.00 / 66.67
G07 Degree by relation incoming exit headway arrival absolute pair difference (40,130 records)	-0.037	-1.15 [-2.83, 0.53]	1.000	100.00 / 100.00
G07 Degree by relation incoming exit headway arrival difference (40,130 records)	-0.037	-1.15 [-2.83, 0.53]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack minimum departure following train (40,130 records)	0.036	-0.09 [-0.31, 0.14]	1.000	100.00 / 66.67
B02 Local edges and density edges 10 min difference (40,130 records)	0.036	1.36 [-1.00, 3.73]	1.000	100.00 / 100.00
H09 Reachable delay totals upstream event delay arrival difference (40,130 records)	0.036	0.72 [-1.01, 2.45]	1.000	100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction arrival difference (38,773 records)	0.036	0.76 [-0.78, 2.29]	1.000	100.00 / 100.00
G07 Degree by relation incoming emergency departure absolute pair difference (40,130 records)	0.036	1.12 [-1.12, 3.36]	1.000	100.00 / 66.67
G18 External-source share source fraction departure absolute pair difference (40,130 records)	0.035	1.03 [-1.15, 3.20]	1.000	100.00 / 66.67
H23 Local clustering undirected clustering arrival absolute pair difference (40,130 records)	0.035	1.04 [0.29, 1.79]	0.104	100.00 / 100.00
H23 Local clustering undirected clustering departure following train (40,130 records)	0.035	0.83 [0.25, 1.41]	0.081	100.00 / 100.00
G15 Active-neighbor dispersion delay range arrival leading train (22,629 records)	0.035	1.75 [-0.78, 4.29]	1.000	100.00 / 66.67
G15 Active-neighbor dispersion delay variance arrival leading train (22,629 records)	0.035	1.63 [-0.44, 3.71]	1.000	100.00 / 66.67
H01 Shortest source distance source edges departure following train (22,528 records)	-0.035	-1.50 [-5.84, 2.83]	1.000	100.00 / 66.67
G13 Upstream-neighbor delay neighbor delay zero fraction departure absolute pair difference (36,410 records)	0.035	1.11 [-0.58, 2.79]	1.000	100.00 / 33.33

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B06 Regional delay variance delay variance 5 min difference (40,130 records)	-0.035	-1.33 [-3.69, 1.03]	1.000	100.00 / 100.00
B05 Regional delay distribution region delay 10 min standard deviation difference (40,130 records)	-0.034	-0.90 [-2.85, 1.04]	1.000	100.00 / 100.00
H23 Local clustering undirected clustering arrival following train (40,130 records)	0.034	1.04 [0.30, 1.77]	0.088	100.00 / 100.00
G14 Downstream-neighbor delay neighbor delay zero fraction departure difference (29,690 records)	0.034	1.28 [0.17, 2.39]	0.301	100.00 / 100.00
H23 Local clustering undirected clustering departure absolute pair difference (40,130 records)	0.034	0.85 [0.28, 1.41]	0.058	100.00 / 100.00
G12 Delay-support distribution incoming support zero fraction arrival absolute pair difference (38,773 records)	-0.034	-1.63 [-3.36, 0.10]	0.721	100.00 / 66.67
G11 Near-tight links slack at most 2 minutes arrival difference (40,130 records)	0.034	1.26 [-0.10, 2.61]	0.753	100.00 / 100.00
B01 Local region size region trains 5 min difference (40,130 records)	0.033	0.92 [-1.07, 2.92]	1.000	100.00 / 100.00
H12 PageRank pagerank arrival following train (40,130 records)	-0.033	-1.07 [-2.42, 0.28]	1.000	100.00 / 66.67
G08 Active degree incoming active arrival difference (40,130 records)	-0.032	-0.81 [-2.23, 0.60]	1.000	100.00 / 66.67
B06 Regional delay variance delay variance 10 min difference (40,130 records)	-0.032	-0.94 [-2.94, 1.06]	1.000	100.00 / 100.00
H16 Local motifs diamonds at source arrival absolute pair difference (40,130 records)	0.031	0.99 [-0.14, 2.12]	0.914	100.00 / 100.00
B01 Local region size region events 10 min difference (40,130 records)	0.031	1.00 [-1.26, 3.27]	1.000	100.00 / 100.00
B05 Regional delay distribution region delay 10 min count difference (40,130 records)	0.031	1.00 [-1.26, 3.27]	1.000	100.00 / 100.00
B04 Boundary connections entering 20 min absolute pair difference (40,130 records)	0.031	0.13 [-1.48, 1.74]	1.000	80.00 / 66.67
B05 Regional delay distribution region delay 5 min standard deviation difference (40,130 records)	-0.031	-0.81 [-3.45, 1.84]	1.000	100.00 / 66.67
B05 Regional delay distribution region delay 10 min median absolute pair difference (40,130 records)	0.031	-1.66 [-3.30, -0.03]	0.531	100.00 / 100.00
B03 Active local density active density 5 min absolute pair difference (38,748 records)	-0.030	-3.51 [-5.48, -1.54]	0.014	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H24 Projection core number core arrival absolute pair difference (40,130 records)	-0.029	-1.23 [-3.24, 0.78]	1.000	100.00 / 66.67
B05 Regional delay distribution region delay 10 min mean absolute pair difference (40,130 records)	0.029	-2.03 [-3.62, -0.45]	0.168	100.00 / 100.00
B06 Regional delay variance delay variance 20 min difference (40,130 records)	0.029	0.45 [-1.62, 2.51]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack zero fraction arrival difference (38,773 records)	0.029	1.05 [-0.28, 2.38]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack minimum arrival leading train (38,990 records)	0.029	-1.01 [-1.37, -0.65]	5.4e-5	100.00 / 100.00
H01 Shortest source distance source minimum time lag arrival following train (18,199 records)	-0.029	-1.25 [-4.77, 2.27]	1.000	100.00 / 66.67
G19 Endpoint contrasts G08 active incoming degree endpoint contrast leading train (40,130 records)	0.028	1.18 [-0.24, 2.60]	1.000	100.00 / 100.00
G08 Active degree incoming active departure absolute pair difference (40,130 records)	0.028	0.29 [-0.81, 1.39]	1.000	93.33 / 33.33
G10 Edge-slack distribution outgoing slack mean arrival following train (40,130 records)	0.028	-0.36 [-0.87, 0.16]	1.000	93.33 / 66.67
G10 Edge-slack distribution outgoing slack median arrival following train (40,130 records)	0.028	-0.36 [-0.87, 0.16]	1.000	93.33 / 66.67
G05 Event degree incoming arrival difference (40,130 records)	-0.028	-1.04 [-2.55, 0.46]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack count arrival difference (40,130 records)	-0.028	-1.04 [-2.55, 0.46]	1.000	100.00 / 100.00
G12 Delay-support distribution incoming support count arrival difference (40,130 records)	-0.028	-1.04 [-2.55, 0.46]	1.000	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay zero fraction arrival difference (34,608 records)	-0.028	-1.01 [-2.54, 0.51]	1.000	100.00 / 100.00
H08 Spatial reachability upstream stations arrival absolute pair difference (40,130 records)	0.028	-1.51 [-3.02, 0.00]	0.567	100.00 / 100.00
H25 Local assortativity local same type fraction arrival leading train (22,629 records)	-0.027	-0.69 [-3.50, 2.12]	1.000	93.33 / 66.67
G13 Upstream-neighbor delay neighbor delay count arrival leading train (40,130 records)	-0.027	-0.66 [-2.04, 0.72]	1.000	100.00 / 100.00
G19 Endpoint contrasts G05 outgoing degree endpoint contrast leading train (40,130 records)	0.027	0.83 [-0.06, 1.72]	0.732	100.00 / 100.00
G10 Edge-slack distribution outgoing slack sum arrival following train (40,130 records)	0.027	-0.06 [-0.47, 0.34]	1.000	73.33 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G06 Cross-train degree incoming cross departure absolute pair difference (40,130 records)	-0.026	-0.83 [-2.55, 0.90]	1.000	100.00 / 100.00
G06 Cross-train degree incoming cross departure difference (40,130 records)	-0.026	-0.83 [-2.55, 0.90]	1.000	100.00 / 100.00
G07 Degree by relation incoming entry headway departure absolute pair difference (40,130 records)	-0.026	-0.83 [-2.55, 0.90]	1.000	100.00 / 100.00
G07 Degree by relation incoming entry headway departure difference (40,130 records)	-0.026	-0.83 [-2.55, 0.90]	1.000	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay count departure absolute pair difference (40,130 records)	-0.026	-0.83 [-2.55, 0.90]	1.000	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay count departure difference (40,130 records)	-0.026	-0.83 [-2.55, 0.90]	1.000	100.00 / 100.00
G06 Cross-train degree incoming cross departure leading train (40,130 records)	0.026	0.83 [-0.90, 2.55]	1.000	100.00 / 100.00
G07 Degree by relation incoming entry headway departure leading train (40,130 records)	0.026	0.83 [-0.90, 2.55]	1.000	100.00 / 100.00
G13 Upstream-neighbor delay neighbor delay count departure leading train (40,130 records)	0.026	0.83 [-0.90, 2.55]	1.000	100.00 / 100.00
G16 Incident-link imbalance pressure balance arrival difference (40,130 records)	-0.026	-0.97 [-2.73, 0.80]	1.000	100.00 / 66.67
H01 Shortest source distance source minimum time lag arrival leading train (19,001 records)	0.026	0.87 [-5.11, 6.85]	1.000	80.00 / 66.67
G05 Event degree incoming departure leading train (40,130 records)	0.026	0.80 [-1.02, 2.62]	1.000	100.00 / 66.67
G10 Edge-slack distribution incoming slack count departure leading train (40,130 records)	0.026	0.80 [-1.02, 2.62]	1.000	100.00 / 66.67
G12 Delay-support distribution incoming support count departure leading train (40,130 records)	0.026	0.80 [-1.02, 2.62]	1.000	100.00 / 66.67
G12 Delay-support distribution outgoing support interquartile range arrival difference (40,130 records)	-0.025	-1.09 [-2.68, 0.50]	1.000	100.00 / 100.00
G12 Delay-support distribution outgoing support standard deviation arrival difference (40,130 records)	-0.025	-1.09 [-2.68, 0.50]	1.000	100.00 / 100.00
G15 Active-neighbor dispersion delay range departure difference (21,016 records)	0.025	0.44 [-1.63, 2.52]	1.000	86.67 / 33.33
H25 Local assortativity local delay absolute difference arrival leading train (22,629 records)	0.025	0.71 [-1.64, 3.06]	1.000	100.00 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G15 Active-neighbor dispersion delay variance departure difference (21,016 records)	0.024	0.00 [-2.28, 2.28]	1.000	46.67 / 66.67
G10 Edge-slack distribution incoming slack standard deviation departure absolute pair difference (40,130 records)	0.024	1.59 [-0.37, 3.54]	1.000	100.00 / 33.33
B05 Regional delay distribution region delay 20 min minimum absolute pair difference (40,130 records)	-0.024	-2.44 [-4.10, -0.78]	0.068	100.00 / 100.00
B05 Regional delay distribution region delay 20 min standard deviation difference (40,130 records)	0.023	0.21 [-1.55, 1.97]	1.000	93.33 / 66.67
G10 Edge-slack distribution incoming slack median departure absolute pair difference (40,130 records)	0.023	1.63 [-0.28, 3.54]	0.991	100.00 / 66.67
G10 Edge-slack distribution incoming slack interquartile range departure absolute pair difference (40,130 records)	0.023	1.57 [-0.40, 3.54]	1.000	100.00 / 33.33
G05 Event degree outgoing departure leading train (40,130 records)	0.023	0.72 [-0.69, 2.12]	1.000	100.00 / 100.00
G06 Cross-train degree outgoing cross departure leading train (40,130 records)	0.023	0.72 [-0.69, 2.12]	1.000	100.00 / 100.00
G07 Degree by relation outgoing station track departure leading train (40,130 records)	0.023	0.72 [-0.69, 2.12]	1.000	100.00 / 100.00
G10 Edge-slack distribution outgoing slack count departure leading train (40,130 records)	0.023	0.72 [-0.69, 2.12]	1.000	100.00 / 100.00
G12 Delay-support distribution outgoing support count departure leading train (40,130 records)	0.023	0.72 [-0.69, 2.12]	1.000	100.00 / 100.00
B09 Boundary pressure imbalance cross boundary pressure balance 5 min leading train (40,130 records)	0.023	2.25 [-0.29, 4.79]	0.887	100.00 / 100.00
G15 Active-neighbor dispersion delay coefficient of variation arrival leading train (22,629 records)	0.022	-0.69 [-2.29, 0.92]	1.000	100.00 / 33.33
G10 Edge-slack distribution outgoing slack 90th percentile arrival following train (40,130 records)	0.022	-0.85 [-1.59, -0.11]	0.307	100.00 / 66.67
G10 Edge-slack distribution incoming slack sum arrival leading train (40,130 records)	0.022	2.74 [0.29, 5.19]	0.346	100.00 / 100.00
G18 External-source share source fraction arrival following train (38,806 records)	0.022	0.68 [0.22, 1.15]	0.066	100.00 / 100.00
G07 Degree by relation incoming emergency arrival following train (40,130 records)	0.022	0.68 [0.22, 1.15]	0.070	100.00 / 100.00
B05 Regional delay distribution region delay 5 min maximum absolute pair difference (40,130 records)	0.022	-1.73 [-3.39, -0.07]	0.477	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution incoming slack maximum departure absolute pair difference (40,130 records)	0.021	1.51 [-0.46, 3.48]	1.000	100.00 / 33.33
G10 Edge-slack distribution outgoing slack maximum arrival following train (40,130 records)	0.021	-0.94 [-1.67, -0.21]	0.159	100.00 / 100.00
G15 Active-neighbor dispersion delay coefficient of variation departure difference (21,016 records)	0.021	-0.06 [-1.63, 1.51]	1.000	73.33 / 66.67
G11 Near-tight links slack at most 1 minute arrival following train (40,130 records)	0.021	0.85 [0.01, 1.69]	0.535	100.00 / 100.00
G07 Degree by relation incoming exit headway arrival leading train (40,130 records)	0.020	0.64 [-0.68, 1.96]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack mean departure absolute pair difference (40,130 records)	0.020	1.55 [-0.39, 3.49]	1.000	100.00 / 33.33
G10 Edge-slack distribution incoming slack 90th percentile departure absolute pair difference (40,130 records)	0.020	1.55 [-0.42, 3.52]	1.000	100.00 / 33.33
G10 Edge-slack distribution incoming slack sum departure absolute pair difference (40,130 records)	0.020	1.64 [-0.33, 3.62]	1.000	100.00 / 66.67
B05 Regional delay distribution region delay 5 min interquartile range difference (40,130 records)	-0.020	-0.29 [-2.29, 1.71]	1.000	86.67 / 66.67
H16 Local motifs fan-in pairs arrival leading train (40,130 records)	-0.019	-0.62 [-1.46, 0.23]	1.000	100.00 / 66.67
G10 Edge-slack distribution incoming slack zero fraction arrival following train (38,806 records)	-0.019	-0.98 [-1.92, -0.05]	0.465	100.00 / 66.67
G10 Edge-slack distribution outgoing slack minimum arrival leading train (40,130 records)	-0.019	-0.82 [-2.45, 0.80]	1.000	100.00 / 66.67
G06 Cross-train degree incoming cross arrival following train (40,130 records)	-0.019	-0.56 [-1.16, 0.05]	0.766	100.00 / 66.67
B08 Community structure community events arrival absolute pair difference (40,130 records)	0.019	0.35 [-1.83, 2.53]	1.000	80.00 / 66.67
H23 Local clustering undirected clustering arrival difference (40,130 records)	0.018	0.56 [0.08, 1.05]	0.293	100.00 / 100.00
G05 Event degree incoming arrival following train (40,130 records)	-0.018	-0.42 [-1.10, 0.25]	1.000	100.00 / 66.67
G10 Edge-slack distribution incoming slack count arrival following train (40,130 records)	-0.018	-0.42 [-1.10, 0.25]	1.000	100.00 / 66.67
G12 Delay-support distribution incoming support count arrival following train (40,130 records)	-0.018	-0.42 [-1.10, 0.25]	1.000	100.00 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
H01 Shortest source distance source edges arrival leading train (19,001 records)	-0.018	-0.96 [-7.36, 5.44]	1.000	93.33 / 66.67
B02 Local edges and density density 10 min difference (39,628 records)	-0.018	0.11 [-1.33, 1.55]	1.000	80.00 / 66.67
G05 Event degree incoming arrival absolute pair difference (40,130 records)	0.018	0.10 [-1.62, 1.82]	1.000	73.33 / 66.67
G10 Edge-slack distribution incoming slack count arrival absolute pair difference (40,130 records)	0.018	0.10 [-1.62, 1.82]	1.000	73.33 / 66.67
G12 Delay-support distribution incoming support count arrival absolute pair difference (40,130 records)	0.018	0.10 [-1.62, 1.82]	1.000	73.33 / 66.67
B05 Regional delay distribution region delay 20 min interquartile range difference (40,130 records)	0.018	0.60 [-1.12, 2.32]	1.000	100.00 / 66.67
G10 Edge-slack distribution incoming slack minimum departure leading train (40,130 records)	-0.018	-0.56 [-0.82, -0.30]	0.002	100.00 / 100.00
G07 Degree by relation incoming station track arrival following train (40,130 records)	-0.017	-0.55 [-1.16, 0.06]	0.835	100.00 / 66.67
G17 Relation-type entropy type entropy arrival leading train (40,130 records)	-0.017	0.42 [-0.31, 1.15]	1.000	100.00 / 66.67
B03 Active local density active density 10 min absolute pair difference (39,628 records)	-0.017	-3.37 [-5.07, -1.67]	0.005	100.00 / 100.00
H23 Local clustering undirected clustering departure difference (40,130 records)	0.017	0.47 [-0.04, 0.97]	0.748	100.00 / 100.00
H23 Local clustering undirected clustering departure leading train (40,130 records)	0.017	0.32 [-0.17, 0.81]	1.000	100.00 / 66.67
G13 Upstream-neighbor delay neighbor delay zero fraction arrival absolute pair difference (34,608 records)	-0.016	-0.58 [-2.27, 1.11]	1.000	93.33 / 66.67
B05 Regional delay distribution region delay 20 min median absolute pair difference (40,130 records)	-0.016	-2.63 [-4.17, -1.10]	0.020	100.00 / 100.00
G07 Degree by relation incoming exit headway arrival following train (40,130 records)	-0.016	-0.49 [-1.14, 0.16]	1.000	100.00 / 66.67
G10 Edge-slack distribution outgoing slack minimum departure leading train (40,130 records)	-0.015	-0.74 [-1.02, -0.45]	2.4e-4	100.00 / 100.00
G10 Edge-slack distribution incoming slack median arrival absolute pair difference (38,773 records)	0.015	1.14 [-0.99, 3.28]	1.000	100.00 / 33.33
B09 Boundary pressure imbalance cross boundary pressure balance 10 min leading train (40,130 records)	-0.015	1.17 [-1.51, 3.86]	1.000	100.00 / 33.33

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G10 Edge-slack distribution incoming slack minimum departure absolute pair difference (40,130 records)	0.015	-0.53 [-0.74, -0.33]	1.5e-4	100.00 / 100.00
G15 Active-neighbor dispersion delay variance arrival absolute pair difference (18,212 records)	0.015	1.30 [-0.95, 3.54]	1.000	100.00 / 66.67
G17 Relation-type entropy type entropy departure leading train (40,130 records)	0.014	0.60 [-1.09, 2.30]	1.000	100.00 / 66.67
H23 Local clustering undirected clustering arrival leading train (40,130 records)	0.014	0.38 [-0.16, 0.92]	1.000	100.00 / 100.00
G19 Endpoint contrasts G05 incoming degree endpoint contrast leading train (40,130 records)	0.013	-0.08 [-1.47, 1.32]	1.000	73.33 / 66.67
H25 Local assortativity local same type fraction arrival absolute pair difference (18,212 records)	0.013	0.95 [-2.30, 4.21]	1.000	93.33 / 66.67
G10 Edge-slack distribution incoming slack mean arrival leading train (38,990 records)	0.013	2.74 [0.27, 5.21]	0.357	100.00 / 100.00
G16 Incident-link imbalance degree balance departure leading train (40,130 records)	-0.013	0.01 [-1.53, 1.56]	1.000	66.67 / 33.33
G19 Endpoint contrasts H10 active betweenness endpoint contrast leading train (40,130 records)	0.012	1.53 [0.61, 2.46]	0.026	100.00 / 100.00
B02 Local edges and density density 5 min difference (38,748 records)	0.012	1.28 [-0.25, 2.81]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack mean arrival absolute pair difference (38,773 records)	0.012	1.13 [-0.72, 2.99]	1.000	100.00 / 66.67
B01 Local region size region events 5 min difference (40,130 records)	-0.012	-0.77 [-2.39, 0.86]	1.000	100.00 / 100.00
B05 Regional delay distribution region delay 5 min count difference (40,130 records)	-0.012	-0.77 [-2.39, 0.86]	1.000	100.00 / 100.00
G17 Relation-type entropy type entropy departure following train (40,130 records)	0.012	1.09 [-0.57, 2.75]	1.000	100.00 / 100.00
B02 Local edges and density density 20 min absolute pair difference (39,938 records)	-0.011	-2.42 [-4.04, -0.80]	0.060	100.00 / 100.00
G07 Degree by relation incoming run arrival leading train (40,130 records)	-0.011	-0.34 [-0.96, 0.27]	1.000	100.00 / 66.67
G07 Degree by relation incoming run arrival following train (40,130 records)	-0.011	-0.34 [-0.96, 0.27]	1.000	100.00 / 66.67
G06 Cross-train degree incoming cross arrival difference (40,130 records)	-0.011	-0.59 [-2.09, 0.91]	1.000	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G07 Degree by relation incoming emergency departure leading train (40,130 records)	0.010	0.31 [-1.84, 2.47]	1.000	80.00 / 33.33
G13 Upstream-neighbor delay neighbor delay sum arrival difference (40,130 records)	-0.010	0.74 [-0.78, 2.27]	1.000	100.00 / 33.33
G18 External-source share source fraction departure leading train (40,130 records)	0.010	0.27 [-1.82, 2.37]	1.000	80.00 / 33.33
G10 Edge-slack distribution incoming slack maximum arrival absolute pair difference (38,773 records)	-0.010	0.82 [-0.72, 2.35]	1.000	100.00 / 66.67
G05 Event degree incoming departure absolute pair difference (40,130 records)	0.009	0.29 [-1.97, 2.54]	1.000	86.67 / 33.33
G10 Edge-slack distribution incoming slack count departure absolute pair difference (40,130 records)	0.009	0.29 [-1.97, 2.54]	1.000	86.67 / 33.33
G12 Delay-support distribution incoming support count departure absolute pair difference (40,130 records)	0.009	0.29 [-1.97, 2.54]	1.000	86.67 / 33.33
G13 Upstream-neighbor delay neighbor delay count arrival absolute pair difference (40,130 records)	0.009	0.23 [-0.96, 1.41]	1.000	100.00 / 66.67
G07 Degree by relation incoming station track arrival leading train (40,130 records)	-0.009	-0.28 [-1.67, 1.11]	1.000	86.67 / 66.67
G15 Active-neighbor dispersion delay range arrival absolute pair difference (18,212 records)	0.009	0.72 [-2.41, 3.84]	1.000	86.67 / 66.67
G10 Edge-slack distribution incoming slack sum arrival absolute pair difference (40,130 records)	0.008	0.97 [-0.86, 2.81]	1.000	100.00 / 66.67
H01 Shortest source distance source edges arrival following train (18,199 records)	0.008	0.29 [-3.54, 4.12]	1.000	80.00 / 66.67
G07 Degree by relation incoming station track arrival absolute pair difference (40,130 records)	0.008	0.26 [-1.32, 1.84]	1.000	86.67 / 66.67
G11 Near-tight links slack at most 1 minute arrival difference (40,130 records)	-0.007	-0.37 [-1.26, 0.51]	1.000	100.00 / 66.67
G19 Endpoint contrasts G05 incoming degree endpoint contrast following train (40,130 records)	-0.007	0.03 [-0.55, 0.62]	1.000	66.67 / 66.67
G03 Activity delay increment dwell delay increment leading train (40,130 records)	-0.007	-0.53 [-2.31, 1.25]	1.000	100.00 / 66.67
G04 Activity-duration increment dwell increment leading train (40,130 records)	-0.007	-0.53 [-2.31, 1.25]	1.000	100.00 / 66.67
B05 Regional delay distribution region delay 10 min interquartile range difference (40,130 records)	0.007	0.28 [-2.24, 2.80]	1.000	86.67 / 66.67

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
B05 Regional delay distribution region delay 10 min minimum absolute pair difference (40,130 records)	-0.006	-2.12 [-3.75, -0.49]	0.155	100.00 / 100.00
B05 Regional delay distribution region delay 20 min mean absolute pair difference (40,130 records)	-0.006	-2.89 [-4.44, -1.33]	0.009	100.00 / 100.00
G10 Edge-slack distribution incoming slack 90th percentile arrival absolute pair difference (38,773 records)	-0.005	0.88 [-0.80, 2.56]	1.000	100.00 / 66.67
G11 Near-tight links slack at most 2 minutes departure difference (40,130 records)	-0.005	-0.32 [-1.68, 1.03]	1.000	100.00 / 100.00
H02 Longest active path longest path arrival difference (40,130 records)	-0.005	-1.46 [-3.18, 0.27]	1.000	100.00 / 100.00
H15 Chain and crossing length longest chain arrival difference (40,130 records)	-0.005	-1.46 [-3.18, 0.27]	1.000	100.00 / 100.00
G17 Relation-type entropy type entropy departure difference (40,130 records)	0.005	0.47 [-0.29, 1.22]	1.000	100.00 / 66.67
B08 Community structure cross community edges arrival following train (40,130 records)	-0.005	-0.17 [-0.84, 0.50]	1.000	93.33 / 66.67
H11 Harmonic closeness incoming harmonic arrival difference (40,130 records)	-0.004	-0.47 [-2.16, 1.21]	1.000	100.00 / 66.67
G19 Endpoint contrasts G05 incoming degree endpoint contrast difference (40,130 records)	-0.004	0.15 [-1.57, 1.87]	1.000	80.00 / 66.67
G15 Active-neighbor dispersion delay coefficient of variation arrival absolute pair difference (18,212 records)	0.003	-0.96 [-3.08, 1.16]	1.000	100.00 / 33.33
G12 Delay-support distribution incoming support interquartile range departure following train (40,130 records)	-0.003	-2.42 [-3.49, -1.35]	0.001	100.00 / 100.00
G12 Delay-support distribution incoming support standard deviation departure following train (40,130 records)	-0.003	-2.41 [-3.48, -1.34]	0.001	100.00 / 100.00
G19 Endpoint contrasts G08 active outgoing degree endpoint contrast following train (40,130 records)	-0.003	0.82 [-0.52, 2.15]	1.000	100.00 / 66.67
G10 Edge-slack distribution incoming slack 90th percentile arrival leading train (38,990 records)	0.002	2.60 [0.36, 4.84]	0.286	100.00 / 100.00
B02 Local edges and density edges 5 min difference (40,130 records)	-0.002	-0.33 [-1.77, 1.10]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack interquartile range arrival absolute pair difference (38,773 records)	-0.002	1.34 [-0.16, 2.84]	0.859	100.00 / 100.00

Continued on the next page.

Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G07 Degree by relation incoming station track arrival difference (40,130 records)	-0.002	-0.08 [-1.35, 1.19]	1.000	80.00 / 66.67
G05 Event degree incoming arrival leading train (40,130 records)	0.001	0.47 [-0.77, 1.70]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack count arrival leading train (40,130 records)	0.001	0.47 [-0.77, 1.70]	1.000	100.00 / 100.00
G12 Delay-support distribution incoming support count arrival leading train (40,130 records)	0.001	0.47 [-0.77, 1.70]	1.000	100.00 / 100.00
G10 Edge-slack distribution incoming slack standard deviation arrival absolute pair difference (38,773 records)	0.001	1.35 [-0.14, 2.85]	0.827	100.00 / 100.00
G10 Edge-slack distribution outgoing slack interquartile range arrival following train (40,130 records)	0.001	-1.82 [-2.91, -0.73]	0.025	100.00 / 100.00
G10 Edge-slack distribution outgoing slack standard deviation arrival following train (40,130 records)	0.001	-1.82 [-2.91, -0.73]	0.025	100.00 / 100.00
G06 Cross-train degree incoming cross arrival leading train (40,130 records)	-0.001	0.14 [-1.23, 1.52]	1.000	66.67 / 66.67
G10 Edge-slack distribution incoming slack standard deviation arrival leading train (38,990 records)	-0.001	2.73 [0.67, 4.80]	0.138	100.00 / 100.00
B09 Boundary pressure imbalance cross boundary pressure balance 20 min leading train (40,130 records)	-0.001	0.92 [-2.10, 3.94]	1.000	100.00 / 33.33
B05 Regional delay distribution region delay 10 min 90th percentile absolute pair difference (40,130 records)	0.001	-2.25 [-3.85, -0.64]	0.095	100.00 / 100.00
G05 Event degree outgoing arrival leading train (40,130 records)	0.001	0.03 [-2.01, 2.06]	1.000	66.67 / 33.33
G06 Cross-train degree outgoing cross arrival leading train (40,130 records)	0.001	0.03 [-2.01, 2.06]	1.000	66.67 / 33.33
G07 Degree by relation outgoing exit headway arrival leading train (40,130 records)	0.001	0.03 [-2.01, 2.06]	1.000	66.67 / 33.33
G10 Edge-slack distribution outgoing slack count arrival leading train (40,130 records)	0.001	0.03 [-2.01, 2.06]	1.000	66.67 / 33.33
G12 Delay-support distribution outgoing support count arrival leading train (40,130 records)	0.001	0.03 [-2.01, 2.06]	1.000	66.67 / 33.33
G14 Downstream-neighbor delay neighbor delay count arrival leading train (40,130 records)	0.001	0.03 [-2.01, 2.06]	1.000	66.67 / 33.33
G10 Edge-slack distribution incoming slack interquartile range arrival leading train (38,990 records)	-0.001	2.62 [0.55, 4.69]	0.182	100.00 / 100.00

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Table S10 continued: Strategy-3 train orders

Indicator and numerical form	Rank corr.	Effect [95% interval]	Adjusted *p*	Direction stable (%)
G06 Cross-train degree incoming cross arrival absolute pair difference (40,130 records)	-0.001	-0.34 [-1.99, 1.30]	1.000	93.33 / 100.00
G08 Active degree outgoing active arrival absolute pair difference (40,130 records)	0.000	-0.18 [-1.63, 1.27]	1.000	93.33 / 66.67
G16 Incident-link imbalance degree balance arrival leading train (40,130 records)	0.000	-0.48 [-2.24, 1.28]	1.000	100.00 / 66.67
G10 Edge-slack distribution incoming slack maximum arrival leading train (38,990 records)	0.000	2.33 [0.30, 4.37]	0.307	100.00 / 100.00

Notes: Codes and definitions follow Table S1. Rank corr.: scenario-weighted Spearman correlation. The response indicates an order reversal. Effects per indicator standard deviation are in percentage points. Brackets contain 95% confidence intervals. Adjusted *p*: Benjamini–Yekutieli p-value; 1.1e-7 denotes 1.1×10^{-7}. Direction stable: unchanged coefficient sign after omitting one start-time group/one blockage location (percent). Pair differences are the following train value minus the leading train value. Record counts include only finite values of the indicated form.

S4 Other catalog items and association availability

Together with Section S1, this section completes the 267-item candidate catalog. The 76 items below have no reported association: 39 have only constant forms where defined, 26 cannot be calculated with the available model and data, and 11 describe operating-state, strategy, or experiment context but are absent from the descriptive-association inventory. The complete estimable results for the other 191 items are retained in Section S3.

An unavailable or constant association is not a zero correlation. Missing or inapplicable values are not replaced by zero. The availability classification below follows the recorded numerical inventory and model coverage, not the preliminary catalog flags. Definitions retain the intended measurement, while references to the present design describe the current experiment. For context items, the absence of an estimated association does not imply that the quantity is irrelevant to model construction.

Table S11: Remaining candidate indicator definitions and association availability

Code	Indicator	Definition	Availability
Disruption conditions			
D06	Decision delay after blockage start	Elapsed time between blockage start and the decision time; zero at the initial decision.	Constant
D07	Elapsed blockage duration at decision	The nonnegative elapsed time since blockage start at the decision time; zero at the initial decision.	Constant
D12	Disrupted facility type	Type of unavailable facility, such as a section track, station track, or station throat; only section running resources are closed here.	Constant
D13	Disruption cause	The recorded cause of the disruption, such as equipment, power, weather, or an accident. No cause records are available.	Unavailable
D14	Number of blocked sections	Number of physical sections closed simultaneously; one in this design.	Constant
D15	Blocked directions	The number and combination of closed directions; both directions of one section are closed here.	Constant
D16	Unavailable running resources	Number of closed directional section-running resources; two here.	Constant
D17	Unavailable station tracks	Number of station tracks directly closed by the disruption; zero here.	Constant
D18	Direct capacity-loss fraction	Unavailable capacity divided by capacity of the same compatible resource pool; the blocked directional running resources lose all capacity here.	Constant
D19	Residual capacity during blockage	The fraction of compatible capacity still usable in a blocked resource pool; zero for the blocked running resources here.	Constant
D21	Number of recovery stages	Number of separate capacity-restoration stages; a single reopening is assumed.	Constant
D22	Capacity-recovery profile	Available resource capacity over the recovery stages; represented here by a single-step recovery indicator.	Constant
D23	Post-reopening restrictions	Additional speed restrictions or headways after reopening; none are imposed here.	Constant
D24	Opposite-direction track use	Whether a train may use the opposite-direction track; this is prohibited here.	Constant
D25	Number of simultaneous disruptions	Number of concurrent disruptions; one in this design.	Constant
D26	Spatial relationship between disruptions	Distance or spatial overlap between multiple disrupted facilities. It is not applicable to the single-disruption design.	Unavailable
D27	Temporal overlap between disruptions	Duration or fraction of overlap between multiple blockage windows. It is not applicable to the single-disruption design.	Unavailable
D28	Overlap with maintenance closures	Overlap of the disruption with maintenance or other resource closures; requires additional closure records.	Unavailable
D29	Overlap with external service constraints	Overlap with connection requirements, boundary acceptance restrictions, or other external service constraints; requires those records.	Unavailable

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Table S11 continued

Code	Indicator	Definition	Availability
E22	Trains still inside the blocked section	Number of trains that have not cleared the section at blockage start; zero because the selected start times have an empty section in both directions.	Constant
E24	Pre-existing delay indicator	Whether any train is already delayed before blockage start; the initial operating state follows the planned timetable here.	Constant
E25	Pre-existing delay distribution	Distribution of train delays before blockage start, distinct from delays in the subsequent reference timetable; zero here.	Constant
I01	Advance warning time	Time between receiving a disruption warning and the disruption start; actual notification records are unavailable.	Unavailable
I02	Detection or reporting delay	Time from the occurrence of a fault until confirmation by the system or dispatcher; requires reporting records.	Unavailable
I03	Instruction or execution delay	Time from a dispatching decision until a train executes the emergency action; requires communication and execution records.	Unavailable
I04	Certainty of reopening time	Whether reopening time is known at the decision; it is assumed known exactly at blockage start.	Constant
I07	Reopening forecast interval width	Width of the prediction interval for reopening time; zero under the deterministic assumption.	Constant
I08	Reopening forecast variability	Variance and distributional shape of the reopening-time forecast; the numerical variance is zero for the deterministic forecast.	Constant
I10	Number of recovery-information updates	Number of updates to recovery information; no dynamic updates are represented here.	Constant
I11	Time since the last information update	Elapsed time since the latest recovery-information update; requires update logs.	Unavailable
I12	Magnitude of forecast revisions	Changes between successive reopening forecasts and their accumulated magnitude; requires forecast histories.	Unavailable
I13	Retrospective reopening forecast error	Actual reopening time minus the prior forecast. This is a retrospective diagnostic, not a permissible input when the actual time is unknown.	Unavailable
I14	Certainty of blockage location and extent	Whether the location and affected resources are known; both are known here. An uncertain setting requires candidate extents and probabilities.	Constant
I15	Certainty of recovery mode and stages	Whether the restoration mode and stages are known; one complete reopening is assumed known here.	Constant
I16	Predicted risk of renewed closure	A forecast of renewed closure after temporary restoration; requires an available prior forecast and a corresponding resource model.	Unavailable
M02	Executed or irrevocable decision status	Whether an arrival, departure, or movement is already executed or irrevocable. This is a hard operating-state condition, not evidence that a freely adjustable decision can safely be fixed.	Context only
M07	Emergency stopping policy	The rule specifying which trains stop and wait, such as all trains or only directly affected trains. The current experiment uses the directly affected-train rule.	Context only
Z03	Weather and environmental conditions	Weather or other environmental conditions relevant to the disruption and train operation; these records are unavailable.	Unavailable
Z04	Repair-resource state	Available repair teams, access conditions, and repair stages; requires actual maintenance-response records.	Unavailable
Z11	Historical disruptions and contingency plans	Frequency and characteristics of comparable past disruptions and plans available at the decision time; requires historical records.	Unavailable
Z12	Damage severity and repair access	Equipment damage, work restrictions, and access for repair; the available inputs specify only the blockage window.	Unavailable
Timetable characteristics			
M01	Decision and activity type	The decision class and activity represented by a record: stopping, dwell duration, running duration, or section order. Decision types define separate analyses.	Context only
M08	Available recovery decisions	The decision classes permitted by each recovery strategy. This is model context rather than an infrastructure property.	Context only

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Table S11 continued

Code	Indicator	Definition	Availability
M09	Reference timetable provenance	The strategy, ordering rule, solution source, and identifier used to obtain the reference timetable.	Context only
M10	Objective and train weights	The objective definition and train-specific weights; the experiment minimizes positive terminal delay with unit train weights.	Context only
M11	Planning endpoint and remaining horizon	The time remaining to the planning endpoint, measured from the decision or relevant activity time. This describes an imposed modeling horizon.	Context only
M12	Time grid and numerical tolerance	The discretization and numerical acceptance settings; these are experiment controls rather than physical mechanisms.	Context only
M13	Representative-solution selection rule	How one solution is chosen when equally good timetables exist, for example by secondary minimum-change criteria or another tie rule. The analysis uses one returned representative per full model.	Context only
M14	Solver seed and initial timetable source	The search seed, encoding choices, and source of the initial timetable, which can influence the returned representative but are not operational causal factors.	Context only
M15	Solution-certification status	The objective bound and acceptance status supporting a returned solution, retained when interpreting adjustment labels. The reported solutions satisfy the prescribed relative-gap tolerance.	Context only
T07	Planned running-time margin	Planned running duration minus its operational minimum; zero in the studied timetable.	Constant
T08	Scheduled dwell margin	Dwell duration at a scheduled stop minus its minimum stopping time; zero in the studied timetable.	Constant
T16	Train objective weight or priority	A train's weight or priority in the objective; all terminal delays have unit weight here.	Constant
T23	Remaining planned buffer	Remaining planned buffer along a train's journey, distinguishing running, dwell, and inter-train headway margins. The recorded numerical forms are constant where available in this design.	Constant
T25	Planned reordering locations	Locations where relative train order changes between consecutive shared directed sections in the original timetable; a merge alone is not classified as overtaking.	Constant
T26	Arrival–departure rank difference	Difference between a train's planned arrival and departure ranks at the same station, keeping direction and parallel occupations distinct.	Constant
Z02	Train length and acceleration performance	Train lengths and acceleration/braking characteristics; additional data and a more detailed movement model are required.	Unavailable
Z09	Dispatching plans and manual priorities	Actual contingency rules or dispatcher priorities, including whether they constrain optimization; requires recorded operating rules.	Unavailable
Z10	Calendar and special operating conditions	Holiday, calendar, or special-service conditions; the synthetic planned timetable has no actual calendar attributes.	Unavailable
Train interactions			
P01	Planned train-pair precedence	Which of two trains precedes the other on a common directed resource. Pairs are oriented by their planned order, so the available numerical form is constant.	Constant
Z05	Passenger demand and connection pressure	Passenger flows and transfer requirements that may affect rescheduling priorities; the present objective does not include passenger allocation.	Unavailable
Z06	Rolling-stock and crew connections	Vehicle and crew circulation requirements between services; these couplings require additional data and model constraints.	Unavailable
Infrastructure utilization			
F08	Section track and direction configuration	Number and direction allocation of section tracks; every section has one running resource in each direction here.	Constant
F09	Minimum entry headway	Minimum separation between same-direction section entries; two minutes throughout the network.	Constant
F10	Minimum exit headway	Minimum separation between same-direction section exits; two minutes throughout the network.	Constant
F11	Station clearance time	Additional track-occupation time after departure or passage; one minute throughout the network.	Constant

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Table S11 continued

Code	Indicator	Definition	Availability
F19	Boundary waiting and acceptance conditions	Rules governing waiting outside the origin boundary and acceptance at destinations; origin waiting is allowed and destination events do not occupy internal station tracks here.	Constant
F20	Station-throat and route conflicts	Incompatible station-throat movements or route settings; these microscopic conflicts are outside the available model.	Unavailable
F21	Platform length and train compatibility	Whether platform lengths and train types permit a train to use a platform; the required compatibility data are unavailable.	Unavailable
F22	Cross-direction sharing conditions	Permitted sharing or switching between directional resources; borrowing is prohibited and no cross-direction resource coupling is modeled here.	Constant
F23	Alternative routes and their costs	Number and cost of rerouting options; train routes are prescribed and are not adjustable here.	Constant
Z01	Section length and speed conditions	Actual section distances, speed limits, and gradients. Only running-time information is available in the present data.	Unavailable
Z07	Boundary and storage capacity	Capacity for trains waiting outside network boundaries or in storage; origin-side waiting space is unrestricted in the present model.	Unavailable
Z08	Detailed station routes and signaling	Microscopic route-setting and signaling conditions; these require data beyond the current headway and track representation.	Unavailable
Delay propagation patterns			
B14	Propagation-source counts and types	Counts of distinct exogenous propagation sources, distinguishing physical disruption and emergency-policy sources and merging duplicated sources.	Constant
H14	Strongly connected component size	Number of nodes in the active strongly connected component containing an event; the active event graph is acyclic, so the available value is one.	Constant

Notes: Constant: every available numerical form is constant within its analysis context. Unavailable: the model or data do not provide a computable form. Context only: recorded as model or experiment context, without an estimate in the descriptive-association inventory. These categories never imply a correlation of zero. The companion complete catalog file preserves original names/definitions and links each estimable item to its numerical results.

S5 Prediction architectures and key parameter settings

Common inputs and training. STAT and ML use the same indicators selected within each training fold: up to 24 local indicators and eight scenario indicators per decision type. Within the training data, numerical forms are ranked by the absolute scenario-weighted Spearman association with a decision change. Constant, unavailable and duplicate numerical forms are excluded. Five additional inputs describe the reference value, lower and upper adjustment bounds, and whether a decrease or increase is permitted. Missing values are replaced by training medians. Inputs are standardized using training means and standard deviations, with a missing-value flag for each input. The resulting local and scenario vectors have 58 and 16 entries, respectively; unused positions are zero-padded.

Before accounting for label ambiguity, training gives equal total weight to each scenario–decision-type combination. Labels identified as ambiguous across available equally good timetables receive half weight. Model selection and probability calibration use only the training part of each outer round. Four inner partitions each use 45 training and 15 validation scenarios; the final model uses five inner partitions of 60 and 15 scenarios. The final transfer models use all 75 training and test scenarios and no transfer scenario.

STAT. Each decision type uses a logistic classifier with squared-coefficient regularization. The inverse regularization strength is selected from $\{0.01, 0.1, 1, 10\}$ by inner-validation logarithmic loss. Separate quantile regressions estimate the magnitude of duration shortening and extension at quantiles 0.1, 0.5 and 0.9. Their regularization coefficient is selected from $\{0.001, 0.01, 0.1\}$ by median quantile loss. Classification uses at most 800 optimization iterations and a convergence tolerance of 10^{-6}. The predicted magnitude is an auxiliary output and does not constrain the rescheduling model.

ML. For each decision type, the local and scenario input branches have 48 and 16 hidden units, respectively, with rectified linear activation. Their concatenated 64-dimensional representation passes through two shared residual blocks. Each block has layers of width 32 and 64, with rectified linear activation and dropout probability 0.1. Separate action heads predict three duration actions or two order actions. Duration heads also estimate three non-decreasing magnitude quantiles for each direction of change. Training combines action cross-entropy with quantile loss weighted by 0.5. Adam uses a learning rate of 0.001, weight decay of 0.001 and batches of 256 decisions. Inner training allows up to 200 epochs and stops after 20 epochs without validation improvement; the median selected epoch count across inner folds is used to refit on all outer-training scenarios.

Calibration and fixing. For each decision type, a logistic calibration of the unchanged probability is fitted to inner-validation predictions. Change directions excluded by operational bounds have zero probability. Both methods fix the highest-ranked 50% of eligible variables, with ties resolved by variable identifier. The two Strategy-3 reference settings use the same trained predictor; only their reference timetable inputs and resulting fixing sets differ.

Table S12: Selected regularization settings and final training lengths

Training set	Strategy-2			Strategy-3		
	STAT inverse regularization	STAT quantile regularization	ML epochs	STAT inverse regularization	STAT quantile regularization	ML epochs
Round 1	1	0.01	62	1	0.1	39
Round 2	1	0.01	50	1	0.1	43
Round 3	0.1	0.01	56	1	0.1	29
Round 4	1	0.01	72	0.1	0.1	58
Round 5	1	0.01	90	0.1	0.1	37
Final transfer model	1	0.01	55	1	0.1	44

Notes: Rounds 1–5 correspond to the five outer training and test partitions. The final models are trained using all 75 scenarios for frozen transfer prediction. STAT inverse regularization applies to action classification; STAT quantile regularization applies to duration magnitude estimates. ML epochs give the selected training length of the final fit for each partition.